# The Periodic Grammar of Technical Analysis - Load, Motion, Constraint, and Commitment Across Recursive Market Worlds

## Load, Motion, Constraint, and Commitment Across Recursive Market Worlds

### From Indicator Folklore to a Protocol-Bound Architecture of Marks, Windows, Structures, Events, Episodes, and Worlds

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# Abstract

Technical analysis is usually presented as a collection of indicators, chart patterns, levels, cycles, and forecasting rules. Moving averages, RSI, MACD, volume profile, candlesticks, support and resistance, Elliott Wave, Fibonacci retracement, and Gann geometry are commonly placed beside one another as if they were comparable tools addressing the same analytical problem.

They are not.

A moving average is primarily a filtered memory construction. MACD compares memory horizons. RSI measures a normalized directional relation under an implicit regime assumption. Volume profile maps accumulated transaction trace across price. Support and resistance convert historical trace into a candidate constraint. A breakout is not an indicator at all, but a boundary interaction seeking market commitment. Elliott Wave attempts to segment higher-order episodes. Gann analysis searches for price–time relations that must survive changes of anchor, scale, and observation protocol.

This article proposes a periodic grammar of technical analysis.

Under a declared observation protocol P, technical-analysis methods are classified according to four recurring market functions:

**Load / Memory**
**Motion / Relation**
**Constraint / Boundary**
**Commitment / Gate**

These functions recur across six levels of market closure:

**Mark**
**Window**
**Structure**
**Event**
**Episode**
**World**

The recurrence supplies the periodic law. A committed trace at one level, together with its unresolved residual, becomes part of the operative market structure observed at the next level:

$\text{Load}_n \rightarrow \text{Motion}_n$ under $\text{Constraint}_n \rightarrow \text{Commitment}_n \rightarrow \text{Ledger}_{n+1} + \text{Residual}_n \rightarrow \text{Load}_{n+1}$. (0.1)

Named indicators are therefore not the elements of technical analysis. They are compounds assembled from recurring observational and closure functions.

Three governance rails run through the entire architecture:

**Residual preservation**
**Cross-frame transport and invariance**
**Ledgered backreaction**

Residual preservation records what an interpretation failed to settle. Cross-frame transport asks whether the claimed structure survives admissible changes of timeframe, scale, anchor, bar construction, or market universe. Ledgered backreaction asks whether an accepted event changes future orders, risk systems, narratives, institutional treatment, or observation protocols.

The framework also distinguishes three advanced constructs that are often incorrectly merged:

$\chi$ = relational feedback signature. (0.2)

$\Xi$ = effective control state. (0.3)

$Z = R + iQ$ = locally justified conjugate state. (0.4)

The signature $\chi$ classifies relations as corrective, critical, or self-confirming. The control state $\Xi$ compresses loading, lock-in, and agitation under a declared protocol. The complex state Z is admitted only when R and Q are independently defensible, dynamically conjugate, phase-relevant, gate-relevant, and empirically superior to an unconstrained two-real-variable alternative.

The CAPM phase construction provides the calibration case. There, Q is derived from a declared valuation geometry and satisfies:

$\partial R/\partial\theta = -Q$. (0.5)

This makes Q the first-order phase exposure of admitted value, but not automatically a loss, realized P&L, gate event, or ledger entry. The full financial sequence remains:

Measurement → Exposure → State Movement → Economic P&L → Gate → Ledger + Residual. (0.6)

That distinction generalizes directly to technical analysis. Divergence is not yet reversal. Overbought is not yet exhaustion. A line crossing is not yet breakout. A local extreme is not yet a wave endpoint. Historical density is not yet future support. Phase exposure is not yet realized consequence.

The result is not a trading system and makes no promise of profitability. It is a protocol-first research architecture for explaining what technical-analysis instruments measure, why they fail, when apparently independent indicators are redundant, how observations become interventions, and which missing instrument families remain to be designed and tested.

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# 0. Reader's Guide

# 0.1 What this article is trying to do

This article asks a deceptively simple question:

> What kind of thing is a technical-analysis method?

The conventional answer is usually practical:

- a trend indicator;
- a momentum indicator;
- a volume indicator;
- a volatility indicator;
- a pattern;
- a support or resistance tool;
- a cycle method;
- a signal generator.

These labels are useful for software menus and trading textbooks, but theoretically they mix objects of very different kinds.

A moving average is a transformation of historical price.

A candlestick is a protocol-defined summary of a time window.

A support level is a historically constructed boundary.

A divergence is a relation between two projected structures.

A breakout is an attempted transition across a boundary.

A closing price is an institutionally privileged trace.

A wave count is an episode-level segmentation.

A regime is a higher-order environment in which lower-level tools change meaning.

The article therefore proposes that technical analysis should be reorganized not around inherited indicator names, but around recurring functional roles.

The central proposal is:

Technical analysis is a historically evolved instrument system for observing load, motion, constraint, and commitment across recursively generated market worlds. (0.7)

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# 0.2 What this article is not claiming

This article is not claiming that technical analysis reliably predicts future prices.

It is not claiming that every chart pattern contains genuine causal information.

It is not claiming that markets literally instantiate quantum mechanics, gauge theory, or a physical field theory.

It is not claiming that the four functional families are metaphysical substances.

It is not claiming that every two-variable market model should be represented by a complex number.

It is not claiming that the proposed periodic table is already empirically complete.

It is not claiming that the recurrence of market functions implies a fixed periodicity in prices.

The word periodic refers to the recurrence of functional roles across increasing levels of closure:

Mark → Window → Structure → Event → Episode → World. (0.8)

It does not mean:

prices repeat according to a universal cycle. (0.9)

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# 0.3 The article's strongest practical correction

The most important correction can be stated in one line:

> A measured relation is not yet a committed event.

This single distinction clarifies many common failures.

RSI may indicate strong directional imbalance, but it does not by itself establish reversal.

MACD divergence may indicate weakening memory expansion, but it does not by itself establish trend failure.

A wick above resistance records attempted displacement, but it does not by itself establish breakout.

A moving-average crossover records a relation between two memory horizons, but it does not by itself establish durable regime change.

A local high may be a candidate wave endpoint, but it does not become one merely because a later count requires it.

A Fibonacci reaction may occur near a ratio, but that does not prove a universal ratio law.

A Gann angle may fit one chart, but that does not establish cross-frame invariance.

The source Technical Analysis framework already distinguishes:

Event ≠ Trace ≠ Ledgered Trace. (0.10)

An event happens.

A trace is recorded.

A ledgered trace changes future admissibility.

That distinction will become one of the foundations of the proposed periodic grammar.

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# 0.4 Why the framework is protocol-first

Technical analysis does not observe "the market in itself."

It observes a declared market object under a declared setup.

A general protocol can be written:

$P = (B, \Delta, h, u)$. (0.11)

where:

$B$ = boundary;
$\Delta$ = observation or aggregation rule;
$h$ = time or state window;
$u$ = admissible intervention family.

For technical analysis, a more explicit form is useful:

$P_{TA} = (\text{Asset}, \text{Universe}, \text{Timeframe}, \text{Scale}, \text{BarRule}, \text{FeatureMap}, \text{GateRule}, \text{ResidualRule})$. (0.12)

This matters because changing the protocol changes the object.

A daily candle and a five-minute candle are not merely the same truth at different magnifications.

A linear-scale trend line and a logarithmic trend line are not necessarily the same boundary.

A cap-weighted breadth measure and an equal-weight breadth measure do not observe the same market field.

A close-based pivot rule and a wick-based pivot rule do not generate the same wave structure.

A support level defined by repeated closes is not identical to a level defined by transaction density.

The protocol-first discipline prevents the analyst from silently changing the market object whenever the original interpretation fails. The gauge-to-market framework makes this point directly: a claim becomes operational only after boundaries, probes, windows, and admissible interventions are fixed.

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# 0.5 Source architecture

The present article synthesizes four related lines of work.

## Technical analysis as self-referential diagnosis

The first line interprets technical analysis as an imperfect language for reading visible traces left by a market that observes itself:

$\text{TechnicalAnalysis}_P = \text{Projection}_P(\text{MarketSelfReference})$. (0.13)

It introduces signature $\chi$, phase, density, selection depth, gate, structural mass, residual, cadence, and cross-frame invariance as deeper characteristics behind familiar indicators.

## Complex completion and secondary time

The second line asks when two real channels earn a complex representation, when phase becomes a meaningful internal coordinate, and when phase-sensitive gates generate persistent trace and a time-bearing world.

### CAPM phase geometry

The third line provides a rigorous financial calibration case:

$$Z = R + iQ. \quad (0.14)$$

$$A^2 = R^2 + Q^2. \quad (0.15)$$

$$R = A \cos \theta. \quad (0.16)$$

$$Q = A \sin \theta. \quad (0.17)$$

$$\partial R/\partial \theta = -Q. \quad (0.18)$$

It then separates exposure, movement, recognition, ledger, and residual.

### Protocol-first market structure

The fourth line separates rich descriptive market state from compiled effective coordinates, and proposes:

$$\Xi = (\rho, \gamma, \tau). \quad (0.19)$$

Here $\rho$ denotes loading, $\gamma$ denotes lock-in, and $\tau$ denotes agitation or dephasing in the source formulation. To avoid collision with internal phase time later in this article, agitation will be written $\nu$:

$$\Xi_{fin} = (\rho, \gamma, \nu). \quad (0.20)$$

The fourth line also distinguishes propagation, transition, confinement, and slow historical basin geometry.

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# Part I — Why Technical Analysis Needs a Grammar

# 1. The Wrong Unit of Analysis

# 1.1 Why the indicator is not the fundamental object

Technical analysis is usually taught through named methods.

The learner encounters:

- simple moving average;
- exponential moving average;
- MACD;
- RSI;
- stochastic oscillator;
- ATR;
- Bollinger Bands;

- VWAP;
- OBV;
- volume profile;
- support and resistance;
- candlestick patterns;
- chart patterns;
- Fibonacci retracement;
- Elliott Wave;
- Gann geometry.

These names are historically understandable. Each method emerged from a particular problem, practitioner tradition, data environment, or market technology.

But named methods are poor theoretical atoms.

They often combine several operations at once.

They frequently share the same inputs.

They may depend upon hidden regime assumptions.

Some measure states.

Some measure relations.

Some construct boundaries.

Some test commitment.

Some describe low-level windows.

Others describe high-level episodes.

Placing them side by side as equivalent “indicators” is therefore comparable to placing temperature, chemical bonds, molecules, reactions, and ecosystems in one undifferentiated list.

The list may be familiar, but the ontology is confused.

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# 1.2 A moving average is not the same kind of thing as a breakout

Consider a moving average:

$$MA_n(t) = \Sigma_{j=0}^{n-1} w_j P_{t-j}. \quad (1.1)$$

with:

$$\Sigma_{j=0}^{n-1} w_j = 1. \quad (1.2)$$

The moving average is a filtered memory construction.

It answers:

> What price history remains visible after a declared weighting rule suppresses some fluctuations and preserves others?

Now consider a breakout.

A breakout is not a smoothing rule. It is not merely a transformed price series. It is an attempted transition across a previously meaningful boundary.

A more complete representation is:

BreakoutCandidate
= BoundaryCrossing

- DirectionalDisplacement
- CommitmentEvidence. (1.3)

A durable breakout may additionally require:

ValidBreakout
= BreakoutCandidate

- CloseGate
- FollowThrough
- AcceptanceOrRetest
  – ExcessResidual. (1.4)

The moving average and the breakout are therefore different logical objects.

The moving average constructs memory.

The breakout tests commitment.

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# 1.3 RSI is not the same kind of thing as support

RSI compares accumulated upward and downward movement over a declared window.

In simplified form:

RS = AverageGain / AverageLoss. (1.5)

RSI = 100 − 100/(1 + RS). (1.6)

RSI is fundamentally relational.

It locates recent directional movement inside a normalized balance between upward and downward displacement.

Support is different.

Support is a candidate boundary whose significance depends on accumulated trace:

SupportStrength
≈ HistoricalReaction

- TransactionDensity
- Positioning
- SharedAttention
- FutureConditionalOrders. (1.7)

RSI may warn that directional pressure is extended.

Support asks where the market may test a historically loaded boundary.

Neither establishes commitment alone.

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# 1.4 A candlestick is not a primitive market fact

A candlestick appears simple because it is familiar.

Yet a candle is already a compiled object.

It depends on:

- start time;
- end time;
- timezone;
- venue;
- price source;
- treatment of outliers;
- treatment of after-hours trades;
- adjustment rules;
- aggregation convention.

A candle may be represented as:

Candle_P = (O_P, H_P, L_P, C_P, V_P). (1.8)

It is therefore a protocol-defined window state.

Its body records displacement between opening and closing gates:

Body_P = C_P − O_P. (1.9)

Its upper and lower shadows preserve attempted but non-final displacement:

UpperWick_P = H_P − max(O_P,C_P). (1.10)

LowerWick_P = min(O_P,C_P) − L_P. (1.11)

The wick is not automatically “rejection.”

It is a residual trace whose interpretation depends on:

- volume;
- location;
- prior trend;
- liquidity;
- higher timeframe;
- following acceptance;
- market regime.

A candle is therefore already a small observational world:

window declaration
→ price projection
→ intrawindow extremes
→ closing gate
→ body and residual. (1.12)

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# 1.5 Elliott Wave is not an indicator in the ordinary sense

Elliott Wave attempts to organize multiple market events into an internally ordered episode.

It depends on:

- declared pivot rules;
- scale;
- timeframe;
- alternation criteria;
- branch selection;
- invalidation;
- higher-order nesting.

Its object is not a single price reading or local transformation.

Its object is an episode-level segmentation.

A provisional wave model may be written:

WaveModel_P
= PivotProjection_P

- SegmentRelation_P

- RegimeSequence_P
- EndpointGates_P
- AlternativeBranchResidual_P. (1.13)

The source Technical Analysis framework interprets impulse-like segments as self-confirming selection and corrective segments as residual digestion or corrective circulation. That makes Elliott Wave a high-order regime-sequencing hypothesis, not a basic indicator.

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# 1.6 Gann is primarily an invariance problem

Gann constructions relate price and time through selected anchors, scales, and geometric conventions.

Their core difficulty is not that they use geometry.

Their core difficulty is that the geometry may not survive reframing.

A candidate Gann claim depends on:

GannClaim_P
= Relation(Price, Time | Anchor, Scale, Calendar, VolatilityConvention). (1.14)

A mature test requires transport:

$T_{P \to P'}(\text{GannClaim}_P) \approx \text{GannClaim}_{P'}$. (1.15)

where P′ may alter:

- anchor;
- linear versus log scale;
- timeframe;
- calendar treatment;
- volatility normalization;
- bar construction.

If the relation disappears under reasonable reframing, it is not a robust market invariant.

It may still be a locally useful convention, but it should not be presented as a universal law.

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# 1.7 Indicators are compounds

The deeper conclusion is:

> Named Technical Analysis methods are compounds assembled from more basic market-observation functions.

Examples:

MovingAverage = MemoryFilter(PriceTrace). (1.16)

MACD = Difference(Memory_fast, Memory_slow). (1.17)

RSI = Normalize(AccumulatedUpMove, AccumulatedDownMove). (1.18)

VWAP = Accumulate(Price × Volume) / Accumulate(Volume). (1.19)

VolumeProfile(p) = Accumulate(Volume | PriceBin = p). (1.20)

Candlestick = WindowProjection + CloseGate + IntrawindowResidual. (1.21)

Breakout = BoundaryTest + CommitmentGate + ResidualAudit. (1.22)

ElliottWave = RecursiveEpisodeSegmentation + BranchGovernance. (1.23)

Gann = CandidatePriceTimeInvariant + TransportTest. (1.24)

This is why the indicator is the wrong fundamental unit.

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# 2. Markets Observe Themselves

# 2.1 Price is both outcome and evidence

In many systems, an observable is mainly an output.

A thermometer reading does not usually become a major causal input into the temperature of the room.

Financial price behaves differently.

Price is an outcome of previous orders, but it also becomes evidence used to generate new orders.

The recursive loop is:

Expectation
→ Order
→ Transaction
→ Price Trace
→ Interpretation
→ Revised Expectation
→ New Order. (2.1)

The source Technical Analysis article states the key principle directly:

Price is not merely an output of market behaviour; price becomes evidence inside the next round of market behaviour.

This transforms the chart from a passive historical image into an active component of market recursion.

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# 2.2 The chart is a trace of market self-reference

A chart records:

- executed transactions;
- institutional closing points;
- volume;
- gaps;
- volatility;
- previous reactions;
- visible boundaries;
- repeated patterns.

But traders, algorithms, risk managers, analysts, journalists, and platforms then interpret those traces.

Their interpretations modify:

- entries;
- exits;
- stop placement;
- hedging;
- leverage;
- liquidity provision;
- fund allocation;
- media emphasis;
- risk limits.

The chart therefore participates in the next round of chart formation.

This gives:

MarketChart_{k+1}
= MarketDynamics(MarketChart_k, ExternalInformation_k, Positioning_k, Interpretation_k). (2.2)

Technical analysis is not outside this loop.

It is one of the mechanisms through which Interpretation_k is produced.

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# 2.3 A signal may become true because it is observed

Suppose many market participants identify the same moving average as support.

Their orders cluster near the moving average.

The line then acquires real market consequence.

The sequence is:

SharedObservation
→ ConditionalOrders
→ LiquidityConcentration
→ MarketReaction
→ StrongerSharedBelief. (2.3)

A descriptive tool becomes a partial causal mechanism.

The same logic applies to:

- previous highs;
- option strikes;
- round numbers;
- VWAP;
- breakout levels;
- Fibonacci zones;
- benchmark prices;
- index rebalancing levels.

The source framework expresses this recursive possibility as:

A technical signal can become true because it is observed. (2.4)

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## 2.4 A signal may fail because it is over-observed

Reflexivity does not always reinforce the signal.

A widely watched level may attract:

- crowded positions;
- predictable stops;
- strategic fading;
- liquidity hunting;
- anticipatory front-running.

The sequence can become:

SharedObservation
→ Crowding
→ FragilePositioning
→ AdversarialTargeting
→ SignalFailure. (2.5)

This gives the opposite principle:

A technical signal can fail because it is over-observed. (2.6)

The same pattern can therefore become:

- self-fulfilling;
- self-cancelling;
- self-exhausting;
- adversarially exploited.

Technical analysis must consequently include observer backreaction.

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## 2.5 From Probe to intervention

A useful operational distinction is:

Probe = observes the market.
Pump = adds or removes loading.
Switch = changes regime or route.
Couple = strengthens or weakens binding. (2.7)

A technical method may begin as a Probe.

Once widely acted upon, it may become another operator.

A moving average may become a Couple by concentrating orders.

A breakout rule may become a Switch by activating systematic strategies.

A volume spike may reveal a Pump.

A benchmark event may combine Switch and Pump.

A regulatory classification may change Constraint and Commitment simultaneously.

Thus:

Probe + Backreaction → Intervention. (2.8)

The protocol-first market-structure framework helps make this transition explicit by distinguishing observation from admissible intervention.

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## 2.6 Why simple backtesting is not enough

A historical signal may appear effective because:

- market participants were not yet watching it;
- transaction costs were lower;
- market structure differed;

- the indicator itself changed behaviour after publication;
- the backtest silently changed anchors or parameters;
- failed signals were retrospectively relabelled;
- the relevant gate was not recorded;
- residual contradictions were discarded.

Therefore:

HistoricalAssociation ≠ StableOperator. (2.9)

A mature test should ask:

1. What protocol generated the signal?
2. What market function did it measure?
3. What gate converted it into an event?
4. What residual remained?
5. Did the signal survive a frame change?
6. Did widespread adoption alter the market relation?
7. Did the interpretation remain stable out of sample?

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# 3. Protocol Before Indicator

## 3.1 The declared market object

A technical-analysis claim should never begin with:

“The market is overbought.”

It should begin with a declaration.

For example:

- which asset?
- which venue?
- which timeframe?
- which price series?
- which adjustment rule?
- which window?
- which baseline?
- which feature?

- which gate?
- which invalidation condition?

The minimal protocol is:

$$P = (B, \Delta, h, u). \quad (3.1)$$

A Technical Analysis implementation may use:

$$P_{TA} = (A,U,T,S,W,\varphi,G,R). \quad (3.2)$$

where:

A = asset or instrument;
U = universe or comparison field;
T = timeframe;
S = scale and normalization;
W = bar or window rule;
φ = feature map;
G = gate and confirmation rule;
R = residual and invalidation rule.

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# 3.2 Boundary B

The boundary specifies what belongs to the observed market object.

Examples:

- one stock;
- one futures contract;
- one index;
- one sector;
- one options surface;
- one exchange;
- one trading session;
- one cross-asset system;
- one funding network.

A stock chart may look strong while its sector breadth weakens.

An index may rise while the median component falls.

A futures contract may appear liquid while collateral conditions tighten outside the chart boundary.

The claim changes when B changes.

Thus:

Claim_B1 ≠ Claim_B2 unless transport is demonstrated. (3.3)

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# 3.3 Observation and aggregation rule Δ

The observation rule decides how raw events become usable data.

Examples:

- last trade;
- midpoint;
- volume-weighted price;
- official close;
- adjusted close;
- time bars;
- tick bars;
- volume bars;
- range bars;
- equal weighting;
- capitalization weighting.

The same transaction stream can produce different chart objects under different Δ.

Therefore:

Chart = Aggregate(RawMarketTrace | Δ). (3.4)

The chart is constructed, not simply found.

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# 3.4 Horizon h

The horizon determines which events are treated as local and which are treated as structural.

Examples:

- one minute;
- one day;
- one quarter;
- one reporting cycle;
- one volatility regime;
- one credit episode;

- one institutional transition.

A five-minute breakdown may occur inside a weekly uptrend.

A daily divergence may occur inside a monthly accumulation structure.

A local pattern may be residual noise in a higher-order world.

Thus:

SignalMeaning = Function(Signal, h). (3.5)

---

## 3.5 Admissible intervention u

The protocol must also declare what actions are possible.

For a retail trader, admissible interventions may include:

- enter;
- exit;
- reduce;
- hedge;
- wait.

For a market maker:

- widen spread;
- reduce size;
- hedge inventory;
- reroute flow.

For a risk manager:

- reduce limits;
- demand collateral;
- change stress assumptions.

For an institution:

- reclassify;
- rebalance;
- recognize impairment;
- change benchmark;
- revise mandate.

The same observation may have different operational meaning under different u.

A technical signal is therefore not fully specified until its action context is declared.

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## 3.6 Feature map φ

The feature map declares what counts as structure.

Examples:

- closing price;
- high–low range;
- volume;
- order imbalance;
- realized volatility;
- breadth;
- open interest;
- transaction density;
- funding spread;
- option skew.

Without a declared feature map, two analysts may use the same word while measuring different objects.

“Momentum” may mean:

- return;
- rate of change;
- moving-average spread;
- MACD;
- RSI;
- price acceleration;
- breadth expansion.

“Volume confirmation” may mean:

- raw volume;
- relative volume;
- dollar volume;
- signed flow;

- open-interest increase;
- volume at price.

A mature claim must state φ.

---

## 3.7 Gate rule G

The gate declares what converts a candidate relation into an accepted event.

Examples:

- intraday touch;
- hourly close;
- daily close;
- weekly close;
- volume threshold;
- breadth threshold;
- retest hold;
- settlement;
- margin trigger;
- legal or accounting recognition.

A line crossing without a gate is only an observation.

A gate may be written:

G_P(X,L) ∈ {Admit, Defer, Reject}. (3.6)

Here:

X = candidate state;
L = existing ledger.

This three-way outcome is preferable to a simple binary rule because many market developments are neither fully accepted nor fully rejected.

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## 3.8 Residual rule R

The residual rule declares what unresolved evidence must remain visible after the gate decision.

Examples:

- weak breadth;
- low volume;

- conflicting higher timeframe;
- missing retest;
- abnormal liquidity;
- branch ambiguity;
- model disagreement;
- event risk;
- unrecognized economic pressure.

A mature output is:

Gate_P(X,L) → (Decision, Trace, Residual, Metadata). (3.7)

The CAPM phase article makes the same discipline explicit. A gate may admit part of an economic change while leaving unresolved damage, liquidity pressure, tail exposure, legal risk, or incomplete settlement:

ε_gate = ΔR_total − ΔR_ledger. (3.8)

Therefore:

Commitment ≠ Exhaustion. (3.9)

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## 3.9 The protocol record

Every technical claim should ideally carry a record such as:

TAClaim =
(Protocol, ObservedState, ClaimedRelation, Boundary, GateRule,
GateStatus, Residual, Invalidation, TransportStatus, Outcome). (3.10)

This makes retrospective relabelling harder.

A failed breakout should remain a failed breakout under the original protocol.

An invalidated wave count should not disappear merely because a new count was drawn.

An RSI reversal signal should not be redefined as “trend confirmation” only after price continued.

A support level should not be moved repeatedly until one reaction appears.

The purpose of declaration is not rigidity.

It is accountability.

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# 4. The Four Functional Families

The central grammar can now be introduced.

Technical-analysis methods repeatedly address four questions.

## Group I — Load / Memory

What consequential structure has accumulated?

## Group II — Motion / Relation

How is that structure changing, coupling, or orienting?

## Group III — Constraint / Boundary

What resists, confines, channels, or compresses movement?

## Group IV — Commitment / Gate

Which possible movement becomes accepted market history?

These four functions are not isolated.

They form a recurrent loop:

Load
→ Motion under Constraint
→ Commitment
→ Ledger + Residual
→ Updated Load. (4.1)

This loop will later generate the six periods of the proto-periodic table.

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# 4.1 Why Load comes first

No market event occurs in an empty system.

Every visible move enters a field already carrying:

- positions;
- memories;
- expectations;
- liquidity;
- institutional rules;
- previous closes;
- prior losses;
- transaction density;
- benchmark relations;

- accumulated residual.

Load does not mean only trading volume.

It means operative structure already present before the next move.

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## 4.2 Why Motion is relational

Motion is never meaningful in isolation.

A price of 100 means little without relation to:

- previous price;
- baseline;
- volatility;
- range;
- volume;
- benchmark;
- support;
- valuation;
- phase;
- market field.

Motion therefore includes:

- difference;
- ratio;
- slope;
- curvature;
- divergence;
- phase;
- feedback orientation.

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## 4.3 Why Constraint is distinct from Load

A heavily loaded market may still be easy to move.

Another market with similar load may be difficult to unwind because of:

- collateral requirements;

- concentrated positioning;
- limited liquidity;
- benchmark anchoring;
- legal restrictions;
- margin structure;
- path dependence.

This is why the protocol-first control coordinates distinguish loading ρ from lock-in γ.

Technical analysis often confuses the two.

High volume at a price level may create structural mass, but whether the level constrains future movement depends on who holds the positions, whether they remain active, and what orders are conditioned on the level.

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## 4.4 Why Commitment is distinct from Motion

A market can move without creating a durable event.

An intraday spike may reverse before the close.

A breakout may fail.

A volatility surge may leave the higher-order regime unchanged.

A price decline may remain unrealized in an accounting ledger.

A candidate wave endpoint may be invalidated.

Commitment requires a gate.

This is the exact distinction established in the CAPM calibration case:

Measurement identifies exposure.

Movement generates consequence.

Gate determines recognition.

Ledger records history.

The next sections will develop the four functional families individually.

# Part II — The Four Functional Families

# 4. Load and Memory: What the Market Carries

## 4.1 Load is more than volume

Every market movement begins inside a structure that already exists.

Before the next trade occurs, the market carries:

- outstanding positions;
- previous transaction prices;
- unrealized gains and losses;
- resting orders;
- institutional reference prices;
- prior closing levels;
- volatility expectations;
- benchmark weights;
- margin requirements;
- narratives;
- remembered successes and failures.

This pre-existing structure will be called **Load**.

Load_P(t) = OperativeStructure carried by the market under protocol P at time t. (4.1)

The word operative is essential.

A historical fact may still exist in a database without influencing present behaviour. It becomes operative load only when it remains capable of affecting:

- expectations;
- orders;
- risk limits;
- liquidity;
- market boundaries;
- gate decisions;
- future interpretation.

Thus:

StoredHistory ≠ OperativeMemory. (4.2)

OperativeMemory = HistoricalTrace that still modifies future admissibility. (4.3)

The original Technical Analysis framework classifies moving averages, VWAP, previous highs and lows, support and resistance, and volume profile as memory instruments because each attempts to show what the market continues to carry from its past.

---

# 4.2 Four forms of market memory

Market memory is not one-dimensional.

At least four forms should be distinguished.

### 4.2.1 Temporal memory

Temporal memory records what persists through elapsed time.

Examples include:

- moving averages;
- rolling highs and lows;
- prior closes;
- average true range;
- recent return history;
- time-decayed volume;
- volatility estimators.

A temporal filter may be written:

$$M_t^{time} = \Sigma_{j=0}^{n-1} w_j X_{t-j}. \quad (4.4)$$

The weights determine what kind of past survives into the present.

Equal weights produce a simple moving average.

Exponentially declining weights produce an exponential moving average.

A long memory retains more history but responds slowly.

A short memory responds quickly but carries less structural depth.

No horizon is intrinsically correct.

The horizon is part of the declared protocol.

---

### 4.2.2 Price-space memory

Price-space memory records where consequential activity accumulated.

Examples include:

- volume profile;
- market profile;
- high-volume nodes;
- low-volume zones;
- gap regions;

- repeated reaction levels;
- anchored VWAP;
- option-strike concentration.

A simple price-conditioned accumulation is:

$$M^{price}(p) = \Sigma_t V_t \cdot 1[p_t \in Bin(p)]. \quad (4.5)$$

This asks:

> How much transaction trace accumulated near price p?

But raw volume at price is not identical to current support or resistance.

It is evidence of historical occupation.

Whether that occupation remains operative depends on:

- who holds the positions;
- whether those positions remain open;
- whether the market regime has changed;
- whether new information has overwritten the old reference;
- whether future orders remain conditioned on the zone.

---

### 4.2.3 Cross-sectional memory

Cross-sectional memory records how widely a market move has been distributed.

Examples include:

- advance–decline statistics;
- percentage above moving averages;
- new highs versus new lows;
- sector participation;
- equal-weight versus capitalization-weighted performance;
- credit-spread participation;
- volatility-surface participation.

A breadth measure may be written:

$$B_t = (1/N) \Sigma_{i=1}^{N} a_{i,t}. \quad (4.6)$$

where $a_{i,t}$ represents whether component i participates in the claimed move.

An index can rise while $B_t$ falls.

This means the visible index trace and the underlying component field no longer carry the same structure.

Thus:

IndexMemory ≠ FieldMemory. (4.7)

Breadth becomes important because the market's apparent state may be supported by only a shrinking subset of components.

---

### 4.2.4 Institutional memory

Institutional memory exists where past decisions remain encoded in rules or repeated behaviour.

Examples include:

- official closing prices;
- benchmark inclusion;
- accounting values;
- margin references;
- previous issuance prices;
- central-bank policy levels;
- institutional VWAP;
- regulatory classifications;
- covenant thresholds.

Institutional memory may persist even when its original economic justification has weakened.

A benchmark level can remain important because systems continue to use it.

This gives:

InstitutionalMemory
= HistoricalTrace

- RepeatedOperationalUse
- Authority
- FutureConditionalAction. (4.8)

Such memory is deeper than visual chart repetition because it is embedded in procedures.

---

# 4.3 Moving averages as declared memory filters

A moving average should not first be interpreted as trend, support, or prediction.

Its primary role is:

> A moving average declares how much of the price past remains visible in the current analytical frame.

For a generic moving average:

$$MA_n(t) = \Sigma_{j=0}^{n-1} w_j P_{t-j}. \quad (4.9)$$

The chosen n and $w_j$ jointly determine the memory protocol.

The moving average suppresses some information and preserves other information.

It removes:

- high-frequency variation;
- local path detail;
- individual gaps;
- intrawindow extremes.

It preserves:

- a weighted price centre;
- broad drift;
- memory orientation;
- approximate persistence.

The output is not the “true trend.”

It is:

$$\text{TrendMemory under filter } F_n. \quad (4.10)$$

This explains why different moving averages can disagree without either being mathematically incorrect.

They retain different histories.

---

# 4.4 Price relative to memory

The relation between current price and a moving average is often more useful than the moving average alone:

$$d_n(t) = P_t - MA_n(t). \quad (4.11)$$

But $d_n$ has no fixed meaning.

A large positive $d_n$ may indicate:

- directional strength;
- temporary extension;

- volatility expansion;
- short covering;
- fundamental repricing;
- low-liquidity displacement.

Interpretation requires a relational regime.

Under corrective feedback:

d_n > 0 may create counter-pressure. (4.12)

Under self-confirming feedback:

d_n > 0 may generate further demand. (4.13)

The moving average supplies Load / Memory.

The distance d_n supplies Motion / Relation.

A later close, reversal, or retest may supply Commitment.

This shows why one indicator can contribute to several analytical stages without becoming all of them.

---

# 4.5 Volume is load, activity, and ambiguity

Volume is often described as confirmation.

That is too simple.

A first approximation is:

Volume ≈ TradeFrequency × AverageTradeSize. (4.14)

Dollar volume adds price:

DollarVolume ≈ TradeFrequency × AverageTradeSize × Price. (4.15)

But volume can arise from many structurally different conditions:

- new position formation;
- liquidation;
- hedging;
- market-making;
- forced selling;
- absorption;
- churn;
- transfer between strong and weak holders;

- mechanical index activity.

Therefore:

HighVolume ≠ StrongCommitment by definition. (4.16)

The Technical Analysis source explicitly treats volume as a mixture of frequency, mass, commitment, and exchange ambiguity.

A better representation is:

Volume_t
= NewLoading_t

- Unloading_t
- Transfer_t
- Hedging_t
- Churn_t
- ForcedFlow_t. (4.17)

The terms are generally not directly observable from ordinary bar volume.

This is why volume is informative but underdetermined.

---

# 4.6 Signed-volume instruments

Methods such as OBV, accumulation/distribution, and Chaikin-style flow attempt to give volume a direction.

A generic signed-volume ledger is:

SV_t = SV_{t−1} + s_t V_t. (4.18)

where s_t is a sign inferred from price behaviour.

For OBV:

s_t = +1 if C_t > C_{t−1};
s_t = −1 if C_t < C_{t−1};
s_t = 0 otherwise. (4.19)

This converts volume into a cumulative directional trace.

But the sign is inferred, not observed directly.

A rising OBV does not prove that informed investors are accumulating.

It shows that volume has been associated with the declared positive-price condition.

Thus:

SignedVolume = Volume interpreted through a price-sign protocol. (4.20)

It remains a compound of Load and Relation.

## 4.7 VWAP as a weighted memory centre

VWAP is:

$$VWAP_T = \Sigma_{t\in T} P_t V_t / \Sigma_{t\in T} V_t. \quad (4.21)$$

It is often treated as a fair-value line.

More precisely, it is a transaction-weighted historical centre under a declared interval T.

Its importance may come from two sources.

First, VWAP summarizes where volume-weighted exchange occurred.

Second, institutional execution systems may actively compare performance against it.

Therefore VWAP can evolve from:

Measurement of transaction memory

into:

Operational benchmark that changes future orders.

The transition is:

$$\text{Probe} \rightarrow \text{SharedReference} \rightarrow \text{ConditionalAction} \rightarrow \text{MarketCoupling}. \quad (4.22)$$

An anchored VWAP strengthens the declaration further because the analyst chooses a specific event as the beginning of relevant memory:

$$AVWAP_{t_0\rightarrow t} = \Sigma_{j=t_0}^{t} P_j V_j / \Sigma_{j=t_0}^{t} V_j. \quad (4.23)$$

The anchor may be:

- earnings;
- a major low;
- a breakout;
- a policy announcement;
- an institutional event.

Anchor choice must remain visible because different anchors generate different memory worlds.

---

## 4.8 Density as organized load

The Technical Analysis source introduces semantic density as a way to describe where market memory, participation, attention, positioning, and consequence concentrate.

A practical market version is:

$$\rho_{mkt}(p;P) \approx \text{Memory}(p)$$

× Participation(p)
× Attention(p)
× Positioning(p)
× Consequence(p). (4.24)

This is not intended as a directly universal multiplicative law.

It is a structural decomposition.

A price zone has greater operative density when:

- many transactions occurred there;
- many observers recognize it;
- many positions depend on it;
- important institutional events occurred there;
- later decisions remain conditioned on it.

A visually obvious line with little consequential trace may have low density.

A visually untidy zone with heavy positioning and institutional relevance may have high density.

This gives:

VisualNeatness ≠ StructuralDensity. (4.25)

---

# 4.9 Structural mass

Density becomes structural mass when accumulated trace creates resistance to change.

Let $M_\ell$ represent the effective inertia of market structure around level $\ell$.

A schematic expression is:

$M_\ell$
= f(TransactionDensity_ℓ, PositionConcentration_ℓ,
InstitutionalUse_ℓ, SharedAttention_ℓ, ResidualExposure_ℓ). (4.26)

A high-mass level may require greater pressure to break.

BreakCandidate_ℓ requires $\lambda_t > M_\ell$ under protocol P. (4.27)

Here $\lambda_t$ denotes effective directional pressure, not a directly universal observable.

The formula expresses a balance:

- pressure attempts movement;
- mass resists reclassification.

Structural mass is not permanent.

It may decay when:

- positions close;

- attention moves;
- new information arrives;
- volatility changes;
- the market repeatedly crosses the level;
- the institutional regime changes.

Therefore:

$M_{\ell} = M_{\ell}(t;P)$. (4.28)

---

## 4.10 The principal failure of memory instruments

Memory instruments fail when old trace is mistaken for current force.

A previous high may remain visible but cease to matter.

A long moving average may continue rising after the regime has changed.

A high-volume node may lose significance after a major declaration event.

An anchored VWAP may become irrelevant when the original anchor no longer governs positioning.

The generic failure is:

$$\text{MemoryLag} = \text{PersistenceOfOldTrace} - \text{CurrentRelevance}. \quad (4.29)$$

This is why memory instruments require Motion, Constraint, and Commitment cross-checks.

---

## 4.11 Load / Memory summary

Load answers:

> What is already present before the next event?

Its major forms are:

- temporal memory;
- price-space density;
- cross-sectional participation;
- institutional trace;
- positioning;
- liquidity;
- residual exposure.

Its representative methods include:

- moving averages;
- prior highs and lows;
- volume;
- OBV;
- VWAP;
- volume profile;
- breadth;
- open interest.

Its characteristic strength is persistence.

Its characteristic weakness is lag.

Load describes what the market carries.

It does not by itself tell us how the market will move.

---

# 5. Motion and Relation: How the Market Changes

## 5.1 Motion is not merely price direction

A price increase is a movement.

But Technical Analysis rarely cares only that price increased.

It asks:

- relative to what?
- at what speed?
- with what acceleration?
- compared with which memory?
- with what breadth?
- under what volatility?
- against what pressure?
- inside which regime?

Motion is therefore relational.

A general relation may be written:

$\Delta_{X,Y}(t) = \text{Relate}[X_t, Y_t \mid P]$. (5.1)

Examples include:

- price versus previous price;
- price versus moving average;
- fast memory versus slow memory;
- price versus volume;
- index versus breadth;
- admitted value R versus conjugate exposure Q;
- present phase versus prior phase.

---

## 5.2 First difference, velocity, and acceleration

The simplest movement measure is return:

r_t = P_t/P_{t−1} − 1. (5.2)

or log return:

g_t = ln(P_t/P_{t−1}). (5.3)

A coarse velocity measure is:

v_t = (P_t − P_{t−k})/k. (5.4)

A coarse acceleration measure is:

a_t = v_t − v_{t−1}. (5.5)

These quantities describe movement but not its cause, durability, or market meaning.

A positive acceleration may occur during:

- trend initiation;
- short covering;
- liquidation reversal;
- low-liquidity gap;
- news repricing;
- terminal euphoria.

Thus:

KinematicDescription ≠ RegimeDiagnosis. (5.6)

---

## 5.3 The relational feedback signature χ

The Technical Analysis source introduces χ to distinguish how realized structure feeds back into future market pressure.

Let:

δλ = change in effective signal pressure;
δs = change in realized market structure. (5.7)

A schematic two-way relation is:

δs = F δλ. (5.8)

δλ′ = χM δs. (5.9)

The sign of χ indicates the feedback orientation.

## Corrective regime

χ < 0. (5.10)

Movement generates opposing pressure.

Examples:

- value buyers respond to price decline;
- profit-taking follows extension;
- range traders fade extremes;
- liquidity providers stabilize displacement.

## Critical or ambiguous regime

χ ≈ 0. (5.11)

The return relation is weak or unstable.

Examples:

- compression;
- indecision;
- regime transition;
- low-conviction drift;
- unstable breakout attempts.

## Self-confirming regime

χ > 0. (5.12)

Movement generates supporting pressure.

Examples:

- trend following;
- momentum allocation;
- stop activation;
- margin cascades;
- narrative reinforcement;
- benchmark chasing.

The same indicator reading changes meaning when χ changes. The source paper identifies wrong-signature use as a principal reason indicators fail.

---

# 5.4 Why RSI changes meaning across regimes

RSI is often interpreted using fixed thresholds:

RSI > 70 → overbought.
RSI < 30 → oversold. (5.13)

But “overbought” is not a complete dynamical statement.

It describes directional imbalance under a declared window.

For RSI to imply reversal, one must add:

Extension generates sufficient corrective pressure. (5.14)

That is a $\chi < 0$ assumption.

Under $\chi > 0$:

High RSI may indicate persistent selection rather than exhaustion. (5.15)

The indicator itself does not determine χ.

Therefore:

RSIReading + CorrectiveRegimeEvidence → ReversalHypothesis. (5.16)

RSIReading without regime evidence → DirectionalCondition only. (5.17)

This is why an oscillator can remain “overbought” throughout a strong trend.

The indicator has not failed mathematically.

The interpretation has applied the wrong relational regime.

---

# 5.5 Moving-average crossover as memory-horizon conflict

Let:

M_f(t) = fast memory;
M_s(t) = slow memory. (5.18)

Define:

D_M(t) = M_f(t) − M_s(t). (5.19)

A crossover occurs when:

D_M(t) changes sign. (5.20)

This means the ordering of the two memory horizons has reversed.

It does not automatically mean the market regime has changed.

The crossover may be:

- late confirmation;
- temporary noise;
- range whipsaw;
- genuine trend selection;
- response to one large gap.

Thus:

MemoryOrderChange ≠ DurableRegimeChange. (5.21)

Commitment requires additional evidence.

---

# 5.6 MACD as memory displacement and curvature

MACD is commonly defined:

MACD_t = EMA_fast(t) − EMA_slow(t). (5.22)

Signal_t = EMA_m(MACD_t). (5.23)

Histogram_t = MACD_t − Signal_t. (5.24)

The components can be reinterpreted:

MACD = displacement between two memory horizons. (5.25)

Histogram = displacement relative to its own filtered memory. (5.26)

The histogram therefore behaves like a change in memory separation.

This is why it can be interpreted as an acceleration-like or curvature-like measure.

But MACD remains derived from price.

It does not independently measure:

- volume commitment;

- liquidity;
- breadth;
- positioning;
- structural mass;
- gate authority.

The source ontology accordingly classifies MACD and related divergences as phase or alignment indicators that may warn early but do not complete the gate.

---

## 5.7 Divergence as a relation, not an event

Suppose price makes a new high while an oscillator does not.

A generic divergence is:

$$Div_t = StructureChange_t - PressureProxyChange_t. \quad (5.27)$$

For example:

$$P_t > P_{t-k}, \text{ but } M_t \leq M_{t-k}. \quad (5.28)$$

This may indicate:

- weakening momentum;
- reduced participation;
- slower memory expansion;
- temporary consolidation;
- indicator saturation;
- changed volatility.

It does not logically imply reversal.

The proper sequence is:

Divergence
→ RelationalWeakening
→ CandidateInstability
→ GateTest
→ PossibleTrendFailure. (5.29)

A reversal requires a failure of the existing structure.

Therefore:

$$\text{Divergence} \neq \text{Reversal}. \quad (5.30)$$

$$\text{Divergence} + \text{StructuralBreak} + \text{GateAcceptance} \rightarrow \text{ReversalEvidence}. \quad (5.31)$$

This is one of the most important category corrections in the article.

## 5.8 Breadth as field coherence

Breadth asks whether a visible market move is supported across the broader component field.

Let $a_{i,t} \in \{-1,0,+1\}$ describe component direction.

Then:

$$\text{Breadth}_t = (1/N)\,\Sigma_i\, a_{i,t}. \quad (5.32)$$

A coherence measure may compare index direction with field participation:

$$C_{\text{field}}(t) = \text{Alignment}(\text{IndexMove}_t, \text{Breadth}_t). \quad (5.33)$$

When both move together, the market field is more coherent.

When price rises while breadth weakens, the index and field diverge.

But breadth also depends on protocol:

- component universe;
- weighting;
- inclusion rules;
- timeframe;
- treatment of unchanged securities.

Thus:

$$\text{Breadth}_P \neq \text{Breadth}_{P'} \text{ by default.} \quad (5.34)$$

Breadth becomes powerful when it supplies an observation channel sufficiently independent from the headline index.

---

## 5.9 Relative strength

Relative strength compares two market traces.

For asset A and benchmark B:

$$RS_{A/B}(t) = P_A(t)/P_B(t). \quad (5.35)$$

Its change is:

$$\Delta RS_{A/B}(t) = RS_{A/B}(t) - RS_{A/B}(t-1). \quad (5.36)$$

A rising asset price with falling relative strength means the asset rises more slowly than the benchmark.

A falling asset price with rising relative strength means the asset is declining less severely than the benchmark.

Relative strength is therefore not absolute momentum.

It is frame-relative motion.

The benchmark is part of the declaration.

Change the benchmark and the relation may change.

---

## 5.10 Phase as a stronger relational structure

Not every relation deserves a phase.

For a genuine complex state:

$Z = R + iQ = A \exp(i\theta)$. (5.37)

the components must support more than coexistence.

The phase article requires:

- independently meaningful R and Q;
- stable conjugacy;
- useful phase ordering;
- gate relevance;
- measurable advantage over a flexible real pair.

The CAPM construction satisfies a particularly strong relation:

$\partial R/\partial\theta = -Q$. (5.38)

The conjugate coordinate Q is therefore the phase exposure coefficient of R.

But ordinary TA proxies do not automatically satisfy this identity.

Volume is not automatically Q.

Momentum is not automatically Q.

Volatility is not automatically Q.

Sentiment is not automatically Q.

A candidate TA conjugate coordinate must earn that role empirically.

---

## 5.11 Motion / Relation summary

Motion answers:

> How does the present state stand relative to another state, memory, field, or phase?

Representative instruments include:

- returns;

- momentum;
- RSI;
- stochastic;
- moving-average crossover;
- MACD;
- relative strength;
- divergence;
- breadth;
- phase estimators.

Its characteristic strength is change detection.

Its characteristic weakness is premature interpretation.

Motion can reveal weakening, acceleration, or misalignment.

It cannot alone determine whether the market has committed to a new history.

---

# 6. Constraint and Boundary: What Resists or Channels Motion

## 6.1 A boundary is a declared transition surface

A market boundary is a region at which the interpretation of the state may change.

Examples include:

- support;
- resistance;
- previous high or low;
- value-area boundary;
- volatility band;
- trend channel;
- gap boundary;
- option strike;
- margin threshold;
- covenant level.

A boundary may be written:

$\partial\Omega_P$ = declared transition surface under protocol P. (6.1)

The boundary separates candidate states:

$X \in \Omega_A$ versus $X \in \Omega_B$. (6.2)

For example:

Price below resistance → old regime remains admissible.
Price accepted above resistance → new regime becomes admissible. (6.3)

But crossing $\partial\Omega_P$ is not automatically sufficient for commitment.

---

# 6.2 Hard and soft constraints

A hard constraint is enforced by an explicit rule.

Examples:

- exchange price limit;
- margin threshold;
- covenant;
- regulatory capital limit;
- option exercise condition;
- trading halt.

A soft constraint arises from distributed behaviour.

Examples:

- support;
- resistance;
- moving average;
- VWAP;
- prior high;
- volume node;
- round number.

Hard constraints possess formal authority.

Soft constraints possess probabilistic influence.

A soft boundary may become stronger when many systems condition orders on it.

Thus:

BoundaryStrength

= FormalAuthority

- StructuralMass
- SharedAttention
- ConditionalOrderDensity. (6.4)

---

# 6.3 Support and resistance as memory converted into constraint

Support and resistance begin as memory.

Repeated reaction writes trace.

Participants then remember the trace.

Orders become conditioned on the level.

Memory thereby becomes constraint.

The sequence is:

HistoricalReaction
→ SharedRecognition
→ ConditionalOrders
→ FutureResistanceOrSupport. (6.5)

This is a transition from Group I to Group III:

Load / Memory → Constraint / Boundary. (6.6)

The level is not powerful because a line exists on the chart.

The line exists because the analyst attempts to represent a deeper field of accumulated consequence.

---

# 6.4 Zones rather than exact lines

Market boundaries are often treated as exact prices.

But the underlying mechanisms are distributed across:

- different entry prices;
- different venues;
- spread;
- slippage;
- volatility;
- heterogeneous memory;

- varying time horizons.

A more realistic representation is a zone:

$\Omega_\ell = [\ell - \delta_-, \ell + \delta_+]$. (6.7)

The widths $\delta_-$ and $\delta_+$ may depend on:

- ATR;
- local liquidity;
- volume profile;
- timeframe;
- historical dispersion.

An exact one-pixel line may create false precision.

---

# 6.5 Volatility bands as adaptive boundaries

Bollinger Bands use a centre and dispersion:

$Middle_t = MA_n(t)$. (6.8)

$Upper_t = MA_n(t) + k\sigma_n(t)$. (6.9)

$Lower_t = MA_n(t) - k\sigma_n(t)$. (6.10)

They construct a volatility-conditioned boundary.

A band touch has no universal meaning.

Under $\chi < 0$:

BandTouch may suggest corrective extension. (6.11)

Under $\chi > 0$:

BandRide may indicate persistent directional selection. (6.12)

Thus the band belongs to Constraint.

The interpretation belongs to Motion / Relation.

A reversal or continuation requires Commitment.

---

# 6.6 Channels and trend boundaries

A trend channel declares an admissible region around an estimated directional path.

A simple form is:

$P_t \approx a + bt + \varepsilon_t$. (6.13)

with boundaries:

Upper_t = a + bt + d_+. (6.14)

Lower_t = a + bt – d_–. (6.15)

The channel assumes that:

- the chosen trend representation is meaningful;
- deviations remain bounded;
- the anchor and scale are legitimate;
- the market regime remains sufficiently stable.

A channel break may indicate:

- acceleration;
- deceleration;
- volatility expansion;
- regime transition;
- estimation failure.

The boundary does not determine which interpretation is correct.

---

# 6.7 Compression and possibility narrowing

Triangles, wedges, bases, and squeezes are commonly described visually.

A deeper interpretation is that the market's admissible movement region narrows.

Let Ω_t represent the currently admissible price or state region.

Compression means:

Measure(Ω_t) decreases through time. (6.16)

But narrowing price range does not necessarily mean possibilities are economically narrowing.

A market can remain quiet while uncertainty grows.

The Technical Analysis source introduces selection depth σ as possibility-suppression depth and distinguishes it from clock time.

Thus:

Δσ ≠ Δt. (6.17)

A long quiet period may produce little selection.

One announcement may eliminate many futures immediately.

A compression pattern becomes structurally meaningful when:

- the region narrows;
- positioning accumulates;
- alternatives become increasingly incompatible;
- a consequential gate approaches.

---

# 6.8 Low-volume and high-volume regions

Volume profile creates two different forms of boundary environment.

### High-volume node

A high-volume node may contain:

- substantial memory;
- many entry prices;
- accepted value;
- structural mass.

Movement through it may be slower because many interests interact.

### Low-volume region

A low-volume region may contain:

- little accepted transaction history;
- weak memory;
- lower structural mass.

Movement may accelerate because less prior structure resists it.

Schematically:

$$\text{VelocityPotential}(p) \propto 1/[M(p) + \varepsilon]. \quad (6.18)$$

This is only a heuristic relation, not a universal market law.

It expresses the idea that lower structural mass may permit faster displacement.

---

# 6.9 Constraint is protocol-relative

A level can be strong for one observer and weak for another.

Examples:

- an intraday VWAP matters to an execution desk;
- a weekly close matters to a long-horizon investor;

- an option strike matters near expiry;
- a covenant matters to a creditor;
- an accounting threshold matters to an institution.

Therefore:

ConstraintStrength = ConstraintStrength(P, Observer, Horizon). (6.19)

There is no boundary without a declared observer context.

This is consistent with the protocol-first finance framework, which emphasizes that loading and lock-in coordinates vary with desk, entity, legal boundary, and horizon.

---

## 6.10 Boundary failure

A boundary fails when the old distinction no longer organizes market behaviour.

Possible failure modes include:

- clean displacement;
- repeated crossing;
- acceptance beyond the zone;
- structural reclassification;
- institutional declaration;
- loss of relevant positioning.

But a single print beyond a line may be insufficient.

The proper question is:

> Did the market merely cross the boundary, or did it recompile the boundary into a new ledger state?

That question leads to Commitment.

---

## 6.11 Constraint / Boundary summary

Constraint answers:

> What limits, channels, resists, or compresses movement?

Representative tools include:

- support and resistance;
- bands;
- channels;

- profile nodes;
- value areas;
- pattern boundaries;
- Fibonacci zones;
- Gann constructions;
- formal thresholds.

Its characteristic strength is localization.

Its characteristic weakness is confusing a drawn boundary with an enforced constraint.

A boundary identifies where a consequential test may occur.

It does not decide whether the test has passed.

---

# 7. Commitment and Gate: How Possibility Becomes History

## 7.1 Movement is not commitment

Markets frequently cross candidate boundaries without creating durable change.

Examples include:

- intraday breakout followed by close below resistance;
- gap that is immediately filled;
- volatility spike without regime change;
- new high with collapsing breadth;
- moving-average crossover that reverses;
- support breach without follow-through.

Movement records that the state changed.

Commitment records that the changed state became consequential.

Thus:

Movement ≠ Commitment. (7.1)

Commitment requires a gate.

---

## 7.2 The gate operator

A gate evaluates a candidate state against declared rules and the existing ledger.

A general form is:

$$G_P(X_t,L_t) \rightarrow (d_t,e_t,r_t,m_t). \quad (7.2)$$

where:

$d_t$ = decision;
$e_t$ = admitted event;
$r_t$ = attached residual;
$m_t$ = gate metadata.

The decision may be:

$$d_t \in \{\text{Admit, Defer, Reject}\}. \quad (7.3)$$

Admit means the candidate enters the ledger.

Defer means evidence remains insufficient.

Reject means the candidate fails under the declared rule.

This three-way structure is important because markets often remain unresolved.

---

# 7.3 The closing price as a micro-gate

The closing price is often privileged because many systems use it to:

- calculate returns;
- mark portfolios;
- determine margin;
- update indicators;
- classify breaks;
- rebalance positions;
- produce official records.

The close is therefore not merely the last observation of a window.

It is an institutional micro-gate.

A candle can be represented:

IntrawindowPath
→ High/Low Attempts
→ ClosingGate
→ AcceptedBody + WickResidual. (7.4)

This does not mean the close is metaphysically superior to all intraday prices.

It means the market has assigned it special ledger authority.

---

# 7.4 Breakout as a compound gate event

A naive breakout rule is:

$P_t > Resistance.$ (7.5)

A more mature breakout model is:

$BreakoutCandidate$
$= BoundaryCross$

- $DirectionalDisplacement.$ (7.6)

Commitment evidence may include:

$C_t > Boundary.$ (7.7)

$RelativeVolume_t > Threshold.$ (7.8)

$Breadth_t$ supports direction. (7.9)

$FollowThrough_{t+1:t+k} > 0.$ (7.10)

Retest holds. (7.11)

Residual remains below tolerance. (7.12)

Therefore:

$ValidBreakout_P$
$= MeaningfulBoundary$

- Displacement
- Participation
- GateAcceptance
- FollowThrough
- $ResidualControl.$ (7.13)

Not every application requires every term.

But the terms should be declared rather than added retrospectively.

---

# 7.5 Gate strength

A binary gate loses information.

A gate-strength measure may be defined schematically:

$S_G$
$= w_1 D$

- $w_2 V$

- $w_3B$
- $w_4C$
- $w_5T$
  $- w_6R$. (7.14)

where:

D = normalized displacement;
V = participation or volume evidence;
B = breadth coherence;
C = close quality;
T = retest or follow-through evidence;
R = residual contradiction.

The weights w_i must be protocol-specific and empirically tested.

The purpose is not to declare a universal breakout equation.

It is to separate:

- evidence supporting admission;
- evidence remaining unresolved.

---

# 7.6 Acceptance versus mere persistence

Price remaining above a level for several bars does not necessarily prove acceptance.

Persistence may result from:

- illiquidity;
- lack of opposing flow;
- delayed information;
- temporary positioning;
- market closure;
- low participation.

A stronger acceptance concept may require:

Acceptance
= Persistence

- TransactionFormation
- ReducedRejection
- CrossFrameSupport. (7.15)

This is why retests are often valued.

A retest asks whether the former boundary has become part of the new load structure.

OldResistance → NewSupport is a ledger hypothesis. (7.16)

It is not automatic.

---

# 7.7 Rejection

Rejection means the candidate state failed to become accepted.

A schematic rejection event is:

BoundaryCross

- ReturnInside
- OpposingCommitment
  → RejectionTrace. (7.17)

A wick alone is insufficient.

A wick becomes stronger rejection evidence when followed by:

- close back inside;
- opposing volume;
- failure of follow-through;
- later movement away from the boundary.

Thus:

Wick = ResidualAttempt. (7.18)

ConfirmedRejection = Wick + GateFailure + OpposingTrace. (7.19)

---

# 7.8 Fakeout and trapped-position residual

A fakeout occurs when a boundary crossing induces commitment by some participants but fails to achieve durable ledger acceptance.

A compact representation is:

Fakeout
= CandidateBreak

- ParticipantCommitment
  – DurableAcceptance. (7.20)

The failed event may leave residual:

- trapped longs;
- trapped shorts;

- stop cascades;
- forced exits;
- distrust of the boundary;
- future reversal fuel.

Thus:

ResidualDebt_{t+1}
= ResidualDebt_t

- NewlyTrappedExposure_t
  – MetabolizedResidual_t. (7.21)

The exact quantity is not directly available from ordinary chart data, but the concept is important.

A failed move does not simply disappear.

It changes the position ledger.

---

# 7.9 Commitment does not eliminate residual

Suppose a daily close confirms a breakout.

The gate admits the event.

Yet residual may remain:

- weekly resistance overhead;
- weak breadth;
- low volume;
- event risk;
- poor liquidity;
- untested retest;
- excessive leverage.

Therefore:

AdmittedEvent ≠ CompleteResolution. (7.22)

The CAPM framework makes this distinction mathematically explicit through the sequence:

Measurement → Exposure → State Movement → Economic P&L → Gate → Ledger + Residual.

It also states that Q is exposure, not realized loss, and that actual phase movement must occur before economic consequence arises.

The same discipline applies to chart signals.

---

# 7.10 Event, trace, and ledgered trace

Three stages should be distinguished.

### Event

Something occurred.

Example:

Price traded above resistance.

### Trace

The event was recorded.

Example:

The chart contains a breakout bar.

### Ledgered trace

The recorded event changes future behaviour.

Example:

Participants reposition, the former resistance becomes a reference, risk systems update, and future orders defend or attack the new level.

Thus:

Event → Record → ConsequentialLedger. (7.23)

The source Technical Analysis paper treats this distinction as central to gate indicators and recursive objectivity.

---

# 7.11 Gate authority

Not all gates carry equal authority.

Potential gate authorities include:

- exchange rules;
- settlement systems;
- accounting standards;
- institutional mandates;
- risk committees;
- market conventions;
- collective trader behaviour.

A daily close has convention-based authority.

A margin call has contractual authority.

A default declaration has legal authority.

A breakout retest has distributed behavioural authority.

Gate metadata should therefore include:

m_t = (Authority, Threshold, Horizon, Evidence, Confidence). (7.24)

This makes two apparently similar events distinguishable.

---

## 7.12 Commitment and self-reference

Once a gate is widely recognized, the admitted event may change the market.

A breakout close may trigger:

- systematic trend entry;
- stop orders;
- analyst upgrades;
- risk-limit changes;
- option hedging;
- media attention.

The event then feeds back into Load and Motion.

The recursive loop is:

GateAdmission
→ LedgerUpdate
→ NewOrders
→ NewLoad
→ NewMotion
→ NextGate. (7.25)

This is the basic world-forming cycle of the periodic grammar.

---

## 7.13 Commitment / Gate summary

Commitment answers:

> Which possible movement became accepted history?

Representative gates include:

- execution;

- close;
- gap hold;
- breakout confirmation;
- VWAP reclaim;
- retest;
- settlement;
- institutional recognition;
- regime declaration.

Its characteristic strength is event formation.

Its characteristic weakness is delay and false closure.

The gate converts possibility into trace.

Residual records what the gate did not settle.

The updated ledger becomes part of the next period's Load.

---

# 8. The Four-Family Cycle

The four functional families can now be expressed as one recurrence:

$\text{Load}_n$
$\rightarrow \text{Motion}_n$
$\rightarrow \text{Constraint}_n$
$\rightarrow \text{Commitment}_n$
$\rightarrow \text{Ledger}_{n+1} + \text{Residual}_n$
$\rightarrow \text{Load}_{n+1}$. (8.1)

This should not be read as a rigid chronological sequence in every case.

Load, Motion, and Constraint interact continuously.

Commitment marks the point at which part of that interaction becomes historically consequential.

The deeper meaning is:

- Load supplies what the market already carries.
- Motion reveals how that structure changes.
- Constraint defines where change becomes consequential.
- Commitment determines what enters the ledger.
- Residual preserves what remains unresolved.
- The ledger and residual together shape the next market world.

The next part will develop the three governance rails that prevent this cycle from becoming another retrospective chart narrative:

**Residual**
**Transport and Invariance**
**Ledger and Backreaction**.

# Part III — The Three Governance Rails

The four-family cycle describes how market structure develops:

Load
→ Motion
→ Constraint
→ Commitment
→ Ledger + Residual
→ Updated Load. (8.2)

But this cycle can still become an elegant retrospective story unless it is governed.

Three rails must therefore run across every period and every functional family:

**Residual preservation**
**Cross-frame transport and invariance**
**Ledgered backreaction**

They answer three different questions.

| Governance rail | Question |
|---|---|
| **Residual** | What did the current interpretation fail to settle? |
| **Transport / Invariance** | Does the claimed structure survive an admissible change of frame? |
| **Ledger / Backreaction** | Did the admitted event alter the system that future analysis observes? |

These rails are not additional indicators.

They regulate how indicators, boundaries, gates, and episode models are allowed to claim validity.

---

# 9. Residual: What Closure Did Not Resolve

# 9.1 Why residual must remain visible

Every analytical model closes only part of the market field.

A moving average suppresses local path detail.

An RSI reading compresses directional history into a bounded relation.

A volume profile aggregates heterogeneous transactions into price bins.

A breakout rule converts a complicated boundary interaction into a gate decision.

A wave count selects one segmentation from several possible branches.

A valuation model projects a claim under a declared discount protocol.

In every case:

VisibleStructure_P + Residual_P = BoundedObservation_P. (9.1)

Residual is the part of the observed field that the current declaration, feature map, model, or gate has not successfully absorbed.

This follows the bounded-observer discipline developed in the wider source architecture:

ObservedReality_T = ExtractableStructure_T + Residual_T. (9.2)

The same discipline prevents technical analysis from treating its chosen representation as the whole market.

---

# 9.2 Residual is not merely error

Residual may contain error, but the two are not identical.

Residual can include:

- unresolved evidence;
- omitted variables;
- branch ambiguity;
- latent positioning;
- conflicting timeframes;
- unrecognized economic consequence;
- measurement uncertainty;
- structural mismatch;
- data quality problems;
- genuine novelty.

A useful decomposition is:

Residual
= MeasurementError

- ModelOmission
- ProtocolExclusion
- UnresolvedAlternative
- UnrecognizedConsequence
- NovelStructure. (9.3)

Not every term can be identified separately.

The purpose of the decomposition is to prevent the analyst from using “noise” as a universal disposal category.

---

# 9.3 Residual is not Q

The CAPM phase construction gives Q a precise role:

$Q = \sqrt{(A^2 - R^2)}$. (9.4)

and:

$\partial R/\partial \theta = -Q$. (9.5)

Q is therefore a declared conjugate coordinate and the first-order phase exposure of R.

Residual is different.

Residual is what remains outside the declared complex closure or outside the gate’s recognized consequence.

A general distinction is:

$Q_P$ = declared conjugate structure inside model P. (9.6)

$\varepsilon_P$ = observed consequence not explained or recognized by model P. (9.7)

Therefore:

$Q_P \neq \varepsilon_P$. (9.8)

Treating every unexplained market effect as Q would destroy the meaning of conjugacy.

Treating every failed Q interpretation as “residual pressure” without preserving the original failure would destroy falsifiability.

The phase framework expressly rejects the use of Q as a universal error bucket and requires reduction to a simpler representation when complexification adds no measurable gain.

---

# 9.4 Gate admission and residual can coexist

A common mistake is to assume that residual exists only when a signal fails.

But a signal can pass its gate while leaving important contradictions unresolved.

Suppose a stock closes above resistance with high volume.

The breakout may be admitted under the declared daily protocol.

Residual may still include:

- declining breadth;
- weekly resistance;

- event risk;
- poor liquidity;
- excessive leverage;
- absent retest;
- option-related distortion.

Thus:

GateAdmission = 1 does not imply Residual = 0. (9.9)

A more complete gate output is:

$G_P(X_t,L_t) \rightarrow (e_t,r_t,m_t)$. (9.10)

where:

$e_t$ = admitted event;
$r_t$ = attached residual;
$m_t$ = authority and gate metadata.

The CAPM recognition framework expresses the same principle:

$\varepsilon_{gate} = \Delta R_{total} - \Delta R_{ledger}$. (9.11)

A recognized event may leave part of the economic movement outside the ledger.

Hence:

Commitment ≠ Exhaustion. (9.12)

---

# 9.5 Residual pressure

Residual becomes dynamically important when unresolved structure influences later movement.

Examples include:

- trapped buyers after a failed breakout;
- unrecognized losses;
- unresolved margin pressure;
- repeated bearish divergence without price failure;
- unfilled gaps;
- abandoned but widely remembered wave counts;
- legal or accounting exposure not yet recognized;
- weak breadth beneath a rising index.

Let $\tilde{R}_t$ denote accumulated unresolved pressure.

A schematic update is:

$\tilde{R}_{t+1} = \tilde{R}_t + r_t - \mu_t$. (9.13)

where:

$r_t$ = newly attached residual;
$\mu_t$ = residual metabolized, resolved, invalidated, or rendered irrelevant.

Residual accumulation does not guarantee later reversal.

It indicates that the current closure carries unresolved tension.

The effect of that tension depends on:

- regime $\chi$;
- structural mass;
- available liquidity;
- future gates;
- institutional response;
- observer attention.

---

# 9.6 Residual debt

When a system repeatedly admits events without adequately carrying their contradictions, it accumulates residual debt.

$\text{ResidualDebt}_n = \Sigma_{k=1}^{n} w_k r_k$. (9.14)

The weights $w_k$ may decay with time or remain high when the residual has continuing institutional consequence.

Examples include:

- repeated failed breakouts near the same level;
- accounting recognition delayed across several periods;
- persistent breadth deterioration;
- successive wave relabeling;
- continually revised support lines;
- risk models repeatedly excluding tail outcomes.

Residual debt is not merely a prediction of collapse.

It is an audit warning:

> The current ledger appears cleaner than the unresolved history from which it was produced.

---

# 9.7 Residual and divergence

Divergence is one important form of residual evidence.

Suppose price continues upward while breadth or momentum weakens.

The primary market structure remains admitted.

The relational contradiction remains residual.

This can be written:

AdmittedTrend_t = 1. (9.15)

RelationalSupport_t ↓. (9.16)

Residual_t = AdmittedTrend_t − SupportingField_t. (9.17)

The correct interpretation is not:

Residual exists, therefore reversal is certain.

It is:

The current trend closure is increasingly dependent on a narrowing support structure. (9.18)

A reversal becomes ledgered only after an appropriate gate fails.

---

# 9.8 Residual and fakeout

A fakeout is especially important because the event may create new residual.

Suppose price crosses a boundary and induces participant commitment.

Then the market returns inside the old region.

The resulting structure is:

Fakeout
= AttemptedTransition

- ParticipantCommitment
  − DurableAcceptance. (9.19)

The residual may include:

- trapped entrants;
- forced exits;
- damaged trust in the level;
- repositioned liquidity;
- increased volatility;
- new narrative tension.

The failed event therefore changes the next market state.

Failure is not absence of history.

Failure can itself be a ledger-writing event.

---

# 9.9 Residual and retrospective relabeling

Some technical methods allow flexible interpretation.

Flexibility is not automatically illegitimate.

Markets are nonlinear, noisy, and multiscale.

But flexibility becomes pathological when a failed interpretation is erased rather than revised.

Examples include:

- moving a support level after every breach;
- changing the wave count without preserving the old count;
- redefining an RSI reversal signal as trend confirmation after continuation;
- choosing a new Fibonacci anchor after the original one fails;
- changing the timeframe until a pattern appears successful.

This produces:

InterpretiveFlexibility – TracePreservation = UnfalsifiableRelabeling. (9.20)

A mature revision rule must preserve:

1. the original declaration;
2. the original gate;
3. the failure condition;
4. the revised interpretation;
5. the reason for revision.

The broader declaration framework defines mature self-revision as trace-preserving, residual-honest, frame-robust, budget-bounded, and non-degenerate. A system that changes its rules whenever contradiction appears is not a mature observer; it is an unstable or dogmatic self-modifier.

---

# 9.10 The residual record

A practical residual record may contain:

ResidualRecord_k =
(Claim_k, Protocol_k, Evidence_k, GateStatus_k,

Contradiction_k, ResidualType_k, Invalidation_k,
Revision_k, LaterOutcome_k). (9.21)

## Claim

What was asserted?

## Protocol

Under which timeframe, boundary, anchor, feature map, and gate?

## Evidence

What supported the interpretation?

## Gate status

Was the claim admitted, deferred, or rejected?

## Contradiction

What evidence remained inconsistent?

## Residual type

Was it:

- measurement;
- model;
- frame;
- gate;
- liquidity;
- positioning;
- branch;
- institutional?

## Invalidation

What future event would weaken or reject the claim?

## Revision

Was the interpretation changed? Why?

## Later outcome

What happened after the declared horizon?

---

## 9.11 Residual honesty

Residual honesty requires that unresolved evidence remain attached to the claim.

A strong technical report should be able to say:

> The breakout passed the daily close and relative-volume gate, but breadth remained weak, the weekly boundary was unresolved, and no retest had occurred.

A weak report says:

> The stock broke out strongly.

The first statement distinguishes admission from exhaustion.

The second statement converts partial closure into total narrative certainty.

Residual honesty is therefore not pessimism.

It is calibrated closure.

---

## 9.12 Residual tolerance

No useful analysis can preserve every unresolved detail.

A residual tolerance must be declared:

$\|r_t\| \le \varepsilon_P$. (9.22)

Here $\varepsilon_P$ is not universal.

It depends on:

- application;
- timeframe;
- transaction cost;
- risk budget;
- data quality;
- decision authority;
- consequence of error.

A short-term trader and a credit committee should not use the same residual tolerance.

The important requirement is:

Residual tolerance must be declared before outcome review. (9.23)

Otherwise the tolerance becomes another retrospective adjustment.

---

# 9.13 Residual as a source of new structure

Residual is not only a burden.

It may reveal that the old feature map is incomplete.

Repeated residual may indicate:

- a missing variable;
- an incorrect boundary;
- an unstable regime;
- a flawed gate;
- an inappropriate timeframe;
- a need for a new indicator family.

The revision chain is:

Residual
→ Pressure on Declaration
→ Admissible Revision
→ New Projection
→ New Gate
→ Improved or Rejected Model. (9.24)

This closely parallels the declared-disclosure sequence:

$\Sigma_0$
→ Declare_P
→ $\hat{O}$_P
→ Gate_P
→ Trace_P + Residual_P
→ Ledger_P
→ Revision. (9.25)

The declaration framework defines time and world formation through precisely this chain of declared projection, gate, trace, residual, and ledger.

---

# 9.14 Residual summary

Residual answers:

> What remains unresolved after the current analytical or market closure?

It must not be confused with:

- noise;
- failure alone;
- Q;

- loss;
- uncertainty in general.

A mature framework preserves residual because:

- admitted events may remain incomplete;
- failed events may create consequential history;
- repeated contradictions should pressure model revision;
- erased failures make technical methods unfalsifiable.

Residual is the first governance rail because no bounded observer achieves complete closure.

---

# 10. Transport and Invariance: Does the Claim Survive Reframing?

## 10.1 The problem of frame dependence

Every technical-analysis claim is made under a protocol.

Change the protocol and the object may change.

Examples:

- daily chart versus weekly chart;
- arithmetic scale versus logarithmic scale;
- time bar versus volume bar;
- adjusted price versus unadjusted price;
- capitalization-weighted index versus equal-weight index;
- close-based pivot versus wick-based pivot;
- one benchmark versus another benchmark;
- one anchor versus another anchor.

Frame dependence is not automatically a defect.

A five-minute structure may genuinely differ from a weekly structure.

The problem arises when an analyst treats a frame-relative claim as universal.

Thus:

ValidUnder_P ≠ ValidUnder_AllFrames. (10.1)

---

## 10.2 Transport

Transport is the rule for carrying a claim from one protocol to another.

Let:

$T_{P \to P'}: Claim_P \to \hat{Claim}_{P'}$. (10.2)

Here $\hat{Claim}_{P'}$ is what the original claim predicts or becomes under the new frame.

The claim observed directly under $P'$ is:

$Claim_{P'}$. (10.3)

Transport consistency requires:

$Distance(\hat{Claim}_{P'}, Claim_{P'}) \le \varepsilon_T$. (10.4)

A small distance means the claim survives the transformation within tolerance.

A large distance means the structure is frame-fragile or the transport rule is inadequate.

---

# 10.3 Invariance is not identical appearance

A structure need not look identical in every frame.

A daily uptrend may appear as one segment inside a weekly range.

A price level may become a zone after volatility normalization.

A moving average period may require rescaling when bar construction changes.

Therefore, invariance does not mean:

Same pixels in every chart. (10.5)

It means:

A governed relation remains equivalent under an admissible transformation. (10.6)

Examples include:

- ordering of major pivots;
- direction of relative performance;
- location of a volatility-normalized boundary;
- persistence of breadth divergence;
- survival of a phase relation;
- consistency of gate outcome.

---

# 10.4 Three forms of invariance

### Structural invariance

The same underlying relation survives a frame change.

Example:

A support zone remains identifiable under reasonable changes in aggregation.

### Dynamical invariance

The same transition law remains approximately valid.

Example:

A corrective relation remains mean-reverting after volatility normalization.

### Ledger invariance

Different observers can align the consequential event and its record.

Example:

Several desks using different local charts agree that the weekly close produced a recognized regime break.

These correspond closely to wider distinctions between structural objectivity, dynamic objectivity, and ledger objectivity.

---

## 10.5 Timeframe transport

Multi-timeframe analysis is often performed by placing several charts beside one another.

That is comparison, not yet transport.

A formal timeframe transport should specify how an object at one scale maps to another.

For example:

T_{5m→1d}(Breakout_5m)
= CandidateIntradayEvent inside DailyWindow. (10.7)

It should not automatically become:

DailyBreakout. (10.8)

Similarly:

T_{1d→1w}(DailyTrend)
= Substructure inside WeeklyEpisode. (10.9)

The higher timeframe may:

- confirm;
- absorb;
- contradict;
- render irrelevant;

- reinterpret the lower-level event.

This prevents the common error:

Lower-level commitment → assumed higher-level commitment. (10.10)

---

## 10.6 Scale transport

A line drawn on arithmetic price has a different geometric meaning from a line drawn on logarithmic price.

Arithmetic scale preserves equal price differences.

Logarithmic scale preserves equal proportional changes.

Let:

$y_{lin} = P$. (10.11)

$y_{log} = \ln P$. (10.12)

A linear trend in $y_{lin}$ is not generally linear in $y_{log}$.

Therefore, any long-horizon geometric claim should declare its price scale.

A structure that survives both scales may have stronger support.

A structure that depends entirely on one undeclared scale should be classified as frame-sensitive.

---

## 10.7 Volatility normalization

A nominal ten-point move has different meaning in low- and high-volatility regimes.

A normalized displacement is:

$\tilde{d}_t = (P_t - \ell_t)/\sigma_t$. (10.13)

where:

$\ell_t$ = reference level;
$\sigma_t$ = declared volatility scale.

A boundary may be invariant in normalized space even when its nominal width changes.

This helps distinguish:

- true structural displacement;
- routine movement caused by volatility expansion.

Transport between raw and normalized frames is especially important for:

- breakouts;
- pattern widths;

- support zones;
- Gann-like slopes;
- phase alignment.

---

## 10.8 Universe transport

An index is a projection of a component universe.

Change the weighting rule and the apparent market structure may change.

Let:

I_t^{cap} = capitalization-weighted index. (10.14)

I_t^{eq} = equal-weight index. (10.15)

A claim of “broad market strength” should survive at least one field-level test:

Direction(I_t^{cap}) ≈ Direction(I_t^{eq}). (10.16)

or:

Direction(I_t^{cap}) ≈ Direction(Breadth_t). (10.17)

If not, the correct claim may be:

Headline index strength with narrowing field participation. (10.18)

This is weaker but more precise.

---

## 10.9 Anchor transport

Anchored methods include:

- anchored VWAP;
- Fibonacci retracement;
- Gann geometry;
- measured moves;
- wave counts;
- event-based returns.

Their conclusions may depend heavily on the chosen anchor.

Let:

C(a) = claim generated from anchor a. (10.19)

Anchor robustness can be tested by:

Robustness(C)
= Proportion of admissible anchors a′ for which C(a′) remains equivalent. (10.20)

An analyst should not search unlimited anchors until one fits.

The admissible anchor family must be declared.

Examples:

- major ledgered pivot;
- earnings event;
- policy event;
- highest-volume reversal;
- legally defined reporting date.

---

## 10.10 Pivot transport and Elliott Wave

Elliott Wave is highly sensitive to pivot extraction.

A pivot rule may depend on:

- minimum reversal size;
- volatility;
- elapsed bars;
- close or wick;
- fractal order.

Let:

Π_pivot^P(Price) = PivotSeries_P. (10.21)

A wave count is then:

W_P = Segment(PivotSeries_P). (10.22)

A robust count should survive reasonable perturbations:

Distance[T_{P→P′}(W_P),W_{P′}] ≤ ε_W. (10.23)

If a minor change in pivot rule completely changes the count, the model has high branch fragility.

That does not make every wave interpretation useless.

It means branch uncertainty must remain in the residual ledger.

---

## 10.11 Gann as a transport test

Gann analysis is best interpreted as a candidate invariance programme.

A price–time relation is only meaningful after declaring:

- price scale;
- time scale;
- anchor;
- volatility convention;
- calendar;
- bar construction.

A candidate relation is:

G_P = g(Price,Time | Scale,Anchor,Calendar). (10.24)

Its burden is:

T_{P→P′}(G_P) ≈ G_{P′}. (10.25)

If the relation survives only one carefully selected visual configuration, it should remain a local descriptive construction rather than a general market law.

The source Technical Analysis framework similarly treats Gann as requiring declared pivots, scales, cadence, selection depth, and cross-frame survival.

---

## 10.12 Cross-indicator agreement is not invariance

Suppose:

- price is above its 20-day moving average;
- MACD is positive;
- the fast moving average is above the slow moving average.

These observations may agree, but they are heavily derived from the same price history.

They do not provide three independent frames.

Their agreement may be represented:

Agreement = SharedInput + SharedFilterFamily. (10.26)

True cross-frame support is stronger when the claim survives:

- price frame;
- volume frame;
- breadth frame;
- volatility-normalized frame;

- higher timeframe;
- position or liquidity frame.

Thus:

IndicatorCount ≠ EvidenceIndependence. (10.27)

---

# 10.13 Frame robustness score

A preliminary frame-robustness score may be written:

$S_T = \Sigma_j w_j I_j$. (10.28)

where:

$I_j = 1$ if the claim survives frame j;
$I_j = 0$ if it fails;
$I_j$ may take an intermediate value for partial survival.

Frames may include:

- timeframe;
- scale;
- volatility normalization;
- universe;
- anchor;
- pivot rule;
- data source.

The weights should reflect relevance rather than convenience.

A higher $S_T$ does not guarantee truth.

It indicates that the claim is less dependent on one narrow representation.

---

# 10.14 Invariance and objectivity

Objectivity does not require an observer-free chart.

It requires that governed relations survive translation between bounded observers.

This can be written:

$Objectivity_{P,P'} \approx Compatibility$

- TransportConsistency
- AccessibleTrace. (10.29)

The formal observer framework develops cross-observer agreement through compatible effects, consistent frame transforms, and accessible records; redundant trace can then support broader consensus.

In markets, analogous conditions include:

- comparable definitions;
- compatible horizons;
- accessible transaction records;
- declared transformations;
- agreement on consequential gates.

---

# 10.15 Transport failure

Transport may fail for several reasons.

### Genuine scale dependence

The structure exists only locally.

### Protocol mismatch

The two observers are measuring different objects.

### Nonlinear aggregation

Lower-level relations do not survive coarse-graining.

### Regime change

The transformation crosses a structural break.

### Observer backreaction

The act of measuring or acting under one frame changes the system.

### Poorly specified mapping

The transport operator itself is inadequate.

Transport failure is therefore informative.

It should not automatically be treated as evidence that the original claim was wrong.

It identifies the limits of the claim.

---

# 10.16 Transport and invariance summary

Transport asks:

What does this claim become under another admissible frame?

Invariance asks:

Which relation survives the transformation?

A mature Technical Analysis framework must declare:

- which frames are admissible;
- how claims are transported;
- what tolerance counts as survival;
- when failure indicates local validity rather than total invalidity.

Transport is the second governance rail because an interpretation confined to one convenient chart frame cannot yet claim robust structure.

---

# 11. Ledger and Backreaction: When Analysis Changes the Market

## 11.1 A trace is more than stored data

A record becomes a ledgered trace when it changes future admissibility.

A database may store every trade.

But only some records alter:

- future orders;
- institutional treatment;
- risk limits;
- legal rights;
- market narratives;
- benchmark composition;
- accounting status.

Thus:

Record = preserved description of an event. (11.1)

LedgeredTrace = preserved event that constrains future action. (11.2)

This distinction is central to the declared-disclosure framework, where time emerges not from mere sequence but from gated trace written into an ordered ledger.

---

## 11.2 The ledger update

A generic market ledger update is:

$$L_{k+1} = U_L(L_k,e_k,r_k). \quad (11.3)$$

where:

$L_k$ = prior ledger;
$e_k$ = admitted event;
$r_k$ = attached residual.

The new ledger does not contain only the event.

It also preserves unresolved qualifications.

This matters because future interpretation should depend on both.

A breakout with strong breadth and a breakout with weak breadth may share the same price event but should not produce identical ledger states.

---

# 11.3 Different market ledgers

Markets contain several overlapping ledgers.

## Transaction ledger

Records executed trades.

## Position ledger

Records who carries exposure, where observable.

## Accounting ledger

Records recognized values, losses, liabilities, and classifications.

## Legal ledger

Records ownership, default, settlement, and enforceable status.

## Risk ledger

Records limits, breaches, stress states, and collateral demands.

## Narrative ledger

Records widely shared interpretations.

## Technical ledger

Records pivots, closes, levels, breakouts, failures, and regimes.

These ledgers may disagree.

A market price can move before an accounting loss is recognized.

A technical breakout can occur before institutional benchmark reclassification.

A legal default can occur after the market has priced severe distress.

Thus:

LedgerTime_transaction
≠ LedgerTime_accounting
≠ LedgerTime_legal
≠ LedgerTime_technical. (11.4)

---

## 11.4 Close as technical ledger write

The official close often acts as a technical ledger event because many processes read it.

A close may update:

- daily return;
- moving averages;
- breakout status;
- margin;
- fund valuation;
- performance reporting;
- systematic signals.

Therefore:

Close_t
→ IndicatorUpdate_t
→ InstitutionalReadout_t
→ NextDayConditionalAction. (11.5)

This is why the close can have consequence beyond being the last trade of the session.

It is a standardized disclosure gate.

---

## 11.5 Ledgered support and resistance

A support level becomes stronger when past reactions enter the active market ledger.

The chain is:

Reaction
→ RecordedLevel
→ SharedInterpretation
→ ConditionalOrders
→ FutureReaction. (11.6)

The level thereby acquires backreaction.

However, repeated testing may either strengthen or weaken it.

It may strengthen when:

- repeated defence confirms commitment;
- more participants recognize it;
- positions build around it.

It may weaken when:

- defending liquidity is consumed;
- residual accumulates;
- participants become trapped;
- repeated tests reduce surprise.

Thus:

RepeatedTest → Strengthening or Depletion. (11.7)

The direction cannot be inferred from repetition alone.

---

## 11.6 Narrative ledger

Market narratives compress scattered events into an actionable history.

Examples:

- “inflation trade”;
- “AI boom”;
- “credit crisis”;
- “policy pivot”;
- “safe-haven rally”;
- “breakout from a multi-year base.”

A narrative selects:

- relevant events;
- causal ordering;
- protagonists;
- boundaries;
- expected future actions.

Narrative_t = Compress(Events_{≤t} | FeatureMap, Authority, Audience). (11.8)

Narratives may become causally effective when they alter allocation.

Thus:

NarrativeLedger
→ SharedExpectation
→ Orders
→ Price
→ NarrativeReinforcement or Failure. (11.9)

Technical analysis often contributes to this narrative ledger by providing visual compression.

---

# 11.7 Backreaction

Backreaction occurs when an observation or ledger entry changes the field being observed.

Let:

$O_t$ = observer interpretation;
$A_t$ = action induced by the interpretation;
$\Sigma_t$ = market field.

Then:

$A_t = \text{Policy}(O_t, L_t)$. (11.10)

$\Sigma_{t+1} = \text{Dynamics}(\Sigma_t, A_t, \text{External}_t)$. (11.11)

The next observation is:

$O_{t+1} = \text{Projection}_P(\Sigma_{t+1})$. (11.12)

Thus the observer is internal to the loop.

This is consistent with the source account of self-referential observers: trace conditions future instrument selection, causing counterfactual branches to diverge through adaptive policy.

---

# 11.8 Passive observation versus active coupling

The operational distinction can be restated:

### Probe

Attempts to observe without materially changing the system.

### Couple

Changes how strongly structures bind.

### Pump

Adds or removes effective loading.

### Switch

Changes the operative regime or routing.

A moving average may begin as Probe.

When enough capital trades around it, the moving average becomes Couple.

A breakout signal may become Switch.

A central-bank announcement may simultaneously act as:

- Switch in policy regime;
- Pump in liquidity;
- Couple in funding conditions.

The PORE framework emphasizes that its effective coordinates are operational and protocol-bound, and that probe backreaction must be tested rather than assumed absent.

---

## 11.9 Crowding as observer load

Suppose many participants follow the same technical signal.

Let N_O denote effective observer crowding.

A simple reflexivity relation is:

BackreactionStrength ∝ N_O × CapitalPerObserver × ActionCoherence. (11.13)

When observer actions align, the signal may become self-confirming.

But high crowding may also increase fragility:

Fragility
∝ Crowding
× SimilarExitRules
× LimitedLiquidity. (11.14)

Therefore the same shared signal can produce:

- stronger initial movement;
- more dangerous later reversal.

A mature system should distinguish signal strength from crowding fragility.

---

## 11.10 Observer-backreaction score

A provisional observer-backreaction score could include:

S_O
= w1Visibility

- $w_2$Adoption
- $w_3$CapitalLinked
- $w_4$ActionSynchrony
  – $w_5$LiquidityCapacity. (11.15)

This is not yet an established market indicator.

It is a research target revealed by the periodic grammar.

Possible proxies include:

- strategy assets under management;
- public indicator popularity;
- option positioning;
- CTA exposure estimates;
- stop clustering;
- benchmark-linked flows;
- search or media attention.

The score should be interpreted cautiously because direct observation of market-wide strategy use is limited.

---

## 11.11 The technical analyst as part of the market

Technical analysis is often written as though an analyst observes from outside.

But a practical analyst may:

- trade;
- publish;
- advise;
- manage risk;
- design algorithms;
- influence others.

The declaration should therefore include observer role:

$O_role \in \{PrivateProbe, Trader, Adviser, Publisher, Institution, Regulator\}$. (11.16)

The same analysis has different backreaction potential in each role.

A private observation may have negligible market effect.

A widely followed institutional signal may alter the market field.

---

## 11.12 Ledger hierarchy

A lower-period event becomes higher-period load only if it survives sufficient closure.

The hierarchy is:

Mark ledger
→ Window ledger
→ Structure ledger
→ Event ledger
→ Episode ledger
→ World ledger. (11.17)

For example:

1. trades occur;
2. a daily candle closes;
3. the candle modifies a support structure;
4. a breakout event is admitted;
5. several events form a trend episode;
6. the episode becomes an institutional regime.

This hierarchy supplies the vertical axis of the future periodic table.

---

## 11.13 Ledger order and time

Clock time orders observations externally.

Ledger order records consequential commitments.

Let t denote calendar time and k denote event order.

Two events may occur far apart in t but adjacent in k.

A long quiet period may contain no major ledger transition.

A crisis may produce several commitments within hours.

Therefore:

$\Delta t$ large does not imply $\Delta k$ large. (11.18)

$\Delta t$ small does not imply $\Delta k$ small. (11.19)

The filtration and declaration sequence defines time as the ordered ledger of gated disclosure rather than as recursion or raw sequence alone.

This distinction will become important when the article later introduces phase time $\tau_i$.

---

## 11.14 Ledger revision

A market ledger is not immutable.

Old interpretations may be revised.

But revision should preserve prior trace.

A mature update is:

$D_{k+1} = U_a(D_k,L_k,R_k)$. (11.20)

where:

$D_k$ = declaration at episode k;
$L_k$ = ledger;
$R_k$ = residual;
$U_a$ = admissible revision.

Admissible revision should be:

- trace-preserving;
- residual-honest;
- frame-robust;
- non-degenerate;
- bounded by declared rules.

This principle comes from the self-revising declaration framework, where observer maturity is defined not by unrestricted self-modification but by admissible revision of the declaration governing future projection.

---

# 11.15 Backreaction pathologies

Several pathologies may occur.

### Self-fulfilling closure

The signal appears valid mainly because observers act on it.

### Crowding collapse

Too many similar positions make the signal unstable.

### Silent ledger rewrite

Failed interpretations are removed from history.

### Residual suppression

Contradictory evidence is not attached to admitted events.

### Authority drift

The gate rule changes without declaration.

### Frame shopping

The analyst moves between frames until confirmation appears.

### Narrative lock-in

A successful interpretation becomes too difficult to revise.

These pathologies convert technical analysis from observation into ungoverned world-making.

---

## 11.16 Ledger and backreaction summary

Ledger asks:

> Which traces remain consequential enough to shape the next market state?

Backreaction asks:

> How does the recorded interpretation alter the field future analysis observes?

The ledger rail is necessary because:

- not every record changes behaviour;
- different institutions maintain different ledgers;
- technical events may acquire market force through shared use;
- observations may become interventions;
- revisions must preserve historical accountability.

The market does not merely leave charts.

Charts, classifications, and interpretations can help produce the next market.

---

# 12. Governance Across the Four Functional Families

The three rails can now be applied to each functional family.

| Functional family | Residual question | Transport question | Ledger question |
|---|---|---|---|
| **Load / Memory** | What relevant history or positioning is omitted? | Does the memory survive another horizon or weighting rule? | Does the remembered structure still affect future action? |
| **Motion / Relation** | What contradictory relation remains? | Does the relation survive normalization or benchmark change? | Has the relation altered positioning or remained diagnostic only? |
| **Constraint / Boundary** | What weakens the claimed boundary? | Does the boundary survive scale, anchor, and timeframe change? | Are future orders actually conditioned on it? |

| Functional family | Residual question | Transport question | Ledger question |
|---|---|---|---|
| **Commitment / Gate** | What remains unresolved after admission? | Do other admissible gates recognize the same event? | Did the event change future admissibility? |

A mature technical claim should therefore answer seven questions:

1. What is loaded?
2. What relation is changing?
3. What constrains the movement?
4. What gate defines commitment?
5. What residual remains?
6. What survives transport?
7. What enters the ledger and backreacts?

In compact form:

MatureTAClaim_P
= Load

- Relation
- Constraint
- Gate
- Residual
- Transport
- LedgerConsequence. (12.1)

The next part will use this governed grammar to construct the six periods of market closure:

**Mark**
**Window**
**Structure**
**Event**
**Episode**
**World**.

# Part IV — The Six Periods of Market Closure

The four functional families recur at increasing levels of organization:

**Load / Memory**
**Motion / Relation**
**Constraint / Boundary**
**Commitment / Gate**

The recurring functions do not act on the same kind of object at every level.

At the lowest level, they act on orders, quotes, and trades.

At the next level, they act on candles and observational windows.

They then act on technical structures, market events, episodes, and finally self-referential institutional worlds.

The six periods are:

**Period 0 — Mark**
**Period 1 — Window**
**Period 2 — Structure**
**Period 3 — Event**
**Period 4 — Episode**
**Period 5 — World**

The periodic-generation rule is:

$\text{Field}_n$
$\rightarrow \text{Load}_n$
$\rightarrow \text{Motion}_n$ under $\text{Constraint}_n$
$\rightarrow \text{Commitment}_n$
$\rightarrow \text{Ledger}_{n+1} + \text{Residual}_n$
$\rightarrow \text{Field}_{n+1}$. (13.1)

The output of one period becomes an input to the next.

Trades become bars.

Bars become structures.

Structures condition events.

Events accumulate into episodes.

Episodes stabilize or destabilize market worlds.

Worlds then reshape the order flow from which later marks arise.

---

# 13. Period 0 — Mark

# 13.1 What is a mark?

A mark is the smallest market occurrence admitted into the chosen data field.

Examples include:

- quote update;
- limit order;
- market order;
- trade;
- cancellation;

- execution;
- bid–ask change;
- order-book depletion;
- auction print.

A mark is not necessarily a transaction.

An order may be submitted but never executed.

A quote may move without a trade.

A cancellation may reveal disappearing liquidity.

A trade may occur without revealing the trader's motive.

Thus:

Mark_P = smallest admitted market record under protocol P. (13.2)

The phrase under protocol P is essential.

A data vendor may record one set of marks.

An exchange may retain a richer order-book history.

A retail chart may display only trades.

A regulator may observe account-level identifiers unavailable to ordinary analysts.

There is no protocol-free mark.

---

## 13.2 A mark is already a projection

It is tempting to treat tick data as raw reality.

But a tick is already a selected and formatted trace.

The protocol may determine:

- which venue is included;
- which trade types are excluded;
- whether corrections are retained;
- whether odd lots are included;
- whether timestamps are exchange- or vendor-generated;
- whether bid and ask updates are synchronized;
- whether hidden orders are visible;
- whether cancelled orders remain available.

Therefore:

RawData_P ≠ TotalMarketReality. (13.3)

RawData_P = ProjectedMarkField_P + UnobservedResidual_P. (13.4)

At Period 0, the analyst is already a bounded observer.

---

# 13.3 Load / Memory at the mark level

Period-0 Load includes immediately available market structure such as:

- displayed bid size;
- displayed ask size;
- queue depth;
- resting orders;
- recent trade direction;
- local inventory;
- hidden-liquidity estimates;
- outstanding conditional orders.

A simplified visible order-book load near price p is:

$\rho_0(p,t)$ = BidDepth(p,t) + AskDepth(p,t). (13.5)

But displayed depth is not total liquidity.

It may omit:

- hidden orders;
- iceberg reserves;
- internalized flow;
- conditional liquidity;
- strategic order cancellation;
- liquidity available only after price movement.

Therefore:

DisplayedLiquidity ≠ ExecutableLiquidity ≠ TotalPotentialLiquidity. (13.6)

Period-0 Load is immediate and local.

It is not yet the historical market memory visible in moving averages or profiles.

---

## 13.4 Motion / Relation at the mark level

Period-0 Motion includes:

- tick displacement;
- spread expansion or contraction;
- order imbalance;
- queue depletion;
- trade-sign sequence;
- short-horizon response to flow.

A simple tick change is:

$$\Delta p_k = p_k - p_{k-1}. \quad (13.7)$$

A visible order imbalance may be written:

$$I_k = (B_k - A_k)/(B_k + A_k). \quad (13.8)$$

where:

$B_k$ = displayed bid depth;
$A_k$ = displayed ask depth.

But positive imbalance does not guarantee price increase.

Possible reasons include:

- spoofed depth;
- hidden sell liquidity;
- aggressive selling into bids;
- rapid order cancellation;
- different queue priority;
- cross-venue activity.

Thus:

OrderImbalance = local relational evidence, not commitment. (13.9)

---

## 13.5 Constraint / Boundary at the mark level

Period-0 constraints include:

- current bid;
- current ask;

- spread;
- queue position;
- tick size;
- exchange price limits;
- trading halts;
- available collateral;
- order-size restrictions.

The bid–ask spread is the smallest visible transaction boundary:

$$Spread_t = Ask_t - Bid_t. \quad (13.10)$$

A market order must cross this boundary to obtain immediate execution.

The spread therefore combines:

- price constraint;
- liquidity condition;
- execution cost;
- local gate threshold.

But the spread is not fixed.

It changes with:

- volatility;
- inventory risk;
- competition;
- information asymmetry;
- order-flow toxicity;
- market stress.

---

## 13.6 Commitment / Gate at the mark level

At Period 0, execution is the clearest commitment event.

An intention to buy is not yet a trade.

A submitted order is not yet an execution.

A displayed quote is not yet accepted exchange.

The gate is:

OrderIntent
→ MatchingRule
→ Execution or Non-Execution. (13.11)

A simplified execution gate is:

Go(Order,Book) → {Executed, PartiallyExecuted, Resting, Cancelled, Rejected}. (13.12)

Each result writes a different trace.

### Executed

The proposed exchange becomes a transaction.

### Partially executed

Part of the proposal becomes history; residual quantity remains.

### Resting

The proposal remains conditional.

### Cancelled

The intention is withdrawn before exchange.

### Rejected

The venue or rule prevents admission.

This already demonstrates:

Commitment is not necessarily binary. (13.13)

---

## 13.7 Mark-level residual

Residual at Period 0 includes:

- unfilled quantity;
- hidden liquidity;
- rejected orders;
- unobserved motive;
- off-venue flow;
- asynchronous timestamps;
- cancelled intentions;
- order-book states lost through aggregation.

A partial fill illustrates the coexistence of commitment and residual:

OrderQuantity = ExecutedQuantity + UnfilledResidual. (13.14)

The transaction enters the ledger.

The remaining quantity may continue to influence later behaviour.

---

## 13.8 Mark-level ledger

The transaction tape is the most visible mark-level ledger.

But the full ledger may also contain:

- order identifiers;
- queue position;
- account identity;
- venue;
- execution condition;
- correction status;
- cancellation history.

Ordinary Technical Analysis usually receives a compressed version of this ledger.

The transition to Period 1 occurs when many marks are aggregated into a declared observational window:

$\{Mark_1, Mark_2, \ldots, Mark_n\}$
$\rightarrow$ WindowState. (13.15)

---

## 13.9 Period-0 summary

| Function | Period-0 form |
|---|---|
| **Load / Memory** | visible depth, resting interest, local inventory |
| **Motion / Relation** | tick change, spread movement, imbalance |
| **Constraint / Boundary** | bid, ask, tick size, price limit |
| **Commitment / Gate** | execution, cancellation, rejection |
| **Residual** | unfilled quantity, hidden liquidity, unobserved motive |
| **Ledger** | trade and order record |

Period 0 is the world of individual marks.

Its closure produces the material from which windows are constructed.

---

# 14. Period 1 — Window

## 14.1 The window as a declared micro-world

A window groups lower-level marks into one observational object.

Examples include:

- one-minute bar;
- hourly candle;
- daily session;
- weekly bar;
- volume bar;
- tick bar;
- range bar;
- event-defined window.

A general window is:

$$W_j = \{Mark_k \mid Mark_k \text{ satisfies } WindowRule_j\}. \quad (14.1)$$

The window rule may be based on:

- elapsed time;
- number of trades;
- cumulative volume;
- price range;
- event boundary;
- institutional reporting cycle.

A candle is therefore not a primitive fact.

It is a compiled summary of marks admitted under $W_j$.

---

# 14.2 OHLCV as compressed disclosure

The standard bar is:

$$Bar_j = (O_j, H_j, L_j, C_j, V_j). \quad (14.2)$$

where:

$O_j$ = first admitted price;
$H_j$ = highest admitted price;
$L_j$ = lowest admitted price;
$C_j$ = final admitted price;
$V_j$ = accumulated volume.

The bar preserves:

- opening state;
- maximum upward excursion;
- maximum downward excursion;
- closing commitment;
- volume.

It discards:

- exact path;
- order sequence;
- internal timing;
- trade-size distribution;
- order-book response;
- number of reversals;
- venue differences.

Therefore:

OHLCV = compressed trace, not complete path. (14.3)

Two different intrawindow paths may produce the same OHLCV bar.

---

## 14.3 Load / Memory at the window level

Window-level Load includes:

- opening position relative to prior close;
- local volume;
- carried gap;
- recent range;
- prior-window memory;
- intrawindow transaction density.

The opening price is already conditioned by previous ledger history:

$O_j = \text{Function}(L_{j-1}, \text{OvernightInformation}, \text{OpeningLiquidity})$. (14.4)

Volume records local exchange load:

$V_j = \Sigma_{k \in W_j} v_k$. (14.5)

But the volume number alone does not reveal whether the window represents:

- accumulation;
- distribution;
- forced liquidation;
- short covering;
- passive transfer;
- churn.

---

## 14.4 Motion / Relation at the window level

Window Motion includes:

- open-to-close displacement;
- high–low excursion;
- gap;
- range position;
- local volatility;
- relation to the prior window.

The body is:

$$Body_j = C_j - O_j. \quad (14.6)$$

The total range is:

$$Range_j = H_j - L_j. \quad (14.7)$$

A normalized close location is:

$$CLV_j = [(C_j - L_j) - (H_j - C_j)]/(H_j - L_j). \quad (14.8)$$

when $H_j \neq L_j$.

$CLV_j$ near +1 indicates a close near the high.

$CLV_j$ near −1 indicates a close near the low.

But close location does not by itself identify:

- informed buying;
- forced selling;
- absorption;
- sustainable continuation.

It is a window-level relation.

---

## 14.5 Constraint / Boundary at the window level

The high and low form local boundaries:

$\Omega_j = [L_j,H_j]$. (14.9)

These boundaries record where the admitted intrawindow path stopped.

But they do not automatically become future support or resistance.

Their later significance depends on:

- volume;
- location in higher-order structure;
- repeated reaction;
- institutional attention;
- later gate behaviour.

A single window therefore creates candidate boundaries, not necessarily operative constraints.

---

## 14.6 Commitment / Gate at the window level

The close is the defining gate of a conventional time bar.

The path may move through many provisional states.

The close selects one terminal state for the window ledger:

IntrawindowPath_j
→ ClosingRule
→ C_j. (14.10)

The close then updates:

- returns;
- moving averages;
- oscillator values;
- breakout status;
- portfolio marks;
- risk reports.

This is why:

Close ≠ merely last observed price. (14.11)

Close = institutionally privileged window trace. (14.12)

---

## 14.7 Body and wick

For an upward-closing candle:

$$\text{Body}_j = C_j - O_j. \quad (14.13)$$

$$\text{UpperWick}_j = H_j - C_j. \quad (14.14)$$

$$\text{LowerWick}_j = O_j - L_j. \quad (14.15)$$

For a downward-closing candle, the definitions adjust using max and min:

$$\text{UpperWick}_j = H_j - \max(O_j, C_j). \quad (14.16)$$

$$\text{LowerWick}_j = \min(O_j, C_j) - L_j. \quad (14.17)$$

A useful structural interpretation is:

Body = displacement retained by the closing gate. (14.18)

Wick = intrawindow displacement not retained at the close. (14.19)

But:

Wick ≠ confirmed rejection. (14.20)

A wick becomes stronger rejection evidence only if later trace supports that interpretation.

---

## 14.8 Gap as interwindow residual

A gap occurs when:

$$O_j \neq C_{j-1}. \quad (14.21)$$

The gap may reflect:

- overnight information;
- illiquidity;
- auction imbalance;
- institutional repricing;
- market closure;
- discontinuous order adjustment.

A gap is both:

- Motion between windows;
- Residual of information not expressed inside the prior window.

Whether the gap becomes committed depends on later acceptance.

A gap that quickly closes and a gap that remains open have different ledger consequences.

## 14.9 Alternative bar constructions

A time bar fixes elapsed time.

A volume bar fixes accumulated volume.

A range bar fixes price displacement.

A tick bar fixes the number of marks.

These methods disclose different internal organizations.

For example:

TimeBar_j = marks occurring during Δt. (14.22)

VolumeBar_j = smallest mark set whose volume reaches V*. (14.23)

RangeBar_j = smallest mark set whose price range reaches R*. (14.24)

A structure that survives several bar constructions is more robust than one visible only under one convenient aggregation.

This is an early form of transport testing.

---

## 14.10 Window-level residual

Window compression leaves residual such as:

- intrawindow sequence;
- hidden liquidity;
- order-book dynamics;
- heterogeneous motives;
- multiple reversals;
- exact timing;
- cross-venue differences.

A bar-level interpretation should therefore not claim more than its representation supports.

For example:

A bullish candle does not prove net informed accumulation. (14.25)

It proves only that the closing price exceeded the opening price under the declared window rule.

---

## 14.11 Window conflict

Different windows may produce conflicting local states.

A daily bullish candle may occur inside:

- a weekly downtrend;
- a monthly range;
- an intraday exhaustion move.

The proper transport is:

$$\text{WindowEvent}_{h_1} \rightarrow \text{Substructure inside Window}_{h_2}. \quad (14.26)$$

not:

$$\text{WindowEvent}_{h_1} = \text{identical event at } h_2. \quad (14.27)$$

This prevents lower-period closure from being promoted automatically to higher-period commitment.

---

## 14.12 From windows to structures

One bar rarely defines a durable market structure.

Structure emerges when relations persist across multiple windows.

The transition is:

$$\{\text{Window}_1, \text{Window}_2, \ldots, \text{Window}_n\} \rightarrow \text{Filter} \rightarrow \text{Relation} \rightarrow \text{AccumulatedTrace} \rightarrow \text{Structure}. \quad (14.28)$$

Moving averages, oscillators, trend lines, profiles, breadth series, and volatility regimes appear mainly at the next period.

---

## 14.13 Period-1 summary

| Function | Period-1 form |
|---|---|
| **Load / Memory** | opening state, bar volume, prior close |
| **Motion / Relation** | return, body, range position, gap |
| **Constraint / Boundary** | high, low, local range, local band |
| **Commitment / Gate** | official close |
| **Residual** | wick, intrawindow path, hidden order dynamics |
| **Ledger** | OHLCV or alternative bar record |

Period 1 is the world of observational windows.

Its closure produces the objects from which conventional Technical Analysis constructs structure.

---

# 15. Period 2 — Structure

## 15.1 The principal domain of conventional Technical Analysis

Period 2 contains most familiar technical indicators.

It includes:

- moving averages;
- trend measures;
- momentum oscillators;
- MACD;
- RSI;
- ATR;
- volatility bands;
- VWAP;
- OBV;
- volume profile;
- breadth;
- support and resistance;
- relative strength.

These methods do not all perform the same function.

But they share one important feature:

> They organize multiple windows into persistent relational structure.

A Period-2 object is therefore:

Structure_P = OrganizedRelation({Window_j} | FeatureMap_P). (15.1)

Structure is more than one window but less than a committed event.

---

## 15.2 Load / Memory at the structure level

Representative Load instruments include:

- moving averages;
- cumulative volume;
- VWAP;

- OBV;
- volume profile;
- breadth history;
- prior highs and lows;
- rolling volatility.

A moving average compiles temporal memory.

Volume profile compiles price-space memory.

Breadth compiles cross-sectional participation.

VWAP compiles transaction-weighted memory.

These are different forms of accumulated trace.

---

## 15.3 Motion / Relation at the structure level

Representative Motion instruments include:

- price return;
- moving-average slope;
- crossover;
- MACD;
- RSI;
- stochastic;
- relative strength;
- divergence;
- breadth coherence;
- phase relation.

They answer:

- Is the state accelerating?
- Is one memory horizon overtaking another?
- Is participation expanding?
- Is the market field aligned?
- Is the current direction corrective or self-confirming?
- Is the visible price trace diverging from supporting pressure?

But these measurements remain diagnostic.

They do not by themselves establish a Period-3 event.

---

## 15.4 Constraint / Boundary at the structure level

Representative structures include:

- support;
- resistance;
- trend channels;
- value areas;
- Bollinger Bands;
- Keltner Channels;
- Donchian boundaries;
- high-volume nodes;
- low-volume zones;
- Fibonacci zones.

These methods declare where a transition may become consequential.

They do not determine whether the transition has been accepted.

Thus:

Period-2 Boundary + Crossing ≠ Period-3 Event by default. (15.2)

A gate is still required.

---

## 15.5 Commitment at the structure level

Period-2 Commitment is weaker than a full breakout event.

It includes:

- sustained memory reordering;
- repeated closes beyond a local average;
- accepted migration from one value region to another;
- confirmed shift in breadth structure;
- stable relative-strength transition.

A structure may be considered admitted when it survives:

- sufficient windows;
- alternative filters;
- relevant frame changes;
- residual audit.

This produces a structural gate:

$G_2(\text{StructureCandidate}) \rightarrow \{\text{Established, Provisional, Rejected}\}$. (15.3)

---

## 15.6 Structure is not signal

A major conceptual correction is:

$\text{Structure} \neq \text{Signal}$. (15.4)

A rising moving average describes memory orientation.

High RSI describes directional imbalance.

A high-volume node describes historical occupation.

A support zone describes a candidate boundary.

These may contribute to a trading decision, but they do not contain a complete action rule.

An action additionally requires:

- objective;
- risk budget;
- admissible intervention;
- gate;
- horizon;
- invalidation.

Thus:

$$\text{TradingDecision} = \text{TechnicalStructure}$$

- DecisionProtocol
- RiskConstraint
- ExecutionRule. (15.5)

---

## 15.7 Functional decomposition of common indicators

| Method | Primary function | Secondary function |
|---|---|---|
| Moving average | Load / Memory | relation to current price |
| MA crossover | Motion / Relation | provisional structural shift |
| MACD | Motion / Relation | memory curvature |
| RSI | Motion / Relation | corrective-condition hypothesis |
| ATR | Motion magnitude | boundary scaling |
| Bollinger Bands | Constraint / Boundary | volatility context |
| VWAP | Load / Memory | institutional reference boundary |
| OBV | Load / Memory | signed directional relation |
| Volume profile | Load + Constraint | density and structural mass |
| Breadth | Motion / Relation | field coherence |
| Support / resistance | Constraint | event-test location |

This decomposition makes redundancy visible.

---

# 15.8 Structural redundancy

Suppose an analyst uses:

- 20-day moving average;
- 50-day moving average;
- MACD;
- moving-average crossover.

These appear to be four confirmations.

But all four derive primarily from filtered price memory.

Their information overlap may be high.

A rough independence principle is:

EvidenceValue
≈ SignalQuality × FunctionalIndependence. (15.6)

Therefore:

Four correlated memory indicators
may provide less evidence than
one memory indicator + one breadth measure + one gate measure. (15.7)

This principle will later support a stronger definition of confirmation.

---

# 15.9 Structural contradiction

Period-2 structures can disagree.

Examples:

- price trend rising while breadth declines;
- price above VWAP while momentum weakens;
- moving averages bullish while volume profile shows rejection;
- RSI strong while relative strength falls;
- support holds locally while higher-timeframe structure deteriorates.

These contradictions should not be forced into one immediate conclusion.

They form a structural residual vector:

$r_2 = (r_{memory}, r_{motion}, r_{breadth}, r_{density}, r_{frame})$. (15.8)

A mature analysis carries this vector toward the event gate.

---

## 15.10 Structure scorecard

A Period-2 scorecard might contain:

StructureCard =
(LoadState, MotionState, ConstraintState,
RegimeSignature, FrameStatus, Residual, Invalidation). (15.9)

For example:

**Load:** rising volume-weighted memory
**Motion:** positive but decelerating
**Constraint:** major resistance overhead
**χ:** mildly self-confirming
**Frame:** daily positive, weekly neutral
**Residual:** declining breadth
**Invalidation:** close below accepted value area

This is more informative than:

“MACD bullish.”

---

## 15.11 From structure to event

A Period-3 event occurs when a Period-2 structure encounters a meaningful boundary and passes or fails a consequential gate.

The transition is:

EstablishedStructure

- BoundaryInteraction
- GateDecision
  → Event. (15.10)

Examples:

- trend meets resistance;
- compression reaches boundary;
- divergence meets support failure;
- price leaves a value area;
- breadth deterioration meets index breakdown;
- moving-average structure meets accepted reversal.

---

## 15.12 Period-2 summary

| Function | Period-2 form |
|---|---|
| **Load / Memory** | MA, VWAP, OBV, profile, breadth history |
| **Motion / Relation** | momentum, MACD, RSI, divergence, relative strength |
| **Constraint / Boundary** | support, resistance, channels, bands, value area |
| **Commitment / Gate** | sustained structural acceptance |
| **Residual** | contradiction across indicators, frames, or fields |
| **Ledger** | recognized technical structure |

Period 2 is the main observatory of conventional Technical Analysis.

It diagnoses the market state.

It does not yet guarantee a committed transition.

---

# 16. Period 3 — Event

## 16.1 What makes a market event?

A market event is not merely movement.

It is a transition whose passage through a gate changes the operative ledger.

A general event is:

$Event_k = Gate_P(StructureBefore, BoundaryInteraction, Evidence)$. (16.1)

Examples include:

- breakout;
- breakdown;
- rejection;
- reversal;
- retest;
- absorption;

- capitulation;
- failed auction;
- value-area migration.

A Period-3 event answers:

> Did the market merely fluctuate, or did it commit to a new relation?

---

## 16.2 Load / Memory at the event level

Event-level Load includes what has accumulated around the transition:

- positions near the boundary;
- stop concentration;
- volume;
- open interest;
- trapped exposure;
- prior failed attempts;
- option positioning;
- narrative attention.

The same price crossing can have different meaning under different event loads.

A lightly loaded crossing may pass easily but lack consequence.

A heavily loaded crossing may trigger:

- stop cascades;
- forced hedging;
- new trend entry;
- sharp rejection.

Thus:

EventImpact
= Function(Boundary, Load, Motion, Gate, Residual). (16.2)

---

## 16.3 Motion / Relation at the event level

Event Motion includes:

- displacement through the boundary;
- acceleration;

- volatility expansion;
- volume response;
- breadth response;
- phase shift;
- feedback-sign change.

A normalized displacement is:

$D_t = (P_t - B_t)/\sigma_t$. (16.3)

where:

$B_t$ = declared boundary;
$\sigma_t$ = volatility scale.

This distinguishes a meaningful break from ordinary noise.

But displacement alone remains insufficient.

---

## 16.4 Constraint at the event level

The event-level boundary must be declared before the event.

Examples:

- prior high;
- value-area edge;
- multiwindow support;
- volatility envelope;
- neckline;
- covenant;
- margin threshold.

A boundary drawn only after the move is not a valid prospective event gate.

Thus:

PredeclaredBoundary is required for prospective event testing. (16.4)

This does not prohibit exploratory analysis.

It distinguishes discovery from validation.

---

## 16.5 Commitment / Gate at the event level

A general gate-strength expression is:

S_G
= w_DD

- w_VV
- w_BB
- w_CC
- w_FF
- w_TT
  − w_RR. (16.5)

where:

D = normalized displacement;
V = participation;
B = breadth coherence;
C = close quality;
F = follow-through;
T = retest acceptance;
R = residual contradiction.

The exact terms and weights depend on the protocol.

The important point is that a gate should preserve both:

- supporting evidence;
- unresolved residual.

---

# 16.6 Continuation event

A continuation event occurs when the existing regime survives a challenge and resumes.

A schematic continuation is:

ExistingTrend

- CorrectiveTest
- BoundaryHold
- RenewedMotion
  → ContinuationEvent. (16.6)

Examples:

- pullback to moving average followed by recovery;
- retest of breakout level;
- consolidation followed by renewed expansion;
- value-area rejection in trend direction.

Continuation is not simply “price went up again.”

It is the reaffirmation of an existing ledgered relation.

---

## 16.7 Reversal event

A reversal event requires more than opposing momentum.

A mature reversal sequence is:

ExistingStructure
→ RelationalWeakening
→ BoundaryFailure
→ OpposingGateAcceptance
→ FollowThrough. (16.7)

Divergence may supply relational weakening.

Support failure may supply boundary failure.

A close and retest may supply gate acceptance.

Thus:

Divergence alone ≠ Reversal. (16.8)

Divergence + StructuralFailure + OpposingCommitment → ReversalEvidence. (16.9)

---

## 16.8 Rejection event

A rejection event occurs when the market attempts to enter a new region but fails to obtain durable acceptance.

A schematic form is:

BoundaryCross

- FailureToHold
- ReturnIntoPriorRegion
- OpposingTrace
  → RejectionEvent. (16.10)

A wick can contribute evidence.

It does not define the event alone.

---

## 16.9 Absorption

Absorption occurs when substantial aggressive flow produces limited displacement because opposing liquidity accepts it.

A schematic signature is:

HighAggressiveFlow + LowPriceDisplacement → CandidateAbsorption. (16.11)

But ordinary OHLCV data may not be sufficient to establish true absorption.

Direct order-flow evidence may be required.

Therefore absorption should remain a protocol-qualified inference.

---

## 16.10 Capitulation

Capitulation refers to a forced or emotionally compressed liquidation event.

Candidate features may include:

- exceptional volume;
- large displacement;
- volatility spike;
- breadth collapse;
- forced-flow evidence;
- rapid reversal or stabilization.

But:

HighVolumeDecline ≠ Capitulation by definition. (16.12)

Capitulation is an interpretation of participant constraint and commitment, not merely bar shape.

---

## 16.11 Fakeout

A fakeout is:

CandidateTransition

- ParticipantCommitment
  – DurableLedgerAcceptance. (16.13)

It may leave:

- trapped positions;
- damaged confidence;
- displaced stops;
- increased opposing pressure.

This makes fakeouts important event types rather than data to discard.

---

## 16.12 Event metadata

A mature event record should include:

$EventRecord_k$ =
(Type, Protocol, Boundary, PreState, Evidence,
GateDecision, GateStrength, Residual, Invalidation, Outcome). (16.14)

The event type may be:

- continuation;
- reversal;
- breakout;
- rejection;
- absorption;
- capitulation;
- fakeout;
- unresolved transition.

---

## 16.13 Event ordering

Calendar time and event order differ.

Let t denote clock time.

Let k denote admitted event order.

Then:

$t_1 < t_2 < t_3$ does not imply that every interval contains a new k. (16.15)

A market may spend months inside one episode with few decisive gates.

A crisis may produce many events within hours.

This distinction prepares the transition to Period 4.

---

## 16.14 From event to episode

An episode is not one event.

It is an ordered sequence of events maintaining enough coherence to be treated as one larger process.

The transition is:

{Event1,Event2,...,Event_m}

- StableRelationalGrammar
  → Episode. (16.16)

Examples:

- trend;
- range;
- squeeze;
- accumulation;
- distribution;
- wave sequence;
- credit deterioration;
- recovery.

---

## 16.15 Period-3 summary

| Function | Period-3 form |
|---|---|
| **Load / Memory** | positions, stops, prior attempts, event participation |
| **Motion / Relation** | displacement, acceleration, phase shift |
| **Constraint / Boundary** | tested or crossed structure |
| **Commitment / Gate** | breakout, retest, rejection, reversal |
| **Residual** | weak breadth, failed follow-through, trapped exposure |
| **Ledger** | admitted transition event |

Period 3 is the level at which Technical Analysis moves from diagnosis to event formation.

---

# 17. Period 4 — Episode

## 17.1 What is an episode?

An episode is an internally ordered sequence of market events.

Examples include:

- trend;
- trading range;
- squeeze;
- accumulation;
- distribution;
- crash;

- recovery;
- bubble;
- wave sequence;
- credit-stress cycle.

A general episode is:

Episode_m = OrderedSet({e_k}) governed by relational law $\mathscr{R}$_m. (17.1)

The relational law need not be perfectly constant.

It must be stable enough that the events belong to one interpretable process.

---

## 17.2 Load / Memory at the episode level

Episode-level Load includes:

- accumulated event history;
- position migration;
- prior gate outcomes;
- unresolved fakeouts;
- narrative development;
- volatility memory;
- institutional response.

The episode carries more than bar history.

It carries the consequences of prior commitments.

A trend after three successful retests is not structurally identical to a trend with no tested support.

A range containing repeated failed breakouts carries different residual from a newly formed range.

---

## 17.3 Motion / Relation at the episode level

Episode Motion includes:

- trend persistence;
- corrective cycles;
- impulse–correction alternation;
- volatility cadence;

- phase progression;
- feedback-sign sequence.

The regime signature $\chi$ may vary through the episode:

$\chi_k < 0 \rightarrow$ corrective segment. (17.2)

$\chi_k \approx 0 \rightarrow$ transitional segment. (17.3)

$\chi_k > 0 \rightarrow$ self-confirming segment. (17.4)

An episode can therefore be represented as:

$\text{EpisodeSignature} = \{\chi_1, \chi_2, \ldots, \chi_m\}$. (17.5)

This is more informative than assigning one fixed label to the entire episode.

---

## 17.4 Constraint at the episode level

Episode constraints include:

- range boundaries;
- trend channel;
- pattern geometry;
- volatility basin;
- institutional mandate;
- funding condition;
- narrative attractor.

A chart pattern is best interpreted as an episode-level boundary system.

For example, a triangle may combine:

- narrowing admissible region;
- accumulated positioning;
- repeated failed commitment;
- approaching transition gate.

The pattern is not a prophecy.

It is a compressed description of how episode constraints are changing.

---

## 17.5 Commitment at the episode level

Episode commitment asks whether a sequence has completed or changed regime.

Examples:

- trend termination;
- range breakout;
- volatility-regime shift;
- accumulation-to-expansion transition;
- distribution-to-decline transition;
- completed wave sequence.

The episode gate is:

$G_4(\text{EpisodeState}, L) \rightarrow \{\text{Continue, Transition, Complete, Fail}\}$. (17.6)

This gate must be stricter than a bar-level close.

An episode transition should survive:

- multiple events;
- frame transport;
- residual audit;
- new-state persistence.

---

# 17.6 Three notions of time

Episode analysis requires separating:

$t$ = calendar time;
$\theta$ = current phase orientation;
$\tau_i$ = accumulated internal phase progress;
$k$ = committed event order. (17.7)

These are not interchangeable.

$\theta$ identifies the present orientation of a conjugate state.

$\tau_i$ records accumulated phase traversal.

$k$ records the order of admitted events.

Therefore:

$\theta \neq \tau_i \neq k \neq t$. (17.8)

The phase framework emphasizes that the same phase angle can be revisited with different direction, branch history, gate history, and residual.

---

# 17.7 Internal phase time

For a validated complex state:

$Z(t) = R(t) + iQ(t) = A(t)e^{i\theta(t)}$. (17.9)

A signed unwrapped phase coordinate may be:

$\tau_i(t) = \text{Unwrap}[\theta(t)]$. (17.10)

A monotone accumulated traversal coordinate may be:

$\tau_i(t) = \int_0^t |d\theta/ds|\, ds$. (17.11)

The first preserves directional phase development.

The second measures total phase activity.

Neither should be used unless the R–Q pair has passed complex-eligibility tests.

---

# 17.8 Selection depth

Selection depth σ measures how many previously admissible futures have been suppressed.

Calendar time and selection depth can diverge:

$\Delta\sigma \neq \Delta t$. (17.12)

A long quiet range may eliminate few possibilities.

One earnings announcement may eliminate many possibilities quickly.

A possible relation is:

$d\sigma/d\tau_i \geq 0$ during genuine selection. (17.13)

But this need not hold universally.

Phase may rotate without materially suppressing alternatives.

Selection depth and phase depth are related but distinct.

---

# 17.9 Elliott Wave as episode segmentation

Elliott Wave belongs naturally at Period 4.

Its most defensible interpretation is not that markets must follow a universal numbered law.

It is that episodes may alternate between:

- self-confirming selection;
- corrective digestion;
- terminal weakening;
- regime transition.

A disciplined wave endpoint requires:

WaveEndpoint
= PivotExtreme

- GateEvidence
- RelationalShift
- DensityContext
- ResidualAudit
- CrossFrameSurvival. (17.14)

An Elliott-style count can be written:

WaveModel_P
= PivotProjection

- SegmentSequence
- χClassification
- EndpointGates
- AlternativeBranchLedger. (17.15)

The alternative branch must remain visible.

Otherwise every failure can be repaired by relabeling.

The Technical Analysis source similarly treats wave structure as recursive impulse–correction segmentation with explicit invalidation and cross-frame requirements.

---

# 17.10 Trend episode

A trend episode is not merely a rising moving average.

It is an ordered structure in which successive events preserve directional admissibility.

A simplified upward trend episode may require:

HigherCommittedLows

- HigherCommittedHighs
- Positiveχ segments
- SuccessfulRetests
- ControlledResidual. (17.16)

A trend may remain active despite local corrective $\chi < 0$ segments.

The higher-order episode depends on whether those corrections destroy or preserve the principal ledger.

---

## 17.11 Range episode

A range episode contains repeated corrective circulation between boundaries.

A simplified range condition is:

$\chi_{local} < 0$ around both extremes. (17.17)

BoundaryAcceptance outside range remains absent. (17.18)

The range ends when one boundary interaction becomes a durable event.

Thus:

RangeEpisode
→ BoundaryTest
→ Breakout or Rejection
→ NewEpisode or RangeContinuation. (17.19)

---

## 17.12 Squeeze episode

A squeeze episode combines:

- declining realized movement;
- narrowing boundaries;
- accumulating positioning;
- unresolved directional selection.

A schematic squeeze state is:

Volatility ↓

- BoundaryDistance ↓
- Load ↑ or remains elevated
- Commitment deferred. (17.20)

A squeeze does not determine direction.

It indicates growing importance of the next gate.

---

## 17.13 Episode residual

Episode residual includes:

- alternate segmentation;
- unresolved higher-timeframe conflict;
- unprocessed fakeouts;

- unclosed gaps;
- narrative contradiction;
- incomplete position unwind;
- unstable phase relation.

A mature episode model should report:

EpisodeResidual_m = Σ_k r_k + BranchResidual_m + FrameResidual_m. (17.21)

An episode with low visible volatility may still carry high residual debt.

---

## 17.14 Episode completion

An episode completes when:

1. its defining relation fails;
2. a boundary is crossed;
3. a new gate is admitted;
4. the new state persists;
5. the old interpretation no longer organizes future events.

Thus:

EpisodeComplete
≠ FinalPriceExtreme alone. (17.22)

EpisodeComplete
= RelationalFailure

- GateTransition
- LedgerReclassification. (17.23)

---

## 17.15 From episode to world

A market world forms when episodes become embedded in a larger institutional environment.

Examples:

- prolonged low-rate world;
- inflationary world;
- credit-crisis world;
- collateral-scarcity world;
- technology-bubble world;

- regulatory world;
- benchmark-dominated world.

The transition is:

EpisodeHistory

- InstitutionalAdoption
- PersistentConstraints
- SharedLedger
- Backreaction
  → MarketWorld. (17.24)

---

## 17.16 Period-4 summary

| Function | Period-4 form |
|---|---|
| **Load / Memory** | event history, positioning history, narrative memory |
| **Motion / Relation** | χ sequence, phase progression, cadence |
| **Constraint / Boundary** | pattern, range, trend basin, funding structure |
| **Commitment / Gate** | episode completion or regime transition |
| **Residual** | alternate branch, unresolved events, frame conflict |
| **Ledger** | ordered market episode |

Period 4 is the level at which Technical Analysis becomes a theory of internally ordered market processes rather than isolated indicators.

---

# 18. Period 5 — World

## 18.1 What is a market world?

A market world is not merely a large trend.

It is a stable environment in which:

- particular states are visible;
- particular relations are expected;
- particular constraints are operative;
- particular gates possess authority;
- particular traces remain consequential;
- participants adapt their behaviour accordingly.

A market world can be represented:

World_P = (X,q,φ,P,L,R,U). (18.1)

where:

X = state space;
q = baseline;
φ = feature map;
P = observation and intervention protocol;
L = ledger;
R = residual structure;
U = admissible update or revision rules.

This follows the broader declaration principle that a world becomes readable only after boundary, observation, horizon, intervention, baseline, feature map, gate, trace, and residual rules are declared.

---

# 18.2 Examples of market worlds

Examples include:

- low-volatility liquidity world;
- inflation world;
- deleveraging world;
- collateral-constrained world;
- growth-stock valuation world;
- commodity-shortage world;
- sovereign-risk world;
- policy-support world;
- indexation-dominated world;
- accounting-impairment world.

The same price may carry different meaning in different worlds.

A high valuation multiple may be ordinary in one discount-rate world and unstable in another.

A credit spread may signal opportunity in one liquidity regime and insolvency risk in another.

---

# 18.3 Load / Memory at the world level

World-level Load includes:

- institutional balance sheets;
- accumulated leverage;
- benchmark composition;
- collateral structure;

- legal obligations;
- accounting classifications;
- policy history;
- dominant narratives;
- long-term positioning.

This is deeper than chart memory.

It is the infrastructure that conditions which chart structures can matter.

A world-level Load may be compressed through protocol-bound coordinates such as:

$\Xi_fin = (\rho,\gamma,\nu)$. (18.2)

where:

$\rho$ = loading or occupancy;
$\gamma$ = lock-in or structural rigidity;
$\nu$ = agitation, turbulence, or dephasing.

This triple is an effective control interface, not a claim about fundamental market ontology. The PORE framework explicitly distinguishes the rich field $\Sigma$ from protocol-compiled effective coordinates $\Xi$.

---

# 18.4 Motion / Relation at the world level

World-level Motion includes:

- reflexive feedback;
- policy transmission;
- balance-sheet expansion or contraction;
- valuation-frame rotation;
- collateral propagation;
- cross-asset phase alignment;
- institutional adaptation.

A world may be:

- corrective;
- self-confirming;
- metastable;
- transitional;
- collapsing;

- recovering.

The signature χ becomes an environmental descriptor rather than merely a local indicator modifier.

For example:

$\chi_{world} > 0$ may describe a broad self-reinforcing leverage or momentum regime. (18.3)

$\chi_{world} < 0$ may describe a stabilizing value or policy-feedback regime. (18.4)

But world-level χ should be inferred from multiple channels, not from one oscillator.

---

## 18.5 Constraint / Boundary at the world level

World-level constraints include:

- monetary policy;
- collateral law;
- accounting rules;
- benchmark mandates;
- capital regulation;
- legal enforceability;
- settlement systems;
- institutional risk tolerance;
- technological infrastructure.

These constraints may be invisible on a price chart until they become active.

Examples:

- margin rules amplify liquidation;
- benchmark rules force rebalancing;
- accounting recognition changes reported solvency;
- collateral eligibility changes funding capacity;
- legal intervention changes transferability.

A world-level boundary therefore has formal and institutional force.

---

## 18.6 Commitment / Gate at the world level

World-level gates include:

- policy decision;

- default;
- legal judgment;
- accounting reclassification;
- index inclusion;
- regulatory intervention;
- collateral call;
- bankruptcy;
- regime declaration;
- structural market closure.

These events do more than alter price.

They change what future actions are admissible.

Thus:

WorldGate
→ InstitutionalLedgerUpdate
→ ChangedActionSpace. (18.5)

This is the deepest form of commitment in the periodic table.

---

## 18.7 A world requires more than dynamics

A system may have complex dynamics without becoming a time-bearing world.

A fuller world requires:

DeclaredState

- InternalOrdering
- Gate
- PersistentTrace
- Residual
- Backreaction. (18.6)

A complex phase alone is insufficient.

A price cycle alone is insufficient.

A recurring pattern alone is insufficient.

Worldhood requires that selected events enter a ledger and alter future admissibility.

This is the central progression of the phase-world framework:

Z
→ θ
→ $\tau_i$
→ Gate
→ Trace
→ Ledger
→ Backreaction. (18.7)

---

# 18.8 CAPM valuation world

The CAPM complex construction provides a disciplined example of a local valuation world.

It declares:

- one cash flow;
- baseline discount protocol;
- CAPM-adjusted discount protocol;
- horizon;
- orientation convention.

It then defines:

A = baseline value. (18.8)

R = admitted CAPM value. (18.9)

$Q = \sqrt{(A^2 - R^2)}$. (18.10)

$Z = R + iQ$. (18.11)

$\partial R/\partial \theta = -Q$. (18.12)

But this local complex geometry does not automatically produce a full world.

A full financial world additionally requires:

- actual phase movement;
- economic consequence;
- recognition gate;
- ledger;
- residual;
- observer backreaction.

The CAPM construction therefore serves as a calibration atom inside the broader world architecture.

---

# 18.9 Multiple overlapping worlds

A financial asset may belong simultaneously to several worlds:

- company operating world;
- sector world;
- equity-index world;
- option-hedging world;
- funding world;
- accounting world;
- regulatory world;
- narrative world.

Each world has:

- different boundaries;
- different observers;
- different gates;
- different ledgers;
- different timescales.

Therefore:

AssetState
= Intersection of multiple protocol-bound worlds. (18.13)

A technical pattern may be valid inside one world and overridden by another.

For example:

- bullish price structure;
- deteriorating credit world;
- supportive policy world;
- restrictive collateral world.

This is not necessarily contradiction.

It may reflect layered world membership.

---

## 18.10 Cross-world transport

A claim should not be copied directly between worlds.

Transport is required:

T_{World_A→World_B}(Claim_A) → Claim^_B. (18.14)

Examples:

- equity breakout translated into credit implications;
- commodity move translated into inflation world;
- volatility regime translated into option-hedging world;
- accounting impairment translated into market-price world.

Different worlds may respond with delay or asymmetry.

Thus:

Event in World_A may become only residual pressure in World_B. (18.15)

It may become a gate in World_B later.

---

## 18.11 World-level backreaction

A world-level interpretation can change institutions.

Examples:

- “systemic crisis” triggers policy action;
- “bubble” changes leverage rules;
- “safe asset” designation changes demand;
- “default risk” changes collateral;
- “breakout market” attracts trend capital.

The sequence is:

WorldInterpretation
→ InstitutionalAction
→ ConstraintChange
→ MarketRepricing
→ RevisedWorldInterpretation. (18.16)

At this level, Technical Analysis is no longer merely chart reading.

It participates in institutional world formation.

---

## 18.12 World revision

Worlds must be revisable.

But revision cannot erase prior trace.

Let D_k be the world declaration at episode k:

D_k = (q_k,φ_k,P_k,Gate_k,TraceRule_k,ResidualRule_k). (18.17)

A mature revision is:

$$D_{k+1} = U_a(D_k,L_k,R_k). \quad (18.18)$$

where $U_a$ is admissible.

The revision should preserve:

- prior claims;
- gate history;
- residual;
- authority;
- reason for change.

The self-revising declaration framework treats mature observerhood as stable, trace-preserving, residual-honest revision rather than arbitrary rule changing.

---

# 18.13 World pathology

World-level pathologies include:

### Dogmatic world

Contradictory residual is suppressed.

### Volatile world

Declarations change too frequently for stable trace.

### Semantic black hole

One interpretation absorbs all evidence and prevents revision.

### Silent ontology drift

The object being analyzed changes without declaration.

### Gate capture

A powerful authority controls which events become official.

### Ledger fragmentation

Different observers cannot align consequential records.

### Reflexive bubble

World interpretation produces the flows that temporarily validate it.

These pathologies are especially important because world-level errors reshape lower-level data.

## 18.14 Period-5 summary

| Function | Period-5 form |
|---|---|
| **Load / Memory** | institutional history, balance sheets, leverage, benchmark depth |
| **Motion / Relation** | reflexivity, policy transmission, valuation-frame dynamics |
| **Constraint / Boundary** | law, collateral, accounting, regulation, infrastructure |
| **Commitment / Gate** | default, policy shift, legal or accounting recognition |
| **Residual** | unresolved systemic risk, competing world interpretation |
| **Ledger** | institutional market world with backreaction |

Period 5 is the level of self-referential, protocol-bound market worlds.

---

# 19. The Periodic Generation Law

The six periods can now be connected.

## 19.1 Mark to Window

Executed and recorded marks are aggregated into windows:

Marks → Bars. (19.1)

## 19.2 Window to Structure

Repeated window relations generate persistent structure:

Bars → Memory, Motion, Density, Boundaries. (19.2)

## 19.3 Structure to Event

A structure encounters and passes or fails a consequential gate:

Structure + BoundaryTest → Event. (19.3)

## 19.4 Event to Episode

Ordered events form a higher-level process:

Events + RelationalGrammar → Episode. (19.4)

## 19.5 Episode to World

Episodes become embedded in persistent institutional rules and shared ledgers:

Episodes + Institutions + Backreaction → World. (19.5)

## 19.6 World to Mark

The world changes the production of new marks:

World
→ Rules, Capital, Expectations, Constraints
→ Orders
→ Trades
→ NewMarks. (19.6)

The complete recursion is:

Mark
→ Window
→ Structure
→ Event
→ Episode
→ World
→ NewMark. (19.7)

This is the article's primary periodicity.

It is not a fixed price cycle.

It is the recurrence of market functions across levels of closure.

---

# 20. The Six-Period, Four-Family Matrix

| Period | Load / Memory | Motion / Relation | Constraint / Boundary | Commitment / Gate |
|---|---|---|---|---|
| **0 Mark** | displayed depth, resting orders, local inventory | tick change, spread, imbalance | bid–ask boundary, tick rule | execution, cancellation, rejection |
| **1 Window** | opening state, bar volume, prior close | return, body, range position, gap | high–low range, local envelope | official close |
| **2 Structure** | MA, VWAP, OBV, profile, breadth | momentum, RSI, MACD, divergence | support, resistance, channels, bands | structural acceptance |
| **3 Event** | positions, stops, participation around test | displacement, acceleration, phase shift | tested or broken level | breakout, retest, rejection, reversal |
| **4 Episode** | event history, positioning history, narrative memory | $\chi$ sequence, phase progress, cadence | range, pattern, trend basin | episode completion, regime transition |
| **5 World** | institutional memory, leverage, benchmark and balance-sheet load | reflexivity, frame transport, valuation dynamics | legal, policy, accounting and collateral structure | recognized institutional state change |

Every cell remains governed by:

**Residual preservation**
**Cross-frame transport**
**Ledgered backreaction**

The next part will turn this matrix into a true **Proto-Periodic Table of Market Observation**, explain why certain methods occupy more than one cell, and show how familiar

Technical Analysis methods can be decomposed into functional compounds rather than treated as indivisible indicators.

# Part V — The Proto-Periodic Table of Market Observation

## 21. How to Read the Table

## 21.1 The table is not a list of indicators

The proposed table should not be read as though each cell contains one preferred trading tool.

Its purpose is more fundamental.

The rows identify increasing levels of market closure:

**Mark**
**Window**
**Structure**
**Event**
**Episode**
**World**

The columns identify recurring market functions:

**Load / Memory**
**Motion / Relation**
**Constraint / Boundary**
**Commitment / Gate**

A cell therefore means:

> How does one recurring market function appear at one particular level of closure?

For example:

- Period-0 Load appears as order-book depth or resting interest.
- Period-1 Load appears as bar volume and carried opening state.
- Period-2 Load appears as moving-average memory, VWAP, OBV, breadth, or volume profile.
- Period-3 Load appears as participation and positioning around an event.
- Period-4 Load appears as accumulated event history and episode memory.
- Period-5 Load appears as leverage, institutional memory, benchmark depth, and balance-sheet structure.

These are not six unrelated objects.

They are six manifestations of the same functional question:

> What consequential structure is already being carried at this level?

---

## 21.2 The four groups are functional families

The four columns should be interpreted as families rather than substances.

### Load / Memory

What has accumulated and remains capable of affecting future behaviour?

### Motion / Relation

How does the current state differ, respond, align, diverge, rotate, or amplify?

### Constraint / Boundary

What channels, resists, confines, compresses, or separates admissible states?

### Commitment / Gate

Which candidate transition becomes accepted trace with future consequence?

The four families form a recurrent grammar:

$\text{Load}_n \rightarrow \text{Motion}_n$ under $\text{Constraint}_n \rightarrow \text{Commitment}_n$. (21.1)

The resulting commitment updates the ledger:

$L_{n+1} = \text{Update}(L_n, e_n, r_n)$. (21.2)

The updated ledger and attached residual become part of the next level's Load:

$\text{Load}_{n+1} = \text{Compile}(L_{n+1}, r_n, \text{Environment}_{n+1})$. (21.3)

This is the table's periodic-generation law.

---

## 21.3 The rows are closure periods

A higher period does not merely contain more data.

It contains a deeper form of closure.

A bar is not simply many trades.

It is a declared window that compresses marks into one closing state.

A technical structure is not simply many bars.

It is a persistent relation extracted through filtering, comparison, or accumulation.

An event is not simply a large structure.

It is a candidate transition that passes or fails a consequential gate.

An episode is not simply many events.

It is an internally ordered sequence governed by a sufficiently stable relational grammar.

A world is not simply a long episode.

It is an institutional environment with persistent boundaries, authoritative gates, shared ledgers, residual governance, and backreaction.

Thus:

More observations ≠ higher period. (21.4)

Deeper commitment structure → higher period. (21.5)

---

# 21.4 The table has an outer protocol shell

No cell has meaning outside a declared protocol.

The general protocol remains:

P = (B,Δ,h,u). (21.6)

where:

B = boundary;
Δ = observation or aggregation rule;
h = time or state window;
u = admissible intervention family.

For Technical Analysis:

P_TA = (Asset,Universe,Timeframe,Scale,BarRule,FeatureMap,GateRule,ResidualRule). (21.7)

The table should therefore be imagined inside an outer shell:

```
                DECLARED PROTOCOL P

          Load      Motion      Constraint      Commitment

Mark
Window
Structure
Event
Episode
World

      Residual · Transport · Ledger · Revision
```

The table does not describe “the market as such.”

It describes a declared market world seen by a bounded observer.

The declaration framework makes the same requirement explicit: boundary, baseline, feature map, observation rule, horizon, gate, trace, and residual must be declared before a field can become a readable world.

---

# 21.5 Every cell has three governance rails

A table cell is incomplete unless it also reports:

### Residual

What did the cell's representation fail to absorb?

### Transport

Does the claimed relation survive a legitimate change of protocol or frame?

### Ledger consequence

Did the observed structure alter future market behaviour?

Therefore, a mature cell record is not simply:

Cell = IndicatorName. (21.8)

It is closer to:

Cell_{p,g,P}
= (Object,Observable,Operator,Gate,Residual,Transport,LedgerEffect). (21.9)

where:

p = period;
g = functional group;
P = protocol.

This cell formalism is introduced here as a synthesis of the four source frameworks. It is not explicitly stated in any single source article.

---

## 21.6 Environment modifiers are not additional columns

Several important variables condition the table without becoming ordinary element groups.

### χ — feedback signature

χ describes whether movement generates:

- corrective pressure;
- weak or ambiguous response;
- self-confirming pressure.

### Ξ — effective control state

A protocol-fixed market control state may be compiled as:

Ξ_fin = (ρ,γ,ν). (21.10)

where:

ρ = loading;

γ = lock-in or rigidity;
ν = agitation, turbulence, or dephasing.

Ξ is an instrument-panel representation of the richer market field, not a claim that the market is fundamentally three-dimensional. The PORE source explicitly separates the rich Σ-layer from the protocol-compiled Ξ-layer.

### Z — optional conjugate state

Where justified:

Z = R + iQ. (21.11)

But Z is available only after the proposed R–Q pair passes a complex-eligibility test.

### Observer role

A private analyst, market maker, regulator, accounting body, and central bank do not possess the same gate authority or intervention capacity.

These constructs behave more like:

- environmental conditions;
- state descriptors;
- algebraic eligibility;
- observer metadata.

They should surround the table rather than expand it into an unreadable grid.

---

# 22. The Element Record

# 22.1 What counts as an “element”?

The word element must be used carefully.

A chemical element is a physical substance defined by atomic number.

Nothing equally literal exists here.

A market-observation element is instead a canonical functional role that recurs across protocols and scales.

It can be defined as:

Element_{p,g}
= minimal reusable market-observation role at period p and functional group g. (22.1)

Examples:

- Mark × Commitment → execution;
- Window × Commitment → close;

- Structure × Constraint → support or resistance;
- Event × Commitment → breakout admission or rejection;
- Episode × Motion → phase progression or χ sequence;
- World × Constraint → legal, policy, accounting, or collateral boundary.

These are not indivisible market particles.

They are minimally useful grammatical units.

The term **proto-periodic table** remains appropriate because the classification is still a proposed research architecture rather than a completed empirical taxonomy.

---

# 22.2 The seven fields of an element record

Every canonical element should contain seven fields.

### 1. Object

What is being observed?

### 2. Observable

What data represent the object?

### 3. Dominant function

Which of the four functional families is primary?

### 4. Closure rule

What makes the object sufficiently real at its period?

### 5. Residual

What remains unresolved?

### 6. Transport test

What frame change should the claim survive?

### 7. Ledger consequence

How does the admitted structure alter future behaviour?

A template is:

```
Element name:
Period:
Functional family:
Declared protocol:
Observed object:
Primary observable:
```

```
Closure rule:
Residual:
Transport test:
Ledger consequence:
Invalidation:
```

This gives Technical Analysis methods a more disciplined description than a formula followed by “buy” or “sell.”

---

## 22.3 Example: the closing price element

**Element name:** Closing Gate
**Period:** Window
**Functional family:** Commitment
**Observed object:** terminal state of a declared window
**Primary observable:** C_j
**Closure rule:** final admitted price under the window convention
**Residual:** intrawindow path, wick, hidden liquidity, later correction
**Transport test:** compare time bars, volume bars, and higher-window closure
**Ledger consequence:** updates returns, indicators, marks, reports, and later gate status
**Invalidation:** vendor correction, auction anomaly, changed window protocol

The closing price is therefore not merely one of four OHLC numbers.

It is the characteristic commitment operator of the Window period.

---

## 22.4 Example: structural support

**Element name:** Loaded Boundary
**Period:** Structure
**Functional family:** Constraint
**Observed object:** historically consequential price region
**Primary observables:** repeated reaction, volume density, anchored references, positioning proxies
**Closure rule:** sufficient persistence and future conditional use
**Residual:** changed participants, stale memory, weak institutional relevance
**Transport test:** timeframe, log scale, volatility normalization, nearby anchor perturbation
**Ledger consequence:** concentrates future orders and risk decisions
**Invalidation:** durable acceptance through the zone or loss of operative relevance

This description avoids treating a horizontal line as an unexplained force.

---

## 22.5 Example: breakout event

**Element name:** Boundary Admission
**Period:** Event
**Functional family:** Commitment
**Observed object:** transition from one structural region to another
**Primary observables:** displacement, close, volume, breadth, follow-through, retest
**Closure rule:** declared gate strength exceeds threshold
**Residual:** weak participation, higher-frame contradiction, absent retest, event risk
**Transport test:** volatility normalization and higher-timeframe survival
**Ledger consequence:** former boundary may become new reference; positions and algorithms

update
**Invalidation:** return into old region with opposing acceptance

The breakout is therefore a compound event built around one dominant element: Event-level Commitment.

---

# 23. Methods as Molecules

## 23.1 Why named methods span several cells

Most Technical Analysis methods cannot be placed in exactly one cell.

A candlestick contains:

- Window Load through local volume;
- Window Motion through body and range position;
- Window Constraint through high and low;
- Window Commitment through close;
- Residual through wicks and lost intrawindow path.

A breakout contains:

- Structure-level Load;
- Structure-level Boundary;
- Event-level Motion;
- Event-level Commitment;
- residual and transport tests.

Elliott Wave contains:

- Structure-level pivot extraction;
- Event-level gate identification;
- Episode-level relational sequencing;
- branch residual;
- cross-frame transport.

Named methods are therefore molecules:

$$\text{Method}_j = \text{Compose}(\text{Element}_1, \text{Element}_2, \ldots, \text{Element}_m \mid P_j). \quad (23.1)$$

The method's **primary placement** should indicate its dominant function and period.

Its secondary placements should show the additional cells required for complete interpretation.

---

# 23.2 Dominant cell versus full composition

For example:

## Moving average

Primary placement:

Structure × Load / Memory.

Composition:

MovingAverage
= WindowPriceTrace

- MemoryFilter
- Structure-level reference. (23.2)

## MACD

Primary placement:

Structure × Motion / Relation.

Composition:

MACD
= TwoFilteredMemories

- DifferenceRelation
- SecondarySignalFilter. (23.3)

## Bollinger Bands

Primary placement:

Structure × Constraint / Boundary.

Composition:

BollingerBands
= FilteredMemory

- DispersionMeasure
- AdaptiveBoundary. (23.4)

## Breakout

Primary placement:

Event × Commitment / Gate.

Composition:

Breakout
= LoadedStructure

- Boundary
- Displacement
- GateEvidence
- ResidualAudit. (23.5)

Primary placement makes the table readable.

Full composition prevents oversimplification.

---

# 23.3 Molecular formula notation

A compact notation can be introduced.

Let:

L_p = Load function at period p;
M_p = Motion function at period p;
C_p = Constraint function at period p;
G_p = Commitment function at period p. (23.6)

Then:

MovingAverage ≈ $L_2$. (23.7)

MACD ≈ Δ($L_2$^fast,$L_2$^slow) → $M_2$. (23.8)

RSI ≈ Normalize[Σ(UpMoves),Σ(DownMoves)] → $M_2$. (23.9)

VolumeProfile ≈ $L_2$ + $C_2$. (23.10)

Candlestick ≈ $L_1$ + $M_1$ + $C_1$ + $G_1$ + $r_1$. (23.11)

Breakout ≈ $L_2$ + $M_3$ + $C_2$ + $G_3$ + $r_3$. (23.12)

ChartPattern ≈ $L_4$ + $M_4$ + $C_4$ + $G_4$ + $r_4$. (23.13)

ElliottWave ≈ Π_pivot + $M_4$ + $G_3/G_4$ + T + r_branch. (23.14)

Gann ≈ $C_2/C_4$ + T_anchor,scale,time + r_frame. (23.15)

These expressions are conceptual decompositions, not established algebraic identities.

Their purpose is to expose role composition.

---

# 23.4 Molecular redundancy

Two methods may appear different while using nearly the same functional ingredients.

For methods A and B, define a qualitative overlap:

Overlap(A,B)
= SharedInputs

- SharedOperators
- SharedHorizon
- SharedRegimeAssumptions. (23.16)

A normalized redundancy score could be proposed:

Redundancy(A,B) ∈ [0,1]. (23.17)

High redundancy means that agreement between A and B provides limited new information.

Examples:

### High likely redundancy

- RSI and stochastic;
- fast/slow moving-average crossover and MACD sign;
- several trend indicators built from the same closing-price history.

### Lower likely redundancy

- price structure and market breadth;
- moving-average memory and option-implied positioning;
- breakout close and volume-profile acceptance;
- momentum divergence and legal/accounting gate.

The score requires empirical construction and should not be assumed from labels alone.

---

# 23.5 Functional independence

A stronger confirmation system should reward support from different families and periods.

Let evidence sources be E1…E_n.

A provisional confirmation score is:

Confirmation
= Σ_i w_i Evidence_i
− λ Σ_{i<j} Redundancy(E_i,E_j)
− μ ResidualBurden. (23.18)

The first term rewards supporting evidence.

The second penalizes duplicated information.

The third penalizes unresolved contradiction.

This formalizes the principle:

> Five indicators are not five confirmations when they are transformed versions of the same price history.

---

# 23.6 A breakout confirmation molecule

A functionally diverse breakout confirmation may contain:

### Load

Relative volume, open interest, participation, or positioning.

### Motion

Normalized displacement, momentum acceleration, or breadth coherence.

### Constraint

A predeclared and transport-tested boundary.

### Commitment

Close, follow-through, or retest acceptance.

### Residual

Higher-timeframe conflict, poor liquidity, event risk, or narrow participation.

A schematic compound is:

$$\text{BreakoutConfidence} = f(L_2, M_3, C_2, G_3, T, r_3). \quad (23.19)$$

This is more mature than:

Three oscillators agree, therefore breakout confirmed. (23.20)

---

# 23.7 Molecular instability

A method may be unstable for several reasons.

### Input instability

Small data changes alter the result.

### Parameter instability

Small window changes alter the result.

### Anchor instability

Small anchor changes alter the result.

### Branch instability

Several interpretations remain equally admissible.

### Regime instability

The method's assumed $\chi$ changes.

### Gate instability

The confirmation rule changes retrospectively.

### Transport instability

The method fails under reasonable reframing.

A molecular atlas should report these instability modes for every major method.

---

# 24. Periodic Homology

# 24.1 The same function reappears at higher scales

The most important evidence for periodicity is functional homology.

A homologous structure performs a related role at different levels even when its material realization changes.

The market table contains four homologous columns.

---

# 24.2 Load homology

| Period | Load / Memory manifestation |
|---|---|
| **Mark** | resting orders, displayed depth, local inventory |
| **Window** | bar volume, opening state, prior close |
| **Structure** | moving average, VWAP, OBV, profile, breadth |
| **Event** | positioning and participation around the transition |
| **Episode** | accumulated event history and unresolved residual |
| **World** | leverage, institutional memory, benchmark and balance-sheet load |

The recurring question is:

What has been carried forward?

---

# 24.3 Motion homology

| Period | Motion / Relation manifestation |
|---|---|
| **Mark** | tick displacement, imbalance, spread movement |
| **Window** | body, return, gap, range position |

| Period | Motion / Relation manifestation |
|---|---|
| **Structure** | momentum, MACD, RSI, divergence, relative strength |
| **Event** | acceleration, displacement, phase shift |
| **Episode** | χ sequence, cadence, internal phase progress |
| **World** | reflexive feedback, valuation-frame movement, policy transmission |

The recurring question is:

How is the current state changing relative to another state or frame?

---

## 24.4 Constraint homology

| Period | Constraint / Boundary manifestation |
|---|---|
| **Mark** | bid, ask, tick size, order limit |
| **Window** | high, low, range, local envelope |
| **Structure** | support, resistance, value area, band, channel |
| **Event** | defended, crossed, or rejected transition surface |
| **Episode** | range, pattern, trend basin, funding condition |
| **World** | law, policy, collateral, accounting, infrastructure |

The recurring question is:

What separates admissible states and resists transition?

---

## 24.5 Commitment homology

| Period | Commitment / Gate manifestation |
|---|---|
| **Mark** | execution, cancellation, rejection |
| **Window** | official close |
| **Structure** | persistent structural acceptance |
| **Event** | breakout, retest, rejection, reversal |
| **Episode** | completion or regime transition |
| **World** | legal, accounting, policy, default, or institutional recognition |

The recurring question is:

What becomes binding history?

---

## 24.6 Vertical inheritance

The periods are not independent layers.

Each higher period inherits the ledgered output of the lower one.

A compact inheritance relation is:

Field_{p+1} = CoarseGrain(Ledger_p,Residual_p | P_{p+1}). (24.1)

This is not simple aggregation.

The next level selects which lower-level traces remain relevant.

Thousands of trades may compress into one candle.

Many candles may compress into one structural level.

Several structural tests may compress into one breakout event.

Many events may compress into one trend episode.

Several episodes may become one institutional regime.

The filtration article similarly distinguishes the full field from viewpoint-selected filtration and defines time through the ordered ledger of selected disclosure rather than through raw recursion.

---

# 24.7 Downward constraint

Causation also runs downward.

A World-level rule changes Episode-level dynamics.

An Episode-level regime changes Event-level meaning.

An Event-level commitment changes Structure-level boundaries.

A Structure-level reference changes Window-level interpretation.

A Window-level gate conditions Mark-level execution.

Thus:

World → Episode → Event → Structure → Window → Mark. (24.2)

The periodic table therefore supports two directions.

### Upward construction

Marks are compiled into worlds.

### Downward governance

Worlds constrain the production and meaning of marks.

The combined architecture is:

Bottom-up trace formation + top-down admissibility. (24.3)

---

# 24.8 The table is recursive, not merely hierarchical

A static hierarchy would place Mark below Window, Window below Structure, and so on.

The market is more recursive.

A World-level interpretation changes Mark-level order flow.

Those marks later update the World.

Therefore:

World_k → Marks_{k+1} → … → World_{k+1}. (24.4)

This closes the self-reference loop.

The market is not merely built from lower levels.

Its higher-level ledgers feed back into the lower-level field.

---

# 25. Empty Cells and Missing Instrument Families

## 25.1 Why empty cells matter

A classification system becomes scientifically useful when it reveals not only where known methods belong, but where important methods are missing.

Some table cells are populated by mature instruments.

Others are represented only by loose practitioner intuition.

The empty or weakly populated cells therefore define a research programme.

---

## 25.2 Mark-level residual instruments

Ordinary charts discard much of the microstructure residual.

A mature Mark-level residual instrument might estimate:

- hidden liquidity;
- cancelled pressure;
- unfilled interest;
- order-book deception;
- cross-venue disagreement;
- execution shortfall.

These instruments exist in specialist microstructure analysis but are rarely integrated with higher-period Technical Analysis.

The missing bridge is:

MarkResidual → WindowInterpretation → EventGate. (25.1)

---

## 25.3 Window-level path instruments

OHLCV bars lose the intrawindow path.

Two bars with identical OHLCV can result from very different sequences.

A path-sensitive window instrument could preserve:

- number of reversals;
- time spent near extremes;
- order of high and low;
- volume concentration through the path;
- closing recovery speed.

This would reduce the residual created by candle compression.

---

## 25.4 Structure-level base-completeness instruments

Most familiar indicators use price alone.

A more complete structure instrument would report whether its observed Base includes:

- price memory;
- transaction density;
- breadth;
- positioning;
- liquidity;
- volatility;
- institutional constraint.

Define a provisional completeness score:

S_base
= Σ_j w_j AvailableChannel_j. (25.2)

A low score would not make the indicator useless.

It would clarify which relevant structures it cannot observe.

---

## 25.5 Event-level gate-strength instruments

Breakout tools often use one threshold:

Price > resistance. (25.3)

A richer gate-strength instrument would combine:

- normalized displacement;

- close quality;
- participation;
- breadth;
- follow-through;
- retest;
- residual burden.

This is one of the clearest underdeveloped regions of the table.

---

# 25.6 Episode-level phase-time instruments

Most episode analysis remains tied to bar count or calendar duration.

The phase framework suggests testing whether episodes align better under accumulated phase progress:

$\tau_i(t) = \int_0^t |\dot{\theta}(s)|\, ds.$ (25.4)

A useful phase-time instrument would test:

$\mathrm{Var}[\mathrm{EpisodePath} \mid \tau_i] < \mathrm{Var}[\mathrm{EpisodePath} \mid t].$ (25.5)

It would also test whether gate hazard concentrates by phase:

$\Pr(\mathrm{Gate} \mid \theta, \mathrm{Controls}) > \Pr(\mathrm{Gate} \mid t, \mathrm{Controls}).$ (25.6)

No mature universal instrument currently occupies this cell.

---

# 25.7 World-level observer-backreaction instruments

Technical Analysis rarely measures how strongly a signal is influencing the market it observes.

A World-level reflexivity instrument might estimate:

- signal visibility;
- strategy adoption;
- capital linked to the signal;
- order synchrony;
- crowding;
- liquidity capacity;
- adversarial exploitation.

A provisional score is:

$S_{reflex}$

= Visibility
× Adoption
× CapitalLinked
× ActionCoherence
÷ LiquidityCapacity. (25.7)

This remains a research hypothesis because the required variables are difficult to observe.

---

# 25.8 Frame-transport instruments

Multi-timeframe analysis is often informal.

A mature transport instrument should explicitly map:

Claim_P → Claim^_{P′}. (25.8)

It should report:

- preserved relation;
- transformed parameters;
- lost structure;
- new residual;
- failure reason.

This would turn “the weekly chart confirms the daily chart” into a reproducible claim.

---

# 25.9 Complex-eligibility instruments

Many financial models introduce imaginary coordinates because complex numbers are suggestive.

A proper eligibility instrument should compare:

## Scalar model

X_t.

## Real-pair model

(X_t,Y_t).

## Complex model

Z_t = R_t + iQ_t.

The complex model earns priority only if it improves:

- compression;
- prediction;

- dynamical simplicity;
- phase alignment;
- gate concentration;
- transport robustness.

If not:

Complex model → Real pair or scalar reduction. (25.9)

This reduction principle is central to the phase article's evidence ladder and rejection rules.

---

## 25.10 Residual-adjusted confirmation

Most confirmation scores count supporting signals.

A mature score should subtract unresolved contradictions:

Confirmation_net
= Confirmation_support
− Residual_burden
− Redundancy_penalty. (25.10)

This combines three insights:

- support matters;
- repeated versions of the same information should not be overcounted;
- unresolved contradictions should remain visible.

---

# 26. What Makes the Table "Proto"?

The table is called proto-periodic because several tasks remain incomplete.

## 26.1 Cell definitions need operational consensus

Different researchers may disagree about:

- period assignment;
- dominant functional group;
- closure rule;
- residual type;
- transport standard.

These disagreements must be measured rather than hidden.

---

## 26.2 Inter-rater reliability must be tested

Independent analysts should classify the same method.

Let κ_class denote agreement beyond chance.

The framework requires:

κ_class > declared minimum threshold. (26.1)

Low agreement would show that the table remains too ambiguous.

---

## 26.3 The periodic law requires empirical testing

The proposed recurrence is structural:

Commitment_p + Residual_p → Load_{p+1}. (26.2)

But its predictive or explanatory value must be tested.

Questions include:

- Do event records improve episode classification?
- Does residual history improve regime detection?
- Do higher-period boundaries predict lower-period gate behaviour?
- Does functional diversification improve confirmation quality?

---

## 26.4 Empty cells may reflect bad theory rather than missing tools

An apparently empty cell may mean:

- the instrument has not yet been invented;
- the function cannot be measured;
- the distinction is unnecessary;
- the proposed period is wrong;
- the table is overstructured.

The framework must allow the table itself to be revised.

---

## 26.5 The table must preserve its failed classifications

The declaration revision rule applies to the table:

Table_{k+1} = U_a(Table_k,EmpiricalLedger_k,Residual_k). (26.3)

Revision should preserve:

- old cell definitions;
- methods moved between cells;
- reasons for movement;
- failed predictions;
- unresolved disputes.

This follows the broader admissible self-revision principle: a mature observer revises its declaration without erasing the trace that forced revision.

---

# Part VI — The Molecular Atlas of Technical Analysis

The periodic table identifies recurring roles.

The molecular atlas now asks:

> How are familiar Technical Analysis methods assembled from those roles?

The next sections will examine:

- moving averages, crossovers, and MACD;
- RSI, stochastic, ATR, and volatility bands;
- volume, OBV, VWAP, and volume profile;
- candlesticks, levels, patterns, and breakouts;
- breadth, Elliott Wave, Fibonacci, and Gann.

Each method will be analyzed through the same fields:

**Primary period**
**Primary functional family**
**Operator composition**
**Regime assumption**
**Gate requirement**
**Residual**
**Transport burden**
**Common category error**.

# Part VI — The Molecular Atlas of Technical Analysis

The periodic table classifies recurring market functions.

The molecular atlas now decomposes familiar methods into combinations of those functions.

For each method, the analysis will identify:

1. primary period;

2. primary functional family;
3. operator composition;
4. implicit regime assumption;
5. commitment requirement;
6. residual;
7. transport burden;
8. common category error.

The purpose is not to replace familiar formulas. It is to show what each formula actually measures, what it omits, and why methods that appear independent may carry almost the same information.

---

# 27. Moving Averages, Crossovers, and MACD

# 27.1 Moving average as declared memory

A moving average is primarily a **Structure-period Load / Memory instrument**.

For a price series $P_t$, a general moving average is:

$$MA_n(t) = \Sigma_{j=0}^{n-1} w_j P_{t-j}. \quad (27.1)$$

with:

$$\Sigma_{j=0}^{n-1} w_j = 1. \quad (27.2)$$

The weights define what part of the past remains visible.

A simple moving average gives equal weight to the declared window.

An exponential moving average gives greater weight to recent observations:

$$EMA_t = \alpha P_t + (1 - \alpha) EMA_{t-1}. \quad (27.3)$$

where:

$$\alpha = 2/(n + 1). \quad (27.4)$$

The primary meaning is not:

The moving average knows the future trend. (27.5)

It is:

The moving average is a declared memory centre produced from historical price. (27.6)

The source Technical Analysis article accordingly classifies moving averages as declared memory filters that measure broad trend memory while missing volume, density, commitment, and early phase change.

---

## 27.2 The moving average as filtration

The moving average performs two operations.

First, it suppresses high-frequency variation.

Second, it preserves a slower component of the price trace.

This can be written:

PriceTrace → Filter_n → MemoryState_n. (27.7)

The resulting line is neither the whole price history nor an independently observed market variable.

It is a transformed projection of price.

Therefore:

MA_n contains no information not ultimately derived from the declared input series. (27.8)

Its usefulness comes from:

- compression;
- noise suppression;
- comparison of memory horizons;
- stable reference construction;
- shared market use.

It does not come from creating new raw information.

---

## 27.3 Price–memory relation

The simplest relational extension is:

$d_n(t) = P_t - MA_n(t)$. (27.9)

A normalized version is:

$\eth_n(t) = [P_t - MA_n(t)]/\sigma_n(t)$. (27.10)

where $\sigma_n$ is a declared volatility scale.

The first quantity measures nominal distance from memory.

The second measures distance relative to recent agitation.

Neither quantity determines whether the price should revert or continue.

That interpretation depends on $\chi$.

Under corrective feedback:

đ_n large and positive may increase reversion pressure. (27.11)

Under self-confirming feedback:

đ_n large and positive may reflect persistent directional selection. (27.12)

Thus:

DistanceFromMemory + RegimeEvidence → Interpretation. (27.13)

DistanceFromMemory alone → RelativePosition only. (27.14)

---

## 27.4 Moving-average slope

A moving-average slope is:

s_n(t) = MA_n(t) − MA_n(t−1). (27.15)

or, over k periods:

s_{n,k}(t) = [MA_n(t) − MA_n(t−k)]/k. (27.16)

The slope converts memory into a Motion / Relation reading.

A positive slope means the filtered memory centre is rising.

It does not prove:

- broad participation;
- sustainable commitment;
- low residual;
- absence of overextension.

The slope may remain positive after the market's internal structure has begun to weaken.

This is ordinary filtration lag.

---

## 27.5 Moving-average crossover

Let:

M_f(t) = fast moving average. (27.17)

M_s(t) = slow moving average. (27.18)

Define the memory spread:

D_M(t) = M_f(t) − M_s(t). (27.19)

A crossover occurs when:

sign[D_M(t)] changes. (27.20)

The correct interpretation is:

> A moving-average crossover is a reversal in the ordering of two declared memory horizons.

It is not automatically:

- trend initiation;
- commitment;
- breakout;
- profitable entry;
- regime change.

The source framework describes the crossover as memory-horizon conflict and delayed regime confirmation, with gate strength and commitment still missing.

---

## 27.6 Why crossover signals lag

Both M_f and M_s are filtered histories.

A crossover occurs only after sufficient new price information has entered the fast memory to overcome the slow memory.

Therefore:

CrossoverTime > InitialMovementTime in many ordinary cases. (27.21)

This delay is not necessarily a flaw.

It is the cost of requiring a stronger memory reordering.

A late signal may be more stable than an early signal.

But it may also enter after much of the displacement has already occurred.

The correct question is not:

Is lag bad? (27.22)

It is:

Does the selected lag improve gate quality enough to justify its delay and cost? (27.23)

---

## 27.7 Whipsaw as wrong-period promotion

In a range, fast and slow memory may repeatedly exchange order.

The crossover system then promotes local Window or Structure fluctuations into apparent Event-level transitions.

This is a category error:

Structure-level relation → falsely promoted to Event-level commitment. (27.24)

Whipsaw becomes more likely when:

- χ remains corrective;
- boundaries are close;
- volatility is high relative to the memory spread;
- no independent commitment gate is required.

A crossover should therefore be distinguished from a crossover-confirmed event:

CrossoverEvent
= Crossover

- BoundaryContext
- CommitmentEvidence
- ResidualControl. (27.25)

---

## 27.8 MACD as a difference between filtered memories

The standard MACD is:

$MACD_t = EMA_{fast}(t) - EMA_{slow}(t)$. (27.26)

The signal line is:

$Signal_t = EMA_m(MACD_t)$. (27.27)

The histogram is:

$Hist_t = MACD_t - Signal_t$. (27.28)

MACD therefore contains a nested structure.

First layer:

price → fast and slow memory. (27.29)

Second layer:

difference between memories. (27.30)

Third layer:

difference between memory displacement and its own filtered history. (27.31)

In operator form:

$MACD = \Delta[F_{fast}(P), F_{slow}(P)]$. (27.32)

$Hist = \Delta[MACD, F_{signal}(MACD)]$. (27.33)

---

## 27.9 MACD as memory curvature

MACD is commonly called a momentum indicator.

That description is useful but incomplete.

Its deeper structure is:

- memory separation;
- change in memory separation;
- acceleration or deceleration of filtered directional structure.

The source article therefore describes MACD as measuring memory curvature and phase acceleration, while missing density, commitment, and the gate itself.

A rising positive histogram may indicate:

- increasing separation between directional memories;
- acceleration of filtered movement.

A falling positive histogram may indicate:

- continued positive memory separation;
- weakening expansion.

The second condition is not yet bearish reversal.

It is relational deceleration.

---

## 27.10 MACD divergence

A bearish divergence may be represented:

$$P_t > P_{t-k}. \quad (27.34)$$

$$MACD_t \leq MACD_{t-k}. \quad (27.35)$$

This means price has advanced while filtered memory acceleration has not confirmed the same degree of advance.

Possible interpretations include:

- weakening directional expansion;
- volatility change;
- mature trend;
- consolidation;
- temporary price spike;
- filter sensitivity.

The disciplined sequence is:

MACD divergence
→ memory-curvature weakening
→ candidate trend instability
→ structural gate test. (27.36)

Therefore:

MACD divergence ≠ trend reversal. (27.37)

---

# 27.11 Moving-average and MACD redundancy

Moving averages, crossovers, and MACD share:

- the same primary price input;
- related filtration families;
- overlapping horizons;
- similar sensitivity to trend and lag.

Their agreement is useful but not highly independent.

A qualitative redundancy relation is:

Redundancy(MA crossover, MACD sign) = high under similar parameters. (27.38)

Combining them may improve presentation or timing.

It should not automatically be counted as multiple independent confirmations.

A stronger cross-check would add a different functional family:

- volume or participation;
- structural boundary;
- breadth;
- close or retest;
- residual audit.

---

# 27.12 Method cards

## Moving average

**Primary period:** Structure
**Primary family:** Load / Memory
**Measures:** filtered historical price memory
**Assumes:** selected horizon remains relevant
**Gate required:** sustained acceptance, crossover, boundary interaction, or retest
**Residual:** volume, breadth, density, positioning, event risk

**Transport burden:** horizon, timeframe, log scale, alternative bar construction
**Common error:** treating lagged memory as an independent forecast

### Moving-average crossover

**Primary period:** Structure
**Primary family:** Motion / Relation
**Measures:** ordering conflict between memory horizons
**Gate required:** structural acceptance and independent commitment evidence
**Residual:** range regime, low participation, higher-timeframe contradiction
**Common error:** treating memory reordering as completed regime transition

### MACD

**Primary period:** Structure
**Primary family:** Motion / Relation
**Measures:** separation and curvature between filtered memories
**Gate required:** price structure, close, volume, or boundary failure
**Residual:** density, participation, liquidity, gate strength
**Common error:** treating divergence as reversal

---

# 28. RSI, Stochastic, ATR, and Volatility Bands

# 28.1 RSI as normalized directional relation

For a declared window n:

$RS_t = AverageGain_n(t)/AverageLoss_n(t)$. (28.1)

$RSI_t = 100 - 100/(1 + RS_t)$. (28.2)

RSI compresses recent directional movement into a bounded scale.

Its primary role is:

> Estimate the recent dominance of positive displacement relative to total directional displacement.

RSI is therefore a Structure-period Motion / Relation instrument.

It does not directly measure:

- valuation;
- transaction density;
- liquidity;
- positioning;
- structural mass;
- gate commitment.

The source table characterizes RSI as a corrective-pressure detector that performs well in range

exhaustion but often makes premature top calls in self-confirming trends.

---

## 28.2 The hidden corrective assumption

The conventional interpretation is:

RSI high → market is overbought → price should decline. (28.3)

But the final arrow requires a feedback assumption.

It assumes that directional extension generates sufficient counter-pressure:

$\chi < 0$. (28.4)

In a self-confirming regime:

$\chi > 0$. (28.5)

continued high RSI may reflect:

- trend persistence;
- institutional allocation;
- forced covering;
- breakout continuation;
- momentum reinforcement.

Thus:

HighRSI + $\chi < 0$ evidence → exhaustion hypothesis. (28.6)

HighRSI + $\chi > 0$ evidence → strength or extension hypothesis. (28.7)

HighRSI without $\chi$ diagnosis → directional imbalance only. (28.8)

---

## 28.3 RSI threshold as a soft boundary

The familiar 70 and 30 thresholds are not natural constants.

They are conventions.

The relevant threshold may vary with:

- asset;
- timeframe;
- volatility;
- trend strength;
- market regime;

- smoothing method.

A protocol-specific threshold is:

RSI_upper = c_upper(P). (28.9)

RSI_lower = c_lower(P). (28.10)

The thresholds should therefore be treated as soft boundaries requiring empirical calibration.

---

# 28.4 RSI divergence

A bearish RSI divergence is:

P_t > P_{t−k}. (28.11)

RSI_t ≤ RSI_{t−k}. (28.12)

This indicates that price advanced with less normalized directional dominance.

It does not prove:

- seller control;
- structural break;
- reversal commitment.

A mature interpretation requires:

RSI divergence

- Constraint interaction
- Gate failure
  → reversal evidence. (28.13)

---

# 28.5 Stochastic oscillator as close location

A common stochastic oscillator is:

%K_t = 100 × [C_t − L_n(t)]/[H_n(t) − L_n(t)]. (28.14)

where:

H_n(t) = highest high over n periods;
L_n(t) = lowest low over n periods.

The oscillator asks:

> Where is the current close located inside the recent range?

It is not identical to RSI.

RSI compares directional gains and losses.

Stochastic compares closing location with range boundaries.

Yet both are strongly conditioned by recent price structure and often share corrective interpretations.

The source table therefore classifies stochastic as a range-exhaustion instrument that can generate false reversal signals when trend-continuation force dominates.

---

## 28.6 Stochastic in range and trend regimes

Inside a corrective range:

High %K near resistance may support an exhaustion hypothesis. (28.15)

Low %K near support may support a recovery hypothesis. (28.16)

Inside a self-confirming trend:

High %K may persist because closes remain near the upper part of the advancing range. (28.17)

Thus the same reading can mean:

- extension requiring correction;
- persistent directional acceptance.

Again, χ determines interpretation.

---

## 28.7 ATR as agitation magnitude

True range may be defined:

$TR_t = \max[H_t - L_t, |H_t - C_{t-1}|, |L_t - C_{t-1}|]$. (28.18)

ATR is a filtered form:

$ATR_n(t) = F_n[TR_t]$. (28.19)

ATR measures realized movement magnitude.

It is primarily:

- Window-to-Structure Motion magnitude;
- a scaling coordinate for boundaries and risk.

It does not determine direction.

It does not tell whether high movement represents:

- healthy trend;
- liquidation;
- event shock;

- instability;
- transition;
- noise.

Therefore:

ATR = agitation magnitude, not market meaning. (28.20)

---

## 28.8 ATR normalization

ATR is especially useful for transport.

A nominal price distance can be normalized:

$d_{ATR}(t) = [P_t - B_t]/ATR_n(t)$. (28.21)

where $B_t$ is a boundary or reference.

This helps compare:

- different assets;
- different volatility regimes;
- different nominal prices;
- different historical episodes.

A breakout of 0.2 ATR and a breakout of 2 ATR should not automatically receive the same displacement interpretation.

---

## 28.9 Volatility bands

Bollinger Bands are commonly defined:

$Middle_t = MA_n(t)$. (28.22)

$Upper_t = MA_n(t) + k\sigma_n(t)$. (28.23)

$Lower_t = MA_n(t) - k\sigma_n(t)$. (28.24)

The method combines:

- Load / Memory through $MA_n$;
- Motion magnitude through $\sigma_n$;
- Constraint / Boundary through the upper and lower bands.

It is therefore a molecular method spanning three functions.

Its primary placement is:

Structure × Constraint.

## 28.10 Band touch is not a reversal signal

A band touch means:

Price has reached a declared distance from filtered memory under the current dispersion estimate. (28.25)

It does not mean:

Price must return immediately. (28.26)

Under $\chi < 0$:

band extension may invite reversion. (28.27)

Under $\chi > 0$:

repeated band contact may indicate directional selection. (28.28)

The relation between price and band requires:

- trend context;
- slope;
- volume;
- breadth;
- boundary acceptance;
- later close.

---

## 28.11 Band width and compression

Band width may be defined:

$BBW_t = [Upper_t - Lower_t]/Middle_t$. (28.29)

or another declared normalization.

Declining band width indicates reduced recent dispersion.

It does not by itself prove accumulating directional energy.

A squeeze interpretation requires additional structure:

Dispersion ↓

- Boundary narrowing
- Load remains or increases
- Commitment deferred. (28.30)

A low-volatility market with no meaningful loading may simply remain inactive.

## 28.12 Method cards

### RSI

**Primary period:** Structure
**Primary family:** Motion / Relation
**Measures:** normalized directional imbalance
**Regime assumption:** reversal reading usually requires $\chi < 0$
**Gate required:** support/resistance break, close, recovery, or reversal structure
**Residual:** volume, breadth, positioning, structural mass
**Transport burden:** timeframe, smoothing, threshold calibration
**Common error:** treating overbought as completed reversal

### Stochastic

**Primary period:** Structure
**Primary family:** Motion / Relation
**Measures:** closing location inside the recent range
**Regime assumption:** most effective interpretation often assumes corrective range behaviour
**Gate required:** boundary rejection or structural break
**Residual:** trend strength, participation, higher-period relation
**Common error:** treating range position as universal exhaustion

### ATR

**Primary period:** Window–Structure
**Primary family:** Motion magnitude
**Measures:** realized agitation
**Gate required:** none for measurement; another method is needed for event meaning
**Residual:** direction, commitment, cause, liquidity
**Common error:** treating volatility increase as directional evidence

### Bollinger Bands

**Primary period:** Structure
**Primary family:** Constraint / Boundary
**Measures:** price location relative to filtered memory and dispersion
**Regime assumption:** touch interpretation depends on $\chi$
**Gate required:** close, rejection, continuation, or acceptance
**Residual:** volume, breadth, structural density
**Common error:** assuming every outer-band contact implies reversion

---

# 29. Volume, OBV, VWAP, and Volume Profile

## 29.1 Volume as a mixed observable

A simple decomposition is:

$$\text{Volume} \approx \text{TradeFrequency} \times \text{AverageTradeSize}. \quad (29.1)$$

Dollar volume adds price:

DollarVolume ≈ TradeFrequency × AverageTradeSize × Price. (29.2)

But volume may represent several different market functions.

The source article summarizes:

Volume = Frequency + Mass + Commitment + ExchangeAmbiguity. (29.3)

High volume can accompany:

- accumulation;
- distribution;
- forced liquidation;
- absorption;
- exhaustion;
- passive rebalancing;
- churn;
- hedging.

Therefore:

HighVolume ≠ Bullish. (29.4)

HighVolume ≠ Bearish. (29.5)

HighVolume = HighRecordedExchange under the declared protocol. (29.6)

Interpretation requires price relation, location, and later gate behaviour.

---

## 29.2 Relative volume

A relative-volume measure is:

RVOL_t = V_t/E[V_t | TimeContext,Protocol]. (29.7)

This is more informative than raw volume when ordinary activity differs by:

- time of day;
- weekday;
- expiry cycle;
- asset;
- market regime.

But expected volume must be declared.

A volume spike relative to an inappropriate baseline can create a false anomaly.

---

## 29.3 Volume as commitment evidence

Volume becomes commitment evidence when it accompanies a meaningful transition.

For example:

BoundaryCross

- HighRelativeVolume
- CloseBeyondBoundary
  → stronger event evidence. (29.8)

But volume can also reflect opposition.

A high-volume bar with little net displacement may suggest:

- absorption;
- two-sided conflict;
- distribution;
- liquidity provision.

Thus the relation:

PriceDisplacement per UnitVolume. (29.9)

may be as important as volume itself.

A simple efficiency measure is:

$E_V(t) = |\Delta P_t|/[V_t + \varepsilon]$. (29.10)

Low $E_V$ with high volume may indicate heavy exchange with limited displacement.

The interpretation remains protocol-dependent.

---

## 29.4 OBV as signed accumulation

OBV is:

$OBV_t = OBV_{t-1} + s_tV_t$. (29.11)

with:

$s_t = +1$ if $C_t > C_{t-1}$;
$s_t = -1$ if $C_t < C_{t-1}$;
$s_t = 0$ otherwise. (29.12)

OBV is not direct order flow.

It is volume signed by a closing-price rule.

Its primary function is Load / Memory, with a secondary directional relation.

The method asks:

> Has cumulative volume been associated more often with positive or negative closing movement?

It does not directly identify:

- buyer identity;
- seller identity;
- informed flow;
- opening versus closing positions;
- absorption.

---

## 29.5 OBV divergence

A bullish OBV divergence may occur when:

Price makes a lower low. (29.13)

OBV does not make a lower low. (29.14)

This may indicate reduced negative volume association.

It is not yet a reversal.

The correct interpretation is:

Price decline lacks equivalent confirmation from the declared signed-volume ledger. (29.15)

The commitment gate remains external.

---

## 29.6 VWAP as volume-weighted ledger centre

VWAP is:

$VWAP_T = \Sigma_{t\in T} P_tV_t/\Sigma_{t\in T} V_t$. (29.16)

VWAP combines:

- price;
- volume;
- accumulation;
- normalization.

It estimates the volume-weighted centre of recorded exchange during T.

The source framework characterizes VWAP as a volume-weighted ledger centre rather than an intrinsic fair value.

This distinction matters.

VWAP says:

Where did volume-weighted exchange occur? (29.17)

It does not necessarily say:

What is the asset economically worth? (29.18)

---

## 29.7 VWAP as self-referential boundary

VWAP may become operationally important because participants use it.

Execution desks compare fills against VWAP.

Algorithms route orders around it.

Traders interpret reclaim or loss of VWAP.

Therefore:

VWAP as Probe
→ benchmark adoption
→ conditional orders
→ VWAP as Couple or Boundary. (29.19)

A descriptive centre becomes part of market structure.

This is a clear example of observer backreaction.

---

## 29.8 Anchored VWAP

Anchored VWAP is:

AVWAP_{a→t} = Σ_{j=a}^{t} P_jV_j/Σ_{j=a}^{t} V_j. (29.20)

The anchor a defines which history is treated as relevant.

Candidate anchors include:

- major high or low;
- earnings release;
- breakout event;
- policy decision;
- large gap;
- institutional issuance.

The method’s quality depends strongly on anchor legitimacy.

Therefore:

No declared anchor → no stable anchored-VWAP claim. (29.21)

Transport should test nearby admissible anchors.

---

## 29.9 Volume profile

A volume profile is:

$VP(p) = \Sigma_t V_t \cdot 1[P_t \in Bin(p)]$. (29.22)

It maps accumulated volume onto price rather than time.

This produces a density-like field.

High-volume regions indicate greater historical occupation.

Low-volume regions indicate less recorded exchange.

The source article interprets volume profile as semantic density across price and warns against treating old density as permanent.

---

## 29.10 High-volume nodes

A high-volume node may indicate:

- accepted value;
- dense historical memory;
- many entry prices;
- strong two-sided exchange;
- structural mass.

But the future effect is ambiguous.

It may:

- attract price;
- slow movement;
- become support;
- become resistance;
- become irrelevant after a regime change.

Thus:

$HistoricalDensity \neq PermanentBoundary$. (29.23)

Current flow and gate behaviour remain necessary.

---

## 29.11 Low-volume regions

A low-volume region may indicate limited historical acceptance.

Price may move rapidly through it because less prior structure is present.

A heuristic is:

ExpectedTraversalSpeed(p) ∝ 1/[VP(p) + ε]. (29.24)

This is not a universal law.

It expresses a testable hypothesis:

Lower historical transaction density may reduce structural resistance under otherwise comparable conditions.

---

## 29.12 Volume profile and structural mass

A stronger level may combine:

- profile density;
- repeated reaction;
- anchored VWAP;
- institutional use;
- shared attention.

A schematic mass score is:

$$M_{level} = w_1 ProfileDensity$$

- $w_2 ReactionHistory$
- $w_3 AnchoredReference$
- $w_4 PositioningProxy$
- $w_5 InstitutionalUse$. (29.25)

The weights must be tested.

The purpose is to distinguish:

line drawn by analyst

from:

boundary supported by multiple load channels.

---

## 29.13 Method cards

### Raw volume

**Primary period:** Window–Event
**Primary family:** Load / Memory
**Measures:** recorded exchange activity
**Gate role:** commitment evidence when attached to a meaningful event
**Residual:** trade motive, participant identity, opening versus closing, hidden liquidity
**Common error:** treating all high volume as directional confirmation

### OBV

**Primary period:** Structure
**Primary family:** Load / Memory
**Measures:** cumulative volume signed by close direction
**Gate required:** price structure and event confirmation
**Residual:** actual order-flow direction, absorption, positioning
**Common error:** treating inferred signed volume as directly observed accumulation

### VWAP

**Primary period:** Structure
**Primary family:** Load / Memory
**Secondary family:** Constraint / Boundary
**Measures:** volume-weighted transaction centre
**Gate required:** reclaim, hold, rejection, or acceptance
**Residual:** motive, hidden flow, anchor sensitivity
**Common error:** treating VWAP as intrinsic fair value

### Volume profile

**Primary period:** Structure
**Primary family:** Load + Constraint
**Measures:** historical transaction density across price
**Gate required:** current acceptance or rejection
**Residual:** participant identity, current positioning, future catalyst
**Common error:** treating historical density as permanent force

---

# 30. Candlesticks, Levels, Patterns, and Breakouts

# 30.1 Candlestick as a micro-ledger

A candle is:

Candle_P = (O,H,L,C,V | Window_P). (30.1)

It is a Period-1 compound containing:

- Load through opening state and volume;
- Motion through body and range;
- Constraint through high and low;
- Commitment through close;

- Residual through wicks and lost path information.

The source article calls candlesticks micro-ledger conflict instruments that observe rejection and acceptance within a window but miss the larger regime.

---

# 30.2 Body and retained displacement

A candle body is:

Body = C − O. (30.2)

Its magnitude relative to range is:

BodyRatio = |C − O|/(H − L). (30.3)

A large body ratio indicates that much of the intrawindow range remained reflected in the close.

This can be interpreted as:

RetainedDisplacement = body accepted by the window gate. (30.4)

It does not prove persistence beyond the window.

---

# 30.3 Wick as residual

The upper wick is:

$W_u = H - \max(O,C)$. (30.5)

The lower wick is:

$W_l = \min(O,C) - L$. (30.6)

The wick records intrawindow excursion not retained at the close.

Therefore:

Wick = attempted displacement residual. (30.7)

But the residual's meaning remains open.

An upper wick may reflect:

- rejection;
- profit-taking;
- thin liquidity;
- data anomaly;
- temporary shock;
- auction structure.

A later gate is required to classify it as consequential rejection.

## 30.4 Candlestick pattern dependence

A hammer-like shape near a major support zone is not equivalent to the same shape in the middle of an unstructured range.

Candlestick meaning depends on:

- location;
- higher-period structure;
- volume;
- prior motion;
- later close;
- regime $\chi$.

Therefore:

$$\text{CandleMeaning} = \text{Shape} \times \text{Location} \times \text{Regime} \times \text{GateFollowThrough}. \quad (30.8)$$

The multiplication sign is conceptual, not a literal calibrated formula.

The main lesson is that shape alone is insufficient.

---

## 30.5 Support and resistance as loaded levels

A support or resistance zone is:

$$\text{Level}_\ell = \text{HistoricalTrace}_\ell + \text{ConditionalAttention}_\ell. \quad (30.9)$$

A stronger construction may include:

$$\text{LevelStrength}_\ell = \text{Density}$$

- ReactionHistory
- InstitutionalReference
- Positioning
- CrossFrameSurvival. (30.10)

A line becomes meaningful when multiple traces point to a similar region.

Arbitrary line drawing occurs when:

- anchors are selected after the move;

- the zone is continually shifted;
- invalidation is absent;
- no independent load or gate evidence exists.

The source table describes support and resistance as ledgered memory and structural mass, with arbitrary line drawing as a principal failure.

---

## 30.6 Chart pattern as compressed episode geometry

A chart pattern is not merely a visible shape.

It is a hypothesis about:

- accumulated load;
- changing motion;
- narrowing or expanding constraints;
- approaching commitment.

For a triangle-like pattern:

UpperBoundary_t ↓ or remains stable. (30.11)

LowerBoundary_t ↑ or remains stable. (30.12)

AdmissibleRegion_t narrows. (30.13)

This indicates compression.

It does not determine direction.

The next gate determines which boundary becomes accepted.

---

## 30.7 Pattern subjectivity

Pattern construction depends on:

- pivot selection;
- boundary placement;
- timeframe;
- scale;
- tolerance;
- minimum touches;
- breakout definition.

A mature pattern protocol should declare:

Pattern_P = (PivotRule,BoundaryRule,Tolerance,GateRule,Invalidation). (30.14)

Without this declaration, pattern analysis can become retrospective visual fitting.

---

## 30.8 Breakout as event-level commitment

A breakout is not:

P_t > Level. (30.15)

It is a candidate transition from one structural region to another.

A fuller formula is:

ValidBreakout
= MeaningfulBoundary

- NormalizedDisplacement
- Participation
- CloseOrAcceptanceGate
- FollowThroughOrRetest
  – ResidualBurden. (30.16)

This formulation synthesizes the source article's emphasis on price, volume, breadth, close, and cross-frame confirmation.

---

## 30.9 Breakout close

A breakout close is:

C_t > UpperBoundary_t. (30.17)

for a bullish event, under the declared window.

But the close should be normalized by:

- boundary width;
- volatility;
- liquidity;
- gap behaviour.

A marginal close above a noisy level is not equivalent to a decisive displacement.

---

## 30.10 Retest

A retest asks whether the new region remains admissible after the initial transition.

A bullish retest sequence is:

Breakout
→ ReturnTowardBoundary
→ HoldAboveOrReclaim
→ RenewedMotion. (30.18)

The retest tests whether:

OldResistance → NewSupport. (30.19)

This conversion is a ledger hypothesis.

It is not guaranteed merely because the breakout occurred.

---

# 30.11 Fakeout

A fakeout is:

BoundaryCross

- ParticipantResponse
  – DurableAcceptance. (30.20)

A stronger fakeout definition may require:

ReturnInsideOldRegion

- OpposingClose
- FailedRetest
- TrappedPositionResidual. (30.21)

The failed event should remain in the ledger.

It may alter later boundary strength and participant behaviour.

---

# 30.12 Pattern completion

A pattern does not complete simply because price touches a target.

Completion should require:

- event gate;
- persistence;
- invalidation of the old geometry;
- ledgered acceptance of the new episode.

Thus:

PatternCompletion
= BoundaryEvent

- NewStatePersistence
- OldPatternNoLongerOrganizesFutureTrace. (30.22)

---

# 30.13 Method cards

### Candlestick

**Primary period:** Window
**Primary family:** Commitment / Gate
**Measures:** intrawindow conflict and closing acceptance
**Residual:** wick, lost path, hidden order flow
**Transport burden:** timeframe and bar construction
**Common error:** treating shape as context-free prophecy

### Support / resistance

**Primary period:** Structure
**Primary family:** Constraint / Boundary
**Measures:** loaded historical transition zone
**Gate required:** acceptance, rejection, or durable crossing
**Residual:** stale memory, changed participants, weak density
**Common error:** arbitrary or retrospectively moved lines

### Chart pattern

**Primary period:** Episode
**Primary family:** Constraint / Boundary
**Measures:** changing episode geometry and compression
**Gate required:** breakout, rejection, or transition
**Residual:** pivot ambiguity, direction uncertainty, fakeout risk
**Common error:** treating visible shape as completed event

### Breakout

**Primary period:** Event
**Primary family:** Commitment / Gate
**Measures:** admission into a new structural region
**Required evidence:** meaningful boundary, displacement, participation, close, follow-through
**Residual:** breadth weakness, untested retest, higher-frame conflict
**Common error:** treating any line crossing as durable commitment

---

# 31. Breadth, Elliott Wave, Fibonacci, and Gann

# 31.1 Breadth as field-wide participation

Breadth observes the component field rather than only the headline price projection.

A simple breadth measure is:

Breadth_t = (1/N)Σ_i a_{i,t}. (31.1)

where a_{i,t} may encode:

- advancing versus declining;
- above versus below moving average;
- new high versus new low;
- positive versus negative return.

Breadth is primarily:

Structure-period Motion / Relation.

It measures whether market movement is coherent across the field.

The source atlas describes breadth as field-wide phase coherence that measures participation quality while often providing early warning rather than final commitment.

---

# 31.2 Breadth divergence

A narrowing market may satisfy:

Index_t ↑. (31.2)

Breadth_t ↓. (31.3)

This means the visible index and the component field are diverging.

It does not prove immediate reversal.

The proper sequence is:

Breadth divergence
→ declining field coherence
→ rising residual
→ price gate required. (31.4)

A headline index can continue rising for a substantial period while participation narrows.

Breadth is therefore often an early-warning instrument.

---

# 31.3 Breadth protocol

Breadth depends on:

- component universe;
- survivorship treatment;
- weighting;
- inclusion changes;
- timeframe;

- unchanged-security treatment.

Thus:

Breadth_P is protocol-relative. (31.5)

An equal-weight breadth result and a cap-weighted breadth result may differ because they answer different questions.

This is not necessarily inconsistency.

It may reveal concentration.

---

# 31.4 Elliott Wave as recursive episode segmentation

Elliott Wave belongs primarily to:

Episode × Motion / Relation.

Its useful core is the attempt to distinguish:

- self-confirming displacement;
- corrective digestion;
- nested scale;
- terminal weakening;
- episode transition.

The source interpretation treats wave analysis as nested selection and correction, while identifying objective pivot validity and retrospective relabeling as major problems.

A disciplined wave model is:

WaveModel_P
= PivotRule

- SegmentRule
- χClassification
- EndpointGate
- AlternativeBranchLedger
- Invalidation. (31.6)

---

# 31.5 Impulse and correction

A possible structural interpretation is:

Impulse segment ≈ sustained $\chi > 0$. (31.7)

Corrective segment ≈ local $\chi < 0$ or residual digestion. (31.8)

This does not prove the traditional count sequence.

It provides a regime-based way to interpret why some segments extend and others rotate.

A wave count becomes stronger when:

- pivots are predeclared;
- momentum and breadth support the segment classification;
- endpoint gates are explicit;
- alternate counts remain recorded;
- the structure survives timeframe transport.

---

## 31.6 Wave endpoint

A local extreme is not automatically a wave endpoint.

A stronger endpoint criterion is:

WaveEndpoint
= PivotExtreme

- RelationalShift
- GateEvidence
- DensityContext
- ResidualAudit
- CrossFrameSurvival. (31.9)

The original count should be invalidated when its declared structural rule fails.

The source text states:

CorrectCounting = DeclaredProtocol + LedgeredPivot + CrossMethodConfirmation. (31.10)

This makes uncertainty explicit rather than eliminating it through hindsight.

---

## 31.7 Fibonacci as ratio-based candidate boundary

A Fibonacci retracement begins from two anchors:

$P_A$ and $P_B$. (31.11)

A candidate retracement level is:

$F_r = P_B - r(P_B - P_A)$. (31.12)

where r may be 0.382, 0.500, 0.618, or another declared ratio.

Its primary classification is:

Structure or Episode × Constraint / Boundary.

The ratio does not create structural mass by itself.

A stronger Fibonacci zone requires alignment with:

- profile density;
- prior support or resistance;
- VWAP;
- candle rejection;
- trend context;
- current volume.

The source article characterizes Fibonacci as a ratio attractor or candidate attention zone that misses actual density and pressure, with arbitrary anchor selection as its major failure.

---

# 31.8 Fibonacci falsification

A disciplined claim should declare:

- anchors;
- ratio;
- zone width;
- gate;
- invalidation.

For example:

Claim: 61.8% retracement zone will act as support. (31.13)

A source-aligned invalidation is:

Fibonacci support claim invalid if price closes below the declared zone and fails to reclaim it. (31.14)

This does not prove that the ratio has universal causal power.

It makes one local claim testable.

---

# 31.9 Fibonacci clustering

Several ratio levels from different anchors may cluster.

A cluster may matter because:

- many analysts observe it;

- different paths generate similar zones;
- the cluster overlaps genuine density;
- orders concentrate around it.

But increasing the number of freely selected anchors raises overfitting risk.

Therefore:

MoreFibonacciLines ≠ MoreEvidence. (31.15)

The admissible anchor family must be limited before outcome review.

---

## 31.10 Gann as candidate price–time invariant

Gann analysis attempts to relate price and time through:

- angles;
- squares;
- cycles;
- proportional movement;
- selected anchors.

Its best placement is not ordinary momentum.

It belongs to:

Episode × Constraint / Boundary

with a heavy Transport / Invariance burden.

A candidate Gann relation is:

G_P = Relation(Price,Time | Anchor,Scale,Calendar,VolatilityConvention). (31.16)

The source atlas explicitly classifies Gann as a candidate price–time invariant that often fails through mystical geometry, scale dependence, and overfitting.

---

## 31.11 Scale problem in Gann

An angle is not invariant under arbitrary rescaling.

If the horizontal and vertical axes are changed, the visual angle changes.

Therefore, a claim such as “one unit of price per one unit of time” requires:

- defined price unit;
- defined time unit;
- defined chart scale;

- defined normalization.

Without these declarations:

VisualAngle ≠ StableMarketInvariant. (31.17)

A stronger test uses normalized coordinates:

$\tilde{p}_t = [P_t - P_{anchor}]/\sigma_P$. (31.18)

$\tilde{t} = [t - t_{anchor}]/\tau_{ref}$. (31.19)

Then the proposed relation can be tested under alternative admissible $\sigma_P$ and $\tau_{ref}$ choices.

---

# 31.12 Gann and cadence

The source framework suggests that Gann cycles may be reinterpreted as cadence hypotheses rather than mystical cosmic schedules.

Relevant cadences may arise from:

- option expiry;
- reporting cycles;
- margin cycles;
- settlement;
- policy meetings;
- portfolio rebalancing;
- institutional review.

This converts the question from:

Does price obey a sacred angle? (31.20)

to:

Does a declared price–time relation survive normalization and align with repeatable event cadence? (31.21)

That is a substantially more testable claim.

---

# 31.13 Gann falsification

A disciplined Gann claim should record:

- anchor;
- scale;
- time convention;

- line drawn before the move;
- expected gate;
- invalidation.

The source framework gives an example:

Gann reversal claim invalid if price closes through the declared level with volume and acceptance. (31.22)

Again, the value lies in making the claim falsifiable.

---

# 31.14 Method cards

## Breadth

**Primary period:** Structure
**Primary family:** Motion / Relation
**Measures:** field-wide participation and coherence
**Gate required:** headline price structure must confirm
**Residual:** universe choice, weighting, single-asset catalyst
**Transport burden:** equal-weight, cap-weight, sector, and timeframe frames
**Common error:** treating early field weakening as completed reversal

## Elliott Wave

**Primary period:** Episode
**Primary family:** Motion / Relation
**Measures:** nested selection–correction segmentation
**Gate required:** endpoint confirmation and regime transition
**Residual:** alternate count, pivot ambiguity, timeframe conflict
**Transport burden:** pivot protocol, scale, timeframe, volatility normalization
**Common error:** retrospective relabeling into hindsight perfection

## Fibonacci

**Primary period:** Structure–Episode
**Primary family:** Constraint / Boundary
**Measures:** candidate ratio-based attention zone
**Gate required:** rejection, support, resistance, close, or reclaim
**Residual:** arbitrary anchors, absent density, regime mismatch
**Transport burden:** nearby anchors, log scale, volatility zone width
**Common error:** treating a ratio as an independently causal force

## Gann

**Primary period:** Episode
**Primary family:** Constraint / Boundary and invariance hypothesis
**Measures:** candidate price–time cadence or relation
**Gate required:** predeclared market reaction and acceptance rule
**Residual:** scale, anchor, calendar, volatility, event-cadence sensitivity
**Transport burden:** exceptionally high
**Common error:** mistaking chart geometry for frame-invariant market law

# 32. The Molecular Atlas Summary

The principal methods can now be located in the periodic grammar.

| Method | Primary cell | Core meaning |
|---|---|---|
| Moving average | Structure × Load | filtered price memory |
| MA crossover | Structure × Motion | memory-horizon reordering |
| MACD | Structure × Motion | memory separation and curvature |
| RSI | Structure × Motion | normalized directional imbalance |
| Stochastic | Structure × Motion | close location within recent range |
| ATR | Window–Structure × Motion | realized agitation magnitude |
| Bollinger Bands | Structure × Constraint | volatility-conditioned boundary |
| Volume | Window–Event × Load | exchange activity and participation |
| OBV | Structure × Load | price-signed volume accumulation |
| VWAP | Structure × Load | transaction-weighted memory centre |
| Volume profile | Structure × Load/Constraint | price-space transaction density |
| Candlestick | Window × Commitment | closing micro-ledger with wick residual |
| Support/ resistance | Structure × Constraint | loaded transition zone |
| Chart pattern | Episode × Constraint | compressed episode geometry |
| Breakout | Event × Commitment | admission beyond a structural boundary |
| Breadth | Structure × Motion | field-wide participation coherence |
| Elliott Wave | Episode × Motion | recursive selection–correction segmentation |
| Fibonacci | Structure/Episode × Constraint | candidate ratio-based zone |
| Gann | Episode × Constraint/ Transport | candidate price–time invariant |

The table exposes three general principles.

# First principle

A method’s formula does not determine its complete interpretation.

Interpretation requires:

- period;
- functional family;
- regime;
- gate;
- residual;
- transport.

# Second principle

Methods derived from the same input and operator family should not be counted as fully independent confirmation.

# Third principle

A Structure-level diagnosis becomes an Event only after a Commitment gate.

These principles transform Technical Analysis from an indicator catalogue into a governed observational grammar.

The next part will introduce the advanced regime and phase architecture:

**$\chi$ as relational signature**
**$\Xi$ as effective control state**
**$Z = R + iQ$ as locally earned complex completion**
**$\tau_i$ as candidate internal phase time**.

# Part VII — Regime, Complex Phase, and Internal Time

The periodic table reorganizes Technical Analysis around:

**Load / Memory**
**Motion / Relation**
**Constraint / Boundary**
**Commitment / Gate**

But these functions do not operate identically in every market condition.

A momentum reading can signal exhaustion in one regime and persistence in another.

A boundary touch can precede reversion in one regime and acceleration in another.

The same quantity of market loading can remain mobile in one environment and become dangerously trapped in another.

A two-variable state can sometimes be treated adequately as an ordinary real pair. In rarer cases, it may earn a complex completion whose phase supplies a useful internal clock.

This part therefore introduces four distinct constructs:

$\chi$ = relational feedback signature. (33.1)

$\Xi$ = effective control state. (33.2)

$Z = R + iQ$ = candidate conjugate state. (33.3)

$\tau_i$ = accumulated internal phase progress. (33.4)

They must not be collapsed into one another.

$\chi$ describes the orientation of feedback.

$\Xi$ describes the effective operating condition.

Z describes a locally validated conjugate geometry.

$\tau_i$ describes progression through that geometry.

# 33. χ as the Relational Regime Signature

## 33.1 Why a regime variable is necessary

Many Technical Analysis failures are not formula failures.

They are **regime-assignment failures**.

RSI may correctly report strong directional imbalance.

The analyst then incorrectly concludes that reversal is imminent.

A Bollinger Band may correctly report large displacement from filtered memory.

The analyst then incorrectly concludes that price must revert.

A moving-average slope may correctly report trend persistence.

The analyst then incorrectly assumes that all future pullbacks will remain corrective.

The missing question is:

> What kind of feedback does the current movement generate?

Does movement create counter-pressure?

Does it create further movement?

Or is the return relation weak, unstable, or transitional?

The signature χ is introduced to classify that feedback orientation.

---

## 33.2 Signal pressure and realized structure

Let:

λ = effective market signal pressure. (33.5)

s = realized market structure. (33.6)

Signal pressure may contain:

- expectations;
- order intention;
- leverage appetite;
- liquidity demand;
- hedging pressure;
- institutional mandate;
- fear of loss;

- fear of missing out;
- forced liquidation;
- algorithmic triggers.

Realized structure may contain:

- price;
- trend;
- volatility;
- support and resistance;
- volume distribution;
- breadth;
- candle structure;
- wave structure;
- accepted gaps;
- moving-average configuration.

The forward relation is:

$\delta\lambda \rightarrow \delta s$. (33.7)

Pressure modifies structure.

But markets are self-referential.

The resulting structure becomes evidence that modifies later pressure:

$\delta s \rightarrow \delta\lambda'$. (33.8)

This two-way loop is the basis of the signed-conjugacy construction in the source Technical Analysis article.

---

# 33.3 The signed-conjugacy operator

The source formulation writes:

$C_{\chi} = [[0,F],[\chi M,0]]$. (33.9)

Here:

F maps signal displacement into structural displacement.
M maps structural displacement back into signal displacement.
$\chi$ records the orientation of the return path.

Applied to a state vector:

$x = [\delta\lambda,\delta s]^{T}$. (33.10)

the operator gives:

$C_{\chi x} = [F\delta s, \chi M\delta\lambda]^T$. (33.11)

Applying the operator twice gives:

$C_{\chi^2} = [[\chi FM, 0], [0, \chi MF]]$. (33.12)

Only under a normalized canonical case such as:

$FM = MF = I$ (33.13)

does this reduce to:

$C_{\chi^2} = \chi I$. (33.14)

This qualification matters.

The sign classification under a normalized canonical case such as:

$FM = MF = I$ (33.13)

does this reduce to:

$C_{\chi^2} = \chi I$. (33.14)

is structurally useful, but the full market operator may contain unequal gains, delays, nonlinearities, and cross-channel coupling.

---

# 33.4 Corrective regime: $\chi < 0$

When:

$\chi < 0$, (33.15)

realized movement tends to generate opposing pressure.

Examples include:

- value buying after decline;
- profit-taking after extension;
- liquidity provision against short-term displacement;
- range trading;
- inventory rebalancing;
- policy stabilization.

The characteristic loop is:

Movement
→ OpposingInterpretation
→ Counter-Order
→ PartialReversal. (33.16)

This regime supports the familiar use of:

- RSI;
- stochastic;
- mean-reversion bands;
- support and resistance;
- overbought and oversold language.

But the corrective interpretation must be diagnosed rather than presumed.

---

## 33.5 Critical or weak-return regime: $\chi \approx 0$

When:

$\chi \approx 0$, (33.17)

realized structure produces little stable return pressure.

Possible conditions include:

- compression;
- low-conviction drift;
- transition;
- unstable liquidity;
- information waiting;
- competing narratives;
- regime uncertainty.

Movement may not generate a reliable counter-move or continuation.

The system can become sensitive to small disturbances.

This regime often produces:

- false breakouts;
- repeated crossovers;
- low-quality oscillator signals;
- changing support and resistance;
- unstable pattern interpretation.

The appropriate interpretation is not necessarily:

Nothing is happening.

It may be:

The return operator has not yet stabilized. (33.18)

---

## 33.6 Self-confirming regime: $\chi > 0$

When:

$\chi > 0$, (33.19)

realized movement tends to generate pressure in the same direction.

Examples include:

- trend following;
- momentum allocation;
- stop activation;
- forced covering;
- margin liquidation;
- benchmark chasing;
- narrative reinforcement;
- reflexive leverage.

The characteristic loop is:

Movement
→ ConfirmingInterpretation
→ Same-DirectionOrder
→ FurtherMovement. (33.20)

This regime explains why:

- high RSI can remain high;
- price can ride an upper band;
- moving-average distance can expand;
- breakouts can accelerate;
- apparent overvaluation can persist.

In such a regime, a conventional corrective indicator may not be mathematically wrong.

Its usual interpretation may be dynamically mismatched.

---

## 33.7 Local algebraic normal forms

Under appropriate normalization, the three signature regions suggest three local normal forms.

### Corrective

$e^2 = -1$. (33.21)

This corresponds to elliptic or ordinary complex-type circulation.

### Critical

$e^2 = 0$. (33.22)

This corresponds to a parabolic or dual-number-like limiting form.

### Self-confirming

$e^2 = +1$. (33.23)

This corresponds to a hyperbolic or split-complex-type amplification.

This article treats these as **local operator normal forms**, not as claims that markets fundamentally consist of three number systems.

The safer proposition is:

> Protocol-fixed market regimes may sometimes be simplified by different local algebraic structures.

The ordinary complex plane is therefore one member of a wider regime family, not the universal geometry of every market state.

---

## 33.8 χ is not an indicator

χ should not be estimated from one reading such as:

- RSI;
- moving-average slope;
- price above a band;
- one breakout;
- one reversal.

It is a regime-level inference.

A possible χ diagnosis may combine:

- response to extension;
- follow-through after breaks;
- autocorrelation;
- momentum persistence;
- breadth response;

- volume efficiency;
- retest behaviour;
- reversal hazard;
- crowding.

A schematic estimator is:

$\hat{\chi}_P = f(\text{ResponseToMovement},\text{FollowThrough},\text{Reversion},\text{Participation},\text{Crowding} \mid P)$. (33.24)

This is a research object, not a mature universal formula.

---

# 33.9 Empirical response test

One direct test is to examine the expected future response after normalized displacement.

Let:

$d_t$ = normalized current displacement. (33.25)

$r_{t\to t+h}$ = subsequent return over horizon h. (33.26)

Then estimate:

$E[r_{t\to t+h} \mid d_t]$. (33.27)

A corrective regime may exhibit:

$d_t \cdot E[r_{t\to t+h} \mid d_t] < 0$. (33.28)

A self-confirming regime may exhibit:

$d_t \cdot E[r_{t\to t+h} \mid d_t] > 0$. (33.29)

The result depends on:

- horizon;
- volatility;
- transaction cost;
- conditioning variables;
- market state.

Therefore χ must always carry a protocol index:

$\chi_{P,h}$. (33.30)

---

# 33.10 χ transition

The most important moment may not be the value of χ but its change.

A regime transition can be written:

$$\chi_t < 0 \rightarrow \chi_t \approx 0 \rightarrow \chi_t > 0. \quad (33.31)$$

This may describe:

range
→ compression
→ breakout trend.

The reverse transition is:

$$\chi_t > 0 \rightarrow \chi_t \approx 0 \rightarrow \chi_t < 0. \quad (33.32)$$

This may describe:

trend
→ exhaustion or distribution
→ corrective circulation.

The transition itself may become a Period-3 or Period-4 gate.

---

## 33.11 χ and the periodic table

χ primarily modifies the **Motion / Relation** column.

But its effect spreads across the other functions.

| Function | $\chi < 0$ | $\chi \approx 0$ | $\chi > 0$ |
|---|---|---|---|
| **Load** | load invites balancing | load remains unresolved | load attracts further loading |
| **Motion** | extension produces counter-motion | motion has unstable continuation | motion reinforces itself |
| **Constraint** | boundaries encourage rotation | boundaries are repeatedly tested | boundaries may fail explosively |
| **Commitment** | reversal gates become more likely | false gates become common | continuation gates become more likely |

This is why χ should be treated as an environmental modifier around the table rather than as a fifth column.

---

## 33.12 χ summary

χ answers:

> Does realized movement generate counter-pressure, weak pressure, or confirming pressure?

It is:

- not an ordinary indicator;
- not globally constant;

- not directly visible;
- not sufficient for commitment;
- not a claim of market ontology.

Its value lies in explaining why the same Technical Analysis instrument changes meaning across regimes.

The source Technical Analysis framework makes this operator-first sor
→ Intrinsic Characteristic
→ Regime Diagnosis
→ Conditional Interpretation. filecite turn23file17

---

# 34. Ξ as the Effective Control State

# 34.1 Rich field versus compiled state

The market contains more information than any compact model can preserve.

Let:

Σ_P = rich logged market field under protocol P. (34.1)

This field may include:

- price;
- volume;
- order flow;
- breadth;
- volatility;
- leverage;
- liquidity;
- collateral;
- positioning;
- legal events;
- policy;
- accounting;
- narrative;
- historical reactions.

A compact control state is compiled from those traces:

Ξ̂_P = C_P(Σ_P). (34.2)

The PORE source defines this as an operational interface rather than an ontology. It explicitly states that Ξ is valid only within a declared boundary, timebase, instrument, and compilatiailure outside that domain should be reported rather than patched through narrative.
filecite turn23file12

---

# 34.2 The source triple

The source PORE formulation uses:

$$\Xi = (\rho,\gamma,\tau). \quad (34.3)$$

with approximate meanings:

ρ = loading, occupancy, or staying power;
γ = closure, binding, lock-in, or leakage resistance;
τ = agitation, recovery, switching, or persistence scale, depending on the specific implementation.

The symbol τ is overloaded across the wider article family.

The phase framework uses $\tau_i$ for accumulated internal phase time.

To prevent collision, this article writes the minimal market dashboard as:

$$\Xi_{fin} = (\rho,\gamma,\nu). \quad (34.4)$$

where:

ρ = effective loading;
γ = effective lock-in;
ν = effective agitation or dephasing.

Recovery and switching times are retained separately:

$\tau_{rec}$ = recovery or recurrence time. (34.5)

$\tau_{sw}$ = switching time. (34.6)

This is a notational refinement introduced for the integrated article.

It does not change the source architecture.

---

# 34.3 Loading ρ

ρ estimates how much operative structure is present.

Possible market proxies include:

- volume;
- open interest;
- leverage;
- transaction density;

- participation;
- balance-sheet exposure;
- position concentration;
- capital committed;
- narrative attention.

A schematic loading coordinate is:

$\hat{\rho}_P = C_\rho(\text{Volume},\text{OpenInterest},\text{Positioning},\text{Density},\text{Participation} \mid P)$. (34.7)

$\rho$ is not identical to any one proxy.

A market can have high volume but low persistent loading if exchange is mostly churn.

A market can have moderate volume but high structural loading if large positions remain trapped.

---

# 34.4 Lock-in $\gamma$

$\gamma$ estimates how difficult it is for the loaded structure to move, unwind, leak, or change state.

Possible proxies include:

- illiquidity;
- position concentration;
- collateral constraints;
- legal restrictions;
- funding maturity;
- institutional mandate;
- transaction cost;
- structural mass;
- path dependence.

A schematic coordinate is:

$\hat{\gamma}_P = C_\gamma(\text{Liquidity},\text{Concentration},\text{Collateral},\text{Rules},\text{ExitCost} \mid P)$. (34.8)

High $\rho$ does not imply high $\gamma$.

A heavily traded market may remain fluid.

A smaller market may be strongly locked because exits are constrained.

This distinction simplifies many Technical Analysis ambiguities.

Volume shows activity or loading.

It does not by itself reveal mobility.

---

## 34.5 Agitation ν

ν estimates the degree of turbulence, churn, dephasing, or unstable movement.

Possible proxies include:

- realized volatility;
- spread instability;
- order cancellation;
- turnover;
- correlation breakdown;
- gap frequency;
- cross-sectional dispersion;
- failed gate frequency.

A schematic coordinate is:

ν̂_P = C_ν(Volatility,Churn,SpreadInstability,GateFailure,Dispersion | P). (34.9)

High ν may mean:

- active adaptation;
- disorder;
- transition;
- information arrival;
- fragile liquidity.

It is not intrinsically bad.

Too little agitation can indicate stagnation or rigid lock-in.

Too much can destroy stable structure.

---

## 34.6 The control-state cube

The three coordinates generate qualitatively different regimes.

### Low ρ, low γ, low ν

Thin and inactive state.

### High ρ, low γ, moderate ν

Loaded but mobile market.

### High ρ, high γ, low ν

Crowded and rigid state that appears stable.

### High ρ, high γ, high ν

Loaded, trapped, and turbulent state with elevated transition risk.

### Low ρ, high γ, low ν

Sparse but institutionally constrained state.

### Low ρ, low γ, high ν

Noisy, weakly anchored state.

The purpose of Ξ is not to label markets poetically.

It is to distinguish operating conditions that ordinary indicators may collapse together.

---

## 34.7 Example: equal volume, different control states

Consider two markets with similar daily volume.

### Market A

- deep liquidity;
- diverse participants;
- low leverage;
- easy exit;
- stable spreads.

### Market B

- concentrated positions;
- high leverage;
- poor exit capacity;
- collateral sensitivity;
- widening spreads.

Raw volume may be similar.

But:

ρ_A ≈ ρ_B. (34.10)

γ_B > γ_A. (34.11)

ν_B > ν_A during stress. (34.12)

The second market carries much greater transition risk.

A volume-only Technical Analysis interpretation misses the difference.

---

# 34.8 Ξ and the four functional families

Ξ does not replace the periodic table.

It compresses part of its operating environment.

### ρ mainly informs Load / Memory

How much structure is present?

### γ mainly informs Constraint / Boundary

How difficult is movement or reclassification?

### ν mainly informs Motion / Relation

How turbulent or dephased is the state?

Commitment remains gate-dependent.

No value of Ξ automatically proves that a market event has occurred.

Thus:

Ξ = operating condition. (34.13)

Gate = event admission. (34.14)

Ledger = historical consequence. (34.15)

---

# 34.9 Ξ and χ are different

χ describes the orientation of feedback.

Ξ describes the effective condition in which feedback operates.

Two markets may have similar Ξ but different χ.

For example:

- both may be highly loaded and moderately rigid;
- one may mean-revert;
- the other may self-amplify.

Conversely, two markets may share $\chi > 0$ but differ in risk because one is lightly loaded and the other is heavily leveraged.

Therefore:

$\chi \neq \Xi$. (34.16)

A richer regime state may be written:

$\text{Regime}_P = (\Xi_P, \chi_P)$. (34.17)

---

# 34.10 PORE intervention channels

The PORE architecture distinguishes four operational channels:

Probe = measure.
Pump = add or remove loading.
Couple = increase or decrease binding.
Switch = change regime or route. (34.18)

In market language:

### Probe

Observe volume, price, breadth, liquidity, or positioning.

### Pump

Add capital, liquidity, leverage, or forced flow.

### Couple

Change collateral, margin, benchmark, or binding rules.

### Switch

Change policy, routing, market regime, legal status, or dominant strategy.

The distinction prevents observation from being confused with intervention.

It also helps diagnose when a Technical Analysis signal ceases to be a passive Probe and becomes part of a Pump, Couple, or Switch.

---

# 34.11 Local controllability

Inside a sufficiently smooth regime, the PORE source proposes:

$\Delta\Xi \approx G\Delta u$. (34.19)

where:

$\Delta u$ = intervention pulse;
$G$ = locally estimated response matrix.

For a market study, this might ask:

- how does liquidity injection alter ρ, γ, and ν?
- how does a margin change alter lock-in?
- how does a circuit breaker alter agitation?
- how does index inclusion alter loading?

This is not ordinary chart Technical Analysis.

It is a possible higher-period control elso warns that jump or Switch regimes should not be forced into smooth local models. filecite turn23file12

---

## 34.12 Ξ estimation burden

Ξ coordinates are compiled rather than directly observed.

A valid implementation must publish:

- boundary;
- data sources;
- proxy definitions;
- normalization;
- estimation window;
- uncertainty;
- jump detection;
- intervention response;
- failure criteria.

A vague statement such as:

“The market has high γ”

is not meaningful without a protocol.

---

## 34.13 Ξ summary

Ξ answers:

> How loaded, locked, and agitated is the declared market system?

It is:

- a compiled control interface;
- not a universal market ontology;

- not identical to χ;
- not a commitment gate;
- not a replacement for the rich field Σ.

Its contribution is compression.

It reduces a high-dimensional operating condition into a small dashboard without claiming that the dashboard contains the whole market.

---

# 35. Z = R + iQ as the Calibration Atom

# 35.1 Why the CAPM construction matters

Technical Analysis often uses words such as:

- hidden pressure;
- latent energy;
- unresolved force;
- imaginary value;
- phase;
- rotation.

These words can become vague.

The CAPM phase construction is important because it provides a case where the complex coordinate is mathematically declared rather than metaphorically assigned.

For one future cash flow:

$A_t = CF_t/(1 + r_{base})^t$. (35.1)

$R_t = CF_t/(1 + r_{CAPM})^t$. (35.2)

$r_{CAPM} = r_{base} + \beta ERP$. (35.3)

The valuation phase is:

$\cos \theta_t = R_t/A_t$. (35.4)

The orthogonal coordinate is:

$Q_t = \sqrt{(A_t^2 - R_t^2)}$. (35.5)

The completed state is:

$Z_t = R_t + iQ_t = A_t \exp(i\theta_t)$. (35.6)

The source article emphasizes that same declared valuation relation rather than introduced as an independent risk score. filecite turn24file2

---

## 35.2 The principal identity

Along a fixed-amplitude phase orbit:

$R = A \cos \theta$. (35.7)

$Q = A \sin \theta$. (35.8)

Differentiating R gives:

$\partial R/\partial \theta = -A \sin \theta$. (35.9)

Therefore:

$\partial R/\partial \theta = -Q$. (35.10)

This gives Q a precise financial identity:

> Q is the magnitude of the first-order dollar exposure of admitted value R to movement in the declared valuation phase.

Q is therefore not merely a geometric remainder.

But neither is it automatically:

- loss;
- volatility;
- VaR;
- expected shortfall;
- opportunity cost;
- a second market price;
- unexplained error.

---

## 35.3 Ordinary sensitivity is preserved

For the ordinary discounted value:

$R = CF/(1 + r)^t$, (35.11)

the required-return sensitivity is:

$dR = -[tR/(1 + r)]dr$. (35.12)

The phase representation gives:

$dR = -Qd\theta$. (35.13)

Therefore:

$Qd\theta = [tR/(1 + r)]dr$. (35.14)

The phase representation does not replace CAPM sensitivity.

It expresses the same local value change in another coordinate.

This is important because a new geometry should preserve mature financial results before claiming additional value.

---

# 35.4 Measurement rotation

Define the rotated measurement:

$M_{\varphi}(Z) = \mathrm{Re}[\exp(i\varphi)Z]$. (35.15)

Then:

$M_{\varphi}(Z) = R\cos\varphi - Q\sin\varphi$. (35.16)

The principal readouts are:

$M_0(Z) = R$. (35.17)

$M_{\pi/2}(Z) = -Q$. (35.18)

$M_{\pi}(Z) = -R$. (35.19)

$M_{3\pi/2}(Z) = Q$. (35.20)

$M_{2\pi}(Z) = R$. (35.21)

This gives the cycle:

$R \to -Q \to -R \to Q \to R$. (35.22)

The cycle is a family of measurement orientations.

It is not automatically a chronological market sequence.

The CAPM source is explicit that the first quarter-turn changes th conjugate phase exposure, while economic consequence requires actual state movement. filecite turn24file2

---

# 35.5 Exposure is not loss

Suppose the phase changes by $\Delta\theta$.

Then:

$R_{new} = R\cos\Delta\theta - Q\sin\Delta\theta$. (35.23)

Therefore:

$\Delta R = R(\cos\Delta\theta - 1) - Q\sin\Delta\theta$. (35.24)

For small $\Delta\theta$:

$\Delta R = -Q\Delta\theta - (R/2)(\Delta\theta)^2 + (Q/6)(\Delta\theta)^3 + O((\Delta\theta)^4)$. (35.25)

The quantity −Q is an exposure coefficient.

Loss or gain requires:

$\Delta\theta \neq 0$. (35.26)

Recognition additionally requires a gate.

Thus:

Reading −Q ≠ RealizedLoss. (35.27)

Exposure + Movement ≠ LedgeredP&L without recognition. (35.28)

The source runtim Exposure
→ State Movement
→ Economic P&L
→ Gate
→ Ledger + Residual. (35.29) filecite turn23file1

---

# 35.6 Why this is the calibration atom

The CAPM construction is a useful calibration object because it supplies:

- declared baseline A;
- admitted value R;
- derived Q;
- exact norm relation;
- exact phase;
- exact first derivative;
- exact measurement cycle;
- ordinary-finance benchmark;
- gate and ledger distinction;
- explicit failure criteria.

It therefore demonstrates what a mature complex financial construction looks like.

A Technical Analysis proposal using R + iQ should be compared against this standard.

---

# 35.7 Candidate Technical Analysis complex states

A Technical Analysis complex state may be proposed as:

$Z_{TA} = R_{TA} + iQ_{TA}$. (35.30)

But the meanings must be declared.

Candidate R_TA might represent:

- accepted price structure;
- confirmed trend;
- realized displacement;
- verified market breadth;
- admitted value region.

Candidate Q_TA might represent:

- independently measured order-pressure channel;
- unresolved queue;
- reactive liquidity;
- latent positioning pressure;
- independently measured breadth-pressure channel;
- verified conjugate risk exposure.

A weak proposal says:

R = visible price.
Q = everything hidden. (35.31)

A strong proposal says:

R and Q are independently defined, measured, coupled, and phase-tested. (35.32)

---

# 35.8 Q-channel array versus one Q

The earlier Technical Analysis programme identified possible channels such as:

Q_volume;
Q_phase;
Q_density;
Q_gate;
Q_residual;
Q_breadth;
Q_volatility;
Q_cadence. (35.33)

These should not be added together automatically.

They first belong to the richer field Σ_P.

A declared complex chart may compile one defensible conjugate coordinate:

Q_P = C_Q(Q_volume,Q_breadth,Q_liquidity,… | P). (35.34)

But the compilation must be justified.

Otherwise Q becomes a convenient label for heterogeneous market variables.

---

# 35.9 Q is not residual

This distinction must be repeated because it is foundational.

Q is internal to the declared conjugate state.

Residual is what remains outside that closure.

Therefore:

ObservedMarketChange
= ModelledComplexChange

- ε. (35.35)

where:

ε = ε_R + iε_Q or another declared residual representation. (35.36)

A large ε may indicate:

- omitted liquidity change;
- protocol break;
- new information;
- model failure;
- institutional intervention;
- path dependence;
- data error.

The residual prevents the elegant complex plane from claiming more explanatory power than it has earned.

---

# 35.10 CAPM phase and the periodic table

The CAPM state belongs primarily to:

World × Motion / Relation.

It describes a protocol-bound valuation world.

But its components interact with several functions.

### A

Declared baseline Load or amplitude.

### R

Admitted valuation structure.

### Q

Conjugate phase exposure.

### θ

Relational orientation.

### Δθ

State movement.

### Gate

Recognition or settlement.

### Ledger

Recorded financial consequence.

### Residual

Economic change not absorbed or recognized.

The calibration atom therefore illustrates the whole periodic grammar inside one mature local model.

---

## 35.11 CAPM phase does not validate all TA phase claims

The existence of one defensible complex financial model does not prove that:

- every momentum indicator is phase;
- volume is reactive power;
- support is impedance;
- every market cycle is complex rotation;
- every hidden pressure is Q.

The legitimate conclusion is narrower:

> Finance can support a genuine complex geometry when a declared valuation construction supplies the required relation.

Other Technical Analysis applications must meet their own evidential burden.

---

## 35.12 Z summary

Z answers:

> Does the declared market state possess a defensible conjugate completion whose phase provides useful structure?

The CAPM construction shows that the answer can sometimes be yes.

But complex notation earns priority only when it does more than store two real values.

The next section defines that burden explicitly.

---

# 36. Complex Eligibility and the Reduction Rule

# 36.1 Algebraic completion is easy

Given two real variables R and Q, one can always write:

Z = R + iQ. (36.1)

This is algebraically valid.

But algebraic validity does not establish modelling priority.

At one instant:

Z = R + iQ (36.2)

contains the same numerical information as:

(R,Q). (36.3)

The stronger question is:

> Does the complex structure reveal a privileged phase, generator, invariance, event alignment, or intervention advantage that a flexible real pair does not?

---

# 36.2 Eligibility Level 0 — Scalar state

The minimal model is:

X_t. (36.4)

A scalar is sufficient when:

- one observable carries the relevant information;
- no stable second channel is required;
- phase adds no value;
- gate prediction does not improve.

Examples may include a simple threshold process or a one-dimensional risk metric.

The framework should remain scalar unless added structure earns its cost.

---

## 36.3 Eligibility Level 1 — Real pair

The next model is:

$X_t = (R_t,Q_t)$. (36.5)

This is appropriate when two real variables matter but no privileged complex geometry has been established.

The real pair permits:

- multivariate regression;
- state-space modelling;
- vector dynamics;
- nonlinear interaction;
- independent scaling.

A real-pair model should be the default benchmark against which complexification is tested.

---

## 36.4 Eligibility Level 2 — Complex state

The complex representation is:

$Z_t = R_t + iQ_t$. (36.6)

It becomes eligible when:

1. R and Q possess independent domain meanings;
2. both can be measured or estimated separately;
3. their scales can be declared;
4. their coupling is stable enough to support phase;
5. the imaginary unit expresses more than decorative notation.

The phase source uses AC power as the calibration case because active and reactive power possess independent measurement, operational meaning, phase relatconsequences. It argues that new domains should meet a similarly disciplined burden. filecite turn23file13

---

## 36.5 Eligibility Level 3 — Phase-dynamical model

A complex state becomes phase-dynamical when:

$\theta_t = \operatorname{atan2}(Q_t,R_t)$ (36.7)

organizes evolution more effectively than the original variables alone.

A candidate local dynamic is:

$$dZ/dt = [g_A(t) + i\omega(t)]Z + \varepsilon(t). \quad (36.8)$$

where:

$g_A$ = amplitude growth or decay;
$\omega$ = phase velocity;
$\varepsilon$ = residual.

Under locally stable amplitude:

$$dZ/d\theta \approx iZ. \quad (36.9)$$

The phase should simplify or regularize the dynamics.

If it does not:

retain the real pair. (36.10)

---

# 36.6 Eligibility Level 4 — Secondary phase time

Phase becomes a candidate internal clock when differently paced calendar episodes align under phase.

Let:

$$\theta(t) = \text{current phase}. \quad (36.11)$$

A signed accumulated coordinate is:

$$\tau_i(t) = \text{Unwrap}[\theta(t)]. \quad (36.12)$$

A monotone traversal coordinate is:

$$\tau_i(t) = \int_0^t |\theta(s)|\, ds. \quad (36.13)$$

A successful internal-time model should satisfy something like:

$$D_{phase} < D_{calendar}. \quad (36.14)$$

where:

$D_{phase}$ = dispersion among episodes aligned by $\tau_i$;
$D_{calendar}$ = dispersion among episodes aligned by $t$.

If phase alignment does not improve episode compression, phase should not be promoted to an internal clock.

---

# 36.7 Eligibility Level 5 — Phase-bearing event model

A stronger model requires consequential gates to concentrate in phase.

Let $\Theta_G$ be a phase region associated with an event.

The hypothesis is:

Pr(Gate = 1 | θ ∈ Θ_G,Controls)

Pr(Gate = 1 | CalendarAge,Controls). (36.15)

Examples might include:

- breakout phase;
- failure phase;
- liquidity-stress phase;
- impairment phase;
- settlement phase.

If gate timing is no more stable in phase than in ordinary state variables, the complex model remains descriptive rather than event-bearing.

---

# 36.8 Eligibility Level 6 — Time-bearing world

The strongest model requires:

- robust phase order;
- gate;
- persistent trace;
- residual;
- ledger;
- backreaction.

The full chain is:

t
→ Z(t)
→ θ(t)
→ $\tau_i(t)$
→ $Gate_k$
→ $Trace_k$
→ $L_{k+1}$
→ Backreaction. (36.16)

A phase-bearing system becomes a time-bearing world only when admitted events alter subsequent dynamics.

The phase source states the strongest candidate condition as:

Uneven parent duration

- independently measurable conjugate state
- robust phasitive gates

- persistent trace
- future constraint
- parent-world backreaction. filecite turn23file9

---

## 36.9 Independent definition test

Reject or revise the complex model when:

Q = everything not explained by R. (36.17)

A valid Q requires:

- prior definition;
- independent proxy;
- stable domain meaning;
- forward relevance.

Unst remain residual.

This is one of the explicit rejection rules in the phase source. filecite turn23file8

---

## 36.10 Scaling robustness test

A complex phase depends on relative scaling.

Suppose:

Z = R + iQ. (36.18)

Rescaling Q gives:

Z_c = R + icQ. (36.19)

The phase becomes:

θ_c = atan2(cQ,R). (36.20)

If small reasonable changes in c radically reorder phase, the phase claim is fragile.

The normalization must therefore be:

- declared;
- economically meaningful;
- transport-tested;
- stable within tolerance.

---

## 36.11 Real-pair benchmark

The complex model must be compared with a flexible real model:

Model_real = f(R,Q,lags,interactions). (36.21)

Model_complex = g(A,θ,lags,phase interactions). (36.22)

Complexification earns priority only when it improves at least one meaningful criterion:

- predictive performance;
- parameter stability;
- episode compression;
- gate localization;
- cross-frame robustness;
- intervention quality;
- interpretability with fewer degrees of freedom.

A beautiful phase portrait is not sufficient.

---

## 36.12 Gate-concentration test

Define:

H_G(θ) = hazard of gate event at phase θ. (36.23)

Compare it with:

H_G(t) = hazard by calendar or episode age. (36.24)

The phase model gains support when:

Var[H_G(θ)] across comparable episodes
< Var[H_G(t)] after appropriate alignment. (36.25)

or when phase-conditioned hazard improves out-of-sample prediction.

The exact statistical test depends on the domain.

---

## 36.13 Intervention test

A phase model becomes more operationally important when phase-aware action improves results.

Let:

Loss_phase = outcome loss under phase-aware intervention. (36.26)

Loss_calendar = outcome loss under calendar-based intervention. (36.27)

A strong result would be:

Loss_phase < Loss_calendar. (36.28)

The phase source proposes this kind of mains such as maintenance, cloud systems, infrastructure, and other logged processes. filecite turn23file2

In finance, possible interventions include:

- hedge timing;
- exposure reduction;
- gate tightening;
- liquidity preparation;
- position sizing;
- review scheduling.

These applications would require careful prospective testing.

---

# 36.14 Failure conditions

The complex interpretation should be rejected, weakened, or reduced when:

1. the norm relation exists only because of arbitrary normalization;
2. Q is an error bucket;
3. phase does not simplify dynamics;
4. episodes do not align better in phase;
5. gates do not concentrate by phase;
6. phase order changes under minor scaling;
7. the complex model does not outperform a real pair;
8. Q adds no diagnostic or intervention value;
9. trace and backreaction are absent despite a claim ofse rejection conditions come directly from the phase framework's evidence discipline. filecite turn23file8

---

# 36.15 The reduction ladder

A mature theory must be able to reduce itself.

The phase source gives the following ladder:

Full phase–gate–ledger world
→ remove backreaction
→ phase-bearing event model

→ remove ledger
→ secondary phase-time model
→ remove gate
→ complex dynamical model
→ remove phase privilege
→ two-real-variable model
→ remove Q
→ one-variable model. (36.29)

The governing rule is:

> Use the least complex model that preserves measurable explanatory, predictive, or intervention gain.

The strongest framework is not the one containing the most advanced mathematics.

It is the one preserving necessary distinctions with the least unsupported structure.

---

# 36.16 Complex eligibility card

Every proposed complex Technical Analysis model should publish:

```
Declared protocol:
Definition of R:
Definition of Q:
Independent Q proxy:
Normalization:
Amplitude A:
Phase θ:
Real-pair benchmark:
Phase-stability test:
Episode-alignment test:
Gate-concentration test:
Transport tests:
Residual:
Intervention test:
Reduction trigger:
```

This card prevents complex notation from becoming an aesthetic overlay.

---

# 36.17 Complex eligibility summary

Complexification is not the starting assumption.

It is an earned modelling status.

The hierarchy is:

Scalar
→ Real pair
→ Complex state
→ Phase dynamics
→ Secondary phase time
→ Phase-bearing event model
→ Time-bearing world. (36.30)

Each step requires new evidence.

No model should advance merely because the previous stage is mathematically elegant.

---

# 37. Phase Time and Time-Bearing Market Episodes

## 37.1 Calendar time is not internal progress

Two market episodes may take the same number of days while developing very differently.

One may traverse:

- accumulation;
- breakout;
- follow-through;
- exhaustion;
- reversal.

Another may remain inside one unresolved range.

Therefore:

Equal $\Delta t$ does not imply equal internal progress. (37.1)

Conversely, two crises may take radically different calendar durations but pass through structurally similar internal stages.

This motivates a candidate secondary time coordinate.

---

## 37.2 Four time variables

The integrated framework distinguishes:

$t$ = parent-world calendar time. (37.2)

$\theta$ = current phase orientation. (37.3)

$\tau_i$ = accumulated internal phase progress. (37.4)

$k$ = committed ledger-event order. (37.5)

These variables answer different questions.

### $t$

How much clock time has elapsed?

### $\theta$

Where is the current state oriented in its conjugate plane?

**$\tau_i$**

How far has the state progressed internally?

**k**

How many consequential events have been committed?

Thus:

$t \neq \theta \neq \tau_i \neq k$. (37.6)

---

# 37.3 Phase orientation

For:

$Z(t) = R(t) + iQ(t)$, (37.7)

the phase is:

$\theta(t) = \text{atan2}(Q(t),R(t))$. (37.8)

The same $\theta$ can be revisited many times.

But the visits may differ in:

- direction;
- amplitude;
- accumulated phase;
- ledger history;
- residual;
- observer state.

Therefore:

Same $\theta \neq$ same historical state. (37.9)

Phase is orientation, not complete identity.

---

# 37.4 Signed phase time

A signed unwrapped phase coordinate is:

$\tau_i^{signed}(t) = \text{Unwrap}[\theta(t)]$. (37.10)

This preserves direction.

A clockwise and counterclockwise traversal produce opposite changes.

This is useful when the direction of phase development matters.

But noise and branch cuts must be handled carefully.

---

## 37.5 Absolute phase depth

A monotone phase-depth coordinate is:

$\tau_i^{abs}(t) = \int_0^t |\theta(s)|\, ds.$ (37.11)

This records total phase traversal irrespective of direction.

It may be useful for:

- accumulated process activity;
- episode ageing;
- repeated cycling;
- degradation;
- stress development.

But it loses directional information.

Therefore the two coordinates should not be confused.

---

## 37.6 Event order k

Suppose an episode contains gates:

$e_1, e_2, \ldots, e_k.$ (37.12)

The ledger index increases only when a gate admits an event:

$k \to k + 1$ after Gate = Admit. (37.13)

A long calendar interval may contain no new k.

A short crisis interval may contain many gate events.

Ledger time is therefore event-selective.

This is consistent with the disclosure framework, where t declared, gated trace rather than through raw recursion alone. filecite turn19file4 filecite turn19file5

---

## 37.7 Selection depth σ

Selection depth records reduction of admissible future branches.

Let $\Omega(t)$ be the set of futures still treated as admissible.

A schematic selection depth is:

$\sigma(t) = -\ln[\text{Measure}(\Omega(t))/\text{Measure}(\Omega(0))].$ (37.14)

This is a conceptual expression; practical applications require a declared branch model.

As alternatives are eliminated:

σ increases. (37.15)

But phase and selection depth are not identical.

A market can rotate repeatedly without eliminating many possibilities.

A sudden event can eliminate many possibilities with little prior phase traversal.

Thus:

$\tau_i \neq \sigma$. (37.16)

A useful study may estimate both.

---

# 37.8 Phase-aligned episodes

Suppose episodes j = 1…N have different calendar durations.

For each episode, estimate:

$Z_j(t) = R_j(t) + iQ_j(t)$. (37.17)

Then transform:

$t \rightarrow \tau_i, j$. (37.18)

Compare episode paths in both coordinates.

A phase-time model gains support when:

$$\text{Dispersion}[\{\text{Episode}_j(\tau_i)\}] < \text{Dispersion}[\{\text{Episode}_j(t)\}]. \quad (37.19)$$

The result should be tested out of sample.

Otherwise phase alignment may merely overfit the episodes used to construct it.

---

# 37.9 Phase-sensitive gate hazard

Let $h_G$ denote gate hazard.

Compare:

$h_G(t \mid X)$ (37.20)

with:

$h_G(\theta \mid X)$ (37.21)

and:

$h_G(\tau_i \mid X)$. (37.22)

Here X contains conventional controls such as:

- volatility;
- volume;
- trend;
- breadth;
- structural mass;
- regime $\chi$.

The phase model becomes useful when $\theta$ or $\tau_i$ contributes stable information beyond X.

---

# 37.10 Market example: breakout development

A breakout episode may progress through:

1. loading;
2. compression;
3. boundary pressure;
4. initial crossing;
5. close;
6. retest;
7. acceptance;
8. expansion;
9. exhaustion or continuation.

These stages may occupy unequal calendar intervals.

A candidate phase model may align different breakouts by internal progression rather than by "day since breakout."

But the phase must be constructed from independently meaningful channels.

It cannot be defined retrospectively from the stage labels it is supposed to predict.

---

# 37.11 Market example: crisis development

A crisis may contain:

- latent leverage;
- liquidity deterioration;
- price decline;

- collateral pressure;
- forced sale;
- institutional recognition;
- intervention;
- recovery.

Visible price may remain stable while Q-like unresolved pressure rises.

A candidate state might be:

R_crisis = verified functioning or admitted market value. (37.23)

Q_crisis = independently estimated unresolved liquidity or collateral pressure. (37.24)

Z_crisis = R_crisis + iQ_crisis. (37.25)

This remains a research proposal.

The Q channel must not be defined merely as “hidden crisis pressure.”

It requires measurable proxies and a real-pair benchmark.

---

# 37.12 Phase and the six periods

Phase can appear at several periods.

### Window phase

Orientation within one local oscillation.

### Structure phase

Relation between price and a conjugate pressure channel.

### Event phase

Location relative to a gate region.

### Episode phase

Accumulated progress through a trend, range, crisis, or recovery.

### World phase

Internal progression of a valuation, policy, collateral, or institutional regime.

Not every period supports a useful complex state.

Complex eligibility remains cell-specific.

---

## 37.13 Phase does not replace the periodic table

The periodic table classifies recurring market functions across closure levels.

Phase is one possible structure inside Motion / Relation.

Therefore:

Periodic grammar = general architecture. (37.26)

Complex phase = locally earned analytic structure. (37.27)

The R → −Q → −R → Q → R cycle should not be imposed on every table cell.

Where justified, it supplies a local phase-valence cycle inside the broader architecture.

---

## 37.14 Time-bearing episode criteria

A market episode qualifies as a candidate phase-time episode when:

1. calendar duration is uneven across comparable cases;
2. R and Q are independently defined;
3. phase order is robust;
4. phase improves episode alignment;
5. event gates concentrate in phase;
6. admitted events enter persistent trace;
7. the trace changes later behaviour;
8. the model outperforms simpler alternatives.

In compact form:

UnevenDuration

- ConjugatePair
- StablePhase
- GateConcentration
- Trace
- Backreaction
  = CandidateTimeBearingEpisode. (37.28)

This is a research criterion, not an established market theorem.

---

## 37.15 Phase-time failure

The phase-time claim fails when:

- phase order depends on arbitrary scaling;
- equivalent episodes do not align;
- gates remain better explained by ordinary variables;
- phase adds no intervention value;
- Q lacks independent meaning;
- backreaction is absent from a claimed world;
- the real-pair model performs equally well.

The model should then be reduced.

---

## 37.16 Phase-time summary

Phase time answers:

> Can irregular calendar evolution be reorganized as a more stable internal progression toward consequential gates?

Its promise is significant.

Its evidential burden is correspondingly high.

A valid phase clock must do more than produce an attractive spiral.

It must improve:

- episode alignment;
- gate prediction;
- transport;
- intervention;
- or explanation under reduced complexity.

---

# 38. The Advanced Architecture in One View

The advanced variables now occupy distinct positions.

| Construct | Primary question | Status |
|---|---|---|
| $\chi$ | What is the feedback orientation? | regime signature |
| $\Xi = (\rho,\gamma,\nu)$ | How loaded, locked, and agitated is the system? | compiled control state |
| $Z = R + iQ$ | Does the state possess a conjugate geometry? | optional complex completion |
| $\theta$ | What is the present conjugate orientation? | phase coordinate |

| Construct | Primary question | Status |
|---|---|---|
| $\tau_i$ | How far has internal phase progressed? | candidate internal time |
| **k** | How many consequential events have entered the ledger? | event order |
| **σ** | How many future alternatives have been suppressed? | selection depth |
| **r** | What remains unresolved? | residual |
| **L** | What history now constrains future action? | ledger |

The integrated runtime is:

Σ_P
→ Compile Ξ_P
→ Diagnose χ_P
→ Construct Z_P only if eligible
→ Estimate θ_P and $\tau_i$,P
→ Test Gate_P
→ Write e_k + r_k into L_{k+1}
→ Observe Backreaction
→ Revise P admissibly. (38.1)

This sequence should not be treated as mandatory in every analysis.

A simpler method may stop at:

Σ_P → Structure diagnosis. (38.2)

A mature framework advances only as far as evidence permits.

The next part will use the periodic table to redefine **confirmation**, identify the table's most important missing cells, and specify an empirical record through which the whole framework can be tested, weakened, or rejected.

# Part VIII — Confirmation, Missing Cells, and Empirical Design

# 39. Confirmation as Functional Independence

# 39.1 The conventional confirmation problem

Technical Analysis often treats confirmation as agreement among indicators.

A common report may say:

- price is above the moving average;
- the fast average is above the slow average;
- MACD is positive;
- RSI is above 50;
- momentum is positive.

This sounds like five separate pieces of evidence.

But most of the evidence may originate from the same closing-price history.

The indicators differ in transformation, smoothing, and normalization, yet their source lineage is highly similar.

Therefore:

IndicatorAgreement ≠ IndependentConfirmation. (39.1)

A better question is:

> How many independently informative market functions support the claim?

---

## 39.2 Source lineage

Every indicator should disclose its source lineage.

A source lineage identifies which raw channels generated the method.

Possible source families include:

- price;
- volume;
- order book;
- breadth;
- open interest;
- options;
- funding;
- collateral;
- accounting;
- legal or policy events;
- narrative attention.

Let S_j denote the source lineage of indicator j.

For example:

S_MA = {closing price}. (39.2)

S_MACD = {closing price}. (39.3)

S_RSI = {closing-price differences}. (39.4)

S_VWAP = {price, volume}. (39.5)

S_Breadth = {component-level price states}. (39.6)

S_OpenInterest = {derivative position ledger}. (39.7)

Methods with identical or strongly overlapping $S_j$ should not be treated as fully independent evidence.

---

## 39.3 Operator lineage

Methods may also share an operator lineage even when their formulas differ.

An operator word describes the sequence of transformations applied to data.

A generic operator word is:

$$w_j = G_j \circ N_j \circ A_j \circ \Delta_j \circ F_j \circ \Pi_j. \quad (39.8)$$

where:

$\Pi$ = projection;
F = filtration;
$\Delta$ = difference or relation;
A = accumulation;
N = normalization;
G = gate or threshold.

Not every method uses every operator.

Examples:

$$\text{MovingAverage} = F(\Pi_{price}). \quad (39.9)$$

$$\text{MACD} = \Delta[F_{fast}(\Pi_{price}),F_{slow}(\Pi_{price})]. \quad (39.10)$$

$$\text{RSI} = N[A(\text{Positive}\Delta\text{Price}),A(\text{Negative}\Delta\text{Price})]. \quad (39.11)$$

$$\text{VWAP} = N[A(\text{Price} \times \text{Volume}),A(\text{Volume})]. \quad (39.12)$$

$$\text{Breakout} = G[\Delta(\text{Price},\text{Boundary})]. \quad (39.13)$$

Two methods may use different formulas while still sharing most of their operator chain.

---

## 39.4 Functional lineage

The periodic grammar adds a third form of lineage.

Each method has a dominant functional family:

- Load;
- Motion;
- Constraint;
- Commitment.

For example:

$$\text{Moving average} \rightarrow \text{Load}. \quad (39.14)$$

MACD → Motion. (39.15)

Support → Constraint. (39.16)

Breakout close → Commitment. (39.17)

Two methods derived from price may nevertheless contribute different functions.

But source overlap still matters.

A moving average and a breakout close are not identical simply because both use price.

The moving average supplies memory.

The close supplies commitment.

Confirmation quality therefore depends on both:

- source independence;
- functional independence.

---

## 39.5 Period independence

Evidence may also come from different closure periods.

For example:

- Window-level bullish candle;
- Structure-level rising breadth;
- Event-level breakout;
- Episode-level trend continuation;
- World-level policy support.

Agreement across periods may strengthen a claim because it shows that the interpretation is not confined to one local object.

But higher-period agreement should not be counted automatically.

The periods must be connected through valid transport.

A daily bullish candle does not become weekly trend confirmation merely because both point upward.

Thus:

CrossPeriodAgreement requires declared transport. (39.18)

---

## 39.6 Failure-mode independence

Two methods are more valuable together when they fail for different reasons.

For example:

### Moving average

May fail through lag or whipsaw.

### Breadth

May fail through universe choice or concentration.

### Volume confirmation

May fail through churn or forced flow.

### Retest gate

May fail through delayed reversal or thin liquidity.

If several methods share the same failure mode, their agreement may collapse simultaneously.

Therefore:

RobustConfirmation
requires not only different signals,
but partially independent failure modes. (39.19)

---

## 39.7 Three dimensions of confirmation independence

A mature confirmation system should evaluate:

I_source = source independence. (39.20)

I_operator = operator independence. (39.21)

I_function = functional independence. (39.22)

I_failure = failure-mode independence. (39.23)

A composite independence score may be written:

I_total = f(I_source,I_operator,I_function,I_failure). (39.24)

The exact function must be empirically defined.

The present article proposes the architecture, not a universal calibration.

---

## 39.8 Pseudo-confirmation

Pseudo-confirmation occurs when many signals appear to agree but are largely transformed versions of the same input.

Examples include:

- price above moving average;

- positive moving-average slope;
- bullish crossover;
- positive MACD;
- positive rate of change.

All may be consequences of the same recent price rise.

Their agreement can still describe trend persistence.

But it should not be interpreted as five independent causal channels.

The proper statement is:

> Several price-memory transformations agree that recent directional structure is positive.

This is more precise than:

> Five indicators confirm the trade.

---

# 39.9 Functionally diverse confirmation

A stronger breakout diagnosis may contain:

## Load evidence

Volume, open interest, participation, positioning, or transaction density.

## Motion evidence

Normalized displacement, acceleration, relative strength, or breadth expansion.

## Constraint evidence

A predeclared boundary with structural mass.

## Commitment evidence

Close, follow-through, retest, settlement, or institutional recognition.

## Residual evidence

Conflicting higher timeframe, liquidity weakness, event risk, or narrow participation.

This yields:

BreakoutConfirmation
= LoadSupport

- MotionSupport
- MeaningfulConstraint

- CommitmentGate
  – ResidualBurden. (39.25)

The terms remain conceptually distinct even when one method contributes to more than one term.

---

# 39.10 Confirmation rank

A practical rank may classify confirmation quality.

### Rank 0 — Single reading

One indicator, one source, no gate.

### Rank 1 — Same-family agreement

Several related indicators agree.

### Rank 2 — Cross-functional agreement

Load, Motion, Constraint, or Commitment evidence aligns.

### Rank 3 — Cross-period agreement

The claim survives transport across closure periods.

### Rank 4 — Residual-controlled agreement

Contradictions are recorded and remain within tolerance.

### Rank 5 — Ledgered confirmation

The event alters subsequent behaviour and the new state persists.

This rank is not a profitability score.

It measures structural closure.

---

# 39.11 Conditional incremental value

The strongest empirical test is not visual agreement but incremental information.

Suppose indicator A is already known.

Indicator B contributes independent value only if:

Information(Y_future | A,B)

Information(Y_future | A). (39.26)

Equivalently, in predictive testing:

Loss(Model[A,B]) < Loss(Model[A]) out of sample. (39.27)

The improvement should remain after:

- transaction costs;
- parameter tuning;
- multiple-testing adjustment;
- regime segmentation;
- walk-forward validation.

If B adds no conditional value, it may be a useful visualization but not independent confirmation.

---

## 39.12 Confirmation and observer crowding

An apparently strong confirmation cluster may increase market crowding.

Suppose many widely used indicators all point in the same direction.

Because they share data and operator lineage, many participants may act simultaneously.

This can produce:

- strong initial follow-through;
- crowded entries;
- shared stops;
- fragile exits.

Therefore:

ConfirmationStrength and CrowdingRisk can rise together. (39.28)

A mature system should report both.

---

## 39.13 Confirmation summary

Confirmation should not be counted by indicator number.

It should be evaluated through:

- source lineage;
- operator lineage;
- functional diversity;
- period transport;
- failure-mode independence;
- residual burden;

- incremental out-of-sample value.

The governing principle is:

> Confirmation is strongest when distinct market functions, derived from partially independent evidence channels, support the same interpretation under compatible protocols and preserve their residuals.

---

# 40. Empty Cells and Missing Instrument Families

# 40.1 Why a periodic table should predict absences

A taxonomy is useful when it classifies known objects.

A periodic architecture becomes more powerful when it also predicts where an object should exist but is missing or immature.

The current table suggests several weakly populated regions.

These absences may indicate:

- historical neglect;
- measurement difficulty;
- data unavailability;
- conceptual confusion;
- or unnecessary theoretical structure.

The empty cells must therefore be treated as hypotheses, not proof that a new indicator must exist.

---

# 40.2 Mark × Residual — unexecuted intention field

Ordinary Technical Analysis primarily records executed trades.

But many market intentions never become transactions.

Missing Mark-level residual instruments include measures of:

- cancelled orders;
- unfilled quantity;
- hidden liquidity;
- queue abandonment;
- cross-venue disagreement;
- execution rejection;
- spoof-like transient depth.

A candidate object is:

IntentResidual_t
= SubmittedInterest_t
− ExecutedInterest_t
− LegitimatelyWithdrawnInterest_t. (40.1)

The terms are difficult to identify cleanly.

Yet the concept matters because executed price alone may omit substantial unrealized pressure.

---

# 40.3 Window × Path — path-bearing candle

The ordinary candle discards the order of intrawindow events.

A path-bearing window instrument could preserve:

- whether the high occurred before the low;
- time spent near each extreme;
- number of reversals;
- volume distribution through the path;
- closing recovery speed;
- order-flow imbalance near the close.

A possible state is:

W_path
= (O,H,L,C,V,OrderHL,ResidenceTime,ReversalCount,VolumePath). (40.2)

This would reduce the ambiguity of candles with identical OHLCV but different internal dynamics.

---

# 40.4 Structure × Load — base-completeness score

Most indicators observe only one narrow base.

A base-completeness instrument would report which important load channels are present.

Let channels include:

- price memory;
- volume;
- breadth;
- positioning;
- liquidity;

- leverage;
- volatility;
- institutional reference.

Define:

BaseCompleteness_P
= Σ_j w_j Coverage_j. (40.3)

The score would not judge whether the indicator is accurate.

It would disclose which load dimensions its interpretation ignores.

For example:

Moving average → strong price-memory coverage, weak positioning coverage. (40.4)

Volume profile → strong transaction-density coverage, weak participant-identity coverage. (40.5)

---

## 40.5 Structure × Relation — relation-without-gate warning

Many indicators generate warnings that are mistakenly promoted to events.

A dedicated warning instrument should explicitly output:

- relation detected;
- gate absent;
- residual unresolved.

For example:

DivergenceStatus
= (RelationalWeakening = 1, ReversalGate = 0, Residual = r). (40.6)

This would institutionalize the distinction:

Warning ≠ Event. (40.7)

Such a tool may be more valuable than another oscillator because it prevents premature closure.

---

## 40.6 Structure × Commitment — structural acceptance score

Between a raw structure and a full event lies the question of whether a structure has become persistent.

A structural acceptance score could measure:

- persistence across windows;
- parameter robustness;
- cross-frame survival;
- breadth coherence;
- reduced contradiction.

For structure S:

$$A_S = w_1\text{Persistence} - w_2\text{Transport} - w_3\text{Participation} - w_4\text{Residual}. \quad (40.8)$$

This would distinguish a temporary moving-average alignment from a genuinely established trend structure.

---

# 40.7 Event × Commitment — residual-bearing gate score

This is one of the clearest missing instrument families.

Most breakout systems return:

$$\text{Breakout} = \text{yes or no}. \quad (40.9)$$

A residual-bearing gate should return:

$$\text{GateOutput} = (\text{Decision}, \text{Strength}, \text{ResidualVector}, \text{Invalidation}, \text{Authority}). \quad (40.10)$$

For example:

Decision = Admit.
Strength = 0.72.
Residual = weak breadth + untested weekly boundary.
Invalidation = close back below level.
Authority = daily market close.

The result preserves partial closure.

---

# 40.8 Event × Transport — event portability score

An event may be strong locally but weak in another frame.

A portability score could test:

- timeframe survival;

- volatility-normalized survival;
- universe coherence;
- alternate bar construction;
- nearby boundary perturbation.

Define:

Portability(e) = Σ_j w_j Survival_j. (40.11)

A breakout visible only on one highly specific chart would receive low portability.

A transition recognized across several admissible frames would receive a higher score.

---

# 40.9 Episode × Load — residual-history state

Episode models often record only successful pivots and transitions.

A more complete state would include accumulated unresolved event history.

Let:

H_res,m = Σ_{k∈Episode_m} D_k r_k. (40.12)

where D_k is a decay or persistence operator.

This state could distinguish:

- clean trend;
- trend built on repeated fakeouts;
- range carrying trapped positions;
- recovery with unresolved funding stress.

The residual-history state may improve episode classification.

---

# 40.10 Episode × Motion — phase-clock instrument

A mature phase-clock instrument would require:

- independently defined R and Q;
- robust phase;
- episode alignment;
- gate concentration;
- real-pair benchmark.

Its output might include:

$\theta_t$ = current phase. (40.13)

$\tau_{i,t}$ = accumulated internal phase. (40.14)

$\omega_t$ = phase velocity. (40.15)

$\alpha_{\theta,t}$ = phase acceleration. (40.16)

PhaseConfidence_t = robustness score. (40.17)

Such an instrument does not yet exist as a mature general TA standard.

The phase framework makes clear that a secondary clock is meaningful only when phase organizes dynamics and consequential gates better than ordinary time.

---

# 40.11 Episode × Commitment — completion certificate

Episode completion is usually declared retrospectively.

A completion certificate would record:

- defining episode relation;
- failure of that relation;
- boundary transition;
- new gate;
- new-state persistence;
- unresolved residual.

A possible output is:

EpisodeCertificate
= (OldGrammarFailed,NewGateAdmitted,Persistence,Residual,BranchStatus). (40.18)

This would be especially useful for:

- wave counts;
- trend endings;
- range breakouts;
- volatility-regime changes.

---

# 40.12 World × Relation — observer-crowding metric

A World-level relation instrument should measure when observation becomes intervention.

Candidate inputs include:

- public signal visibility;

- assets following similar rules;
- option and futures positioning;
- benchmark-linked capital;
- stop concentration;
- social attention;
- liquidity capacity.

A provisional crowding ratio is:

CrowdingRatio
= EstimatedSignalLinkedCapital/LiquidityCapacity. (40.19)

A high ratio may indicate:

- stronger self-confirmation initially;
- higher later fragility;
- adversarial exploitation risk.

This is difficult to measure but structurally important.

---

# 40.13 World × Constraint — cross-ledger rigidity map

Market worlds contain multiple constraints:

- legal;
- accounting;
- collateral;
- regulatory;
- benchmark;
- operational.

A cross-ledger rigidity map could show where the same market object is locked differently across institutions.

Let:

Γ_world = {γ_legal,γ_accounting,γ_collateral,γ_market,γ_policy}. (40.20)

A price signal may appear flexible while the legal or collateral state remains rigid.

This can explain delayed or abrupt transitions.

---

# 40.14 World × Commitment — ledger reconciliation

## gate

Different ledgers may recognize events at different times.

A reconciliation gate would compare:

- market-price recognition;
- accounting recognition;
- legal recognition;
- risk-system recognition;
- policy recognition.

Define:

RecognitionVector_k
= (g_market,g_accounting,g_legal,g_risk,g_policy). (40.21)

A fully reconciled world event would require sufficient alignment among the relevant components.

This is especially valuable for:

- default;
- impairment;
- crisis;
- restructuring;
- regime transition.

---

# 40.15 Missing cells as falsifiable predictions

The existence of an empty cell does not guarantee that filling it will improve prediction.

Each proposed family should be tested against:

- simpler alternatives;
- data availability;
- operational usefulness;
- stability;
- cost;
- redundancy.

A missing cell may be removed if:

- it cannot be defined reliably;

- it duplicates another cell;
- it adds no value;
- it has no stable empirical object.

The table must remain revisable rather than defend every empty position.

---

# 41. The Technical Analysis Record

## 41.1 Why a standard record is necessary

Technical Analysis often fails through poor memory rather than poor imagination.

Analysts remember successful interpretations.

Failed signals disappear.

Anchors move.

Wave counts change.

Thresholds are altered.

Contradictory evidence is forgotten.

A standard record converts analysis into an auditable sequence.

The broader declared-disclosure framework treats mature observation as a chain of:

Declaration
→ Projection
→ Gate
→ Trace
→ Residual
→ Ledger
→ Admissible revision. (41.1)

The same discipline can be applied to Technical Analysis.

---

## 41.2 Core record

A proposed record is:

$$TARecord_k = (P_k,p_k,g_k,S_k,w_k,\chi_k,\Xi_k,G_k,e_k,r_k,T_k,L_k,I_k,O_k). \quad (41.2)$$

where:

$P_k$ = protocol;
$p_k$ = period;
$g_k$ = functional group;
$S_k$ = source lineage;
$w_k$ = operator word;

$\chi_k$ = regime assumption;
$\Xi_k$ = effective control state, where used;
$G_k$ = gate rule;
$e_k$ = event or claim;
$r_k$ = residual;
$T_k$ = transport tests;
$L_k$ = ledger consequence;
$I_k$ = invalidation;
$O_k$ = later outcome.

---

# 41.3 Protocol block

The protocol block should record:

- asset;
- market;
- venue;
- universe;
- timeframe;
- scale;
- data source;
- bar construction;
- feature map;
- horizon;
- admissible action;
- transaction-cost assumption.

Without this block, later reproduction is unreliable.

---

# 41.4 Claim block

The claim block should state one clear proposition.

Examples:

- price has entered a self-confirming trend;
- resistance has become support;
- breadth divergence warns of trend fragility;
- the current move is corrective;
- the episode has completed;

- the phase has entered a high gate-hazard region.

The claim should avoid combining several propositions into one vague narrative.

---

# 41.5 Source and operator lineage block

The record should show:

### Source lineage

Which raw data channels were used?

### Operator lineage

How were the channels transformed?

For example:

Source = closing price. (41.3)

Operator = EMA_fast − EMA_slow. (41.4)

This makes redundancy analysis possible later.

---

# 41.6 Functional classification block

The record should specify:

PrimaryPeriod = Structure. (41.5)

PrimaryFamily = Motion / Relation. (41.6)

SecondaryFamilies = Load memory + Commitment warning. (41.7)

This prevents a relational indicator from being silently treated as a gate.

---

# 41.7 Regime block

The regime block records the assumed $\chi$.

For example:

$\chi$_assumed = corrective. (41.8)

Evidence:

- repeated mean reversion;
- negative response after normalized extension;
- stable boundaries;
- weak follow-through.

The record should distinguish:

χ_assumed from χ_estimated. (41.9)

A failed interpretation may result from wrong regime classification rather than indicator failure.

---

## 41.8 Gate block

The gate block should record:

- event threshold;
- authority;
- decision;
- gate strength;
- timing;
- defer condition;
- rejection condition.

For example:

Gate = daily close above resistance with RVOL > 1.5 and breadth > threshold. (41.10)

Decision = Admit. (41.11)

A gate specified after the outcome should be labelled retrospective.

---

## 41.9 Residual block

The residual block should record:

- conflicting evidence;
- missing data;
- alternative branch;
- higher-timeframe disagreement;
- liquidity risk;
- institutional uncertainty;
- model residual;
- measurement uncertainty.

Residual severity may be classified:

0 = negligible;
1 = minor;

2 = material;
3 = major;
4 = model-threatening. (41.12)

The scale must be defined by protocol.

---

## 41.10 Invalidation block

Every claim should include a condition under which it weakens or fails.

Examples:

- close below reclaimed level;
- breadth makes new low;
- phase order reverses;
- alternate wave branch becomes dominant;
- volatility-normalized boundary is breached;
- gate fails within declared horizon.

Invalidation differs from loss control.

A stop-loss is an action rule.

Invalidation is a knowledge rule.

They may coincide, but they need not.

---

## 41.11 Transport block

Transport tests should include the relevant subset of:

- higher timeframe;
- lower timeframe;
- log scale;
- volatility normalization;
- alternate bar construction;
- benchmark change;
- component-universe change;
- nearby anchor perturbation;
- data-vendor comparison.

The result may be:

Survives.
Partially survives.
Fails.
Not applicable. (41.13)

---

## 41.12 Outcome block

The later outcome should be measured at the originally declared horizon.

Possible fields include:

- price return;
- maximum favourable excursion;
- maximum adverse excursion;
- gate persistence;
- residual resolution;
- episode classification;
- transaction-cost-adjusted result;
- later revision.

The outcome should not be selected after seeing which horizon looks best.

---

## 41.13 Revision block

When a claim is revised, record:

$$D_{k+1} = U_a(D_k,L_k,R_k). \quad (41.14)$$

The revision should include:

- old claim;
- new claim;
- evidence forcing revision;
- residual carried forward;
- whether the old claim failed;
- whether the protocol changed.

The self-revising declaration framework requires revision to remain trace-preserving, residual-honest, frame-robust, budget-bounded, and non-degenerate.

---

## 41.14 Example record

## Claim

A daily breakout from a twelve-week range has been admitted.

## Protocol

Daily adjusted close, log scale, twelve-week boundary, volume and breadth confirmation.

## Period

Event.

## Primary family

Commitment / Gate.

## Sources

Price, volume, component breadth.

## Operator word

Boundary projection → normalized displacement → close gate → confirmation filter.

## χ assumption

Transition from critical to self-confirming.

## Gate

Daily close 1.2 ATR above resistance, relative volume above 1.5, breadth positive.

## Residual

Weekly resistance remains 3% overhead; no retest; options expiry within four sessions.

## Transport

Survives log scale and equal-weight breadth; weekly gate deferred.

## Invalidation

Daily close back inside range followed by failed reclaim.

## Outcome horizon

Twenty trading sessions.

This record is much more informative than:

“Bullish breakout confirmed.”

---

# 41.15 Database schema

A practical database could include:

```
record_id
timestamp
asset
universe
protocol_version
period
primary_family
secondary_families
source_lineage
operator_word
claim
regime_assumption
xi_state
gate_rule
gate_decision
gate_strength
residual_types
residual_severity
transport_tests
invalidation
action_taken
outcome_horizon
outcome
revision_id
reviewer
```

This turns Technical Analysis into a cumulative research ledger.

---

# 41.16 Observer and authority metadata

The record should include observer role:

- private trader;
- research analyst;
- risk manager;
- algorithm;
- investment committee;
- regulator;
- accounting authority.

It should also include gate authority.

A private interpretation and an official market or legal declaration do not have the same world-forming force.

The formal observer framework similarly treats adaptive policy, accessible trace, compatible mapping, and record redundancy as conditions of cross-observer agreement rather than assuming one universal observer.

---

## 41.17 The record as anti-hindsight architecture

The record reduces several pathologies:

- anchor moving;
- timeframe shopping;
- indicator substitution;
- wave relabeling;
- hidden residual;
- outcome-horizon selection;
- silent regime reassignment;
- forgotten false positives.

It does not eliminate bias.

It makes bias auditable.

---

# 42. Empirical Research Programme

## 42.1 Test 1 — Classification reliability

Independent analysts should classify the same methods and signals.

Questions include:

- Do they assign the same period?
- Do they identify the same dominant family?
- Do they agree on source lineage?
- Do they agree on whether a gate occurred?

Let $\kappa$ denote agreement beyond chance.

A viable grammar should produce:

$\kappa > \kappa_{min}$ under a declared coding protocol. (42.1)

If agreement remains low, definitions must be revised.

---

## 42.2 Test 2 — Redundancy graph

Construct a graph where each method is a node.

Edges represent:

- shared source;
- shared operator;
- shared horizon;
- correlated errors;
- conditional information overlap.

A redundancy matrix may be:

R_ij = Redundancy(Method_i,Method_j). (42.2)

The study should test whether portfolios of low-redundancy evidence outperform equal-sized collections of highly redundant indicators in:

- classification;
- event detection;
- calibration;
- robustness.

---

# 42.3 Test 3 — Functional-diversity confirmation

Compare two confirmation systems.

### System A

Counts the number of bullish indicators.

### System B

Requires support across Load, Motion, Constraint, and Commitment, with residual penalty.

The hypothesis is:

CalibrationError_B < CalibrationError_A. (42.3)

or:

FalseGateRate_B < FalseGateRate_A. (42.4)

The result must be tested across regimes and markets.

---

# 42.4 Test 4 — Period-transition validity

The central periodic law claims:

Commitment_p + Residual_p → Load_{p+1}. (42.5)

This can be tested by asking:

- Do mark-level execution residuals improve window classification?
- Do window-level closes improve structure persistence estimates?
- Do event histories improve episode segmentation?
- Do episode histories improve world-regime identification?

If lower-period committed trace does not improve higher-period modelling, the proposed inheritance rule is weakened.

---

## 42.5 Test 5 — Gate strength and fakeout

Estimate whether stronger gate scores predict more persistent transitions.

Let S_G be gate strength.

Let F be fakeout indicator.

The expected relation is:

Pr(F = 1 | S_G high) < Pr(F = 1 | S_G low). (42.6)

But the relation may be nonlinear.

Very strong visible gates may attract crowding and later fragility.

Therefore observer-backreaction variables should be included.

---

## 42.6 Test 6 — Residual burden

Test whether residual burden predicts:

- failure;
- delayed continuation;
- higher volatility;
- revision;
- weaker transport;
- larger drawdown.

For residual score R_b:

OutcomeRisk = f(R_b,GateStrength,χ,Ξ,Controls). (42.7)

If residual recording adds no incremental value, the residual architecture may be too subjective or poorly measured.

---

## 42.7 Test 7 — Cross-frame survival

For each claim, compute survival under:

- timeframe transformation;
- scale transformation;
- volatility normalization;
- anchor perturbation;
- bar-rule change.

Test whether higher transport scores predict:

- greater persistence;
- lower false-positive rates;
- stronger cross-observer agreement.

A negative result may show that local frame-specific methods are more useful than broad invariance for some tasks.

---

## 42.8 Test 8 — Regime signature χ

Estimate whether corrective and self-confirming interpretations improve conditionally when χ is diagnosed first.

For RSI, compare:

Model_1 = Outcome ~ RSI. (42.8)

Model_2 = Outcome ~ RSI + χ + RSI×χ. (42.9)

The framework predicts that the interaction term matters.

A similar test can be applied to:

- band touches;
- moving-average distance;
- support reactions;
- breakouts;
- divergence.

---

## 42.9 Test 9 — Ξ control state

Test whether compiled loading, lock-in, and agitation improve regime diagnosis beyond ordinary price and volume.

Compare:

Model_base = f(Price,Volume,Volatility). (42.10)

Model_Ξ = f(Price,Volume,Volatility,ρ,γ,ν). (42.11)

The Ξ model should be penalized for extra complexity.

The PORE framework itself requires protocol declaration, jump detection, residual tracking, and comparison with simpler baselines rather than narrative repair.

---

# 42.10 Test 10 — Complex eligibility

For every candidate R–Q model, compare:

Model_scalar.
Model_realpair.
Model_complex. (42.12)

Evaluate:

- predictive loss;
- calibration;
- stability;
- compression;
- phase robustness;
- gate localization;
- intervention performance.

Complexification earns promotion only if:

Benefit_complex – ComplexityPenalty > Benefit_realpair. (42.13)

---

# 42.11 Test 11 — Phase-time alignment

Collect comparable episodes of unequal duration.

Align them by:

- calendar time;
- event count;
- phase $\theta$;
- accumulated phase $\tau_i$;
- selection depth $\sigma$.

Compare trajectory dispersion and gate predictability.

The phase-time hypothesis gains support when:

Dispersion_$\tau_i$ < min(Dispersion_t,Dispersion_k). (42.14)

and when the result survives out-of-sample testing.

---

## 42.12 Test 12 — Observer backreaction

Identify signals with different levels of public adoption.

Test whether greater adoption changes:

- initial continuation;
- crowding;
- reversal severity;
- liquidity;
- stop clustering;
- signal decay.

A possible non-monotonic relation is:

Adoption ↑
→ early effectiveness ↑
→ crowding ↑
→ later fragility ↑. (42.15)

This is a difficult causal problem because adoption and expected effectiveness may be jointly determined.

---

## 42.13 Test 13 — Residual-honest versus retrospective databases

Compare two research databases.

### Conventional database

Stores only final signal labels.

### Residual-honest database

Stores original claims, protocols, failures, residuals, and revisions.

Test whether the second produces:

- lower apparent but more stable performance;
- better calibration;
- more reproducible conclusions;
- less hindsight bias;
- more accurate regime failure diagnosis.

The declaration architecture predicts that trace-preserving revision should improve long-run objectivity even when it makes short-run results look less impressive.

---

# 43. Falsification Programme

## 43.1 Why the framework must risk failure

A framework that can reinterpret every outcome cannot become a research programme.

The periodic grammar must therefore specify conditions under which it should be:

- revised;
- reduced;
- or rejected.

Its concepts should not be protected through increasingly elaborate narrative.

---

## 43.2 Falsifier 1 — unstable classification

The four functional families are weakened if independent analysts cannot distinguish:

- Load;
- Motion;
- Constraint;
- Commitment.

If most methods occupy arbitrary cells depending on the analyst, the table lacks operational meaning.

---

## 43.3 Falsifier 2 — arbitrary period assignment

The six periods are weakened if:

- Window and Structure cannot be distinguished;
- Structure and Event collapse into one category;
- Episode and World depend only on duration;
- period transitions cannot be operationalized.

The row system must capture closure depth rather than subjective importance.

---

## 43.4 Falsifier 3 — no periodic inheritance

The periodic law is weakened if lower-period committed traces do not become useful higher-

period load.

If:

EventHistory adds no value to EpisodeModel, (43.1)

and:

EpisodeHistory adds no value to WorldModel, (43.2)

then the proposed recurrence may be decorative rather than explanatory.

---

## 43.5 Falsifier 4 — functional diversity adds no value

The confirmation thesis is weakened if:

many same-source indicators

perform as well as or better than

functionally diverse evidence

after equal complexity and cost.

The framework should accept that result.

---

## 43.6 Falsifier 5 — residual recording adds no value

Residual governance is weakened if residual fields:

- cannot be coded reliably;
- do not improve calibration;
- do not explain failure;
- do not improve revision;
- merely provide narrative excuses.

Residual must remain measurable enough to matter.

---

## 43.7 Falsifier 6 — transport cannot be defined

The invariance rail is weakened if no practical transport map can be defined between relevant frames.

Some methods may then need to be classified as explicitly local rather than transportable.

The framework should not force universal invariance.

---

## 43.8 Falsifier 7 — $\chi$ does not modify indicator meaning

The regime-signature hypothesis is weakened if interaction tests show that RSI, bands, crossovers, and divergence behave similarly regardless of diagnosed feedback orientation.

$\chi$ would then add complexity without explanatory gain.

---

## 43.9 Falsifier 8 — $\Xi$ does not improve diagnosis

The control-state interface is weakened if $\rho$, $\gamma$, and $\nu$ cannot be estimated consistently or do not improve upon ordinary variables.

$\Xi$ should then remain a conceptual dashboard rather than an empirical model.

---

## 43.10 Falsifier 9 — complex model fails real-pair benchmark

A complex TA model should be reduced when:

- phase is scale-fragile;
- Q lacks independent meaning;
- gate concentration fails;
- episode alignment does not improve;
- a real-pair model performs equally well.

The correct response is not to invent a deeper imaginary axis.

It is to return to the simpler model.

---

## 43.11 Falsifier 10 — phase time adds no internal ordering

The phase-time hypothesis fails if $\tau_i$:

- does not align episodes;
- does not improve gate prediction;
- does not survive transport;
- does not improve intervention;
- depends on retrospective construction.

In that case, phase remains a descriptive coordinate, not a clock.

---

## 43.12 Falsifier 11 — observer backreaction is negligible

Some markets or signals may have little reflexive effect.

If signal adoption does not materially alter market behaviour, the analysis should remain at the Probe level.

Not every observation becomes a Pump, Switch, or Couple.

---

## 43.13 Falsifier 12 — simpler taxonomy wins

The entire periodic grammar should be weakened if a simpler classification produces:

- equal or better explanatory clarity;
- higher classification reliability;
- equal predictive value;
- lower complexity;
- easier falsification.

The framework's purpose is not to maximize conceptual richness.

It is to preserve useful distinctions.

---

# 44. The Research Contract

The empirical programme can be summarized as a contract.

A proposed Technical Analysis method should declare:

1. its protocol;
2. its source lineage;
3. its operator lineage;
4. its primary period;
5. its primary functional family;
6. its regime assumption;
7. its gate;
8. its residual;
9. its transport burden;
10. its invalidation;
11. its simpler benchmark;
12. its revision rule.

In compact form:

MethodValidity_P
= ClassificationReliability

- IncrementalEvidence
- GateDiscipline
- ResidualHonesty
- TransportRobustness
- BenchmarkSuperiority
- TracePreservingRevision. (44.1)

This is a demanding standard.

That is intentional.

Technical Analysis does not need more visually persuasive labels.

It needs stronger interfaces between observation, commitment, memory, and revision.

---

# 45. Provisional Conclusion of Part VIII

The proto-periodic table becomes scientifically useful only when it changes how evidence is recorded and tested.

Its practical implications are:

- do not count transformed copies of the same data as independent confirmation;
- separate relation warnings from commitment gates;
- preserve residual after both success and failure;
- test whether claims survive admissible frame changes;
- record the original protocol before revising it;
- compare complex models with simpler real alternatives;
- treat missing cells as research hypotheses rather than guaranteed discoveries.

The central empirical transition is:

Indicator Folklore
→ Functional Classification
→ Protocol Declaration
→ Gate Testing
→ Residual Ledger
→ Cross-Frame Audit
→ Admissible Revision. (45.1)

The next part will address the framework's limits, clarify what it does not solve, and draw the final distinction between a **Periodic Grammar of Technical Analysis** and a universal theory of market prediction.

# Part IX — Limits, Scope, and Scientific Status

# 46. What the Framework Does Not Solve

# 46.1 Classification is not prediction

The periodic grammar explains what kind of market function a method performs.

It does not prove that the method predicts profitable future returns.

For example:

MovingAverage = filtered memory. (46.1)

RSI = normalized directional relation. (46.2)

VolumeProfile = price-space transaction density. (46.3)

Breakout = boundary-transition gate. (46.4)

These classifications may improve conceptual clarity even when the methods possess little or no standalone predictive value.

Therefore:

BetterClassification ≠ ProfitableForecast. (46.5)

The framework separates four questions that are often merged:

1. What does the method measure?
2. Does it measure that object reliably?
3. Does the object contain forward information?
4. Can that information be converted into net economic value?

A method may succeed at the first two and fail at the last two.

The source Technical Analysis article makes the same distinction by treating Technical Analysis as a family of imperfect diagnostic projections rather than a complete science of prophecy.

---

# 46.2 Diagnostic validity is not trading profitability

Suppose an indicator correctly diagnoses:

- increasing trend persistence;
- declining breadth;
- rising structural mass;
- weakening gate quality;
- increasing residual burden.

That diagnosis may still fail to produce profitable trades because of:

- transaction costs;
- slippage;
- delay;
- position sizing;
- stop placement;
- opportunity cost;
- tax;
- market impact;
- portfolio interaction;
- rare large losses.

A trading result can be written schematically:

NetTradingValue
= DiagnosticEdge
– TransactionCost
– Slippage
– MarketImpact
– ModelError
– TimingError
– RiskCost. (46.6)

The periodic grammar primarily addresses DiagnosticEdge and ModelError.

It does not automatically solve the remaining terms.

---

# 46.3 A strong gate can still be too late

Commitment gates reduce premature closure.

But stronger confirmation often arrives after part of the movement has occurred.

This creates a general trade-off:

EarlyGate → lower delay + higher false-admission risk. (46.7)

LateGate → higher confidence + lower remaining opportunity. (46.8)

No universal gate removes this trade-off.

The proper gate depends on:

- decision horizon;
- transaction cost;

- consequence of false admission;
- consequence of delayed admission;
- liquidity;
- risk budget.

Thus:

OptimalGate = Function(Application,ErrorCost,DelayCost,Protocol). (46.9)

A research framework can make the trade-off visible.

It cannot eliminate it.

---

## 46.4 Residual honesty does not remove uncertainty

Recording residual improves accountability.

It does not make uncertainty disappear.

A complete record may still contain:

- unknown participant motives;
- hidden liquidity;
- uncertain causal structure;
- unstable regime;
- incomplete data;
- novel events.

Residual honesty means:

Uncertainty is attached to the claim. (46.10)

It does not mean:

Uncertainty has been solved. (46.11)

The value of residual governance is epistemic discipline, not omniscience.

---

## 46.5 Cross-frame survival is not universal truth

A structure that survives several frames is stronger than one that appears in only one convenient representation.

But even a transport-robust claim may remain:

- local;
- temporary;

- protocol-bound;
- contingent on market institutions.

For example, a level may survive:

- daily and weekly frames;
- linear and logarithmic scales;
- price and volume-profile analysis.

It may still fail after:

- policy change;
- earnings shock;
- legal event;
- liquidity collapse;
- structural market reform.

Therefore:

CrossFrameRobustness ≠ TimelessLaw. (46.12)

It means only that the claim is less dependent on one narrow representation.

---

## 46.6 The table does not prove causality

The framework distinguishes functions and closure levels.

But association among:

- volume;
- price;
- breadth;
- boundary;
- gate;
- later movement

does not automatically identify causal direction.

For example:

Volume may contribute to a breakout. (46.13)

Expected breakout may attract volume. (46.14)

External news may produce both. (46.15)

A causal study requires:

- intervention;
- natural experiment;
- structural model;
- instrumental variable;
- randomized mechanism;
- or another credible identification strategy.

The table can help identify which variables and transitions require investigation.

It does not replace causal inference.

---

## 46.7 The framework does not resolve market efficiency

The periodic grammar does not by itself establish whether prices fully incorporate information.

It does not prove:

- persistent exploitable inefficiency;
- complete informational efficiency;
- universal behavioural repetition;
- universal failure of chart methods.

It reframes the issue.

Instead of asking only:

Does this pattern predict return? (46.16)

it first asks:

What market function does the pattern claim to observe? (46.17)

What gate turns the observation into an event? (46.18)

What residual and cost remain? (46.19)

Prediction and efficiency questions remain empirical after the object has been clarified.

---

## 46.8 Self-reference does not guarantee predictability

A market that observes itself may exhibit repeated structures.

But self-reference can also reduce predictability.

An observed signal may become:

- self-fulfilling;
- self-negating;
- crowded;
- adversarially exploited;
- institutionally altered.

Thus:

SelfReference → PatternFormation and PatternInstability. (46.20)

The fact that a market uses its own traces does not mean its future becomes simple.

It means the observer must be included in the model.

---

## 46.9 Institutional worlds are not reducible to charts

Period 5 includes:

- law;
- accounting;
- collateral;
- regulation;
- benchmark rules;
- institutional authority.

These objects may affect price before they are visible in ordinary chart structure.

A purely chart-based implementation cannot fully observe them.

Therefore:

ChartWorld ⊂ FinancialWorld. (46.21)

The periodic grammar broadens Technical Analysis toward institutional context.

It does not imply that every world-level variable can be reconstructed from price and volume alone.

---

## 46.10 The framework does not make metaphysical claims

Gauge language, phase language, complex numbers, and world formation are used structurally and operationally.

The article does not claim:

- markets are literal quantum fields;
- traders are particles;
- technical levels are physical gauge potentials;
- breakouts are quantum measurements;
- phase rotation proves quantum behaviour in finance.

The gauge-to-market source explicitly proposes a middle layer between literal ontology and loose metaphor: protocol-fixed operational translation.

The safe claim is:

> Certain mathematical and systems concepts may provide disciplined role grammars for market structure when their operational meanings and limits are declared.

---

# 47. The Status of the Periodic Analogy

# 47.1 Why call it periodic?

The table is periodic because four functions recur across increasing levels of closure:

Load_p
→ Motion_p under Constraint_p
→ Commitment_p
→ Ledger_{p+1} + Residual_p. (47.1)

The committed output of one period becomes part of the next period's load.

Examples:

Trade commitment → bar load. (47.2)

Closing trace → structural memory. (47.3)

Breakout event → episode history. (47.4)

Episode history → institutional regime memory. (47.5)

The recurrence is structural.

It is not a claim of regular calendar repetition.

---

# 47.2 Why the analogy is not yet a true periodic law

The chemical periodic table has a mature physical basis.

Its organization is supported by:

- atomic number;
- electronic structure;

- recurring chemical properties;
- predictive gaps;
- independent laboratory verification.

The Technical Analysis table currently has:

- a proposed closure-depth axis;
- a proposed functional-family axis;
- method decompositions;
- missing-cell hypotheses;
- empirical test requirements.

It does not yet possess:

- a unique market equivalent of atomic number;
- universally agreed cell boundaries;
- proven transition theorems;
- stable quantitative valence rules;
- broad independent validation.

Therefore, the proper term is:

Proto-Periodic Table of Market Observation. (47.6)

not:

Completed Periodic Law of Markets. (47.7)

---

## 47.3 What would strengthen the periodic claim?

The analogy would become stronger if studies showed that:

1. independent analysts reliably assign periods and groups;
2. the same four functions recur across markets and protocols;
3. lower-period commitments consistently become higher-period load;
4. empty cells predict useful new instrument families;
5. functional diversity improves confirmation;
6. higher-period structures constrain lower-period events;
7. the table predicts characteristic failure modes.

Only then should the framework claim more than a disciplined grammar.

---

## 47.4 The table may evolve

The present 6 × 4 structure is a provisional minimum.

Possible future findings include:

- Mark and Window should merge;
- World should divide into institutional and reflexive layers;
- Load should separate from Memory;
- Constraint should divide into soft and authoritative forms;
- another functional family is genuinely required;
- some current cells are redundant.

Revision is allowed.

But revision must preserve trace:

Table_{n+1} = U_adm(Table_n, Evidence_n, Residual_n). (47.8)

A revised table should record:

- what changed;
- why it changed;
- which evidence failed;
- which unresolved problems remain.

This follows the source framework's principle that mature revision must preserve past trace rather than rewrite it silently.

---

# 48. The Limits of the Complex and Quantum Analogies

## 48.1 Two real variables do not automatically form a complex state

Any ordered pair can be written:

(R,Q) ↔ R + iQ. (48.1)

At one instant, no new numerical information has been created.

A complex state gains modelling priority only when it supplies:

- meaningful phase;
- stable generator;
- useful rotation;

- gate alignment;
- dynamical simplification;
- or improved intervention.

Otherwise:

(R,Q) should remain a real pair. (48.2)

The phase source explicitly requires reduction when a complex model does not outperform a flexible two-real-variable model.

---

# 48.2 Q cannot mean everything invisible

A weak market complexification says:

R = visible market state. (48.3)

Q = hidden market pressure. (48.4)

This is insufficient.

“Hidden pressure” may include:

- volume;
- liquidity;
- leverage;
- breadth;
- sentiment;
- legal risk;
- residual;
- volatility;
- event risk.

These variables do not necessarily form one conjugate coordinate.

A valid Q requires:

- prior definition;
- independent estimation;
- stable meaning;
- forward relevance;
- declared scaling.

Unexplained change remains residual:

$\varepsilon = \text{ObservedChange} - \text{ModelledChange}$. (48.5)

The phase source directly rejects models where Q is defined as everything not explained by R.

---

# 48.3 The CAPM geometry is a calibration case, not universal proof

The CAPM construction supplies:

$A^2 = R^2 + Q^2$. (48.6)

$R = A \cos \theta$. (48.7)

$Q = A \sin \theta$. (48.8)

$\partial R / \partial \theta = -Q$. (48.9)

This makes Q a precise phase-exposure coordinate under that declared valuation protocol.

But the construction does not prove that:

- momentum and volume are conjugate;
- price and sentiment are conjugate;
- every technical residual is imaginary;
- all markets traverse a circular phase orbit.

The CAPM source itself notes that, in a static one-period model, Q is algebraically derived from A and R; its practical value must arise through dynamic attribution, multi-horizon structure, protocol comparison, gate diagnosis, communication, or intervention. Otherwise it remains an optional reparameterization.

---

# 48.4 Measurement rotation is not market movement

Multiplication by i gives:

$iZ = -Q + iR$. (48.10)

This changes the measurement orientation.

It does not necessarily change the financial state.

The rigorous distinction is:

Passive rotation = change of readout. (48.11)

Active rotation = change of state. (48.12)

Economic consequence requires active movement:

$\text{Exposure} \times \text{Movement} \rightarrow \text{EconomicP\&L}$. (48.13)

A recognition gate is then required before the consequence enters a ledger.

The same caution applies to Technical Analysis.

Looking at the market through another indicator is not the same as the market changing.

---

## 48.5 Complex behaviour is not specifically quantum behaviour

Complex numbers appear in many mature non-quantum domains.

A complex market state may display:

- phase;
- rotation;
- quadrature;
- interference-like combination;
- contextual projection.

These features do not establish specifically quantum properties such as:

- Born-rule probability;
- Bell inequality violation;
- no-cloning;
- irreducible coherent superposition;
- physical wavefunction collapse.

The CAPM source explicitly separates the complex financial architecture from these specifically quantum residues.

Therefore:

ComplexFinance ≠ QuantumFinance by definition. (48.14)

Complex geometry may be useful without making a quantum ontological claim.

---

## 48.6 Phase is not automatically time

A phase variable becomes an internal clock only when it improves internal ordering.

A legitimate phase-time model should show:

D_phase < D_calendar. (48.15)

or improved gate hazard:

Prediction(Gate | Phase)

Prediction(Gate | CalendarTime, Controls). (48.16)

Without such gains:

Phase = descriptive orientation. (48.17)

not:

Phase = secondary time. (48.18)

The phase framework distinguishes two-channel descriptions, complex completion, phase-bearing dynamics, secondary ordering, phase-sensitive events, and full ledgered worlds as separate evidence levels.

---

## 48.7 Phase-time worlds remain a research programme

The strongest candidate world requires:

ConjugateState

- StablePhaseOrder
- PhaseSensitiveGate
- PersistentTrace
- FutureConstraint
- Backreaction. (48.19)

The existing source framework presents this as a graded research programme, not a completed universal discipline. It states that the proper claim is not that a new calculus has already been discovered, but that a cross-domain programme has been identified in which complex phase may act as a portable internal-ordering operator.

The present article adopts the same restraint.

---

# 49. The Framework's Proper Scientific Status

## 49.1 A conceptual architecture

At its current stage, the Periodic Grammar of Technical Analysis is primarily a conceptual architecture.

It provides:

- typed categories;
- closure levels;
- method decompositions;
- governance rails;
- missing-cell hypotheses;
- empirical tests;

- falsification conditions.

It does not yet provide:

- one universally calibrated model;
- one validated trading system;
- one theorem proving the six periods;
- one established numerical table.

This distinction should be stated explicitly.

---

## 49.2 A measurement programme

The framework becomes scientific only when its concepts are translated into records.

Required objects include:

- declared protocol;
- indicator source lineage;
- operator lineage;
- functional classification;
- gate metadata;
- residual;
- invalidation;
- transport tests;
- outcome;
- revision.

The source Technical Analysis article already proposes residual fields, invalidation fields, outcome fields, and preservation of original claims so that fakeouts, regime effects, pivot stability, and density reactions can be tested rather than remembered selectively.

The present article extends that record into a periodic classification system.

---

## 49.3 A comparative programme

The framework should be compared with simpler alternatives.

Examples include:

- ordinary indicator categories;
- feature-based machine learning;

- regime-switching models;
- real-vector state models;
- direct event classifiers;
- causal market-microstructure models.

The periodic grammar earns its place only if it improves at least one of:

- explanatory clarity;
- coding reliability;
- failure diagnosis;
- evidence independence;
- event calibration;
- transport;
- revision quality.

Otherwise it should be reduced.

---

## 49.4 A compiler rather than a predictor

The framework's strongest initial role may be as a **semantic compiler**.

It takes a loose claim such as:

"The breakout looks strong because MACD and RSI confirm it."

and compiles it into:

- Structure-level memory relation;
- two price-derived Motion indicators;
- Event-level boundary crossing;
- missing Load independence;
- unspecified Commitment gate;
- unresolved breadth and timeframe residual;
- likely confirmation redundancy.

Thus:

LooseChartNarrative
→ TypedMarketClaim. (49.1)

This resembles the wider runtime-kernel principle that broad natural-language requirements become more useful when compiled into explicit boundary, operator, gate, trace, and residual structures.

## 49.5 A research ontology, not a final ontology

The table proposes useful objects for research:

- marks;
- windows;
- structures;
- events;
- episodes;
- worlds;
- load;
- motion;
- constraint;
- commitment.

These categories should be treated as operational research objects.

They are not final declarations about what financial reality fundamentally is.

Different scientific programmes may use different ontologies.

The periodic grammar should be judged by whether it improves disciplined observation.

---

# Part X — From Indicator Folklore to Market-Observation Science

# 50. The Central Reconstruction

## 50.1 The old organization

The inherited organization of Technical Analysis is largely method-based:

Moving averages
RSI
MACD
Volume
Candlesticks
Support and resistance
Patterns
Fibonacci
Elliott Wave
Gann. (50.1)

This organization is historically understandable.

But it mixes:

- data transformations;
- memory structures;
- relational diagnostics;
- boundaries;
- gates;
- events;
- episode models;
- invariance hypotheses.

The resulting confusion produces recurring category errors.

---

## 50.2 The new organization

The proposed reconstruction begins with four recurring functions:

Load
Motion
Constraint
Commitment. (50.2)

These functions recur across six closure periods:

Mark
Window
Structure
Event
Episode
World. (50.3)

The resulting matrix is governed by:

Residual
Transport
Ledger
AdmissibleRevision. (50.4)

The full grammar is:

Protocol
→ Period
→ FunctionalRole
→ Operator
→ Gate
→ Trace
→ Residual
→ Transport
→ LedgerConsequence
→ Revision. (50.5)

## 50.3 The table’s central law

The central law is:

Committed trace at period p becomes part of the operative load at period p + 1. (50.6)

More fully:

Load_p
→ Motion_p under Constraint_p
→ Commitment_p
→ Trace_p + Residual_p
→ Ledger_{p+1}
→ Load_{p+1}. (50.7)

This law explains the recurrence.

- executions become trade history;
- trade history becomes candles;
- candles become structural memory;
- structural memory becomes event context;
- events become episode history;
- episodes become institutional worlds;
- worlds generate new orders and marks.

Thus the final recursion is:

Mark
→ Window
→ Structure
→ Event
→ Episode
→ World
→ NewMark. (50.8)

---

## 50.4 Technical Analysis as a trace science

The chart should no longer be treated as a crystal ball.

It is better understood as a compressed trace of remembered conflict.

Price records where exchange occurred.

Volume records how much recorded exchange occurred.

Candles record window closure and unretained excursion.

Moving averages record filtered memory.

Support and resistance record candidate loaded boundaries.

Breakouts record attempted transition into a new ledger state.

Episodes record ordered histories of commitment and failure.

Market worlds record the institutional conditions under which those traces remain consequential.

The source Technical Analysis article states its final position succinctly:

Technical analysis is useful when treated as diagnostic projection and dangerous when treated as prophecy.

---

# 50.5 The mature analyst

The mature analyst is not the person who accumulates the greatest number of indicators.

The mature analyst asks:

- What has been declared?
- What is being measured?
- Which closure period does the claim occupy?
- Which functional family is primary?
- What regime assumption is being made?
- What gate converts warning into event?
- What residual remains?
- Does the claim survive another frame?
- What has entered the ledger?
- What would force revision?

This produces:

MatureAnalysis
= Signal

- FunctionalMeaning
- Gate
- Residual
- Invalidation
- Transport
- LedgerEffect. (50.9)

The source article gives a closely related requirement:

Analysis = Signal + Confirmation + Residual + InvalidationRule.

The periodic grammar expands this into a full market-observation contract.

---

# 51. Twelve Final Principles

# Principle 1 — An indicator is a projection

No indicator is the market.

Indicator_P = Projection of one market characteristic under protocol P. (51.1)

---

# Principle 2 — Named methods are compounds

Moving averages, MACD, RSI, VWAP, breakouts, Elliott Wave, and Gann are constructed from more basic observational roles.

Method = Compose(Load,Motion,Constraint,Commitment | Period,Protocol). (51.2)

---

# Principle 3 — Period is closure depth, not timeframe

A five-minute bar and a weekly bar are both Window-period objects.

A breakout is an Event-period object.

A trend or wave is an Episode-period object.

Thus:

Timeframe ≠ Period. (51.3)

---

# Principle 4 — A relation is not an event

Divergence is not reversal.

Overbought is not exhaustion.

Crossing is not breakout.

A high is not a completed wave endpoint.

Relation + Gate → Event. (51.4)

---

# Principle 5 — Boundary is not gate

Support or resistance identifies where a consequential test may occur.

The gate determines whether the test enters the ledger.

Boundary ≠ Commitment. (51.5)

---

## Principle 6 — Commitment does not exhaust residual

A breakout can be admitted while breadth remains weak.

A loss can be recognized while legal risk remains unresolved.

Commitment ≠ Exhaustion. (51.6)

This distinction is made explicit in the CAPM recognition framework.

---

## Principle 7 — Confirmation requires independence

Several price-derived indicators may be one evidence lineage.

Stronger confirmation combines distinct functions, sources, frames, and failure modes.

IndicatorCount ≠ IndependentEvidence. (51.7)

---

## Principle 8 — Regime changes meaning

The same reading may imply correction under $\chi < 0$ and continuation under $\chi > 0$.

IndicatorMeaning = Function(Reading,χ,Protocol). (51.8)

---

## Principle 9 — Complexification must be earned

Writing R + iQ is easy.

Establishing conjugacy, phase utility, gate alignment, and superiority to a real pair is difficult.

ComplexNotation ≠ ComplexDynamics. (51.9)

---

## Principle 10 — Phase is not yet time

Phase becomes an internal clock only when it aligns unequal episodes and consequential gates better than calendar time.

PhaseOrientation ≠ HistoricalCommitment. (51.10)

The phase source states this directly: historical commitment requires a phase-sensitive gate and persistent trace.

---

## Principle 11 — Observation can become intervention

A widely used technical signal may alter orders, liquidity, constraints, and later price.

Probe + Adoption + Capital → Backreaction. (51.11)

---

## Principle 12 — Revision must preserve trace

A failed claim should not disappear.

A mature method records:

OriginalClaim

- Failure
- Residual
- RevisionReason
- NewDeclaration. (51.12)

Without trace preservation, flexibility becomes unfalsifiability.

---

# 52. Final Synthesis

Technical Analysis has often occupied an unstable intellectual position.

It is too persistent to dismiss as mere historical accident.

It is too inconsistent to accept as a unified predictive science.

Its methods frequently point toward real market structures:

- memory;
- feedback;
- density;
- constraint;
- commitment;
- cadence;
- residual;
- observer convergence.

Yet the methods are commonly misused because their logical roles are not distinguished.

A moving average is treated as a forecast.

An oscillator is treated as a reversal.

A boundary crossing is treated as a committed breakout.

A wave count is treated as completed history.

A geometric relation is treated as a universal law.

A complex coordinate is treated as hidden truth.

The periodic grammar corrects these errors by reorganizing the field around typed observational acts.

Its central claim is:

> Technical Analysis is a historically evolved instrument system for observing load, motion, constraint, and commitment across recursively generated market worlds.

Its central table is:

| Period | Load / Memory | Motion / Relation | Constraint / Boundary | Commitment / Gate |
|---|---|---|---|---|
| **Mark** | liquidity and resting interest | tick relation | bid–ask and execution limits | execution |
| **Window** | bar volume and carried state | return and range | high–low envelope | close |
| **Structure** | MA, VWAP, profile, breadth | momentum, RSI, MACD, divergence | levels, bands, channels | structural acceptance |
| **Event** | positioning and participation | displacement and phase shift | tested boundary | breakout, retest, rejection |
| **Episode** | accumulated event history | $\chi$ sequence and internal progress | pattern and basin | episode transition |
| **World** | institutional and balance-sheet memory | reflexivity and frame dynamics | law, collateral, policy, accounting | recognized state change |

Its three governance rails are:

Residual
Transport
LedgeredBackreaction. (52.1)

Its advanced state variables are:

$\chi$ = feedback signature. (52.2)

$\Xi$ = effective loading–lock-in–agitation state. (52.3)

$Z = R + iQ$ when conjugacy is earned. (52.4)

$\theta$ = phase orientation. (52.5)

$\tau_i$ = candidate internal phase time. (52.6)

$k$ = committed event order. (52.7)

$\sigma$ = selection depth. (52.8)

Its reduction discipline is:

World model
→ event model
→ phase-time model

→ complex dynamics
→ real pair
→ scalar. (52.9)

The correct level is the least complex model that preserves measurable gain.

---

# 53. Conclusion — The Periodic Grammar of Technical Analysis

A market does not merely move.

It carries history.

It moves relative to that history.

It encounters boundaries formed from that history.

It commits some movements into new history.

It leaves other possibilities unresolved.

The resulting ledger changes the field in which the next movement occurs.

In one line:

Market
→ Observation
→ Interpretation
→ Commitment
→ Ledger
→ ChangedMarket. (53.1)

Technical Analysis lives inside this loop.

Its methods are neither isolated prophecies nor meaningless decorations.

They are partial instruments distributed across a recursive architecture of memory, relation, boundary, and commitment.

The deepest practical correction is therefore not a new indicator.

It is a new discipline of interpretation:

Do not ask only whether an indicator is bullish or bearish.

Ask what it measures.

Ask which period it occupies.

Ask which regime gives the reading meaning.

Ask what boundary is being tested.

Ask what gate creates the event.

Ask what residual remains.

Ask whether the claim survives another frame.

Ask whether the admitted event changed the market that future analysis will observe.

Then Technical Analysis can begin to move beyond folklore.

Not into certainty.

Not into prophecy.

But into a more honest science of bounded market observation.

The final recursive law is:

Load_p
→ Motion_p under Constraint_p
→ Commitment_p
→ Trace_p + Residual_p
→ Ledger_{p+1}
→ Load_{p+1}. (53.2)

And the final methodological law is:

Declare
→ Project
→ Diagnose
→ Gate
→ Record
→ AuditResidual
→ Transport
→ Revise. (53.3)

The article's main argument is now complete. The appendices can next formalize the symbol system, expanded 6 × 4 table, method cards, residual-ledger schema, transport tests, complex-eligibility checklist, and CAPM calibration atom.

# Appendix A — Symbol Dictionary, Units, and Protocol Declarations

## A.1 Purpose

The Periodic Grammar combines several previously separate vocabularies:

- Technical Analysis indicators and market structures;
- protocol-bound observation;
- gate, trace, residual, and ledger;
- relational regime signature $\chi$;
- effective control coordinates $\Xi$;
- complex completion $Z = R + iQ$;
- phase orientation $\theta$;

- internal phase time $\tau_i$;
- event order k;
- selection depth σ.

Because several symbols have been used differently across earlier articles, this appendix fixes the notation used in the present paper.

No symbol should be interpreted independently of its protocol.

The basic discipline is:

Meaning(Symbol) = Meaning(Symbol | P, Period, FunctionalRole, Units). (A.1)

A numerical value without these declarations is not yet a reproducible market object.

---

# A.2 The observation protocol

The minimal protocol is:

P = (B,Δ,h,u). (A.2)

where:

| Symbol | Meaning | Technical Analysis interpretation |
|---|---|---|
| B | boundary | asset, venue, portfolio, index, sector, institution, or market system included |
| Δ | observation or aggregation rule | close, midpoint, time bar, volume bar, equal weighting, profile construction |
| h | time or state window | one minute, one session, one episode, one reporting regime |
| u | admissible intervention family | observe, enter, exit, hedge, rebalance, classify, regulate |

The declaration framework uses P to prevent claims about “the system itself” from being confused with claims made under one bounded observational setup.

For Technical Analysis, the expanded protocol is:

P_TA = (A,U,V,T,S,W,φ,G,R). (A.3)

where:

| Symbol | Meaning |
|---|---|
| A | asset or primary instrument |
| U | comparison universe |
| V | venue or data source |
| T | timeframe |
| S | scale and normalization |
| W | bar or window-construction rule |
| φ | feature map |
| G | gate rule |
| R | residual and invalidation rule |

The use of R inside P_TA as a residual-rule label should not be confused with R as the real component of a complex state. In implementations, the residual rule should preferably be coded as `Rule_res`.

A software-safe version is:

P_TA = (Asset,Universe,Venue,Timeframe,Scale,WindowRule,FeatureMap,GateRule,ResidualRule). (A.4)

---

# A.3 The declared world

A protocol-bound world may be written:

World_P = (X,q,φ,P,L,$\mathscr{R}$). (A.5)

where:

| Symbol | Meaning |
|---|---|
| X | declared state space |
| q | baseline or reference condition |
| φ | feature map |
| P | observation and intervention protocol |
| L | current ledger |
| $\mathscr{R}$ | residual register |

The broader declaration framework distinguishes the undeclared possibility field from the field made readable under a declared baseline, feature map, boundary, observation rule, horizon, and intervention family. Projection, gate, trace, and residual become meaningful only after these declarations.

---

# A.4 The six periods

Let p denote closure period:

p ∈ {0,1,2,3,4,5}. (A.6)

| p | Period | Canonical object | Closure condition |
|---|---|---|---|
| 0 | Mark | quote, order, trade, cancellation | admitted individual market record |
| 1 | Window | candle, bar, session | declared aggregation plus terminal window gate |
| 2 | Structure | MA, momentum, level, profile, breadth | persistent relation across windows |
| 3 | Event | breakout, rejection, reversal, retest | meaningful structural transition passes or fails a gate |
| 4 | Episode | trend, range, squeeze, wave sequence | ordered event sequence with stable relational grammar |
| 5 | World | valuation, policy, funding, accounting, institutional regime | persistent protocol, authoritative gates, ledger, and backreaction |

Period is not timeframe.

A five-minute candle and a monthly candle are both Window-period objects.

A five-minute breakout may be an Event-period object.

A multi-year policy regime may be a World-period object.

Therefore:

Period ≠ ClockDuration. (A.7)

Period = ClosureDepth. (A.8)

---

# A.5 The four functional groups

Let g denote functional group:

g ∈ {L,M,C,G}. (A.9)

where:

| Code | Functional family | Central question |
|---|---|---|
| L | Load / Memory | What consequential structure is already carried? |
| M | Motion / Relation | How is the state changing or relating to another state? |
| C | Constraint / Boundary | What channels, resists, or separates admissible states? |
| G | Commitment / Gate | What becomes accepted trace and future history? |

The letter G is used for Gate.

To avoid ambiguity, the wider framework's residual register will be written $\mathscr{R}$ rather than R, and structural inertia will be written M_str rather than M when necessary.

The recurring functional law is:

L_p → M_p under C_p → G_p → Ledger_{p+1} + $\mathscr{R}$_p. (A.10)

The output of period p becomes part of the Load at period p + 1:

L_{p+1} = Compile(Ledger_{p+1},$\mathscr{R}$_p,Environment_{p+1}). (A.11)

This 6 × 4 organization is the principal synthesis introduced by the present article. It is not asserted as an already established law in the source papers.

---

# A.6 Cell notation

A table cell is written:

E_{p,g}. (A.12)

where:

p = closure period;
g = functional family.

Examples:

$E_{0,G}$ = Mark-level Commitment. (A.13)

$E_{1,G}$ = Window-level Commitment. (A.14)

$E_{2,C}$ = Structure-level Constraint. (A.15)

$E_{3,G}$ = Event-level Commitment. (A.16)

$E_{4,M}$ = Episode-level Motion. (A.17)

$E_{5,C}$ = World-level Constraint. (A.18)

A fully declared cell is:

$E_{p,g}^{P,O}$. (A.19)

where:

P = protocol;
O = observer or authority.

This notation allows the same nominal object to differ across observers.

For example:

$E_{1,G}^{P_{exchange}}$ = official exchange close. (A.20)

$E_{1,G}^{P_{private}}$ = analyst-defined session close. (A.21)

The two gates may produce similar prices while carrying different authority.

---

# A.7 The market field Σ

Σ denotes the richer market field available under the protocol:

$\Sigma_P$ = complete logged or modelled field accessible under P. (A.22)

It may include:

- price;
- volume;
- order flow;
- breadth;
- liquidity;
- positioning;
- leverage;
- collateral;
- accounting;
- law;

- policy;
- narrative;
- prior gate history.

No indicator observes all of Σ_P.

A method produces a projection:

V_j = Ô_j(Σ_P). (A.23)

where Ô_j is the method's projection operator.

The Technical Analysis source emphasizes that moving averages, RSI, volume, wave counts, and Gann constructions are partial instruments rather than the market itself.

---

# A.8 Projection operator Ô

Ô denotes a bounded projection:

Ô_P: Σ_P → V_P. (A.24)

Examples include:

Ô_MA(Σ_P) = moving-average memory. (A.25)

Ô_RSI(Σ_P) = normalized directional relation. (A.26)

Ô_VP(Σ_P) = volume density across price. (A.27)

Ô_Breadth(Σ_P) = cross-component participation. (A.28)

Projection selects visible structure while leaving residual:

Σ_P → V_P + $\mathscr{R}$_P. (A.29)

The projection symbol does not imply physical quantum measurement in this article.

It denotes bounded observational extraction.

---

# A.9 Source lineage S

Let S_j denote the source lineage of method j.

Examples:

S_MA = {Close}. (A.30)

S_MACD = {Close}. (A.31)

S_RSI = {Close differences}. (A.32)

S_VWAP = {Price,Volume}. (A.33)

S_Breadth = {Component price states}. (A.34)

S_Options = {Option prices, strikes, expiries, open interest}. (A.35)

Source overlap is one component of confirmation redundancy.

---

# A.10 Operator word w

A method's operator word is:

$w_j = O_{j,n} \circ O_{j,n-1} \circ \ldots \circ O_{j,1}$. (A.36)

Common primitives include:

| Symbol | Operator |
|---|---|
| Π | projection |
| F | filtration or smoothing |
| Δ | difference |
| A | accumulation |
| N | normalization |
| D | density mapping |
| B | boundary construction |
| G | gate |
| T | frame transport |
| U | update or revision |

Examples:

$w_{MA} = F \circ \Pi_{price}$. (A.37)

$w_{MACD} = \Delta \circ (F_{fast}, F_{slow}) \circ \Pi_{price}$. (A.38)

$w_{RSI} = N \circ A \circ \Delta \circ \Pi_{price}$. (A.39)

$w_{VWAP} = N \circ A \circ \Pi_{\{price,volume\}}$. (A.40)

$w_{Breakout} = G \circ \Delta \circ (\Pi_{price}, B_{boundary})$. (A.41)

The operator-word notation is introduced in the present article to support method decomposition and redundancy analysis.

---

# A.11 Regime signature χ

χ denotes the orientation of the return path between realized structure and future signal pressure.

The principal classification is:

$\chi < 0 \rightarrow$ corrective feedback. (A.42)

$\chi \approx 0 \rightarrow$ weak, critical, or unstable feedback. (A.43)

$\chi > 0 \rightarrow$ self-confirming feedback. (A.44)

The source Technical Analysis framework defines χ through a signed conjugacy operator:

$C_\chi = [[0,F],[\chi M,0]]$. (A.45)

Under canonical normalization:

$C_{\chi^2} = \chi I$. (A.46)

The source uses χ to explain why oscillators perform differently in mean-reverting and self-confirming regimes.

χ must carry a protocol and horizon:

$\chi = \chi_{P,h}$. (A.47)

It should not be interpreted as one permanent property of an asset.

---

# A.12 Effective control state Ξ

The source protocol-first finance framework defines:

$\Xi = (\rho,\gamma,\tau)$. (A.48)

where:

ρ = effective loading or occupancy;
γ = effective lock-in or constraint strength;
τ = agitation, turbulence, or dephasing.

To avoid collision with phase-time notation, the present article uses:

$\Xi_{fin} = (\rho,\gamma,\nu)$. (A.49)

where ν replaces the source article's agitation symbol τ.

Separate timescales are:

$\tau_{rec}$ = recovery or recurrence time. (A.50)

$\tau_{sw}$ = switching time. (A.51)

The renaming is editorial and local to this integrated article.

It does not alter the original Ξ framework.

---

# A.13 Loading ρ

ρ measures effective loaded structure under P.

Possible units depend on the implementation:

- currency value;
- contracts;
- normalized participation;

- leverage ratio;
- open interest;
- density score;
- dimensionless compiled index.

A generic compiled estimator is:

$\hat{\rho}_P = C_\rho(\text{Volume},\text{OpenInterest},\text{Leverage},\text{Density},\text{Participation} \mid P)$. (A.52)

$\rho$ is not identical to volume.

Volume is one possible proxy channel.

---

# A.14 Lock-in $\gamma$

$\gamma$ measures effective resistance to exit, movement, transfer, or reclassification.

Possible proxies include:

- bid–ask cost;
- market depth;
- collateral requirement;
- concentration;
- legal restriction;
- maturity mismatch;
- structural mass;
- institutional rigidity.

A generic estimator is:

$\hat{\gamma}_P = C_\gamma(\text{Liquidity},\text{Concentration},\text{Collateral},\text{Rules},\text{ExitCost} \mid P)$. (A.53)

The Gauge Grammar finance mapping similarly associates high $\gamma$ with accounting lock, funding rigidity, margin, collateral, and institutional constraints.

---

# A.15 Agitation $\nu$

$\nu$ measures effective turbulence, churn, or dephasing.

Possible proxies include:

- realized volatility;
- spread instability;
- cancellation intensity;

- correlation breakdown;
- gate failure;
- cross-sectional dispersion.

A generic estimator is:

$\hat{\nu}_P = C_\nu(\text{Volatility},\text{Churn},\text{SpreadInstability},\text{GateFailure},\text{Dispersion} \mid P)$. (A.54)

$\nu$ measures operating condition.

It does not itself determine direction or commitment.

---

# A.16 Complex state Z

A candidate complex state is:

$Z = R + iQ$. (A.55)

where:

$R$ = declared real-side or admitted component;
$Q$ = declared conjugate component;
$i^2 = -1$.

The polar representation is:

$Z = A \exp(i\theta)$. (A.56)

with:

$A = \sqrt{(R^2 + Q^2)}$. (A.57)

$\theta = \text{atan2}(Q,R)$. (A.58)

This representation becomes analytically privileged only when:

- R and Q possess independent operational meanings;
- their normalization is defensible;
- their coupling supports phase;
- phase improves explanation, prediction, comparison, or control.

The phase source explicitly requires a complex model to outperform a flexible real-pair alternative and rejects models where Q is merely everything not explained by R.

---

# A.17 CAPM calibration variables

For one declared future cash flow:

$A_t = CF_t/(1 + r_{base})^t$. (A.59)

$R_t = CF_t/(1 + r)^t$. (A.60)

$r = r_{base} + \beta ERP$. (A.61)

$Q_t = \sqrt{(A_t^2 - R_t^2)}$. (A.62)

$Z_t = R_t + iQ_t$. (A.63)

The principal identity is:

$\partial R/\partial \theta = -Q$. (A.64)

The measurement cycle is:

$R \rightarrow -Q \rightarrow -R \rightarrow Q \rightarrow R$. (A.65)

The CAPM source defines Q as the conjugate valuation-phase coordinate and first-order phase exposure generated by the same declared valuation geometry as R. It also distinguishes measurement rotation from actual state movement and ledger recognition.

---

# A.18 Q versus residual $\mathscr{R}$

Q is internal to a declared complex state.

$\mathscr{R}$ is what remains outside the current closure.

Therefore:

$Q \neq \mathscr{R}$. (A.66)

A model residual may be written:

$\varepsilon_t = Z_{observed,t} - Z_{model,t}$. (A.67)

or, in a non-complex model:

$\varepsilon_t = X_{observed,t} - X_{model,t}$. (A.68)

A complex model should not absorb every unexplained term into Q.

The phase source requires unexplained content to remain residual.

---

# A.19 Phase θ

θ denotes current orientation in the conjugate plane:

$\theta(t) = \text{atan2}[Q(t),R(t)]$. (A.69)

θ is an angular coordinate.

It is not:

- calendar duration;
- event count;
- complete historical state;

- proof of internal time.

The same θ may be revisited under different:

- direction;
- amplitude;
- branch;
- ledger;
- residual.

---

# A.20 Internal phase time $\tau_i$

Two forms are distinguished.

### Signed unwrapped phase

$\tau_i^{signed}(t) = \mathrm{Unwrap}[\theta(t)]$. (A.70)

### Absolute phase depth

$\tau_i^{abs}(t) = \int_0^t |\theta(s)|\, ds$. (A.71)

The first preserves direction.

The second is monotone but discards direction.

A phase coordinate earns interpretation as internal time only if differently paced episodes align better in phase than in clock time and consequential gates show stable phase dependence.

---

# A.21 Calendar time t

t denotes external clock or parent-world duration.

Typical units include:

- seconds;
- minutes;
- trading sessions;
- days;
- quarters;
- years.

Calendar time remains valid even when internal phase time is introduced.

The two answer different questions.

---

# A.22 Event order k

k denotes the order of committed ledger events:

$k \to k + 1$ when Gate_P = Admit. (A.72)

A long clock interval may contain no new event.

A short crisis interval may contain many.

Thus:

$\Delta t \neq \Delta k$. (A.73)

The phase source distinguishes t, θ, $\tau_i$, and k as four different coordinates.

---

# A.23 Selection depth σ

σ denotes the degree to which previously admissible future branches have been suppressed.

A conceptual definition is:

$\sigma(t) = -\ln[\text{Measure}(\Omega_t)/\text{Measure}(\Omega_0)]$. (A.74)

where $\Omega_t$ is the set of futures still treated as admissible.

This is not yet a standard empirical market quantity.

Any implementation must declare:

- how possible futures are represented;
- how their measure is estimated;
- what constitutes elimination.

σ should therefore be reported as a model-dependent coordinate.

---

# A.24 Gate G

A gate maps a candidate state and existing ledger into an admission decision:

$G_P(X,L) \to (d,e,r,m)$. (A.75)

where:

| Symbol | Meaning |
|---|---|
| d | decision |
| e | admitted event |
| r | attached residual |
| m | gate metadata |

Decision values may be:

d ∈ {Admit,Defer,Reject}. (A.76)

Gate metadata may contain:

m = (Authority,Threshold,Horizon,Evidence,Confidence). (A.77)

The declared-disclosure framework treats the gate as the operator converting projection into committed trace rather than leaving it as a merely visible possibility.

---

# A.25 Gate strength S_G

A conceptual gate-strength model is:

S_G = w_DD + w_VV + w_BB + w_CC + w_FF + w_TT − w_RR. (A.78)

where:

D = normalized displacement;
V = participation or volume;
B = breadth coherence;
C = close quality;
F = follow-through;
T = retest evidence;
R = residual burden.

This formula is introduced as a research template.

The weights are not established constants.

They must be calibrated under a declared protocol.

---

# A.26 Trace e

e denotes the event admitted into trace:

e_k = AdmittedEvent_k. (A.79)

Examples include:

- execution;
- close;
- breakout;
- default;
- policy decision;
- accounting recognition.

An event can occur without becoming an authoritative ledgered trace for every observer.

---

# A.27 Residual r and residual register $\mathscr{R}$

r_k denotes residual attached to one gate event.

$\mathscr{R}$_k denotes the accumulated residual register.

An update may be:

$\mathscr{R}$_{k+1} = D_$\mathscr{R}$($\mathscr{R}$_k) + r_k − m_k. (A.80)

where:

D_$\mathscr{R}$ = decay or persistence operator;
m_k = metabolized, resolved, or invalidated residual.

The Technical Analysis source proposes residual records containing type, severity, resolution status, and later outcome, and emphasizes that failed traces must remain preserved.

---

# A.28 Ledger L

L denotes the ordered record of consequential commitments:

L_{k+1} = Update(L_k,e_k,r_k). (A.81)

A ledger differs from a log.

A log stores data.

A ledgered trace changes future:

- observation;
- interpretation;
- admissible action;
- constraint;
- responsibility.

The filtration and declaration articles define experienced time through the ordered ledger of selected and gated disclosure rather than through raw recursion alone.

---

# A.29 Transport T

Transport maps a claim between frames:

T_{P→P′}: Claim_P → Claim$\hat{}$_{P′}. (A.82)

The directly observed claim in the destination frame is:

Claim_{P′}. (A.83)

Transport consistency requires:

Distance[Claim$\hat{}$_{P′},Claim_{P′}] ≤ ε_T. (A.84)

Admissible transformations may include:

- timeframe;
- scale;
- volatility normalization;
- bar construction;
- benchmark;
- anchor;
- component universe;
- data source.

The Technical Analysis source defines stronger signals as those surviving admissible changes of projection and specifically identifies scale, pivot, timeframe, breadth, volume, and close as relevant cross-checks.

---

# A.30 Invalidation I

I denotes a predeclared knowledge-failure rule:

I_P(Claim) → {Valid,Weakened,Invalidated}. (A.85)

Invalidation is not necessarily the same as a trading stop.

A stop is an action rule.

Invalidation is an epistemic rule.

A database should preserve:

- original claim;
- invalidation condition;
- invalidation time;
- relabeling;
- reason for revision.

The source Technical Analysis schema expressly requires preservation of the original claim when a method is relabelled after failure.

---

# A.31 Revision U_a

An admissible revision is:

D_{k+1} = U_a(D_k,L_k,$\mathscr{R}$_k). (A.86)

where D_k is the current declaration.

A mature revision should be:

- trace-preserving;
- residual-honest;
- frame-robust;
- budget-bounded;
- non-degenerate.

The self-revising declaration source defines mature observerhood through admissible revision rather than unrestricted rule changing.

---

# A.32 Method signature

A complete method signature is:

Method_j = (P,p,g,S,w,χ,Ξ,Z,G,$\mathscr{R}$,T,I,L,U). (A.87)

where:

| Symbol | Meaning |
|---|---|
| P | protocol |
| p | period |
| g | dominant functional group |
| S | source lineage |
| w | operator word |
| χ | regime assumption or estimate |
| Ξ | effective control state, where used |
| Z | complex state, where eligible |
| G | gate |
| $\mathscr{R}$ | residual |
| T | transport tests |
| I | invalidation |
| L | ledger consequence |
| U | revision rule |

Not every method needs every field.

A simple moving average may leave Ξ and Z empty.

A claimed phase-clock model must populate them.

---

# A.33 Units and dimensional discipline

Every reported quantity should declare units.

Examples:

| Quantity | Possible units |
|---|---|
| Price | currency per unit |

| Quantity | Possible units |
|---|---|
| Return | dimensionless or percent |
| Volume | shares, contracts, currency value |
| Open interest | contracts |
| Volatility | return per √time or declared realized scale |
| $\rho$ | native loading units or normalized index |
| $\gamma$ | cost, rigidity index, inverse mobility, or normalized score |
| $\nu$ | volatility or normalized turbulence score |
| R, Q, A | same units within one complex state |
| $\theta$ | radians |
| $\tau_i$ | radians, turns, or normalized phase depth |
| k | event count |
| $\sigma$ | dimensionless information or branch-reduction measure |
| S_G | dimensionless calibrated score |

A valid complex state requires:

$$\text{Units}(R) = \text{Units}(Q) = \text{Units}(A). \quad (A.88)$$

If R and Q have different native units, a declared conversion or metric is required before writing:

$$Z = R + iQ. \quad (A.89)$$

Without such a declaration, the complex phase is dimensionally arbitrary.

---

# A.34 Scaling declaration

For a normalized complex state:

$$\tilde{R} = (R - \mu_R)/s_R. \quad (A.90)$$

$$\tilde{Q} = (Q - \mu_Q)/s_Q. \quad (A.91)$$

$$\tilde{Z} = \tilde{R} + i\tilde{Q}. \quad (A.92)$$

The scales s_R and s_Q must be declared.

Phase fragility should be tested under admissible scaling perturbations:

$$s_Q \to cs_Q. \quad (A.93)$$

If small changes in c radically alter phase order, the complex interpretation is weak.

This is one of the explicit failure conditions in the phase source.

---

# A.35 Symbol-collision rules

The following collisions are prohibited in the present article.

**R**

Use R for the real or admitted component.

Use $\mathscr{R}$ for residual register.

### M

Use M for Motion only as a group code in table headings.

Use M_str for structural mass.

Use M_map for a return mapping inside the χ operator.

### G

Use G for Gate.

Use G_gap only for a Fenchel–Young or health gap in cross-domain extensions.

### τ

Use $\tau_i$ for internal phase time.

Use τ_rec for recovery time.

Use τ_sw for switching time.

Use ν for agitation.

### P

Use P for protocol.

Use P_price only when referring to market price.

These rules prevent the integrated notation from inheriting ambiguities across its source articles.

---

# A.36 Minimal declaration header

Every empirical study should begin with:

```
Declared asset or system:
Boundary B:
Observation rule Δ:
Window h:
Admissible interventions u:
Data source:
Scale:
Feature map ϕ:
Primary period:
Primary functional family:
Regime assumption χ:
Gate rule:
Residual rule:
Transport tests:
Invalidation:
Outcome horizon:
```

A claim without this header should be treated as exploratory rather than reproducible.

---

# Appendix B — Expanded Proto-Periodic Table of Market Observation

## B.1 Reading convention

Each cell contains six items:

1. canonical function;
2. representative observables;
3. typical methods;
4. closure condition;
5. residual;
6. principal transport test.

The table is intentionally broader than a list of named indicators.

Some cells contain mature tools.

Others contain research targets.

---

## B.2 Period 0 — Mark

## B.2.1 $E_{0,L}$ — Mark-Level Load

**Canonical function**

Measure immediately available market interest before the next execution.

**Representative observables**

- displayed bid size;
- displayed ask size;
- queue depth;
- resting limit orders;
- local dealer inventory;
- open marketable interest.

**Typical methods**

- order-book depth;
- queue imbalance;

- depth-weighted price;
- liquidity heat map.

**Closure condition**

The order or quote is admitted into the selected mark data stream.

**Residual**

- hidden liquidity;
- iceberg reserve;
- off-venue interest;
- conditional orders;
- false or transient displayed interest.

**Transport test**

Compare:

- venues;
- depth levels;
- consolidated versus local books;
- different sampling intervals.

---

# B.2.2 $E_{0,M}$ — Mark-Level Motion

**Canonical function**

Measure change between individual market marks.

**Representative observables**

- tick return;
- spread change;
- queue depletion;
- order imbalance;
- trade-sign sequence.

**Typical methods**

- microprice;
- order-flow imbalance;
- signed tick rule;

- short-horizon impact model.

**Closure condition**

A mark-to-mark relation is calculable under synchronized timestamps.

**Residual**

- asynchronous feeds;
- hidden trades;
- cancelled liquidity;
- uncertain trade initiation;
- cross-venue routing.

**Transport test**

Compare:

- tick time versus event time;
- venue-local versus consolidated data;
- alternative trade-sign algorithms.

---

# B.2.3 $E_{0,C}$ — Mark-Level Constraint

**Canonical function**

Define immediate execution boundaries.

**Representative observables**

- best bid;
- best ask;
- spread;
- tick size;
- price limit;
- queue priority;
- order-size restriction.

**Typical methods**

- quoted spread;
- effective spread;
- depth curves;

- exchange-limit monitoring.

**Closure condition**

The constraint is enforceable under venue rules.

**Residual**

- hidden priority;
- internalization;
- discretionary orders;
- latency;
- changing matching logic.

**Transport test**

Compare across:

- venues;
- order types;
- normal versus auction sessions;
- stress versus ordinary conditions.

---

# B.2.4 E0,G — Mark-Level Commitment

**Canonical function**

Convert order intention into an execution, cancellation, rejection, or resting state.

**Representative observables**

- trade;
- partial fill;
- cancellation;
- order rejection;
- execution condition.

**Typical methods**

- transaction record;
- fill-ratio analysis;
- execution-quality statistics.

**Closure condition**

The matching or venue rule issues a recorded outcome.

**Residual**

- unfilled quantity;
- participant motive;
- hidden counterparty state;
- later correction.

**Transport test**

Compare:

- execution venues;
- reported versus actual fills;
- account versus public tape;
- partial versus complete execution.

---

# B.3 Period 1 — Window

# B.3.1 $E_{1,L}$ — Window-Level Load

**Canonical function**

Compile the marks carried into and through a declared window.

**Representative observables**

- opening price;
- volume;
- gap;
- prior close;
- local transaction density;
- intrawindow participation.

**Typical methods**

- OHLCV;
- session volume;
- relative volume;
- volume bar.

**Closure condition**

The window terminates under its declared rule.

**Residual**

- intrawindow path;
- participant identity;
- hidden liquidity;
- venue heterogeneity.

**Transport test**

Compare:

- time bars;
- volume bars;
- tick bars;
- range bars;
- alternative session boundaries.

---

# B.3.2 $E_{1,M}$ — Window-Level Motion

**Canonical function**

Measure displacement and excursion within one window.

**Representative observables**

- open-to-close body;
- high–low range;
- close location;
- gap;
- realized intrawindow volatility.

**Typical methods**

- candle body ratio;
- range expansion;
- true range;
- close-location value.

**Closure condition**

The closing state and intrawindow extremes are known.

**Residual**

- sequence of high and low;
- number of reversals;
- residence time;
- order-flow path.

**Transport test**

Compare:

- alternative bar construction;
- normalized versus nominal range;
- local versus higher timeframe.

---

# B.3.3 $E_{1,C}$ — Window-Level Constraint

**Canonical function**

Define the admissible range observed within the window.

**Representative observables**

- high;
- low;
- local volatility envelope;
- auction boundary;
- opening range.

**Typical methods**

- high–low range;
- opening-range boundary;
- local Donchian channel;
- session value area.

**Closure condition**

The window range is finalized.

**Residual**

- whether extremes were strongly defended;
- whether they were data artifacts;

- whether later windows preserve them.

**Transport test**

Compare:

- adjacent windows;
- alternative session definitions;
- volatility-normalized ranges.

---

# B.3.4 $E_{1,G}$ — Window-Level Commitment

**Canonical function**

Select the terminal state written into the bar or session ledger.

**Representative observables**

- official close;
- settlement price;
- auction close;
- final bar value.

**Typical methods**

- closing-price classification;
- close-above-level rule;
- close-location analysis.

**Closure condition**

The authoritative window-ending rule is applied.

**Residual**

- wick;
- intrawindow excursion;
- post-close trading;
- later price correction;
- auction distortion.

**Transport test**

Compare:

- official close;

- continuous-session close;
- settlement;
- after-hours price;
- higher-window close.

The source Technical Analysis article emphasizes that daily, weekly, and five-minute candles are distinct declared ledgers rather than merely magnified views of one truth.

---

# B.4 Period 2 — Structure

# B.4.1 $E_{2,L}$ — Structure-Level Load

**Canonical function**

Compile persistent market memory across windows.

**Representative observables**

- moving averages;
- VWAP;
- OBV;
- cumulative volume;
- volume profile;
- breadth history;
- prior highs and lows.

**Typical methods**

- SMA;
- EMA;
- VWAP;
- anchored VWAP;
- OBV;
- volume profile;
- advance–decline line.

**Closure condition**

The structure persists across the declared minimum number of windows or meets a stability criterion.

**Residual**

- positioning;
- hidden liquidity;
- participant identity;
- leverage;
- institutional rules;
- stale memory.

**Transport test**

Compare:

- parameter windows;
- timeframe;
- anchor;
- universe;
- alternative weighting.

The source Technical Analysis framework classifies moving averages as memory filters, VWAP as a volume-weighted ledger centre, and volume profile as density across price.

---

# B.4.2 $E_{2,M}$ — Structure-Level Motion

**Canonical function**

Measure persistent relations among price, memory, participation, and comparative fields.

**Representative observables**

- momentum;
- RSI;
- stochastic;
- MACD;
- moving-average spread;
- relative strength;
- breadth divergence.

**Typical methods**

- RSI;
- MACD;
- rate of change;

- stochastic;
- relative-strength line;
- breadth oscillator.

**Closure condition**

The relational reading remains stable enough to be treated as structure rather than one-window noise.

**Residual**

- wrong χ regime;
- missing volume or density;
- higher-frame contradiction;
- gate absence.

**Transport test**

Compare:

- smoothing;
- timeframe;
- benchmark;
- volatility normalization;
- component universe.

The source identifies WrongSignature, MissingVariable, WeakGate, HiddenResidual, and ProtocolOverfit as major causes of indicator failure.

---

# B.4.3 E2,C — Structure-Level Constraint

**Canonical function**

Identify historically loaded transition zones.

**Representative observables**

- support;
- resistance;
- value-area boundary;
- volatility band;
- channel;
- profile node;

- Fibonacci zone.

**Typical methods**

- horizontal levels;
- Bollinger Bands;
- Keltner Channels;
- Donchian Channels;
- volume-profile nodes;
- anchored VWAP zones.

**Closure condition**

The boundary is predeclared or supported by a reproducible construction.

**Residual**

- arbitrary anchor;
- stale memory;
- changed volatility;
- absent position data;
- weak institutional relevance.

**Transport test**

Compare:

- linear and log scale;
- timeframe;
- volatility-normalized width;
- nearby anchor perturbation;
- price and profile frames.

---

# B.4.4 $E_{2,G}$ — Structure-Level Commitment

**Canonical function**

Determine whether a candidate structure has become sufficiently persistent to be used as the basis of later event analysis.

**Representative observables**

- repeated closes;
- stable memory ordering;

- persistent relative-strength shift;
- sustained value migration;
- breadth structure.

**Typical methods**

- trend-state classifier;
- structural-acceptance score;
- persistence filter;
- moving-average regime filter.

**Closure condition**

The structure survives the declared persistence and transport thresholds.

**Residual**

- late confirmation;
- crowded positioning;
- unresolved higher-period boundary;
- low participation.

**Transport test**

Compare:

- alternative filters;
- nearby parameter settings;
- higher timeframe;
- independent data channels.

This cell is less mature than the other Structure-period cells and is one of the proposed instrument-development areas.

---

# B.5 Period 3 — Event

# B.5.1 $E_{3,L}$ — Event-Level Load

**Canonical function**

Measure the structure accumulated around a consequential transition.

**Representative observables**

- relative volume;

- open-interest change;
- option positioning;
- stop concentration;
- prior failed attempts;
- crowding;
- event participation.

**Typical methods**

- event-volume study;
- positioning dashboard;
- options open-interest map;
- prior-attempt ledger.

**Closure condition**

The event context is defined before or at the gate.

**Residual**

- hidden positioning;
- off-balance-sheet exposure;
- unknown stop distribution;
- participant motive.

**Transport test**

Compare:

- derivatives and cash markets;
- venues;
- participant categories;
- alternative event windows.

---

# B.5.2 $E_{3,M}$ — Event-Level Motion

**Canonical function**

Measure displacement and relational change during a candidate transition.

**Representative observables**

- normalized boundary displacement;

- acceleration;
- volatility expansion;
- breadth expansion;
- phase shift;
- volume efficiency.

**Typical methods**

- ATR-normalized break;
- event momentum;
- breadth thrust;
- phase-change score.

**Closure condition**

The move exceeds the declared noise or tolerance region.

**Residual**

- gap distortion;
- temporary short covering;
- thin liquidity;
- unstable volatility estimate.

**Transport test**

Compare:

- nominal versus volatility-normalized movement;
- cash versus derivative markets;
- local versus higher timeframe.

---

# B.5.3 $E_{3,C}$ — Event-Level Constraint

**Canonical function**

Identify the boundary actively tested by the event.

**Representative observables**

- prior high or low;
- value-area edge;
- trend boundary;

- covenant;
- margin threshold;
- option strike;
- policy threshold.

**Typical methods**

- breakout boundary;
- neckline;
- range edge;
- liquidation threshold;
- collateral trigger.

**Closure condition**

The boundary is meaningful and declared before outcome classification.

**Residual**

- zone width;
- disputed anchor;
- temporary rule suspension;
- competing boundaries.

**Transport test**

Compare:

- alternative boundary definitions;
- higher timeframe;
- institutional versus chart boundary;
- scale normalization.

---

# B.5.4 $E_{3,G}$ — Event-Level Commitment

**Canonical function**

Admit, defer, or reject the transition.

**Representative observables**

- close;
- volume;

- breadth;
- follow-through;
- retest;
- settlement;
- institutional recognition.

**Typical methods**

- breakout confirmation;
- reversal gate;
- rejection gate;
- retest gate;
- fakeout classification.

**Closure condition**

The event passes the predeclared gate rule.

**Residual**

- weak breadth;
- absent retest;
- event risk;
- higher-period contradiction;
- trapped positions.

**Transport test**

Compare:

- daily and weekly gates;
- alternative bar rules;
- price, volume, breadth, and institutional frames.

The Technical Analysis source argues that a touch is not a confirmed break and that breakout quality should be cross-checked across price, volume, breadth, and close.

---

# B.6 Period 4 — Episode

# B.6.1 $E_{4,L}$ — Episode-Level Load

**Canonical function**

Carry forward the accumulated history of events, commitments, failures, and unresolved residual.

**Representative observables**

- successful and failed breakouts;
- pivot history;
- position migration;
- volatility memory;
- narrative memory;
- accumulated residual debt.

**Typical methods**

- episode ledger;
- regime-history state;
- cumulative fakeout register;
- trend-quality history.

**Closure condition**

Events are assigned to one episode under a declared segmentation rule.

**Residual**

- omitted events;
- alternate segmentation;
- decayed but still active memory;
- unobserved position migration.

**Transport test**

Compare:

- pivot rule;
- episode start point;
- timeframe;
- volatility-normalized segmentation.

---

# B.6.2 $E_4$,M — Episode-Level Motion

**Canonical function**

Describe internal progression through a trend, range, squeeze, wave sequence, crisis, or

recovery.

**Representative observables**

- $\chi$ sequence;
- phase $\theta$;
- accumulated phase $\tau_i$;
- cadence;
- selection depth $\sigma$;
- impulse–correction order.

**Typical methods**

- Elliott-like wave sequence;
- trend-stage model;
- phase clock;
- regime-switching sequence;
- internal progress index.

**Closure condition**

The sequence shows a sufficiently stable relational grammar.

**Residual**

- branch ambiguity;
- phase-scaling sensitivity;
- alternate wave count;
- nonstationary $\chi$;
- calendar mismatch.

**Transport test**

Compare:

- pivot algorithms;
- phase scaling;
- timeframe;
- episode families;
- real-pair benchmark.

The phase source requires differently paced episodes to align better in phase than in clock time before phase is promoted to an internal clock.

## B.6.3 $E_{4,C}$ — Episode-Level Constraint

**Canonical function**

Define the basin or geometry within which the episode develops.

**Representative observables**

- trading range;
- trend channel;
- triangle;
- wedge;
- volatility basin;
- funding envelope;
- narrative attractor.

**Typical methods**

- chart pattern;
- trend channel;
- accumulation or distribution range;
- Gann candidate geometry;
- volatility-regime envelope.

**Closure condition**

The episode boundary system remains stable under the declared segmentation.

**Residual**

- subjective pivot selection;
- pattern overfit;
- anchor sensitivity;
- scale dependence;
- unmodelled institutional constraint.

**Transport test**

Compare:

- log scale;
- volatility normalization;

- alternative anchors;
- timeframe;
- event-defined versus visually defined boundaries.

---

# B.6.4 $E_4$,G — Episode-Level Commitment

**Canonical function**

Determine whether the episode continues, completes, fails, or transitions.

**Representative observables**

- defining-relation failure;
- boundary transition;
- new-state persistence;
- wave endpoint;
- volatility-regime change;
- trend termination.

**Typical methods**

- episode completion certificate;
- trend-transition model;
- wave-end gate;
- range-to-trend gate.

**Closure condition**

The old episode grammar no longer organizes future events and a new grammar gains persistence.

**Residual**

- alternative branch;
- false transition;
- unresolved higher-world conditions;
- incomplete position unwinding.

**Transport test**

Compare:

- higher timeframe;
- independent episode model;

- cross-market confirmation;
- institutional ledger.

---

# B.7 Period 5 — World

# B.7.1 $E_{5,L}$ — World-Level Load

### Canonical function

Measure institutional, balance-sheet, legal, policy, and narrative structure carried across episodes.

### Representative observables

- leverage;
- balance-sheet exposure;
- benchmark weights;
- collateral encumbrance;
- legal obligations;
- policy history;
- valuation conventions;
- institutional memory.

### Typical methods

- $\Xi$ loading coordinate $\rho$;
- balance-sheet map;
- funding exposure map;
- benchmark concentration;
- institutional-positioning study.

### Closure condition

The structure persists across multiple episodes and alters available actions.

### Residual

- off-balance-sheet exposure;
- unreported leverage;
- informal guarantees;
- legal ambiguity;

- hidden concentration.

**Transport test**

Compare:

- desk and enterprise views;
- market and accounting values;
- local and systemic boundaries;
- alternative legal entities.

The Gauge Grammar finance mapping warns that local desk truth may not become enterprise truth when funding, collateral, legal, accounting, and risk frames are not connected.

---

# B.7.2 $E_{5,M}$ — World-Level Motion

**Canonical function**

Describe reflexive, institutional, policy, and valuation-frame dynamics.

**Representative observables**

- leverage feedback;
- funding transmission;
- policy response;
- cross-asset contagion;
- observer crowding;
- narrative reinforcement;
- world-level $\chi$.

**Typical methods**

- reflexivity model;
- funding-stress propagation;
- observer-backreaction score;
- regime-state model;
- cross-market transport analysis.

**Closure condition**

The relation persists across episodes and changes institutional behaviour.

**Residual**

- causal ambiguity;

- hidden intervention;
- changing authority;
- non-observed adoption;
- policy discontinuity.

**Transport test**

Compare:

- markets;
- institutions;
- legal frames;
- accounting frames;
- policy jurisdictions.

---

# B.7.3 $E_5,C$ — World-Level Constraint

**Canonical function**

Define authoritative structures limiting action and transition.

**Representative observables**

- law;
- accounting;
- collateral;
- margin;
- capital regulation;
- benchmark rules;
- market infrastructure;
- policy mandate.

**Typical methods**

- constraint map;
- cross-ledger rigidity vector;
- regulatory threshold analysis;
- collateral network model;
- $\gamma$ coordinate.

**Closure condition**

The constraint is enforceable or repeatedly governs behaviour.

**Residual**

- legal interpretation;
- emergency discretion;
- informal practice;
- regulatory arbitrage;
- jurisdictional conflict.

**Transport test**

Compare:

- legal and economic ownership;
- accounting and liquidity value;
- entity and consolidated levels;
- domestic and international regimes.

---

# B.7.4 $E_{5,G}$ — World-Level Commitment

**Canonical function**

Convert a contested or evolving state into authoritative institutional history.

**Representative observables**

- default;
- impairment;
- policy decision;
- legal judgment;
- index inclusion;
- regulatory classification;
- collateral call;
- bankruptcy;
- restructuring.

**Typical methods**

- recognition vector;

- cross-ledger reconciliation gate;
- institutional event classifier;
- world-transition certificate.

**Closure condition**

The relevant authority writes the event into an enforceable or operational ledger.

**Residual**

- appeal;
- delayed accounting consequence;
- market disagreement;
- incomplete settlement;
- political reversal;
- unresolved systemic effects.

**Transport test**

Compare:

- market;
- legal;
- accounting;
- risk;
- policy ledgers.

A world-level event may be recognized at different times by different institutions.

The recognition vector is:

$$\text{RecognitionVector}_k = (g_{market}, g_{accounting}, g_{legal}, g_{risk}, g_{policy}). \quad (B.1)$$

---

# B.8 Compact 6 × 4 Table

| Period | Load / Memory | Motion / Relation | Constraint / Boundary | Commitment / Gate |
|---|---|---|---|---|
| **0 Mark** | depth, resting interest, inventory | tick change, spread, imbalance | bid, ask, tick and venue limits | execution, cancellation, rejection |
| **1 Window** | opening state, bar volume, prior close | body, return, range, gap | high, low, local envelope | official close or settlement |

| Period | Load / Memory | Motion / Relation | Constraint / Boundary | Commitment / Gate |
|---|---|---|---|---|
| **2 Structure** | MA, VWAP, OBV, profile, breadth | RSI, MACD, momentum, divergence | support, resistance, bands, channels | structural acceptance |
| **3 Event** | positioning, stops, event participation | displacement, acceleration, phase shift | actively tested boundary | breakout, retest, rejection, reversal |
| **4 Episode** | event history, residual history, narrative memory | χ sequence, cadence, phase progress | range, pattern, trend or funding basin | episode completion or transition |
| **5 World** | leverage, balance sheets, benchmark and institutional memory | reflexivity, policy and valuation-frame motion | law, accounting, collateral, regulation | default, policy, legal or accounting recognition |

---

# B.9 The table's outer rails

The table should always be published with four outer rails:

# Residual rail

What was omitted, contradicted, or left unresolved?

# Transport rail

Does the claim survive another admissible frame?

# Ledger rail

Did the event alter future admissibility?

# Revision rail

Can the declaration change without erasing its previous trace?

The full table runtime is:

Declare_P
→ LocatePeriod_p
→ IdentifyFunction_g
→ Project
→ Diagnoseχ
→ TestConstraint
→ ApplyGate
→ WriteTrace + Residual
→ Transport
→ ObserveBackreaction
→ ReviseAdmissibly. (B.2)

This expanded table completes the article's central classificatory object.

The next appendix can provide the **Molecular Method Cards** for each major Technical Analysis method in a standardized, copy-ready format.

# Appendix C — Molecular Method Cards

## C.1 Purpose

The periodic table identifies recurring functional roles.

The molecular cards describe how familiar Technical Analysis methods combine those roles into practical instruments.

Each card uses the same fields:

```
Method:
Primary period:
Primary family:
Secondary roles:
Source lineage:
Operator word:
Observed object:
Declared parameters:
Regime dependence:
Gate requirement:
Residual:
Transport tests:
Invalidation:
Common category error:
Research status:
```

The cards are not trading recommendations.

They are standardized descriptions of what each method measures, what must be added before it can support an event claim, and what remains unresolved.

---

## C.2 Moving Average

## Method

Simple, exponential, weighted, or otherwise declared moving average.

## Primary period

**Structure**

## Primary family

**Load / Memory**

## Secondary roles

- Motion, through slope;
- Relation, through price distance;

- soft Constraint, when widely used as support or resistance.

# Source lineage

Price, usually closing price.

S_MA = {P_close}. (C.1)

# Operator word

w_MA = F_n ∘ Π_price. (C.2)

where F_n is the declared smoothing operator.

# Core formula

MA_n(t) = Σ_{j=0}^{n−1} w_jP_{t−j}. (C.3)

Σ_{j=0}^{n−1} w_j = 1. (C.4)

For an exponential moving average:

EMA_t = αP_t + (1 − α)EMA_{t−1}. (C.5)

# Observed object

A filtered memory centre of historical price.

The method removes part of the local path and preserves a slower representation of price history.

# Declared parameters

- price field;
- window n;
- weight function;
- timeframe;
- adjustment rule;
- bar construction;
- scale.

# Regime dependence

A price far above its moving average may indicate:

- extension under χ < 0;
- persistent directional selection under χ > 0;

- unstable separation under $\chi \approx 0$.

The moving average does not diagnose $\chi$ by itself.

## Gate requirement

A moving average requires an external gate before supporting an Event-level claim.

Possible gates include:

- sustained close above or below;
- crossover;
- successful retest;
- boundary break;
- volume and breadth confirmation.

## Residual

- volume;
- breadth;
- positioning;
- liquidity;
- transaction density;
- institutional constraint;
- filter lag;
- higher-period conflict.

## Transport tests

- nearby window lengths;
- simple versus exponential weights;
- timeframe;
- log versus arithmetic scale;
- time bars versus volume bars;
- adjusted versus unadjusted price.

## Invalidation

The moving average itself is not usually invalidated.

A claim based on it is invalidated when its declared relation fails.

Example:

“Price has established an upward memory regime” is weakened if price closes below the average, the average turns down, and the structure fails its declared persistence gate.

## Common category error

Treating filtered historical memory as an independent predictor of future trend.

## Research status

Mature as a filter.

Conditional as a market-state classifier.

Insufficient as a standalone commitment gate.

The source Technical Analysis framework treats moving averages as declared memory filters rather than direct forecasts.

---

# C.3 Moving-Average Crossover

## Method

Fast and slow moving-average crossover.

## Primary period

**Structure**

## Primary family

**Motion / Relation**

## Secondary roles

- Load, because both inputs are memories;
- provisional Commitment, when used as a regime filter.

## Source lineage

Price.

S_Cross = {P_close}. (C.6)

## Operator word

$w_{Cross} = G_{sign} \circ \Delta \circ (F_{fast}, F_{slow}) \circ \Pi_{price}$. (C.7)

# Core formula

$D_M(t) = M_{fast}(t) - M_{slow}(t)$. (C.8)

A crossover occurs when:

$\text{sign}[D_M(t)] \neq \text{sign}[D_M(t-1)]$. (C.9)

# Observed object

A reversal in the ordering of two declared memory horizons.

# Declared parameters

- fast window;
- slow window;
- smoothing rule;
- price field;
- timeframe;
- persistence threshold;
- crossover tolerance.

# Regime dependence

Crossovers perform differently across:

- self-confirming trends;
- corrective ranges;
- volatile transition zones.

In a range, repeated crossovers may reflect memory-horizon oscillation rather than regime transition.

# Gate requirement

A crossover should be supported by at least one independent event condition:

- boundary break;
- breadth expansion;
- volume support;
- follow-through;

- retest;
- volatility-normalized displacement.

## Residual

- lag;
- whipsaw;
- shared-input redundancy;
- range conditions;
- late entry;
- higher-timeframe contradiction.

## Transport tests

- nearby fast and slow parameters;
- alternate smoothing rules;
- higher timeframe;
- volatility normalization;
- out-of-sample regime.

## Invalidation

A crossover claim is invalidated when the memory ordering reverses again before the declared persistence horizon or when the associated structural gate fails.

## Common category error

Treating memory-order reversal as completed Event- or Episode-level regime change.

## Research status

Useful as a delayed structure classifier.

Highly vulnerable to wrong-period promotion.

---

# C.4 MACD

## Method

Moving Average Convergence Divergence.

# Primary period

**Structure**

# Primary family

**Motion / Relation**

# Secondary roles

- Load, through its moving-average inputs;
- phase-warning function, through changes in memory separation.

# Source lineage

Price.

S_MACD = {P_close}. (C.10)

# Operator word

w_MACD
= Δ ∘ (F_fast,F_slow) ∘ Π_price. (C.11)

w_Hist
= Δ ∘ [w_MACD,F_signal(w_MACD)]. (C.12)

# Core formulas

MACD_t = EMA_fast(t) − EMA_slow(t). (C.13)

Signal_t = EMA_m(MACD_t). (C.14)

Histogram_t = MACD_t − Signal_t. (C.15)

# Observed object

Separation between two price-memory horizons and the change of that separation relative to its own filtered history.

# Declared parameters

- fast EMA;
- slow EMA;
- signal EMA;
- timeframe;
- price field;

- normalization.

## Regime dependence

Positive MACD may describe persistent positive memory ordering.

Falling positive histogram may indicate deceleration without establishing reversal.

The method is especially sensitive to whether χ remains self-confirming.

## Gate requirement

MACD should be paired with:

- structural boundary;
- close;
- breadth;
- volume;
- retest;
- or explicit reversal gate.

## Residual

- volume;
- liquidity;
- breadth;
- structural density;
- crowding;
- event authority;
- filter lag.

## Transport tests

- parameter perturbation;
- normalized MACD versus raw MACD;
- timeframe;
- benchmark-relative MACD;
- alternate price field.

## Invalidation

A MACD divergence claim is invalidated when price continues with renewed MACD expansion or when the declared reversal gate never occurs within the stated horizon.

## Common category error

Treating divergence as an already committed reversal.

## Research status

Mature as a filtered relational indicator.

Incomplete as a gate.

The source article interprets MACD as memory curvature or phase acceleration rather than a complete event signal.

---

# C.5 Relative Strength Index

## Method

RSI.

## Primary period

**Structure**

## Primary family

**Motion / Relation**

## Secondary roles

- soft Constraint through upper and lower thresholds;
- candidate corrective-pressure signal.

## Source lineage

Price differences.

S_RSI = {ΔP}. (C.16)

## Operator word

w_RSI
= N ∘ A ∘ Split_sign ∘ Δ ∘ Π_price. (C.17)

## Core formulas

$RS_t = AverageGain_n(t)/AverageLoss_n(t)$. (C.18)

$RSI_t = 100 - 100/(1 + RS_t)$. (C.19)

# Observed object

Normalized directional dominance over the declared lookback.

# Declared parameters

- lookback n;
- smoothing convention;
- price source;
- timeframe;
- upper and lower thresholds;
- divergence rule.

# Regime dependence

The conventional reversal interpretation usually assumes:

$\chi < 0$. (C.20)

Under $\chi > 0$, persistent high RSI may represent continued directional selection.

# Gate requirement

A reversal claim should require:

- support or resistance interaction;
- structural failure;
- opposing close;
- breadth reversal;
- follow-through;
- or another independent Commitment channel.

# Residual

- trend persistence;
- participation;
- liquidity;
- density;

- positioning;
- volatility regime;
- threshold arbitrariness.

## Transport tests

- threshold calibration;
- lookback perturbation;
- timeframe;
- alternative smoothing;
- trend-conditioned versus range-conditioned performance.

## Invalidation

An RSI exhaustion claim is invalidated if price remains accepted in the same direction, RSI re-expands, and no declared reversal gate occurs.

## Common category error

Overbought = reversal. (C.21)

Oversold = reversal. (C.22)

The correct statement is:

Overbought or oversold = directional condition under a declared normalization. (C.23)

## Research status

Useful for relational diagnosis.

Interpretation strongly regime-dependent.

The source framework identifies RSI as a corrective-pressure detector whose usual reversal reading may fail in self-confirming trends.

---

# C.6 Stochastic Oscillator

## Method

Stochastic %K and related smoothed versions.

## Primary period

**Structure**

# Primary family

**Motion / Relation**

# Secondary roles

- relation to recent range boundaries;
- corrective-regime warning.

# Source lineage

Price, high, low, and close.

S_Stoch = {H,L,C}. (C.24)

# Operator word

w_Stoch
= N ∘ Relate(C,RollingHigh,RollingLow). (C.25)

# Core formula

%K_t = 100 × [C_t − L_n(t)]/[H_n(t) − L_n(t)]. (C.26)

# Observed object

Location of the current close inside the recent high–low range.

# Declared parameters

- lookback;
- smoothing;
- high–low definition;
- threshold;
- timeframe.

# Regime dependence

Most useful when repeated range rotation supports $\chi < 0$.

Persistent high readings may be normal inside a strong upward regime.

# Gate requirement

- boundary rejection;

- close back inside range;
- momentum turn;
- structural break;
- independent participation evidence.

## Residual

- trend continuation;
- gap behaviour;
- volatility expansion;
- range instability;
- higher-period direction.

## Transport tests

- timeframe;
- range length;
- smoothed versus unsmoothed;
- volatility-conditioned thresholds.

## Invalidation

A reversal claim fails when the close remains near the same range extreme and price continues beyond the declared horizon.

## Common category error

Treating position inside a recent range as a universal exhaustion law.

## Research status

Mature as a normalized range-location measure.

Conditional as a reversal tool.

---

# C.7 Average True Range

## Method

ATR.

# Primary period

**Window–Structure**

# Primary family

**Motion magnitude**

# Secondary roles

- scaling coordinate for Constraint;
- normalization tool for transport.

# Source lineage

High, low, and prior close.

S_ATR = {H,L,C_{t−1}}. (C.27)

# Operator word

w_ATR = F_n ∘ MaxRange ∘ Π_{H,L,C}. (C.28)

# Core formulas

TR_t = max[H_t − L_t,|H_t − C_{t−1}|,|L_t − C_{t−1}|]. (C.29)

ATR_n(t) = F_n(TR_t). (C.30)

# Observed object

Recent realized movement magnitude, including gaps.

# Declared parameters

- smoothing rule;
- lookback;
- session definition;
- adjusted or unadjusted prices;
- annualization, where used.

# Regime dependence

ATR rises in many different conditions:

- trend;

- crisis;
- news;
- illiquidity;
- transition;
- liquidation.

The method does not identify direction or cause.

## Gate requirement

None for volatility measurement.

An external directional or Commitment method is required for trading interpretation.

## Residual

- direction;
- cause;
- liquidity;
- implied volatility;
- order-flow source;
- future persistence.

## Transport tests

ATR is itself useful for transport:

d_ATR = Distance/ATR. (C.31)

Compare across assets, regimes, and timeframes.

## Invalidation

A claim such as “volatility regime has expanded” is weakened if ATR elevation is temporary and fails the declared persistence threshold.

## Common category error

Treating high volatility as bearish or low volatility as bullish.

## Research status

Mature as a realized-movement measure.

Not a directional indicator.

# C.8 Bollinger Bands

# Method

Moving average plus dispersion bands.

# Primary period

**Structure**

# Primary family

**Constraint / Boundary**

# Secondary roles

- Load through the moving-average centre;
- Motion magnitude through dispersion;
- compression diagnosis through band width.

# Source lineage

Price.

S_BB = {P}. (C.32)

# Operator word

w_BB = Boundary_k ∘ (F_n,Dispersion_n) ∘ Π_price. (C.33)

# Core formulas

Middle_t = MA_n(t). (C.34)

Upper_t = MA_n(t) + kσ_n(t). (C.35)

Lower_t = MA_n(t) – kσ_n(t). (C.36)

# Observed object

Price location relative to a filtered centre and recent dispersion.

# Declared parameters

- window;
- centre definition;

- dispersion definition;
- multiplier k;
- timeframe;
- price field.

## Regime dependence

Under $\chi < 0$, outer-band extension may precede reversion.

Under $\chi > 0$, repeated band contact may represent sustained directional selection.

## Gate requirement

- close back inside;
- rejection candle;
- continuation close;
- breadth confirmation;
- volatility expansion;
- structural retest.

## Residual

- non-normal returns;
- regime shift;
- volume;
- breadth;
- structural density;
- tail behaviour.

## Transport tests

- alternative dispersion estimator;
- nearby k;
- timeframe;
- log-price construction;
- ATR or Keltner alternative.

## Invalidation

A mean-reversion claim is invalidated when price remains accepted beyond the band and the band itself expands in the same direction.

## Common category error

Every outer-band contact implies reversion.

## Research status

Mature as an adaptive boundary construction.

Regime-dependent as an event interpretation.

---

# C.9 Raw and Relative Volume

## Method

Bar volume, dollar volume, or relative volume.

## Primary period

**Window–Event**

## Primary family

**Load / Memory**

## Secondary roles

- event participation;
- Commitment evidence;
- potential structural mass.

## Source lineage

Trades and reported quantity.

S_Volume = {TradeQuantity,TradePrice where relevant}. (C.37)

## Operator word

w_Volume = A ∘ Π_trade. (C.38)

w_RVOL = N_baseline ∘ w_Volume. (C.39)

## Core formulas

$V_t = \Sigma_{k \in Window_t} v_k$. (C.40)

$DollarVolume_t = \Sigma_{k \in Window_t} p_k v_k$. (C.41)

$RVOL_t = V_t / E[V_t \mid Context]$. (C.42)

# Observed object

Recorded exchange activity inside the declared window.

# Declared parameters

- venue coverage;
- time-of-day baseline;
- trade exclusions;
- adjusted or unadjusted volume;
- window;
- comparison period.

# Regime dependence

High volume can accompany:

- new loading;
- liquidation;
- absorption;
- transfer;
- rebalancing;
- churn.

The interpretation depends on displacement, location, and later gate behaviour.

# Gate requirement

Volume becomes commitment evidence only when attached to a meaningful event.

# Residual

- participant identity;
- opening versus closing;
- hidden liquidity;
- derivatives linkage;

- off-venue volume;
- motive.

## Transport tests

- volume versus dollar volume;
- venue-local versus consolidated;
- time-of-day normalization;
- cash versus derivative activity.

## Invalidation

A claim of strong commitment based on volume is weakened if price fails to hold, breadth does not confirm, and the move reverses despite the activity.

## Common category error

High volume = directional commitment.

## Research status

Mature as an activity measure.

Underdetermined as a directional interpretation.

The source framework describes volume as a mixture of frequency, mass, commitment, and exchange ambiguity.

---

# C.10 On-Balance Volume

## Method

OBV.

## Primary period

**Structure**

## Primary family

**Load / Memory**

## Secondary roles

Directional relation through price-signed accumulation.

## Source lineage

Closing direction and volume.

S_OBV = {sign(ΔC),V}. (C.43)

## Operator word

w_OBV = A ∘ Multiply[Sign(ΔC),V]. (C.44)

## Core formula

OBV_t = OBV_{t−1} + s_tV_t. (C.45)

## Observed object

Cumulative volume associated with positive or negative closing-price direction.

## Declared parameters

- sign rule;
- volume source;
- timeframe;
- treatment of unchanged closes;
- corporate actions.

## Regime dependence

OBV can trend with price in directional markets.

Divergence may reveal weakening signed-volume association, but not necessarily reversal.

## Gate requirement

- price boundary failure;
- close;
- retest;
- breadth;
- event confirmation.

## Residual

- actual aggressor side;

- participant identity;
- absorption;
- opening versus closing positions;
- hidden activity.

## Transport tests

- alternative volume sources;
- signed-volume method;
- timeframe;
- accumulation horizon.

## Invalidation

An OBV divergence claim fails if price continues and OBV later re-aligns without the declared event gate.

## Common category error

Treating inferred price-signed volume as directly observed accumulation or distribution.

## Research status

Useful cumulative trace.

Interpretation limited by sign inference.

---

# C.11 VWAP and Anchored VWAP

## Method

Volume-weighted average price.

## Primary period

**Structure**

## Primary family

**Load / Memory**

## Secondary roles

- Constraint / Boundary;

- institutional reference;
- observer-backreaction channel.

## Source lineage

Price and volume.

S_VWAP = {P,V}. (C.46)

## Operator word

w_VWAP = Divide[A(PV),A(V)]. (C.47)

## Core formula

VWAP_T = Σ_{t∈T}P_tV_t/Σ_{t∈T}V_t. (C.48)

Anchored form:

AVWAP_{a→t} = Σ_{j=a}^{t}P_jV_j/Σ_{j=a}^{t}V_j. (C.49)

## Observed object

Transaction-weighted centre of recorded exchange over the declared interval.

## Declared parameters

- anchor;
- session;
- venue coverage;
- price convention;
- volume convention;
- reset rule.

## Regime dependence

VWAP may act as:

- neutral transaction centre;
- soft support or resistance;
- institutional benchmark;
- trend reference.

Its behavioural significance increases when many participants use it.

## Gate requirement

- reclaim;
- hold;
- rejection;
- repeated close;
- retest;
- volume-supported acceptance.

## Residual

- anchor choice;
- hidden flow;
- participant identity;
- off-venue activity;
- economic value versus transaction centre.

## Transport tests

- nearby admissible anchors;
- session versus multi-session;
- alternative venues;
- higher timeframe;
- volume-profile overlap.

## Invalidation

A VWAP-support claim fails when price remains accepted below the declared VWAP and later retests are rejected.

## Common category error

VWAP = intrinsic fair value.

The correct claim is:

VWAP = volume-weighted transaction memory under T. (C.50)

## Research status

Mature as a transaction benchmark.

Self-referential when widely adopted.

The source article classifies VWAP as a volume-weighted ledger centre rather than intrinsic value.

---

# C.12 Volume Profile

# Method

Volume accumulated by price bin.

# Primary period

**Structure**

# Primary family

**Load / Memory**

# Secondary roles

Constraint through structural density and value-area boundaries.

# Source lineage

Price and volume.

$S_{VP} = \{P,V\}$. (C.51)

# Operator word

$w_{VP} = Density_{price} \circ A \circ \Pi_{\{P,V\}}$. (C.52)

# Core formula

$VP(p) = \Sigma_t V_t \cdot 1[P_t \in Bin(p)]$. (C.53)

# Observed object

Historical transaction density across price.

# Declared parameters

- bin width;
- interval;
- volume source;

- value-area rule;
- session or anchor;
- price convention.

## Regime dependence

High-density zones may:

- attract;
- slow;
- support;
- resist;
- or become irrelevant after regime transition.

Low-density zones may permit faster traversal.

## Gate requirement

Current acceptance or rejection is required before historical density becomes an Event-level claim.

## Residual

- participant identity;
- position retention;
- opening versus closing;
- hidden derivatives exposure;
- changed market regime;
- bin sensitivity.

## Transport tests

- bin width;
- anchored interval;
- timeframe;
- cash and derivatives;
- volume versus time-at-price.

## Invalidation

A high-volume node fails as support if price becomes durably accepted below it and current flow no longer reacts to the zone.

## Common category error

Historical density = permanent future force.

## Research status

Mature as a historical density map.

Conditional as a boundary predictor.

The source Technical Analysis framework treats profile as semantic or structural density across price.

---

# C.13 Candlestick

## Method

OHLC or OHLCV candle.

## Primary period

**Window**

## Primary family

**Commitment / Gate**

## Secondary roles

- Load through opening state and volume;
- Motion through body;
- Constraint through high and low;
- Residual through wicks.

## Source lineage

Marks aggregated into a declared window.

S_Candle = {O,H,L,C,V}. (C.54)

## Operator word

w_Candle = WindowAggregate ∘ Π_marks. (C.55)

## Core state

Candle_P = (O,H,L,C,V). (C.56)

Body = C − O. (C.57)

UpperWick = H − max(O,C). (C.58)

LowerWick = min(O,C) − L. (C.59)

## Observed object

Compressed intrawindow movement whose terminal state is selected by the close.

## Declared parameters

- window;
- session;
- venue;
- after-hours treatment;
- price source;
- volume inclusion.

## Regime dependence

The same shape can mean different things in:

- trend;
- range;
- low liquidity;
- high volatility;
- major structural boundary.

## Gate requirement

The candle’s own close is a Window gate.

A higher-period event requires later follow-through or structural confirmation.

## Residual

- full path;
- sequence of high and low;

- residence time;
- order-flow direction;
- hidden liquidity;
- later reversal.

## Transport tests

- timeframe;
- alternative bar construction;
- regular versus extended session;
- path-sensitive reconstruction.

## Invalidation

A candlestick reversal claim fails if later bars do not confirm and the apparent rejection is reabsorbed.

## Common category error

Candle shape = context-free market intention.

## Research status

Mature as compressed window trace.

Weak as standalone episode prediction.

The source framework characterizes candlesticks as micro-ledgers of attempted and retained displacement.

---

# C.14 Support and Resistance

## Method

Horizontal or zone-based loaded boundary.

## Primary period

**Structure**

## Primary family

**Constraint / Boundary**

## Secondary roles

- Load through accumulated memory;
- Commitment through later acceptance or rejection.

## Source lineage

Price reactions, volume density, institutional references, and sometimes positioning.

S_Level = {PriceHistory,ReactionHistory,Density,Reference}. (C.60)

## Operator word

w_Level = BoundaryConstruct ∘ Accumulate ∘ DetectReaction. (C.61)

## Core representation

Level_ℓ
= HistoricalReaction_ℓ

- TransactionDensity_ℓ
- SharedAttention_ℓ
- ConditionalOrders_ℓ. (C.62)

The sum is conceptual rather than a calibrated universal formula.

## Observed object

A price region whose historical trace may influence future behaviour.

## Declared parameters

- anchor;
- zone width;
- minimum reactions;
- timeframe;
- wick or close rule;
- density threshold;
- decay.

## Regime dependence

A level may produce:

- corrective rotation under $\chi < 0$;
- repeated testing under $\chi \approx 0$;
- explosive failure under $\chi > 0$.

## Gate requirement

- hold;
- rejection;
- close through;
- retest;
- value migration;
- institutional recognition.

## Residual

- stale memory;
- changed participants;
- weak position retention;
- multiple nearby levels;
- volatility expansion;
- arbitrary drawing.

## Transport tests

- log scale;
- higher timeframe;
- zone perturbation;
- volume profile;
- anchored VWAP;
- institutional reference.

## Invalidation

A support claim fails after durable acceptance below the zone and unsuccessful reclaim under the declared gate.

## Common category error

A line drawn on a chart possesses causal force by itself.

## Research status

Conceptually strong when linked to operative memory and conditional orders.

Highly vulnerable to retrospective construction.

---

# C.15 Breakout

## Method

Transition beyond a declared structural boundary.

## Primary period

**Event**

## Primary family

**Commitment / Gate**

## Secondary roles

- Load around the boundary;
- Motion through displacement;
- Constraint through the tested level;
- residual through fakeout and contradiction.

## Source lineage

Price, volume, breadth, volatility, and optional positioning.

$S_{Breakout} = \{P,V,Breadth,Volatility,Positioning\}$. (C.63)

## Operator word

$w_{Breakout}$
$= G_{event} \circ Confirm \circ \Delta(Price,Boundary)$. (C.64)

## Core condition

BreakoutCandidate
= BoundaryCross + DirectionalDisplacement. (C.65)

A fuller form is:

ValidBreakout
= MeaningfulBoundary

- NormalizedDisplacement
- Participation
- CloseOrAcceptance
- FollowThroughOrRetest
  – ResidualBurden. (C.66)

# Observed object

Candidate admission from one structural region into another.

# Declared parameters

- boundary;
- zone width;
- displacement threshold;
- close rule;
- relative-volume rule;
- breadth rule;
- retest rule;
- horizon.

# Regime dependence

Continuation is more likely when $\chi$ becomes positive.

False breaks may increase in critical or unstable $\chi \approx 0$ conditions.

Crowding can strengthen the first move while increasing later fragility.

# Gate requirement

The breakout is itself a gate construct.

The gate must be specified before outcome review.

# Residual

- higher-frame resistance;
- weak breadth;
- low liquidity;

- no retest;
- event risk;
- crowded positioning;
- derivative mismatch.

## Transport tests

- timeframe;
- volatility normalization;
- log scale;
- volume and breadth frames;
- alternate bar construction;
- cash versus derivative markets.

## Invalidation

A bullish breakout is invalidated by durable return inside the prior region, failed reclaim, and opposing acceptance.

## Common category error

Any print beyond a line = breakout.

## Research status

Central Event-level method.

Quality depends primarily on gate definition and residual handling.

The source Technical Analysis framework defines a strong breakout through boundary, price, volume, breadth, close, and cross-frame confirmation rather than touch alone.

---

# C.16 Relative Strength and Breadth

## Method

Asset-relative performance and component-field participation.

## Primary period

**Structure**

## Primary family

**Motion / Relation**

## Secondary roles

- field coherence;
- transport between asset and universe frames.

## Source lineage

Asset prices, benchmark prices, and component prices.

$S_{RS} = \{P_{asset}, P_{benchmark}\}$. (C.67)

$S_{Breadth} = \{P_i \mid i \in Universe\}$. (C.68)

## Operator word

$w_{RS} = Ratio \circ \Pi_{\{asset,benchmark\}}$. (C.69)

$w_{Breadth} = Aggregate \circ Classify \circ \Pi_{components}$. (C.70)

## Core formulas

$RS_{A/B}(t) = P_A(t)/P_B(t)$. (C.71)

$Breadth_t = (1/N)\Sigma_i a_{i,t}$. (C.72)

## Observed object

Relative motion of one asset against a frame and participation coherence across a component field.

## Declared parameters

- benchmark;
- universe;
- weighting;
- inclusion rules;
- timeframe;
- component classification.

## Regime dependence

Narrow breadth may coexist with rising headline prices for an extended period.

Broad participation may support self-confirming continuation.

## Gate requirement

- headline price break;
- breadth thrust;
- component confirmation;
- structural failure;
- cross-frame acceptance.

## Residual

- benchmark choice;
- universe concentration;
- constituent changes;
- survivorship;
- sector-specific catalysts;
- cap weighting.

## Transport tests

- equal-weight versus capitalization-weighted;
- sectors;
- alternative benchmarks;
- timeframe;
- international versus local universe.

## Invalidation

A breadth-divergence reversal claim fails if breadth later re-expands and no price gate fails.

## Common category error

Breadth warning = immediate headline reversal.

## Research status

Strong independent relation channel when universe construction is sound.

The source article treats breadth as cross-component coherence and an early warning rather than a final gate.

---

# C.17 Chart Pattern

# Method

Triangle, wedge, head-and-shoulders, base, flag, range, or related geometric episode pattern.

# Primary period

**Episode**

# Primary family

**Constraint / Boundary**

# Secondary roles

- Load through accumulated event history;
- Motion through compression or expansion;
- Commitment through breakout or failure.

# Source lineage

Pivots, price, volume, and sometimes volatility.

S_Pattern = {PivotSeries,Price,Volume,Volatility}. (C.73)

# Operator word

w_Pattern
= BoundaryFit ∘ Segment ∘ PivotDetect ∘ Π_price. (C.74)

# Observed object

Changing episode geometry and the narrowing, widening, or reorganization of admissible movement.

# Declared parameters

- pivot rule;
- minimum touches;
- tolerance;
- scale;
- timeframe;
- volume condition;

- breakout rule;
- invalidation.

## Regime dependence

Compression patterns often correspond to $\chi \approx 0$ before transition.

Continuation patterns require the prior regime to reassert itself.

Reversal patterns require failure of the prior episode grammar.

## Gate requirement

- breakout;
- close;
- retest;
- volume;
- new-state persistence.

## Residual

- subjective pivots;
- alternate boundary;
- false break;
- direction uncertainty;
- scale dependence;
- higher-world event risk.

## Transport tests

- log scale;
- pivot perturbation;
- timeframe;
- volatility-normalized boundaries;
- independent pattern coder.

## Invalidation

The pattern is invalidated when its declared geometry fails without producing the expected gate or when the opposite boundary gains acceptance.

## Common category error

Recognizable shape = completed prediction.

## Research status

Useful as episode compression.

High subjectivity unless pivot and boundary rules are formalized.

---

# C.18 Elliott Wave

## Method

Recursive impulse–correction episode segmentation.

## Primary period

**Episode**

## Primary family

**Motion / Relation**

## Secondary roles

- Commitment through wave endpoints;
- Constraint through alternate-count rules;
- residual through branch ambiguity.

## Source lineage

Pivoted price, momentum, breadth, volume, and sometimes ratio relations.

$S_{EW} = \{PivotSeries, Price, Momentum, Breadth, Volume\}$. (C.75)

## Operator word

$w_{EW}$
$= RecursiveSegment$
$\circ\ Classify_{\chi}$
$\circ\ PivotDetect$
$\circ\ \Pi_{market}$. (C.76)

## Observed object

Nested alternation between self-confirming displacement and corrective digestion.

## Declared parameters

- pivot rule;
- wave-degree rule;
- alternation rule;
- endpoint rule;
- invalidation;
- alternate-branch policy;
- timeframe.

## Regime dependence

Impulse-like segments may correspond to sustained $\chi > 0$.

Corrective segments may correspond to $\chi < 0$ or residual metabolism.

Terminal phases may show divergence without immediate reversal.

## Gate requirement

A wave endpoint requires:

PivotExtreme

- RelationalShift
- GateEvidence
- DensityContext
- ResidualAudit
- CrossFrameSurvival. (C.77)

## Residual

- alternate count;
- pivot sensitivity;
- degree ambiguity;
- retrospective relabeling;
- ratio overfit;
- timeframe conflict.

## Transport tests

- pivot algorithm;
- timeframe;
- volatility-normalized segmentation;
- alternate analyst;
- phase and real-pair models.

## Invalidation

The count is invalidated when a declared structural rule fails.

The original count must remain recorded.

## Common category error

A convenient retrospective count = objective market history.

## Research status

Potentially useful as recursive episode grammar.

Weak when alternate branches and invalidations are not preserved.

The source framework insists that wave counts should use declared pivots, ledgered endpoints, cross-method confirmation, and explicit residual branches.

---

# C.19 Fibonacci Retracement

## Method

Ratio-based candidate support or resistance zone.

## Primary period

**Structure–Episode**

## Primary family

**Constraint / Boundary**

## Secondary roles

- observer convention;
- shared-attention effect;
- confluence tool.

## Source lineage

Selected anchor prices.

$S_{Fib} = \{P_A, P_B\}$. (C.78)

## Operator word

$w_{Fib} = \mathrm{RatioMap} \circ \mathrm{AnchorSelect} \circ \Pi_{price}$. (C.79)

## Core formula

$F_r = P_B - r(P_B - P_A)$. (C.80)

## Observed object

Candidate ratio-based zone between declared anchors.

## Declared parameters

- anchors;
- ratio family;
- zone width;
- scale;
- timeframe;
- confluence rule;
- gate;
- invalidation.

## Regime dependence

A ratio zone may work through:

- shared attention;
- overlap with density;
- overlap with prior boundary;
- episode correction structure.

The ratio alone does not establish causal force.

## Gate requirement

- rejection;

- close;
- reclaim;
- support hold;
- volume response;
- confluence with independent structure.

## Residual

- arbitrary anchors;
- multiple available ratios;
- absent density;
- pattern overfit;
- higher-period contradiction.

## Transport tests

- nearby admissible anchors;
- log scale;
- volatility-normalized zone;
- different timeframe;
- comparison with non-Fibonacci ratios.

## Invalidation

A support claim is invalidated by close below the declared zone and failure to reclaim within the stated horizon.

## Common category error

Ratio appearance = universal natural law of market motion.

## Research status

Useful as a candidate attention or confluence map.

Requires stronger null comparisons.

The source Technical Analysis article describes Fibonacci levels as candidate ratio attractors that remain vulnerable to arbitrary anchor selection.

---

# C.20 Gann Geometry

# Method

Price–time angles, squares, cycles, or proportional constructions.

# Primary period

**Episode**

# Primary family

**Constraint / Boundary**

# Secondary roles

- cadence hypothesis;
- transport and invariance test;
- possible phase-time comparison.

# Source lineage

Price, time, anchor, scale, and calendar convention.

S_Gann = {P,t,Anchor,Scale,Calendar}. (C.81)

# Operator word

w_Gann
= GeometricRelation
∘ Normalize
∘ AnchorSelect
∘ Π_{price,time}. (C.82)

# Observed object

Candidate price–time relation or cadence.

# Declared parameters

- anchor;
- price unit;
- time unit;
- chart scale;
- calendar;
- volatility normalization;

- gate;
- invalidation.

## Regime dependence

A relation may be meaningful only within one episode or cadence regime.

Institutional cycles such as reporting, expiry, margin, or policy review may provide real timing structure.

## Gate requirement

- predeclared reaction;
- close;
- reversal;
- acceptance through line;
- event-cadence alignment.

## Residual

- arbitrary scale;
- visual-angle dependence;
- anchor sensitivity;
- multiple-testing;
- calendar mismatch;
- volatility change.

## Transport tests

Gann carries an unusually heavy burden:

- arithmetic versus logarithmic price;
- alternate chart aspect ratio;
- normalized coordinates;
- alternate admissible anchors;
- trading-day versus calendar-day time;
- volatility scaling.

## Invalidation

A reversal claim is invalidated by close through the declared relation with participation and later acceptance.

## Common category error

Visual chart angle = frame-invariant market law.

## Research status

Best treated as a candidate invariance and cadence programme rather than accepted geometric law.

The source framework classifies Gann as a price–time invariant search whose scale dependence and overfitting must be audited.

---

# C.21 CAPM Complex Valuation State

## Method

Complex completion of baseline and risk-adjusted discounted value.

## Primary period

**World**

## Primary family

**Motion / Relation**

## Secondary roles

- Load through amplitude A;
- Commitment through recognition gate;
- ledger through realized and recognized consequence.

## Source lineage

Cash flow, base discount rate, beta, equity risk premium, and horizon.

$S_{CAPM\text{-}Z} = \{CF, r_{base}, \beta, ERP, t\}$. (C.83)

## Operator word

$w_{CAPM\text{-}Z}$
= ComplexComplete
∘ CompareDiscountProtocols
∘ Discount
∘ DeclareCashFlow. (C.84)

# Core formulas

$r_{CAPM} = r_{base} + \beta ERP$. (C.85)

$A_t = CF_t/(1 + r_{base})^t$. (C.86)

$R_t = CF_t/(1 + r_{CAPM})^t$. (C.87)

$Q_t = \sqrt{(A_t^2 - R_t^2)}$. (C.88)

$Z_t = R_t + iQ_t$. (C.89)

$\partial R/\partial \theta = -Q$. (C.90)

# Observed object

A protocol-bound valuation state in which R is admitted value and Q is its conjugate phase exposure.

# Declared parameters

- cash flow;
- horizon;
- base discount protocol;
- CAPM protocol;
- beta;
- ERP;
- sign convention;
- recognition gate.

# Regime dependence

The static construction does not itself diagnose dynamic market regime.

Dynamic use requires actual movement in $\theta$, updated inputs, and a recognition protocol.

# Gate requirement

Economic movement must be distinguished from ledger recognition:

Measurement
→ Exposure
→ StateMovement
→ EconomicP&L
→ RecognitionGate
→ Ledger + Residual. (C.91)

## Residual

- model misspecification;
- cash-flow revision;
- liquidity;
- legal risk;
- accounting treatment;
- tail risk;
- incomplete recognition.

## Transport tests

- alternative baseline;
- alternative factor model;
- horizon;
- cash-flow scenarios;
- real-pair benchmark;
- dynamic versus static use.

## Invalidation

The complex representation should be reduced if it adds no explanatory, predictive, diagnostic, or intervention value beyond A and R or an ordinary real pair.

## Common category error

−Q = realized or recognized loss. (C.92)

The correct interpretation is:

−Q = quarter-turn readout or phase-exposure coefficient under the declared geometry. (C.93)

## Research status

Mature as an internally coherent calibration construction.

Its practical superiority remains an empirical question.

The CAPM source makes the exposure–movement–gate distinction explicit.

---

# C.22 Candidate Technical Analysis Complex State

# Method

Proposed market state:

Z_TA = R_TA + iQ_TA. (C.94)

# Primary period

Usually **Episode** or **World**

# Primary family

**Motion / Relation**

# Secondary roles

Potential gate-phase and internal-time structure.

# Source lineage

Must be declared case by case.

# Operator word

w_ZTA
= ComplexComplete
∘ Normalize
∘ Couple
∘ (Π_R,Π_Q). (C.95)

# Observed object

Candidate conjugate market state.

# Required declaration

- independent R meaning;
- independent Q meaning;
- units;
- normalization;
- amplitude;
- phase;
- coupling law;
- residual;

- real-pair benchmark.

## Regime dependence

The ordinary complex form is most naturally associated with corrective or rotational regimes.

Self-confirming regimes may require a hyperbolic normal form rather than $i^2 = -1$.

## Gate requirement

The phase should improve:

- event alignment;
- gate hazard;
- episode compression;
- intervention timing;
- or cross-frame transport.

## Residual

Everything not absorbed by the declared R–Q closure must remain $\varepsilon$ or $\mathscr{R}$.

Q must not become the error bucket.

## Transport tests

- scaling perturbation;
- timeframe;
- alternate Q proxy;
- real-pair comparison;
- episode family;
- out-of-sample gate prediction.

## Invalidation

The complex model should be reduced if:

- phase order is unstable;
- Q lacks independent meaning;
- real-pair performance is equal;
- phase does not improve gates or episode alignment;
- scaling changes reverse conclusions.

## Common category error

Adding i creates hidden financial information.

## Research status

Research hypothesis.

Not yet a general mature Technical Analysis instrument.

The phase framework requires complex structures to earn priority over two-real-variable descriptions through operational gain and stable phase meaning.

---

# C.23 Cross-Method Comparison Matrix

| Method | Primary period | Primary family | Gate strength inside method | Principal residual |
|---|---|---|---|---|
| Moving average | Structure | Load | weak | lag and omitted market channels |
| MA crossover | Structure | Motion | weak–moderate | whipsaw and shared-input dependence |
| MACD | Structure | Motion | weak | volume, breadth, and gate absence |
| RSI | Structure | Motion | weak | wrong $\chi$ regime |
| Stochastic | Structure | Motion | weak | persistent trend |
| ATR | Window–Structure | Motion magnitude | none | direction and cause |
| Bollinger Bands | Structure | Constraint | weak | regime dependence |
| Volume | Window–Event | Load | conditional | motive and position identity |
| OBV | Structure | Load | weak | inferred sign |
| VWAP | Structure | Load | moderate when adopted | anchor and motive |
| Volume profile | Structure | Load/Constraint | weak | stale density |
| Candlestick | Window | Commitment | window-only | lost path |
| Support/ resistance | Structure | Constraint | none until tested | arbitrary or stale boundary |
| Breakout | Event | Commitment | strong if declared | fakeout and higher-frame conflict |
| Breadth | Structure | Motion | weak | universe and weighting |
| Chart pattern | Episode | Constraint | none until break | pivot subjectivity |
| Elliott Wave | Episode | Motion | endpoint-dependent | branch ambiguity |
| Fibonacci | Structure–Episode | Constraint | none until reaction | anchor arbitrariness |
| Gann | Episode | Constraint/ Transport | none until reaction | scale and anchor fragility |
| CAPM complex state | World | Motion/Relation | recognition external | model and recognition residual |

| Method | Primary period | Primary family | Gate strength inside method | Principal residual |
|---|---|---|---|---|
| TA complex state | Episode–World | Motion/Relation | phase-dependent | eligibility failure |

# C.24 Confirmation Bundles

# C.24.1 Weak confirmation bundle

```
Price above MA
MACD positive
RSI above 50
Momentum positive
```

These methods are mainly price-derived Motion and Memory transformations.

The bundle says:

Recent price structure is directionally positive under several related transformations. (C.96)

It does not yet establish:

- independent loading;
- meaningful boundary;
- commitment;
- low residual.

---

# C.24.2 Stronger event bundle

```
Predeclared resistance
ATR-normalized displacement
Relative volume
Breadth expansion
Daily close
Retest hold
Residual audit
```

The bundle covers:

- Load;
- Motion;
- Constraint;
- Commitment;
- residual.

Its strength comes from functional diversity, not indicator count.

---

## C.24.3 Stronger episode bundle

```
Event ledger
χ sequence
Stable pivot protocol
Phase or cadence estimate
Higher-frame transport
Episode-completion gate
Alternative branch residual
```

This bundle supports an Episode-level claim.

A single oscillator cannot substitute for it.

---

## C.24.4 Stronger world bundle

```
Institutional loading
Funding and collateral constraints
Market-price state
Accounting or legal recognition
Policy response
Cross-ledger residual
Observer backreaction
```

This bundle supports a World-level claim.

Chart data alone are insufficient.

---

## C.25 Method-Selection Rule

A method should be selected according to the missing function in the current analysis.

If Load is missing:

use volume, profile, positioning, breadth, or institutional exposure.

If Motion is missing:

use return, momentum, relative strength, divergence, χ, or phase.

If Constraint is missing:

declare support, resistance, range, volatility boundary, legal threshold, or collateral rule.

If Commitment is missing:

declare close, retest, settlement, recognition, or episode-completion gate.

If Residual is missing:

record contradiction, branch uncertainty, hidden positioning, frame conflict, and invalidation.

If Transport is missing:

test timeframe, scale, anchor, normalization, universe, or observer frame.

The governing selection principle is:

Choose the method that supplies missing information, not the method that repeats the strongest existing signal. (C.97)

---

# C.26 Molecular Card Conclusion

The molecular atlas shows that familiar Technical Analysis methods are not competing answers to one question.

They occupy different locations in a larger observation grammar.

A moving average remembers.

RSI compares directional displacement.

ATR measures agitation.

A band constructs a boundary.

Volume records exchange activity.

VWAP compiles a transaction centre.

A candle closes a window.

Support and resistance localize a transition.

A breakout admits or rejects that transition.

Breadth checks field coherence.

Elliott Wave segments episodes.

Gann searches for transported price–time structure.

CAPM complex geometry calibrates what a genuine conjugate financial state must look like.

The resulting rule is:

MethodMeaning
= Period

- FunctionalRole
- Protocol
- Regime
- Gate
- Residual
- Transport. (C.98)

The next appendix can formalize the **Residual Ledger and Admissible Revision Schema**, including copy-ready tables for recording signals, failures, relabeling, fakeouts, and cross-frame audits.

# Appendix D — Residual Ledger and Admissible Revision Schema

## D.1 Purpose

A Technical Analysis framework becomes auditable only when it records more than signals and outcomes.

An ordinary backtest may store:

- signal time;
- direction;
- entry;
- exit;
- profit or loss.

That is insufficient for the Periodic Grammar.

A mature record must also preserve:

- the original protocol;
- the claimed market characteristic;
- the closure period;
- the functional family;
- the gate required;
- evidence missing at the time;
- residual contradiction;
- invalidation;
- later relabeling;
- cross-frame survival;
- the reason for revision.

The minimum research record is therefore:

TestRecord
= Protocol

- OriginalClaim
- Projection
- Gate
- Residual

- Invalidation
- Outcome
- Revision. (D.1)

The source Technical Analysis framework already proposes protocol, signal, gate, residual, invalidation, and outcome fields, and expressly requires the original claim to remain preserved when later relabeling occurs.

The present appendix extends that schema into a full residual-bearing and revision-governed ledger.

---

# D.2 Why Outcome-Only Databases Are Inadequate

## D.2.1 Successful-pattern memory

A database that stores only successful patterns cannot distinguish:

- a genuinely robust method;
- a heavily relabelled method;
- a method with many erased failures;
- a method whose success depends on one frame;
- a method that works only under one $\chi$ regime.

Suppose ten apparent breakouts occur.

Three continue.

Four fail.

Three remain ambiguous.

If only the three successful cases are retained, the historical ledger creates a false impression:

ObservedSuccessRate = 100%. (D.2)

The true declared-protocol result was:

ObservedSuccessRate = 3/10. (D.3)

More importantly, the seven non-successes may contain information about:

- weak gates;
- poor boundaries;
- crowding;
- insufficient breadth;
- wrong timeframe;

- regime transition;
- residual accumulation.

Erasing failure prevents diagnosis.

The source Technical Analysis article states:

LearningQuality ↑ when FailedTrace is preserved. (D.4)

---

## D.2.2 Hindsight compression

After an outcome is known, analysts tend to compress the past into one coherent story.

Examples include:

- “The breakout was obviously weak.”
- “That was not a real Wave 3.”
- “The correct Fibonacci anchor was the later low.”
- “The signal was really continuation, not reversal.”
- “The weekly chart always overruled the daily chart.”

These statements may sometimes be reasonable revisions.

But without the original record, one cannot tell whether the interpretation was genuinely available at the time.

Therefore:

FinalNarrative ≠ OriginalInformationSet. (D.5)

A residual ledger must preserve both.

---

## D.2.3 Protocol drift

A claim may appear successful because its protocol changes after the signal.

Examples:

- the timeframe changes;
- the support line moves;
- the volume threshold changes;
- the required close changes;
- the outcome horizon changes;
- the wave degree changes;
- the benchmark changes.

This is protocol drift:

$P_k \rightarrow P'_k$ after outcome observation. (D.6)

Protocol drift is not always illegitimate.

A model may require revision.

But the revision must be declared as a new protocol rather than silently substituted for the old one.

---

# D.3 Log, Trace, Ledger, and Residual

## D.3.1 Log

A log stores what happened.

$Log_k = StoredRecord(Event_k)$. (D.7)

Examples include:

- a candle;
- indicator value;
- trade print;
- analyst note;
- saved chart.

A log may remain passive.

---

## D.3.2 Trace

A trace is a stored record that changes later interpretation or action.

$Trace_k$
$= Record_k$

- $FutureConsequence_k$. (D.8)

The declaration source states:

Log = stored record.
Trace = stored record that bends future projection.

Examples include:

- a failed breakout that alters later trust in the level;
- a margin breach that changes future limits;
- a wave invalidation that changes later pivot selection;

- a prior crisis that changes liquidity behaviour.

---

# D.3.3 Ledger

A ledger is the ordered structure carrying consequential traces.

L_{k+1} = UpdateLedger(L_k,e_k,r_k,m_k). (D.9)

where:

e_k = admitted event;
r_k = attached residual;
m_k = gate metadata.

A mature ledger should preserve:

- what was admitted;
- who or what admitted it;
- when it was admitted;
- under which protocol;
- what remained unresolved.

---

# D.3.4 Residual

Residual is what remains after projection and gate.

The declaration source identifies residual as potentially containing:

- unobserved structure;
- missing evidence;
- unresolved contradiction;
- unselected alternatives;
- model limitation;
- boundary leakage;
- observer disagreement;
- ambiguity;
- future option value;
- unpaid cost;
- residual risk.

For Technical Analysis:

Residual_k
= UnresolvedEvidence_k

- FailedConfirmation_k
- Contradiction_k
- Ambiguity_k. (D.10)

This follows the source Technical Analysis schema.

Residual is not necessarily error.

It may become later structure:

Residual_today → Structure_tomorrow. (D.11)

---

# D.4 The Full Technical-Analysis Disclosure Cycle

The complete record cycle is:

D_k
→ Σ_k
→ V_k
→ G_k
→ e_k + r_k
→ L_{k+1}
→ Outcome_k
→ Revision_k. (D.12)

where:

D_k = declared protocol and interpretation;
Σ_k = accessible market field;
V_k = projected technical structure;
G_k = gate;
e_k = admitted event;
r_k = residual;
L_{k+1} = updated ledger.

The next declaration is:

D_{k+1} = U_a(D_k,L_{k+1},$\mathscr{R}$_{k+1}). (D.13)

The source declaration framework uses essentially this recursive unit:

Declaration
→ Projection
→ Gate
→ Trace update
→ Residual update
→ Declaration revision.

The present appendix applies that runtime specifically to Technical Analysis.

---

# D.5 The Seven Linked Ledgers

A practical implementation should separate seven ledgers rather than compress everything into one table.

## D.5.1 Protocol ledger

Records what world was declared.

## D.5.2 Projection ledger

Records the indicators, structures, and operator outputs.

## D.5.3 Claim ledger

Records what the analyst asserted.

## D.5.4 Gate ledger

Records whether the claim passed, failed, or remained pending.

## D.5.5 Residual ledger

Records unresolved evidence and contradictions.

## D.5.6 Outcome ledger

Records later market behaviour under the original horizon.

## D.5.7 Revision ledger

Records how and why the declaration changed.

The linked structure is:

ProtocolID
→ ProjectionID
→ ClaimID
→ GateID
→ ResidualID
→ OutcomeID
→ RevisionID. (D.14)

Keeping the ledgers separate permits:

- one claim to have several residuals;
- one signal to face several gates;
- one protocol to generate several claims;
- one revision to affect multiple later claims.

# D.6 Protocol Ledger

# D.6.1 Required fields

| Field | Meaning |
|---|---|
| `protocol_id` | unique immutable identifier |
| `protocol_version` | version number |
| `asset_id` | instrument or market object |
| `universe_id` | comparison or component universe |
| `boundary_rule` | what is included |
| `timeframe_rule` | daily, weekly, intraday, event-defined |
| `price_scale` | linear, logarithmic, normalized |
| `bar_rule` | time, volume, tick, range, event |
| `feature_map` | observed indicators and structures |
| `gate_rule` | confirmation rule |
| `residual_rule` | residual classification rule |
| `invalidation_rule` | claim-failure rule |
| `outcome_horizon` | evaluation horizon |
| `observer_role` | trader, analyst, system, committee, regulator |
| `created_time` | protocol declaration time |
| `retrospective_flag` | whether declared after the event |

The source Technical Analysis schema similarly requires boundary, timeframe, scale, bar construction, feature map, gate, residual, and invalidation fields.

---

# D.6.2 Protocol immutability

After a signal is issued, its protocol record should not be overwritten.

Let:

$P_k^0$ = original protocol. (D.15)

A later revision creates:

$P_k^1$ = revised protocol. (D.16)

The database should preserve:

$P_k^0 \neq$ overwritten by $P_k^1$. (D.17)

Instead:

$P_k^1.\text{parent_protocol_id} = P_k^0.\text{protocol_id}$. (D.18)

This preserves the genealogy of the analysis.

---

# D.7 Projection Ledger

## D.7.1 Purpose

The projection ledger records what was actually visible under the declared feature map.

It should not contain conclusions.

It contains measured or derived states.

## D.7.2 Representative fields

| Field | Meaning |
|---|---|
| `projection_id` | unique projection record |
| `protocol_id` | governing protocol |
| `timestamp` | observation time |
| `period_class` | Mark, Window, Structure, Event, Episode, World |
| `primary_family` | Load, Motion, Constraint, Commitment |
| `source_lineage` | raw input sources |
| `operator_word` | transformations applied |
| `price_state` | price-related output |
| `volume_state` | volume-related output |
| `breadth_state` | breadth output |
| `boundary_state` | declared levels or zones |
| `phase_state` | θ or phase proxy, where eligible |
| `xi_state` | ρ, γ, ν, where used |
| `data_quality` | completeness and reliability |
| `missing_variables` | known unobserved channels |

The distinction is:

Projection = what the method produced. (D.19)

Claim = what the analyst inferred. (D.20)

This separation is critical.

---

# D.8 Claim Ledger

## D.8.1 Claim fields

| Field | Meaning |
|---|---|
| `claim_id` | unique immutable claim |
| `projection_id` | supporting projection |
| `signal_type` | breakout, divergence, reversal, wave endpoint, etc. |
| `direction` | bullish, bearish, neutral, non-directional |
| `claim_text` | exact proposition |
| `claimed_characteristic` | memory, density, phase, gate, etc. |
| `regime_assumption` | $\chi < 0$, $\chi \approx 0$, $\chi > 0$, unknown |
| `confidence_pre_gate` | confidence before gate |

| Field | Meaning |
|---|---|
| `cross_checks_required` | evidence still required |
| `original_claim_preserved` | true by default |
| `claim_time` | timestamp |
| `claim_authority` | private, model, committee, official |

A claim should contain one testable proposition.

Weak claim:

“The chart looks bullish.” (D.21)

Stronger claim:

“Under protocol P, price has crossed the twelve-week resistance zone, but Event-level breakout admission remains pending daily-close and breadth confirmation.” (D.22)

---

# D.8.2 Claim formula

The source schema proposes:

Signal
= Method

- Direction
- Characteristic
- MissingVariableAudit. (D.23)

The present article extends this:

Claim
= Projection

- Interpretation
- Period
- FunctionalRole
- RegimeAssumption
- RequiredGate
- MissingVariableAudit. (D.24)

---

# D.9 Gate Ledger

# D.9.1 Gate output

A gate should not return only true or false.

The general output is:

$G_P(X,L,\mathscr{R}) \rightarrow (d,\alpha,e,r,m)$. (D.25)

where:

d = categorical decision;
α = admission fraction or strength;
e = admitted event;
r = attached residual;
m = gate metadata.

The decision may be:

$d \in \{Admit,PartiallyAdmit,Defer,Reject,Ambiguous\}$. (D.26)

The CAPM gate framework likewise permits Admit, Partially Admit, Defer, and Reject, with an admission fraction $0 \leq \alpha \leq 1$.

---

# D.9.2 Gate fields

| Field | Meaning |
|---|---|
| `gate_id` | unique gate record |
| `claim_id` | tested claim |
| `gate_time` | time of test |
| `gate_authority` | close, exchange, model, committee, law |
| `close_confirmation` | true, false, not applicable |
| `volume_confirmation` | true, false, pending |
| `vwap_confirmation` | true, false, pending |
| `breadth_confirmation` | true, false, pending |
| `retest_confirmation` | true, false, pending |
| `followthrough_confirmation` | true, false, pending |
| `gate_strength_score` | calibrated score |
| `admission_fraction` | 0 to 1 |
| `gate_status` | accepted, partial, failed, pending, ambiguous |
| `gate_rule_version` | exact rule used |
| `gate_notes` | context |

The source Technical Analysis schema uses these same confirmation categories and defines GateStrength from close quality, volume expansion, VWAP acceptance, follow-through, retest success, and breadth confirmation.

---

# D.9.3 Gate strength

A conceptual score is:

$S_G$
$= w_{CC}$

- $w_{VV}$

- w_WW
- w_FF
- w_TT
- w_BB
  – w_RR. (D.27)

where:

C = close quality;
V = volume expansion;
W = VWAP or value acceptance;
F = follow-through;
T = retest;
B = breadth;
R = residual conflict.

The score must be calibrated prospectively.

A gate score assembled after seeing the outcome is not a valid gate test.

---

# D.10 Residual Ledger

# D.10.1 Core fields

| Field | Meaning |
|---|---|
| `residual_id` | unique residual record |
| `claim_id` | related claim |
| `gate_id` | related gate, where applicable |
| `residual_type` | classification |
| `residual_source` | price, volume, breadth, frame, institution, model |
| `residual_severity` | low, medium, high, critical |
| `residual_description` | plain-language statement |
| `residual_open_time` | when identified |
| `residual_expected_horizon` | when it may resolve |
| `residual_resolved` | true or false |
| `residual_resolution_time` | when resolved |
| `residual_resolution_type` | confirmed, dissipated, invalidated, superseded |
| `residual_outcome` | effect on original claim |
| `carry_forward` | whether attached to later episodes |
| `parent_residual_id` | prior residual genealogy |

These fields closely extend the source Technical Analysis residual schema.

---

# D.10.2 Residual type ontology

A practical ontology may include the following major classes.

### R1 — Missing confirmation

Examples:

- weak volume;
- absent breadth;
- no retest;
- no follow-through.

### R2 — Direct contradiction

Examples:

- price breakout with breadth deterioration;
- bullish structure with worsening relative strength;
- Wave 3 claim with weak momentum.

### R3 — Frame conflict

Examples:

- daily breakout beneath weekly resistance;
- linear-scale support absent on log scale;
- cap-weighted strength with equal-weight weakness.

### R4 — Boundary uncertainty

Examples:

- arbitrary anchor;
- wide zone;
- multiple competing levels;
- poor pivot definition.

### R5 — Regime uncertainty

Examples:

- unknown $\chi$;
- transition between corrective and self-confirming behaviour;
- unstable volatility state.

### R6 — Data and measurement residual

Examples:

- incomplete volume;
- asynchronous data;
- survivorship;
- vendor disagreement;
- corporate-action distortion.

## R7 — Positioning and liquidity residual

Examples:

- hidden leverage;
- trapped positions;
- thin depth;
- derivative mismatch;
- funding stress.

## R8 — Institutional residual

Examples:

- pending legal action;
- accounting non-recognition;
- regulatory uncertainty;
- policy decision.

## R9 — Branch residual

Examples:

- alternate wave count;
- alternative pattern;
- competing regime model;
- unresolved scenario.

## R10 — Model residual

Examples:

- real-pair model and complex model disagree;
- phase scaling instability;
- unexplained forecast error;

- parameter instability.

A residual record may carry more than one type.

---

# D.10.3 Residual severity

A four-level severity scale may be used.

| Level | Meaning |
|---|---|
| Low | unlikely to change the claim materially |
| Medium | may affect timing or confidence |
| High | may invalidate the claim or gate |
| Critical | the protocol or interpretation may be unusable |

A numerical mapping can be:

Severity ∈ {1,2,3,4}. (D.28)

But ordinal values should not be added mechanically without a declared weighting scheme.

---

# D.10.4 Residual state

Each residual may have a lifecycle state:

Open
→ Monitoring
→ Resolved
→ Dissipated
→ Invalidated
→ Superseded. (D.29)

### Resolved

Later evidence directly settles the issue.

### Dissipated

The residual loses operative relevance without direct settlement.

### Invalidated

The residual defeats the original claim.

### Superseded

A new declaration absorbs or replaces the original issue while preserving trace.

---

# D.11 Residual Accumulation and Dissipation

# D.11.1 Accumulation equation

Let $\mathscr{R}_k$ be the accumulated residual register.

A general update is:

$\mathscr{R}_{k+1} = D_{\mathscr{R}}(\mathscr{R}_k) + r_k - q_k.$ (D.30)

where:

$D_{\mathscr{R}}$ = persistence or decay operator;
$r_k$ = new residual;
$q_k$ = resolved or dissipated residual.

The CAPM gate framework similarly proposes:

$E_{k+1} = E_k + \varepsilon_k - Dissipation_k - Resolution_k.$ (D.31)

---

# D.11.2 Residual is not permanently cumulative

Not every contradiction becomes future crisis.

Residual may dissipate through:

- time;
- new liquidity;
- hedging;
- position closure;
- clarification;
- new information;
- successful retest;
- changed institutional relevance.

Thus:

$ResidualAccumulation \neq InevitableFailure.$ (D.32)

A mature ledger records both accumulation and release.

---

# D.11.3 Residual debt

Define residual debt:

$RD_k = \Sigma_{j \leq k} \omega_{k-j} s_j r_j.$ (D.33)

where:

$\omega_{k-j}$ = persistence weight;

s_j = severity weight.

RD_k is not automatically a directional signal.

It is an audit measure of unresolved burden carried by the current interpretation.

---

# D.11.4 Residual alarm

A simple alarm may be:

$\|\mathscr{R}_{k+1}\| - \|\mathscr{R}_k\| > \varepsilon_{\mathscr{R}}$ for N consecutive updates. (D.34)

A concealment alarm may be:

DeclaredResidual_{k+1} < ObservedContradiction_{k+1} − ε_hidden. (D.35)

The declaration source similarly treats residual growth or hidden residual as evidence that the declaration requires repair.

---

# D.12 False Closure

# D.12.1 Definition

False closure occurs when a gate entry is treated as proof that all relevant uncertainty has disappeared.

In financial recognition:

ε_gate = ΔR_econ − ΔR_admitted. (D.36)

The CAPM gate framework warns that a ledger entry is evidence of commitment, not proof of total resolution.

For Technical Analysis:

ε_TA
= TotalContradiction
− ContradictionRecorded. (D.37)

False closure occurs when:

ε_TA is materially positive but omitted. (D.38)

---

# D.12.2 Examples

### Breakout false closure

Daily close above resistance is recorded as complete trend transition despite:

- weekly resistance;
- weak breadth;

- absent retest;
- low liquidity.

### Wave false closure

One local high is recorded as completed Wave 5 despite:

- no structural reversal;
- no breadth confirmation;
- alternate count;
- continued positive $\chi$.

### Fibonacci false closure

Reaction near 61.8% is recorded as proof of ratio law despite:

- many nearby levels;
- arbitrary anchors;
- no prospective declaration.

### Complex-state false closure

Q is treated as total hidden risk despite large unexplained residual.

---

# D.13 Fakeout Ledger

# D.13.1 Fakeout definition

A fakeout is not simply a breakout that later loses money.

It is:

Fakeout
= CandidateTransition

- ParticipantCommitment
  – DurableAcceptance. (D.39)

A fakeout record should identify:

- what boundary was crossed;
- which gate initially admitted the event;
- who or what likely committed;
- when acceptance failed;
- what residual the failure created.

## D.13.2 Fakeout fields

| Field | Meaning |
|---|---|
| `fakeout_id` | unique record |
| `original_event_id` | breakout or transition event |
| `boundary_id` | tested boundary |
| `initial_gate_status` | admitted, partial, ambiguous |
| `initial_gate_strength` | score |
| `failure_time` | when acceptance failed |
| `return_inside_old_region` | true or false |
| `failed_reclaim` | true or false |
| `opposing_acceptance` | true or false |
| `trapped_position_proxy` | estimated trapped exposure |
| `residual_created` | linked residual IDs |
| `later_event` | continuation, reversal, unresolved |

---

## D.13.3 Fakeout classification

### Type A — weak-gate fakeout

The original gate lacked sufficient evidence.

### Type B — crowding fakeout

Strong visible confirmation attracted excessive same-direction positioning.

### Type C — higher-frame rejection

Local admission failed at a higher-period boundary.

### Type D — catalyst reversal

An external event invalidated the accepted state.

### Type E — liquidity fakeout

Thin or discontinuous liquidity created apparent displacement.

These types may overlap.

---

# D.14 Divergence Ledger

## D.14.1 Divergence as residual warning

Divergence should not be entered directly as reversal.

It should be entered as:

RelationalWarning. (D.40)

A divergence record contains:

| Field | Meaning |
|---|---|
| `divergence_id` | unique record |
| `price_structure` | higher high, lower low, etc. |
| `comparison_channel` | RSI, MACD, breadth, volume |
| `divergence_direction` | bullish or bearish |
| `comparison_period` | anchors used |
| `gate_required` | reversal or continuation gate |
| `gate_status` | pending, accepted, rejected |
| `residual_status` | open or resolved |
| `outcome_horizon` | original horizon |
| `later_outcome` | reversal, continuation, no event |

The key rule is:

DivergenceRecord
= RelationalWeakening

- GatePending
- ResidualOpen. (D.41)

---

# D.15 Pivot and Wave Ledger

# D.15.1 Pivot record

Every candidate pivot should carry:

PivotRecord
= PivotClaim

- Evidence
- ResidualLabel
- InvalidationRule. (D.42)

This follows the source Technical Analysis framework.

Candidate labels include:

- confirmed;
- provisional;
- ambiguous;
- failed;
- absorbed;

- exhausted;
- untested;
- contradicted by volume;
- contradicted by breadth;
- contradicted by higher timeframe;
- awaiting close.

---

# D.15.2 Countable endpoint

A countable wave endpoint requires:

CountableWaveEndpoint
= Extreme

- Gate
- PhaseShift
- DensityContext
- ResidualAudit
- CrossFrameSurvival. (D.43)

This is the source rule for reducing arbitrary wave counting.

---

# D.15.3 Wave revision fields

| Field | Meaning |
|---|---|
| `wave_model_id` | model version |
| `parent_wave_model_id` | prior count |
| `pivot_protocol_id` | pivot rule |
| `original_count` | preserved count |
| `alternative_counts` | branches |
| `invalidation_rule` | exact failure condition |
| `invalidation_time` | if triggered |
| `new_count` | revised interpretation |
| `revision_reason` | evidence |
| `original_trace_preserved` | required true |
| `branch_residual` | unresolved alternatives |

---

# D.15.4 Relabeling rule

Relabeling is admissible only when:

1. the original count remains visible;

2. the invalidation rule is recorded;
3. new evidence is identified;
4. the new protocol is versioned;
5. the change is not merely outcome fitting.

Thus:

ValidWaveRevision
⇒ OriginalCountPreserved. (D.44)

The source Technical Analysis schema states:

ValidClaim requires OriginalTracePreserved.

---

# D.16 Fibonacci and Gann Revision Ledger

# D.16.1 Anchor record

For anchor-dependent methods, the ledger must preserve:

- original anchor;
- anchor-selection rule;
- time chosen;
- alternative admissible anchors;
- scale;
- derived levels;
- invalidation.

A new anchor should never overwrite the old anchor.

---

# D.16.2 Fibonacci record

| Field | Meaning |
|---|---|
| `fib_id` | unique model |
| `anchor_a` | first pivot |
| `anchor_b` | second pivot |
| `anchor_validity` | score |
| `ratio_set` | declared ratios |
| `zone_width` | tolerance |
| `confluence` | profile, VWAP, support, etc. |
| `gate_rule` | reaction required |
| `invalidation` | close through and failed reclaim |
| `reanchor_flag` | whether later reanchored |

| Field | Meaning |
|---|---|
| `reanchor_reason` | exact reason |

A Fibonacci support claim without predeclared anchors and invalidation is not auditable.

The source framework similarly requires anchor declaration and treats failure below the zone plus failed reclaim as an invalidation example.

---

# D.16.3 Gann record

| Field | Meaning |
|---|---|
| `gann_id` | unique model |
| `anchor` | declared pivot |
| `price_unit` | vertical normalization |
| `time_unit` | horizontal normalization |
| `calendar_rule` | trading or calendar time |
| `scale_rule` | linear or log |
| `volatility_normalization` | if used |
| `line_declared_time` | before or after move |
| `gate_rule` | required reaction |
| `invalidation` | accepted close through relation |
| `transport_results` | alternate-frame survival |

The source Gann audit requires anchor validity, scale survival, time cadence, density confluence, gate evidence, and residual.

---

# D.17 Cross-Frame Audit Ledger

# D.17.1 Purpose

A claim should not be labelled invariant merely because it looks similar on two charts.

The audit must record:

- source frame;
- destination frame;
- transport rule;
- expected transformed claim;
- observed destination claim;
- distance or discrepancy;
- residual created.

---

# D.17.2 Transport record

| Field | Meaning |
|---|---|
| transport_id | unique record |
| claim_id | source claim |
| source_protocol_id | P |
| target_protocol_id | P′ |
| transport_operator | mapping rule |
| expected_target_claim | T_{P→P′}(Claim_P) |
| observed_target_claim | Claim_{P′} |
| distance_metric | comparison measure |
| tolerance | ε_T |
| transport_status | survives, partial, fails |
| transport_residual | discrepancy explanation |

The transport condition is:

Dist[T_{P→P′}(Claim_P),Claim_{P′}] ≤ ε_T. (D.45)

---

# D.17.3 Standard frame set

The minimum frame audit may include:

- lower timeframe;
- higher timeframe;
- arithmetic price;
- logarithmic price;
- volatility-normalized price;
- time bar;
- volume bar;
- alternate benchmark;
- equal-weight universe;
- nearby anchor;
- alternate data source.

Not every claim requires every frame.

The relevant set must be declared prospectively.

---

# D.17.4 Failure interpretation

A transport failure may mean:

- the original claim was false;

- the claim was only local;
- the transport operator was poor;
- aggregation destroyed the relation;
- the protocol objects were not comparable;
- a regime changed between frames.

Therefore:

TransportFailure ≠ AutomaticClaimFailure. (D.46)

It weakens any universal or cross-frame claim.

The declaration source similarly states that failure across admissible frames downgrades a law-like claim to a local regularity.

---

# D.18 Outcome Ledger

# D.18.1 Required outcome fields

| Field | Meaning |
|---|---|
| `outcome_id` | unique record |
| `claim_id` | original claim |
| `outcome_window` | original horizon |
| `return_after_n` | return after horizon |
| `max_favorable_move` | best move after signal |
| `max_adverse_move` | worst move after signal |
| `close_after_n` | closing state |
| `volatility_after` | later realized volatility |
| `gate_persistence` | whether admitted event survived |
| `fakeout` | true or false |
| `continuation` | true or false |
| `reversal` | true or false |
| `regime_change` | true or false |
| `residual_resolution` | how residual evolved |
| `transaction_cost_result` | where applicable |
| `notes` | contextual record |

These fields follow the source Technical Analysis outcome schema.

---

# D.18.2 Outcome horizon integrity

The horizon must be defined before the outcome.

Let:

$h_0$ = original evaluation horizon. (D.47)

A later exploratory horizon $h_1$ may be added, but it must not replace $h_0$.

The database should preserve:

Outcome($h_0$) and Outcome($h_1$). (D.48)

not:

Outcome = whichever horizon looks best. (D.49)

---

# D.18.3 Outcome is not the only validity test

A method may produce a correct diagnosis but poor trading outcome.

A claim may also be correct at one period and irrelevant at another.

Therefore outcome should be recorded at several levels:

- projection accuracy;
- gate accuracy;
- event persistence;
- episode classification;
- economic return.

These are distinct.

---

# D.19 Admissible Revision

# D.19.1 Revision operator

Let D_k denote the complete declaration.

Revision is:

D_{k+1} = U_a(D_k,L_{k+1},$\mathscr{R}$_{k+1}). (D.50)

The source declaration framework defines admissible revision as a map from declaration, ledger, and residual into another admissible declaration.

---

# D.19.2 Admissibility conditions

A revised declaration should satisfy six conditions.

### Well-formed

All required fields are defined.

### Trace-preserving

The original claim and protocol remain recoverable.

### Residual-honest

Contradictions are not hidden by revision.

### Frame-robust

The revised claim does not depend on unexplained frame shopping.

### Budget-bounded

The revision does not add unlimited complexity.

### Non-degenerate

The revised claim remains falsifiable.

Define the admissible declaration family:

$\mathcal{D}$_adm
= {D | WellFormed(D) ∧ TracePreserving(D) ∧ ResidualHonest(D) ∧ FrameRobust(D) ∧ BudgetBounded(D) ∧ NonDegenerate(D)}. (D.51)

This is the source declaration criterion adapted to Technical Analysis.

---

# D.19.3 Revision validity

A revision is admissible only when:

D_k ∈ $\mathcal{D}$_adm. (D.52)

D_{k+1} ∈ $\mathcal{D}$_adm. (D.53)

U_a(D_k,L_k,$\mathscr{R}$_k) = D_{k+1}. (D.54)

A change may still occur outside $\mathcal{D}$_adm, but it should not be classified as disciplined self-revision.

---

# D.20 Types of Admissible Revision

# D.20.1 Parameter revision

Example:

RSI threshold changes from 70 to 75 after a documented calibration study.

Requirements:

- original threshold preserved;
- new evidence declared;

- no rewriting of earlier outcomes.

---

## D.20.2 Feature-map revision

Example:

Breadth is added because repeated price-only breakout failures reveal a missing field channel.

The revision is:

$\varphi_{k+1} = \varphi_k \cup \{\text{Breadth}\}$. (D.55)

The prior price-only results remain in the ledger.

---

## D.20.3 Boundary revision

Example:

A narrow support line becomes a volatility-normalized zone.

This is admissible when:

- the old line is preserved;
- the reason is measurement resolution;
- the new zone is not fitted solely to rescue a failed claim.

---

## D.20.4 Regime revision

Example:

The interpretation changes from corrective $\chi < 0$ to self-confirming $\chi > 0$.

The record should identify:

- evidence of sign change;
- transition time;
- affected indicator interpretations;
- prior claims invalidated.

---

## D.20.5 Gate revision

Example:

A close-only breakout gate is replaced by close + breadth + retest.

This may improve reliability but increase delay.

Both versions should continue to be evaluated.

---

## D.20.6 Period revision

Example:

A supposed Event-level reversal is downgraded to Structure-level divergence warning.

This is a legitimate category correction.

The original Event claim must remain visible as a classification error.

---

## D.20.7 Model-class reduction

Example:

A complex R + iQ model is reduced to a real pair because phase adds no value.

Reduction is not failure of intellectual ambition.

It is successful model governance.

---

# D.21 Inadmissible Revision Patterns

## D.21.1 Revision by amnesia

The original signal is deleted.

RevisionByAmnesia ⇒ Inadmissible. (D.56)

---

## D.21.2 Revision by timeframe shopping

The analyst moves to a new timeframe only after failure.

No transport rule is recorded.

---

## D.21.3 Revision by anchor migration

A Fibonacci or Gann anchor is moved repeatedly until a level fits.

---

## D.21.4 Revision by semantic drift

The original claim changes meaning.

Example:

“RSI predicts reversal” becomes “RSI merely confirmed strength” after continuation.

## D.21.5 Revision by gate inflation

Extra confirmation requirements are added only after the first gate fails.

---

## D.21.6 Revision by residual burial

Contradictory evidence is removed from the revised report.

---

## D.21.7 Revision by unlimited complexity

Every failed case produces another exception.

Let complexity be C(D).

A suspicious revision satisfies:

C(D_{k+1}) − C(D_k) ≫ InformationGain. (D.57)

This produces an unfalsifiable method.

---

## D.21.8 Revision by outcome selection

The analyst changes the target variable or outcome horizon after observing results.

---

# D.22 Revision Record

## D.22.1 Required fields

| Field | Meaning |
|---|---|
| `revision_id` | unique revision |
| `old_protocol_id` | original protocol |
| `new_protocol_id` | revised protocol |
| `old_claim_id` | original claim |
| `new_claim_id` | revised claim |
| `revision_type` | parameter, feature, boundary, regime, gate, period, reduction |
| `triggering_evidence` | evidence forcing change |
| `triggering_residual_ids` | relevant residuals |
| `old_invalidation_status` | whether old claim failed |
| `trace_preserved` | required |
| `residual_carried_forward` | required where still open |
| `complexity_change` | added or reduced complexity |
| `expected_improvement` | declared before new test |
| `approval_authority` | analyst, model, committee |
| `revision_time` | timestamp |

---

## D.22.2 Revision note template

```
Revision ID:
Original protocol:
Original claim:
Original gate:
Original residual:
Invalidation status:
New evidence:
Revision type:
New protocol:
New claim:
Residual carried forward:
Expected improvement:
New falsification condition:
```

---

# D.23 Revision Quality Score

A provisional revision-quality score may be defined:

Q_rev
= w_TTracePreservation

- w_RResidualHonesty
- w_FFrameRobustness
- w_EEvidenceStrength
- w_PPredictiveImprovement
  – w_CComplexityGrowth
  – w_HHindsightDependence. (D.58)

This is a research template, not an established formula.

Its purpose is to make clear that a revision is better when it:

- preserves accountability;
- responds to evidence;
- controls complexity;
- improves future performance;
- does not merely repair the past.

---

# D.24 Complete Signal Record

A complete Technical Analysis signal can be written:

SignalRecord_k
= {
Protocol_k,
Projection_k,
Claim_k,

Gate_k,
Residual_k,
Transport_k,
Outcome_k,
Revision_k
}. (D.59)

Its minimum validity conditions are:

ProtocolDeclared = true. (D.60)

OriginalClaimPreserved = true. (D.61)

InvalidationDeclared = true. (D.62)

ResidualRecorded = true. (D.63)

OutcomeHorizonFixed = true. (D.64)

RevisionVersioned = true. (D.65)

The source article states the same discipline more sharply:

NoInvalidation → NoDiscipline.

---

# D.25 Copy-Ready Residual Table

| Residual ID | Claim ID | Type | Severity | Description | Open Time | Gate Effect | Frame | Status | Resol |
|---|---|---|---|---|---|---|---|---|---|
| R-001 | C-001 | weak breadth | high | index breakout lacked component participation | To | weakens admission | equal-weight | open | — |
| R-002 | C-001 | no retest | medium | former resistance not yet tested | To | keeps gate partial | daily | monitoring | — |
| R-003 | C-001 | weekly boundary | high | weekly resistance remains overhead | To | prevents World/ Episode promotion | weekly | open | — |

This table is illustrative.

The actual classification must follow the declared residual ontology.

---

# D.26 Copy-Ready Gate Table

| Gate ID | Claim ID | Authority | Close | Volume | VWAP | Breadth | Retest | Follow-Through | Strength | Statu |
|---|---|---|---|---|---|---|---|---|---|---|
| G-001 | C-001 | daily close | 1 | 1 | 1 | 0 | pending | 1 | 0.68 | partial admitt |

The gate should retain pending and failed evidence rather than convert missing values into implicit success.

---

# D.27 Copy-Ready Revision Table

| Revision ID | Old Claim | New Claim | Trigger | Revision Type | Old Trace Preserved | Residual Carried | Complexity Change | Inva |
|---|---|---|---|---|---|---|---|---|
| V-001 | confirmed breakout | provisional breakout | weak breadth and weekly resistance | gate downgrade | yes | yes | none | close rang recla |

---

# D.28 Example — Full Breakout Ledger

# D.28.1 Original declaration

**Protocol**

Daily log-scale chart, twelve-week resistance zone, adjusted close, relative volume, equal-weight breadth, twenty-session outcome horizon.

**Claim**

Price is attempting an Event-level bullish breakout.

**Period**

Event.

**Primary family**

Commitment / Gate.

**χ assumption**

Critical regime may be changing toward self-confirming.

---

# D.28.2 Projection

- price closed 1.1 ATR above the boundary;
- relative volume = 1.7;
- VWAP reclaimed;
- breadth positive but below declared threshold;
- weekly resistance remains 2.5% overhead.

---

# D.28.3 Gate

Close confirmation = true.
Volume confirmation = true.
VWAP confirmation = true.
Breadth confirmation = false.
Retest confirmation = pending.
Follow-through confirmation = pending.

Gate status:

Partially Admit. (D.66)

---

## D.28.4 Residual

R1: weak breadth, severity high.
R2: no retest, severity medium.
R3: weekly resistance, severity high.
R4: options expiry, severity medium.

---

## D.28.5 Initial ledger entry

The event enters the ledger as:

Provisional Daily Breakout. (D.67)

It does not enter as:

Confirmed Multi-Timeframe Trend Transition. (D.68)

---

## D.28.6 Later outcome A — successful acceptance

Suppose:

- breadth expands;
- retest holds;
- weekly close clears resistance;
- residual R1, R2, and R3 resolve.

The ledger is updated:

Provisional Daily Breakout
→ Confirmed Episode Transition. (D.69)

The original partial status remains visible.

---

## D.28.7 Later outcome B — fakeout

Suppose:

- price returns inside the range;
- reclaim fails;
- breadth deteriorates;
- trapped-position proxy rises.

The ledger is updated:

Provisional Daily Breakout
→ Failed Breakout / Fakeout. (D.70)

New residual is created:

TrappedLongResidual. (D.71)

The original claim is not deleted.

---

# D.29 Example — RSI Reversal Revision

# D.29.1 Original claim

RSI above 75 near resistance indicates likely reversal.

# D.29.2 Hidden assumption

$\chi < 0$. (D.72)

# D.29.3 Outcome

Price breaks resistance, volume expands, RSI remains above 70, and trend continues.

# D.29.4 Inadmissible response

“RSI was actually confirming trend strength.”

This silently changes the original claim.

# D.29.5 Admissible revision

The record should say:

1. the original corrective-regime assumption failed;
2. $\chi$ was likely positive;
3. the reversal claim was invalidated;
4. a new regime-conditioned RSI protocol will be tested prospectively.

The revised rule may be:

RSI reversal interpretation allowed only when $\hat{\chi} < \chi_{threshold}$ and boundary rejection gate is present. (D.73)

The old result remains in the failure ledger.

---

# D.30 Example — Elliott Wave Relabeling

## D.30.1 Original count

Wave 5 completed at pivot $P_5$.

## D.30.2 Required gate

- terminal divergence;
- pivot close;
- breadth weakening;
- lower-degree reversal;
- cross-frame survival.

## D.30.3 Outcome

Price continues upward and the supposed endpoint is exceeded.

## D.30.4 Inadmissible response

$P_5$ is silently relabelled as Wave 3.

## D.30.5 Admissible response

The record states:

- original Wave 5 claim invalidated;
- endpoint gate failed;
- original count preserved;
- new Wave 3 interpretation created as a new branch;
- reason for revision documented;
- branch uncertainty carried forward.

This converts Elliott Wave from retrospective artwork into a falsifiable episode ledger.

---

# D.31 Cross-Observer Audit

# D.31.1 Why multiple observers matter

Technical Analysis classifications may differ because observers use:

- different data;
- different timeframes;
- different gate authority;
- different admissible actions;
- different expertise.

Cross-observer agreement should therefore be recorded rather than presumed.

The formal observer source treats agreement as depending on compatible measurements, consistent frame mapping, and accessible records.

---

# D.31.2 Observer record

| Field | Meaning |
|---|---|
| `observer_id` | human, model, desk, committee |
| `observer_role` | analyst, trader, risk, regulator |
| `protocol_access` | data and frame available |
| `claim_id` | claim assessed |
| `classification` | period and family |
| `gate_decision` | decision |
| `residual_assessment` | residual view |
| `confidence` | calibrated confidence |
| `agreement_status` | agree, partial, disagree |
| `mapping_note` | frame-translation explanation |

---

# D.31.3 Consensus does not erase dissent

A consensus record should preserve:

- majority view;
- minority view;
- protocol differences;
- unresolved disagreement.

Consensus ≠ ResidualZero. (D.74)

An expert minority may be wrong.

It may also identify the residual the majority ignored.

---

# D.32 Database Relationships

A relational schema may use:

```
PROTOCOL
  1 ─── n PROJECTION

PROJECTION
  1 ─── n CLAIM

CLAIM
  1 ─── n GATE_TEST
  1 ─── n RESIDUAL
  1 ─── n TRANSPORT_TEST
  1 ─── n OUTCOME
  1 ─── n REVISION

REVISION
  n ─── 1 OLD_PROTOCOL
  n ─── 1 NEW_PROTOCOL
  n ─── 1 OLD_CLAIM
  n ─── 1 NEW_CLAIM
```

This structure allows one claim to remain stable while its residual and outcome history expands.

---

# D.33 Minimal SQL-Style Schema

```
CREATE TABLE ta_claim (
    claim_id              VARCHAR(64) PRIMARY KEY,
    protocol_id           VARCHAR(64) NOT NULL,
    projection_id         VARCHAR(64) NOT NULL,
    claim_time            TIMESTAMP NOT NULL,
    period_class          VARCHAR(16) NOT NULL,
    primary_family        VARCHAR(24) NOT NULL,
    signal_type           VARCHAR(32) NOT NULL,
    direction             VARCHAR(16),
    claim_text            TEXT NOT NULL,
    claimed_characteristic TEXT,
    regime_assumption     VARCHAR(32),
    original_claim_preserved BOOLEAN NOT NULL DEFAULT TRUE,
    invalidation_condition TEXT NOT NULL,
    outcome_horizon       VARCHAR(64) NOT NULL
);

CREATE TABLE ta_residual (
    residual_id           VARCHAR(64) PRIMARY KEY,
    claim_id              VARCHAR(64) NOT NULL,
    gate_id               VARCHAR(64),
    residual_type         VARCHAR(32) NOT NULL,
    residual_source       VARCHAR(32),
    residual_severity     INTEGER NOT NULL,
    residual_description  TEXT NOT NULL,
    residual_open_time    TIMESTAMP NOT NULL,
    residual_resolved     BOOLEAN NOT NULL DEFAULT FALSE,
    residual_resolution_time TIMESTAMP,
    residual_resolution_type VARCHAR(32),
    carry_forward         BOOLEAN NOT NULL DEFAULT TRUE,
    parent_residual_id    VARCHAR(64)
);

CREATE TABLE ta_revision (
    revision_id           VARCHAR(64) PRIMARY KEY,
    old_protocol_id       VARCHAR(64) NOT NULL,
```

```
    new_protocol_id       VARCHAR(64) NOT NULL,
    old_claim_id          VARCHAR(64) NOT NULL,
    new_claim_id          VARCHAR(64),
    revision_type         VARCHAR(32) NOT NULL,
    triggering_evidence   TEXT NOT NULL,
    trace_preserved       BOOLEAN NOT NULL,
    residual_carried_forward BOOLEAN NOT NULL,
    complexity_change     REAL,
    expected_improvement  TEXT,
    new_invalidation      TEXT NOT NULL,
    revision_time         TIMESTAMP NOT NULL
);
```

These tables are conceptual implementation templates.

Production systems require:

- referential constraints;
- version control;
- access control;
- timezone discipline;
- immutable audit fields;
- data lineage;
- review status.

---

# D.34 Research Metrics Derived from the Ledger

# D.34.1 Fakeout rate

FakeoutRate
= NumberOfFakeouts/NumberOfAdmittedBreakoutCandidates. (D.75)

Condition by gate strength:

FakeoutRate(S_G high) versus FakeoutRate(S_G low). (D.76)

---

# D.34.2 Residual-adjusted success

RawSuccessRate
= SuccessfulOutcomes/TotalClaims. (D.77)

Residual-adjusted success may penalize unresolved high-severity contradictions:

AdjustedSuccess
= RawSuccess
– λMeanResidualBurden. (D.78)

The exact metric requires calibration.

---

## D.34.3 Relabeling rate

RelabelRate
= RelabeledClaims/InvalidatedClaims. (D.79)

A high rate is not automatically bad.

But a high rate combined with low original-trace preservation indicates poor discipline.

---

## D.34.4 Revision improvement

RevisionGain
= Loss_old,out-of-sample − Loss_new,out-of-sample. (D.80)

A revision should be judged prospectively.

Improved fit to the same historical data is insufficient.

---

## D.34.5 Frame survival

FrameSurvivalRate
= SurvivingTransportTests/ApplicableTransportTests. (D.81)

This should be reported with:

- frame set;
- tolerance;
- failure type.

---

## D.34.6 Residual resolution time

For residual r_j:

T_resolve,j = t_resolution − t_open. (D.82)

The distribution of T_resolve may reveal:

- quickly dissipating noise;
- persistent structural contradiction;
- long-latency institutional risk.

---

## D.34.7 Original-trace preservation

TracePreservationRate
= RevisionsWithOriginalTrace/RevisionsTotal. (D.83)

For a mature system, the target should be close to one.

# D.35 Operational Workflow

The complete workflow is:

## Step 1 — Declare

Create immutable protocol version.

## Step 2 — Project

Calculate indicators and structures without writing conclusions into the projection fields.

## Step 3 — Claim

Record one testable proposition.

## Step 4 — Gate

Apply predeclared confirmation and authority rules.

## Step 5 — Attach residual

Record missing, contradictory, and ambiguous evidence.

## Step 6 — Transport

Test relevant alternative frames.

## Step 7 — Observe outcome

Use the original horizon.

## Step 8 — Invalidate where required

Do not protect the claim through semantic drift.

## Step 9 — Revise

Create a new version while retaining old protocol, claim, residual, and failure.

## Step 10 — Retest prospectively

A revision earns status only through later evidence.

The workflow can be written:

Declare
→ Project

→ Claim
→ Gate
→ Trace + Residual
→ Transport
→ Outcome
→ Invalidate
→ Revise
→ Retest. (D.84)

---

# D.36 Residual-Honest Writing Template

A mature analysis can be written as:

> Under protocol P, method M projects characteristic C at period p. The current evidence supports claim H, but the required gate is only partially satisfied. Residuals $r_1 \ldots r_n$ remain open. The claim survives frames $F_1$ and $F_2$ but fails or remains unresolved in $F_3$. Invalidation occurs under condition I. Any later reinterpretation will preserve this original claim and residual record.

This sentence architecture is deliberately less dramatic than ordinary chart commentary.

Its value lies in making analytical closure proportionate to evidence.

---

# D.37 Strong and Weak Ledger Entries

## Weak ledger entry

> Bullish breakout confirmed.

Problems:

- protocol absent;
- boundary unspecified;
- gate unclear;
- residual erased;
- timeframe authority unknown;
- invalidation missing.

## Strong ledger entry

> Under the daily log-scale protocol, price closed 1.1 ATR above the predeclared twelve-week resistance zone with relative volume of 1.7 and VWAP acceptance. Equal-weight breadth remained below its confirmation threshold, no retest had occurred, and weekly resistance remained unresolved. The event is therefore recorded as a partially admitted daily breakout, not yet an Episode-level trend transition. It is invalidated by a close back inside the range followed by a failed reclaim.

The second entry creates a usable research object.

## D.38 Why the Residual Ledger Matters

The residual ledger performs five functions.

## Epistemic function

It prevents partial observation from being mistaken for total truth.

## Statistical function

It makes failed and ambiguous cases available for testing.

## Governance function

It prevents silent relabeling and protocol drift.

## Dynamical function

It preserves unresolved pressure that may influence future gates.

## Developmental function

It reveals which missing variables or cells require new instruments.

The source declaration framework states that residual honesty means recording what was not observed, not decided, assumed, contradictory, capable of reopening the gate, and still awaiting evidence.

That principle becomes operational here.

---

## D.39 Appendix D Conclusion

A Technical Analysis method should not be judged only by how often its final label resembles the later chart.

It should be judged by the integrity of its whole disclosure process:

- Was the protocol declared?
- Was the original claim preserved?
- Was the gate explicit?
- Was residual attached?
- Was invalidation obeyed?
- Did the claim survive transport?
- Was later revision trace-preserving?

- Did the revision improve future performance rather than merely repair the past?

The governing equation is:

MatureTechnicalRecord
= OriginalDeclaration

- VisibleProjection
- ExplicitGate
- PersistentTrace
- ResidualHonesty
- CrossFrameAudit
- AdmissibleRevision. (D.85)

And the governing prohibition is:

RevisionWithoutTrace

- ResidualSuppression
- OutcomeFitting
  = UnfalsifiableTechnicalNarrative. (D.86)

The next appendix can formalize the **Transport and Invariance Test Suite**, including timeframe transport, scale transport, anchor perturbation, bar-construction transport, universe transport, and cross-ledger reconciliation.

# Appendix E — Transport and Invariance Test Suite

# E.1 Purpose

A Technical Analysis claim is usually born inside one frame:

- one timeframe;
- one price scale;
- one bar construction;
- one benchmark;
- one anchor;
- one data source;
- one observer role.

The claim may be useful inside that frame.

But it should not be promoted into a general market structure until its dependence on the frame has been tested.

The central transport question is:

What should this claim become when the observation protocol changes?

The central invariance question is:

Which part of the claim remains stable after that change?

The source Technical Analysis framework defines stronger signals as those surviving admissible changes such as daily-to-weekly aggregation, linear-to-log scale, raw-to-ATR-normalized price, time-to-volume bars, price-to-volume-profile analysis, index-to-breadth analysis, and candle-to-higher-timeframe structure.

This appendix turns that principle into an explicit test suite.

---

# E.2 Transport Is Not Mere Recalculation

Suppose a claim is made under protocol P:

C_P = Claim observed under P. (E.1)

A second protocol P′ is introduced.

One may calculate the method again under P′:

C_{P′} = Claim observed directly under P′. (E.2)

But transport requires an additional object:

T_{P→P′}(C_P) = expected form of the original claim in P′. (E.3)

The transport test compares:

T_{P→P′}(C_P)

with:

C_{P′}. (E.4)

A claim survives when:

Dist[T_{P→P′}(C_P),C_{P′}] ≤ ε_T. (E.5)

where:

Dist = declared discrepancy measure;
ε_T = transport tolerance.

Without T_{P→P′}, the analyst is merely comparing two separate charts.

Transport asks how the first claim should transform into the second.

---

# E.3 Invariance Does Not Mean Visual Identity

A daily trend and a weekly trend need not look identical.

A horizontal level in nominal price may become a zone after volatility normalization.

A breakout event on a five-minute chart may become one intraday subevent inside a daily candle.

A volume-profile support zone may occupy a wider region than a single closing-price line.

Therefore:

Invariance ≠ identical picture. (E.6)

The stronger definition is:

InvariantStructure
= relation preserved after the relevant transformation. (E.7)

Possible preserved relations include:

- directional ordering;
- pivot order;
- boundary identity;
- event status;
- phase ordering;
- participation coherence;
- gate outcome;
- ledger consequence.

---

# E.4 Exact, Approximate, and Covariant Survival

Three types of survival should be distinguished.

## E.4.1 Exact invariance

The claim remains numerically unchanged:

$T_{P\to P'}(C_P) = C_{P'}$. (E.8)

This is rare in market analysis.

---

## E.4.2 Approximate invariance

The claim changes slightly but remains within tolerance:

$\mathrm{Dist}[T_{P\to P'}(C_P), C_{P'}] \le \varepsilon_T$. (E.9)

Most empirical market invariance will be approximate.

---

## E.4.3 Covariance

The claim changes in a predictable way.

For example:

PriceLevel_P = £100. (E.10)

After a stock split:

PriceLevel_{P′} = £50. (E.11)

The numerical value changes, but the economic boundary remains equivalent after the declared corporate-action transform.

Covariance means:

C_{P′} = T_{P→P′}(C_P). (E.12)

The claim is not invariant in value.

Its transformation law is invariant.

This distinction is essential for:

- corporate actions;
- currency translation;
- logarithmic transformation;
- volatility normalization;
- different bar frequencies.

---

# E.5 Four Levels of Objectivity

The transport suite distinguishes four levels.

# E.5.1 Frame-local appearance

A pattern appears under one protocol.

Appearance_P. (E.13)

No transport claim is yet made.

---

# E.5.2 Local trace regularity

The pattern repeats inside the same protocol.

Regularity_P. (E.14)

This is stronger than one appearance but remains frame-local.

---

## E.5.3 Cross-frame structure

The relation survives several admissible protocols:

Structure = ∩*{j=1}^{n} SurvivingRelation*{P_j}. (E.15)

---

## E.5.4 Ledger objectivity

Several observer classes recognize and act upon the relation, making it consequential.

LedgerObjectivity
= AgreementAcrossObserverProtocols

- FutureConditionalUse. (E.16)

The source article distinguishes operational objectivity from metaphysical observer-independence and defines Technical Objectivity through cross-protocol survival.

The wider filtration framework likewise defines law as what survives admissible filtrations and objectivity as invariance across viewpoints.

---

# E.6 Admissible Versus Adversarial Transformations

Not every transformation is a fair test.

An admissible transformation should preserve the object being studied while changing the observation frame.

Examples include:

- daily to weekly aggregation;
- arithmetic to logarithmic scale;
- raw to ATR-normalized displacement;
- time bars to volume bars;
- capitalization-weighted to equal-weight index;
- one reasonable pivot rule to another;
- one nearby event anchor to another admissible anchor.

An inadmissible transformation changes the object itself.

Examples include:

- comparing different assets without a declared mapping;
- changing the outcome horizon after failure;
- moving the boundary to include the later result;
- replacing an event study with a different event class;

- switching to an unrelated benchmark because it confirms the claim.

Therefore:

Admissibility
= ObjectPreservation

- DeclaredTransformation
- PredefinedTolerance. (E.17)

---

# E.7 The Transport Test Record

Every transport test should record:

| Field | Meaning |
|---|---|
| `transport_id` | unique test |
| `claim_id` | original claim |
| `source_protocol` | P |
| `target_protocol` | P′ |
| `transformation_type` | timeframe, scale, anchor, universe, etc. |
| `transport_operator` | T_{P→P′} |
| `expected_target_form` | predicted transformed claim |
| `observed_target_form` | actual claim under P′ |
| `distance_metric` | discrepancy rule |
| `tolerance` | ε_T |
| `status` | survives, partial, fails, indeterminate |
| `transport_residual` | explanation of mismatch |
| `effect_on_claim` | unchanged, weakened, localized, invalidated |

This record should be created before the target-frame outcome is interpreted whenever prospective testing is possible.

---

# E.8 Transport Status Vocabulary

A standard status vocabulary is useful.

## Survives

The relation remains within tolerance.

## Covariantly survives

The relation changes according to the declared transformation law.

## Partially survives

The central relation remains, but boundaries, strength, timing, or authority change.

## Local only

The claim is valid under P but is not expected to survive P′.

## Fails

The target-frame relation materially contradicts the transported claim.

## Indeterminate

Data, mapping, or tolerance are inadequate.

## Non-comparable

P and P′ do not refer to sufficiently equivalent objects.

This vocabulary prevents every mismatch from being labelled “wrong.”

---

# E.9 Test Suite Overview

The minimum transport suite contains:

1. timeframe transport;
2. closure-period transport;
3. linear–log scale transport;
4. corporate-action transport;
5. volatility normalization;
6. currency and numeraire transport;
7. bar-construction transport;
8. price-field transport;
9. anchor perturbation;
10. pivot-rule transport;
11. universe and benchmark transport;
12. venue and data-vendor transport;
13. price–volume–breadth transport;
14. cash–derivatives transport;
15. observer-role transport;
16. gate-authority transport;

17. cross-ledger reconciliation;
18. regime transport;
19. complex-state scaling transport;
20. phase-time transport.

Not every method requires every test.

The required subset should be declared by method class.

---

# E.10 Timeframe Transport

# E.10.1 The central problem

A signal may exist on a five-minute chart and disappear on a daily chart.

This does not necessarily make the five-minute signal false.

It may mean that the signal is a lower-level substructure.

A correct transport must preserve closure depth.

For example:

$T_{5m→1d}(Breakout_{5m})$
= intraday Event candidate inside Daily Window. (E.18)

It should not automatically become:

Daily Breakout. (E.19)

---

# E.10.2 Aggregation rule

Suppose lower-period bars are:

$W_{1},W_{2},...,W_{n}$. (E.20)

The higher-timeframe bar is:

$W_H = Aggregate(W_1,...,W_n)$. (E.21)

For OHLCV:

$O_H = O_1$. (E.22)

$H_H = \max_j H_j$. (E.23)

$L_H = \min_j L_j$. (E.24)

$C_H = C_n$. (E.25)

$V_H = \Sigma_j V_j$. (E.26)

A lower-frame event may disappear because the higher-frame close rejects it.

This is not data loss alone.

It is a higher-authority gate outcome.

---

# E.10.3 Timeframe transport classes

### Preserved event

The lower-frame event remains accepted after aggregation.

### Absorbed event

The event becomes one internal fluctuation inside the higher window.

### Rejected event

The higher-frame close returns to the prior region.

### Reclassified event

The lower event becomes a wick, gap, or failed attempt in the higher frame.

### Escalated event

Repeated lower-frame events create a higher-period structural transition.

---

# E.10.4 Test procedure

1. Declare the source timeframe.
2. Declare the destination timeframe.
3. Specify whether the claim concerns Window, Structure, Event, or Episode.
4. Define the expected transformed object.
5. Compare gate status.
6. Record lost detail.
7. Record new higher-frame residual.

---

# E.10.5 Example

Source claim:

Five-minute bullish breakout above £100. (E.27)

Expected daily transform:

Daily range contains accepted intraday trade above £100; daily Event gate remains pending. (E.28)

Observed daily close:

£99.40. (E.29)

Transport result:

The intraday event becomes an upper-wick residual in the daily ledger.

It does not survive as a daily breakout.

---

# E.11 Closure-Period Transport

# E.11.1 Why timeframe and period differ

A daily candle is a Window-period object.

A daily moving average is a Structure-period object.

A daily breakout is an Event-period object.

A daily trend lasting months is an Episode-period object.

Therefore transport must test both:

- clock aggregation;
- closure-depth promotion.

A claim should not move upward in period merely because it persists for several bars.

---

# E.11.2 Promotion conditions

A lower-period object may be promoted only after a new closure condition is met.

### Window → Structure

Requires persistence or stable relational extraction.

### Structure → Event

Requires meaningful boundary interaction and gate.

### Event → Episode

Requires ordered event sequence and stable episode grammar.

### Episode → World

Requires institutional embedding, persistent constraints, shared ledger, and backreaction.

Thus:

Persistence alone ≠ period promotion. (E.30)

Promotion requires:

NewClosureRuleSatisfied = true. (E.31)

---

## E.11.3 Demotion

A claim may also be downgraded.

Examples:

- “confirmed reversal” becomes “momentum divergence”;
- “new trend” becomes “local breakout”;
- “regime change” becomes “temporary volatility event.”

Demotion is not merely rhetorical weakening.

It corrects the object’s closure depth.

---

# E.12 Linear–Log Scale Transport

## E.12.1 Two price geometries

Arithmetic price uses:

y_lin = P. (E.32)

Logarithmic price uses:

y_log = ln P. (E.33)

Arithmetic scale preserves equal nominal differences.

Log scale preserves equal proportional differences.

A movement from £10 to £20 equals +£10.

A movement from £100 to £110 also equals +£10.

But proportionally:

£10 → £20 = +100%. (E.34)

£100 → £110 = +10%. (E.35)

Long-horizon geometric claims can therefore change materially between scales.

---

## E.12.2 Transport operator

The mapping is:

$T_{lin \to log}(P) = \ln P.$ (E.36)

A linear boundary:

$P(t) = a + bt$ (E.37)

becomes:

$\ln P(t) = \ln(a + bt),$ (E.38)

which is generally not linear.

A trend line that exists only because of arithmetic scaling should not be called scale-invariant.

---

# E.12.3 Appropriate claims

## Likely scale-sensitive

- trend-line slope;
- visual angle;
- Gann geometry;
- long-horizon channels;
- pattern symmetry.

## More likely covariant

- pivot order;
- percentage return;
- rank ordering;
- ratio-based levels;
- close above a transformed boundary.

---

# E.12.4 Test criterion

A geometric claim survives when its economic relation remains equivalent after transformation.

For a support zone:

$T_{lin \to log}(Zone_{lin}) \approx Zone_{log}.$ (E.39)

For a slope claim:

$Direction[Slope_{lin}] = Direction[Slope_{log}]$ (E.40)

may be a weak survival criterion.

Stronger tests should compare normalized curvature and boundary interaction.

---

## E.12.5 Gann burden

A visible angle is especially fragile because chart aspect ratio and axis units change its appearance.

The source framework treats a Gann level as a candidate invariant rather than a law and requires survival across log scale, volatility normalization, density context, cadence, and gate evidence.

Thus:

ScreenAngle ≠ MarketInvariant. (E.41)

---

# E.13 Corporate-Action Transport

## E.13.1 Why it matters

Splits, dividends, rights issues, consolidations, and spin-offs change recorded prices.

A chart pattern may appear to break if unadjusted data are compared with adjusted data.

Let c_t be a cumulative adjustment factor.

Then:

P_adj,t = c_tP_raw,t. (E.42)

Historical levels should transform under the same factor.

---

## E.13.2 Boundary covariance

A raw historical level ℓ_raw becomes:

ℓ_adj = c_tℓ_raw. (E.43)

A support claim survives when:

P_raw relative to ℓ_raw

and:

P_adj relative to ℓ_adj

produce equivalent relations.

A level that is not adjusted consistently is a data artefact, not a market failure.

---

## E.13.3 Volume adjustment

Volume may require inverse adjustment for splits:

$V_{adj,t} = V_{raw,t}/c_t$ (E.44)

under the relevant convention.

The exact rule depends on the data provider.

It must be documented.

---

# E.14 Volatility-Normalization Transport

## E.14.1 Purpose

A nominal price move should be interpreted relative to current market agitation.

Let $B_t$ be a boundary.

Raw displacement is:

$d_t = P_t - B_t$. (E.45)

Volatility-normalized displacement is:

$\eth_t = (P_t - B_t)/\sigma_t$. (E.46)

Possible $\sigma_t$ include:

- ATR;
- realized volatility;
- robust range estimator;
- implied volatility;
- local standard deviation.

---

## E.14.2 Transport law

The transport is:

$T_{raw \to vol}(d_t) = d_t/\sigma_t$. (E.47)

A raw breakout survives when its normalized displacement remains material.

For example:

$d_t > 0$ (E.48)

but:

$\eth_t = 0.08$ (E.49)

may indicate only marginal movement relative to noise.

A nominally smaller break with:

đ_t = 2.1 (E.50)

may be structurally stronger.

---

## E.14.3 Pattern transport

Pattern widths and stop distances should also be normalized:

PatternWidth_norm = PatternWidth/σ. (E.51)

This permits comparison across:

- assets;
- regimes;
- historical periods;
- nominal price scales.

---

## E.14.4 Failure modes

Volatility normalization can fail when:

- σ_t jumps after the event;
- volatility estimate is lagged;
- tails dominate;
- market liquidity changes discontinuously;
- one estimator is chosen to rescue the claim.

The volatility estimator is part of the protocol.

---

# E.15 Currency and Numeraire Transport

## E.15.1 The hidden frame

An asset price is always expressed in a numeraire.

Let:

P_t^{GBP} = asset price in pounds. (E.52)

P_t^{USD} = asset price in dollars. (E.53)

With exchange rate X_t^{USD/GBP}:

P_t^{USD} = P_t^{GBP} × X_t^{USD/GBP}. (E.54)

A trend in one currency may weaken or reverse in another.

## E.15.2 Relative-value transport

A claim such as:

“Asset A is in a strong uptrend”

should specify the numeraire.

The transported claim may become:

Trend(A/GBP) ≠ Trend(A/USD). (E.55)

This does not imply contradiction.

It may reveal currency exposure.

---

## E.15.3 Real versus nominal price

A longer-horizon claim may also be transported into inflation-adjusted form:

P_real,t = P_nominal,t/CPI_t. (E.56)

A nominal high may not be a real high.

World-level Technical Analysis should declare whether its claimed boundary is nominal, real, or relative.

---

# E.16 Bar-Construction Transport

## E.16.1 Alternative windows

Common bar rules include:

TimeBar_h. (E.57)

VolumeBar_V*. (E.58)

TickBar_N*. (E.59)

RangeBar_R*. (E.60)

EventBar_E*. (E.61)

Each partitions the same mark field differently.

---

## E.16.2 Why transport matters

A candle pattern visible on a time chart may disappear on a volume chart.

This may indicate that the pattern depends on irregular activity distribution rather than a stable market relation.

Conversely, a structure surviving several bar rules may be more robust.

The source framework explicitly includes time bars to volume bars among admissible cross-frame transformations.

---

## E.16.3 Test procedure

1. Hold asset and data source constant.
2. Define approximately comparable information budgets.
3. reconstruct bars;
4. recalculate pivots and structures;
5. transport the original claim;
6. compare event order and gate status.

---

## E.16.4 Information-budget matching

A fair comparison should avoid matching a one-minute bar with an arbitrarily large volume bar.

Possible matching rules include:

E[Duration(VolumeBar)] ≈ Duration(TimeBar). (E.62)

or:

E[Volume(TimeBar)] ≈ VolumeBarThreshold. (E.63)

The matching rule must be declared.

---

## E.16.5 Path-sensitive result

A candlestick signal that fails under alternative bar construction may be classified:

WindowProtocolDependent. (E.64)

This does not make it useless.

It limits its claim.

---

# E.17 Price-Field Transport

## E.17.1 Multiple price observables

A market may provide:

- last trade;

- midpoint;
- bid;
- ask;
- settlement;
- official close;
- VWAP;
- typical price;
- mark price.

These are not interchangeable.

Let:

P_last,t; P_mid,t; P_close,t; P_settle,t; P_VWAP,t. (E.65)

A support break based on last trade may not survive in settlement price.

---

## E.17.2 Gate authority

The choice depends on the gate.

Examples:

- margin may use settlement;
- chart indicators may use close;
- execution uses bid and ask;
- derivatives may use mark price;
- institutional benchmarking may use VWAP.

Thus:

CorrectPriceField = Function(Claim,Authority,Protocol). (E.66)

---

## E.17.3 Transport test

For claim C based on P_close:

T_{close→settle}(C_close) (E.67)

should specify what equivalent event settlement would need to show.

If no legitimate mapping exists, the claim is price-field local.

---

# E.18 Anchor Perturbation Test

## E.18.1 Anchor-dependent methods

Anchor sensitivity affects:

- anchored VWAP;
- Fibonacci;
- Gann;
- measured moves;
- event studies;
- phase episode start;
- trend channels.

Let a be the original anchor.

The claim is:

C(a). (E.68)

---

## E.18.2 Admissible anchor set

Define:

$\mathcal{A}$_adm = {a′ | a′ satisfies the predeclared anchor rule}. (E.69)

Examples include:

- all pivots exceeding two ATR;
- earnings announcement;
- highest-volume reversal;
- legally recognized event date;
- confirmed episode boundary.

The analyst may not search unlimited anchors.

---

## E.18.3 Robustness score

Anchor robustness is:

S_anchor
= (1/|$\mathcal{A}$_adm|)Σ_{a′∈$\mathcal{A}$_adm} I[C(a′) ≈ C(a)]. (E.70)

A softer version uses average discrepancy:

D_anchor
= (1/|$\mathcal{A}$_adm|)Σ_{a′∈$\mathcal{A}$_adm} Dist[C(a′),C(a)]. (E.71)

High S_anchor or low D_anchor supports robustness.

---

# E.18.4 Influence curve

An anchor influence curve can show:

I(a′) = Dist[C(a′),C(a)]. (E.72)

A sharp jump after tiny anchor changes indicates fragile geometry.

---

# E.18.5 Fibonacci null comparison

A Fibonacci zone should be compared with:

- nearby non-Fibonacci ratios;
- random admissible ratios;
- ordinary prior support;
- volume-density zones.

If the Fibonacci ratio performs no better than generic nearby levels, the special-ratio claim is weakened.

The source framework requires predeclared anchors, zones rather than exact points, confluence, gate evidence, and invalidation for Fibonacci claims.

---

# E.19 Pivot-Rule Transport

# E.19.1 Why pivots matter

Wave counts, patterns, channels, and episode boundaries depend on pivot extraction.

A pivot operator is:

Π_pivot^P(Price) = {p1,p2,…,p_n}. (E.73)

Different rules may use:

- percentage reversal;
- ATR reversal;
- fractal bars;
- local extrema;
- close-only pivots;

- volume confirmation.

---

## E.19.2 Transport test

Let P and P′ differ only in pivot rule.

The original segmentation is:

W_P = Segment[Π_pivot^P(Price)]. (E.74)

The transported segmentation is:

T_{P→P′}(W_P). (E.75)

The directly observed target segmentation is:

W_{P′}. (E.76)

Compare:

Dist[T_{P→P′}(W_P),W_{P′}]. (E.77)

---

## E.19.3 Pivot correspondence

A pivot correspondence relation may be:

Match(p_i^P,p_j^{P′}) = 1 (E.78)

when:

|Price_i − Price_j| ≤ ε_price (E.79)

and:

|Time_i − Time_j| ≤ ε_time. (E.80)

One may then calculate:

PivotSurvivalRate
= MatchedMajorPivots/MajorPivots_P. (E.81)

---

## E.19.4 Wave-count robustness

A wave count should report:

- stable pivots;
- unstable pivots;
- branch changes;
- degree changes;
- endpoint survival.

A count that completely changes under small pivot perturbation should be labelled:

HighBranchFragility. (E.82)

The source framework explicitly states that wave counts should survive objective pivot rules, momentum, volume, invalidation, and cross-timeframe structure rather than being preserved through unrestricted relabelling.

---

# E.20 Universe and Benchmark Transport

# E.20.1 Index versus field

An index is a projection of a component universe.

Let:

$I_{cap,t}$ = capitalization-weighted index. (E.83)

$I_{eq,t}$ = equal-weight index. (E.84)

$B_t$ = breadth measure. (E.85)

A claim of broad market strength should transport across at least one field representation.

---

# E.20.2 Concentration decomposition

Index return may be written:

$R_{index,t} = \Sigma_i w_{i,t} r_{i,t}$. (E.86)

A small number of large weights can dominate.

Define effective participation:

$N_{eff} = 1/\Sigma_i w_i^2$. (E.87)

A rising index with falling $N_{eff}$ or breadth may indicate narrowing participation.

---

# E.20.3 Transport statuses

### Coherent

Cap-weighted, equal-weight, and breadth frames broadly align.

### Concentrated

Headline index rises while equal-weight or breadth weakens.

### Sector-local

Strength is confined to one industry group.

### Universe-sensitive

The claim changes materially when constituent rules change.

---

# E.20.4 Benchmark-relative transport

Relative strength is:

RS_{A/B} = P_A/P_B. (E.88)

Changing B changes the frame.

A claim should declare whether it survives:

- market benchmark;
- sector benchmark;
- currency benchmark;
- risk-free or inflation benchmark.

There is no benchmark-free relative strength.

---

# E.21 Venue and Data-Vendor Transport

# E.21.1 Why data frames differ

Different feeds may differ in:

- timestamp;
- trade inclusion;
- odd-lot treatment;
- corporate-action adjustment;
- volume coverage;
- outlier filtering;
- session boundary;
- settlement.

A chart signal may therefore be vendor-dependent.

---

# E.21.2 Test procedure

1. Match timestamps and timezone.
2. reconcile corporate actions;

3. align session definitions;
4. compare raw fields;
5. recalculate the method;
6. identify whether the difference is market or data residual.

---

## E.21.3 Data discrepancy

For price field X:

$\delta_{vendor,t} = X_t^{VendorA} - X_t^{VendorB}$. (E.89)

A method is data-robust when:

$Dist[Method(X^A),Method(X^B)] \le \varepsilon_{data}$. (E.90)

Otherwise it should carry a data-source residual.

---

# E.22 Price–Volume–Breadth Transport

## E.22.1 Independent projections

Price, volume, and breadth observe different aspects of the market.

A price breakout claim may be transported into:

- volume participation frame;
- breadth coherence frame;
- transaction-density frame;
- closing-gate frame.

The source framework states that a breakout is stronger when confirmed across price, volume, breadth, and close.

---

## E.22.2 Translation rules

A price claim:

$C_{price}$ = Price accepted above boundary. (E.91)

may imply the following target-frame expectations:

$T_{price \to volume}(C_{price})$
= participation should not materially contract. (E.92)

$T_{price \to breadth}(C_{price})$
= component support should be compatible with the claimed scope. (E.93)

T_{price→profile}(C_price)
= transaction density should migrate or develop beyond the old zone. (E.94)

T_{price→close}(C_price)
= the relevant closing gate should preserve the displacement. (E.95)

These are not identities.

They are declared confirmation expectations.

---

## E.22.3 Scope matching

A single-stock breakout does not necessarily require broad-index breadth.

The breadth frame must match the claim scope.

Examples:

- stock claim → sector or peer breadth;
- index claim → constituent breadth;
- global risk-on claim → cross-asset breadth.

Thus:

CorrectCrossFrame = Function(ClaimScope). (E.96)

---

# E.23 Cash–Derivatives Transport

## E.23.1 Related but non-identical worlds

A market may be observed through:

- spot;
- futures;
- options;
- swaps;
- ETFs;
- credit instruments.

These markets contain different:

- participants;
- funding;
- leverage;
- settlement;

- maturity;
- optionality.

A cash-price signal may not transport directly into derivatives.

---

## E.23.2 Futures basis

Let:

Basis_t = F_t − S_t. (E.97)

where:

F_t = futures price;
S_t = spot price.

A spot breakout with weakening basis may indicate different participation from a spot and futures breakout occurring together.

---

## E.23.3 Options frame

Option-related transport may examine:

- implied volatility;
- skew;
- open interest;
- gamma concentration;
- strike distribution.

A price boundary near large option positioning may carry different event dynamics.

But options open interest should not automatically be treated as deterministic support or resistance.

It is a Load and Constraint proxy requiring current hedging and expiry context.

---

## E.23.4 Credit–equity transport

A bullish equity claim may be transported into credit:

T_{Equity→Credit}(BullishEquity)
= credit spread should remain compatible with improving enterprise condition. (E.98)

If equity rises while credit deteriorates, the result is a cross-world residual.

It may reflect:

- equity optionality;

- leverage;
- speculative flow;
- different investor horizons;
- data timing.

---

# E.24 Observer-Role Transport

# E.24.1 Different observers see different objects

Possible observers include:

- retail trader;
- market maker;
- execution desk;
- portfolio manager;
- risk committee;
- lender;
- accountant;
- regulator;
- central bank.

Each observer has:

- different data;
- different boundary;
- different admissible action;
- different gate authority;
- different ledger.

Therefore:

Claim^{ObserverA}_P ≠ Claim^{ObserverB}_P by default. (E.99)

---

# E.24.2 Role mapping

A support zone may mean:

### Trader

Potential entry or stop location.

### Market maker

Inventory and order-flow concentration.

### Risk manager

Exposure threshold.

### Lender

Collateral-value boundary.

### Accountant

No direct recognition unless an accounting rule is triggered.

The same price region is transported into different functional roles.

---

## E.24.3 Observer-compatible agreement

Cross-observer agreement requires:

- compatible objects;
- declared mapping;
- accessible trace;
- aligned gate definitions.

A formal compatibility test is:

Compat(O_A,O_B | Claim) = 1 (E.100)

only when both observers can meaningfully assess equivalent aspects of the claim.

The formal observer framework treats agreement as dependent on compatible effects, consistent frame transformations, and accessible records rather than assuming one universal observer. The present use is an operational market adaptation of that principle.

---

# E.25 Gate-Authority Transport

## E.25.1 Gates are not equal

A five-minute close, daily close, weekly close, exchange settlement, accounting recognition, and legal judgment possess different authority.

Let:

G_A = gate under authority A. (E.101)

The same candidate event may receive:

G_daily = Admit. (E.102)

G_weekly = Defer. (E.103)

G_accounting = NotApplicable. (E.104)

G_legal = NotApplicable. (E.105)

This is not necessarily inconsistency.

The gates operate on different objects.

---

## E.25.2 Authority hierarchy

A possible hierarchy is:

PrivateInterpretation
< ModelClassification
< InstitutionalCommittee
< ExchangeSettlement
< ContractualTrigger
< AccountingRecognition
< LegalJudgment. (E.106)

This ordering is context-dependent rather than universal.

Higher formal authority does not always mean greater predictive accuracy.

It means greater power to alter future admissibility.

---

## E.25.3 Gate transport record

| Field | Meaning |
|---|---|
| Source gate | original authority |
| Target gate | destination authority |
| Object equivalence | whether both gates assess the same state |
| Expected translation | what the source event implies for target |
| Target status | admit, defer, reject, not applicable |
| Ledger effect | changed action space |
| Residual | unresolved inter-gate difference |

---

# E.26 Cross-Ledger Reconciliation

## E.26.1 Multiple ledgers

A market event may appear in:

- transaction ledger;

- market-price ledger;
- risk ledger;
- accounting ledger;
- legal ledger;
- policy ledger;
- narrative ledger.

Recognition times differ.

Let:

$$g_{mkt,k};$$
$$g_{risk,k};$$
$$g_{acct,k};$$
$$g_{legal,k};$$
$$g_{policy,k}. \quad (E.107)$$

Define the recognition vector:

$$\mathbb{G}_k = (g_{mkt,k}, g_{risk,k}, g_{acct,k}, g_{legal,k}, g_{policy,k}). \quad (E.108)$$

---

# E.26.2 Reconciliation state

Possible states include:

### Aligned

Relevant ledgers recognize the event.

### Market-leading

Price recognizes before institutions.

### Institution-leading

Legal, policy, or risk gate changes before price fully responds.

### Fragmented

Ledgers materially disagree.

### Delayed

Recognition is expected but not yet written.

### Contested

Authorities or observers disagree about the event itself.

---

## E.26.3 Reconciliation score

A provisional score is:

S_recon
= Σ_j w_jg_j
− λDisagreement(𝔾). (E.109)

The weights depend on the claim.

For default, legal and contractual gates may dominate.

For a short-term breakout, market and technical gates may dominate.

---

## E.26.4 Ledger mismatch as residual

Define:

r_ledger
= ExpectedRecognitionVector
− ObservedRecognitionVector. (E.110)

This residual may predict:

- delayed repricing;
- litigation;
- accounting adjustment;
- liquidity stress;
- narrative conflict.

But it is not automatically directional.

---

# E.27 Regime Transport

## E.27.1 χ is horizon-dependent

A market can be:

- corrective intraday;
- self-confirming daily;
- corrective monthly.

Therefore:

χ_{5m} ≠ χ_{1d} ≠ χ_{1m} by default. (E.111)

A regime label must be transported with its horizon.

## E.27.2 Sign transport

A lower-frame χ sequence may aggregate into a higher-frame relation.

For example:

$\{\chi_1<0,\chi_2<0,\chi_3>0,\chi_4>0,\ldots\}$ (E.112)

may produce a higher-level self-confirming episode if positive segments dominate consequential gates.

But simple averaging may be inappropriate.

A gate-weighted aggregate is:

$\chi_H = \Sigma_k w_k \chi_k$, (E.113)

where $w_k$ reflects event consequence rather than duration alone.

This is a proposed research construction.

---

## E.27.3 Regime-conflict record

| Frame | χ estimate | Evidence | Confidence |
|---|---|---|---|
| Intraday | negative | repeated mean reversion | medium |
| Daily | positive | breakout follow-through | high |
| Weekly | near zero | unresolved range | medium |

The correct interpretation may be:

Intraday corrections inside a daily self-confirming movement within a weekly critical range.

This is more precise than declaring one universal regime.

---

# E.28 Ξ Control-State Transport

## E.28.1 Protocol dependence

The effective state:

$\Xi_P = (\rho_P,\gamma_P,\nu_P)$ (E.114)

is compiled under P.

Changing:

- entity boundary;
- data source;
- horizon;

- observer;
- intervention family

may change all three coordinates.

---

## E.28.2 Local-to-systemic transport

A trading desk may observe:

Ξ_desk = (ρ_d,γ_d,ν_d). (E.115)

The enterprise may observe:

Ξ_ent = (ρ_e,γ_e,ν_e). (E.116)

High local liquidity does not guarantee low enterprise lock-in if:

- funding is concentrated;
- collateral is encumbered;
- legal entities cannot transfer resources;
- accounting restrictions apply.

Thus:

T_{desk→enterprise}(Ξ_desk) ≠ Ξ_desk. (E.117)

The transport must include funding, legal, accounting, and risk connections.

---

## E.28.3 Cross-frame repair

When the local and systemic states disagree, the failure may be a connection problem rather than a local-variable problem.

The Gauge Grammar source explicitly identifies cross-frame repair as repairing the connection or invariance failure between frames and recommends using the lowest intervention level capable of closing the relevant residual.

---

# E.29 Complex-State Scaling Transport

## E.29.1 The scaling problem

For:

Z = R + iQ, (E.118)

the phase is:

θ = atan2(Q,R). (E.119)

Rescale Q:

$Q' = cQ$. (E.120)

Then:

$\theta' = \operatorname{atan2}(cQ,R)$. (E.121)

The phase changes unless $c = 1$ or a compatible metric is supplied.

---

## E.29.2 Scaling family

Define an admissible scaling family:

$\mathcal{C}_{adm} = \{c \mid c \text{ represents a defensible normalization perturbation}\}$. (E.122)

Phase robustness is:

$S_{phase\text{-}scale} = 1 - \text{NormalizedDispersion}[\{\theta_c \mid c \in \mathcal{C}_{adm}\}]$. (E.123)

A high score means phase ordering is relatively stable.

A low score indicates that the complex geometry depends heavily on arbitrary units.

---

## E.29.3 Unit transport

If R and Q have different native units, a metric is required.

Let:

$\tilde{R} = R/s_R$. (E.124)

$\tilde{Q} = Q/s_Q$. (E.125)

$\tilde{Z} = \tilde{R} + i\tilde{Q}$. (E.126)

The scales $s_R$ and $s_Q$ must derive from:

- domain theory;
- calibration;
- risk equivalence;
- empirical normalization;
- or operational conversion.

They should not be selected solely to create an attractive circle.

---

## E.29.4 Real-pair comparison

Every scaling transport should be compared with the real pair:

$X = (R,Q)$. (E.127)

If conclusions change dramatically with complex normalization but the real-pair model remains stable, the real pair may be preferable.

---

# E.30 Phase-Order Transport

## E.30.1 Phase order versus phase value

The exact phase value may shift across normalization.

The order of events may remain.

Suppose:

$\theta(t_1) < \theta(t_2) < \theta(t_3)$. (E.128)

A weaker invariance requirement is:

$\theta'(t_1) < \theta'(t_2) < \theta'(t_3)$. (E.129)

This preserves phase order even if angular spacing changes.

---

## E.30.2 Circular-order problem

Because phase is periodic:

$\theta$ and $\theta + 2\pi$ represent the same orientation. (E.130)

Transport must compare unwrapped phase or circular order.

Possible measures include:

- circular correlation;
- phase-locking value;
- rank correlation of unwrapped phase;
- event-order consistency.

---

## E.30.3 Direction preservation

A valid transport should record whether:

$\operatorname{sgn}(\theta') = \operatorname{sgn}(\theta)$. (E.131)

The same phase traversed in the opposite direction is not the same episode state.

---

# E.31 Phase-Time Transport

## E.31.1 Calendar-to-phase mapping

The candidate internal time is:

$\tau_i(t) = \text{Unwrap}[\theta(t)]$ (E.132)

or:

$\tau_i(t) = \int_0^t |\dot{\theta}(s)| ds.$ (E.133)

Transport from clock time to phase time is:

$T_{t \to \tau_i}: \text{Episode}(t) \to \text{Episode}(\tau_i).$ (E.134)

---

## E.31.2 Episode alignment test

For episodes $j = 1 \ldots N$:

$D_t = \text{Dispersion}[\{X_j(t_{norm})\}].$ (E.135)

$D_\tau = \text{Dispersion}[\{X_j(\tau_{i,norm})\}].$ (E.136)

Phase-time gains support when:

$D_\tau < D_t$ (E.137)

out of sample and after complexity adjustment.

---

## E.31.3 Gate-hazard transport

Compare:

$h_G(t \mid \text{Controls})$ (E.138)

with:

$h_G(\tau_i \mid \text{Controls}).$ (E.139)

A useful phase clock should improve:

- localization;
- calibration;
- stability;
- intervention timing.

If not, $\tau_i$ remains a descriptive coordinate.

---

## E.31.4 Phase-time failure

Transport fails when:

- phase order changes under minor scaling;
- unwrapping is ambiguous;
- comparable episodes do not align;
- gate concentration is absent;
- event count k performs equally well;
- the real pair explains the same dynamics.

The phase framework treats phase-bearing dynamics, secondary ordering, phase-sensitive events, and full time-bearing worlds as separate evidence levels rather than automatic consequences of complex notation.

---

# E.32 Cross-Method Transport

# E.32.1 Same claim, different instruments

A support claim may be represented by:

- prior reaction;
- moving average;
- VWAP;
- volume profile;
- options positioning.

The question is whether these methods transport into one common structural object.

Let:

C_MA; C_VWAP; C_Profile; C_Options. (E.140)

A common latent boundary claim B* is supported when:

T_{MA→B*}(C_MA)
≈ T_{VWAP→B*}(C_VWAP)
≈ T_{Profile→B*}(C_Profile)
≈ T_{Options→B*}(C_Options). (E.141)

This is stronger than simple numerical proximity.

The mechanisms should also be compatible.

---

# E.32.2 False confluence

Several lines may cluster by chance.

Confluence is weaker when:

- anchors are unconstrained;
- many indicators are searched;
- tolerance is wide;
- only successful clusters are reported.

A confluence study should control the search space.

---

# E.33 Transport of Support and Resistance

## E.33.1 Source claim

Support zone under P:

$$B_P = [\ell_P - \delta_P^-, \ell_P + \delta_P^+]. \quad (E.142)$$

---

## E.33.2 Timeframe transform

The destination zone may widen:

$$\delta_{P'} > \delta_P \quad (E.143)$$

because higher timeframes aggregate greater variation.

The centre may remain approximately stable.

---

## E.33.3 Volatility transform

Normalized zone:

$$\tilde{B}_P = [(\ell - \delta^- - \mu)/\sigma, (\ell + \delta^+ - \mu)/\sigma]. \quad (E.144)$$

The nominal zone can change while the normalized boundary remains equivalent.

---

## E.33.4 Density transform

A price-line support claim may transport into volume profile as:

Historical transaction density should be locally elevated or a value boundary should align. (E.145)

Absence of density does not automatically reject support, because support may arise from:

- institutional rule;
- option strike;
- narrative attention;

- prior close;
- thin-liquidity discontinuity.

The mismatch becomes residual.

---

# E.34 Transport of Breakouts

## E.34.1 Source event

A source-frame breakout is:

e_P = G_P[Price crosses B_P]. (E.146)

---

## E.34.2 Required transports

A strong breakout study should test:

- higher timeframe;
- normalized displacement;
- volume;
- breadth;
- retest;
- profile migration;
- derivative participation;
- gate authority.

---

## E.34.3 Breakout transport vector

Define:

$\mathcal{T}$_break
= (t_higher, t_vol, t_volume, t_breadth, t_retest, t_profile, t_deriv). (E.147)

Each component may be:

1 = survives;
0.5 = partial;
0 = fails;
NA = not applicable.

A weighted score is:

S_break,T
= Σ_j w_jt_j / Σ_j w_j. (E.148)

This score measures cross-frame support, not expected return.

---

# E.34.4 Interpretation classes

### Local breakout

Price-frame event only.

### Participatory breakout

Price and volume support.

### Field-coherent breakout

Price, volume, and breadth support.

### Structurally accepted breakout

Retest and profile migration support.

### Multi-ledger transition

Institutional or cross-market ledgers also update.

These classes correspond to increasing closure depth.

---

# E.35 Transport of Divergence

# E.35.1 Source relation

A bearish divergence is:

Price makes higher high. (E.149)

Indicator fails to make higher high. (E.150)

---

# E.35.2 Relevant transports

- alternate momentum indicator;
- breadth;
- volume;
- higher timeframe;
- normalized price;
- structural gate.

---

## E.35.3 Divergence survival

A divergence is stronger when the weakening relation survives several partially independent channels.

But even a highly transported divergence remains a warning until an Event gate fails.

Thus:

CrossFrameDivergence ≠ ReversalEvent. (E.151)

It means:

Relational residual is broad rather than indicator-specific. (E.152)

---

# E.36 Transport of Elliott Wave

## E.36.1 Required dimensions

A wave interpretation should be tested across:

- pivot rule;
- timeframe;
- volatility normalization;
- momentum;
- volume;
- breadth;
- alternate count;
- endpoint gate.

---

## E.36.2 Wave transport vector

$\mathcal{T}$_wave
= (PivotSurvival, DegreeStability, SegmentOrder, χConsistency, EndpointGate, BranchResidual). (E.153)

A robust wave claim should not require identical minor pivots.

It should preserve:

- major segment order;
- episode grammar;
- endpoint conditions;
- invalidation.

## E.36.3 Endpoint transport

A candidate Wave 5 endpoint should transport into:

- terminal momentum weakening;
- breadth weakening or climax;
- boundary failure;
- lower-degree reversal;
- higher-frame compatibility.

The source framework expresses the endpoint requirement as terminal structure plus phase weakening plus gate failure and keeps the label provisional without those confirmations.

---

# E.37 Transport of Fibonacci Claims

## E.37.1 Required tests

A Fibonacci level should be tested under:

- admissible anchor perturbation;
- zone width;
- log scale;
- volatility normalization;
- non-Fibonacci ratio controls;
- volume profile;
- prior reaction;
- current gate.

---

## E.37.2 Invariance claim

The strongest legitimate claim is not:

61.8% is a universal law. (E.154)

It is:

Under declared anchors and tolerance, a ratio-derived zone coincides with independently observed structural density and passes a market-reaction gate. (E.155)

This is a local confluence claim.

---

# E.38 Transport of Gann Claims

# E.38.1 Required tests

A Gann claim carries the heaviest transport burden.

It should declare:

- anchor;
- price unit;
- time unit;
- calendar;
- aspect ratio;
- linear or log scale;
- volatility normalization;
- event cadence;
- gate;
- invalidation.

---

# E.38.2 Normalized coordinate system

Define:

$\tilde{p}_t = [P_t - P_a]/s_P$. (E.156)

$\tilde{t} = [t - t_a]/s_t$. (E.157)

A candidate line is:

$\tilde{p} = m\tilde{t} + b$. (E.158)

The relation should survive reasonable changes in $s_P$ and $s_t$.

Otherwise the visual angle is an artefact.

---

# E.38.3 Cadence transport

A time point should be compared with actual market cadence:

- expiry;
- reporting;
- policy meeting;

- settlement;
- rebalancing;
- margin review.

A geometric date with no cadence or gate evidence remains a weak hypothesis.

---

# E.39 Transport of Complex CAPM Geometry

## E.39.1 Protocol transport

The CAPM state depends on:

- cash flow;
- horizon;
- base discount rate;
- beta;
- ERP;
- valuation baseline.

Changing any of these changes:

A; R; Q; θ. (E.159)

The transport should preserve the valuation logic.

---

## E.39.2 Horizon transport

For horizon t:

A_t = CF_t/(1+r_base)^t. (E.160)

R_t = CF_t/(1+r_CAPM)^t. (E.161)

Q_t = √(A_t²−R_t²). (E.162)

Transport from t to t′ is not a simple rotation.

It requires recalculating the discount relation.

A phase comparison across horizons should therefore distinguish:

- state transport;
- protocol recomputation;
- actual market evolution.

---

## E.39.3 Baseline transport

Changing r_base changes A and R.

The claim should test whether the qualitative phase relation remains robust under defensible baseline alternatives.

If Q changes dramatically because the baseline is arbitrary, its practical interpretation weakens.

---

## E.39.4 Model transport

CAPM may be compared with:

- multifactor discounting;
- cost-of-equity alternatives;
- scenario discount rates;
- market-implied rates.

The complex construction is local to its declared valuation protocol.

Survival across models would support a broader valuation-phase claim.

Failure would localize it to CAPM.

---

# E.40 Law, Regularity, and Local Pattern

The transport suite supports a hierarchy.

## Appearance

Observed once under one frame.

## Local pattern

Repeated under the same frame.

## Protocol regularity

Stable under parameter variation inside one protocol family.

## Cross-frame invariant

Survives admissible transformations.

## Ledger regularity

Repeatedly enters consequential shared trace.

# Candidate law

Relation survives broad admissible declarations, independent observers, and repeatable gates.

This hierarchy can be written:

Appearance
→ LocalPattern
→ ProtocolRegularity
→ CrossFrameInvariant
→ LedgerRegularity
→ CandidateLaw. (E.163)

The wider filtration framework states that law is not mere repetition but the relation surviving many admissible disclosures.

The 成界之學 source similarly distinguishes pattern from law through cross-declaration invariance and interprets stable cross-frame coherence as a deeper structural achievement.

---

# E.41 Invariance Failure Taxonomy

Transport failures should be classified rather than merely counted.

# E.41.1 Representation failure

The same object was encoded differently without correct transformation.

Example:

split-adjusted and raw price levels compared directly.

---

# E.41.2 Scale failure

The claim depends on arithmetic versus logarithmic scale.

---

# E.41.3 Aggregation failure

The relation disappears after bar or timeframe aggregation.

---

# E.41.4 Anchor failure

Small changes in anchor destroy the claim.

---

# E.41.5 Pivot failure

The episode model changes under reasonable pivot rules.

---

## E.41.6 Universe failure

The claim depends on capitalization concentration or benchmark choice.

---

## E.41.7 Gate failure

The source-frame event is not recognized by the destination-frame gate.

---

## E.41.8 Authority failure

Observers disagree because they possess different powers to recognize the event.

---

## E.41.9 Regime failure

The relation was transported across a genuine regime transition.

---

## E.41.10 Data failure

The mismatch arises from data quality or vendor conventions.

---

## E.41.11 Model failure

The proposed invariant does not exist.

The 成界之學 material similarly distinguishes representation, rule, projection, gate, trace, residual, revision, and invariance errors.

---

## E.42 Transport Residual

A transport test should produce a residual:

$$r_T = C_{P'} - T_{P \to P'}(C_P). \quad \text{(E.164)}$$

This symbolic subtraction may represent:

- numerical difference;
- classification mismatch;
- gate disagreement;
- branch disagreement;
- authority mismatch.

Transport residual should be decomposed where possible:

r_T
= r_scale

- r_timeframe
- r_anchor
- r_data
- r_regime
- r_authority
- r_model. (E.165)

The components may overlap.

The decomposition exists to guide diagnosis, not to imply exact additivity.

---

# E.43 Transport Tolerance

# E.43.1 Why tolerance is required

Market data are noisy.

Exact equality is usually unrealistic.

A tolerance may depend on:

- volatility;
- data error;
- timeframe;
- boundary width;
- event importance;
- method precision.

---

# E.43.2 Boundary tolerance

For a price zone:

ε_B = kσ_t (E.166)

or:

ε_B = declared percentage of price. (E.167)

---

# E.43.3 Timing tolerance

For pivots or events:

ε_time = m bars (E.168)

or a phase window.

---

## E.43.4 Classification tolerance

For categorical claims:

- exact match;
- compatible match;
- partial match;
- conflict.

The tolerance rule must be fixed before observing the destination result.

---

# E.44 Transport Robustness Score

A general score may be:

S_T
= [Σ_j w_js_j]/[Σ_j w_j] – λR_T. (E.169)

where:

s_j = survival score for transformation j;
w_j = relevance weight;
R_T = normalized transport residual.

Possible s_j values:

1 = survives;
0.5 = partial;
0 = fails.

This score is a research template.

It should not be treated as an established universal index.

---

# E.45 Essential Versus Optional Transport Tests

## E.45.1 Moving average

Essential:

- timeframe;
- parameter;

- bar construction;
- volatility normalization.

Optional:

- universe;
- venue.

---

## E.45.2 RSI or stochastic

Essential:

- timeframe;
- parameter;
- χ regime;
- structural gate.

Optional:

- breadth;
- benchmark.

---

## E.45.3 Support or resistance

Essential:

- timeframe;
- log scale;
- volatility width;
- profile or prior-reaction frame.

Optional:

- options;
- institutional reference.

---

## E.45.4 Breakout

Essential:

- higher timeframe;
- normalized displacement;

- volume;
- breadth;
- close;
- retest.

Optional:

- derivatives;
- cross-ledger recognition.

---

## E.45.5 Elliott Wave

Essential:

- pivot rule;
- timeframe;
- alternate branch;
- endpoint gate;
- momentum and breadth.

---

## E.45.6 Fibonacci

Essential:

- anchor perturbation;
- zone tolerance;
- non-Fibonacci control;
- gate.

---

## E.45.7 Gann

Essential:

- log scale;
- normalized units;
- anchor;
- calendar;
- cadence;

- gate.

---

## E.45.8 Complex phase model

Essential:

- unit scaling;
- Q proxy;
- real-pair benchmark;
- phase order;
- episode alignment;
- gate concentration.

---

# E.46 Minimal Prospective Transport Protocol

A prospective study should follow this sequence.

## Step 1 — Declare source claim

Record C_P before testing target frames.

## Step 2 — Select admissible transformations

Define P′1…P′n.

## Step 3 — Define transport operators

Specify T_{P→P′j}.

## Step 4 — Define tolerances

Specify ε_j.

## Step 5 — Generate expected target forms

Do this before inspecting the target result where feasible.

## Step 6 — Observe destination claims

Calculate C_{P′j}.

## Step 7 — Score survival

Record exact, covariant, partial, local, failed, or indeterminate.

## Step 8 — Record residual

Explain the mismatch without erasing it.

## Step 9 — Revise claim scope

The claim may remain:

- universal candidate;
- cross-frame regularity;
- protocol-family regularity;
- frame-local pattern;
- invalidated.

---

# E.47 Copy-Ready Transport Matrix

| Test | Source frame | Target frame | Expected transported claim | Observed claim | Tolerance | Status | Residual |
|---|---|---|---|---|---|---|---|
| Timeframe | Daily | Weekly | daily breakout becomes weekly candidate | weekly close below resistance | one weekly bar | fails | higher-frame rejection |
| Scale | Linear | Log | major support zone remains aligned | zone shifts slightly but overlaps | 0.5 ATR | survives | minor scale residual |
| Volume | Price | Relative volume | breakout should show elevated participation | RVOL 1.8 | RVOL > 1.5 | survives | none material |
| Breadth | Index | Equal-weight | broad breakout should preserve component support | equal-weight index flat | threshold 0.5% | partial | concentration |
| Retest | Initial event | Later test | old resistance should hold as support | close below zone | one daily close | fails | fakeout risk |

---

# E.48 Copy-Ready Method Transport Card

```
Claim ID:
Method:
Source protocol:
Target protocol:
Transformation type:
Transport operator:
Expected target form:
Observed target form:
Distance metric:
Tolerance:
Transport status:
```

```
Residual:
Effect on original claim:
Revision required:
```

---

# E.49 Example — Multi-Frame Breakout Audit

# E.49.1 Original claim

Daily price has broken above a twelve-week resistance zone.

# E.49.2 Source protocol

- daily adjusted close;
- log scale;
- ATR normalization;
- equal-weight breadth;
- relative volume;
- twenty-session horizon.

# E.49.3 Timeframe transport

Expected weekly form:

Weekly candle should preserve a close beyond the resistance zone or at least avoid decisive rejection.

Observed:

Weekly close remains marginally below the zone.

Status:

Partial survival.

Residual:

Daily commitment has not yet become weekly commitment.

---

# E.49.4 Scale transport

Expected:

Zone remains materially aligned on log scale.

Observed:

Zone overlaps within 0.3 ATR.

Status:

Survives.

---

## E.49.5 Volume transport

Expected:

Relative participation above baseline.

Observed:

RVOL = 1.6.

Status:

Survives.

---

## E.49.6 Breadth transport

Expected:

At least 60% of relevant components participate.

Observed:

47%.

Status:

Fails.

Residual:

Headline concentration.

---

## E.49.7 Profile transport

Expected:

Transaction density should begin migrating above the old value area.

Observed:

Most volume remains below the boundary.

Status:

Partial.

Residual:

Price has crossed before value acceptance.

---

## E.49.8 Final classification

The event should be recorded as:

Daily price breakout with scale and volume support, but incomplete weekly, breadth, and value-area acceptance. (E.170)

It should not yet be recorded as:

Broad multi-timeframe Episode transition. (E.171)

---

# E.50 Example — Gann Audit

## E.50.1 Original claim

Price should reverse at a declared price–time line.

## E.50.2 Required records

- anchor declared before event;
- arithmetic scale;
- price unit;
- time unit;
- line equation;
- expected reaction;
- invalidation.

## E.50.3 Log-scale transport

The line no longer intersects the same structural region.

Status:

Fails.

## E.50.4 Volatility transport

After normalization, the apparent angle is not stable.

Status:

Fails.

## E.50.5 Cadence transport

The date does not align with any declared event cycle.

Status:

Fails.

## E.50.6 Price-density transport

The level coincides with a major volume-profile node.

Status:

Survives partially.

## E.50.7 Gate

No rejection close occurs.

Status:

Reject.

## E.50.8 Final classification

The Gann line is not admitted as a transported price–time invariant.

The density zone remains independently relevant.

This demonstrates why a failed geometric theory may still contain one useful local boundary.

---

# E.51 Example — Wave Transport Audit

## E.51.1 Original claim

A five-wave upward episode has completed.

## E.51.2 Pivot transport

ATR-based and percentage-based pivot rules agree on major pivots 1–4 but disagree on the final high.

Status:

Partial.

## E.51.3 Timeframe transport

Daily count appears complete.

Weekly structure remains inside an active advancing episode.

Status:

Local only.

## E.51.4 Momentum transport

MACD and RSI show terminal weakening.

Status:

Survives.

## E.51.5 Breadth transport

Breadth reaches a new high.

Status:

Fails.

## E.51.6 Gate transport

No downside structural break has occurred.

Status:

Defer.

## E.51.7 Final classification

Candidate terminal divergence at daily episode level; Wave 5 completion remains provisional.

The source framework similarly requires terminal structure, phase weakening, and gate failure before treating a Wave 5 top as complete.

---

# E.52 Example — Complex Phase Audit

## E.52.1 Candidate state

$Z_t = R_t + iQ_t$. (E.172)

$R_t$ = accepted trend structure.
$Q_t$ = independently estimated breadth-pressure channel.

## E.52.2 Scaling test

Three defensible Q normalizations are used.

Observed:

Phase order is stable, but angular spacing varies.

Status:

Order survives; metric phase partially survives.

## E.52.3 Timeframe test

Daily and weekly phase orders disagree.

Status:

Frame-local.

## E.52.4 Real-pair benchmark

The real-pair model predicts gates as accurately as the complex model.

Status:

Complex priority fails.

## E.52.5 Episode alignment

Phase alignment does not reduce dispersion relative to event count.

Status:

Phase-time claim fails.

## E.52.6 Final model status

Retain:

(R,Q) as a real pair. (E.173)

Reject:

privileged complex phase and internal clock. (E.174)

This is a successful reduction, not a failed research process.

---

# E.53 Cross-Frame Repair

Transport failure may suggest repair.

But repair should occur at the lowest effective level.

## Level 1 — Clarify

Correct missing protocol or units.

## Level 2 — Local repair

Correct one indicator, boundary, or gate.

## Level 3 — Cross-frame repair

Repair the mapping between frames.

## Level 4 — Regime repair

Change $\chi$ or $\Xi$ diagnosis.

## Level 5 — Protocol revision

Change boundary, timeframe, feature map, or gate.

## Level 6 — Governance escalation

Move to a higher-authority observer or broader system.

The Gauge Grammar source explicitly presents this hierarchy and describes cross-frame repair as repairing the connection or invariance failure rather than automatically replacing the local model.

---

# E.54 When Not to Demand Invariance

Some market objects are genuinely local.

Examples include:

- microstructure imbalance;
- one-session auction effect;
- expiry-specific option pressure;
- one-jurisdiction legal rule;
- one-desk funding constraint;
- one-time liquidity shock.

Demanding universal invariance would erase real locality.

The correct label is:

ValidLocalStructure_P. (E.175)

not:

FailedUniversalLaw. (E.176)

The transport suite should therefore distinguish:

- local validity;
- family-level validity;
- broad invariance.

Scientific discipline does not require every useful claim to become universal.

It requires the scope to be honest.

---

# E.55 When Invariance Can Become Self-Fulfilling

A structure recognized across many frames may attract more observers.

For example:

- weekly resistance;
- high-volume node;
- VWAP cluster;
- option strike;
- widely reported technical level.

Cross-frame visibility may increase:

- conditional orders;
- media attention;
- algorithmic response;
- risk-management use.

Therefore:

ObservedInvariance
→ ObserverConvergence
→ IncreasedLedgerObjectivity. (E.177)

But the same convergence can create:

- crowding;
- stop concentration;
- adversarial attack;
- violent failure.

Thus:

CrossFrameRecognition
can strengthen both BoundaryMass and FailureConsequence. (E.178)

The source Technical Analysis article emphasizes this recursive duality: a signal can become true because it is observed, or fail because it becomes over-observed.

---

# E.56 Transport and the Periodic Table

Every period has characteristic transport problems.

| Period | Principal transport burden |
|---|---|
| **Mark** | venue, feed, latency, trade-sign rule |
| **Window** | bar construction, session boundary, price field |
| **Structure** | timeframe, parameter, scale, benchmark |
| **Event** | gate authority, volume, breadth, higher frame |
| **Episode** | pivot rule, segmentation, phase, branch |
| **World** | observer role, legal entity, accounting, policy, ledger |

Every functional family also has characteristic tests.

| Family | Principal transport question |
|---|---|
| **Load** | Does the remembered or loaded structure survive another measurement channel? |
| **Motion** | Does direction or feedback survive normalization and benchmark change? |
| **Constraint** | Does the boundary survive scale, anchor, and institutional frame? |
| **Commitment** | Do other relevant gates recognize an equivalent event? |

---

# E.57 The Transport-Invariance Ladder

A claim may advance through the following ladder.

## Level 0 — Untested appearance

One frame, one observation.

## Level 1 — Parameter robustness

Survives minor parameter variation.

## Level 2 — Protocol-family robustness

Survives related bar, scale, or anchor choices.

## Level 3 — Cross-functional support

Survives price, volume, breadth, or density translation.

## Level 4 — Cross-period support

Survives promotion into a higher closure level.

## Level 5 — Cross-observer support

Compatible observers recognize the relation.

## Level 6 — Cross-ledger consequence

The relation enters multiple consequential ledgers.

## Level 7 — Candidate invariant

The relation survives broad admissible declarations and repeatable gates.

This ladder is a grading system, not a claim that every method should reach Level 7.

---

# E.58 Transport Failure as Information

A failed transport test can reveal:

- hidden frame dependence;
- concentration;
- stale memory;
- wrong anchor;
- unstable phase;
- local-only validity;
- institutional mismatch;
- data-quality problem;
- real regime transition.

Therefore:

TransportFailure
= DiagnosticEvent, not merely negative score. (E.179)

The failed mapping should enter the residual ledger and may motivate a revised declaration.

---

# E.59 Transport-Informed Revision

A revision triggered by transport failure should answer:

1. Which relation failed?
2. Which frame exposed the failure?
3. Was the source claim false or only local?
4. Was the mapping incorrect?
5. Did the object itself change?
6. What residual remains?
7. What simpler claim survives?

Example:

Original claim:

“This is a major market breakout.”

Transport result:

Only cap-weighted price confirms; breadth and equal-weight frames fail.

Admissible revision:

“This is a concentrated headline-index breakout, not yet a broad market breakout.”

The revised claim is narrower and better supported.

---

# E.60 Minimum Publication Standard

A published Technical Analysis claim using the language of robustness, objectivity, invariance, or cross-frame confirmation should state:

- source protocol;
- target protocols;
- transformation rules;
- tolerances;
- survival results;
- failed frames;
- residual;
- effect on claim scope.

A statement such as:

“The level works on multiple timeframes”

is insufficient without defining:

- which timeframes;
- what counts as the same level;
- zone tolerance;
- whether closes or wicks are used;
- whether price is adjusted;
- whether the test was prospective.

---

# E.61 Appendix E Conclusion

A market structure is not made objective by removing observers.

It becomes more operationally objective when its trace can be transported across compatible observers, frames, and ledgers without losing the relation that made the claim meaningful.

The core transport equation is:

$T_{P \to P'}(C_P) \approx C_{P'}$. (E.180)

The core invariance rule is:

Invariant
= Relation preserved across admissible transformations. (E.181)

The core scope rule is:

FailureToTransport
→ localize, weaken, revise, or reject the claim. (E.182)

The core governance rule is:

Transport residual must remain in the ledger. (E.183)

The complete audit is:

Declare source frame
→ Define target frame
→ Specify transformation
→ Predict transported form
→ Observe destination claim
→ Measure discrepancy
→ Record residual
→ Revise claim scope. (E.184)

This turns cross-frame confirmation from a visual intuition into a reproducible research operation.

The next appendix can formalize the **Complex-Eligibility and Phase-Time Validation Checklist**, including real-pair benchmarks, dimensional tests, scaling robustness, phase-order stability, episode alignment, gate-hazard concentration, and reduction triggers.

# Appendix F — Complex Eligibility and Phase-Time Validation Checklist

## F.1 Purpose

A complex market model can always be written syntactically:

$Z = R + iQ$. (F.1)

But syntax alone does not establish that the market state is naturally complex.

At one instant:

$Z \leftrightarrow (R,Q)$. (F.2)

The complex notation contains no more raw numerical information than the ordered real pair.

A complex representation earns priority only when the additional structure of complex

arithmetic provides a stable and operationally useful organization of the state.

Possible gains include:

- a meaningful amplitude;
- a robust phase;
- a stable rotational generator;
- improved episode alignment;
- concentration of event gates by phase;
- simpler dynamical laws;
- better transport across comparable cases;
- improved intervention timing.

The source phase framework requires complex models to be compared with scalar and flexible two-real-variable alternatives, and it rejects complexification when Q is merely an error bucket or when phase fails to improve explanation, prediction, or control.

This appendix provides a staged validation procedure.

---

# F.2 The Evidence Ladder

A proposed model should pass through seven levels.

| Level | Model status | Required evidence |
|---|---|---|
| 0 | Scalar | one variable sufficiently describes the task |
| 1 | Real pair | two independently meaningful variables improve the task |
| 2 | Complex-eligible pair | units, scaling, and coupling justify R + iQ |
| 3 | Phase-bearing dynamics | phase simplifies or stabilizes the dynamics |
| 4 | Secondary phase time | internal episode order is clearer in $\tau_i$ than in t |
| 5 | Phase-sensitive event model | consequential gates concentrate by phase |
| 6 | Time-bearing world | gates write trace that changes later dynamics |

The model should advance only one level at a time.

Failure at a higher level does not automatically invalidate the lower levels.

For example:

- phase time may fail;
- the complex dynamical model may still be useful;
- if complex dynamics also fail, the real pair may remain useful.

The reduction ladder is:

TimeBearingWorld
→ PhaseSensitiveEventModel

→ SecondaryPhaseTime
→ ComplexDynamics
→ RealPair
→ Scalar. (F.3)

---

# F.3 Validation Rule

For every proposed complex state, the burden of proof is:

ComplexModelGain

RealPairGain + ComplexityPenalty. (F.4)

The comparison should include:

- predictive performance;
- calibration;
- parameter stability;
- interpretability;
- transport robustness;
- gate localization;
- intervention value;
- computational cost.

A complex model should not be preferred merely because its phase portrait is visually compelling.

---

# F.4 Stage 0 — Declare the Scientific Task

Before defining R or Q, specify what the model is intended to improve.

Possible tasks include:

- classify trend versus correction;
- predict breakout persistence;
- align crisis episodes;
- estimate gate hazard;
- identify valuation exposure;
- improve hedge timing;
- compare protocols;
- detect regime transition.

The task declaration is:

Task_P = (Target,Horizon,LossFunction,DecisionContext). (F.5)

where:

Target = what is being estimated;
Horizon = when success is evaluated;
LossFunction = how error is measured;
DecisionContext = how the result will be used.

Without a declared task, complex-model superiority cannot be tested.

---

# F.5 Stage 1 — Define R Independently

# F.5.1 Requirement

R must have a domain meaning independent of the proposed complex notation.

Valid examples may include:

- admitted discounted value;
- accepted price structure;
- realized physical output;
- verified system performance;
- confirmed participation state.

The definition must state:

R = ObservableOrEstimate(Data,Protocol). (F.6)

It should be measurable without first calculating Q or θ.

---

# F.5.2 R declaration card

```
Name of R:
Domain meaning:
Data source:
Units:
Observation protocol:
Estimation operator:
Uncertainty:
Update frequency:
Gate relevance:
```

---

# F.5.3 Failure conditions

R fails independence when:

- it is defined using the desired phase result;

- it changes meaning across episodes;
- it is a narrative label rather than a measurable variable;
- its units are unspecified;
- it cannot be reconstructed from the declared data.

---

# F.6 Stage 2 — Define Q Independently

# F.6.1 Requirement

Q must also possess an independent domain meaning.

A weak definition is:

Q = everything invisible or unexplained. (F.7)

A stronger definition is:

Q = independently measured channel with a stable coupling to R. (F.8)

Possible candidate Q channels in market research might include:

- independently estimated liquidity pressure;
- breadth-pressure coordinate;
- verified reactive-order-flow channel;
- collateral stress;
- conjugate valuation exposure;
- independently measured phase-lag channel.

The phase source explicitly rejects defining Q as the unexplained remainder of R. Unexplained content must remain residual.

---

# F.6.2 Q declaration card

```
Name of Q:
Domain meaning:
Independent data source:
Units:
Estimation operator:
Relationship to R:
Expected sign convention:
Uncertainty:
Known omitted variables:
```

---

# F.6.3 Independence tests

### Data independence

Does Q use a genuinely different data channel?

### Construction independence

Can Q be estimated without using the later target?

### Semantic independence

Does Q retain its meaning when the model fails?

### Forward independence

Does Q contribute conditional information beyond R?

A statistical test is:

Information(Y | R,Q) > Information(Y | R). (F.9)

or:

Loss(Model[R,Q]) < Loss(Model[R]). (F.10)

out of sample.

---

## F.6.4 Residual separation

The observed outcome should be written:

$Y_{observed} = Y_{model}(R,Q) + \varepsilon$. (F.11)

where ε remains residual.

The model is invalidly closed if:

$Q := \varepsilon$. (F.12)

because the proposed conjugate coordinate then expands whenever the model fails.

---

# F.7 Stage 3 — Dimensional Compatibility

## F.7.1 Equal-unit requirement

A direct complex sum requires:

Units(R) = Units(Q). (F.13)

If R and Q have different native units, a conversion or normalization is required.

Let:

$\tilde{R} = (R - \mu_R)/s_R$. (F.14)

$\tilde{Q} = (Q - \mu_Q)/s_Q$. (F.15)

Then:

$\tilde{Z} = \tilde{R} + i\tilde{Q}$. (F.16)

The scales s_R and s_Q must be economically or operationally justified.

---

## F.7.2 Valid scaling sources

Possible defensible scales include:

- common currency units;
- risk-equivalent units;
- physical conversion law;
- theoretically derived metric;
- stable empirical calibration;
- protocol-defined normalization;
- unit-variance scaling used only for statistical comparison.

The chosen scale should be declared before testing phase outcomes.

---

## F.7.3 Invalid scaling practices

The following weaken the model:

- choosing s_Q to make the orbit circular;
- choosing scales after observing gate locations;
- changing normalization from episode to episode;
- using incompatible units without a metric;
- suppressing scale sensitivity in the report.

---

# F.8 Stage 4 — Norm and Amplitude Test

## F.8.1 Candidate amplitude

The ordinary complex amplitude is:

$A = \sqrt{(R^2 + Q^2)}$. (F.17)

A valid model should explain what A means.

Possible meanings include:

- total declared valuation amplitude;

- conserved or slowly varying system capacity;
- combined state magnitude;
- normalized state intensity.

A should not be introduced merely because complex arithmetic supplies it.

---

# F.8.2 Amplitude questions

1. Is A independently meaningful?
2. Is A constant, slowly varying, or dynamic?
3. What changes A?
4. What changes phase while approximately preserving A?
5. Does A predict anything beyond R and Q separately?
6. Does the norm survive admissible rescaling?

---

# F.8.3 Circular versus noncircular geometry

If:

$R^2 + Q^2 = A^2$ (F.18)

is theoretically derived, circular geometry may be justified.

If the relation exists only after arbitrary normalization, a more general metric may be appropriate:

$A^2 = [R,Q]M[R,Q]^T$. (F.19)

where M is a positive-definite metric.

A general ellipse may be more honest than forcing a circle.

The metric must itself be estimated and validated.

---

# F.8.4 CAPM calibration

In the CAPM construction:

$A_t = CF_t/(1 + r_{base})^t$. (F.20)

$R_t = CF_t/(1 + r_{CAPM})^t$. (F.21)

$Q_t = \sqrt{(A_t^2 - R_t^2)}$. (F.22)

Therefore:

$A_t^2 = R_t^2 + Q_t^2$. (F.23)

Here the norm relation is not introduced through statistical scaling.

It follows from the declared valuation geometry.

This is one reason the CAPM model serves as a calibration atom.

---

# F.9 Stage 5 — Coupling and Generator Test

# F.9.1 Why a real pair is not yet complex

Two real variables become meaningfully complex only when their evolution supports a privileged coupling.

The canonical complex generator is:

$$J = [[0,-1],[1,0]]. \quad (F.24)$$

with:

$$J^2 = -I. \quad (F.25)$$

For state:

$$x = [R,Q]^T, \quad (F.26)$$

the generator gives:

$$Jx = [-Q,R]^T. \quad (F.27)$$

A phase-bearing state should approximately satisfy:

$$dx/d\theta \approx Jx. \quad (F.28)$$

equivalently:

$$dR/d\theta \approx -Q. \quad (F.29)$$

$$dQ/d\theta \approx R. \quad (F.30)$$

---

# F.9.2 Empirical generator test

Estimate local derivatives:

$$\dot{R}_t = dR/dt. \quad (F.31)$$

$$\dot{Q}_t = dQ/dt. \quad (F.32)$$

For a phase velocity $\omega_t$:

$$\dot{R}_t \approx -\omega_t Q_t. \quad (F.33)$$

$$\dot{Q}_t \approx \omega_t R_t. \quad (F.34)$$

Estimate residuals:

$\varepsilon_{R,t} = \dot{R}_t + \omega_t Q_t$. (F.35)

$\varepsilon_{Q,t} = \dot{Q}_t - \omega_t R_t$. (F.36)

A complex rotational model gains support when these residuals are:

- small;
- stable;
- out-of-sample;
- lower than those of plausible alternatives.

---

# F.9.3 Amplitude-changing dynamics

A more general model is:

$dZ/dt = [g_A(t) + i\omega(t)]Z + \varepsilon(t)$. (F.37)

where:

$g_A(t)$ = amplitude growth or decay;
$\omega(t)$ = phase velocity;
$\varepsilon(t)$ = residual.

In real form:

$\dot{R} = g_A R - \omega Q + \varepsilon_R$. (F.38)

$\dot{Q} = \omega R + g_A Q + \varepsilon_Q$. (F.39)

This separates:

- radial growth;
- angular motion;
- unexplained dynamics.

---

# F.9.4 Competing generators

The signed-regime framework warns that not every market relation is rotational.

Other local normal forms include:

### Elliptic

$J^2 = -I$. (F.40)

### Parabolic

$N^2 = 0$. (F.41)

### Hyperbolic

$K^2 = +I$. (F.42)

A self-confirming market may be better represented by a hyperbolic generator than by ordinary complex rotation.

Therefore the test should compare:

- elliptic;
- parabolic;
- hyperbolic;
- unrestricted real-vector dynamics.

A failed complex model may indicate the wrong algebra rather than the absence of structure.

---

# F.10 Stage 6 — Phase Definition

# F.10.1 Phase coordinate

For eligible R and Q:

$\theta = \operatorname{atan2}(Q,R)$. (F.43)

The phase should possess an operational interpretation.

Examples include:

- valuation orientation;
- relationship between accepted and conjugate state;
- internal cycle position;
- stress progression;
- conversion between paired channels.

---

# F.10.2 Branch declaration

Because phase is periodic:

$\theta \equiv \theta + 2\pi n$. (F.44)

The implementation must declare:

- principal branch;
- unwrapping rule;
- discontinuity treatment;

- zero-amplitude handling;
- missing-data handling.

At:

$A \approx 0$, (F.45)

phase becomes unstable or undefined.

Such intervals should be flagged rather than silently interpolated.

---

# F.10.3 Phase confidence

A phase-confidence score may combine:

$S_\theta$
$= w_A$AmplitudeReliability

- $w_S$ScalingRobustness
- $w_G$GeneratorFit
- $w_O$OrderStability
  – $w_R$ResidualBurden. (F.46)

This is a research template.

Phase should not be reported without uncertainty.

---

# F.11 Stage 7 — Scaling-Robustness Test

# F.11.1 Perturbation family

Let:

$Q_c = cQ$. (F.47)

Then:

$\theta_c = \text{atan2}(cQ,R)$. (F.48)

Choose a declared admissible family:

$c \in \mathcal{C}_{adm}$. (F.49)

The phase model should be tested under all c in that family.

---

# F.11.2 Three robustness levels

### Metric robustness

Angular values remain close:

$|\theta_c - \theta_1| \le \varepsilon_\theta$. (F.50)

### Order robustness

Relative phase ordering remains stable:

$\mathrm{Rank}(\theta_c(t)) \approx \mathrm{Rank}(\theta_1(t))$. (F.51)

### Gate robustness

Phase-based event classification remains stable:

$\mathrm{GatePhaseClass}_c(t) = \mathrm{GatePhaseClass}_1(t)$. (F.52)

A model may fail metric robustness while preserving order or gate robustness.

That distinction should be reported.

---

## F.11.3 Phase-scale score

One possible score is:

$$S_{scale} = 1 - \mathrm{Mean}_c[\mathrm{Dist}_{circular}(\theta_c,\theta_1)]/\pi. \quad (F.53)$$

where:

$0 \le S_{scale} \le 1$. (F.54)

The exact score is implementation-dependent.

---

# F.12 Stage 8 — Real-Pair Benchmark

## F.12.1 Required benchmark

The complex model should be compared with:

$\mathrm{Model}_{pair} = f(R,Q,R_{lags},Q_{lags},RQ,\mathrm{Controls})$. (F.55)

The complex model is:

$\mathrm{Model}_{complex} = g(A,\theta,A_{lags},\theta_{lags},\mathrm{Controls})$. (F.56)

Both models should have comparable flexibility.

A weak benchmark such as linear R-only regression is insufficient.

---

## F.12.2 Comparison criteria

Compare:

- out-of-sample loss;
- calibration;
- parameter count;
- parameter stability;
- sensitivity to missing data;
- robustness across regimes;
- interpretation;
- computation;
- gate prediction.

Complex priority requires at least one meaningful gain without unacceptable losses elsewhere.

---

## F.12.3 Nested information principle

If:

A and θ are invertible functions of R and Q, (F.57)

then any gain from the complex model must arise from:

- inductive bias;
- reduced parameterization;
- stable phase structure;
- easier transport;
- better event alignment;
- more suitable regularization.

It cannot arise from additional raw information at one instant.

---

## F.12.4 Decision rule

Promote the complex model only when:

ΔPerformance

- ΔStability
- ΔInterpretability
- ΔOperationalValue

  ComplexityPenalty. (F.58)

Otherwise retain the real pair.

---

# F.13 Stage 9 — Phase-Order Stability

## F.13.1 Order rather than exact angle

A phase model may remain useful even when exact angles vary across normalizations.

The weaker requirement is preservation of event ordering.

Suppose events occur at:

$\theta_1,\theta_2,\ldots,\theta_n$. (F.59)

Under admissible transport:

$\theta'_1,\theta'_2,\ldots,\theta'_n$. (F.60)

Phase order survives when the circular ordering remains equivalent.

---

## F.13.2 Order metrics

Possible metrics include:

- circular rank correlation;
- Kendall-style order agreement after unwrapping;
- phase-locking value;
- sequence-edit distance;
- gate-order consistency.

Define:

$S_{order} = \text{Agreement}(\text{Order}_\theta,\text{Order}_{\theta'})$. (F.61)

A model with low $S_{order}$ should not support an internal clock.

---

## F.13.3 Direction test

The model should also preserve traversal direction:

$\operatorname{sgn}(\dot{\theta}_t) = \operatorname{sgn}(\dot{\theta}'_t)$ (F.62)

within tolerance.

The same phase reached in opposite directions may represent different market states.

---

# F.14 Stage 10 — Episode Definition

## F.14.1 Comparable episodes

Phase-time validation requires a family of comparable episodes.

Examples include:

- breakouts from similar ranges;
- credit-stress episodes;
- valuation-compression episodes;
- liquidity crises;
- trend reversals;
- accumulation-to-expansion sequences.

The episode family must be defined without using the phase result being tested.

---

## F.14.2 Episode boundaries

For episode j:

E_j = [t_start,j,t_end,j]. (F.63)

The start and end rules must be declared.

Possible rules include:

- structural gate;
- institutional event;
- volatility transition;
- pivot criterion;
- legal or accounting recognition.

Retrospective hand-selection weakens the test.

---

## F.14.3 Episode inclusion criteria

The record should state:

- required initial state;
- minimum data quality;
- asset class;
- market regime;
- boundary type;

- exclusion conditions.

A phase clock cannot be validated by combining structurally unrelated episodes.

---

# F.15 Stage 11 — Calendar Alignment Benchmark

## F.15.1 Normalized calendar time

For episode j:

$u_j(t) = [t - t_{start,j}]/[t_{end,j} - t_{start,j}]$. (F.64)

Then:

$u_j \in [0,1]$. (F.65)

This provides a basic calendar-normalized benchmark.

---

## F.15.2 Event-count benchmark

A second benchmark is event order:

$k_j = 0,1,\ldots,K_j$. (F.66)

Normalize:

$\kappa_j = k_j/K_j$. (F.67)

The phase clock must be compared with both:

- normalized calendar time;
- normalized event count.

If event order performs equally well, complex phase may be unnecessary.

---

# F.16 Stage 12 — Internal Phase Time

## F.16.1 Signed internal time

$\tau_{i,j}^{signed}(t) = \mathrm{Unwrap}[\theta_j(t)]$. (F.68)

This retains direction and cumulative turns.

---

## F.16.2 Absolute phase depth

$\tau_{i,j}^{abs}(t) = \int_{t_{start,j}}^{t} |\dot{\theta}_j(s)| ds$. (F.69)

This is monotone but loses direction.

---

# F.16.3 Normalized phase time

For cross-episode comparison:

$$\upsilon_j(t) = [\tau_{i,j}(t) - \tau_{i,j}(t_{start})] / [\tau_{i,j}(t_{end}) - \tau_{i,j}(t_{start})]. \quad (F.70)$$

Then:

$$\upsilon_j \in [0,1]. \quad (F.71)$$

This normalization should not be used prospectively unless the episode endpoint is already independently defined.

For online use, a rolling or expected-total-phase model is required.

---

# F.17 Stage 13 — Episode Alignment Test

# F.17.1 Dispersion comparison

Let $X_j$ be an episode state vector.

Calculate:

$$D_{calendar} = \text{Dispersion}[\{X_j(u)\}]. \quad (F.72)$$

$$D_{event} = \text{Dispersion}[\{X_j(\kappa)\}]. \quad (F.73)$$

$$D_{phase} = \text{Dispersion}[\{X_j(\upsilon)\}]. \quad (F.74)$$

The phase-time model gains support when:

$$D_{phase} < \min(D_{calendar}, D_{event}). \quad (F.75)$$

The comparison must be out of sample or cross-validated.

---

# F.17.2 Possible dispersion measures

- mean squared trajectory distance;
- dynamic time-warping distance;
- Wasserstein distance;
- functional principal-component variance;
- event-alignment error;
- gate-time variance.

The metric should be selected before inspecting the result.

---

## F.17.3 Cross-validation

Possible procedures include:

- leave-one-episode-out;
- train/test by date;
- train/test by asset;
- train/test by regime;
- nested cross-validation for normalization choices.

A phase clock validated on the same episodes used to define R and Q is vulnerable to circularity.

---

## F.17.4 Null models

Compare with:

- random monotone clocks;
- event-count clocks;
- cumulative volatility;
- cumulative volume;
- cumulative absolute return;
- flexible learned monotone warping.

The phase clock must outperform plausible alternative internal clocks.

---

# F.18 Stage 14 — Gate-Hazard Concentration

## F.18.1 Event hazard

Let G_t be an indicator that a consequential gate occurs.

The calendar hazard is:

h_t = Pr(G_t = 1 | CalendarAge,Controls). (F.76)

The phase hazard is:

h_θ = Pr(G_t = 1 | θ_t,Controls). (F.77)

The internal-time hazard is:

h_τ = Pr(G_t = 1 | $\tau_i$,t,Controls). (F.78)

## F.18.2 Incremental phase value

Phase contributes useful information when:

Information(G | θ,Controls)

Information(G | Controls). (F.79)

or:

Loss[h_θ] < Loss[h_t] (F.80)

out of sample.

## F.18.3 Phase-hazard concentration

A simple concentration score is:

C_G
= Var_θ[h_G(θ)]/Mean_θ[h_G(θ)]. (F.81)

A higher value may indicate that gates cluster in particular phase regions.

But concentration alone is insufficient.

The clusters must:

- replicate;
- survive scaling;
- survive episode selection;
- improve prospective prediction.

## F.18.4 Circular hazard model

A circular regression may use:

logit Pr(G_t = 1)
= $\beta_0$ + $\beta_1$cos θ_t + $\beta_2$sin θ_t + $\beta^T$X_t. (F.82)

where X_t contains conventional controls.

The phase terms add value only if:

$\beta_1$ or $\beta_2$ remains stable and predictive out of sample. (F.83)

## F.18.5 Gate classes

Phase may predict different gate types separately:

- breakout;
- rejection;
- impairment;
- default;
- settlement;
- policy intervention;
- recovery;
- episode completion.

Combining all event types into one target may hide useful phase structure.

---

# F.19 Stage 15 — Selection-Depth Comparison

# F.19.1 Selection depth

Let Ω_t be the set of futures still treated as admissible.

A conceptual selection depth is:

σ_t = −ln[Measure(Ω_t)/Measure(Ω_0)]. (F.84)

Phase time and selection depth are different.

A system may rotate without suppressing many alternatives.

A single gate may eliminate many alternatives with little phase traversal.

---

# F.19.2 Comparative test

Compare:

Gate prediction from $\tau_i$.
Gate prediction from σ.
Gate prediction from $(\tau_i,\sigma)$. (F.85)

Possible outcomes include:

### Phase dominates

Internal rotation organizes the episode.

### Selection depth dominates

Branch elimination matters more than phase.

### Joint model dominates

Rotation and closure carry distinct information.

### Neither dominates

Simpler state variables are sufficient.

---

# F.20 Stage 16 — Ledger Test

# F.20.1 A phase-sensitive event is not yet a world

Even if gates cluster by phase, the model has not yet established a time-bearing world.

The admitted event must enter persistent trace:

L_{k+1} = Update(L_k,e_k,r_k). (F.86)

The trace must then alter subsequent conditions.

---

# F.20.2 Ledger criteria

A valid ledger should preserve:

- gate time;
- gate authority;
- event type;
- phase at admission;
- residual;
- later rule changes;
- consequences for future state.

A mere event list is insufficient if later dynamics ignore it.

---

# F.20.3 Ledger dependence test

Test whether future dynamics depend on ledger state after controlling for current observable state.

Let X_t be the current state.

Let L_t be the prior event ledger.

The test is:

Information(X_{t+h} | X_t,L_t)

Information(X_{t+h} | X_t). (F.87)

If L_t adds no information, the process may be approximately Markovian in X_t.

The claimed historical world is weakened.

---

# F.21 Stage 17 — Backreaction Test

# F.21.1 Definition

Backreaction exists when recorded interpretation or gate recognition changes the system being observed.

Let:

A_t = action induced by gate or ledger. (F.88)

Then:

Σ_{t+1} = F(Σ_t,A_t,External_t). (F.89)

The next complex state becomes:

Z_{t+1} = Project_Z(Σ_{t+1}). (F.90)

---

# F.21.2 Possible market backreaction channels

- portfolio rebalancing;
- margin call;
- accounting recognition;
- hedge activation;
- benchmark inclusion;
- legal default;
- public signal adoption;
- policy response;
- liquidity withdrawal;
- collateral reclassification.

---

# F.21.3 Backreaction test

Compare similar gates with differing recognition or intervention strength.

Estimate:

ΔFutureDynamics
= f(GateRecognition,ObserverAction,Controls). (F.91)

A strong world-forming model requires stable evidence that:

Gate → Trace → ChangedFutureDynamics. (F.92)

Without this, the model remains a phase-bearing event model rather than a time-bearing world.

---

# F.22 Stage 18 — Intervention Test

## F.22.1 Why intervention matters

A model may be descriptively elegant but operationally irrelevant.

A stronger test asks whether phase-aware action improves outcomes.

Let policy $\pi_{phase}$ use $\theta$ or $\tau_i$.

Let policy $\pi_{base}$ use ordinary variables or calendar time.

Compare:

$Loss(\pi_{phase}) < Loss(\pi_{base})$. (F.93)

---

## F.22.2 Possible financial interventions

- hedge activation;
- exposure reduction;
- position sizing;
- liquidity preparation;
- review frequency;
- gate tightening;
- stop adjustment;
- scenario escalation.

These applications must include:

- transaction cost;
- market impact;
- latency;
- regulatory limits;
- risk budget.

---

## F.22.3 Intervention leakage

Actions based on the phase model may change the phase dynamics themselves.

Therefore prospective evaluation should record:

- model adoption;
- capital linked to policy;
- crowding;
- feedback;
- post-adoption degradation.

A phase rule that works only before it becomes widely used may be observer-sensitive rather than invariant.

---

# F.23 Stage 19 — Residual Audit

# F.23.1 Complex residual

For model:

$$dZ/dt = [g_A + i\omega]Z + \varepsilon, \quad (F.94)$$

the residual is:

$$\varepsilon = \varepsilon_R + i\varepsilon_Q. \quad (F.95)$$

The residual should be analyzed by:

- magnitude;
- direction;
- persistence;
- regime;
- gate proximity;
- data quality;
- institutional event.

---

# F.23.2 Residual ratio

Define:

$$\eta_\varepsilon = \|\varepsilon\|/[\|[g_A + i\omega]Z\| + \delta]. \quad (F.96)$$

A large $\eta_\varepsilon$ indicates that the proposed phase dynamics explain little of the local movement.

The threshold must be calibrated.

## F.23.3 Structured residual

Residual may contain systematic structure.

Test whether:

$$E[\varepsilon_t \mid X_t] \neq 0. \quad (F.97)$$

or:

$$\text{Autocorrelation}(\varepsilon_t) \neq 0. \quad (F.98)$$

or:

GateHazard depends on $\varepsilon_t$. (F.99)

Systematic residual indicates:

- missing variable;
- wrong generator;
- wrong regime;
- wrong scaling;
- protocol change.

---

## F.23.4 Residual honesty

A successful phase model should still report:

- episodes that do not align;
- gates outside predicted phase regions;
- unstable branches;
- periods with undefined phase;
- failed interventions;
- alternative real-pair explanations.

Complex elegance does not eliminate residual governance.

---

# F.24 Stage 20 — Reduction Triggers

The model must be reduced when one or more declared triggers occur.

# Trigger R1 — Q independence fails

Q cannot be measured independently.

Reduce to:

R + residual. (F.100)

---

## Trigger R2 — Dimensional coherence fails

No defensible conversion between R and Q exists.

Retain:

heterogeneous real-vector model. (F.101)

---

## Trigger R3 — Generator fit fails

The rotational coupling does not outperform unrestricted real dynamics.

Retain:

real pair. (F.102)

---

## Trigger R4 — Phase scaling fails

Minor normalization changes reorder phase materially.

Retain:

real pair or ordinal state classes. (F.103)

---

## Trigger R5 — Episode alignment fails

$\tau_i$ does not outperform calendar or event-count alignment.

Retain:

complex dynamics without phase-time claim. (F.104)

---

## Trigger R6 — Gate concentration fails

Consequential events do not depend incrementally on phase.

Retain:

secondary ordering or descriptive phase only. (F.105)

---

## Trigger R7 — Ledger dependence fails

Past admitted events do not alter future dynamics beyond current state.

Retain:

phase-sensitive event model without world claim. (F.106)

---

## Trigger R8 — Backreaction fails

Recognition does not change the system.

Retain:

external-observer phase model. (F.107)

---

## Trigger R9 — Real-pair benchmark matches

Complex representation supplies no measurable gain.

Retain:

(R,Q). (F.108)

---

## Trigger R10 — Scalar benchmark matches

Q supplies no incremental value.

Retain:

R. (F.109)

---

## F.25 Complex-Eligibility Scorecard

A provisional scorecard can organize the evidence.

| Criterion | 0 | 1 | 2 |
|---|---|---|---|
| Independent R definition | absent | partial | strong |
| Independent Q definition | absent | proxy only | strong |
| Unit compatibility | absent | normalized | theoretically grounded |
| Stable coupling | absent | local | replicated |
| Generator fit | poor | comparable | superior |
| Scaling robustness | poor | order survives | metric survives |
| Real-pair superiority | none | small | material |
| Episode alignment | none | in-sample | out-of-sample |
| Gate concentration | none | descriptive | predictive |
| Ledger dependence | absent | partial | strong |
| Backreaction | absent | plausible | demonstrated |
| Intervention value | none | simulated | prospective |

Maximum score:

$S_{max} = 24$. (F.110)

The score is not a universal scientific threshold.

A suggested classification is:

0–5 → scalar or descriptive pair.
6–10 → useful real pair.
11–15 → complex-eligible research model.
16–19 → phase-bearing model.
20–22 → phase-sensitive event model.
23–24 → candidate time-bearing world. (F.111)

These ranges are provisional and should not replace criterion-specific judgment.

---

# F.26 Mandatory Pass Conditions

Regardless of total score, several criteria are mandatory.

# For complex eligibility

- independent R;
- independent Q;
- dimensional compatibility;
- real-pair benchmark.

# For phase dynamics

- stable phase;
- coupling or generator evidence;
- scaling robustness.

# For phase time

- comparable episodes;
- out-of-sample alignment gain;
- superiority over event-count and other clocks.

# For phase-sensitive events

- predeclared gate;
- phase-conditioned hazard gain;
- transport robustness.

## For a time-bearing world

- persistent ledger;
- future-state dependence;
- backreaction.

A high total score cannot compensate for a missing mandatory condition.

---

# F.27 Complex-Eligibility Decision Tree

```
Are R and Q independently defined?
│
├─ No → Keep scalar/vector model; record unexplained content as residual.
│
└─ Yes
   │
   ├─ Do R and Q have compatible units or a defensible metric?
   │  ├─ No → Use heterogeneous real vector.
   │  └─ Yes
   │     │
   │     ├─ Does a complex generator simplify or stabilize dynamics?
   │     │  ├─ No → Retain real pair.
   │     │  └─ Yes
   │     │     │
   │     │     ├─ Is phase robust to scaling and frame changes?
   │     │     │  ├─ No → Retain real pair or local phase description.
   │     │     │  └─ Yes
   │     │     │     │
   │     │     │     ├─ Does phase align episodes better than t or k?
   │     │     │     │  ├─ No → Complex dynamics only.
   │     │     │     │  └─ Yes
   │     │     │     │     │
   │     │     │     │     ├─ Do gates concentrate by phase?
   │     │     │     │     │  ├─ No → Secondary phase time only.
   │     │     │     │     │  └─ Yes
   │     │     │     │     │     │
   │     │     │     │     │     ├─ Does trace alter future dynamics?
   │     │     │     │     │     │  ├─ No → Phase-sensitive event model.
   │     │     │     │     │     │  └─ Yes → Candidate time-bearing world.
```

---

# F.28 Prospective Validation Protocol

## Step 1 — Freeze the protocol

Declare:

- data;
- R;
- Q;
- normalization;
- episodes;

- gates;
- horizon;
- benchmark models;
- outcome metrics.

## Step 2 — Separate training and test periods

All normalization and phase-construction choices should be fitted only on training data.

## Step 3 — Fit scalar baseline

Estimate:

Model_R. (F.112)

## Step 4 — Fit real-pair baseline

Estimate:

Model_RQ. (F.113)

## Step 5 — Fit complex model

Estimate:

Model_Z. (F.114)

## Step 6 — Test phase robustness

Apply:

- scale perturbation;
- timeframe transport;
- Q-proxy alternatives;
- episode variation.

## Step 7 — Test episode alignment

Compare:

t; k; $\tau_i$; σ; cumulative volatility; cumulative volume.

## Step 8 — Test gate hazard

Evaluate phase terms beyond conventional controls.

## Step 9 — Test ledger dependence

Determine whether past gates alter future conditional dynamics.

## Step 10 — Test intervention

Compare prospective policies where ethically and operationally appropriate.

## Step 11 — Preserve failures

Every failed episode and unstable phase branch enters the residual ledger.

## Step 12 — Apply reduction rule

Use the lowest model level supported by the evidence.

---

# F.29 Copy-Ready Complex Model Registration

```
Model ID:
Research task:
Target:
Outcome horizon:
Loss function:

Protocol:
Asset or system:
Boundary:
Data source:
Timeframe:
Episode definition:

R definition:
R units:
R estimator:
R uncertainty:

Q definition:
Q units:
Q estimator:
Q independent data:
Q uncertainty:

Normalization:
Metric:
Amplitude definition:
Phase definition:
Branch rule:
Unwrapping rule:

Scalar benchmark:
Real-pair benchmark:
Alternative generators:
Null clocks:

Gate definition:
Gate authority:
Gate outcome:
Ledger rule:
```

```
Backreaction hypothesis:

Scaling tests:
Frame tests:
Episode-alignment metric:
Gate-hazard metric:
Intervention metric:

Residual ontology:
Reduction triggers:
Prospective test period:
```

---

# F.30 Copy-Ready Validation Results Table

| Criterion | Test | Result | Status | Residual |
|---|---|---|---|---|
| R independence | independent measurement audit | passed | strong | measurement noise |
| Q independence | conditional-information test | partial | provisional | shared price input |
| Unit compatibility | declared risk-equivalent scaling | passed | acceptable | scale sensitivity |
| Generator fit | complex versus vector dynamics | complex slightly better | provisional | regime variation |
| Scaling robustness | $c \in [0.8,1.2]$ | order stable | partial | angle variation |
| Episode alignment | $\tau_i$ versus t and k | $\tau_i$ better in sample only | weak | overfit risk |
| Gate hazard | circular regression | no out-of-sample gain | failed | phase not event-bearing |
| Ledger dependence | lagged ledger test | not tested | open | data unavailable |
| Backreaction | adoption study | not tested | open | causal identification |
| Model status | reduction rule | real pair retained | final | complex claim rejected |

---

# F.31 CAPM Calibration Audit

# F.31.1 Task

Represent the difference between baseline and CAPM-adjusted valuation as a phase geometry.

---

# F.31.2 Declared variables

$A_t = CF_t/(1 + r_{base})^t$. (F.115)

$R_t = CF_t/(1 + r_{CAPM})^t$. (F.116)

$Q_t = \sqrt{(A_t^2 - R_t^2)}$. (F.117)

$Z_t = R_t + iQ_t$. (F.118)

---

## F.31.3 Independence status

R is independently defined by CAPM-adjusted discounting.

Q is not independently measured from separate market data.

It is algebraically derived from A and R.

Therefore:

CAPM Q is structurally meaningful inside the declared geometry, but not an independent empirical channel. (F.119)

This limits some claims but does not invalidate the internal identity.

---

## F.31.4 Dimensional status

A, R, and Q are all monetary values.

Therefore:

Units(A) = Units(R) = Units(Q). (F.120)

Dimensional compatibility passes.

---

## F.31.5 Generator status

The geometry gives:

$R = A \cos \theta$. (F.121)

$Q = A \sin \theta$. (F.122)

$\partial R/\partial \theta = -Q$. (F.123)

$\partial Q/\partial \theta = R$. (F.124)

The canonical complex generator is exact along the fixed-A orbit.

---

## F.31.6 Economic movement status

The model must distinguish measurement orientation from state movement.

The exact update is:

$R_{new} = R \cos \Delta\theta - Q \sin \Delta\theta$. (F.125)

The value change is:

$\Delta R = R(\cos \Delta\theta - 1) - Q \sin \Delta\theta$. (F.126)

For small $\Delta\theta$:

$\Delta R \approx -Q\Delta\theta$. (F.127)

Therefore Q is exposure, not automatically loss.

The CAPM source explicitly distinguishes phase-exposure readout, actual state movement, economic consequence, and recognition gate.

---

# F.31.7 Phase-time status

The static one-period construction does not by itself establish:

- a recurring dynamical episode;
- an internal clock;
- phase-conditioned gate hazard;
- backreaction.

Therefore the correct model status is:

Complex calibration geometry. (F.128)

not automatically:

Time-bearing financial world. (F.129)

Dynamic extension requires additional evidence.

---

# F.31.8 CAPM audit result

| Criterion | Status |
|---|---|
| R definition | strong |
| Q definition | algebraically derived |
| unit compatibility | exact |
| norm relation | exact |
| generator relation | exact |
| real-pair superiority | not automatically established |
| phase time | unestablished |
| gate concentration | requires external recognition protocol |
| ledger | externally supplied |
| backreaction | possible but not intrinsic to static model |

The CAPM state is therefore an excellent calibration atom for complex eligibility, while remaining a limited proof of broader market phase-time claims.

---

# F.32 Hypothetical TA Complex-State Audit

# F.32.1 Proposed state

$R_t$ = accepted normalized price trend. (F.130)

$Q_t$ = normalized breadth pressure. (F.131)

$Z_t = R_t + iQ_t$. (F.132)

---

## F.32.2 Initial appeal

The model attempts to represent:

- visible price trend;
- field-wide supporting or opposing participation;
- phase relation between headline price and market breadth.

---

## F.32.3 Independence test

R uses index price.

Q uses component-level breadth.

Source independence is moderate.

However, both ultimately derive from component prices.

Therefore:

Q is partially independent, not fully independent. (F.133)

---

## F.32.4 Dimensional test

R and Q are normalized dimensionless scores.

The normalization must be fixed and transport-tested.

---

## F.32.5 Generator test

Suppose estimated dynamics show:

$\dot{R} \approx -\omega Q$ only during range regimes. (F.134)

$\dot{Q} \approx \omega R$ poorly during trending regimes. (F.135)

The ordinary complex generator is local rather than universal.

This may indicate:

- elliptic behaviour under $\chi < 0$;
- hyperbolic behaviour under $\chi > 0$.

---

## F.32.6 Phase test

Phase is robust under small scaling changes in range regimes but unstable in trends.

The correct claim is:

Local corrective-regime complex eligibility. (F.136)

not:

Universal market complex state. (F.137)

---

## F.32.7 Episode alignment

Suppose phase aligns range rotations better than calendar time.

But breakout events do not concentrate at a stable phase.

The model earns:

Secondary phase ordering for corrective episodes. (F.138)

It does not earn:

Phase-sensitive breakout clock. (F.139)

---

## F.32.8 Final classification

The model should remain:

A local phase-bearing range model with explicit regime boundary. (F.140)

It should switch or reduce when $\chi$ becomes positive.

This example shows why complex eligibility can be local and regime-conditioned.

---

# F.33 Hyperbolic Alternative Audit

## F.33.1 Self-confirming relation

Suppose two channels amplify each other:

$\dot{R} \approx \kappa Q$. (F.141)

$\dot{Q} \approx \kappa R$. (F.142)

Then:

$d^2R/dt^2 \approx \kappa^2 R$. (F.143)

The generator satisfies:

$K^2 = +I$. (F.144)

This is hyperbolic rather than elliptic.

---

## F.33.2 State representation

A split-complex representation may use:

$Z_h = R + jQ$. (F.145)

with:

$j^2 = +1$. (F.146)

The invariant becomes:

$A_h^2 = R^2 - Q^2$. (F.147)

This may be more suitable for self-amplifying regimes.

But it carries the same evidential burden:

- independent variables;
- stable coupling;
- real-pair benchmark;
- gate relevance;
- transport.

---

## F.33.3 Regime-switching algebra

A broader model may use:

$e_\chi^2 = \chi$. (F.148)

Then:

$\chi < 0 \rightarrow$ elliptic.
$\chi = 0 \rightarrow$ parabolic.
$\chi > 0 \rightarrow$ hyperbolic. (F.149)

This formulation is conceptually attractive.

It remains a research hypothesis until an empirical operator can estimate $\chi$ and demonstrate that the changing algebra improves modelling.

---

# F.34 Online Phase-Time Estimation

## F.34.1 Retrospective versus online clocks

A phase clock may appear successful retrospectively because the full episode is known.

A practical clock must operate online.

At time t, it may use only:

Information_{≤t}. (F.150)

No future endpoint or later normalization may enter the estimate.

---

## F.34.2 Online state

The online state may be:

State_t
= (A_t,θ_t,$\tau_i$,t,θ̇_t,S_θ,t,Branch_t,Residual_t). (F.151)

The record should include uncertainty.

---

## F.34.3 Branch revision

When phase unwrapping changes, the original branch must remain preserved.

A branch revision is:

Branch_{t+1} = Revise(Branch_t,NewEvidence,Residual_t). (F.152)

Silent phase relabelling is the complex-model equivalent of silent Elliott Wave relabelling.

---

## F.34.4 Online gate forecast

A phase-based forecast should be issued prospectively:

Pr(Gate within h | θ_t,$\tau_i$,t,X_t). (F.153)

Later evaluation must use the original forecast and horizon.

---

# F.35 Uncertainty Quantification

## F.35.1 State uncertainty

If R and Q are estimates:

R̂_t ± u_R,t. (F.154)

Q̂_t ± u_Q,t. (F.155)

then phase uncertainty depends on both.

A first-order approximation is:

$$\mathrm{Var}(\hat{\theta}) \approx [Q^2\mathrm{Var}(\hat{R}) + R^2\mathrm{Var}(\hat{Q}) - 2RQ\mathrm{Cov}(\hat{R},\hat{Q})]/(R^2 + Q^2)^2. \quad (F.156)$$

This approximation weakens near:

$$R^2 + Q^2 \approx 0. \quad (F.157)$$

---

# F.35.2 Confidence region

The state should ideally be represented as a confidence ellipse in the R–Q plane.

Phase uncertainty is then the angular span subtended by the ellipse.

A wide angular span means phase classification is unreliable.

---

# F.35.3 Gate uncertainty

A phase gate should report:

$$\Pr(\theta \in \Theta_G \mid \mathrm{Data}). \quad (F.158)$$

not merely:

$$\theta \in \Theta_G. \quad (F.159)$$

This is especially important when Q is indirectly estimated.

---

# F.36 Multiple Q Channels

# F.36.1 Rich field

A market may contain several candidate conjugate channels:

$$Q_{\mathrm{volume}};\ Q_{\mathrm{breadth}};\ Q_{\mathrm{liquidity}};\ Q_{\mathrm{positioning}};\ Q_{\mathrm{volatility}};\ Q_{\mathrm{gate}};\ Q_{\mathrm{cadence}}. \quad (F.160)$$

These should initially remain separate:

$$q_t = [Q_1, Q_2, \ldots, Q_m]^T. \quad (F.161)$$

---

# F.36.2 Compilation burden

A single Q may be compiled:

$$Q^* = w^T q. \quad (F.162)$$

But the weights must be justified by:

- theory;
- calibration;
- predictive stability;
- factor analysis;
- operational conversion.

A compiled Q that changes composition across episodes does not provide a stable phase coordinate.

---

# F.36.3 Multicomplex temptation

Adding more imaginary axes is not automatically helpful.

A higher-dimensional state may be better represented as:

$x_t \in \mathbb{R}^n$. (F.163)

Before introducing:

- quaternions;
- Clifford algebras;
- multiple complex planes;

the model should show why ordinary multivariate geometry is insufficient.

The reduction principle remains:

Use the minimum algebra required by the operator structure. (F.164)

---

# F.37 Phase Gate Design

# F.37.1 Gate region

A phase-sensitive gate may be defined:

$\Theta_G = [\theta_a,\theta_b]$. (F.165)

The gate also requires conventional conditions:

$$
\begin{aligned}
&G_t \\
&= 1[\theta_t \in \Theta_G] \\
&\times 1[Load_t \geq L^*] \\
&\times 1[ConstraintTest_t = true] \\
&\times 1[Residual_t \leq r^*]. \quad (F.166)
\end{aligned}
$$

This prevents phase from becoming the only criterion.

---

## F.37.2 Soft gate

A probabilistic gate may use:

$$\Pr(G_t = 1) = \text{logistic}(\beta_0 + \beta_1 \cos \theta_t + \beta_2 \sin \theta_t + \beta^T X_t). \quad (F.167)$$

The phase terms should be tested for:

- significance;
- stability;
- calibration;
- out-of-sample gain.

---

## F.37.3 Gate transport

The phase gate should survive:

- nearby phase boundaries;
- alternative normalization;
- alternate Q proxy;
- comparable episode families;
- higher timeframe.

A gate that exists only under one narrowly tuned $\theta$ interval is likely overfit.

---

# F.38 Phase-Time Versus Oscillator Cycles

## F.38.1 Oscillation is insufficient

A bounded oscillator such as RSI or stochastic has a repeating numerical range.

This does not automatically create a phase clock.

An oscillator may:

- saturate;
- dwell near an extreme;
- jump discontinuously;
- depend on one price channel;
- lack a conjugate coordinate.

Therefore:

BoundedOscillator ≠ ComplexPhase. (F.168)

---

## F.38.2 Hilbert-transform phase

Signal-processing methods may construct an analytic signal:

$Z_a(t) = x(t) + iH[x(t)]$, (F.169)

where H is a Hilbert transform.

This produces a mathematical phase.

But the economic meaning of the imaginary component must still be justified.

The analytic signal can be useful for local oscillation analysis without implying an independently real market variable.

It should be classified as:

SignalProcessingPhase. (F.170)

not automatically:

DomainConjugatePhase. (F.171)

---

## F.38.3 Distinction

| Phase type | Q source | Interpretation |
|---|---|---|
| Geometric CAPM phase | derived from declared valuation geometry | financial phase exposure |
| Analytic-signal phase | Hilbert transform of one signal | signal-processing orientation |
| Two-channel empirical phase | independently measured Q | candidate domain conjugacy |
| Narrative cycle phase | assigned stages | qualitative episode ordering |

These phase types have different evidential status.

They should not be mixed.

---

# F.39 Phase-Time Validation Against Flexible Warping

## F.39.1 The strongest null model

A phase clock should be compared with a flexible learned monotone warping:

$\tau_{flex} = f_j(t)$, (F.172)

where $f_j$ is fit to align episodes.

If a highly flexible warping outperforms phase, that is unsurprising.

The relevant question is whether phase provides comparable alignment with:

- fewer parameters;
- interpretable structure;
- stable transport;
- predictive gates.

---

## F.39.2 Complexity-adjusted gain

Define:

Gain_adj
= AlignmentGain
– λParameterCount
– μInstability
– νRetrospectiveDependence. (F.173)

Phase time earns priority when Gain_adj exceeds the alternatives.

---

# F.40 Publication Claims Ladder

The language used in a paper should match the evidence level.

## Level 0 claim

“R and Q are two useful real variables.”

## Level 1 claim

“The variables admit a convenient complex representation.”

## Level 2 claim

“The complex representation supports stable local phase dynamics.”

## Level 3 claim

“Phase improves internal ordering of comparable episodes.”

## Level 4 claim

“Consequential events show reproducible phase dependence.”

## Level 5 claim

“Phase-sensitive gates write trace that changes subsequent dynamics.”

Each statement is stronger than the previous one.

A paper should not jump from Level 1 to Level 5.

---

# F.41 Red-Flag Vocabulary

The following phrases require immediate audit:

- “hidden energy”;
- “imaginary market force”;
- “quantum-like collapse”;
- “phase transition”;
- “internal clock”;
- “conjugate pressure”;
- “market wavefunction”;
- “complex risk field.”

For each phrase, ask:

1. What is measured?
2. What are the units?
3. What operator defines the relation?
4. What simpler model was rejected?
5. What gate is predicted?
6. What residual remains?

Without answers, the phrase should be treated as metaphor.

---

# F.42 Minimal Rejection Report

When the complex model fails, the paper should report:

```
The proposed R–Q pair remained useful as a two-variable state.
However, phase ordering was unstable under admissible scaling,
episode alignment did not improve beyond event count,
and phase-conditioned gates did not outperform the real-pair benchmark.
The complex-priority and phase-time claims are therefore rejected.
The retained model is the real pair (R,Q), with residual ε.
```

This is a scientifically successful outcome.

It identifies the model level actually supported by evidence.

# F.43 Minimal Acceptance Report

A stronger acceptance report may state:

```
R and Q were independently defined and dimensionally aligned.
A locally stable rotational generator was replicated out of sample.
Phase order survived normalization and timeframe perturbations.
Comparable episodes aligned more tightly under τᵢ than under
calendar time, cumulative volatility, or event count.
The declared gate hazard concentrated in a stable phase region.
Ledgered gate events altered subsequent transition probabilities.
The model is therefore classified as a candidate phase-sensitive,
time-bearing episode model under protocol P.
```

Even this statement should remain protocol-bound.

---

# F.44 Appendix F Conclusion

Complex notation is easy.

Complex eligibility is demanding.

A valid complex market model must pass through a sequence of increasingly strong claims:

Independent variables
→ dimensional compatibility
→ stable coupling
→ useful phase
→ robust internal order
→ phase-sensitive gates
→ persistent trace
→ backreaction. (F.174)

The minimum complex model is:

$Z = R + iQ$. (F.175)

The minimum phase-dynamical model is:

$dZ/dt = [g_A + i\omega]Z + \varepsilon$. (F.176)

The minimum phase-time test is:

$D_{phase} < \min(D_{calendar}, D_{event})$. (F.177)

The minimum phase-gate test is:

Information(Gate | Phase,Controls)

Information(Gate | Controls). (F.178)

The minimum world test is:

Gate
→ Trace
→ Ledger

→ ChangedFutureDynamics. (F.179)

The governing reduction law is:

Use the least complex representation that preserves demonstrated explanatory, predictive, transport, or intervention gain. (F.180)

And the governing prohibition is:

No independent Q

- no stable generator
- no phase advantage
  = no privileged complex model. (F.181)

The next appendix can present the **CAPM Calibration Atom in full derivational form**, including exact rotation equations, sensitivity reconciliation, haircut accumulation, gate residual, and the distinction between measurement orientation, economic movement, recognition, and ledgered consequence.

# Appendix G — The CAPM Calibration Atom in Full Derivational Form

## G.1 Purpose

The periodic framework uses the CAPM complex valuation state as its principal calibration atom because the construction begins from familiar finance and produces an exact conjugate geometry.

It supplies:

- a declared baseline valuation;
- an ordinary CAPM-discounted valuation;
- a dimensionally compatible conjugate coordinate;
- an exact phase-sensitivity identity;
- a closed quarter-turn operator;
- an exact distinction between measurement and movement;
- a recognition gate;
- a ledger residual.

The construction does **not** modify CAPM.

It reorganizes the relationship between two valuations of the same cash flow:

1. valuation under a declared baseline discount rate;
2. valuation under the CAPM required return.

The source paper's principal result is:

$\partial R/\partial \theta = -Q$. (G.1)

Q is therefore the magnitude of first-order dollar exposure to movement in the declared valuation phase. The same source distinguishes this mathematical exposure from economic P&L, recognition, settlement, and ledger history.

---

# G.2 The Finance-First Declaration

Consider one future cash flow $CF_t$ payable at horizon t.

Define the CAPM required return:

$r = r_{base} + \beta ERP$. (G.2)

where:

$r_{base}$ = declared baseline rate;
$\beta$ = CAPM beta;
ERP = equity risk premium.

The construction requires two present values of the **same** cash flow.

# Baseline-discounted amplitude

$A_t = CF_t/(1 + r_{base})^t$. (G.3)

# CAPM-admitted value

$R_t = CF_t/(1 + r)^t$. (G.4)

When:

$r \geq r_{base}$, (G.5)

and the cash flow is positive:

$0 < R_t \leq A_t$. (G.6)

The ratio is:

$R_t/A_t = [(1 + r_{base})/(1 + r)]^t$. (G.7)

This ratio defines the valuation phase.

---

# G.3 The Valuation Phase

Define:

$\cos \theta_t = R_t/A_t$. (G.8)

Therefore:

$\theta_t = \arccos(R_t/A_t)$. (G.9)

Substituting the discount formulas:

$\theta_t = \arccos\{[(1 + r_{base})/(1 + r)]^t\}$. (G.10)

Under the positive-value convention:

$0 \le \theta_t \le \pi/2$. (G.11)

The orthogonal coordinate is:

$Q_t = \sqrt{(A_t^2 - R_t^2)}$. (G.12)

Equivalently:

$R_t = A_t \cos \theta_t$. (G.13)

$Q_t = A_t \sin \theta_t$. (G.14)

The completed state is:

$Z_t = R_t + iQ_t$. (G.15)

or:

$Z_t = A_t \exp(i\theta_t)$. (G.16)

The norm identity is:

$|Z_t|^2 = R_t^2 + Q_t^2 = A_t^2$. (G.17)

All three coordinates A, R, and Q have monetary units.

This dimensional compatibility is one reason the construction is stronger than an arbitrary pairing of price with an unrelated standardized indicator.

---

# G.4 The Conjugate Risk Theorem

# Theorem G.1 — CAPM Phase Delta

Along a fixed-amplitude valuation orbit:

$\partial R/\partial\theta = -Q$. (G.18)

# Proof

From:

$R = A \cos \theta$, (G.19)

holding A fixed:

$\partial R/\partial\theta = -A \sin \theta$. (G.20)

Since:

$Q = A \sin \theta$, (G.21)

it follows that:

$$\partial R/\partial \theta = -Q. \quad (G.22)$$

■

Define the signed CAPM Phase Delta:

$$\Delta_\theta \equiv \partial R/\partial \theta. \quad (G.23)$$

Then:

$$\Delta_\theta = -Q. \quad (G.24)$$

The units are:

$$\text{Units}(Q) = \text{currency per radian}. \quad (G.25)$$

Because radians are dimensionless in conventional dimensional analysis, Q retains monetary units.

Its operational interpretation is:

> For a small positive movement in valuation phase, the first-order change in admitted value is −Q times the phase movement.

Thus:

$$dR = -Q d\theta. \quad (G.26)$$

Q is not merely the unused side of a right triangle.

It is also the tangent exposure of R along the declared phase orbit.

---

# G.5 Ordinary Required-Return Sensitivity

The ordinary discounted value is:

$$R = CF/(1 + r)^t. \quad (G.27)$$

Differentiate with respect to r:

$$\partial R/\partial r = -tCF/(1 + r)^{t+1}. \quad (G.28)$$

Since:

$$R = CF/(1 + r)^t, \quad (G.29)$$

the result becomes:

$$\partial R/\partial r = -tR/(1 + r). \quad (G.30)$$

Therefore:

$$dR = -[tR/(1 + r)]dr. \quad (G.31)$$

This is conventional required-return sensitivity.

The phase representation gives:

$dR = -Qd\theta$. (G.32)

Equating the two descriptions of the same local value change:

$-Qd\theta = -[tR/(1 + r)]dr$. (G.33)

Hence:

$Qd\theta = [tR/(1 + r)]dr$. (G.34)

The phase construction does not alter first-order CAPM economics.

It changes the risk coordinate through which the sensitivity is expressed.

---

# G.6 The Rate-to-Phase Conversion

From:

$\cos \theta = [(1 + r_base)/(1 + r)]^t$, (G.35)

differentiate with respect to r while holding r_base, CF, and t fixed:

$-\sin \theta \cdot d\theta/dr$
$= -[t/(1 + r)]\cos \theta$. (G.36)

Therefore:

$d\theta/dr = [t \cos \theta]/[(1 + r)\sin \theta]$. (G.37)

Using:

$\cos \theta = R/A$ (G.38)

and:

$\sin \theta = Q/A$, (G.39)

gives:

$d\theta/dr = tR/[(1 + r)Q]$. (G.40)

Multiplying by Q:

$Q \cdot d\theta/dr = tR/(1 + r)$. (G.41)

This reproduces (G.34).

---

# G.6.1 Beta sensitivity

Because:

$dr/d\beta = ERP$, (G.42)

the ordinary beta sensitivity is:

$$\partial R/\partial \beta = -[tR/(1 + r)]ERP. \quad (G.43)$$

The phase sensitivity is:

$$\partial \theta/\partial \beta = [tR/((1 + r)Q)]ERP. \quad (G.44)$$

Therefore:

$$\partial R/\partial \beta = -Q \cdot \partial \theta/\partial \beta. \quad (G.45)$$

---

## G.6.2 ERP sensitivity

Because:

$$dr/dERP = \beta, \quad (G.46)$$

it follows that:

$$\partial R/\partial ERP = -[tR/(1 + r)]\beta. \quad (G.47)$$

and:

$$\partial \theta/\partial ERP = [tR/((1 + r)Q)]\beta. \quad (G.48)$$

Therefore:

$$\partial R/\partial ERP = -Q \cdot \partial \theta/\partial ERP. \quad (G.49)$$

---

## G.6.3 Coordinate singularity near Q = 0

When:

$$r \to r_base, \quad (G.50)$$

then:

$$R \to A, \quad (G.51)$$

$$\theta \to 0, \quad (G.52)$$

$$Q \to 0. \quad (G.53)$$

The derivative:

$$d\theta/dr = tR/[(1 + r)Q] \quad (G.54)$$

becomes large as Q approaches zero.

This does not mean ordinary value sensitivity becomes infinite.

The product remains finite:

$$Q \cdot d\theta/dr = tR/(1 + r). \quad (G.55)$$

The apparent divergence belongs to the phase coordinate near the zero-angle boundary.

It is not an economic singularity in R.

---

# G.7 Radial and Angular Change

The fixed-A theorem isolates angular change.

More generally:

$R = A \cos \theta$. (G.56)

Differentiation gives:

$dR = \cos \theta dA - A \sin \theta d\theta$. (G.57)

Therefore:

$dR = (R/A)dA - Qd\theta$. (G.58)

Similarly:

$Q = A \sin \theta$. (G.59)

so:

$dQ = \sin \theta dA + A \cos \theta d\theta$. (G.60)

Therefore:

$dQ = (Q/A)dA + Rd\theta$. (G.61)

These equations separate two channels.

# Radial channel

dA changes the size of the valuation state.

Possible drivers include:

- cash-flow revision;
- baseline-rate change;
- horizon change;
- baseline-protocol change.

# Angular channel

dθ changes the valuation orientation.

Possible drivers include:

- beta change;
- ERP change;

- risk-adjusted discount change relative to the baseline.

The decomposition is:

Total valuation movement
= Radial movement

- Angular movement. (G.62)

For the admitted value:

dR = RadialContribution − PhaseContribution. (G.63)

---

# G.8 Q Is Not the Scalar Haircut

Define the ordinary CAPM discount haircut:

H = A − R. (G.64)

Since:

R = A cos θ, (G.65)

the haircut is:

H = A(1 − cos θ). (G.66)

This is a same-axis scalar difference.

By contrast:

Q = A sin θ (G.67)

is an orthogonal coordinate.

Therefore:

Q ≠ H. (G.68)

Differentiate the haircut while holding A fixed:

dH/dθ = A sin θ. (G.69)

Hence:

dH/dθ = Q. (G.70)

Integrating from the zero-phase baseline:

H(θ) = ∫₀^θ Q(φ)dφ. (G.71)

Thus:

- Q is a marginal phase exposure;
- H is its accumulated same-axis valuation effect.

The source paper makes this distinction explicit and warns against treating Q as A − R.

## G.8.1 Exact relationship between Q and H

Because:

$R = A - H$, (G.72)

the norm identity gives:

$Q^2 = A^2 - (A - H)^2$. (G.73)

Therefore:

$Q^2 = 2AH - H^2$. (G.74)

Equivalently:

$H = A - \sqrt{(A^2 - Q^2)}$. (G.75)

For small phase or small H relative to A:

$Q^2 \approx 2AH$. (G.76)

Therefore:

$H \approx Q^2/(2A)$. (G.77)

This explains why Q can be much larger than the scalar haircut at small or moderate angles.

Q is first-order in $\theta$:

$Q \approx A\theta$. (G.78)

The haircut is second-order near $\theta = 0$:

$H \approx A\theta^2/2$. (G.79)

---

## G.8.2 Opportunity-cost interpretation

Q becomes a marginal opportunity-cost coordinate only when A is explicitly defined as the value of the best foregone alternative.

Without that additional declaration, the safer interpretation is:

> Q is the marginal phase sensitivity of the gap between baseline valuation and admitted valuation.

It is not automatically an actual paid cost.

---

## G.9 The Quarter-Turn Operator

Represent the valuation state as:

$x = [R,Q]^T$. (G.80)

Define:

$J = [[0,-1],[1,0]]$. (G.81)

Then:

$Jx = [-Q,R]^T$. (G.82)

Applying the operator twice:

$J^2x = [-R,-Q]^T$. (G.83)

Therefore:

$J^2 = -I$. (G.84)

Applying it four times:

$J^4 = I$. (G.85)

The sequence of real-axis readouts is:

$R$
$\to -Q$
$\to -R$
$\to Q$
$\to R$. (G.86)

This is the closed financial measurement cycle established in the source construction.

---

# G.10 Rotated Financial Measurements

Define the measurement family:

$M_\varphi(Z) = \mathrm{Re}[\exp(i\varphi)Z]$. (G.87)

Since:

$Z = R + iQ$, (G.88)

and:

$\exp(i\varphi) = \cos\varphi + i\sin\varphi$, (G.89)

the real measurement becomes:

$M_\varphi(Z) = R\cos\varphi - Q\sin\varphi$. (G.90)

The principal readouts are:

$M_0(Z) = R$. (G.91)

$M_{\pi/2}(Z) = -Q$. (G.92)

$M_\pi(Z) = -R$. (G.93)

$M_{3\pi/2}(Z) = Q$. (G.94)

$M_{2\pi}(Z) = R.$ (G.95)

The first quarter-turn asks for the conjugate phase exposure.

The second quarter-turn reverses the signed valuation orientation.

The third reads the opposite-signed phase exposure.

The fourth restores the original mark.

---

## G.10.1 Measurement cycle, not historical sequence

The sequence:

$R \to -Q \to -R \to Q \to R$ (G.96)

does not mean that the asset must chronologically pass through these four economic states.

It is first a cycle of measurement orientation.

Changing the measurement question does not itself change the asset.

This distinction is essential:

$\text{MeasurementRotation} \neq \text{StateEvolution}.$ (G.97)

---

# G.11 Passive Rotation and Active Rotation

## G.11.1 Passive rotation

A passive rotation leaves Z unchanged and changes the readout:

Z fixed. (G.98)

$M_0(Z) \to M_{\varphi}(Z).$ (G.99)

No economic P&L is created merely by rotating the measurement frame.

The observer asks a different question about the same state.

---

## G.11.2 Active rotation

An active rotation changes the valuation state:

$Z_{new} = \exp(i\Delta\theta)Z.$ (G.100)

The measurement axis remains fixed.

Expanding:

$$Z_{new} = (\cos\Delta\theta + i\sin\Delta\theta)(R + iQ).$$ (G.101)

Therefore:

$R_{new} = R \cos \Delta\theta - Q \sin \Delta\theta$. (G.102)

$Q_{new} = R \sin \Delta\theta + Q \cos \Delta\theta$. (G.103)

An active phase movement can generate economic value change.

The source paper explicitly separates passive measurement rotation from active state rotation and from ledger commitment.

---

# G.12 Exact Finite Value Change

From (G.102):

$\Delta R = R_{new} - R$. (G.104)

Therefore:

$\Delta R = R(\cos \Delta\theta - 1) - Q \sin \Delta\theta$. (G.105)

This is the exact fixed-amplitude value change.

The corresponding conjugate-coordinate change is:

$\Delta Q = R \sin \Delta\theta + Q(\cos \Delta\theta - 1)$. (G.106)

---

# G.12.1 Taylor expansion

Using:

$\cos \Delta\theta = 1 - (\Delta\theta)^2/2 + (\Delta\theta)^4/24 - \ldots$ (G.107)

and:

$\sin \Delta\theta = \Delta\theta - (\Delta\theta)^3/6 + \ldots$, (G.108)

the admitted-value change becomes:

$\Delta R$
$= -Q\Delta\theta$
$- (R/2)(\Delta\theta)^2$

- $(Q/6)(\Delta\theta)^3$
- $(R/24)(\Delta\theta)^4$
- $O((\Delta\theta)^5)$. (G.109)

To first order:

$\Delta R \approx -Q\Delta\theta$. (G.110)

The first-order relation identifies exposure.

The higher-order terms describe curvature and finite-movement effects.

# G.13 Exposure Is Not Loss

The measurement:

$M_{\pi/2}(Z) = -Q$ (G.111)

answers:

> What is the signed first-order response of admitted value to a unit positive phase movement?

It does not mean that a loss of Q has occurred.

Three objects must remain distinct:

Exposure = $-Q$. (G.112)

Movement = $\Delta\theta$. (G.113)

Economic consequence = $\Delta R$. (G.114)

For small movements:

Economic consequence ≈ Exposure × Movement. (G.115)

Therefore:

$-Q$ alone ≠ EconomicP&L. (G.116)

The source runtime is:

Measurement
→ Exposure
→ State Movement
→ Economic P&L
→ Gate
→ Ledger + Residual. (G.117)

---

# G.14 Recognition Gate

Economic change is not necessarily recognized immediately by every institution.

Define a recognition gate:

$G_P(\Delta R, X, L) \rightarrow d$. (G.118)

where:

P = declared recognition protocol;
X = current financial state;
L = prior ledger;
d = gate decision.

A basic decision set is:

$d \in \{\text{Admit}, \text{Defer}, \text{Reject}\}$. (G.119)

A richer gate may permit:

$d \in \{\text{Admit}, \text{PartiallyAdmit}, \text{Defer}, \text{Reject}\}$. (G.120)

Examples include:

- official close;
- settlement;
- margin call;
- option exercise;
- covenant breach;
- default threshold;
- impairment test;
- collateral revaluation;
- regulatory-capital rule;
- legal judgment.

The gate does not create the underlying economic movement.

It decides whether and how that movement becomes an operative institutional event.

---

# G.15 Partial Admission

Let:

$\Delta R_{\text{total}}$ = total economic change. (G.121)

Let:

$0 \le \alpha \le 1$ (G.122)

be the admitted fraction under the gate.

Then:

$\Delta R_{\text{ledger}} = \alpha \Delta R_{\text{total}}$. (G.123)

Possible interpretations include:

$\alpha = 1 \rightarrow$ fully admitted. (G.124)

$0 < \alpha < 1 \rightarrow$ partially admitted. (G.125)

$\alpha = 0 \rightarrow$ deferred or rejected. (G.126)

The value of $\alpha$ may depend on:

- accounting rule;
- settlement status;
- legal enforceability;
- risk policy;
- valuation convention;
- collateral eligibility.

---

# G.16 Gate Residual

Define:

$\varepsilon_{gate} = \Delta R_{total} - \Delta R_{ledger}$. (G.127)

Under the simple proportional admission rule:

$\varepsilon_{gate} = (1 - \alpha)\Delta R_{total}$. (G.128)

The gate residual may represent:

- unrecognized economic damage;
- pending settlement;
- unresolved legal exposure;
- liquidity pressure;
- model disagreement;
- remaining tail risk;
- delayed accounting effect.

Therefore:

Commitment ≠ Exhaustion. (G.129)

A gate may write a consequential trace while leaving substantial residual.

---

# G.17 Ledger Update

When a consequence is admitted:

$L_{k+1} = \mathrm{Update}(L_k, \mathrm{Trace}_k, \varepsilon_{gate}, k)$. (G.130)

A simplified additive expression is:

$L_{k+1} = L_k + \mathrm{Trace}(\Delta R_{ledger})$. (G.131)

But a mature ledger should also preserve:

- recognition authority;
- admission fraction;
- date;
- protocol;
- residual;
- later revision.

The state after recognition is not completely described by current market value alone.

It also carries the history of how the value became recognized.

---

# G.17.1 Backreaction

A ledger event can alter future financial conditions.

Examples include:

- recognized loss changes capital;
- covenant breach changes borrowing rights;
- margin call forces asset sales;
- default changes collateral;
- impairment changes reported equity;
- settlement changes cash and legal ownership.

The backreaction sequence is:

EconomicMovement
→ Gate
→ LedgerUpdate
→ ChangedConstraints
→ ChangedFutureValuation. (G.132)

Therefore:

Z_{k+1} = F(Z_k,L_{k+1},ExternalState_k). (G.133)

The ledger is not merely a passive history.

It may become part of the next valuation state.

---

# G.18 The Four Operations

| Operation | What changes? | Creates economic P&L? | Creates ledger trace? |
|---|---|---|---|
| **Measurement** | readout orientation | no | no |

| Operation | What changes? | Creates economic P&L? | Creates ledger trace? |
|---|---|---|---|
| **rotation** | | | |
| **State movement** | Z, θ, A, R, or Q | potentially | not automatically |
| **Recognition gate** | institutional status | recognizes consequence | yes when admitted |
| **Ledger update** | usable financial history | records consequence | yes |

The distinctions are:

Measurement identifies exposure. (G.134)

Movement generates economic consequence. (G.135)

Gate decides recognition. (G.136)

Ledger preserves consequential history. (G.137)

Multiplication by i performs the first operation.

It does not automatically perform the other three.

---

# G.19 Signed Long–Short Interpretation

In a linear signed-position space:

Long mark = R. (G.138)

Long phase exposure = −Q. (G.139)

Short mark = −R. (G.140)

Short phase exposure = Q. (G.141)

This gives the four-node cycle:

Long mark
→ Long conjugate exposure
→ Short mark
→ Short conjugate exposure
→ Long mark. (G.142)

But the interpretation is local to a signed tradable-position space.

It does not imply that:

- a company is the negative of its shareholders;
- an asset must chronologically become its own short;
- applying J twice causes two losses.

The half-turn:

$J^2Z = -Z$ (G.143)

reverses signed orientation.

It is not a double economic-loss operator.

---

# G.20 Multi-Period Cash-Flow Extension

Consider cash flows CF_t for t = 1...T.

For each term:

A_t = CF_t/(1 + r_base,t)^t. (G.144)

R_t = CF_t/(1 + r_t)^t. (G.145)

Q_t = √(A_t² − R_t²). (G.146)

Z_t = R_t + iQ_t = A_t exp(iθ_t). (G.147)

The total complex value is:

Z = Σ_{t=1}^{T} Z_t. (G.148)

Therefore:

R = Σ_{t=1}^{T} R_t. (G.149)

Q = Σ_{t=1}^{T} Q_t. (G.150)

The source framework describes this as a phase-exposure term structure rather than a simple pile of scalar cash flows.

---

# G.20.1 Composite norm

The composite norm is:

|Z|² = R² + Q². (G.151)

Expanding:

|Z|²
= Σ_t A_t²

- 2Σ_{s<t} A_sA_t cos(θ_s − θ_t). (G.152)

Therefore:

|Z|² ≠ Σ_t A_t² (G.153)

unless the cross terms happen to vanish or satisfy a special relation.

Cash-flow terms with similar phases reinforce one another geometrically.

Terms with different phases alter the composite amplitude and orientation.

This is a geometric aggregation result.

It is not evidence of physical quantum interference.

---

## G.20.2 Term-specific movements

If individual cash-flow phases move separately:

$dR = -\Sigma_t Q_t d\theta_t$ (G.154)

when the $A_t$ are fixed.

With radial changes included:

$dR = \Sigma_t[(R_t/A_t)dA_t - Q_t d\theta_t]$. (G.155)

This produces a phase-exposure term structure:

$Q_{term} = (Q_1, Q_2, \ldots, Q_T)$. (G.156)

A scalar total Q compresses that structure and may hide horizon-specific exposure.

---

## G.20.3 Common measurement rotation

For a common readout rotation $\varphi$:

$M_\varphi(Z) = \mathrm{Re}[\exp(i\varphi)Z]$. (G.157)

Therefore:

$dM_\varphi(Z)/d\varphi \,|_{\varphi=0} = -Q$. (G.158)

The total conjugate coordinate remains the first derivative of the aggregate real readout under a common measurement rotation.

---

## G.21 Worked Numerical Example

Assume:

CF = £100. (G.159)

t = 5 years. (G.160)

$r_{base}$ = 4%. (G.161)

$\beta$ = 1.2. (G.162)

ERP = 5%. (G.163)

The CAPM required return is:

$r = 4\% + 1.2 \times 5\% = 10\%$. (G.164)

The baseline value is:

$A = 100/(1.04)^5 = £82.193$. (G.165)

The CAPM-admitted value is:

R = 100/(1.10)^5 = £62.092. (G.166)

The conjugate coordinate is:

Q = √(82.193² − 62.092²) = £53.854. (G.167)

The phase is:

θ = arccos(62.092/82.193) = 0.7145 radians. (G.168)

or approximately:

θ = 40.94°. (G.169)

The scalar haircut is:

H = A − R = £20.101. (G.170)

Therefore:

Q = £53.854 (G.171)

while:

H = £20.101. (G.172)

This illustrates numerically:

Q ≠ H. (G.173)

---

# G.21.1 One-percentage-point increase in required return

Increase the required return from 10% to 11%.

The new admitted value is:

R_new = 100/(1.11)^5 = £59.345. (G.174)

The exact economic change is:

ΔR = 59.345 − 62.092 = −£2.747. (G.175)

The new phase is:

θ_new = 0.7641 radians. (G.176)

Therefore:

Δθ = 0.04961 radians. (G.177)

Using the exact rotation equation:

ΔR
= R(cos Δθ − 1) − Q sin Δθ

= −£2.747. (G.178)

The first-order phase approximation is:

$\Delta R \approx -Q\Delta\theta$. (G.179)

Therefore:

$\Delta R \approx -53.854 \times 0.04961 = -£2.672$. (G.180)

The difference between −£2.672 and −£2.747 is the finite-movement curvature omitted by the first-order approximation.

---

## G.21.2 Ordinary sensitivity approximation

At r = 10%:

$\partial R/\partial r = -5 \times 62.092/1.10$. (G.181)

Therefore:

$\partial R/\partial r = -£282.237$ per unit rate. (G.182)

For:

dr = 0.01, (G.183)

the ordinary first-order estimate is:

$dR \approx -282.237 \times 0.01 = -£2.822$. (G.184)

The phase and ordinary approximations are both local.

They differ slightly from the exact finite change because the one-percentage-point movement is not infinitesimal and the sensitivity changes along the path.

---

## G.22 What Q Is

Within the declared CAPM geometry, Q is:

1. the orthogonal coordinate completing A and R;
2. $A \sin \theta$;
3. $\sqrt{(A^2 - R^2)}$;
4. the magnitude of first-order phase exposure;
5. $-\partial R/\partial\theta$;
6. the marginal growth rate of the haircut H with respect to $\theta$;
7. part of a four-step signed measurement cycle;
8. algebraically determined by A and R in the static one-cash-flow construction.

---

## G.23 What Q Is Not Automatically

Q is not automatically:

- beta;
- the equity risk premium;
- volatility;
- expected loss;
- realized loss;
- Value at Risk;
- Expected Shortfall;
- duration;
- convexity;
- option premium;
- market price;
- scalar haircut A − R;
- opportunity cost;
- model residual;
- unrecognized accounting loss.

Those interpretations require separate definitions and tests.

The source paper explicitly limits Q in this manner.

---

## G.24 Static Identity Versus Empirical Value

In the static one-period model:

$$Q = \sqrt{(A^2 - R^2)}. \quad (G.185)$$

Therefore Q contains no raw numerical information beyond A and R.

Its value must arise from what the completed representation enables.

Candidate gains include:

- dynamic phase attribution;
- multi-horizon exposure;
- comparison among protocols;
- clearer sensitivity communication;

- recognition-gate diagnosis;
- residual comparison;
- intervention design.

If no such gain appears:

Q remains an optional reparameterization. (G.186)

The source article makes this empirical limit explicit.

---

# G.25 Five Levels of CAPM Claim

# Level 1 — Mathematical identities

$A^2 = R^2 + Q^2$. (G.187)

$\partial R/\partial\theta = -Q$. (G.188)

$J^2 = -I$. (G.189)

These follow from the declared representation.

---

# Level 2 — Financial construction

A is the same cash flow discounted under the baseline protocol.

R is the same cash flow discounted under CAPM.

Q completes the declared norm.

These statements depend on the valuation declarations.

---

# Level 3 — Measurement interpretation

$M_0$ reads the mark.

$M_{\pi/2}$ reads conjugate exposure.

$M_\pi$ reverses signed valuation orientation.

These depend on the chosen sign and orientation convention.

---

# Level 4 — Empirical hypotheses

Examples include:

- Q improves cross-asset risk comparison;
- phase improves multi-horizon attribution;

- gate residual predicts later financial consequence;
- complex representation improves intervention.

These require testing.

---

# Level 5 — Broader foundational hypotheses

Examples include:

- scalar valuation may not exhaust the effective measurement state;
- movement becomes financial history only through gate and trace;
- observer and recognition protocol matter to valuation worlds.

These are research implications rather than established consequences of CAPM alone.

---

# G.26 Relation to the Proto-Periodic Table

The CAPM calibration atom can be mapped onto the four functional families.

| Functional family | CAPM manifestation |
|---|---|
| **Load / Memory** | cash flow, baseline A, horizon, valuation inputs |
| **Motion / Relation** | θ, Q, rate sensitivity, phase movement |
| **Constraint / Boundary** | baseline protocol, CAPM filter, recognition rules |
| **Commitment / Gate** | settlement, impairment, margin, accounting or legal recognition |

Its period depends on use.

## Structure-period use

Static valuation relation and sensitivity.

## Event-period use

A change in required return or phase produces an economic valuation event.

## World-period use

Recognition rules, ledgers, institutions, and backreaction govern how the valuation becomes operative history.

Thus the CAPM atom is not the whole periodic table.

It is a mature local example showing how one declared scalar output can possess a rigorous conjugate completion.

---

# G.27 Why the CAPM Atom Matters for Technical Analysis

Any proposed Technical Analysis complex state should be asked to meet a comparable burden.

It should declare:

1. What plays the role of A?
2. What exactly is R?
3. How is Q defined?
4. Do R and Q have compatible units?
5. What produces the phase?
6. What is the generator?
7. What movement creates economic consequence?
8. What gate creates commitment?
9. What enters the ledger?
10. What remains residual?
11. What simpler real-pair model was rejected?

Without these declarations:

$Z_{TA} = R_{TA} + iQ_{TA}$ (G.190)

is notation, not yet an earned complex market model.

---

# G.28 CAPM Calibration Card

```
Valued claim:
Cash flow:
Horizon:
Baseline rate:
CAPM beta:
Equity risk premium:
CAPM required return:

Baseline amplitude A:
Admitted value R:
Conjugate coordinate Q:
Phase θ:
Scalar haircut H:

Required-return exposure:
Phase exposure:
Finite phase movement:
Economic value change:

Recognition gate:
Admission fraction:
Ledgered consequence:
Gate residual:
Ledger authority:

Radial drivers:
Angular drivers:
```

```
Model residual:
Backreaction:
```

---

# G.29 Final Calibration Sequence

The complete CAPM runtime is:

Cash Flow
→ Baseline Valuation A
→ CAPM Filter
→ Admitted Value R
→ Declared Phase θ
→ Conjugate Exposure Q
→ Measurement Rotation
→ Actual State Movement
→ Economic P&L
→ Recognition Gate
→ Ledger Trace + Residual
→ Updated Financial World. (G.191)

Its central theorem is:

$\partial R/\partial\theta = -Q$. (G.192)

Its central sensitivity equivalence is:

$-Qd\theta = -[tR/(1 + r)]dr$. (G.193)

Its central haircut relation is:

$H = \int_0^{\theta} Q(\varphi)d\varphi$. (G.194)

Its central measurement cycle is:

$R \to -Q \to -R \to Q \to R$. (G.195)

Its central governance distinction is:

Measurement ≠ Movement ≠ Recognition ≠ Ledger. (G.196)

Its central residual rule is:

$\varepsilon_{gate} = \Delta R_{total} - \Delta R_{ledger}$. (G.197)

And its central limitation is:

Exact complex reconstruction does not by itself establish empirical superiority, internal phase time, or quantum ontology.

The next appendix can separate the entire article into **static identities, protocol-dependent constructions, empirical hypotheses, and falsifiable research claims**, preventing mathematical truths from being confused with untested market propositions.

# Appendix H — Claim Typology, Evidence Levels, and Falsification Boundaries

# H.1 Purpose

The article combines several kinds of statement:

- mathematical identities;
- protocol declarations;
- model constructions;
- interpretive definitions;
- empirical hypotheses;
- engineering proposals;
- cross-domain analogies;
- philosophical extensions.

These statements do not possess the same evidential status.

For example:

$A^2 = R^2 + Q^2$ (H.1)

may be an exact identity inside a declared CAPM construction.

But:

Phase aligns market episodes better than calendar time (H.2)

is an empirical hypothesis.

And:

Technical Analysis possesses a proto-periodic grammar (H.3)

is a proposed classificatory synthesis.

Confusing these levels produces two opposite errors.

### Overclaiming

A definition or algebraic identity is presented as though it proves a market law.

### Underclaiming

A useful formal construction is dismissed merely because its strongest empirical extensions have not yet been demonstrated.

The purpose of this appendix is to separate the claim types explicitly.

---

# H.2 The Seven Claim Classes

The article uses seven principal claim classes.

| Code | Claim class | Status |
|---|---|---|
| **I** | mathematical identity | true given the stated definitions |
| **D** | protocol declaration or definition | fixed by the observer or modelling protocol |
| **C** | model construction | logically follows from declared components |
| **T** | interpretive translation | assigns operational meaning to a formal object |
| **H** | empirical hypothesis | requires observation and testing |
| **E** | engineering proposal | specifies a tool, record, or intervention architecture |
| **A** | analogy or cross-domain extension | proposes structural similarity without identity |

A statement may belong to more than one class.

For example:

$Q = \sqrt{(A^2 - R^2)}$ (H.4)

is an identity after A and R are declared.

Its interpretation as:

Q = valuation-phase exposure (H.5)

is a translation supported by:

$\partial R/\partial \theta = -Q.$ (H.6)

Its usefulness for risk management is an empirical and engineering question.

---

# H.3 Claim-Tagging Convention

For clarity, later publications based on this framework may tag claims as follows:

```
[I] Identity
[D] Declaration
[C] Construction
[T] Interpretation
[H] Empirical hypothesis
[E] Engineering proposal
[A] Analogy
```

Examples:

```
[I] ∂R/∂θ = −Q.

[D] The daily close is selected as the Window-level commitment gate.

[C] A breakout score combines displacement, volume, breadth, and retest.

[H] Higher breakout-gate strength predicts lower fakeout probability.

[E] Store every failed breakout with its residual and later relabelling.

[A] Market Constraint plays a role analogous to a boundary condition.
```

The tags do not determine whether a claim is important.

They determine what kind of support it requires.

---

# H.4 Class I — Mathematical Identities

## H.4.1 Definition

A mathematical identity follows exactly from previously declared definitions.

It does not require market data for its internal validity.

Its truth is conditional on the construction being used correctly.

---

## H.4.2 CAPM norm identity

Given:

$Q = \sqrt{(A^2 - R^2)}$, (H.7)

it follows that:

$A^2 = R^2 + Q^2$. (H.8)

This is an exact identity inside the declared valuation geometry.

It does not establish that Q is independently observed.

It does not establish that the representation improves prediction.

---

## H.4.3 CAPM phase identity

Given:

$R = A \cos \theta$, (H.9)

and:

$Q = A \sin \theta$, (H.10)

then:

$\partial R/\partial \theta = -Q$. (H.11)

and:

$\partial Q/\partial \theta = R$. (H.12)

These follow exactly from differentiation.

The CAPM source uses this relation to define Q as the conjugate phase-exposure coordinate.

---

## H.4.4 Quarter-turn identity

For:

$J = [[0,-1],[1,0]]$, (H.13)

it follows that:

$J^2 = -I$. (H.14)

and:

$J^4 = I$. (H.15)

Therefore the real-axis readout cycle is:

$R \to -Q \to -R \to Q \to R$. (H.16)

This is an operator identity.

It does not show that the market chronologically travels through the four readouts.

---

# H.4.5 Haircut identity

Given:

$H = A - R$, (H.17)

$R = A \cos \theta$, (H.18)

and:

$Q = A \sin \theta$, (H.19)

then:

$dH/d\theta = Q$. (H.20)

and:

$H(\theta) = \int_0^\theta Q(\varphi)d\varphi$. (H.21)

This distinguishes the scalar haircut H from the marginal phase coordinate Q.

---

# H.4.6 Exact active rotation

If:

$Z_{new} = \exp(i\Delta\theta)Z$, (H.22)

then:

$R_{new} = R \cos \Delta\theta - Q \sin \Delta\theta$. (H.23)

$Q_{new} = R \sin \Delta\theta + Q \cos \Delta\theta$. (H.24)

Therefore:

$\Delta R = R(\cos \Delta\theta - 1) - Q \sin \Delta\theta$. (H.25)

These are exact algebraic consequences of the declared active rotation.

They do not prove that actual market dynamics obey a fixed-amplitude rotation.

---

## H.4.7 Indicator formulas

Formulas such as:

MACD_t = EMA_fast(t) − EMA_slow(t) (H.26)

and:

RSI_t = 100 − 100/(1 + RS_t) (H.27)

are identities relative to their declared calculation rules.

Their empirical interpretations are not identities.

For example:

RSI > 70 (H.28)

may be a calculation result.

The statement:

RSI > 70 implies imminent reversal (H.29)

is an empirical hypothesis and often a regime-dependent one.

---

## H.4.8 Limits of identity claims

An identity proves only internal consistency.

It does not prove:

- empirical adequacy;
- unique interpretation;
- predictive value;
- causal status;
- economic materiality;
- transport robustness;
- intervention value.

Therefore:

IdentityTruth ≠ ModelSuperiority. (H.30)

---

# H.5 Class D — Protocol Declarations and Definitions

## H.5.1 Definition

A declaration fixes the world in which a claim will be evaluated.

Examples include:

- selected asset;
- observation boundary;
- timeframe;
- bar rule;
- feature map;
- gate;
- residual rule;
- invalidation;
- observer authority.

A declaration is not discovered in the same sense as an empirical law.

It is chosen, but its consequences can be tested.

The declaration framework treats boundary, baseline, feature map, observation rule, horizon, gate, trace, and residual as necessary for producing a readable world.

---

## H.5.2 Period declaration

The present article declares six closure periods:

Mark
Window
Structure
Event
Episode
World. (H.31)

This is a proposed taxonomy.

It is not a mathematical theorem.

Its validity depends on whether the classification proves:

- operationally distinct;
- reproducible;
- empirically useful;

- revisable.

---

## H.5.3 Functional-family declaration

The article declares four functional families:

Load / Memory
Motion / Relation
Constraint / Boundary
Commitment / Gate. (H.32)

This is likewise a proposed minimal grammar.

It should be retained only if independent analysts can use it reliably and if it improves method comparison.

---

## H.5.4 Gate declaration

A rule such as:

Breakout admitted only after a daily close 1 ATR beyond resistance with RVOL above 1.5 (H.33)

is a declaration.

Whether this rule improves breakout classification is an empirical question.

The daily close does not become a natural universal gate merely because it is declared.

It becomes the authoritative gate for that protocol.

---

## H.5.5 Residual declaration

A residual ontology such as:

R1 = missing confirmation;
R2 = direct contradiction;
R3 = frame conflict;
R4 = boundary uncertainty (H.34)

is a classification declaration.

Its usefulness depends on:

- coding reliability;
- explanatory value;
- predictive value;
- revision usefulness.

---

## H.5.6 Observer declaration

The same market state may be evaluated by:

- trader;
- exchange;
- lender;
- accountant;
- regulator;
- court.

The declaration must identify whose gate and ledger are relevant.

The formal observer source similarly treats observation, accessible trace, and cross-observer compatibility as protocol-dependent rather than assuming one universal observer.

---

# H.6 Class C — Model Constructions

# H.6.1 Definition

A construction combines declared objects into a formal model.

A construction can be logically coherent even when its empirical value remains uncertain.

---

# H.6.2 Complex valuation construction

The CAPM model constructs:

$$Z = R + iQ \quad \text{(H.35)}$$

from:

A = baseline-discounted value;
R = CAPM-discounted value;
$$Q = \sqrt{(A^2 - R^2)}. \quad \text{(H.36)}$$

The construction is internally coherent.

Its static Q is algebraically derived rather than independently measured.

The source explicitly acknowledges that its practical value must arise from dynamic attribution, comparison, communication, gate analysis, or intervention rather than from new one-period information.

---

# H.6.3 Periodic-table construction

The present article constructs a 6 × 4 matrix:

Cell(p,g)
= closure period p × functional family g. (H.37)

This matrix is not directly stated in any one source.

It is a synthesis derived from:

- protocol-bound projection;
- gate and ledger formation;
- residual governance;
- recursive world formation;
- Technical Analysis method roles.

Its scientific status is therefore:

Proposed classificatory construction. (H.38)

---

# H.6.4 Molecular-method construction

The article constructs named methods as compounds:

$$\text{Method}_j = \text{Compose}(E_{p_1,g_1},E_{p_2,g_2},\ldots,E_{p_n,g_n} \mid P). \quad (H.39)$$

For example:

$$\text{Breakout} \approx \text{StructureConstraint}$$

- EventMotion
- EventLoad
- EventCommitment
- ResidualAudit. (H.40)

This is a formal decomposition proposed by the present article.

It should be tested through:

- classification reliability;
- information decomposition;
- failure analysis;
- confirmation studies.

---

# H.6.5 Residual-bearing gate construction

The article proposes:

$$\text{GateOutput} = (\text{Decision},\text{Strength},\text{Residual},\text{Invalidation},\text{Authority}). \quad (H.41)$$

This extends a binary signal into a richer engineering object.

Its value is not guaranteed by definition.

It becomes useful if it improves:

- calibration;
- auditability;
- failure diagnosis;
- later revision.

---

# H.6.6 Ξ construction

The source protocol-first framework constructs an effective state:

$\Xi = (\rho,\gamma,\tau)$. (H.42)

The present article relabels the agitation coordinate as ν:

$\Xi_fin = (\rho,\gamma,\nu)$. (H.43)

to avoid collision with internal phase time $\tau_i$.

This is an editorial adaptation, not a claim that the source definition was incorrect.

The status of Ξ remains:

Protocol-compiled effective control state. (H.44)

not:

fundamental ontology of markets.

---

# H.6.7 Signed-regime construction

The source Technical Analysis framework introduces:

$C_\chi = [[0,F],[\chi M,0]]$. (H.45)

and, under canonical normalization:

$C_\chi^2 = \chi I$. (H.46)

This construction distinguishes:

$\chi < 0 \rightarrow$ corrective;
$\chi \approx 0 \rightarrow$ critical;
$\chi > 0 \rightarrow$ self-confirming. (H.47)

Whether empirical market data are well described by these local normal forms remains a testable hypothesis.

---

# H.7 Class T — Interpretive Translations

## H.7.1 Definition

An interpretive translation assigns a domain meaning to a mathematical or operational object.

It must be justified by the construction but is not always uniquely determined by it.

---

## H.7.2 Q as phase exposure

The interpretation:

Q = magnitude of first-order valuation-phase exposure (H.48)

is supported by:

$\partial R/\partial \theta = -Q$. (H.49)

This is stronger than calling Q “hidden risk.”

The derivative gives Q a precise local meaning.

But Q remains specific to the declared valuation geometry.

---

## H.7.3 Moving average as memory

A moving average is interpreted as a declared memory filter.

This follows from its weighted historical construction:

$MA_n(t) = \Sigma_j w_j P_{t-j}$. (H.50)

The interpretation does not imply that the moving average literally stores all market memory.

It stores one filtered price history.

The source Technical Analysis article makes this operator-first interpretation explicit.

---

## H.7.4 Candlestick wick as residual

The article interprets a wick as:

intrawindow excursion not retained by the close. (H.51)

This is a disciplined structural translation.

It does not prove why the excursion failed to remain.

The wick may reflect:

- rejection;

- thin liquidity;
- profit-taking;
- data error;
- temporary shock.

Causal interpretation remains open.

---

## H.7.5 Support as loaded boundary

Support and resistance are interpreted as historically loaded transition zones.

This is stronger than treating them as arbitrary lines, but weaker than claiming they are objective physical forces.

Their future effect remains conditional on:

- current participants;
- density;
- attention;
- orders;
- regime;
- gate behaviour.

---

## H.7.6 Phase as internal orientation

For:

$Z = R + iQ$, (H.52)

$\theta = \operatorname{atan2}(Q,R)$ (H.53)

is interpreted as orientation in the declared conjugate plane.

This does not yet justify calling $\theta$ time.

Internal-time status requires episode-alignment and gate-hazard evidence.

The phase source explicitly separates complex state, phase dynamics, secondary time, phase-bearing events, and time-bearing worlds.

---

## H.7.7 Ledger as time-bearing history

The declaration and filtration sources interpret time-bearing history as ordered, gated trace rather than raw recursion alone.

The present article translates that principle into market terms:

Event
→ Gate
→ Ledger
→ changed future admissibility. (H.54)

This is a structural interpretation.

Its strength in any specific market system requires empirical demonstration of backreaction.

---

# H.8 Class H — Empirical Hypotheses

## H.8.1 Definition

An empirical hypothesis can be supported, weakened, or rejected by observation.

It should declare:

- variables;
- protocol;
- target;
- horizon;
- benchmark;
- falsification rule.

---

## H.8.2 Functional-diversity hypothesis

The article proposes:

Functionally diverse confirmation outperforms the same number of highly redundant indicators. (H.55)

A possible test is:

Loss(DiverseConfirmation)
< Loss(RedundantConfirmation). (H.56)

after controlling for:

- method count;
- complexity;
- transaction cost;
- tuning.

This remains unproven until tested.

---

## H.8.3 Periodic-inheritance hypothesis

The article proposes:

Committed trace at period p becomes useful Load at period p + 1. (H.57)

Examples include:

- executions improving bar interpretation;
- window closes improving structural memory;
- event histories improving episode models;
- episode histories improving world-regime models.

The hypothesis fails if these records do not improve higher-period explanation or prediction.

---

## H.8.4 Residual-burden hypothesis

The proposed hypothesis is:

Higher unresolved residual burden predicts lower event persistence or greater revision probability. (H.58)

Possible outcomes include:

Pr(Failure | Residual high)

Pr(Failure | Residual low). (H.59)

The result may depend on residual type.

Some residuals may represent productive optionality rather than danger.

---

## H.8.5 Gate-strength hypothesis

The article proposes:

Stronger predeclared gate evidence predicts lower fakeout probability. (H.60)

A basic test is:

Pr(Fakeout | S_G high)
< Pr(Fakeout | S_G low). (H.61)

But the relation may be non-monotonic if highly visible signals create crowding.

---

## H.8.6 χ interaction hypothesis

The article proposes that indicator meaning changes with feedback regime.

For RSI:

Outcome
$= \beta_0 + \beta_1 RSI + \beta_2 \chi + \beta_3 RSI \times \chi + \varepsilon$. (H.62)

The hypothesis predicts:

$\beta_3 \neq 0$. (H.63)

A similar test applies to:

- band touches;
- moving-average distance;
- divergence;
- support reactions.

---

## H.8.7 Ξ incremental-value hypothesis

The proposed test is:

Model_base = f(Price,Volume,Volatility). (H.64)

Model_Ξ = f(Price,Volume,Volatility,ρ,γ,ν). (H.65)

The hypothesis is:

Loss(Model_Ξ) < Loss(Model_base) (H.66)

after complexity adjustment.

If not, Ξ remains a conceptual dashboard rather than a superior empirical model.

---

## H.8.8 Complex-priority hypothesis

For a candidate R–Q pair:

Model_complex outperforms a comparably flexible real-pair model. (H.67)

Possible criteria include:

- lower prediction error;
- greater parameter stability;
- better gate localization;
- better episode compression;
- clearer intervention.

If no gain appears, the complex-priority hypothesis fails.

---

## H.8.9 Phase-time hypothesis

The proposed hypothesis is:

D_phase < min(D_calendar,D_event). (H.68)

where:

D_phase = dispersion after phase alignment;
D_calendar = dispersion after calendar alignment;
D_event = dispersion after event-count alignment.

This is one of the central empirical requirements of the phase framework.

---

## H.8.10 Phase-sensitive gate hypothesis

The proposed hypothesis is:

Information(Gate | θ,Controls)

Information(Gate | Controls). (H.69)

or:

Loss(GateModel_phase)
< Loss(GateModel_base). (H.70)

Without such improvement, phase should not be called event-bearing.

---

## H.8.11 Ledger-backreaction hypothesis

A market world is time-bearing only if admitted events alter future dynamics.

The hypothesis is:

Information(X_{t+h} | X_t,L_t)

Information(X_{t+h} | X_t). (H.71)

If ledger history contributes no incremental information, the world-forming claim is weakened.

---

## H.8.12 Transport hypothesis

The article proposes that stronger claims survive more admissible frames.

A possible empirical claim is:

Higher transport score predicts greater event persistence. (H.72)

This may fail for genuinely local effects.

The correct test must distinguish local methods from methods making broad invariance claims.

---

# H.9 Class E — Engineering Proposals

## H.9.1 Definition

An engineering proposal specifies how to build, record, diagnose, or intervene.

It can be useful even before a full scientific theory is confirmed.

Its evaluation criteria include:

- usability;
- reliability;
- auditability;
- error reduction;
- decision quality;
- cost.

---

## H.9.2 Residual ledger

The article proposes storing:

- original claim;
- gate;
- residual;
- invalidation;
- outcome;
- revision.

This is an engineering architecture.

Its value should be tested by whether it reduces:

- hindsight bias;
- silent relabelling;
- protocol drift;
- repeated failure.

The source Technical Analysis article already supports preserving original claims, residuals, invalidations, and later outcomes.

---

## H.9.3 Method cards

The Molecular Method Cards are engineering documentation objects.

They standardize:

- period;
- functional family;
- source lineage;
- operator word;
- regime assumption;
- gate;
- residual;
- transport;
- invalidation.

Their success should be judged by classification reliability and practical clarity.

---

## H.9.4 Transport audit

The article proposes a formal transport record:

$T_{P\to P'}(C_P) \approx C_{P'}$. (H.73)

This converts “multiple timeframe confirmation” into a reproducible engineering procedure.

It does not guarantee that transport will succeed.

---

## H.9.5 Complex-model registration

The complex-eligibility card requires:

- R definition;
- Q definition;
- units;
- normalization;
- benchmark;
- gate;
- phase test;
- reduction trigger.

This is a governance tool designed to prevent decorative complexification.

---

## H.9.6 Episode completion certificate

The article proposes recording:

EpisodeCertificate
= OldGrammarFailed

- NewGateAdmitted
- Persistence
- Residual
- BranchStatus. (H.74)

This is an engineering proposal for controlling retrospective episode relabelling.

---

# H.10 Class A — Analogies and Cross-Domain Extensions

## H.10.1 Definition

An analogy asserts that two domains share a structural role or relation.

It does not assert that their underlying substances or causal mechanisms are identical.

---

## H.10.2 Gauge analogy

The protocol-first finance source translates gauge concepts into market-language roles such as:

- propagation;
- switching;
- binding;
- basin history.

It explicitly presents these as operational translations rather than literal physical ontology.

Therefore:

MarketConstraint analogous to gauge binding (H.75)

does not imply:

Markets are physical gauge fields. (H.76)

---

## H.10.3 Quantum-observer analogy

Projection, gate, and trace may resemble quantum-measurement grammar.

But the present framework does not establish:

- Born probabilities;
- physical superposition;
- nonlocal entanglement;
- Bell inequality violation;
- quantum wavefunction ontology.

The CAPM source separates complex geometry from specifically quantum residues.

---

## H.10.4 Periodic-table analogy

The table is called proto-periodic because functional families recur across closure levels.

This does not imply a market equivalent of:

- atomic number;
- electron shell;
- chemical valence;
- physical element.

The analogy is organizational and generative.

Its strength depends on whether it predicts useful missing instrument families.

---

## H.10.5 Grammar analogy

Technical Analysis methods are treated as grammatical compounds.

For example:

Load + Motion + Constraint + Gate → Event. (H.77)

This does not imply that markets literally speak a language.

It means the framework distinguishes:

- typed roles;
- valid combinations;
- incomplete statements;
- malformed event claims.

---

## H.11 Claims Explicitly Not Established

The present article does **not** establish the following propositions.

## H.11.1 Universal market periodicity

It does not prove that all markets obey a universal six-period cycle.

---

## H.11.2 Guaranteed profitability

It does not prove that the proposed classification generates profitable trading rules.

---

## H.11.3 Universal complex market ontology

It does not prove that market reality is fundamentally complex-valued.

---

## H.11.4 Universal Q coordinate

It does not establish one common Q applicable across:

- valuation;
- technical analysis;
- liquidity;
- breadth;
- sentiment;
- accounting;
- law.

---

## H.11.5 Phase as universal clock

It does not establish that every market episode possesses an internal phase time.

---

## H.11.6 Quantum market ontology

It does not establish that markets are quantum systems in the physical sense.

---

## H.11.7 Causal superiority of Technical Analysis

It does not prove that chart-based methods identify causal mechanisms better than:

- econometrics;
- market microstructure;

- fundamental analysis;
- institutional analysis;
- machine learning.

---

## H.11.8 Uniqueness of the 6 × 4 table

It does not prove that no other classification could perform equally well or better.

---

## H.12 Claim Audit of the Central Periodic Law

The article's central recursive relation is:

Load_p
→ Motion_p under Constraint_p
→ Commitment_p
→ Trace_p + Residual_p
→ Ledger_{p+1}
→ Load_{p+1}. (H.78)

This statement contains several claim classes.

## Declaration component

The four functional families and six periods are declared categories.

## Construction component

The sequence is a proposed grammar connecting them.

## Interpretive component

Committed trace is interpreted as higher-period load.

## Empirical component

The recurrence should improve classification and prediction across actual market records.

## Engineering component

The sequence guides database and workflow design.

Therefore the central law should currently be labelled:

Proposed recursive market-observation law. (H.79)

not:

Proven universal law of financial markets. (H.80)

## H.13 Claim Audit of χ

The statement:

$\chi < 0$ corresponds to corrective feedback (H.81)

is a model interpretation.

The statement:

ordinary complex algebra may locally represent $\chi < 0$ (H.82)

is a mathematical analogy based on a normalized operator satisfying:

$e^2 = -1$. (H.83)

The statement:

RSI reversals work better when $\chi < 0$ (H.84)

is an empirical hypothesis.

The statement:

$\chi$ can be estimated reliably in real time (H.85)

is an engineering and empirical question.

These four statements must not be merged.

---

## H.14 Claim Audit of Ξ

The state:

$\Xi = (\rho, \gamma, \nu)$ (H.86)

is a protocol-compiled construction.

The interpretations are:

$\rho$ = loading;
$\gamma$ = lock-in;
$\nu$ = agitation. (H.87)

The statement:

Ξ provides a useful control dashboard (H.88)

is an engineering hypothesis.

The statement:

Ξ improves crisis prediction beyond price and volume (H.89)

is an empirical hypothesis.

The statement:

markets fundamentally consist of three variables (H.90)

is not asserted.

The source PORE framework likewise presents Ξ as a protocol-relative operational interface compiled from a richer field rather than a final ontology.

---

# H.15 Claim Audit of the CAPM Complex State

## Identity

$A^2 = R^2 + Q^2$. (H.91)

## Construction

A and R are values of the same cash flow under two declared discount protocols.

## Interpretation

Q is conjugate phase exposure because:

$\partial R/\partial \theta = -Q$. (H.92)

## Empirical hypothesis

The complex representation improves dynamic risk attribution, communication, comparison, or intervention.

## Unsupported extension

All financial risk is Q.

The first three are supported inside the source construction.

The fourth requires testing.

The fifth is rejected by the source's own limitations.

---

# H.16 Claim Audit of Phase Time

## Identity

$\theta = \operatorname{atan2}(Q,R)$. (H.93)

## Construction

$\tau_i = \text{Unwrap}(\theta)$ (H.94)

or:

$\tau_i = \int |\theta| dt.$ (H.95)

# Interpretation

$\tau_i$ measures accumulated phase progress.

# Empirical hypothesis

$\tau_i$ aligns comparable episodes better than calendar time.

# Strong world claim

Phase-sensitive gates create persistent trace that changes future dynamics.

The identity and construction are immediate once R and Q are supplied.

The strong world claim requires the entire gate–ledger–backreaction chain.

---

# H.17 Claim Audit of Confirmation Independence

The article proposes:

IndicatorCount ≠ IndependentEvidence. (H.96)

This is partly definitional because indicators may share inputs and operators.

The stronger statement:

Functionally diverse evidence improves event calibration (H.97)

is empirical.

A valid test must control for:

- number of indicators;
- parameter count;
- feature richness;
- regime;
- cost;
- multiple testing.

The claim may fail in some applications if one highly informative source dominates all others.

---

# H.18 Claim Audit of Residual Governance

The statement:

Every bounded model leaves residual (H.98)

is a modelling principle.

The statement:

Recording residual improves scientific honesty (H.99)

is a methodological proposition.

The statement:

Residual burden predicts later failure (H.100)

is an empirical hypothesis.

The statement:

Residual always becomes future crisis (H.101)

is not supported.

Residual may:

- dissipate;
- resolve;
- remain irrelevant;
- become structure;
- trigger revision.

---

# H.19 Evidence-Burden Ladder

The burden of evidence rises through seven levels.

# Level 1 — Internal consistency

Are the equations and definitions coherent?

# Level 2 — Operational clarity

Can another observer reconstruct the variables?

# Level 3 — Measurement reliability

Are the variables estimated consistently?

# Level 4 — Incremental empirical value

Does the model add information beyond simpler alternatives?

## Level 5 — Cross-frame robustness

Does the relation survive admissible transformations?

## Level 6 — Gate and intervention value

Does the model improve consequential decisions?

## Level 7 — Ledgered world formation

Do admitted events alter subsequent system dynamics?

This ladder can be written:

Consistency
→ Reproducibility
→ Reliability
→ IncrementalValue
→ Transport
→ Intervention
→ Backreaction. (H.102)

A claim should not be promoted past the highest level actually supported.

---

# H.20 Theorem, Model, and Hypothesis Language

The following vocabulary should be used carefully.

## Theorem

Use only when a statement follows deductively from stated assumptions.

Example:

Given $R = A \cos \theta$ and $Q = A \sin \theta$, $\partial R/\partial \theta = -Q$. (H.103)

---

## Proposition

Use for a formal consequence whose assumptions are explicit but whose domain application may remain limited.

---

## Construction

Use when defining a model object.

Example:

$Z_{TA} = R_{TA} + iQ_{TA}$. (H.104)

---

## Interpretation

Use when assigning domain meaning.

Example:

Q_TA is interpreted as independently measured breadth pressure. (H.105)

---

## Hypothesis

Use when the statement requires empirical testing.

Example:

Phase predicts gate hazard. (H.106)

---

## Framework

Use for an organized set of definitions, constructions, and research tests.

---

## Law

Reserve for a relation showing broad, repeatable, cross-protocol survival.

The filtration source similarly treats law as what survives admissible disclosures rather than as mere repeated appearance.

---

# H.21 Mandatory Claim Header for Future Studies

A study applying this framework should publish:

```
Claim:
Claim class:
Source or derivation:
Protocol:
Variables:
Units:
Assumptions:
Benchmark:
Evidence:
Residual:
Falsification condition:
Highest supported evidence level:
Unsupported stronger interpretation:
```

Example:

```
Claim:
Phase improves breakout-event alignment.

Claim class:
[H] Empirical hypothesis.
```

```
Protocol:
Daily equity-index breakouts from twelve-week ranges.

Variables:
R = normalized accepted price structure.
Q = independently measured breadth pressure.

Benchmark:
Calendar time, event count, and real-pair model.

Falsification:
No out-of-sample alignment or gate-hazard gain.

Highest supported level:
Not yet established.
```

---

# H.22 Claim Downgrading Rules

A claim should be downgraded when its evidence fails.

# From law to regularity

When cross-frame survival fails.

# From event to warning

When the commitment gate is absent.

# From phase time to phase description

When episode alignment fails.

# From complex state to real pair

When complex priority fails.

# From causal explanation to association

When intervention or identification is absent.

# From world model to event model

When ledger backreaction is not demonstrated.

# From general framework to local model

When the relation survives only one protocol family.

Downgrading is not rhetorical retreat.

It is evidence alignment.

---

## H.23 Claim Promotion Rules

A claim may be promoted only after passing the next evidential gate.

## Warning → Event

Requires Commitment gate.

## Event → Episode transition

Requires persistent new grammar.

## Local pattern → Cross-frame regularity

Requires transport.

## Real pair → Complex state

Requires stable conjugate structure and benchmark gain.

## Complex state → Phase time

Requires episode-alignment gain.

## Phase time → Phase-sensitive event model

Requires gate-hazard concentration.

## Event model → Time-bearing world

Requires ledger dependence and backreaction.

The promotion chain is:

Description
→ Relation
→ Gate
→ Trace
→ Persistence
→ Transport
→ Backreaction. (H.107)

---

## H.24 Source-Derived Claims Versus Present Synthesis

To prevent attribution confusion, the article's claims can be divided into two groups.

## H.24.1 Directly source-derived foundations

The source articles support:

- protocol-relative projection;
- declaration before observation;
- gate, trace, residual, and ledger distinction;
- admissible self-revision;
- moving averages as memory filters;
- $\chi$ as feedback signature;
- breakout confirmation through close, volume, breadth, and related evidence;
- Q as CAPM phase exposure;
- distinction between measurement, movement, recognition, and ledger;
- phase-time validation through episode alignment and gates;
- complex-model reduction to simpler real models when evidence fails.

These claims are grounded in the cited sources.

---

## H.24.2 Present article's principal synthesis

The following are introduced or substantially reorganized by the present article:

- the six closure periods;
- the four-column proto-periodic table;
- the periodic inheritance law;
- named methods as molecular compounds;
- confirmation independence across source, operator, function, and failure mode;
- the standard Method Card;
- the seven-ledger database architecture;
- the empty-cell research programme;
- the unified $\chi$–$\Xi$–Z–$\tau_i$ hierarchy;
- the explicit claim-tagging system.

These should be presented as proposed synthesis rather than as conclusions already established by the source papers.

---

## H.25 Claim Matrix for the Article's Major Propositions

| Proposition | Claim class | Current status | Required next evidence |
|---|---|---|---|

| Proposition | Claim class | Current status | Required next evidence |
|---|---|---|---|
| $A^2 = R^2 + Q^2$ in CAPM construction | I | exact | none beyond correct declaration |
| $\partial R/\partial \theta = -Q$ | I/T | exact and interpretable | practical value still empirical |
| Moving average is a memory filter | T | strongly supported by construction | none for basic interpretation |
| RSI meaning changes with $\chi$ | H | plausible and source-motivated | regime-conditioned tests |
| Breakout requires a gate beyond crossing | D/T/H | strong methodological claim | gate-comparison studies |
| Six periods form distinct closure levels | D/C/H | proposed synthesis | inter-rater and empirical validation |
| Four functions recur across periods | C/H | proposed synthesis | cross-market classification |
| Commitment_p becomes Load_{p+1} | H | central hypothesis | period-transition tests |
| Functional diversity improves confirmation | H | unproven | controlled comparison |
| Residual burden predicts failure | H | unproven | prospective residual database |
| $\Xi$ improves regime diagnosis | H | unproven | benchmark comparison |
| TA supports stable complex states | H | local candidate only | independent Q and generator tests |
| Phase aligns episodes better than time | H | unproven generally | out-of-sample alignment |
| Phase-sensitive gates create time-bearing worlds | H/A | strong research claim | ledger and backreaction evidence |
| Technical Analysis is a periodic grammar | C/H | conceptual framework | reliability, prediction, and missing-cell tests |

# H.26 Falsification Boundary of the Whole Framework

The framework should be substantially revised or rejected if the following results occur consistently.

# Classification failure

Independent analysts cannot reliably distinguish periods or functional families.

# Redundancy failure

Functional decomposition adds no useful information beyond ordinary indicator labels.

# Periodic failure

The same functional groups do not recur meaningfully across closure levels.

## Inheritance failure

Lower-period commitments do not improve higher-period models.

## Gate failure

Explicit gates do not improve calibration or failure diagnosis.

## Residual failure

Residual records cannot be coded reliably or add no value.

## Transport failure

Cross-frame audit proves too ambiguous to operationalize.

## Complex failure

Complex models consistently fail to outperform real pairs.

## Phase-time failure

Internal phase does not improve episode order or event hazard.

## Engineering failure

The resulting database and cards are too costly or complex to improve decisions.

A framework surviving only through reinterpretation after every such failure would become unfalsifiable.

---

# H.27 The Minimum Defensible Article Claim

Even if the strongest empirical hypotheses fail, one modest claim may remain defensible:

> Technical Analysis methods can be clarified by separating what they remember, how they measure motion, which boundaries they construct, and what gates convert a reading into consequential trace.

This claim is primarily classificatory and methodological.

It does not require:

- universal market periodicity;
- complex phase;
- internal time;
- predictive superiority.

The stronger periodic, complex, and world-forming claims remain optional layers requiring additional evidence.

---

# H.28 The Maximum Research Claim

The strongest research programme proposed by the article is:

> Markets may sometimes form protocol-bound, recursively layered observational worlds in which loaded traces generate motion under constraints, gates convert selected transitions into history, residuals preserve non-closure, and some independently measurable conjugate states may support phase-sensitive internal ordering whose ledgered events backreact on future market dynamics.

This statement combines:

- protocol construction;
- recursive closure;
- residual governance;
- complex eligibility;
- phase time;
- ledger backreaction.

It should be presented as a programme of testable hypotheses, not as an established universal theory.

---

# H.29 Appendix H Conclusion

The article's scientific discipline depends less on how advanced its mathematics appears than on whether it keeps distinct kinds of claim separate.

The governing distinctions are:

Identity ≠ Interpretation. (H.108)

Definition ≠ Discovery. (H.109)

Construction ≠ Empirical Superiority. (H.110)

Analogy ≠ Ontology. (H.111)

Phase ≠ Time. (H.112)

Gate ≠ Total Resolution. (H.113)

Trace ≠ Universal Law. (H.114)

Revision ≠ Permission to Erase Failure. (H.115)

The complete claim discipline is:

Declare the object
→ identify the claim class
→ state the assumptions
→ provide the correct evidence
→ preserve residual
→ define falsification
→ limit the conclusion to the highest level actually supported. (H.116)

The next appendix can provide a **Full Research Protocol and Benchmarking Blueprint**, turning the article into a staged empirical programme for indicator classification, breakout gates, χ estimation, residual prediction, complex eligibility, and phase-time validation.

# Appendix I — Full Research Protocol and Benchmarking Blueprint

## I.1 Purpose

The Periodic Grammar becomes scientifically meaningful only when its claims are translated into a staged empirical programme.

The programme must test several questions separately:

1. Can observers classify Technical Analysis methods reliably?
2. Does the four-family decomposition reveal genuine informational differences?
3. Do explicit gates outperform simple line-crossing rules?
4. Does χ improve regime-conditioned interpretation?
5. Does residual recording predict failure, delay, or revision?
6. Do transport-robust claims survive better than frame-local claims?
7. Does Ξ add value beyond ordinary price, volume, and volatility?
8. Do candidate complex states outperform real-pair models?
9. Does phase align episodes and gates better than calendar time?
10. Do ledgered interpretations measurably backreact on later market dynamics?

These questions should not be tested as one indivisible theory.

The proper sequence is:

Classification
→ Measurement
→ Gate
→ Residual
→ Transport
→ Regime
→ Complex eligibility
→ Phase time
→ Ledger backreaction. (I.1)

A failure at one stage should constrain later claims.

It should not be repaired by adding more abstract structure.

---

# I.2 Programme Architecture

The research programme contains five layers.

## Layer 1 — Taxonomic validation

Test whether:

- the six periods are distinguishable;
- the four functional families are distinguishable;
- named methods can be decomposed consistently.

## Layer 2 — Diagnostic validation

Test whether the classifications improve:

- signal interpretation;
- redundancy detection;
- failure diagnosis;
- event labelling.

## Layer 3 — Event validation

Test whether:

- explicit gates improve event persistence;
- residual burden predicts fakeout or revision;
- transport robustness improves calibration.

## Layer 4 — Advanced-state validation

Test:

- $\chi$;
- $\Xi$;
- complex R–Q states;
- phase ordering;
- internal phase time.

## Layer 5 — World validation

Test whether:

- ledger history changes later transition probabilities;
- signal adoption produces backreaction;
- institutional gates create persistent market consequences.

The layers form an evidence hierarchy:

Taxonomy
≺ Diagnosis
≺ Event Model
≺ Phase Model
≺ Time-Bearing World. (I.2)

The symbol ≺ means “requires less evidence than.”

---

# I.3 Research-Programme Registration

Every study should be registered before the primary analysis.

The registration should declare:

```
Study ID:
Research question:
Claim class:
Primary hypothesis:
Secondary hypotheses:
Asset class:
Market universe:
Sample period:
Data sources:
Protocol family:
Unit of observation:
Outcome horizon:
Training period:
Validation period:
Test period:
Primary benchmark:
Statistical method:
Multiple-testing rule:
Transaction-cost assumptions:
Residual ontology:
Transport tests:
Reduction rule:
Stopping rule:
```

The registration should also distinguish:

- exploratory analysis;
- confirmatory analysis;
- post-hoc interpretation.

A result discovered during exploration may motivate a future confirmatory test.

It should not be relabelled as predeclared evidence.

---

# I.4 Data Architecture

# I.4.1 Required data layers

A full implementation may contain six data layers.

### Layer A — Mark data

- trades;
- quotes;
- bid–ask spread;
- depth;
- cancellations;
- venue.

### Layer B — Window data

- OHLCV;
- settlement;
- session definitions;
- alternative bar constructions.

### Layer C — Structure data

- moving averages;
- oscillators;
- profiles;
- breadth;
- support and resistance;
- volatility bands.

### Layer D — Event data

- breakouts;
- retests;
- rejections;
- reversals;

- gaps;
- earnings;
- policy announcements.

### Layer E — Episode data

- trends;
- ranges;
- squeezes;
- crises;
- recoveries;
- wave or regime segments.

### Layer F — World data

- leverage;
- open interest;
- funding;
- collateral;
- legal recognition;
- accounting events;
- policy changes;
- benchmark membership.

Not every study requires all six layers.

The data burden should match the claim.

A Structure-period moving-average study does not require legal data.

A World-period default study cannot rely on candles alone.

---

# I.4.2 Immutable raw layer

The raw data layer should be immutable.

Derived features should be versioned separately:

RawData
→ CleaningVersion
→ FeatureVersion
→ ProtocolVersion
→ ClaimVersion. (I.3)

This prevents later feature changes from silently altering historical signals.

---

## I.4.3 Time discipline

Every timestamp should declare:

- timezone;
- daylight-saving treatment;
- market-session convention;
- event publication time;
- data-availability time;
- revision time.

The analysis must use information as it was available at the decision time.

For an event released at t_e:

Feature_t may include Event only when t ≥ t_e + DataLatency. (I.4)

This avoids look-ahead bias.

---

## I.4.4 Corporate-action discipline

Price and volume data should preserve:

- raw series;
- adjusted series;
- adjustment factor;
- adjustment date;
- provider rule.

All transported levels should use consistent adjustment.

---

# I.5 Sampling Design

## I.5.1 Cross-sectional sampling

The study should include markets with different:

- liquidity;
- volatility;
- institutional structure;

- participant concentration;
- trading hours;
- market maturity.

Possible classes include:

- large-cap equities;
- small-cap equities;
- indices;
- sovereign bonds;
- corporate credit;
- foreign exchange;
- commodities;
- listed futures;
- liquid cryptocurrencies.

The purpose is not to assume universality.

It is to determine where the grammar transports and where it remains local.

---

## I.5.2 Temporal sampling

The sample should include:

- quiet regimes;
- trends;
- ranges;
- crises;
- recoveries;
- policy transitions;
- structural market changes.

A model tested only in one persistent bull market cannot support a general regime claim.

---

## I.5.3 Event sampling

For event studies, define the event universe before outcome review.

For breakouts:

BreakoutUniverse
= all predeclared boundary-crossing candidates satisfying minimum data quality. (I.5)

Do not select only visually clean examples.

---

## I.5.4 Survivorship control

The sample should include:

- delisted securities;
- bankrupt entities;
- index removals;
- failed funds;
- discontinued contracts.

Otherwise apparent signal performance may partly reflect survival.

---

## I.6 Workstream 1 — Classification Reliability

## I.6.1 Question

Can independent analysts use the periodic grammar consistently?

---

## I.6.2 Test objects

Analysts classify:

- named methods;
- individual signals;
- market episodes;
- institutional events.

Each object receives:

- primary period;
- primary family;
- secondary family;
- gate status;
- residual type.

---

## I.6.3 Analyst groups

Use at least three groups where feasible:

1. experienced Technical Analysis practitioners;
2. quantitative researchers;
3. trained but initially neutral coders.

This tests whether the grammar depends on prior doctrinal commitment.

---

## I.6.4 Training protocol

Coders receive:

- symbol dictionary;
- cell definitions;
- positive examples;
- boundary cases;
- classification exercises;
- adjudication rules.

The test set should remain hidden during training.

---

## I.6.5 Reliability metrics

For nominal classification, use:

- Cohen's κ for two raters;
- Fleiss' κ for multiple raters;
- Krippendorff's α for missing or mixed data;
- confusion matrices.

For hierarchical classification, measure:

- exact agreement;
- period-only agreement;
- family-only agreement;
- adjacent-period disagreement;
- severe disagreement.

A provisional minimum is:

$\kappa \geq 0.60$ for exploratory use. (I.6)

$\kappa \geq 0.75$ for stronger operational use. (I.7)

These thresholds are conventions, not natural laws.

---

## I.6.6 Failure analysis

When coders disagree, classify the disagreement:

- period ambiguity;
- functional overlap;
- protocol omission;
- method compound;
- insufficient data;
- genuinely disputed interpretation.

The purpose is not merely to maximize $\kappa$.

It is to determine where the table needs refinement.

---

## I.6.7 Falsification condition

The taxonomy is materially weakened when:

- agreement remains low after training;
- disagreements are not concentrated in identifiable boundary cases;
- alternative simpler taxonomies produce higher reliability.

---

# I.7 Workstream 2 — Molecular Decomposition and Redundancy

## I.7.1 Question

Do source lineage, operator lineage, and functional lineage explain indicator redundancy better than conventional category labels?

---

## I.7.2 Feature families

Construct a broad feature library containing:

- moving averages;

- crossovers;
- MACD;
- RSI;
- stochastic;
- ATR;
- volatility bands;
- volume;
- OBV;
- VWAP;
- volume profile;
- breadth;
- support and resistance;
- breakout variables;
- candlestick features.

Each feature receives:

- source lineage S_j;
- operator word w_j;
- period p_j;
- family g_j.

---

# I.7.3 Empirical redundancy

Estimate several forms of redundancy.

## Correlation redundancy

R_corr,ij = |Corr(X_i,X_j)|. (I.8)

## Conditional-information redundancy

R_info,ij
= 1 − I(Y;X_i,X_j)/[I(Y;X_i) + I(Y;X_j) + ε]. (I.9)

## Error redundancy

R_error,ij = Corr(e_i,e_j). (I.10)

## Regime redundancy

R_regime,ij
= similarity of performance across χ or volatility regimes. (I.11)

---

# I.7.4 Predicted redundancy

Construct a grammar-derived score:

R_grammar,ij
= w_SS_source

- w_OO_operator
- w_HH_horizon
- w_GG_family
- w_PP_period. (I.12)

The hypothesis is:

R_grammar,ij predicts R_empirical,ij. (I.13)

---

# I.7.5 Benchmark taxonomies

Compare the periodic grammar with:

- trend versus momentum versus volatility categories;
- price versus volume categories;
- unsupervised feature clustering;
- purely statistical correlation clustering;
- no taxonomy.

---

# I.7.6 Confirmation portfolio test

Construct equal-size evidence bundles.

### Bundle A — high redundancy

Several price-derived trend indicators.

### Bundle B — functional diversity

One Load, one Motion, one Constraint, and one Commitment feature.

Compare:

- calibration;
- false-positive rate;

- event persistence;
- turnover;
- transaction-cost-adjusted performance;
- stability across regimes.

The hypothesis is:

Performance_B,adjusted > Performance_A,adjusted. (I.14)

---

## I.7.7 Ablation

Remove one family at a time:

Model_full. (I.15)

Model_−Load. (I.16)

Model_−Motion. (I.17)

Model_−Constraint. (I.18)

Model_−Commitment. (I.19)

Measure which family contributes incremental value for each task.

---

# I.8 Workstream 3 — Gate Validation

## I.8.1 Question

Does separating boundary crossing from Commitment improve event classification?

The source Technical Analysis framework treats a touch or crossing as insufficient and requires gate evidence such as close, volume, breadth, follow-through, or retest.

---

## I.8.2 Candidate-event universe

For breakouts, collect every case where:

Price_t crosses predeclared Boundary_t. (I.20)

This creates the candidate universe.

Do not require later success for inclusion.

---

## I.8.3 Gate variants

Compare:

### $G_0$ — crossing only

Price crosses boundary.

### $G_1$ — closing gate

Relevant window closes beyond boundary.

### $G_2$ — close + volume

Add relative-volume threshold.

### $G_3$ — close + volume + breadth

Add participation coherence.

### $G_4$ — close + volume + breadth + retest

Add durable acceptance.

### $G_5$ — residual-adjusted gate

Subtract conflicting higher-frame and liquidity evidence.

---

# I.8.4 Outcomes

Possible outcome definitions include:

- survival beyond boundary for h periods;
- maximum adverse excursion;
- return after h;
- failed reclaim;
- fakeout;
- episode transition;
- value-area migration.

No single outcome should define all gate quality.

---

# I.8.5 Gate calibration

For predicted probability $p_i$ and event outcome $y_i$, evaluate:

- Brier score;
- log loss;
- reliability curve;

- expected calibration error;
- precision–recall;
- time-to-failure.

A stronger gate should improve calibration, not merely reduce signal count.

---

# I.8.6 Delay–accuracy frontier

For each gate, measure:

Delay_g = GateTime_g – InitialCrossTime. (I.21)

Accuracy_g = 1 – FalseAdmissionRate_g. (I.22)

Plot the Pareto frontier:

Accuracy versus Delay. (I.23)

This reveals whether added confirmation justifies later entry.

---

# I.8.7 Crowding interaction

Test whether highly visible gate bundles exhibit:

- stronger early continuation;
- greater later reversal;
- larger shared-stop effects.

A possible model is:

Outcome
$= \beta_0 + \beta_1 \text{GateStrength} + \beta_2 \text{Crowding}$

- $\beta_3 \text{GateStrength} \times \text{Crowding} + \varepsilon$. (I.24)

---

# I.9 Workstream 4 — Estimating χ

# I.9.1 Question

Can the corrective, critical, and self-confirming feedback signatures be estimated reliably and used to condition indicator interpretation?

---

# I.9.2 Response-based estimator

Let d_t be normalized displacement from memory or boundary.

Let r_{t→t+h} be later return.

Estimate:

$\hat{\chi}_{t,h} = \text{sign}\{d_t \cdot E[r_{t\to t+h} \mid State_t]\}$. (I.25)

Interpretation:

$\hat{\chi} < 0$ → corrective response.
$\hat{\chi} \approx 0$ → weak or unstable response.
$\hat{\chi} > 0$ → continuation response.

This is one possible estimator, not a definitive χ formula.

---

# I.9.3 State-space estimator

Model:

$r_{t+1} = a_{s_t} r_t + b_{s_t} X_t + \varepsilon_t$. (I.26)

where $s_t$ is a latent regime.

Then:

$a_{s_t} < 0$ may indicate corrective feedback. (I.27)

$a_{s_t} \approx 0$ may indicate weak persistence. (I.28)

$a_{s_t} > 0$ may indicate self-confirming persistence. (I.29)

The mapping depends on horizon and model specification.

---

# I.9.4 Event-response estimator

Estimate the average response after:

- band touch;
- RSI extreme;
- support test;
- breakout;
- divergence.

For event type e:

$\hat{\chi}_e = \text{sign}[E[DirectionalContinuation_e - Reversal_e]]$. (I.30)

---

# I.9.5 Indicator interaction tests

For RSI:

Outcome
$= \beta_0 + \beta_1 RSI + \beta_2 \hat{\chi} + \beta_3 RSI \times \hat{\chi} + Controls + \varepsilon.$ (I.31)

For band displacement:

Outcome
$= \gamma_0 + \gamma_1 BandDistance + \gamma_2 \hat{\chi}$

- $\gamma_3 BandDistance \times \hat{\chi} + Controls + \varepsilon.$ (I.32)

The framework predicts non-zero interaction terms.

---

# I.9.6 Real-time constraint

$\chi$ must be estimated using information available at t.

A retrospective regime label may be useful for explanation.

It does not validate a prospective regime-conditioned rule.

---

# I.9.7 Benchmark models

Compare $\chi$-conditioned models with:

- volatility regime;
- trend filter;
- moving-average state;
- hidden Markov model;
- machine-learned regime classifier;
- no regime conditioning.

---

# I.9.8 Failure condition

$\chi$ is weakened when:

- estimates are unstable;
- different estimators disagree without explainable mapping;
- interaction terms fail out of sample;
- conventional regime variables perform equally well with less complexity.

---

# I.10 Workstream 5 — Residual-Burden Validation

# I.10.1 Question

Does explicit residual recording improve event calibration, revision quality, and failure diagnosis?

The source framework requires residuals, invalidations, and original claims to remain preserved rather than being erased after failure.

---

## I.10.2 Residual coding

Code residuals using the ontology:

- missing confirmation;
- direct contradiction;
- frame conflict;
- boundary uncertainty;
- regime uncertainty;
- data residual;
- liquidity and positioning;
- institutional residual;
- branch residual;
- model residual.

---

## I.10.3 Human versus automated coding

Compare:

- analyst-coded residuals;
- rule-based residuals;
- machine-classified residuals;
- hybrid adjudication.

Measure:

- inter-rater agreement;
- stability;
- predictive value;
- coding cost.

---

## I.10.4 Residual-burden score

A simple score is:

$RB_k = \Sigma_j s_j p_j d_j$. (I.33)

where:

$s_j$ = severity;
$p_j$ = persistence;
$d_j$ = directional relevance or claim impact.

The components must be declared.

The score should not replace the underlying residual vector.

---

# I.10.5 Outcomes

Test whether RB predicts:

- fakeout;
- delayed continuation;
- adverse excursion;
- revision;
- gate failure;
- transport failure;
- episode reclassification.

A basic hazard model is:

$h_{failure}(t) = h_0(t)\exp(\beta_{RRB} + \beta_{GS} G + \beta_{\chi\chi} + \beta^T X)$. (I.34)

The hypothesis is:

$\beta_R > 0$. (I.35)

But some residual types may reduce rather than increase directional risk.

Effects should be estimated separately by type.

---

# I.10.6 Residual-resolution study

Measure:

$T_{resolve} = t_{resolution} - t_{open}$. (I.36)

Classify resolution as:

- confirmed;
- dissipated;

- invalidated;
- superseded;
- unresolved.

Test whether different residual types have distinct resolution-time distributions.

---

## I.10.7 Residual-ledger experiment

Compare two analyst groups.

### Group A

Uses ordinary signal and outcome records.

### Group B

Uses residual, invalidation, transport, and revision fields.

Measure:

- hindsight bias;
- relabelling rate;
- calibration;
- consistency;
- false certainty;
- model improvement after revision.

---

# I.11 Workstream 6 — Transport and Invariance

## I.11.1 Question

Do claims surviving admissible frame changes exhibit greater persistence or objectivity?

---

## I.11.2 Transformation set

Test the relevant subset of:

- timeframe;
- log scale;
- volatility normalization;
- bar construction;

- price field;
- benchmark;
- universe;
- anchor;
- pivot rule;
- data vendor;
- cash versus derivatives;
- observer role;
- gate authority.

---

## I.11.3 Transport score

For claim i:

S_T,i = Σ_jw_js_{ij}/Σ_jw_j. (I.37)

where:

s_{ij} = 1, 0.5, or 0 for survival, partial survival, or failure.

Also retain the vector:

$\mathcal{T}$_i = (s_{i1},s_{i2},...,s_{in}). (I.38)

The vector may be more informative than the scalar score.

---

## I.11.4 Outcome test

Estimate:

Outcome_i
= α + β_TS_T,i + β_GS_G,i + β_RRB_i + Controls + ε_i. (I.39)

The hypothesis is:

β_T > 0 (I.40)

for methods making cross-frame claims.

Local methods should be evaluated separately.

---

## I.11.5 Transport versus signal scarcity

Higher transport requirements may sharply reduce signal count.

Measure:

SignalFrequency(S_T threshold). (I.41)

Calibration(S_T threshold). (I.42)

EconomicValue(S_T threshold). (I.43)

The optimal threshold depends on decision cost.

---

## I.11.6 Failure decomposition

When transport fails, classify:

- representation failure;
- scale failure;
- aggregation failure;
- anchor failure;
- pivot failure;
- universe failure;
- gate failure;
- authority failure;
- regime failure;
- model failure.

This converts failed invariance into diagnostic evidence.

---

# I.12 Workstream 7 — Ξ Control-State Validation

## I.12.1 Question

Does the compiled state:

Ξ = (ρ,γ,ν) (I.44)

improve regime or transition diagnosis beyond ordinary market variables?

The source PORE architecture treats Ξ as a protocol-relative operational interface compiled from a richer field rather than a fundamental ontology.

---

## I.12.2 Candidate ρ proxies

Possible loading proxies include:

- volume;

- dollar volume;
- open interest;
- leverage;
- transaction density;
- participation;
- position concentration;
- narrative attention.

---

## I.12.3 Candidate γ proxies

Possible lock-in proxies include:

- spread;
- market depth;
- concentration;
- collateral encumbrance;
- funding maturity;
- margin requirements;
- legal restrictions;
- exit cost.

---

## I.12.4 Candidate ν proxies

Possible agitation proxies include:

- realized volatility;
- intraday range;
- cancellation rate;
- spread instability;
- cross-sectional dispersion;
- correlation breakdown;
- failed-gate frequency.

---

## I.12.5 Compilation methods

Compare:

- theory-weighted index;
- principal components;
- factor analysis;
- supervised latent state;
- Bayesian state-space model;
- autoencoder or nonlinear representation.

The final Ξ coordinates should retain interpretable monotonicity.

---

# I.12.6 Benchmark comparison

Compare:

Model_base
= f(Return,Volume,Volatility,Breadth). (I.45)

Model_Ξ
= f(Return,Volume,Volatility,Breadth,ρ,γ,ν). (I.46)

Evaluate:

- transition prediction;
- crisis hazard;
- fakeout;
- liquidity failure;
- regime duration;
- intervention response.

---

# I.12.7 Protocol transport

Test Ξ across:

- desk versus enterprise;
- cash versus derivatives;
- local versus systemic boundary;
- daily versus weekly horizon.

If coordinates change meaning across frames, they must remain frame-indexed:

Ξ_P rather than Ξ. (I.47)

---

## I.12.8 Failure condition

Ξ should be reduced when:

- coordinates cannot be estimated consistently;
- signs or meanings drift;
- conventional variables perform equally well;
- the compiled index conceals important residual channels.

---

# I.13 Workstream 8 — Complex Eligibility

## I.13.1 Question

Does a candidate R–Q state earn complex priority over a real-pair model?

---

## I.13.2 Candidate construction

For each proposal, register:

- R definition;
- Q definition;
- units;
- normalization;
- amplitude;
- phase;
- generator;
- residual;
- gate;
- benchmark.

---

## I.13.3 Model hierarchy

Estimate:

$M_0$ = scalar model. (I.48)

$M_1$ = real-pair model. (I.49)

$M_2$ = complex amplitude–phase model. (I.50)

$M_3$ = regime-switching elliptic/hyperbolic model. (I.51)

The phase framework requires comparison with flexible two-real-variable alternatives and reduction when complexification adds no value.

---

# I.13.4 Generator tests

For elliptic dynamics:

$\dot{R} = g_AR - \omega Q + \varepsilon_R$. (I.52)

$\dot{Q} = \omega R + g_AQ + \varepsilon_Q$. (I.53)

For hyperbolic dynamics:

$\dot{R} = g_AR + \kappa Q + \varepsilon_R$. (I.54)

$\dot{Q} = \kappa R + g_AQ + \varepsilon_Q$. (I.55)

Compare:

- fit;
- stability;
- interpretability;
- regime localization;
- out-of-sample loss.

---

# I.13.5 Scaling tests

Apply admissible scaling:

$Q_c = cQ$. (I.56)

Measure:

- exact-angle stability;
- phase-order stability;
- gate-class stability;
- model-performance stability.

A fragile complex model should remain a real pair.

---

# I.13.6 Benchmark decision

Complex priority is supported only when:

$\text{Loss}(M_2) + \text{Penalty}(M_2)$

< Loss($M_1$) + Penalty($M_1$). (I.57)

or when $M_2$ provides a clearly superior operational structure not captured by loss alone.

---

# I.14 Workstream 9 — Phase-Time Validation

## I.14.1 Question

Does phase provide a better internal ordering of uneven market episodes than calendar time or event count?

---

## I.14.2 Episode families

Possible episode families include:

- range-to-breakout;
- trend-to-reversal;
- credit deterioration;
- liquidity crisis;
- volatility squeeze;
- post-earnings adjustment;
- policy-cycle transition.

Each family should be tested separately before pooling.

---

## I.14.3 Candidate clocks

Compare:

t = calendar time. (I.58)

u = normalized episode age. (I.59)

k = event count. (I.60)

v = cumulative volume. (I.61)

a = cumulative absolute return. (I.62)

σ = selection depth. (I.63)

$\tau_i$ = internal phase time. (I.64)

A phase clock must outperform plausible non-phase clocks.

---

# I.14.4 Alignment metrics

Possible primary metrics include:

- mean trajectory dispersion;
- dynamic-time-warping distance;
- functional variance;
- gate-time variance;
- event-order error.

Define:

$$D_{clock}(c) = \text{mean cross-episode dispersion under clock } c. \quad (I.65)$$

The phase hypothesis is:

$$D_{clock}(\tau_i) < \min_{c \neq \tau_i} D_{clock}(c). \quad (I.66)$$

---

# I.14.5 Prospective phase estimation

At time t, use only data available up to t.

Do not normalize by a future episode endpoint unless the study is explicitly retrospective.

Online state:

$$S_t = (R_t, Q_t, A_t, \theta_t, \tau_i, t, \dot{\theta}_t, \text{PhaseConfidence}_t). \quad (I.67)$$

---

# I.14.6 Gate-hazard model

For gate G:

$$\text{logit } \Pr(G_{t+h}=1) = \beta_0 + \beta_1 \cos\theta_t + \beta_2 \sin\theta_t$$

- $\beta_3 \tau_i, t + \beta^T X_t.$ (I.68)

Test whether phase terms improve:

- log loss;
- calibration;
- discrimination;
- stability;
- transport.

## I.14.7 Phase-time falsification

The claim fails when:

- phase is scale-fragile;
- episode alignment is in-sample only;
- event count performs equally well;
- gate hazard does not improve;
- phase branches are unstable;
- the real-pair model explains the same structure.

---

# I.15 Workstream 10 — Ledger Dependence and Backreaction

## I.15.1 Question

Do admitted events and public interpretations alter future market dynamics?

---

## I.15.2 Ledger dependence

Let X_t be the current observable state.

Let L_t be the historical ledger.

Compare:

M_current = f(X_t). (I.69)

M_ledger = f(X_t,L_t). (I.70)

Ledger dependence is supported when:

Loss(M_ledger) < Loss(M_current). (I.71)

after complexity adjustment.

---

## I.15.3 Candidate ledger variables

- prior failed breakouts;
- prior retests;
- trapped-position estimates;
- default history;

- margin events;
- accounting recognition;
- policy intervention;
- benchmark inclusion;
- public signal adoption.

---

# I.15.4 Adoption and crowding

Estimate:

$$\text{Adoption}_t = \text{PublicVisibility} \times \text{EstimatedCapitalLinked} \times \text{ActionCoherence}. \quad (I.72)$$

A crowding ratio is:

$$\text{CrowdingRatio}_t = \text{EstimatedSignalLinkedCapital}/\text{LiquidityCapacity}. \quad (I.73)$$

Test whether adoption has a nonlinear effect:

$$\text{Outcome} = \alpha + \beta_1 \text{Adoption} + \beta_2 \text{Adoption}^2 + \text{Controls} + \varepsilon. \quad (I.74)$$

A possible pattern is:

low adoption → weak effect;
moderate adoption → stronger self-confirmation;
high adoption → fragility.

---

# I.15.5 Identification problem

Adoption is endogenous.

Signals become popular partly because they appeared useful.

Possible identification strategies include:

- rule publication dates;
- index-methodology changes;
- platform default changes;
- regulatory changes;
- exogenous media exposure;
- staggered institutional adoption.

Causal claims should not be based on correlation alone.

---

# I.15.6 World-formation criterion

The strongest claim requires:

Gate
→ Trace
→ ChangedAction
→ ChangedFutureTransitionProbability. (I.75)

Without ChangedAction or ChangedFutureTransitionProbability, the result is a recorded event but not demonstrated world-forming backreaction.

---

# I.16 Benchmark Families

# I.16.1 Naive benchmarks

- buy and hold;
- random signal;
- unconditional event rate;
- previous-state persistence;
- simple moving-average filter.

---

# I.16.2 Statistical benchmarks

- logistic regression;
- regularized linear model;
- random forest;
- gradient boosting;
- hidden Markov model;
- state-space model;
- survival model.

---

# I.16.3 Market-structure benchmarks

- price and volume only;
- price, volume, and volatility;
- price, breadth, and liquidity;

- conventional regime-switching model.

---

## I.16.4 Flexible real-vector benchmark

For complex-state studies:

$X_t = (R_t,Q_t,Controls_t)$. (I.76)

Use interactions and nonlinearities sufficient to prevent an unfairly weak comparison.

---

## I.16.5 Flexible internal-clock benchmark

For phase-time studies:

$\tau_{flex,j} = f_j(t;\theta_{parameters})$. (I.77)

Compare phase time with flexible monotone warping while penalizing complexity.

---

# I.17 Statistical Validation Standards

## I.17.1 Train, validation, and test split

Use chronological splits:

Training < Validation < Test. (I.78)

Random shuffling is inappropriate when dependence and regime evolution matter.

---

## I.17.2 Walk-forward testing

For period m:

Train on [1,m].
Test on [m+1,m+h]. (I.79)

Then roll forward.

This approximates repeated real-time use.

---

## I.17.3 Nested model selection

All parameter tuning should occur inside the training and validation folds.

The final test set should remain untouched until model specification is frozen.

---

## I.17.4 Multiple testing

The programme contains many:

- indicators;
- parameters;
- markets;
- horizons;
- transformations.

Control false discovery using:

- Bonferroni where appropriate;
- Benjamini–Hochberg;
- White's reality check;
- deflated Sharpe ratio;
- hierarchical testing;
- holdout replication.

No single method is universally preferred.

The correction should match the dependence structure and claim type.

---

## I.17.5 Dependence-robust inference

Use methods appropriate for:

- serial correlation;
- heteroskedasticity;
- cross-sectional dependence;
- clustered events;
- overlapping horizons.

Possible tools include:

- block bootstrap;
- stationary bootstrap;
- cluster-robust errors;
- Newey–West adjustments;
- permutation tests preserving dependence.

---

## I.17.6 Rare-event metrics

For infrequent gates or crises, accuracy is misleading.

Use:

- precision;
- recall;
- F1;
- precision–recall area;
- calibration;
- expected cost;
- time-to-event loss.

---

## I.17.7 Economic evaluation

Where trading applications are studied, include:

- spread;
- commissions;
- market impact;
- borrow cost;
- funding;
- latency;
- turnover;
- position limits;
- capacity.

Gross predictive success is not sufficient.

---

# I.18 Ablation Programme

## I.18.1 Functional ablation

Remove each family:

Δ_Load
= Performance_full − Performance_−Load. (I.80)

Δ_Motion

= Performance_full − Performance_−Motion. (I.81)

$\Delta$_Constraint
= Performance_full − Performance_−Constraint. (I.82)

$\Delta$_Commitment
= Performance_full − Performance_−Commitment. (I.83)

---

# I.18.2 Governance ablation

Remove:

- residual;
- transport;
- ledger;
- invalidation.

Measure the change in:

- calibration;
- false certainty;
- revision quality;
- reproducibility.

---

# I.18.3 Advanced-variable ablation

Compare:

Base.
Base + $\chi$.
Base + $\Xi$.
Base + R,Q.
Base + $\theta$.
Base + $\tau_i$.
Base + ledger history. (I.84)

This shows whether each layer contributes incremental value.

---

# I.18.4 Protocol ablation

Remove or alter:

- breadth;
- volume;
- retest;

- higher timeframe;
- anchor rule.

This identifies which gate components matter.

---

# I.19 Model Complexity and Reduction

## I.19.1 Complexity budget

Define a complexity cost:

$$C_{model} = \lambda_1 Parameters$$

- $\lambda_2 DataChannels$
- $\lambda_3 TuningChoices$
- $\lambda_4 RuntimeCost$
- $\lambda_5 InterpretiveBurden.$ (I.85)

The final evaluation is:

$$Utility_{adj} = EmpiricalUtility - C_{model}.$$ (I.86)

---

## I.19.2 Reduction rule

Reduce the model when:

$$\Delta Utility_{adj} \leq 0.$$ (I.87)

The allowed reductions include:

Time-bearing world
→ phase-sensitive event model
→ phase-time model
→ complex dynamics
→ real pair
→ scalar. (I.88)

This reduction ladder follows the phase framework's requirement that claims be weakened when stronger evidence does not survive.

---

# I.20 Benchmarking the Six Periods

## I.20.1 Mark → Window

Test whether mark-level variables improve:

- candle classification;
- close quality;
- wick interpretation;
- short-horizon gate strength.

Model:

WindowOutcome
= f(WindowOHLCV,MarkHistory). (I.89)

Compare with:

WindowOutcome
= f(WindowOHLCV). (I.90)

---

## I.20.2 Window → Structure

Test whether window-level traces improve:

- trend persistence;
- support formation;
- profile stability;
- moving-average regime.

---

## I.20.3 Structure → Event

Test whether structural memory and boundaries improve event classification beyond raw price change.

---

## I.20.4 Event → Episode

Test whether ordered event histories improve:

- trend segmentation;
- wave classification;
- range-break evolution;
- crisis staging.

---

## I.20.5 Episode → World

Test whether episode history improves:

- institutional-regime recognition;

- policy response;
- leverage transition;
- accounting or legal gate timing.

---

## I.20.6 Downward constraint

Also test the reverse influence:

World → Episode → Event → Structure → Window → Mark. (I.91)

Examples include:

- policy world constraining price events;
- collateral world constraining liquidity marks;
- institutional regime changing breakout success.

The periodic model predicts bidirectional recursion rather than one-way aggregation.

---

# I.21 Minimum Viable Study

A first empirical paper should avoid testing the whole framework.

A defensible minimum study is:

## Title

**Do Functionally Diverse Breakout Gates Reduce False Admission? A Residual-Honest Test of Price, Volume, Breadth, Close, and Retest Evidence**

## Scope

- one liquid equity-index universe;
- daily data;
- twenty years;
- predeclared horizontal and range boundaries;
- all boundary-crossing candidates.

## Compared models

1. crossing only;
2. close only;
3. close + volume;

4. close + volume + breadth;
5. close + volume + breadth + retest;
6. residual-adjusted gate.

## Primary outcome

Fakeout within twenty sessions.

## Secondary outcomes

- persistence;
- maximum adverse excursion;
- delay;
- transaction-cost-adjusted return;
- episode promotion.

## Core tests

- gate calibration;
- delay–accuracy frontier;
- residual burden;
- higher-timeframe transport;
- redundancy control.

This study would test a central practical claim without requiring $\chi$, $\Xi$, complex phase, or world formation.

---

# I.22 Second-Stage Study

A second study could test $\chi$.

## Title

**When Overbought Means Strength: Regime-Conditioned Interpretation of RSI and Volatility-Band Extension**

## Scope

- several liquid asset classes;
- daily and intraday horizons;
- prospectively estimated $\chi$.

## Primary hypotheses

1. RSI reversal performance depends on χ.
2. Band-touch reversal performance depends on χ.
3. χ-conditioned models outperform simple trend filters.

## Benchmarks

- moving-average regime;
- volatility regime;
- hidden Markov model;
- no regime conditioning.

---

# I.23 Third-Stage Study

A third study could test residual governance.

## Title

**Do Recorded Contradictions Predict Signal Failure? A Prospective Residual-Ledger Study of Breakouts and Divergences**

## Primary hypotheses

- residual burden predicts fakeout;
- frame conflict predicts delayed failure;
- residual-honest revision improves calibration;
- preserved failed traces reduce hindsight relabelling.

---

# I.24 Fourth-Stage Study

A fourth study could test complex eligibility.

## Title

**Real Pair or Complex State? Testing Price Acceptance and Breadth Pressure as a Candidate Market Phase System**

## Required comparisons

- scalar price model;

- real pair;
- complex amplitude–phase model;
- regime-switching elliptic/hyperbolic model.

## Mandatory outcomes

- scaling robustness;
- generator fit;
- phase-order stability;
- gate prediction;
- reduction decision.

The study must be publishable even if the conclusion is:

Retain the real pair.

---

# I.25 Fifth-Stage Study

A fifth study could test phase time.

## Title

**Can Phase Become a Clock? Aligning Unequal Breakout Episodes by Conjugate Market State**

## Required benchmarks

- normalized calendar time;
- event count;
- cumulative volume;
- cumulative volatility;
- cumulative absolute return;
- flexible monotone warping.

## Primary outcome

Out-of-sample episode dispersion.

## Secondary outcome

Gate-hazard calibration.

---

# I.26 Full-Scale Consortium Study

A mature multi-institution programme could contain:

### Team 1 — Taxonomy

Classification and method ontology.

### Team 2 — Market microstructure

Mark and Window periods.

### Team 3 — Event gates

Breakouts, retests, fakeouts, and residuals.

### Team 4 — Regime dynamics

$\chi$ and $\Xi$.

### Team 5 — Complex systems

R–Q states, phase, and alternative generators.

### Team 6 — Institutional worlds

Accounting, legal, policy, and collateral ledgers.

### Team 7 — Reproducibility

Data lineage, code, preregistration, and independent replication.

The teams should share:

- common schemas;
- claim tags;
- immutable protocols;
- reduction rules;
- failed-result publication.

---

# I.27 Reproducibility Package

Every empirical publication should provide, where legally possible:

- protocol registration;
- raw-data provenance;
- cleaning code;

- feature code;
- gate code;
- residual-coding guide;
- transport operators;
- model specifications;
- test-period predictions;
- failed models;
- revision history.

The reproducibility package should preserve:

Model_v1
→ Failure_v1
→ Revision_v2
→ Test_v2. (I.92)

The self-revising declaration framework requires revisions to preserve prior trace rather than rewriting the record.

---

# I.28 Negative-Result Publication

The programme should publish negative results such as:

- no classification reliability;
- no gate improvement;
- no residual predictiveness;
- no $\chi$ interaction;
- no transport value;
- no complex advantage;
- no phase-time alignment;
- no ledger backreaction.

These results are theoretically informative.

For example:

No complex advantage
→ retain real pair. (I.93)

No phase alignment
→ phase remains descriptive. (I.94)

No periodic inheritance

→ revise the row architecture. (I.95)

A framework that suppresses these outcomes cannot become self-correcting.

---

# I.29 Evidence Dashboard

A full programme may maintain a dashboard.

| Component | Evidence status | Reliability | Incremental value | Transport | Current model status |
|---|---|---|---|---|---|
| Four families | proposed | untested | untested | untested | conceptual |
| Six periods | proposed | untested | untested | untested | conceptual |
| Breakout gate | source-motivated | moderate | pending | pending | empirical target |
| Residual ledger | source-motivated | pending | pending | applicable | engineering prototype |
| $\chi$ | formal proposal | pending | pending | horizon-sensitive | research model |
| $\Xi$ | protocol interface | pending | pending | frame-dependent | research dashboard |
| CAPM Z | exact construction | exact internally | practical gain pending | protocol-relative | calibration atom |
| TA complex state | hypothetical | untested | untested | mandatory | not established |
| Phase time | research hypothesis | untested | untested | mandatory | not established |
| World backreaction | plausible | case-dependent | untested | observer-dependent | high-burden claim |

This dashboard should be updated as evidence accumulates.

---

# I.30 Decision Rules

# I.30.1 Promote

Promote a claim when it:

- replicates;
- survives transport;
- beats the benchmark;
- remains calibrated;
- preserves residual;
- passes the next evidence gate.

# I.30.2 Hold

Hold the claim at its current level when:

- evidence is promising but incomplete;
- results are local;
- transport is unresolved;
- sample size is inadequate.

## I.30.3 Downgrade

Downgrade when:

- stronger interpretation fails;
- the relation remains useful at a lower level.

## I.30.4 Reject

Reject when:

- mandatory conditions fail;
- simpler models dominate;
- no stable operational object remains.

## I.30.5 Revise

Revise when:

- the failure identifies a correctable protocol or measurement problem;
- the original trace remains preserved;
- the new model is tested prospectively.

---

# I.31 The Full Benchmarking Matrix

| Research object | Primary benchmark | Mandatory transport | Falsification trigger |
|---|---|---|---|
| Period classification | simpler taxonomy | analyst and market class | low inter-rater reliability |
| Functional family | conventional indicator group | source and operator lineage | no incremental clarity |
| Gate strength | crossing-only rule | timeframe and breadth | no calibration gain |
| Residual burden | gate-only model | frame and residual type | no predictive or revision value |
| $\chi$ | trend and volatility regime | horizon and asset | no interaction gain |

| Research object | Primary benchmark | Mandatory transport | Falsification trigger |
|---|---|---|---|
| $\Xi$ | price–volume–volatility model | local/systemic boundary | no incremental value |
| Complex state | flexible real pair | scaling and Q proxy | no superiority |
| Phase time | calendar and event clocks | episode family and frame | no alignment gain |
| Ledger dependence | current-state model | observer and authority | no future-state gain |
| Backreaction | non-adoption control | market and period | no causal effect |

# I.32 Research Ethics and Use Limits

## I.32.1 Financial use

The framework is not investment advice.

Empirical performance in one sample does not guarantee future profitability.

---

## I.32.2 Market impact

A published signal may alter the behaviour it measures.

Researchers should report:

- adoption;
- capacity;
- crowding;
- deterioration after publication.

---

## I.32.3 Automated intervention

Automated use should include:

- position limits;
- kill switches;
- liquidity checks;
- model-confidence thresholds;
- human escalation;
- residual alarms.

A complex or phase-based model should not receive greater authority merely because its mathematics appears sophisticated.

## I.32.4 Institutional consequences

World-level models affecting:

- lending;
- collateral;
- risk limits;
- impairment;
- regulatory decisions

require higher validation standards than exploratory chart classification.

---

# I.33 Research Programme Summary

The complete empirical programme is:

```
1. Validate the taxonomy.
2. Measure indicator redundancy.
3. Test functionally diverse confirmation.
4. Compare gate architectures.
5. Record residual prospectively.
6. Test transport and invariance.
7. Estimate χ and test regime interactions.
8. Compile Ξ and compare with simpler controls.
9. Register candidate R–Q states.
10. Compare scalar, real-pair, and complex models.
11. Test phase-order robustness.
12. Compare phase time with alternative clocks.
13. Test phase-sensitive gate hazard.
14. Test ledger dependence.
15. Identify observer backreaction.
16. Reduce unsupported claims.
17. Preserve every failed model and revision.
```

In equation form:

ResearchProgress
= ClassificationReliability
× BenchmarkSuperiority
× ResidualHonesty
× TransportRobustness
× ProspectiveReplication. (I.96)

The multiplicative form is conceptual.

It emphasizes that a severe failure in one dimension can materially weaken the whole claim.

---

# I.34 Appendix I Conclusion

The Periodic Grammar should not be validated by showing that familiar chart examples can be redescribed in its vocabulary.

It must demonstrate that the vocabulary changes research outcomes.

The first evidential test is:

Can observers classify methods consistently? (I.97)

The second is:

Does the classification expose redundancy and category errors? (I.98)

The third is:

Do explicit gates and residuals improve event calibration? (I.99)

The fourth is:

Do regime and transport variables improve conditional interpretation? (I.100)

Only after those stages should the programme ask:

Does a market state earn complex phase? (I.101)

Does phase earn the status of internal time? (I.102)

Do phase-sensitive gates create a time-bearing world? (I.103)

The governing scientific sequence is:

Concept
→ Operational Definition
→ Measurement
→ Benchmark
→ Prospective Test
→ Transport
→ Replication
→ Revision. (I.104)

And the governing reduction rule remains:

When the strongest claim fails, retain the strongest simpler claim that survives. (I.105)

The next appendix can present a **Worked End-to-End Case Study**, applying the entire grammar to one hypothetical breakout—from Mark and Window through Structure, Event, Episode, World, residual ledger, transport audit, χ diagnosis, and optional complex-state reduction.

# Appendix J — Worked End-to-End Case Study: From Mark to Market World

## J.1 Purpose and Status

This appendix applies the entire Periodic Grammar to one **fictional breakout case**.

The example is deliberately synthetic. It is designed to show how the framework operates, not to describe a real security or recommend a transaction.

The case follows one market object through:

Mark
→ Window
→ Structure
→ Event
→ Episode
→ World. (J.1)

It also demonstrates:

- protocol declaration;
- functional classification;
- χ regime diagnosis;
- Ξ control-state estimation;
- event gating;
- residual recording;
- cross-frame transport;
- admissible revision;
- optional complex-state audit;
- model reduction;
- ledger backreaction.

The source Technical Analysis framework treats an indicator as a protocol-bound projection of market self-reference rather than as the market itself. It also requires explicit confirmation, residual, invalidation, and cross-frame testing.

---

# J.2 The Fictional Market Object

The instrument is:

**Northbridge Energy plc**

Ticker:

NBE. (J.2)

The comparison universe is:

**Forty listed renewable-infrastructure and power-network companies.**

The market has traded for approximately twelve weeks inside a range:

£94.50 ≤ P ≤ £100.00. (J.3)

The upper boundary:

B_daily = £100.00 (J.4)

has been tested twice and rejected twice.

A higher-timeframe resistance zone exists at:

B_weekly = [£102.00,£102.30]. (J.5)

NBE is also under review for possible inclusion in a fictional large-cap benchmark:

**Albion 100 Index.**

The review result has not yet been announced at the beginning of the case.

---

# J.3 Declared Protocol

# J.3.1 Protocol record

The Technical Analysis protocol is:

P_NBE
= (Asset,Universe,Venue,Timeframe,Scale,BarRule,FeatureMap,GateRule,ResidualRule). (J.6)

The declared fields are:

| Field | Declaration |
|---|---|
| Asset | Northbridge Energy plc |
| Universe | Forty renewable-infrastructure peers |
| Venue | Primary exchange consolidated tape |
| Primary timeframe | Daily |
| Higher timeframe | Weekly |
| Scale | Logarithmic for long-horizon structure |
| Bar rule | Regular-session OHLCV |
| Volatility scale | 14-day ATR |
| Boundary | £99.80–£100.00 resistance zone |
| Primary event | Bullish breakout |
| Primary outcome horizon | Twenty sessions |
| Invalidation | Close inside old range followed by failed reclaim |
| Residual rule | Record missing breadth, retest, weekly confirmation, and institutional uncertainty |

The event gate requires:

1. close beyond the resistance zone;
2. displacement of at least 0.75 ATR;
3. relative volume above 1.50;
4. close above session VWAP;
5. peer breadth above 65%;
6. follow-through or successful retest;

7. no decisive weekly rejection.

Not all conditions must pass on the first day.

The gate may return:

Admit;
Partially Admit;
Defer;
Reject. (J.7)

A crossing is therefore only a candidate event.

The source framework likewise distinguishes a price crossing from a mature breakout and specifically asks whether the move closed above resistance, carried volume, showed VWAP and value acceptance, attracted breadth, produced follow-through, and survived a retest.

---

# J.4 The Pre-Breakout Field

# J.4.1 Twelve-week range

Before the candidate breakout:

RangeLow = £94.50. (J.8)

RangeHigh = £100.00. (J.9)

RangeWidth = £5.50. (J.10)

The range contains:

- two failed attempts above £100;
- declining realized volatility;
- rising volume near the upper half of the range;
- repeated closing acceptance above £98;
- increasing peer-sector attention.

The prior failed attempts matter because they are already part of the episode ledger.

They may create:

- scepticism;
- trapped short positions;
- breakout buyers waiting for confirmation;
- concentrated stops above £100;
- concentrated stops below the latest higher low.

Thus the range is not an empty geometric box.

It is a memory-bearing constraint.

---

## J.4.2 Structural indicators

Immediately before the candidate event:

$EMA_{20}$ = £98.70. (J.11)

$EMA_{50}$ = £97.95. (J.12)

AnchoredVWAP = £98.85. (J.13)

VolumeProfilePOC = £97.20. (J.14)

VolumeProfileVAH = £99.95. (J.15)

$RSI_{14}$ = 67. (J.16)

MACD > 0. (J.17)

$ATR_{14}$ = £0.76. (J.18)

These readings do not yet establish a breakout.

They describe:

- positive price memory;
- proximity to a loaded boundary;
- moderate directional dominance;
- compressed volatility;
- accepted trading near the top of the range.

---

# J.5 Period 0 — Mark Formation

## J.5.1 Opening order field

On the candidate-breakout day, the market opens at:

O = £99.55. (J.19)

Immediately before the first break of £100:

BestBid = £99.96. (J.20)

BestAsk = £99.98. (J.21)

DisplayedAskDepth at £100.00 = 420,000 shares. (J.22)

DisplayedBidDepth at £99.90–£99.96 = 610,000 shares. (J.23)

The local Mark-period classification is:

### Load

Resting orders around £100.

### Motion

Queue depletion and upward tick sequence.

### Constraint

The £100 offer and price increment.

### Commitment

Individual executions above £100.

---

# J.5.2 First trade above the boundary

At 10:14:03:

TradePrice = £100.04. (J.24)

TradeSize = 85,000 shares. (J.25)

This execution is a Mark-level commitment.

It proves that at least one transaction occurred above the boundary.

It does **not** prove:

- daily acceptance;
- structural breakout;
- episode transition;
- institutional regime change.

Therefore:

ExecutionAboveBoundary
≠ BreakoutEvent. (J.26)

The Mark-level event becomes an input to the Window ledger.

---

# J.5.3 Repeated marks

During the next thirty minutes:

- 3.2 million shares trade;
- 72% of trade volume occurs at or above £100;
- the offer repeatedly rebuilds;

- price reaches £100.42;
- price then returns to £99.88.

This path contains both:

- attempted upward commitment;
- unresolved selling pressure.

The return below £100 creates a Mark-level residual.

At this stage, the correct statement is:

> The market has tested and temporarily crossed the boundary, but higher-level acceptance remains unresolved.

---

# J.6 Period 1 — The Breakout Window

# J.6.1 Daily candle

The session closes with:

O = £99.55. (J.27)

H = £101.10. (J.28)

L = £99.30. (J.29)

C = £100.88. (J.30)

V = 17.4 million shares. (J.31)

Twenty-day average volume is:

$\bar{V}_{20}$ = 10.0 million shares. (J.32)

Relative volume is:

RVOL = 17.4/10.0 = 1.74. (J.33)

The daily close is:

C − B_daily = £0.88 (J.34)

above the upper boundary.

Normalized displacement is:

d_ATR = (100.88 − 100.00)/0.76 = 1.16 ATR. (J.35)

The close-location value is:

CLV = (C − L)/(H − L). (J.36)

Therefore:

CLV = (100.88 − 99.30)/(101.10 − 99.30) = 0.878. (J.37)

The close occurred near the upper end of the day's range.

Session VWAP was:

VWAP = £100.41. (J.38)

Therefore:

C > VWAP. (J.39)

---

# J.6.2 Window-period decomposition

## Load

- high session volume;
- opening memory;
- prior range;
- active orders around £100.

## Motion

- 1.16 ATR displacement beyond the boundary;
- strong close location;
- positive intraday path.

## Constraint

- resistance zone £99.80–£100.00;
- intraday high £101.10;
- weekly resistance above £102.

## Commitment

- official daily close at £100.88.

## Residual

- intraday return below £100 before recovery;
- breadth below full confirmation;
- no later retest yet;
- weekly gate still open.

The daily close creates a stronger ledger entry than the first trade above £100.

But the daily window still cannot prove an Episode transition.

---

# J.7 Period 2 — Structure Reclassification

## J.7.1 Structure before the close

Before the breakout day, the dominant structural statement was:

NBE is inside a twelve-week range beneath £100. (J.40)

After the close, the structural statement becomes:

NBE has closed above the daily range boundary, while higher-frame acceptance remains unresolved. (J.41)

This is a revision of structure, not yet a declaration of a new market world.

---

## J.7.2 Moving-average structure

Price is now:

$C/EMA_{20} - 1 = 2.21\%$. (J.42)

and:

$C/EMA_{50} - 1 = 2.99\%$. (J.43)

The moving averages remain positively ordered:

$EMA_{20} > EMA_{50}$. (J.44)

This supplies Load and memory context.

It is not independent breakout confirmation because price, $EMA_{20}$, $EMA_{50}$, MACD, and RSI all derive largely from the price record.

---

## J.7.3 Volume-profile structure

The prior value-area high was:

VAH = £99.95. (J.45)

The close above VAH suggests attempted value migration.

However, most historical transaction density remains below £100.

The profile therefore says:

Price has crossed beyond prior value, but new density has not yet been built. (J.46)

This distinction matters.

Price displacement is immediate.

Value migration requires later trading and acceptance.

---

## J.7.4 Breadth structure

Among the forty peer companies:

23 close above their twenty-day moving average. (J.47)

Therefore:

Breadth = 23/40 = 57.5%. (J.48)

The gate threshold is:

Breadth* = 65%. (J.49)

Breadth therefore does not pass.

The correct interpretation is not:

Breadth is bearish. (J.50)

It is:

Breadth does not yet support a broad sector transition at the declared threshold. (J.51)

The breakout remains more concentrated than the headline price alone suggests.

---

# J.8 Diagnosing χ

## J.8.1 Prior corrective signature

During the twelve-week range, extensions toward £100 repeatedly produced opposing pressure.

For the preceding twenty comparable upper-range tests:

Average three-session return after extension = −0.42%. (J.52)

This suggests a locally corrective relation:

$\chi_{prior} < 0$. (J.53)

Price extension tended to generate:

- selling;
- profit-taking;
- liquidity provision;
- return toward the centre of the range.

Oscillator-based reversal interpretations were therefore more plausible during the established range.

---

## J.8.2 Breakout-day ambiguity

The candidate breakout shows:

- strong displacement;
- high volume;
- strong close;
- incomplete breadth;
- no retest;
- unresolved weekly boundary.

The regime should not immediately be labelled:

$\chi > 0$. (J.54)

The more disciplined classification is:

$\chi_{To} \approx 0$ with positive-transition evidence. (J.55)

The prior corrective loop has weakened.

But self-confirming continuation has not yet been demonstrated.

---

## J.8.3 Simplified χ score

For illustration, define a hypothetical normalized χ score:

$\hat{\chi}$
$= 0.30F_{follow}$

- $0.25F_{break}$
- $0.20F_{breadth}$
- $0.15F_{volume}$
  $- 0.10F_{reversion}$. (J.56)

This is an illustrative diagnostic, not a universal formula.

On the first breakout day:

$F_{follow} = 0$. (J.57)

$F_{break} = 1$. (J.58)

$F_{breadth} = 0.35$. (J.59)

$F_{volume} = 0.80$. (J.60)

$F_{reversion} = 0.40$. (J.61)

Therefore:

$\hat{\chi}_{To}$
$= 0.30(0) + 0.25(1) + 0.20(0.35) + 0.15(0.80) - 0.10(0.40)$. (J.62)

Hence:

$\hat{\chi}_{To} = 0.40$. (J.63)

Under the case-study calibration:

$\hat{\chi} < -0.25 \rightarrow$ corrective.
$-0.25 \leq \hat{\chi} \leq 0.45 \rightarrow$ critical or transitional.
$\hat{\chi} > 0.45 \rightarrow$ self-confirming. (J.64)

Thus:

$\hat{\chi}_{To} = 0.40 \rightarrow$ transitional. (J.65)

The exact thresholds are hypothetical.

The important point is that regime status is inferred from a response structure, not from one indicator.

The source framework similarly distinguishes corrective, critical, and self-confirming signatures and warns that the same oscillator reading changes meaning across those regimes.

---

# J.9 Compiling Ξ

# J.9.1 Control-state estimate

The case uses normalized coordinates:

$\Xi_{NBE} = (\rho,\gamma,\nu)$. (J.66)

where:

ρ = loading;
γ = lock-in;
ν = agitation.

The estimated values on the breakout day are:

$\rho = 0.79$. (J.67)

$\gamma = 0.62$. (J.68)

$\nu = 0.58$. (J.69)

These numbers are illustrative compiled scores, not directly observed natural constants.

---

# J.9.2 Loading ρ

The high loading score reflects:

- RVOL = 1.74;

- rising sector attention;
- active option positioning near £100;
- prior failed breakout history;
- possible benchmark-inclusion speculation.

Interpretation:

The boundary carries substantial participant attention and accumulated interest. (J.70)

---

## J.9.3 Lock-in γ

The moderate lock-in score reflects:

- strong volume-profile density below £100;
- short positions established during the range;
- institutional portfolios not yet benchmark-forced;
- adequate but not exceptional liquidity.

Interpretation:

Positions exist, but the system is not yet fully trapped or institutionally locked. (J.71)

---

## J.9.4 Agitation ν

The moderate agitation score reflects:

- rising intraday range;
- expanding volume;
- temporary return below £100;
- stable but increasing spread activity.

Interpretation:

The system is moving out of compression, but it is not yet in disorder. (J.72)

---

## J.9.5 What Ξ does not prove

Ξ does not prove:

- breakout success;
- direction;
- phase;

- institutional recognition.

It describes the operating condition in which the event gate is being tested.

---

# J.10 Period 3 — Candidate Breakout Event

## J.10.1 Gate components

Define the raw gate score:

$$S_G = 0.20C + 0.15V + 0.10W + 0.15B + 0.15F + 0.15T + 0.10H. \quad (J.73)$$

where:

C = close confirmation;
V = volume confirmation;
W = VWAP acceptance;
B = breadth confirmation;
F = follow-through;
T = retest;
H = higher-timeframe confirmation.

Each component takes a value from 0 to 1.

The classification is:

$S_G < 0.40 \rightarrow$ Defer or Reject. (J.74)

$0.40 \le S_G < 0.70 \rightarrow$ Partially Admit. (J.75)

$S_G \ge 0.70 \rightarrow$ Admit. (J.76)

Residual burden is reported separately rather than hidden inside the score.

---

## J.10.2 Breakout-day gate

On $T_0$:

$C = 1.00$. (J.77)

$V = 1.00$. (J.78)

W = 1.00. (J.79)

B = 0.00. (J.80)

F = 0.00. (J.81)

T = 0.00. (J.82)

H = 0.00. (J.83)

Therefore:

S_G,To
= 0.20 + 0.15 + 0.10
= 0.45. (J.84)

The event is recorded as:

**Partially Admitted Daily Breakout.**

It is not recorded as:

**Confirmed Episode Transition.**

---

# J.10.3 Gate statement

The correct ledger statement is:

> NBE closed 1.16 ATR above the daily resistance zone on 1.74 relative volume and above VWAP. Peer breadth remained below threshold, no retest or follow-through had yet occurred, and weekly resistance remained unresolved. The daily breakout is partially admitted.

This statement is longer than:

“Breakout confirmed.”

It is also more useful.

---

# J.11 Initial Residual Ledger

# J.11.1 Residual records

| ID | Residual | Severity | Status | Gate effect |
|---|---|---|---|---|
| R-001 | Breadth below 65% | High | Open | prevents broad-field confirmation |
| R-002 | No retest | Medium | Open | prevents durable acceptance claim |
| R-003 | Weekly resistance at £102.00–£102.30 | High | Open | prevents higher-frame promotion |
| R-004 | Benchmark-review outcome unknown | Medium | Open | institutional world remains unresolved |
| R-005 | Two prior false breaks | Medium | Open | raises fakeout prior |

| ID | Residual | Severity | Status | Gate effect |
|---|---|---|---|---|
| R-006 | New value density above £100 not yet formed | Medium | Open | value migration remains incomplete |

The residual register is:

$\mathscr{R}$_T0 = {R-001,R-002,R-003,R-004,R-005,R-006}. (J.85)

The source framework defines residual as unresolved evidence, failed confirmation, contradiction, and ambiguity, and requires the original claim and invalidation to remain preserved.

---

# J.11.2 Invalidation

The predeclared invalidation condition is:

I_breakout
= DailyCloseInsideOldRange

- FailedReclaimWithinThreeSessions. (J.86)

More specifically:

C_t < £99.80 (J.87)

followed by:

H_{t+1:t+3} < £100.00. (J.88)

If both occur, the partially admitted breakout is relabelled:

FailedBreakout / Fakeout. (J.89)

The original partial-admission record remains in the ledger.

---

# J.12 Transport Audit at T0

# J.12.1 Daily to weekly

Expected transported claim:

The daily breakout should become a weekly breakout candidate, not yet a weekly commitment. (J.90)

Observed state:

The weekly bar has not closed.

Status:

Defer. (J.91)

Residual:

R-003 remains open.

---

## J.12.2 Linear to logarithmic scale

The resistance zone remains approximately aligned under log scaling.

Distance between transformed boundaries:

0.18 ATR. (J.92)

Status:

Survives. (J.93)

---

## J.12.3 Raw to volatility-normalized price

Raw displacement:

£0.88. (J.94)

Normalized displacement:

1.16 ATR. (J.95)

Status:

Survives. (J.96)

The break is not merely a tiny nominal crossing.

---

## J.12.4 Price to volume frame

Price crosses above the boundary.

RVOL is 1.74.

Status:

Survives. (J.97)

But high volume does not identify whether the activity is:

- genuine accumulation;
- short covering;
- churn;
- event speculation.

That ambiguity remains residual.

---

## J.12.5 Price to breadth frame

The price claim implies that a broad sector transition should show substantial peer participation.

Observed breadth is 57.5%.

Required breadth is 65%.

Status:

Fails or remains partial. (J.98)

Residual:

Concentration.

---

## J.12.6 Price to profile frame

The price closes above the old value-area high.

But most transaction density remains below £100.

Status:

Partial survival. (J.99)

Residual:

Value has not yet migrated.

---

## J.12.7 Transport summary

| Frame | Status | Interpretation |
|---|---|---|
| Log scale | Survives | boundary is not a linear-scale artefact |
| ATR normalization | Survives | displacement is material |
| Volume | Survives | participation is elevated |
| Breadth | Partial/fails | move is concentrated |
| Profile | Partial | price leads value migration |
| Weekly | Deferred | higher-frame gate not closed |

The result is not “confirmed” or “unconfirmed” in one undifferentiated sense.

It is a vector of cross-frame outcomes.

The source framework defines technical objectivity through cross-protocol survival and treats failure under timeframe, scale, breadth, anchor, or other admissible transformations as a reason to weaken or localize the claim.

---

## J.13 Follow-Through and Retest

## J.13.1 Session T1

The next session records:

O = £100.92. (J.100)

H = £101.46. (J.101)

L = £100.52. (J.102)

C = £101.18. (J.103)

RVOL = 1.22. (J.104)

Peer breadth = 62.5%. (J.105)

Price remains above the old boundary.

Follow-through is positive but breadth remains below threshold.

Update:

F = 0.65. (J.106)

B = 0.50. (J.107)

R-001 remains open.

---

## J.13.2 Session T2 — retest

On T2:

O = £101.10. (J.108)

H = £101.22. (J.109)

L = £100.05. (J.110)

C = £100.64. (J.111)

The old resistance zone is tested.

The low remains above £100.00 by £0.05 and the close returns above VWAP.

The retest is classified:

RetestStatus = Pass, low confidence. (J.112)

R-002 changes from:

Open → Provisionally Resolved. (J.113)

The close is not spectacular.

But the test contributes stronger evidence of acceptance than a second momentum indicator would.

---

## J.13.3 Session T4

On T4:

C = £101.74. (J.114)

RVOL = 1.43. (J.115)

Peer breadth = 70.0%. (J.116)

New transaction density has begun accumulating in:

[£100.40,£101.50]. (J.117)

The breadth residual resolves:

R-001 → Resolved. (J.118)

The profile residual weakens:

R-006 → Monitoring. (J.119)

---

# J.14 Gate Reassessment

At T4:

C = 1.00. (J.120)

V = 0.85. (J.121)

W = 1.00. (J.122)

B = 1.00. (J.123)

F = 1.00. (J.124)

T = 0.85. (J.125)

H = 0.00. (J.126)

Therefore:

S_G,T4
= 0.20

- 0.15(0.85)
- 0.10
- 0.15
- 0.15
- 0.15(0.85). (J.127)

Hence:

$S_{G,T4} = 0.855$. (J.128)

The daily event gate now returns:

Admit. (J.129)

The ledger changes from:

Partially Admitted Daily Breakout (J.130)

to:

Daily Breakout with Retest and Breadth Acceptance. (J.131)

Weekly promotion remains deferred.

---

# J.15 χ Transition

# J.15.1 Follow-through evidence

By T4:

- price remains above the boundary;
- retest has held;
- breadth has expanded;
- volume remains above ordinary levels;
- new value is forming above the old range.

The previous corrective response has weakened.

The simplified χ score becomes:

$F_{follow} = 1.00$. (J.132)

$F_{break} = 1.00$. (J.133)

$F_{breadth} = 0.85$. (J.134)

$F_{volume} = 0.65$. (J.135)

$F_{reversion} = 0.10$. (J.136)

Therefore:

$\hat{\chi}_{T4}$
$= 0.30(1.00)$

- $0.25(1.00)$
- $0.20(0.85)$
- $0.15(0.65)$
  $- 0.10(0.10)$. (J.137)

Hence:

$\hat{\chi}_{T4} = 0.808$. (J.138)

Under the illustrative thresholds:

$\hat{\chi}_{T4} > 0.45$. (J.139)

The regime is now classified:

$\chi > 0$, self-confirming, with moderate confidence. (J.140)

---

## J.15.2 Interpretation shift

Before the breakout:

High RSI near resistance
→ possible corrective exhaustion. (J.141)

After accepted breakout and $\chi$ transition:

High RSI
→ possible continuation strength. (J.142)

The indicator has not changed.

Its regime meaning has changed.

---

# J.16 Period 4 — Episode Transition

## J.16.1 Weekly gate

At the end of the week:

WeeklyClose = £102.42. (J.143)

WeeklyResistanceUpper = £102.30. (J.144)

Therefore:

WeeklyClose – WeeklyResistanceUpper = £0.12. (J.145)

The weekly close is only marginally above the zone.

But it is accompanied by:

- sector breadth of 72.5%;
- weekly volume at 1.48 of average;
- continued acceptance above £100;
- new profile density above the old range.

The weekly gate returns:

Partially Admit → Admit after one additional week of persistence. (J.146)

---

# J.16.2 Old and new episode grammars

## Old episode grammar

Range rotation:

UpperBoundaryTest
→ Selling
→ ReturnTowardValue. (J.147)

## New episode grammar

Breakout acceptance:

BoundaryCross
→ Close
→ Retest
→ BreadthExpansion
→ HigherFrameAcceptance. (J.148)

The episode transition occurs when the old grammar no longer organizes the sequence and the new grammar persists.

Therefore:

Episode$_0$ = TwelveWeekRange. (J.149)

Episode$_1$ = AcceptedUpwardExpansion. (J.150)

---

# J.16.3 Episode ledger

The episode ledger becomes:

$L_{episode,0}$
= {FailedBreak$_1$,FailedBreak$_2$,RangeCompression}. (J.151)

After the successful transition:

$L_{episode,1}$
= $L_{episode,0}$

- {DailyBreak,RetestHold,BreadthExpansion,WeeklyAcceptance}. (J.152)

The prior failed breaks are not deleted.

They become part of the new episode's load.

This demonstrates the periodic inheritance rule:

Commitment_Event + Residual_Event
→ Load_Episode. (J.153)

---

# J.17 The Institutional World Candidate

## J.17.1 Benchmark-review residual

Throughout the breakout, NBE has been under review for entry into the Albion 100 Index.

Before the official decision:

BenchmarkInclusion = possible but uncommitted. (J.154)

This possibility influences:

- speculation;
- option positioning;
- anticipatory buying;
- analyst coverage.

But it is not yet a World-level committed event.

It remains:

R-004 = Institutional residual. (J.155)

---

## J.17.2 Official index announcement

On T12, the index committee announces:

NBE will enter the Albion 100 after the close on T20. (J.156)

This event has:

- identified authority;
- formal rule;
- effective date;
- benchmark consequence;
- expected passive rebalancing.

The market had already partly anticipated the possibility.

But the announcement changes the state from:

possible inclusion (J.157)

to:

declared future inclusion. (J.158)

This is a World-level gate.

---

## J.17.3 Recognition vector

Define:

$\mathbb{G}$_inclusion
= (g_market,g_index,g_fund,g_accounting,g_legal). (J.159)

Immediately after announcement:

g_market = Admit. (J.160)

g_index = AdmitFutureEffective. (J.161)

g_fund = Prepare. (J.162)

g_accounting = NotApplicable. (J.163)

g_legal = NotApplicable. (J.164)

The World event has been recognized by the index authority and market, but the passive-fund implementation has not yet occurred.

---

# J.18 World-Level Commitment

## J.18.1 Effective inclusion

At the closing auction on $T_{20}$:

AuctionVolume = 49 million shares. (J.165)

NormalDailyVolume = 10 million shares. (J.166)

The stock is formally added to the benchmark.

Passive and benchmark-relative funds rebalance.

The World ledger is updated:

L_world,T20
= L_world,T19

- BenchmarkInclusionEvent. (J.167)

This event changes future admissibility because certain portfolios must now:

- own or benchmark against NBE;
- track its index weight;
- manage index-related risk;
- participate in future rebalances.

---

## J.18.2 World-period decomposition

### Load

- benchmark-linked ownership;
- increased passive demand;
- new institutional monitoring;
- larger derivative interest.

### Motion

- rebalancing flows;
- benchmark-relative trading;
- index-arbitrage activity;
- changed peer comparison.

### Constraint

- index methodology;
- tracking mandates;
- rebalance dates;
- fund limits.

### Commitment

- official inclusion;
- effective-date auction;
- benchmark ledger update.

### Residual

- crowded post-inclusion positioning;
- temporary demand;
- possible post-event reversal;
- future index-weight changes.

The event becomes World-level not because price rose.

It becomes World-level because an authoritative institutional rule changes the future action space.

---

# J.19 Backreaction

# J.19.1 Changes after inclusion

Over the following month, the fictional case records:

MedianSpread_before = 8 basis points. (J.168)

MedianSpread_after = 5 basis points. (J.169)

AverageDailyVolume_before = 10 million. (J.170)

AverageDailyVolume_after = 14 million. (J.171)

Institutional ownership rises.

Option and futures activity increases.

These changes alter the future microstructure in which Technical Analysis operates.

The world event therefore backreacts downward:

World
→ Episode
→ Event
→ Structure
→ Window
→ Mark. (J.172)

---

# J.19.2 Downward consequences

### World → Episode

Benchmark membership supports a new institutional ownership episode.

### Episode → Event

Future index reviews and rebalances become predictable event gates.

### Event → Structure

VWAP, profile, and liquidity structure change.

### Structure → Window

Average volume and intraday range change.

### Window → Mark

Depth, spread, and execution behaviour change.

Thus the full cycle is recursive rather than merely bottom-up.

---

# J.20 Residual Evolution

# J.20.1 Residual updates

| Residual | $T_0$ | $T_4$ | Weekly gate | $T_{12}$ announcement | $T_{20}$ effective inclusion |
|---|---|---|---|---|---|
| Breadth below threshold | Open | Resolved | Resolved | — | — |
| No retest | Open | Resolved | Resolved | — | — |
| Weekly resistance | Open | Open | Resolved | — | — |
| Benchmark uncertainty | Open | Open | Open | Resolved | Resolved |
| Prior fakeout history | Open | Carried | Carried | Carried | Historical load |
| No new value density | Open | Monitoring | Resolved | — | — |
| Post-inclusion crowding | — | — | — | Open | High |
| Temporary passive-flow distortion | — | — | — | Open | Monitoring |

Residual does not simply fall to zero.

Old residual resolves or becomes historical load.

New residual appears at higher closure levels.

---

# J.20.2 Residual debt

A conceptual residual burden is:

RB_t = Σ_js_jp_j. (J.173)

where:

s_j = severity;
p_j = persistence.

At $T_0$, RB is high because several important confirmations remain absent.

At $T_4$, RB falls.

At $T_{20}$, technical-breakout residual is low, but institutional-crowding residual rises.

This demonstrates:

ResidualResolution at one period
can create NewResidual at another period. (J.174)

---

# J.21 Optional Complex-State Audit

# J.21.1 Candidate pair

Suppose a researcher proposes:

R_TA = standardized price-acceptance score. (J.175)

Q_TA = standardized peer-breadth pressure. (J.176)

At T0:

R_TA = 1.16. (J.177)

Q_TA = 0.32. (J.178)

Both variables are dimensionless standardized scores.

The candidate complex state is:

Z_TA = 1.16 + 0.32i. (J.179)

Its amplitude is:

A_TA = √(1.16² + 0.32²). (J.180)

Therefore:

A_TA = 1.203. (J.181)

Its phase is:

θ_TA = atan2(0.32,1.16). (J.182)

Therefore:

θ_TA = 0.269 radians. (J.183)

or:

θ_TA = 15.4°. (J.184)

---

## J.21.2 What has actually been established?

The construction establishes that:

- two standardized variables exist;
- they can be stored as an ordered pair;
- they can be written in complex notation;
- a mathematical phase can be calculated.

It does not establish:

- conjugacy;
- rotational dynamics;
- a stable generator;
- phase-sensitive gates;
- internal phase time;

- model superiority.

---

## J.21.3 Q independence

Q_TA uses peer-component data rather than NBE price alone.

This gives some source independence.

But it is not yet proven that breadth is the natural conjugate of price acceptance.

Therefore:

Q independence = partial. (J.185)

Conjugacy = unestablished. (J.186)

---

## J.21.4 Generator test

To justify ordinary complex dynamics, the researcher would need evidence such as:

$\dot{R} \approx -\omega Q$. (J.187)

$\dot{Q} \approx \omega R$. (J.188)

No such evidence has yet been supplied.

The breakout sequence instead appears largely self-confirming after $T_4$.

A hyperbolic or ordinary real-vector model may be more appropriate than an elliptic rotation.

---

## J.21.5 Phase-time test

Only one episode has been studied.

No cross-episode comparison exists.

Therefore it is impossible to test whether:

$D_{phase} < D_{calendar}$. (J.189)

There is also no evidence that breakout gates concentrate around a stable $\theta$ across other episodes.

---

## J.21.6 Reduction decision

The candidate complex model is reduced to:

$X_{TA} = (R_{TA}, Q_{TA})$. (J.190)

The retained interpretation is:

- R_TA measures price acceptance;

- Q_TA measures peer participation;
- both contribute separately to gate assessment.

The rejected interpretation is:

- θ_TA is an established internal market clock.

This follows the phase framework's reduction rule: if Q is not demonstrably conjugate, if phase adds no stable simplification, or if a flexible real pair performs equally well, retain the real pair.

---

# J.22 Admissible Revision Sequence

## J.22.1 Declaration $D_0$

Before the breakout:

$D_0$ = RangeBoundCorrectiveRegime. (J.191)

Claim:

Extensions near £100 are likely to rotate back into the range.

---

## J.22.2 Declaration $D_1$

After $T_0$:

$D_1$ = PartiallyAdmittedDailyBreakout. (J.192)

Reason:

- close;
- normalized displacement;
- volume;
- VWAP acceptance.

Residual carried:

- breadth;
- retest;
- weekly gate;
- institutional uncertainty.

---

## J.22.3 Declaration $D_2$

After $T_4$:

$D_2$ = AdmittedDailyBreakoutWithSelfConfirmingTransition. (J.193)

Reason:

- retest;
- follow-through;
- breadth;
- value migration;
- $\chi$ transition.

---

## J.22.4 Declaration $D_3$

After weekly acceptance:

$D_3$ = EpisodeTransitionFromRangeToExpansion. (J.194)

Reason:

The old range grammar no longer organizes the event sequence.

---

## J.22.5 Declaration $D_4$

After benchmark announcement and effective inclusion:

$D_4$ = InstitutionallyRewrittenMarketWorld. (J.195)

Reason:

An authoritative external ledger changes future portfolio obligations and market microstructure.

---

## J.22.6 Revision operator

The update sequence is:

$D_{k+1} = U_a(D_k, L_{k+1}, \mathscr{R}_{k+1})$. (J.196)

Each revision preserves:

- old declaration;
- triggering evidence;
- residual carried;
- new invalidation.

The declaration framework requires revisions to remain trace-preserving, residual-honest, frame-robust, budget-bounded, and non-degenerate.

---

# J.23 The Full Six-Period Chain

| Period | Object in the case | Commitment | Output carried upward |
|---|---|---|---|
| **Mark** | trades above £100 | executions | raw transaction trace |
| **Window** | daily OHLCV bar | close at £100.88 | breakout window record |
| **Structure** | range, VWAP, profile, breadth | accepted price structure | loaded event context |
| **Event** | breakout gate | daily admission after retest | event history |
| **Episode** | range-to-expansion sequence | weekly and persistent transition | new episode grammar |
| **World** | Albion 100 inclusion | official index decision and rebalance | new institutional action space |

The recursive upward law is:

MarkCommitment
→ WindowLoad. (J.197)

WindowCommitment
→ StructureLoad. (J.198)

StructureCommitment
→ EventLoad. (J.199)

EventCommitment
→ EpisodeLoad. (J.200)

EpisodeCommitment
→ WorldLoad. (J.201)

The downward backreaction is:

WorldConstraint
→ EpisodeDynamics
→ EventProbability
→ StructureFormation
→ WindowDistribution
→ MarkBehaviour. (J.202)

---

# J.24 Functional Chain Across the Case

## J.24.1 Load

Load appears as:

- resting orders;
- daily volume;
- range memory;
- prior fakeouts;

- benchmark speculation;
- passive-fund obligation.

## J.24.2 Motion

Motion appears as:

- tick displacement;
- candle range;
- momentum;
- $\chi$ transition;
- episode expansion;
- institutional rebalancing.

## J.24.3 Constraint

Constraint appears as:

- offer at £100;
- range boundary;
- weekly resistance;
- index methodology;
- fund-tracking mandate.

## J.24.4 Commitment

Commitment appears as:

- execution;
- daily close;
- retest acceptance;
- weekly close;
- official inclusion;
- effective rebalance.

The same four functions recur.

Their objects and authorities change with period.

---

# J.25 Counterfactual Failed Branch

A good framework should also specify what would have happened if the breakout failed.

Suppose the path after $T_0$ had instead been:

$T_1$ close = £99.72. (J.203)

$T_2$ high = £99.94. (J.204)

$T_2$ close = £99.38. (J.205)

The invalidation condition would pass:

CloseInsideOldRange = true. (J.206)

FailedReclaim = true. (J.207)

The ledger would update:

PartiallyAdmittedDailyBreakout
→ FailedBreakout. (J.208)

$\chi$ would remain:

$\chi < 0$ or $\chi \approx 0$. (J.209)

New residual would be:

TrappedLongPressure. (J.210)

The event would not be promoted to Episode.

The benchmark review would remain a separate World-level possibility rather than being used retrospectively to rescue the failed breakout claim.

This counterfactual matters because the original gate was designed to allow failure.

A framework that could reinterpret both paths as success would be unfalsifiable.

---

# J.26 Comparison with Conventional Commentary

# Conventional statement

> NBE broke resistance on strong volume. MACD and RSI confirm the breakout. The next target is higher resistance.

# Periodic-grammar statement at $T_0$

> Under the daily protocol, NBE produced a 1.16 ATR close above a predeclared resistance zone on 1.74 relative volume and above VWAP. The event is only partially admitted because peer breadth remains below threshold, no retest or follow-through has occurred, weekly resistance remains unresolved, and new transaction density has not yet formed above the range. The prior regime was corrective; current $\chi$ is transitional rather than conclusively self-confirming. Invalidation requires a close inside the old range followed by failed reclaim.

The second statement does not guarantee a better trade.

It creates a better research record.

---

## J.27 The Case as a Disclosure Runtime

The whole case can be represented as:

$\Sigma_0$
→ Declare_P
→ Project_P
→ Diagnose$\chi$
→ Estimate$\Xi$
→ TestBoundary
→ ApplyGate
→ WriteTrace + Residual
→ Transport
→ ReviseDeclaration
→ ObserveBackreaction. (J.211)

In the source declaration architecture, a field becomes auditable only after boundary, observation rule, horizon, feature map, gate, trace, residual, and revision are declared.

The market case makes that architecture concrete.

---

## J.28 What the Case Demonstrates

The case demonstrates twelve distinctions.

### 1. Trade above resistance ≠ breakout

The first execution is Mark-level commitment only.

### 2. Close ≠ complete acceptance

The daily close is a stronger gate, but breadth, retest, and weekly structure remain unresolved.

### 3. Price indicators are not independent confirmation

EMA, MACD, and RSI share price lineage.

### 4. Breadth adds a different field channel

Breadth tests whether the move extends beyond one headline instrument.

### 5. Volume is meaningful only in context

High volume supports participation but does not reveal motive by itself.

## 6. χ changes interpretation

The same high RSI can mean exhaustion under corrective circulation or strength under self-confirming selection.

## 7. Ξ describes conditions, not event truth

High loading and moderate lock-in do not replace the breakout gate.

## 8. Residual survives commitment

The daily breakout can be admitted while weekly and institutional residual remain open.

## 9. Higher-period promotion requires new closure

A daily breakout becomes an Episode transition only after retest, breadth, persistence, and higher-frame acceptance.

## 10. A World event requires authority and backreaction

Index inclusion changes portfolio obligations and future microstructure.

## 11. Complex notation must be earned

The price–breadth pair does not gain a privileged phase merely because it can be written as R + iQ.

## 12. Revision must preserve old claims

The range interpretation, partial breakout, admitted event, episode transition, and world update remain distinct ledger entries.

---

# J.29 End-to-End Case Record

```
Case ID:
NBE-BREAK-001

Initial declaration:
Twelve-week corrective range beneath £100.

Primary boundary:
£99.80–£100.00.

Initial Mark event:
Execution at £100.04.

Window event:
Daily close at £100.88.

Normalized displacement:
1.16 ATR.

Relative volume:
```

```
1.74.

VWAP status:
Close above VWAP.

Breadth status:
57.5%, below 65% threshold.

Initial gate:
Partially Admit.

Initial χ:
Transitional, χ ≈ 0 with positive evidence.

Initial Ξ:
ρ = 0.79, γ = 0.62, ν = 0.58.

Initial residual:
Weak breadth, no retest, weekly resistance,
benchmark uncertainty, prior fakeout history,
incomplete value migration.

Invalidation:
Daily close inside old range followed by failed reclaim.

Later retest:
Passed provisionally.

Later breadth:
70%, passed.

Daily gate revision:
Admit.

Weekly gate:
Passed after persistence.

Episode revision:
Range → accepted expansion.

World gate:
Official Albion 100 inclusion.

World backreaction:
Higher volume, tighter spread, new benchmark-linked demand.

Complex-state audit:
Price acceptance and breadth retained as real pair.
Complex priority and phase-time claim rejected.

Final unresolved residual:
Post-inclusion crowding and temporary passive-flow distortion.
```

---

# J.30 Final Interpretation

The completed case is not simply:

Price broke resistance and continued upward. (J.212)

Its full structure is:

Resting interest accumulated at a remembered boundary.

Executions crossed the boundary.

A daily window closed beyond it.

The close altered structural memory.

The event passed additional gates.

The event sequence changed the episode grammar.

An institutional authority later committed a higher-level state change.

That World event altered future liquidity and participant obligations.

Residual remained at every stage.

The end-to-end chain is:

$$
\begin{aligned}
&\text{Load}_0 \\
&\rightarrow \text{Motion}_0 \text{ under Constraint}_0 \\
&\rightarrow \text{Commitment}_0 \\
&\rightarrow \text{Load}_1 \\
&\rightarrow \text{Motion}_1 \text{ under Constraint}_1 \\
&\rightarrow \text{Commitment}_1 \\
&\rightarrow \dots \\
&\rightarrow \text{WorldCommitment} \\
&\rightarrow \text{NewMarketField.} \quad \text{(J.213)}
\end{aligned}
$$

Or, in the article's compressed form:

$$
\begin{aligned}
&\text{Mark} \\
&\rightarrow \text{Window} \\
&\rightarrow \text{Structure} \\
&\rightarrow \text{Event} \\
&\rightarrow \text{Episode} \\
&\rightarrow \text{World} \\
&\rightarrow \text{NewMark.} \quad \text{(J.214)}
\end{aligned}
$$

The case therefore illustrates the central thesis:

> A market event becomes increasingly real not merely by moving farther, but by surviving progressively stronger gates, entering progressively broader ledgers, and changing the conditions under which later observations and actions occur.

The next appendix will convert the framework into a compact **Practitioner's Field Manual**, including one-page diagnostic sheets for ordinary indicators, breakouts, episode transitions, complex-state proposals, and institutional World events.

# Appendix K — Practitioner's Field Manual

# K.1 Purpose

This field manual compresses the Periodic Grammar into a sequence that can be used during actual analysis.

Its purpose is not to tell the practitioner what to buy or sell.

Its purpose is to prevent five common analytical failures:

1. treating one projection as the whole market;
2. promoting a relation into an event before a gate occurs;
3. counting correlated indicators as independent confirmation;
4. erasing residual and failed traces;
5. escalating a simple two-variable relationship into complex phase or internal time without sufficient evidence.

The operating principle is:

TechnicalAnalysis_P = Projection_P(MarketSelfReference). (K.1)

A technical method observes one characteristic under one protocol. It does not reveal the entire market. The source framework therefore recommends declaring the protocol, identifying the method's intrinsic function, finding its missing variables, testing independent cross-references, auditing residual, and defining invalidation.

---

# K.2 The One-Minute Diagnostic

Before trusting any chart interpretation, answer seven questions.

# Question 1 — What is the protocol?

State:

- asset;
- market boundary;
- timeframe;
- price scale;
- bar rule;
- feature map;
- gate;
- residual rule.

P_TA
= (Asset,Boundary,Timeframe,Scale,BarRule,FeatureMap,GateRule,ResidualRule). (K.2)

---

# Question 2 — Which period is being observed?

Choose one:

- Mark;

- Window;
- Structure;
- Event;
- Episode;
- World.

Do not confuse timeframe with period.

A daily candle is a Window.

A daily moving average is a Structure.

A daily breakout is an Event.

A multi-month trend is an Episode.

---

# Question 3 — Which function is primary?

Choose one:

- Load / Memory;
- Motion / Relation;
- Constraint / Boundary;
- Commitment / Gate.

Most errors begin when a method is assigned the wrong role.

Examples:

Moving average → Load.
RSI → Motion.
Resistance → Constraint.
Confirmed breakout → Commitment. (K.3)

---

# Question 4 — What is missing?

Common missing channels include:

- volume;
- breadth;
- liquidity;
- positioning;
- higher timeframe;
- institutional authority;

- retest;
- residual;
- invalidation.

A method's value is partly defined by what it does **not** measure.

---

# Question 5 — Has the relevant gate occurred?

A relation is not yet an event.

Ask:

- Was there a close?
- Was the displacement material?
- Was there participation?
- Was there acceptance?
- Was there follow-through?
- Was there a retest?
- Did a higher authority recognize the event?

---

# Question 6 — What residual remains?

Residual may contain:

- missing confirmation;
- direct contradiction;
- timeframe conflict;
- weak volume;
- failed retest;
- alternate wave count;
- institutional uncertainty;
- hidden positioning.

The source Technical Analysis framework defines residual as unresolved evidence, failed confirmation, contradiction, and ambiguity.

---

# Question 7 — What would invalidate the claim?

An interpretation without invalidation is not yet disciplined.

State the exact condition that would:

- weaken;
- invalidate;
- or downgrade the claim.

The complete one-minute test is:

Declare
→ LocatePeriod
→ IdentifyFunction
→ FindMissingVariables
→ TestGate
→ RecordResidual
→ DefineInvalidation. (K.4)

---

# K.3 The Six-Period Pocket Map

| Period | Object | Key question | Typical error |
|---|---|---|---|
| **Mark** | quote, order, execution | What was individually admitted? | treating one print as broad acceptance |
| **Window** | candle, bar, session | What did the aggregation retain at closure? | treating candle shape as complete intention |
| **Structure** | MA, RSI, VWAP, level, profile | What persistent relation or memory exists? | treating structure as an event |
| **Event** | breakout, rejection, retest | Has a meaningful transition passed a gate? | treating crossing as commitment |
| **Episode** | trend, range, squeeze, wave | What event grammar persists? | retrospective segmentation |
| **World** | policy, accounting, collateral, law | What authoritative state changes future action? | reducing institutional reality to chart price |

The upward inheritance rule is:

Commitment_p + Residual_p → Load_{p+1}. (K.5)

The downward constraint rule is:

World → Episode → Event → Structure → Window → Mark. (K.6)

The first explains how history accumulates.

The second explains how higher-order institutions shape lower-order market behaviour.

---

# K.4 The Four-Function Pocket Map

# K.4.1 Load / Memory

Ask:

> What structure has already accumulated and remains operative?

Possible evidence:

- moving averages;
- volume;
- VWAP;
- profile;
- prior highs and lows;
- prior failed breakouts;
- open interest;
- institutional exposure.

Warning:

Load is not direction.

High volume may represent:

- accumulation;
- liquidation;
- transfer;
- absorption;
- churn.

The source framework explicitly treats volume as a mixture of frequency, mass, commitment, and exchange ambiguity rather than a pure directional variable.

---

# K.4.2 Motion / Relation

Ask:

> How is the current state changing relative to another state?

Possible evidence:

- returns;
- momentum;
- RSI;
- MACD;
- divergence;
- relative strength;

- breadth;
- χ.

Warning:

Motion is not commitment.

Divergence warns.

It does not itself reverse the market.

---

# K.4.3 Constraint / Boundary

Ask:

> What resists, channels, separates, or defines admissible movement?

Possible evidence:

- support;
- resistance;
- volatility bands;
- channel;
- profile boundary;
- option strike;
- collateral threshold;
- legal rule.

Warning:

Boundary is not gate.

A level identifies where a consequential test may occur.

It does not decide the result.

---

# K.4.4 Commitment / Gate

Ask:

> What converts possibility into trace?

Possible gates include:

- execution;
- close;

- settlement;
- retest;
- follow-through;
- default;
- impairment;
- policy decision;
- legal judgment.

Warning:

Commitment does not eliminate residual.

The CAPM source distinguishes exposure, actual movement, economic consequence, gate recognition, ledger entry, and residual as separate stages.

---

# K.5 Indicator Triage Sheet

Use the following sheet whenever an indicator appears important.

```
Indicator:

Primary period:
Primary function:

Raw source:
Operator transformation:
Output units:

What it directly measures:
What it only approximates:
What it does not measure:

Assumed χ regime:
Required boundary:
Required event gate:

Independent confirmation channel:
Residual:
Transport tests:
Invalidation:

Permitted claim level:
Warning / Structure / Event / Episode / World
```

The last field is essential.

A Structure-period indicator should not be allowed to issue a World-level conclusion without additional evidence.

---

# K.6 Moving-Average Field Sheet

# Direct measurement

Filtered price memory.

$MA_n(t) = \Sigma_j w_j P_{t-j}$. (K.7)

# Primary role

Load / Memory.

# Secondary readings

- slope → Motion;
- price distance → Relation;
- widely watched average → possible soft Constraint.

# It does not directly measure

- participation;
- liquidity;
- positioning;
- institutional recognition;
- future direction.

# Required gate for trend-change claim

At minimum:

- persistent ordering change;
- meaningful boundary;
- close;
- follow-through or retest;
- preferably an independent participation channel.

# Main residual

- lag;
- whipsaw;
- range regime;
- same-source confirmation.

# Permitted statement

Price memory has become positively ordered under the declared filters.

## Overclaim

The moving-average crossover proves a new bull market.

---

# K.7 Oscillator Field Sheet

Applies to:

- RSI;
- stochastic;
- related bounded directional indicators.

## Direct measurement

Normalized directional relation or range location.

## Primary role

Motion / Relation.

## Required regime question

Is $\chi$:

- negative;
- near zero;
- positive?

Under:

$\chi < 0$, (K.8)

extreme readings may support corrective interpretation.

Under:

$\chi > 0$, (K.9)

extreme readings may represent persistent selection.

## Required gate

For reversal:

- boundary rejection;
- opposing close;

- structural break;
- breadth reversal;
- follow-through.

## Main residual

- persistent trend;
- wrong regime;
- no event gate;
- threshold arbitrariness.

## Permitted statement

> Directional dominance is elevated and may become unstable if the corrective regime remains active.

## Overclaim

> Overbought means the market must fall.

---

# K.8 Divergence Field Sheet

## Direct observation

One observable continues while another weakens.

Example:

Price higher high.
Momentum lower high. (K.10)

## Primary role

Motion / Relation.

## Status

Warning.

Not event.

## Required record

```
Price relation:
Comparison channel:
Anchor pair:
```

```
Direction:
Regime:
Boundary:
Required reversal gate:
Time horizon:
Residual:
Invalidation:
```

# Stronger divergence

A divergence becomes more important when it survives:

- alternate momentum method;
- breadth;
- volume;
- higher timeframe;
- independent pivot rule.

But:

CrossFrameDivergence ≠ Reversal. (K.11)

It means the relational weakening is broader.

A gate remains required.

---

# K.9 Volume Field Sheet

# Direct measurement

Recorded exchange activity.

V_t = Σ_kv_k. (K.12)

# Primary role

Load.

# Questions to ask

- Was volume high relative to the correct baseline?
- Was the baseline adjusted for time of day?
- Did volume produce displacement?
- Did price retain the displacement?
- Was the activity broad or concentrated?
- Was it cash, derivative, or auction volume?

- Did new value form afterward?

## Possible interpretations

- commitment;
- absorption;
- liquidation;
- short covering;
- churn;
- transfer.

## Required companion variables

At least:

- price location;
- displacement;
- close;
- later acceptance.

## Overclaim

> High volume confirms direction.

## Better statement

> Elevated participation accompanied the event, but motive and position transfer remain unresolved.

---

# K.10 VWAP and Profile Field Sheet

## VWAP

Direct role:

Volume-weighted transaction memory.

$VWAP_T = \Sigma P_t V_t / \Sigma V_t$. (K.13)

Do not call it intrinsic value unless a separate valuation argument is supplied.

## Volume profile

Direct role:

Historical transaction density across price.

$VP(p) = \Sigma_t V_t \cdot 1[P_t \in Bin(p)]$. (K.14)

# Questions

- What is the anchor?
- Has the anchor been changed?
- Is price merely crossing the reference or becoming accepted?
- Is new density forming?
- Is the frame local, session-based, or multi-session?
- Does the result survive another bin width?

# Event gate

- reclaim;
- hold;
- rejection;
- repeated close;
- value migration.

# Main residual

- anchor dependence;
- participant identity;
- position retention;
- stale historical density.

---

# K.11 Candlestick Field Sheet

# Direct object

Window-level compressed trace.

$Candle_P = (O,H,L,C,V)$. (K.15)

# Functional decomposition

Load → opening state and volume.
Motion → body and range.
Constraint → high and low.

Commitment → close.
Residual → unretained excursion and lost path. (K.16)

## Questions

- What bar rule created the candle?
- Did the high occur before or after the low?
- Was the candle produced by continuous flow or a gap?
- Is the wick meaningful relative to ATR?
- Did the next window confirm the close?
- Does the shape survive another bar construction?

## Permitted statement

> The window closed near its high after rejecting part of the lower excursion.

## Overclaim

> Buyers controlled the entire session.

The OHLC candle does not preserve enough path information to support that conclusion by itself.

---

# K.12 Support and Resistance Field Sheet

## Direct object

A candidate loaded boundary.

## Evidence sources

- repeated reaction;
- transaction density;
- prior close;
- VWAP;
- option strike;
- institutional reference;
- shared attention.

## Structural mass

A provisional level-mass expression is:

M_level
≈ ReactionHistory

- Density
- SharedAttention
- ConditionalOrders. (K.17)

The source framework describes structural mass as the inertia of a ledgered structure and warns that a heavily traded or widely watched level requires stronger pressure to break.

# Questions

- Was the level declared before the event?
- Is it a line or a zone?
- Does it survive log scale?
- Does it survive timeframe change?
- Is it supported by independent density?
- What gate defines a break?
- How does the level decay?

# Break condition

Boundary cross alone is insufficient.

A mature break may require:

BoundaryCross

- Close
- Participation
- Acceptance
- ResidualControl. (K.18)

---

# K.13 Breakout Field Manual

# K.13.1 Step 1 — Declare the boundary

Record:

- construction;
- zone width;

- timeframe;
- scale;
- age;
- number of prior tests;
- structural mass.

A boundary drawn after the break is not valid prospective evidence.

---

## K.13.2 Step 2 — Measure displacement

Raw displacement:

$$d = P - B. \quad (K.19)$$

Normalized displacement:

$$\tilde{d} = (P - B)/ATR. \quad (K.20)$$

Ask whether the break is material relative to ordinary variation.

---

## K.13.3 Step 3 — Audit participation

Use the relevant subset of:

- volume;
- dollar volume;
- open interest;
- breadth;
- peer participation;
- derivative confirmation.

Do not treat several price-derived momentum indicators as participation.

---

## K.13.4 Step 4 — Apply the close gate

State:

- intraday print;
- bar close;
- daily close;
- weekly close;

- settlement.

These gates have different authority.

---

## K.13.5 Step 5 — Test acceptance

Look for:

- follow-through;
- successful retest;
- profile migration;
- continued close beyond boundary;
- failure of the old region to reclaim price.

---

## K.13.6 Step 6 — Record residual

Typical breakout residuals include:

- weak breadth;
- no retest;
- higher-frame resistance;
- low liquidity;
- event risk;
- trapped positions;
- absent value migration.

---

## K.13.7 Step 7 — Classify the event

### Candidate

Crossing only.

### Partially admitted

Close and some supporting evidence.

### Admitted

Declared gate passes.

### Structurally accepted

Retest, persistence, or value migration.

### Episode transition

The old event grammar is replaced by a persistent new grammar.

### World transition

An institutional authority changes future action space.

---

# K.13.8 Breakout statement template

> Under protocol P, price crossed boundary B by đ normalized units. The relevant close gate [passed / failed / remains pending]. Participation evidence is [state]. Acceptance evidence is [state]. Residuals are [list]. The event is classified as [candidate / partial / admitted / structurally accepted]. Invalidation occurs under [condition].

---

# K.14 Fakeout Field Manual

# Definition

Fakeout
= CandidateTransition

- ParticipantCommitment
  – DurableAcceptance. (K.21)

# Required questions

- Was the initial gate weak or strong?
- Did the market return to the old region?
- Did a reclaim attempt fail?
- Was there higher-frame rejection?
- Did crowding increase?
- Who may now be trapped?
- Did the failed event create a stronger opposite boundary?

# Classification

### Weak-gate fakeout

The event should never have been fully admitted.

### Higher-frame fakeout

Local commitment failed at a stronger boundary.

### Crowding fakeout

Broad confirmation created excessive same-direction positioning.

### Liquidity fakeout

Thin market conditions produced misleading displacement.

### Catalyst reversal

A later external event invalidated the transition.

## Required ledger update

Do not delete the original breakout claim.

Update:

Candidate or Admitted Breakout
→ Failed Breakout

- TrappedPositionResidual
- NewBoundaryMemory. (K.22)

---

# K.15 Episode Field Manual

# K.15.1 Episode definition

An Episode is an ordered sequence of events organized by a relatively stable grammar.

Examples:

- trend;
- range;
- squeeze;
- wave sequence;
- crisis;
- recovery.

# K.15.2 Episode declaration

Record:

```
Episode type:
Start gate:
Event sequence:
```

```
Dominant χ:
Primary boundaries:
Load history:
Residual history:
Alternative branch:
Completion gate:
```

# K.15.3 Completion test

An episode is not completed merely because:

- an oscillator diverges;
- a Fibonacci level is reached;
- five waves can be counted;
- price touches a trend line.

A completion certificate should require:

OldGrammarFailed

- NewGateAdmitted
- NewStatePersistence
- ResidualAudit
- BranchStatus. (K.23)

# K.15.4 Wave endpoint checklist

- Is the pivot objectively defined?
- Is there momentum or breadth phase shift?
- Has a structural gate failed?
- Does the endpoint survive another timeframe?
- Is there an alternate count?
- Is the original count preserved?

---

# K.16 World-Event Field Manual

# K.16.1 World-level examples

- default;
- impairment;
- margin trigger;
- policy decision;

- index inclusion;
- legal judgment;
- regulatory classification;
- restructuring.

## K.16.2 Necessary components

### Authority

Who can commit the event?

### Rule

What declared rule applies?

### Ledger

Where is the event written?

### Consequence

What future action becomes required, forbidden, or newly available?

### Backreaction

How does the event alter later market dynamics?

## K.16.3 Recognition vector

$$\mathbb{G}_k = (g_{market}, g_{accounting}, g_{legal}, g_{risk}, g_{policy}). \quad (K.24)$$

Use it when several authorities may recognize the same economic condition at different times.

## K.16.4 Warning

Market price anticipation is not identical to institutional recognition.

A market may price a likely default before legal default.

A legal default may occur after the price has already adjusted.

The ledgers remain distinct.

---

## K.17 χ Regime Sheet

## K.17.1 Purpose

χ describes the orientation of the feedback relation.

$\chi < 0 \rightarrow$ corrective. (K.25)

$\chi \approx 0 \rightarrow$ critical or unstable. (K.26)

$\chi > 0 \rightarrow$ self-confirming. (K.27)

## K.17.2 Diagnostic questions

- Do extensions produce opposing or reinforcing flow?
- Do breakouts hold or repeatedly fail?
- Do oscillator extremes revert or persist?
- Does breadth expand with price?
- Does volume create displacement or absorption?
- Are boundaries stable or being rewritten?

## K.17.3 Required horizon

Always index $\chi$ by protocol and horizon:

$\chi = \chi_{P,h}$. (K.28)

An asset may be:

- corrective intraday;
- self-confirming daily;
- critical weekly.

## K.17.4 Indicator translation

| Reading | $\chi < 0$ | $\chi \approx 0$ | $\chi > 0$ |
|---|---|---|---|
| High RSI | possible extension | unstable | possible persistent strength |
| Band touch | reversion candidate | ambiguous | trend ride |
| Divergence | stronger warning | uncertain | may persist |
| Breakout | higher fakeout risk | transition test | continuation candidate |
| Retest | rotation | decisive | acceptance reinforcement |

$\chi$ is not itself a trade signal.

It conditions the meaning of other projections.

---

## K.18 Ξ Control-State Sheet

## State

$\Xi = (\rho,\gamma,\nu)$. (K.29)

where:

ρ = loading;
γ = lock-in;
ν = agitation.

# Questions for ρ

- How much participation or exposure is loaded?
- Is there leverage?
- Is there accumulated position density?
- Is attention concentrated?

# Questions for γ

- How hard is exit or reclassification?
- Is liquidity limited?
- Is collateral constrained?
- Are legal or institutional rules binding?

# Questions for ν

- How unstable is the current operating condition?
- Are spreads widening?
- Are cancellations or failed gates increasing?
- Is volatility or dispersion rising?

# Prohibition

Do not infer direction directly from Ξ.

A highly loaded, locked, agitated state may break in more than one direction.

Ξ describes the operating condition, not the event outcome.

---

# K.19 Confirmation Independence Sheet

# K.19.1 Source independence

Do the methods use different raw channels?

Examples:

- price;

- volume;
- breadth;
- liquidity;
- options;
- accounting;
- policy.

## K.19.2 Operator independence

Do they use different transformations?

Examples:

- smoothing;
- accumulation;
- normalization;
- density;
- gate;
- relative comparison.

## K.19.3 Functional independence

Do they cover different functions?

- Load;
- Motion;
- Constraint;
- Commitment.

## K.19.4 Failure-mode independence

Can they fail for different reasons?

## K.19.5 Quick redundancy warning

The following may largely repeat one price movement:

- price above MA;
- positive MA slope;
- positive MACD;

- RSI above 50;
- positive momentum.

This is useful agreement about price structure.

It is not five independent sources of evidence.

## K.19.6 Preferred confirmation bundle

For a breakout:

Load → participation.
Motion → normalized displacement.
Constraint → meaningful boundary.
Commitment → close and acceptance.
Residual → contradiction register.
Transport → higher frame and independent channel. (K.30)

---

# K.20 Transport Pocket Checklist

A claim should be challenged by the relevant subset of:

- higher timeframe;
- lower timeframe;
- log scale;
- ATR normalization;
- alternate bar construction;
- alternate benchmark;
- equal-weight breadth;
- nearby anchor;
- alternative pivot rule;
- data vendor;
- cash versus derivative market;
- observer authority.

The source Technical Analysis framework defines stronger signals as those surviving admissible changes of projection and specifically includes timeframe, scale, volatility normalization, bar construction, volume profile, breadth, and higher-timeframe structure.

# Result vocabulary

- survives;

- covariantly survives;
- partially survives;
- local only;
- fails;
- indeterminate;
- non-comparable.

## Transport statement template

> Claim C was observed under P. Under transformation $T_{P \to P'}$, the expected target form was $\hat{C}$. The observed target claim was $C'$. The claim [survives / partially survives / fails] within tolerance $\varepsilon$. The resulting transport residual is $r_T$.

---

# K.21 Residual Pocket Checklist

## K.21.1 Missing confirmation

- volume;
- breadth;
- close;
- retest;
- follow-through.

## K.21.2 Direct contradiction

- price rises while breadth falls;
- breakout occurs while value remains below boundary;
- Wave 3 is claimed with weak directional expansion.

## K.21.3 Frame conflict

- daily bullish;
- weekly resistance unresolved;
- log-scale boundary differs;
- equal-weight index fails.

## K.21.4 Boundary uncertainty

- arbitrary anchor;

- broad zone;
- competing levels;
- unstable pivot.

## K.21.5 Regime uncertainty

- $\chi$ not known;
- corrective and reinforcing evidence coexist.

## K.21.6 Institutional residual

- legal recognition pending;
- accounting gate pending;
- policy uncertainty;
- settlement incomplete.

## K.21.7 Model residual

- unexplained movement;
- parameter instability;
- complex phase scaling failure;
- alternate model performs equally well.

The mature analyst does not ask:

> How can I remove all residual?

The better question is:

> Which residual can be resolved, which must be carried, and which invalidates the current claim?

---

# K.22 Invalidation Pocket Checklist

A valid invalidation should be:

- observable;
- protocol-specific;
- time-bounded;
- recorded before outcome;
- distinct from a trading stop where necessary.

# Examples

## Breakout

Close back inside old range plus failed reclaim.

## Support

Durable acceptance below zone and failed recovery.

## Divergence

No reversal gate within declared horizon and indicator re-expansion.

## Wave endpoint

Declared wave rule breached.

## Phase clock

No out-of-sample alignment gain over calendar time or event count.

## Complex state

Real pair performs equally well and phase is scale-fragile.

The governing rule is:

NoInvalidation → NoDiscipline. (K.31)

---

# K.23 Complex-State Proposal Sheet

Use this sheet before writing:

Z = R + iQ. (K.32)

```
Research task:

R:
Meaning:
Units:
Independent estimator:

Q:
Meaning:
Units:
Independent estimator:

Why are R and Q conjugate?
What is the norm A?
What is the phase θ?
What generator couples them?
What residual remains?

Scalar benchmark:
Real-pair benchmark:
```

```
Scaling test:
Timeframe test:
Episode-alignment test:
Gate-hazard test:

Reduction trigger:
```

# Automatic rejection conditions

Reject complex priority when:

- Q means everything unexplained by R;
- R and Q use incompatible units without a metric;
- phase depends strongly on arbitrary scaling;
- no stable generator exists;
- episode alignment does not improve;
- gates do not concentrate by phase;
- the real pair performs equally well.

The phase source states that a real pair should remain primary when R and Q are merely correlated, phase is scale-dependent, rotational structure is absent, or a flexible real model performs equally well.

---

# K.24 Phase-Time Proposal Sheet

# Four distinct coordinates

t = external duration. (K.33)

$\theta$ = complex-state orientation. (K.34)

$\tau_i$ = accumulated phase depth. (K.35)

k = committed event order. (K.36)

These must not be collapsed.

The source phase framework states that phase progression becomes historical commitment only when a phase-sensitive gate produces persistent trace; it distinguishes parent duration, phase orientation, internal phase depth, and ledger-event order.

# Minimum tests

1. comparable episodes exist;
2. R and Q are independently defensible;
3. phase order is stable;

4. $\tau_i$ aligns episodes better than t;
5. $\tau_i$ outperforms event count k;
6. event gates concentrate in phase;
7. results survive out of sample;
8. trace alters later dynamics for a world-level claim.

## Minimal phase-time claim

Phase provides a useful secondary ordering of episode progress.

## Stronger event claim

Consequential gates occur in stable phase regions.

## Strongest world claim

Phase-sensitive gates write persistent trace that changes subsequent dynamics.

Do not jump directly from Z to the strongest claim.

---

# K.25 CAPM Complex-Geometry Sheet

## Core definitions

$r_{CAPM} = r_{base} + \beta ERP$. (K.37)

$A_t = CF_t/(1 + r_{base})^t$. (K.38)

$R_t = CF_t/(1 + r_{CAPM})^t$. (K.39)

$Q_t = \sqrt{(A_t^2 - R_t^2)}$. (K.40)

$Z_t = R_t + iQ_t$. (K.41)

$\partial R/\partial \theta = -Q$. (K.42)

## Meaning

R = admitted value.
Q = conjugate valuation-phase exposure.
A = declared baseline amplitude.
$\theta$ = valuation orientation.

## Do not confuse

$Q \neq$ haircut $A - R$. (K.43)

$Q \neq$ realized loss. (K.44)

Q ≠ residual. (K.45)

# Runtime

Measurement
→ Exposure
→ State Movement
→ Economic P&L
→ Gate
→ Ledger + Residual. (K.46)

The source CAPM article explicitly states that −Q identifies exposure and that economic consequence requires actual phase movement, after which a gate determines recognition.

---

# K.26 Claim-Level Control Sheet

Use this table to stop claim inflation.

| Evidence available | Maximum permitted claim |
|---|---|
| One indicator reading | local projection |
| Several related indicators | same-lineage structure |
| Independent functional channels | multi-channel diagnosis |
| Explicit gate | Event |
| Persistence and ordered events | Episode |
| Authority, ledger, and future constraint | World |
| R–Q pair only | two-variable state |
| stable generator and phase utility | complex dynamical state |
| better episode alignment | secondary phase time |
| phase-sensitive gates | phase-bearing event model |
| trace plus backreaction | time-bearing world |

The principle is:

ClaimStrength ≤ EvidenceClosure. (K.47)

---

# K.27 Analysis Writing Templates

# K.27.1 Structure-level template

> Under protocol P, method M projects a [Load / Motion / Constraint] structure at period Structure. The reading is [state]. Its interpretation assumes χ = [state]. It does not yet establish an Event because gate G has not occurred. Missing variables are [list]. Invalidation is [condition].

---

# K.27.2 Event-level template

> Price or state crossed boundary B by normalized displacement đ. Gate evidence consists of [close, volume, breadth, retest, follow-through]. The event is [candidate /

partially admitted / admitted]. Residuals are [list]. The claim survives [frames] and fails or remains pending under [frames]. Invalidation is [condition].

---

## K.27.3 Episode-level template

The prior episode grammar was [old sequence]. Evidence now supports [new sequence]. The transition gate is [event]. The new grammar has persisted for [declared condition]. Alternative branch [branch] remains residual. Completion or failure occurs under [condition].

---

## K.27.4 World-level template

Authority A applied rule R and wrote event e into ledger L. The event changes future admissibility by [consequence]. Market, accounting, legal, risk, and policy recognition are [vector]. Remaining residual includes [list]. Backreaction is evaluated through [future-state measures].

---

## K.27.5 Complex-model template

Variables R and Q are independently defined as [meanings] with compatible units under protocol P. The candidate complex state is $Z = R + iQ$. Generator evidence is [result]. Phase survives [tests] but fails [tests]. The complex model [does / does not] outperform the real-pair benchmark. The retained model level is [level].

---

## K.28 Daily Research Workflow

## Before the market or study window

1. Freeze protocol.
2. Declare boundaries.
3. Record expected gates.
4. Record known residual.
5. Specify invalidation.
6. Avoid changing anchors after the event begins.

## During the event

1. Record marks and windows.
2. Do not promote period prematurely.
3. Distinguish crossing from close.
4. Record missing confirmation as pending.

5. Update residual without deleting the original state.

## After the gate

1. Record admission status.
2. Test relevant frames.
3. Observe follow-through or failure.
4. preserve fakeouts;
5. update episode classification only after persistence.

## After outcome

1. Evaluate the original horizon.
2. Do not choose the best later horizon.
3. record maximum favourable and adverse movement;
4. apply invalidation;
5. revise only through a new version.

---

# K.29 The Ten Prohibitions

## Prohibition 1

Do not call an indicator the market.

## Prohibition 2

Do not call a relation an event.

## Prohibition 3

Do not call a crossing a confirmed breakout without a gate.

## Prohibition 4

Do not count shared-input indicators as independent evidence.

## Prohibition 5

Do not delete failed traces.

The source article states that a system recording only successful patterns cannot learn.

## Prohibition 6

Do not change timeframe, anchor, or horizon silently after failure.

## Prohibition 7

Do not use Q as a container for everything unexplained.

## Prohibition 8

Do not call phase time unless phase improves internal ordering.

## Prohibition 9

Do not call a chart event a World transition without authority, ledger, and changed future action.

## Prohibition 10

Do not retain a more complex model when a simpler model performs equally well.

---

# K.30 The Ten Positive Rules

## Rule 1

Declare before interpreting.

## Rule 2

Classify the closure period.

## Rule 3

Identify the primary function.

## Rule 4

Search for the missing function.

## Rule 5

Apply a predeclared gate.

## Rule 6

Record residual explicitly.

## Rule 7

Attack the claim through admissible transport.

## Rule 8

Preserve original claims and failures.

## Rule 9

Promote only when a new closure condition is met.

## Rule 10

Reduce unsupported complexity.

The source Technical Analysis framework summarizes mature practice as cross-frame survival, residual honesty, and gate discipline.

---

# K.31 Compact Practitioner Dashboard

| Category | Current state | Evidence | Missing | Gate | Residual | Invalidation |
|---|---|---|---|---|---|---|
| Load | | | | — | | |
| Motion | | | | — | | |
| Constraint | | | | | | |
| Commitment | | | | | | |
| χ | | | | — | | |
| Ξ | | | | — | | |
| Transport | | | | — | | |
| Episode | | | | | | |
| World | | | | | | |

A blank cell should remain blank or be marked unknown.

It should not be replaced by an inferred value merely to complete the dashboard.

---

# K.32 Model-Status Labels

Every advanced model should display one of the following labels.

## Descriptive

Useful representation only.

## Diagnostic

Improves interpretation or classification.

## Predictive

Improves out-of-sample prediction.

## Gate-bearing

Improves consequential event detection.

## Intervention-bearing

Improves decision or control.

## World-forming

Demonstrates trace and backreaction.

A complex model may be:

Diagnostic but not Predictive. (K.48)

A phase model may be:

Predictive but not WorldForming. (K.49)

A simple breakout gate may be:

GateBearing without ComplexPhase. (K.50)

These labels prevent one success dimension from being mistaken for total validation.

---

# K.33 The Minimal Complete Analysis

A minimal but complete Technical Analysis statement contains:

1. protocol;
2. period;
3. primary function;
4. observed state;
5. regime assumption;
6. gate status;
7. residual;
8. transport result;
9. invalidation;
10. current claim level.

In compressed form:

Analysis
= P

- Period
- Function
- Projection
- χ
- Gate
- Residual
- Transport
- Invalidation
- ClaimLevel. (K.51)

This may appear elaborate.

But most fields can be expressed in one sentence once the discipline becomes familiar.

---

# K.34 One-Sentence Example

> Under the daily log-scale protocol, the asset has produced a Structure-level positive price-memory and momentum state near a predeclared resistance zone; the interpretation assumes a transitional χ, but the Event gate remains pending because breadth and retest are absent, the weekly boundary remains residual, the claim survives ATR normalization but not yet the higher timeframe, and it is invalidated by a close below the reclaimed value zone followed by failed recovery.

This sentence is not a prediction.

It is an auditable state declaration.

---

# K.35 Practitioner's Final Decision Tree

```
What am I looking at?
│
├─ A single print or order
│   └─ Mark
│
├─ A bar or candle
│   └─ Window
│
├─ A persistent indicator, level, or relation
│   └─ Structure
│
├─ A tested transition
│   └─ Event
│
├─ An ordered sequence of events
```

```
│   └─ Episode
│
└─ An authoritative institutional state
    └─ World

What function does it perform?
│
├─ Carries history → Load
├─ Measures change → Motion
├─ Defines a limit → Constraint
└─ Writes consequence → Commitment

Has a gate occurred?
│
├─ No → Keep as structure or warning
└─ Yes
    │
    ├─ Residual material? → Partial or qualified admission
    └─ Residual controlled? → Admit under declared protocol

Does it survive another frame?
│
├─ No → Localize or weaken
└─ Yes → Increase transport status

Does it require complex phase?
│
├─ No → Use scalar or real pair
└─ Yes
    │
    ├─ Stable generator and scaling? No → Real pair
    │
    └─ Yes
        │
        ├─ Episode alignment gain? No → Complex dynamics only
        └─ Yes
            │
            ├─ Gate concentration? No → Secondary phase time
            └─ Yes
                │
                ├─ Trace and backreaction? No → Phase-bearing event model
                └─ Yes → Candidate time-bearing world
```

---

# K.36 Appendix K Conclusion

The field manual reduces the full theory to one practical discipline:

> Never allow a market claim to become stronger than the closure that has actually occurred.

A Mark is not a Window.

A Window is not a Structure.

A Structure is not an Event.

An Event is not an Episode.

An Episode is not automatically a World.

A pair of variables is not automatically complex.

A phase is not automatically time.

A gate is not total resolution.

A ledger entry is not the disappearance of residual.

The complete practitioner runtime is:

Declare
→ Project
→ Classify
→ Diagnose
→ CrossCheck
→ Gate
→ RecordTrace
→ PreserveResidual
→ Transport
→ Invalidate
→ Revise
→ Reduce. (K.52)

The final professional rule is the same rule stated by the source Technical Analysis article:

> Trust not the signal that looks beautiful, but the signal that survives declared challenge.

# Appendix L — Proto-Eight Crosswalk: From Market Observation to Actuation Grammar

## L.1 Purpose

The Periodic Grammar developed in this article has so far used two principal dimensions:

1. **six closure periods**—Mark, Window, Structure, Event, Episode, and World;
2. **four observational functions**—Load, Motion, Constraint, and Commitment.

Proto-Eight Dynamics introduces a third dimension.

It asks not merely:

> What kind of market object is being observed?

It asks:

> Through which primitive operation does the system move, transfer, retain, select, or commit that object?

The eight Proto-Eight roles are:

$$A_8 = \{\text{Gradient},\text{Gate},\text{Boundary},\text{Exchange},\text{Trigger},\text{Guidance},\text{Memory},\text{Focus}\}. \quad (L.1)$$

They are organized into four complementary dyads:

$$D_1 = \text{Gradient} \times \text{Gate}. \quad (L.2)$$

$D_2$ = Boundary × Exchange. (L.3)

$D_3$ = Trigger × Guidance. (L.4)

$D_4$ = Memory × Focus. (L.5)

The Proto-Eight engineering sources map these roles respectively to 乾, 坤, 艮, 兌, 震, 巽, 坎, and 離, and treat them as practical system levers rather than merely symbolic labels.

The relationship between the present article and Proto-Eight can therefore be summarized as:

- **the six periods locate closure depth;**
- **the four functions classify what the observation contributes;**
- **the eight primitives describe how the system actuates and transforms.**

These dimensions overlap, but they are not identical.

---

# L.2 Observation Grammar Versus Actuation Grammar

The four-family grammar asks:

| Family | Question |
|---|---|
| Load / Memory | What consequential structure is already carried? |
| Motion / Relation | How is the state changing? |
| Constraint / Boundary | What limits or channels the change? |
| Commitment / Gate | What becomes accepted history? |

Proto-Eight asks a different set of questions:

| Primitive | Question |
|---|---|
| Gradient | What potential difference creates pressure to move? |
| Gate | What qualifies passage or admission? |
| Boundary | What separates, buffers, or contains regions? |
| Exchange | What crosses the interface? |
| Trigger | What initiates movement? |
| Guidance | What steers movement after initiation? |
| Memory | What is retained from earlier movement? |
| Focus | What is selected for present attention or rendering? |

The distinction is:

ObservationGrammar classifies the function of evidence. (L.6)

ActuationGrammar classifies the primitive operation generating or regulating change. (L.7)

A moving average, for example, is primarily a **Load** instrument in the four-family table.

Under Proto-Eight, it is primarily a **Memory** operator.

A breakout close is primarily a **Commitment** object.

Under Proto-Eight, it is the action of a **Gate**.

A price catalyst may appear as Event-level Motion.

Under Proto-Eight, it functions as a **Trigger**.

The two classifications are compatible because they answer different questions.

---

# L.3 The Eight Proto-Eight Roles

# L.3.1 Gradient — 乾

## Engineering meaning

Gradient is the potential difference capable of producing directed flow.

The P8D source begins with a two-tank model:

- one side contains productive capacity;
- the other contains reachable demand;
- their imbalance produces a potential gradient;
- the resulting flow becomes throughput.

Its compact growth model treats throughput as depending on enablement, fit, retention, the availability of both sides, and a gate responsive to the capacity–demand gradient.

## Market translation

Possible financial gradients include:

- bid–ask imbalance;
- demand–supply imbalance;
- price–value difference;
- spot–futures basis;
- yield spread;
- funding differential;
- relative-strength differential;
- price distance from memory;
- pressure across a support or resistance zone.

A gradient is not yet motion.

It is the structured possibility of motion.

## Failure mode

Gradient without an effective gate can produce:

- pressure without realized flow;
- crowding;
- backlog;
- latent volatility;
- repeated failed attempts.

---

# L.3.2 Gate — 坤

## Engineering meaning

Gate qualifies what may pass into the receiving region.

A gate may:

- admit;
- reject;
- defer;
- partially admit;
- redirect.

## Market translation

Examples include:

- trade execution;
- official close;
- settlement;
- breakout confirmation;
- successful retest;
- margin trigger;
- impairment;
- default recognition;
- policy decision;
- legal judgment.

## Failure mode

Gate without gradient may become:

- ossified procedure;
- empty formalism;
- excessive filtering;
- low throughput;
- false rejection.

Gradient without Gate creates uncontrolled pressure.

Gate without Gradient creates sterile closure.

Their dyad converts potential into qualified flow.

---

# L.3.3 Boundary — 艮

## Engineering meaning

Boundary defines where one region ends and another begins.

It can also serve as:

- buffer;
- damper;
- interface;
- firebreak;
- locality constraint.

## Market translation

Examples include:

- bid and ask;
- tick size;
- high and low;
- support and resistance;
- volatility band;
- range edge;
- collateral limit;
- margin threshold;
- legal entity;
- policy mandate.

### Failure mode

Boundary without exchange can produce:

- isolation;
- trapped inventory;
- poor discovery;
- accumulated pressure;
- fragmentation.

---

## L.3.4 Exchange — 兌

### Engineering meaning

Exchange is the transfer of matter, value, information, or obligation through an interface.

The Proto-Eight sources pair Boundary and Exchange because a healthy interface must neither block everything nor allow uncontrolled leakage. The pair is treated as a buffer–exchange mechanism capable of damping variability while preserving useful circulation.

### Market translation

Examples include:

- execution;
- volume;
- order flow;
- turnover;
- capital transfer;
- rebalancing;
- collateral exchange;
- information dissemination;
- cross-component participation.

### Failure mode

Exchange without boundary may create:

- leakage;
- noise reinjection;
- unstable feedback;

- contagion;
- poor attribution.

The dyad therefore asks:

> What must remain separate, and what must be allowed to cross?

---

# L.3.5 Trigger — 震

## Engineering meaning

Trigger supplies the impulse that initiates movement.

Possible trigger variables include:

- threshold crossing;
- pulse amplitude;
- pulse duration;
- activation frequency;
- ignition condition.

## Market translation

Examples include:

- order imbalance;
- price crossing;
- earnings announcement;
- policy release;
- stop activation;
- volatility shock;
- margin breach;
- index announcement;
- catalyst.

## Failure mode

Trigger without Guidance often produces:

- spike and collapse;
- overshoot;

- false breakout;
- attention burst without retention;
- repeated activation fatigue.

---

# L.3.6 Guidance — 巽

## Engineering meaning

Guidance steers an initiated flow toward a viable route.

The Proto-Eight source pairs short trigger pulses with a guidance field that bends orientation and attempts to preserve phase lock. It explicitly distinguishes ignition from continuing on the intended route.

## Market translation

Possible guidance structures include:

- trend direction;
- VWAP;
- channel;
- liquidity route;
- benchmark constraint;
- policy corridor;
- algorithmic execution path;
- follow-through;
- dominant order-flow direction.

## Failure mode

Guidance without Trigger produces:

- planning without movement;
- unused route capacity;
- inert strategy;
- slow drift.

Trigger without Guidance produces chaotic movement.

The dyad distinguishes:

Starting from continuing. (L.8)

---

# L.3.7 Memory — 坎

## Engineering meaning

Memory retains the consequence of prior interaction.

The Proto-Eight sources model Memory as a retention well whose main engineering variables include:

- capture;
- forgetting;
- resurfacing;
- dwell time;
- leakage.

## Market translation

Examples include:

- moving average;
- volume profile;
- anchored VWAP;
- OBV;
- prior high or low;
- failed-breakout history;
- event ledger;
- balance sheet;
- accounting record;
- legal precedent.

## Failure mode

Memory without Focus may produce:

- information overload;
- stale structure;
- indiscriminate retention;
- path dependence without relevance;
- semantic or analytical black hole.

---

# L.3.8 Focus — 離

## Engineering meaning

Focus selects and sharpens the currently relevant region of a larger retained field.

The Proto-Eight source treats Focus as a lens associated with:

- selective rendering;
- passband;
- lens stiffness;
- recall latency;
- concentration.

Memory retains.

Focus determines what part of memory becomes operational now.

## Market translation

Examples include:

- selected timeframe;
- chosen benchmark;
- watched indicator;
- active resistance zone;
- event window;
- analyst attention;
- market narrative;
- regulatory review focus.

## Failure mode

Focus without Memory may produce:

- myopia;
- recency bias;
- isolated signal worship;
- repeated rediscovery;
- no learning from failed traces.

Memory without Focus cannot act efficiently.

---

# L.4 The Four Dyads

## L.4.1 Gradient × Gate — Potential Becomes Qualified Throughput

The first dyad answers:

> What creates pressure, and what allows that pressure to become realized flow?

In markets:

Gradient
= imbalance, spread, relative pressure, or valuation difference. (L.9)

Gate
= execution, close, settlement, or recognition rule. (L.10)

Examples include:

- order imbalance × execution;
- price pressure × breakout close;
- economic loss × accounting recognition;
- funding stress × margin gate;
- anticipated default × contractual default.

The dyad prevents two category errors.

### Category error A

Pressure is treated as already realized.

### Category error B

Formal recognition is treated as though it created the underlying pressure.

The correct sequence is:

PotentialDifference
→ CandidateFlow
→ QualificationGate
→ AdmittedThroughput + Residual. (L.11)

---

## L.4.2 Boundary × Exchange — Separation Becomes Governed Circulation

The second dyad answers:

> Where is the interface, and what may cross it?

Examples include:

- bid–ask boundary × executed trade;
- range boundary × breakout volume;
- legal entity × capital transfer;
- collateral boundary × funding exchange;
- sector boundary × breadth participation.

The boundary supplies structure.

Exchange supplies circulation.

A market with no boundaries cannot localize responsibility or value.

A market with no exchange cannot discover or transfer value.

The operative relation is:

HealthyInterface
= SufficientSeparation

- SufficientExchange
  – Leakage
  – Blockage. (L.12)

This is an engineering template rather than an established universal market equation.

---

# L.4.3 Trigger × Guidance — Ignition Becomes Directed Progress

The third dyad answers:

> What starts the movement, and what keeps it on a coherent route?

Examples include:

| Trigger | Guidance |
|---|---|
| resistance crossing | follow-through |
| earnings surprise | analyst and capital-flow revision |
| stop activation | liquidity route |
| volatility shock | policy response |
| index announcement | benchmark-rebalancing schedule |
| initial order pulse | execution algorithm |

A trigger may be strong but brief.

Guidance determines whether the resulting movement:

- persists;

- dissipates;
- fragments;
- overshoots;
- reaches a new gate.

The dyad is especially important for distinguishing:

BreakoutAttempt from BreakoutAcceptance. (L.13)

---

## L.4.4 Memory × Focus — Retained History Becomes Selective Present

The fourth dyad answers:

> What has been retained, and which part becomes relevant now?

Examples include:

| Memory | Focus |
|---|---|
| prior range | active resistance zone |
| full price history | selected moving-average window |
| transaction history | current volume-profile node |
| event ledger | current episode branch |
| balance sheet | current covenant test |
| legal history | present dispute |
| policy history | current mandate |

The dyad prevents:

- total-history overload;
- attention without context;
- indicator monoculture;
- repeated failure without learning.

It converts stored history into an operational lens.

---

# L.5 Crosswalk to the Four Functional Families

The Proto-Eight roles do not map one-to-one onto the four families.

A more accurate relationship is many-to-many:

$\varphi_P: A_8 \to \mathcal{P}(\{\text{Load},\text{Motion},\text{Constraint},\text{Commitment}\})$. (L.14)

where $\mathcal{P}$ denotes the set of possible functional combinations under protocol P.

The principal crosswalk is:

| Proto-Eight role | Load | Motion | Constraint | Commitment |
|---|---|---|---|---|
| Gradient | Secondary | **Primary** | — | — |
| Gate | — | — | Secondary | **Primary** |
| Boundary | Secondary | — | **Primary** | — |
| Exchange | — | **Primary** | Secondary | Secondary |
| Trigger | — | **Primary** | — | Secondary |
| Guidance | — | **Primary** | Secondary | — |
| Memory | **Primary** | — | Secondary | — |
| Focus | Secondary | — | **Primary** | Secondary |

“Primary” identifies the most common role.

“Secondary” identifies a frequent contextual role.

The mapping may change by protocol.

For example:

- execution is Exchange as actuation but Commitment as observation;
- VWAP is Memory as actuation but Load as observation;
- a close is Gate as actuation and Commitment as observation;
- a trend channel is Guidance as actuation and Constraint as observation.

Thus:

ActuationRole ≠ ObservationalFunction. (L.15)

---

# L.6 Proto-Eight Across the Six Periods

## L.6.1 Compact 6 × 8 atlas

| Period | Gradient | Gate | Boundary | Exchange | Trigger | Guidance | Memory | |
|---|---|---|---|---|---|---|---|---|
| **Mark** | queue or price imbalance | execution | bid, ask, tick | trade | marketable order | routing | queue state | s fi |
| **Window** | open–close pressure | close | high–low range | volume | gap or impulse | intrawindow path | prior close | b |
| **Structure** | price–memory distance | structural acceptance | level, band, channel | profile and breadth | crossover or test | trend, VWAP | MA, OBV, profile | ir |
| **Event** | pressure across boundary | break, rejection, retest | tested zone | event participation | catalyst or crossing | follow-through | prior attempts | e p |
| **Episode** | regime or basin pressure | completion or transition | pattern or attractor basin | event succession | defining event | χ or phase progression | event ledger | s |

| Period | Gradient | Gate | Boundary | Exchange | Trigger | Guidance | Memory | |
|---|---|---|---|---|---|---|---|---|
| **World** | funding, policy, or valuation pressure | legal, accounting, policy recognition | law, collateral, regulation | capital transfer and rebalancing | default or formal decision | mandate and institutional route | balance sheet and ledger | a o |

The atlas shows that the eight roles recur at every closure depth.

But the meaning of each role changes.

A Mark-level Gate is execution.

A Window-level Gate is a close.

An Event-level Gate is a breakout or rejection rule.

A World-level Gate may be legal or accounting recognition.

This recurrence is one reason the framework is called periodic.

---

# L.7 A Local Eightfold Market Runtime

The Primordial BaGua arrangement should not be treated as one fixed chronological sequence.

The Chinese source characterizes it as a symmetric pre-time structure or tension configuration rather than an ordinary temporal procession.

Nevertheless, one protocol may project the eight roles into an operational runtime.

For a Technical Analysis event, one useful ordering is:

Memory_k
→ Focus_k
→ Boundary_k
→ Gradient_k
→ Trigger_k
→ Guidance_k
→ Exchange_k
→ Gate_k
→ Memory_{k+1} + Residual_k. (L.16)

This is not asserted as the universal order of the eight trigrams.

It is a market-analysis runtime derived from their engineering functions.

---

# L.7.1 Memory

Prior traces preserve:

- range;
- volume density;
- failed breaks;

- trend;
- institutional events.

## L.7.2 Focus

The observer selects:

- timeframe;
- boundary;
- benchmark;
- feature map;
- event class.

## L.7.3 Boundary

A meaningful transition zone is declared.

## L.7.4 Gradient

Pressure accumulates across or around that boundary.

## L.7.5 Trigger

A crossing, catalyst, or threshold initiates motion.

## L.7.6 Guidance

Follow-through, flow, trend, or policy routes the motion.

## L.7.7 Exchange

Transactions, participation, or institutional transfer express the movement.

## L.7.8 Gate

A close, retest, settlement, or authoritative decision admits or rejects the event.

## L.7.9 New memory

The admitted trace and unresolved residual become future Load.

The cycle then repeats under a revised world.

---

# L.8 Worked Proto-Eight Breakout Crosswalk

Consider a breakout from a twelve-week range.

## Memory

The market retains:

- prior resistance;
- failed attempts;
- profile density;
- trapped positions.

## Focus

The analyst declares:

- the upper range boundary;
- daily timeframe;
- relevant breadth universe;
- breakout gate.

## Boundary

The resistance zone separates:

- prior value region;
- candidate higher region.

## Gradient

Pressure appears as:

- repeated upper-range closes;
- positive order imbalance;
- rising relative strength;
- compression beneath resistance.

## Trigger

Price crosses the zone appears as:

- repeated upper-range closes;
- positive order imbalance;
- rising relative strength;

- compression beneath resistance.

## Trigger

Price crosses the zone.

This is only ignition.

## Guidance

The move receives or fails to receive:

- follow-through;
- VWAP support;
- trend alignment;
- breadth expansion.

## Exchange

Volume and executed flow transfer ownership beyond the old boundary.

## Gate

The event is judged by:

- close;
- normalized displacement;
- participation;
- retest;
- higher-frame acceptance.

## New memory

If admitted, the old resistance may become:

- new support;
- new volume density;
- event history;
- reference for later risk.

If rejected, the failure becomes:

- fakeout memory;
- trapped-position residual;

- stronger future boundary.

The same market episode can therefore produce different Memory_{k+1} depending on the Gate outcome.

---

# L.9 Proto-Eight and Confirmation Independence

Proto-Eight also clarifies why several indicators may fail to provide independent confirmation.

Suppose a breakout analysis uses:

- price above moving average;
- MACD positive;
- RSI above 50;
- momentum positive.

These methods mostly observe:

- Memory;
- Gradient;
- Guidance;

through one price source.

The analysis may still lack:

- Exchange;
- Gate;
- independent Boundary;
- residual control.

A more complete eight-role bundle might include:

| Role | Evidence |
|---|---|
| Memory | prior range and profile |
| Focus | declared event protocol |
| Boundary | predeclared resistance zone |
| Gradient | normalized pressure |
| Trigger | actual crossing |
| Guidance | follow-through and route stability |
| Exchange | volume and breadth |
| Gate | close and retest |

The principle is:

ConfirmationCompleteness
≠ IndicatorCount. (L.17)

A stronger confirmation bundle covers missing operational roles.

---

# L.10 Proto-Eight and χ

Proto-Eight roles behave differently across χ regimes.

## Corrective regime: χ < 0

Typical pattern:

Gradient
→ Trigger
→ Boundary response
→ opposing Exchange
→ Gate rejection
→ Memory of reversion. (L.18)

## Critical regime: χ ≈ 0

Typical pattern:

Gradient accumulates, but Guidance and Gate remain unstable.

The system may alternate among:

- failed triggers;
- partial exchanges;
- uncertain boundaries;
- weak commitments.

## Self-confirming regime: χ > 0

Typical pattern:

Gradient
→ Trigger
→ Guidance
→ reinforcing Exchange
→ Gate admission
→ stronger Memory and future Gradient. (L.19)

The eight roles therefore help explain χ operationally.

χ is not an additional ninth primitive.

It describes the relational orientation through which the eight roles couple.

---

# L.11 Proto-Eight and Ξ

The effective control state:

$\Xi = (\rho,\gamma,\nu)$ (L.20)

may also be interpreted through Proto-Eight.

## Loading ρ

Primarily influenced by:

- Gradient;
- Memory;
- Exchange.

## Lock-in γ

Primarily influenced by:

- Boundary;
- Gate;
- Memory.

## Agitation ν

Primarily influenced by:

- Trigger;
- Exchange;
- unstable Guidance.

A tentative compilation is:

$\rho = C_\rho(\text{Gradient},\text{Memory},\text{Exchange})$. (L.21)

$\gamma = C_\gamma(\text{Boundary},\text{Gate},\text{Memory})$. (L.22)

$\nu = C_\nu(\text{Trigger},\text{Exchange},\text{GuidanceFailure})$. (L.23)

These are research constructions.

They do not appear as established equations in the P8D source.

They show how the actuation grammar might supply interpretable inputs to the effective-state interface.

---

# L.12 Proto-Eight and 成界之學

## L.12.1 Two different theoretical jobs

Proto-Eight and 成界之學 should not be treated as interchangeable.

Proto-Eight explains:

> What primitive functions allow a system to generate, route, exchange, retain, and select movement?

成界之學 explains:

> How does an open field of possibility become a bounded, declared, gated, traced, residual-bearing, and time-carrying world?

The latter source describes a progression from an undeclared possibility field through:

- declaration;
- filtration;
- projection;
- gate;
- trace;
- residual;
- ledgered revision.

Its central concern is not merely producing motion but making events become history, history become order, and order become a world. filecite turn32file15

The relationship is therefore:

ProtoEight = ActuationGrammar. (L.24)

PeriodicTable = ObservationAndClosureTaxonomy. (L.25)

成界之學 = WorldFormationGrammar. (L.26)

---

# L.12.2 Combined runtime

A combined architecture is:

UndeclaredField $\Sigma_0$
→ DeclareProtocol P
→ SelectFocus
→ ActivateProtoEightRoles
→ GenerateCandidateMovement
→ ProjectIntoPeriodicCells
→ ApplyGate
→ WriteTrace + Residual
→ UpdateLedger
→ TestInvariance
→ ReviseDeclaration. (L.27)

In compact form:

Field
→ Actuation
→ Observation
→ Commitment
→ History
→ World. (L.28)

Proto-Eight supplies neither the whole observer nor the whole world.

It supplies the actuation vocabulary inside the larger world-forming runtime.

---

# L.13 Necessary but Not Sufficient

The Chinese comparison source concludes that the structural logic associated with 先天八卦 is a **necessary but not sufficient** condition for recursively generating an ordered and prosperous world.

Its stated reason is that the eightfold topolognme and recursion;

- observer compatibility;
- projection;
- gate;
- trace;
- residual governance;
- environmental coupling. filecite turn32file7 filecite turn32file18

In symbolic form:

ProtoEight alone
≠ RichWorld. (L.29)

A broader condition is:

RichWorld_P
= Seed

- ActuationGrammar
- Energy
- Recursion
- Observer
- Gate
- Ledger
- ResidualGovernance
- Invariance. (L.30)

This expresses the source's internal conclusion.

However, the stronger statement that the eight roles are universally necessary for every market, organization, or world has not yet been empirically established.

The scientifically cautious formulation for this article is:

> Proto-Eight is a candidate minimal actuation repertoire whose necessity should be tested by ablation, comparison, and cross-domain transport.

This preserves the source's theoretical proposition while distinguishing it from demonstrated empirical law.

---

# L.14 The Necessity Test

For a target task Y, define the full eight-role model:

$M_8 = f(Gd,Gt,B,E,T,Gn,M,F)$. (L.31)

where:

Gd = Gradient;
Gt = Gate;
B = Boundary;
E = Exchange;
T = Trigger;
Gn = Guidance;
M = Memory;
F = Focus.

Remove one role at a time:

$M_{-a} = M_8$ without role a. (L.32)

The incremental contribution of role a is:

$\Delta_a = Loss(M_{-a}) - Loss(M_8)$. (L.33)

A role is empirically necessary for the declared task only if:

$\Delta_a > \varepsilon_a$ (L.34)

across:

- out-of-sample periods;
- admissible protocols;
- relevant markets;
- comparable simpler models.

The eight roles may be necessary for one task but not another.

For example:

- Focus may be essential for observer interpretation;

- it may not be required in a narrow automated execution model;
- World-level analysis may require Memory and Gate;
- one Mark-level prediction may not.

Necessity must therefore remain protocol-relative:

Necessary(a | Task,P). (L.35)

---

# L.15 Dyad-Synergy Tests

The four dyads generate specific testable hypotheses.

# L.15.1 Gradient × Gate

Hypothesis:

A gradient predicts realized throughput more strongly when an effective gate is available.

Model:

Flow
$= \beta_0 + \beta_1 \text{Gradient} + \beta_2 \text{Gate}$

- $\beta_3 \text{Gradient} \times \text{Gate} + \varepsilon$. (L.36)

Prediction:

$\beta_3 > 0$. (L.37)

---

# L.15.2 Boundary × Exchange

Hypothesis:

Boundary significance depends on actual exchange and transfer around it.

Model:

BoundaryPersistence
$= \gamma_0 + \gamma_1 \text{BoundaryMass} + \gamma_2 \text{Exchange}$

- $\gamma_3 \text{BoundaryMass} \times \text{Exchange} + \varepsilon$. (L.38)

A boundary with no current exchange may be stale.

Exchange with no boundary may be unstructured churn.

---

# L.15.3 Trigger × Guidance

Hypothesis:

Trigger events persist more often when Guidance is coherent.

Model:

$$\text{EventPersistence} = \delta_0 + \delta_1 \text{TriggerStrength} + \delta_2 \text{Guidance}$$

- $\delta_3 \text{TriggerStrength} \times \text{Guidance} + \varepsilon$. (L.39)

Prediction:

$\delta_3 > 0$. (L.40)

This is directly relevant to distinguishing breakout attempts from sustained transitions.

---

# L.15.4 Memory × Focus

Hypothesis:

Retained information improves decisions only when the observation lens selects relevant history.

Model:

$$\text{DecisionQuality} = \eta_0 + \eta_1 \text{MemoryDepth} + \eta_2 \text{FocusQuality}$$

- $\eta_3 \text{MemoryDepth} \times \text{FocusQuality} + \varepsilon$. (L.41)

Too little Memory creates amnesia.

Too much unfiltered Memory creates overload.

---

# L.16 Failure-Signature Predictions

Each missing or imbalanced role predicts a characteristic failure.

| Missing or excessive role | Predicted failure |
|---|---|
| Gradient absent | no reason for directional flow |
| Gradient excessive, Gate weak | pressure, overshoot, unstable flow |
| Gate excessive | ossification and false rejection |
| Boundary absent | leakage and poor attribution |
| Boundary excessive | isolation and trapped pressure |
| Exchange absent | no discovery or transfer |
| Exchange excessive | churn, contagion, noise |
| Trigger absent | latent structure without activation |
| Trigger excessive | spike-and-crash, fatigue |
| Guidance absent | misrouting and failed continuation |
| Guidance excessive | rigid routing and adaptation failure |
| Memory absent | repeated mistakes and no compounding |
| Memory excessive | stale path dependence |
| Focus absent | analytic overload |
| Focus excessive | monoculture and blindness |

These predictions make the crosswalk falsifiable.

If missing-role diagnoses do not correspond to distinctive failure modes, the eightfold decomposition loses value.

---

# L.17 P8D Growth Variables and Financial Translation

The small P8D model emphasizes several ordinary growth variables:

- capacity;
- demand;
- fit;
- enablement;
- retention;
- buffer;
- friction;
- throughput.

Its central practical claim is that sustainable growth requires more tor capacity. It depends on matching both sides, preserving retention and buffers, reducing harmful friction, and reinvesting successful throughput. filecite turn34file0

A financial-market translation is:

| P8D growth variable | Market interpretation |
|---|---|
| Capacity | liquidity, capital, market-making ability, production capacity |
| Demand | willing buyers, investment demand, reachable capital |
| Fit | compatibility between asset, mandate, valuation, and investor need |
| Enablement | market access, credit, settlement, trust, regulation |
| Retention | position persistence, investor holding, recurring liquidity |
| Buffer | cash, collateral, inventory, capital headroom |
| Friction | spread, transaction cost, legal barrier, latency |
| Throughput | executed volume, financed activity, realized transaction flow |

This does not mean the original organizational-growth equations can be copied into market analysis without recalibration.

It means the actuation vocabulary supplies candidate variables for a market-flow model.

---

# L.18 Proto-Eight Does Not Replace the Four-Family Table

It may appear that eight roles should replace four functional families.

That would be a category mistake.

The four families are deliberately broader.

For example:

- Memory is one primitive actuation role;
- Load is the observational family containing any consequentially carried structure.

Load may include:

- Memory;
- gradient potential;
- position inventory;
- institutional obligation.

Similarly:

- Gate is one primitive;
- Commitment is the broader observational status of something becoming accepted trace.

Commitment may be implemented by:

- Gate;
- Exchange;
- Trigger plus authority;
- multi-stage recognition.

Therefore:

ProtoEight refines the operational mechanism. (L.42)

FourFamilyGrammar preserves the higher-level observational classification. (L.43)

---

## L.19 Proto-Eight Does Not Replace χ, Ξ, or Z

The roles must also remain distinct from the article's advanced coordinates.

## Proto-Eight

What primitive operation is active?

### χ

What is the feedback orientation among active operations?

### Ξ

What effective loading, lock-in, and agitation state has been compiled?

## Z = R + iQ

Does an eligible conjugate state exist?

### $\tau_i$

Does phase define useful internal ordering?

The complete hierarchy is:

ProtoEightRoles
→ CouplingSignature χ
→ EffectiveState Ξ
→ OptionalComplexState Z
→ OptionalPhaseTime $\tau_i$
→ Gate
→ Ledger. (L.44)

None of the later layers follows automatically from the earlier one.

---

# L.20 Eightfold Roles and Complex Eligibility

The Proto-Eight octet does not imply eight complex dimensions.

The roles are functional categories.

They do not automatically form:

- four complex numbers;
- an eight-dimensional Hilbert space;
- a Clifford algebra;
- a physical quantum state.

A candidate conjugate pairing must still satisfy the criteria in Appendix F.

For example, one might hypothesize:

$Z_1$ = Gradient + iGatePressure. (L.45)

$Z_2$ = Boundary + iExchangePressure. (L.46)

$Z_3$ = Trigger + iGuidance. (L.47)

$Z_4$ = Memory + iFocus. (L.48)

But these expressions are not yet justified.

Many of the paired quantities:

- have different units;
- are not orthogonal;

- may not support rotational dynamics;
- may be control complements rather than mathematical conjugates.

The four dyads are operational complements.

They should not be called complex conjugate pairs without additional evidence.

---

# L.21 Eightfold Roles and the Primordial BaGua

The engineering sources map:

| Trigram | Proto-Eight role |
|---|---|
| 乾 | Gradient |
| 坤 | Gate |
| 艮 | Boundary nce |
| 坎 | Memory |
| 離 | Focus |

This mapping is the declared engineering interpretation used by the Proto-Eight series. filecite turn33file6 filecite turn33file11

The present article adopts the role names as a systems vocabulary.

It does not independently establish that:

- this is the only valid interpretation of the classical trigrams;
- the ancient texts encoded modern market microstructure;
- the mapping is a historically intended scientific theory;
- all eight roles are universal natural laws.

The legitimate claim is narrower:

> The eightfold engineering mapping provides a compact and potentially testable actuation grammar that can be cross-referenced with Technical Analysis.

Historical, philological, and classical philosophical claims require separate scholarship.

---

# L.22 The Strong and Weak Interpretations

## Weak interpretation

Proto-Eight is a useful checklist of system functions.

This requires only practical usefulness.

## Intermediate interpretation

The four dyads recur across multiple markets and improve diagnosis.

This requires cross-domain evidence.

## Strong interpretation

The octet is a minimal necessary actuation grammar for self-organizing systems.

This requires ablation and universality tests.

## Strongest interpretation

The Primordial BaGua is the uniquely optimal topology for geneds.

This stronger exclusivity claim appears rhetorically in parts of the Chinese source discussion, but it is not empirically demonstrated there. filecite turn32file18

The present article does not rely on that strongest proposition.

Its research claim is:

CandidateMinimality(P8D) remains testable. (L.49)

---

# L.23 Full Integrated Architecture

The article's complete architecture can now be written as four nested grammars.

## Layer 1 — Closure depth

$P_6$
= {Mark,Window,Structure,Event,Episode,World}. (L.50)

## Layer 2 — Observational function

$G_4$
= {Load,Motion,Constraint,Commitment}. (L.51)

## Layer 3 — Actuation role

$A_8$
= {Gradient,Gate,Boundary,Exchange,Trigger,Guidance,Memory,Focus}. (L.52)

## Layer 4 — Governance rails

$R_3$
= {Residual,Transport,Ledger}. (L.53)

A protocol-bound market object can therefore be indexed:

X_{p,g,a}^{P}. (L.54)

where:

p = closure period;
g = observational function;
a = actuation role;
P = declared protocol.

The notation does not imply that all:

$$6 \times 4 \times 8 = 192 \quad \text{(L.55)}$$

combinations are distinct or empirically necessary.

It is a typing system.

Its purpose is to prevent one object from being described ambiguously.

---

# L.24 Example of Full Typing

Consider a daily breakout close.

Its full type may be:

$$X_{Event,Commitment,Gate}^{P_{daily}}. \quad \text{(L.56)}$$

The volume around the breakout may be:

$$X_{Event,Load,Exchange}^{P_{daily}}. \quad \text{(L.57)}$$

The resistance zone may be:

$$X_{Structure,Constraint,Boundary}^{P_{daily}}. \quad \text{(L.58)}$$

The prior failed attempts may be:

$$X_{Episode,Load,Memory}^{P_{daily}}. \quad \text{(L.59)}$$

The chosen twelve-week resistance window may be:

$$X_{Structure,Constraint,Focus}^{P_{daily}}. \quad \text{(L.60)}$$

The same market episode is not one cell.

It is a compound of typed objects.

---

# L.25 Proto-Eight Method Card Extension

The Method Card from Appendix C can be expanded:

```
Method:
Protocol:
Closure period:
Observational family:
Proto-Eight role:
Supporting dyad:
Source lineage:
Operator word:
Regime assumption:
Gate:
```

```
Residual:
Transport:
Invalidation:
```

Example:

```
Method:
Breakout close.

Closure period:
Event.

Observational family:
Commitment.

Proto-Eight role:
Gate.

Supporting dyads:
Gradient × Gate;
Boundary × Exchange;
Trigger × Guidance.

Source lineage:
Price, volume, breadth, boundary history.

Residual:
Higher-frame resistance, no retest, crowding.

Invalidation:
Close inside old region plus failed reclaim.
```

This makes the method's actuation dependencies visible.

---

# L.26 Research Programme

# L.26.1 Classification reliability

Can independent coders assign the eight roles consistently?

Measure:

- exact agreement;
- dyad agreement;
- role confusion;
- overlap with four-family classification.

---

# L.26.2 Incremental explanatory value

Compare:

$\text{Model}_4$ = period + four-family features. (L.61)

$\text{Model}_{4+8}$ = period + four-family + Proto-Eight roles. (L.62)

Test:

$\text{Loss}(\text{Model}_{4+8}) < \text{Loss}(\text{Model}_4)$. (L.63)

If not, the eight-role refinement may be unnecessary for that task.

---

## L.26.3 Dyad interaction

Test the four interaction hypotheses from Section L.15.

The dyads gain status when their interaction terms replicate across markets.

---

## L.26.4 Missing-role diagnosis

For failed events, ask whether one or more roles were absent.

Examples:

- Trigger without Guidance;
- Boundary without Exchange;
- Gradient without Gate;
- Memory without Focus.

Test whether these labels predict distinctive failures.

---

## L.26.5 Cross-period recurrence

Determine whether each role reappears reliably from Mark to World.

If a role has no meaningful interpretation at several periods, the claim of periodic recurrence weakens.

---

## L.26.6 Cross-domain transport

Compare:

- equity;
- credit;
- derivatives;
- organizational finance;
- accounting;
- legal recognition;
- policy.

The role definitions should remain stable even when their observables change.

---

# L.27 Falsification Conditions

The Proto-Eight crosswalk should be revised or rejected if:

1. coders cannot distinguish the roles reliably;
2. role assignments merely restate ordinary indicator categories;
3. dyad interactions add no empirical value;
4. missing-role diagnoses do not predict characteristic failures;
5. the roles do not recur across periods;
6. a smaller role set performs equally well;
7. the classical labels add confusion without operational benefit;
8. the crosswalk cannot survive protocol changes.

A successful engineering metaphor is not exempt from falsification.

---

# L.28 Compact Integrated Runtime

The whole article can now be condensed into one extended runtime:

Declare_P
→ Focus relevant Memory
→ Locate Boundary
→ Estimate Gradient
→ Observe Trigger
→ Diagnose Guidance
→ Measure Exchange
→ Apply Gate
→ Write Trace + Residual
→ Update Period
→ Test Transport
→ Diagnose χ and Ξ
→ Construct Z only if eligible
→ Revise Admissibly. (L.64)

This runtime combines:

- Proto-Eight actuation;
- periodic closure;
- gate and ledger;
- residual governance;
- complex-eligibility discipline.

---

# L.29 Appendix L Conclusion

Proto-Eight Dynamics does not replace the Periodic Grammar.

It explains what moves inside it.

The six periods answer:

> At what depth has closure occurred?

The four families answer:

> What functional contribution does the observation make?

The eight roles answer:

> Through what primitive operation does the system generate, route, transfer, retain, select, or commit change?

成界之學 answers:

> How do those operations become a declared, traced, residual-bearing, and time-carrying world?

The integrated hierarchy is:

Proto-Eight generates actuation. (L.65)

The four families classify observation. (L.66)

The six periods classify closure depth. (L.67)

The gate creates commitment. (L.68)

The residual preserves non-closure. (L.69)

The ledger creates consequential history. (L.70)

Transport tests objectivity. (L.71)

Admissible revision preserves learning. (L.72)

The resulting architecture is:

Potential
→ Directed Actuation
→ Bounded Observation
→ Gated Commitment
→ Residual-Bearing Trace
→ Recursive Market World. (L.73)

The Proto-Eight contribution can therefore be stated precisely:

> It supplies an eight-role actuation grammar nested inside the article's six-period, four-family, ledger-governed architecture—without requiring the trigrams to be treated as chronological stages, literal physical entities, or already-proven universal laws.

# Appendix M — Empty Cells and the Instrument-Design Laboratory

## M.1 Purpose

A periodic table becomes scientifically useful when it does more than classify existing objects.

It should also reveal:

- overcrowded regions;
- redundant instruments;
- weakly measured functions;
- missing transitions;
- candidate new instrument families.

Traditional Technical Analysis has developed many instruments for Structure-level Motion:

- momentum;
- RSI;
- stochastic;
- MACD;
- rate of change;
- divergence.

It has also developed many Structure-level Constraints:

- support;
- resistance;
- channels;
- volatility bands;
- Fibonacci zones.

By comparison, it has fewer mature instruments for:

- residual-bearing commitment;
- cross-frame transport;
- episode progress;
- episode completion;
- observer crowding;
- institutional ledger reconciliation.

This imbalance helps explain why Technical Analysis is often rich in signals but poor in disciplined closure.

The source Technical Analysis framework already asks what each indicator measures, what it fails to measure, which independent method could observe the missing characteristic, what residual remains, and what would invalidate the claim.

The present appendix extends that logic:

> A missing measurement function should be treated as an instrument-design problem rather than filled by interpretive intuition.

---

# M.2 Existing Instrument Versus Missing Instrument

Let the proto-periodic table be:

$\mathcal{T} = P_6 \times G_4$. (M.1)

where:

$P_6$
= {Mark,Window,Structure,Event,Episode,World}. (M.2)

$G_4$
= {Load,Motion,Constraint,Commitment}. (M.3)

For each cell E_{p,g}, define instrument maturity:

$\mu_{p,g} \in [0,1]$. (M.4)

where:

$\mu \approx 0 \rightarrow$ almost no operational instrument;
$\mu \approx 0.5 \rightarrow$ proxies exist but closure is weak;
$\mu \approx 1 \rightarrow$ mature, reproducible instrument family.

Instrument maturity is not the same as popularity.

A popular indicator may still have:

- vague units;
- weak gates;
- unstable transport;
- poor falsifiability.

A less familiar instrument may be mature if it possesses:

- declared variables;
- reliable measurement;
- known error;
- clear scope;

- prospective validation.

---

# M.3 Provisional Maturity Map

| Period | Load | Motion | Constraint | Commitment |
|---|---|---|---|---|
| **Mark** | High | High | High | High |
| **Window** | High | High | High | High |
| **Structure** | High | Very high | Very high | Medium |
| **Event** | Medium | High | High | Medium–high |
| **Episode** | Low–medium | Medium | Medium | Low |
| **World** | Medium outside TA | Low inside TA | Medium outside TA | Low inside TA |

This table is a research judgement, not a measured result.

Its main message is qualitative:

> Technical Analysis becomes progressively less instrumentally complete as it moves from visible chart structure toward episode closure and institutional world formation.

The weakest regions are not necessarily those with the fewest names.

Elliott Wave, cycle analysis, and Gann provide many names for Episode-level objects.

Their difficulty is that the relevant:

- pivot;
- branch;
- gate;
- residual;
- transport;
- completion condition

are often insufficiently controlled.

---

# M.4 Instrument Design Principles

A candidate instrument should satisfy ten requirements.

# M.4.1 Declared protocol

The instrument must state:

$$P = (\text{Asset},\text{Boundary},\text{Timeframe},\text{Scale},\text{BarRule},\text{FeatureMap},\text{GateRule},\text{ResidualRule}). \quad (M.5)$$

The declaration framework requires boundary, observation rule, horizon, intervention family,

baseline, feature map, gate, trace, and residual before a field becomes auditable.

---

## M.4.2 Typed location

The instrument must identify:

- closure period;
- primary functional family;
- Proto-Eight actuation role.

A candidate object may be written:

Instrument_j
∈ E_{p,g,a}^{P}. (M.6)

---

## M.4.3 Measurable input

The instrument must use data that can be reconstructed.

---

## M.4.4 Defined output

The output must possess:

- units;
- range;
- direction;
- uncertainty.

---

## M.4.5 Distinct informational role

It should not merely reproduce an existing indicator under a new name.

---

## M.4.6 Gate relation

The instrument should state whether it:

- diagnoses a candidate state;
- strengthens a gate;
- constitutes a gate;
- audits a completed gate.

---

## M.4.7 Residual output

Every instrument should report what remains unresolved.

---

## M.4.8 Transport requirement

Its output should be tested under relevant frame changes.

---

## M.4.9 Prospective falsification

The instrument should be able to fail.

---

## M.4.10 Simpler benchmark

It must be compared with the simplest plausible alternative.

The design law is:

NewInstrumentValue
= MissingFunctionCoverage

- IncrementalInformation
- GateUtility
- TransportRobustness
  – ComplexityCost. (M.7)

---

## M.5 The Instrument Design Card

```
Instrument name:

Target table cell:
Closure period:
Primary functional family:
Proto-Eight role:

Scientific task:
Input data:
Source lineage:
Operator word:
Output:
Units:
Uncertainty:

Gate relationship:
Residual output:
Transport tests:
Benchmark:
Invalidation:
Prospective outcome:
Reduction rule:
```

The Proto-Eight engineering source already proposes operational experiments organized around gradient, gate, boundary, exchange, trigger, guidance, memory, and focus, with short-cycle logs, explicit controls, failure smells, stop rules, and measurable KPIs.

The present design card translates that experiment discipline into market-observation instruments.

---

# M.6 Candidate Instrument 1 — Unexecuted Intention Field

## M.6.1 Target cell

**Mark × Load**

## M.6.2 Missing problem

Ordinary trade data records executed transactions.

It does not fully record:

- cancelled orders;
- withdrawn depth;
- hidden liquidity;
- rejected marketable interest;
- repeated non-executed intention.

Yet these non-executed states may shape later price behaviour.

---

## M.6.3 Proposed object

Define the Unexecuted Intention Field:

UIF_t(p)
= SubmittedLiquidity_t(p)
− ExecutedLiquidity_t(p)
− PersistentDisplayedLiquidity_t(p). (M.8)

A simplified mark-level scalar is:

UIF_t
= Cancellations_t

- Rejections_t
- UnfilledMarketableQuantity_t
- WithdrawnDepth_t. (M.9)

The quantities should be normalized by available depth or ordinary event-time activity.

## M.6.4 Proto-Eight roles

Primary:

- Memory;
- Gradient.

Secondary:

- Gate;
- Exchange.

---

## M.6.5 Interpretation

High UIF may indicate:

- frustrated demand;
- spoof-like transient interest;
- liquidity avoidance;
- adverse-selection fear;
- failed urgency;
- hidden pressure.

It is not intrinsically bullish or bearish.

---

## M.6.6 Required decomposition

Separate:

UIF_buy. (M.10)

UIF_sell. (M.11)

UIF_cancel. (M.12)

UIF_reject. (M.13)

UIF_hidden_estimate. (M.14)

A single aggregated score may conceal opposing mechanisms.

---

## M.6.7 Gate relation

UIF does not constitute commitment.

It measures load that failed to become execution.

---

## M.6.8 Main residual

- participant identity;
- strategic cancellation;
- cross-venue routing;
- hidden order type;
- latency.

---

## M.6.9 Falsification

The instrument loses value if UIF adds no information about:

- near-term liquidity movement;
- fill probability;
- spread change;
- later execution;
- price response

beyond ordinary order-flow imbalance.

---

# M.7 Candidate Instrument 2 — Path-Bearing Candle

## M.7.1 Target cell

**Window × Motion**

## M.7.2 Missing problem

An OHLC candle preserves:

- opening price;
- high;
- low;
- close.

It does not preserve:

- whether the high preceded the low;

- time spent near each region;
- number of reversals;
- path efficiency;
- intrawindow commitment sequence.

Two windows can have identical OHLC values but very different internal histories.

---

# M.7.3 Proposed object

Define the Path-Bearing Candle:

PBC_t
= (O,H,L,C,V,π_t,η_t,n_rev,t_H,t_L). (M.15)

where:

π_t = signed path sequence;
η_t = path efficiency;
n_rev = reversal count;
t_H = time of high;
t_L = time of low.

---

# M.7.4 Path efficiency

Let the total travelled distance be:

D_path
= Σ_{k=1}^{N}|P_k − P_{k−1}|. (M.16)

Let net displacement be:

D_net = |C − O|. (M.17)

Define:

η_path = D_net/(D_path + ε). (M.18)

Then:

η_path ≈ 1 → efficient directional movement.
η_path ≈ 0 → high internal rotation or churn.

---

# M.7.5 Directional path ordering

Define:

π_HL = sign(t_L − t_H). (M.19)

Then:

$\pi_{HL} > 0 \rightarrow$ high occurred before low.
$\pi_{HL} < 0 \rightarrow$ low occurred before high.

The same candle shape can therefore be divided into distinct path classes.

---

## M.7.6 Proto-Eight roles

- Trigger;
- Guidance;
- Exchange;
- Memory.

---

## M.7.7 Gate relation

The close remains the Window gate.

The PBC explains how that gate was reached.

---

## M.7.8 Benchmark

Compare with:

- OHLCV alone;
- candle-body ratio;
- close-location value;
- realized volatility.

---

## M.7.9 Falsification

The extra path variables should be rejected if they add no value for:

- next-window persistence;
- fakeout;
- retest quality;
- liquidity classification;
- event-gate confidence.

---

## M.8 Candidate Instrument 3 — Structural Base Completeness Score

# M.8.1 Target cell

**Structure × Load**

# M.8.2 Missing problem

Chart analysts often describe a base as:

- mature;
- underdeveloped;
- accumulated;
- incomplete;
- ready to break.

These descriptions usually lack a standard measurement model.

---

# M.8.3 Proposed dimensions

Define base completeness through five channels:

1. duration;
2. participation;
3. value density;
4. volatility compression;
5. failed-boundary history.

Let:

$B_{duration} \in [0,1]$. (M.20)

$B_{density} \in [0,1]$. (M.21)

$B_{participation} \in [0,1]$. (M.22)

$B_{compression} \in [0,1]$. (M.23)

$B_{boundary} \in [0,1]$. (M.24)

Then:

$$BCS = w_1 B_{duration}$$

- $w_2 B_{density}$
- $w_3 B_{participation}$
- $w_4 B_{compression}$

- $w_5 B$_boundary. (M.25)

The weights must be calibrated prospectively.

---

## M.8.4 Interpretation

BCS measures how much structured load has accumulated inside a declared region.

It does not determine breakout direction.

---

## M.8.5 Proto-Eight roles

- Memory;
- Boundary;
- Exchange;
- Focus.

---

## M.8.6 Gate relation

A high BCS identifies a loaded Structure.

A later Trigger and Gate are still required.

---

## M.8.7 Residual

- hidden positioning;
- stale participants;
- options exposure;
- external catalyst;
- range-selection subjectivity.

---

## M.8.8 Benchmark

Compare with:

- range duration alone;
- Bollinger Band width;
- ATR compression;
- volume profile alone.

---

## M.8.9 Falsification

BCS fails if it does not improve prediction or classification of:

- boundary reaction;
- breakout magnitude;
- post-break value migration;
- false-break probability.

---

# M.9 Candidate Instrument 4 — Structural Acceptance Score

## M.9.1 Target cell

**Structure × Commitment**

## M.9.2 Missing problem

Technical Analysis contains many ways to identify a structure.

It contains fewer disciplined ways to say when a structure has become sufficiently persistent to organize later interpretation.

Examples include:

- accepted uptrend;
- accepted value migration;
- established relative-strength regime;
- confirmed support.

These phrases are often used without a common closure rule.

---

## M.9.3 Proposed score

Let:

$P_{persist}$ = persistence of relation.
$P_{transport}$ = cross-frame survival.
$P_{density}$ = compatible value formation.
$P_{participation}$ = independent participation.
$P_{residual}$ = unresolved contradiction.

Define:

SAS
= $w_1 P_{persist}$

- $w_2 P_transport$
- $w_3 P_density$
- $w_4 P_participation$
  – $w_5 P_residual$. (M.26)

---

## M.9.4 Acceptance states

SAS may produce:

- Candidate;
- Provisional;
- Accepted;
- Degraded;
- Invalidated.

This state vocabulary is more informative than a permanent binary label.

---

## M.9.5 Proto-Eight roles

- Gate;
- Memory;
- Focus;
- Guidance.

---

## M.9.6 Distinction from breakout gate

A breakout gate admits one Event.

SAS determines whether the resulting relation has persisted as Structure.

Therefore:

EventAdmission ≠ StructuralAcceptance. (M.27)

A breakout may occur without establishing durable support.

---

## M.9.7 Falsification

SAS should be rejected if an ordinary persistence rule performs equally well.

---

# M.10 Candidate Instrument 5 — Pre-Gate Commitment

# Density

## M.10.1 Target cell

**Event × Load**

## M.10.2 Missing problem

Before a visible event, market participants may already have committed through:

- positioning;
- open interest;
- option structures;
- repeated tests;
- anticipatory volume;
- public narrative;
- stop placement.

This pre-event load affects both:

- the probability of transition;
- the severity of failure.

---

## M.10.3 Proposed state

Define:

PGCD
= $w_1$Positioning

- $w_2$OpenInterest
- $w_3$BoundaryVolume
- $w_4$RepeatedTests
- $w_5$Attention
  – $w_6$ExitCapacity. (M.28)

Each component must be normalized under the protocol.

---

## M.10.4 Interpretation

High PGCD may produce:

### Productive commitment

Enough participation exists to support transition.

### Crowded fragility

Too many participants depend on the same transition.

Therefore the relationship may be nonlinear:

$$\text{EventPersistence} = \alpha + \beta_1 \text{PGCD} + \beta_2 \text{PGCD}^2 + \varepsilon. \quad \text{(M.29)}$$

A possible pattern is:

$$\beta_1 > 0. \quad \text{(M.30)}$$

$$\beta_2 < 0. \quad \text{(M.31)}$$

Moderate commitment may help.

Extreme commitment may create fragility.

---

## M.10.5 Proto-Eight roles

- Gradient;
- Memory;
- Exchange;
- Focus.

---

## M.10.6 Gate relation

PGCD measures event load before the gate.

It must not be confused with gate success.

---

## M.10.7 Falsification

PGCD fails if it does not improve:

- event persistence;
- fakeout prediction;
- post-failure reversal magnitude;
- trapped-position estimates.

---

# M.11 Candidate Instrument 6 — Residual-Bearing Gate Score

## M.11.1 Target cell

**Event × Commitment**

## M.11.2 Missing problem

Most confirmation scores add positive evidence.

They often fail to preserve contradiction as a first-class output.

A score of 8/10 does not show whether the missing two points were:

- unimportant;
- pending;
- directly contradictory;
- high-severity higher-frame resistance.

---

## M.11.3 Two-output design

The proposed gate should output:

RBG_t = (A_t,R_t). (M.32)

where:

A_t = admission strength;
R_t = residual-burden vector.

The scalar admission score is:

A_t
= w_CC

- w_VV
- w_BB
- w_TT
- w_FF
- w_PP. (M.33)

The residual vector is:

R_t
= (r_frame,r_liquidity,r_positioning,r_boundary,r_regime,r_institution). (M.34)

---

## M.11.4 Admission matrix

| Admission Residual | Classification |
|---|---|

| Admission | Residual | Classification |
|---|---|---|
| Low | Low | weak or irrelevant event |
| Low | High | unresolved failed candidate |
| High | Low | strongly admitted event |
| High | High | admitted but fragile event |

The final category is especially important.

A strong visible gate can coexist with high hidden fragility.

---

## M.11.5 Proto-Eight roles

- Gate;
- Boundary;
- Exchange;
- Memory.

---

## M.11.6 Advantage over one scalar

The two-output system prevents:

HighGateStrength → ResidualAssumedZero. (M.35)

The declaration framework explicitly requires trace and residual to be written together rather than treating commitment as total resolution.

---

## M.11.7 Falsification

The residual-bearing version must outperform:

- binary gate;
- scalar confirmation score;
- crossing-only rule

in calibration and failure diagnosis.

---

# M.12 Candidate Instrument 7 — Event Portability Index

## M.12.1 Target

**Event × Transport rail**

Transport is not one of the four table columns, but it governs every Event claim.

# M.12.2 Missing problem

Analysts often describe events as:

- confirmed on several timeframes;
- broadly supported;
- scale-independent;
- robust.

These descriptions usually lack a standardized transformation audit.

---

# M.12.3 Transport vector

Define:

$\mathcal{T}$_e
= (t_time,t_scale,t_vol,t_bar,t_breadth,t_profile,t_deriv,t_authority). (M.36)

Each component may be:

1 = survives;
0.5 = partially survives;
0 = fails;
NA = not applicable.

---

# M.12.4 Portability score

EPI
= Σ_jw_jt_j/Σ_jw_j. (M.37)

The scalar should always be published with the vector.

---

# M.12.5 Portability classes

## P0 — Frame-local

Survives only source protocol.

## P1 — Parameter-robust

Survives nearby settings.

## P2 — Protocol-family robust

Survives scale, bar, or timeframe variation.

## P3 — Cross-functional

Survives price, volume, breadth, or density translation.

### P4 — Cross-authority

Recognized by several relevant gates.

---

## M.12.6 Proto-Eight roles

- Guidance;
- Boundary;
- Gate;
- Focus.

---

## M.12.7 Falsification

EPI loses status if:

- coders cannot reproduce it;
- weights determine the conclusion;
- higher EPI does not predict broader event persistence;
- local methods are unfairly penalized.

The source Technical Analysis article defines mature practice through declared protocol, missing-variable audit, invariance tests, residual recording, and invalidation rather than visual confidence alone.

---

# M.13 Candidate Instrument 8 — Residual-History State

## M.13.1 Target cell

**Episode × Load**

## M.13.2 Missing problem

An episode is not determined only by successful events.

It also carries:

- failed breakouts;
- unrepaired divergences;
- unresolved higher-frame conflicts;
- trapped participants;

- abandoned narratives;
- delayed institutional gates.

Most episode classifiers ignore this residual history.

---

## M.13.3 Proposed state vector

Define:

RHS_k
= (N_failed,N_unresolved,S_residual,T_residual,C_branch,C_frame). (M.38)

where:

N_failed = number of failed gates;
N_unresolved = open residual count;
S_residual = severity-weighted burden;
T_residual = mean persistence;
C_branch = branch complexity;
C_frame = cross-frame contradiction.

---

## M.13.4 Decay operator

Residual history may decay:

RHS_{k+1}
= D_RRHS_k + r_{k+1} − q_{k+1}. (M.39)

where:

D_R = persistence operator;
r = new residual;
q = resolved residual.

---

## M.13.5 Interpretation

High RHS may indicate:

- unstable trend;
- repeated false closure;
- potential squeeze;
- unresolved structural debt;
- high revision probability.

It does not determine the direction of the next move.

---

## M.13.6 Proto-Eight roles

- Memory;
- Boundary;
- Gate;
- Focus.

---

## M.13.7 Falsification

RHS should be rejected if event history without residual labels performs equally well.

---

# M.14 Candidate Instrument 9 — Phase-Cadence Progress Meter

## M.14.1 Target cell

**Episode × Motion**

## M.14.2 Missing problem

Analysts often ask:

- Is the trend young or mature?
- Is the squeeze near release?
- Has the correction completed?
- Is the crisis entering recovery?

Calendar age alone is often inadequate.

Named pattern stages are often subjective.

---

## M.14.3 Conservative design

The instrument should begin as a real multivariate progress model:

PCM_t
= f(EventOrder_t,Cadence_t,VolatilityState_t,χ_t,ResidualHistory_t). (M.40)

It should not begin by assuming complex phase.

---

## M.14.4 Progress components

### Event depth

$\kappa_t = k_t/K_{expected}$. (M.41)

### Cadence depth

$c_t$ = accumulated event cadence relative to reference episodes.

### Selection depth

$\sigma_t$ = estimated reduction of admissible branches.

### Regime sequence

$\chi_{1:t}$ = ordered feedback-signature history.

The combined progress state is:

$\Pi_t = (\kappa_t, c_t, \sigma_t, \chi_{1:t})$. (M.42)

---

# M.14.5 Optional phase promotion

Only if a valid R–Q state exists may the model introduce:

$\theta_t = \operatorname{atan2}(Q_t, R_t)$. (M.43)

$\tau_{i,t} = \text{Unwrap } \theta_t$. (M.44)

Complex phase is therefore a possible refinement, not the instrument's starting assumption.

---

# M.14.6 Proto-Eight roles

- Trigger;
- Guidance;
- Memory;
- Focus.

---

# M.14.7 Falsification

The progress meter fails if it does not improve:

- episode alignment;
- gate timing;
- completion classification;
- cross-episode comparison

beyond calendar age and event count.

---

# M.15 Candidate Instrument 10 — Episode Completion Certificate

## M.15.1 Target cell

**Episode × Commitment**

## M.15.2 Missing problem

Episode endings are frequently identified retrospectively.

Examples include:

- “the bull trend ended there”;
- “Wave 5 completed at that high”;
- “the accumulation phase ended at the breakout”;
- “the crisis ended when volatility peaked.”

These statements often mix:

- extreme point;
- warning;
- gate;
- later confirmation.

---

## M.15.3 Certificate structure

Define:

ECC
= (OldGrammarFailure,NewGate,Persistence,Transport,Residual,BranchStatus). (M.45)

A certificate is issued only when:

OldGrammarFailure = true. (M.46)

NewGateAdmitted = true. (M.47)

MinimumPersistence = true. (M.48)

ResidualDeclared = true. (M.49)

---

## M.15.4 Certificate levels

### ECC-0 — Candidate endpoint

An extreme or warning exists.

### ECC-1 — Local endpoint

A lower-frame gate has failed.

### ECC-2 — Episode completion

The prior grammar is no longer operative.

### ECC-3 — Regime transition

A stable new grammar has formed.

### ECC-4 — World-recognized transition

Institutional ledgers also change.

---

## M.15.5 Example

A bearish divergence alone gives:

ECC-0. (M.50)

A daily structural break may give:

ECC-1. (M.51)

A weekly failed reclaim and persistent lower structure may give:

ECC-2 or ECC-3. (M.52)

A credit downgrade, covenant breach, or accounting impairment may promote the event toward:

ECC-4. (M.53)

---

## M.15.6 Proto-Eight roles

- Gate;
- Guidance;
- Memory;
- Boundary.

---

## M.15.7 Falsification

ECC fails if simpler episode segmentation produces:

- equal reliability;
- equal timing;
- equal transport;
- lower complexity.

---

# M.16 Candidate Instrument 11 — Observer Crowding and Reflexivity Index

## M.16.1 Target cell

**World × Motion**

## M.16.2 Missing problem

A signal's behaviour may change as more observers:

- notice it;
- publish it;
- automate it;
- allocate capital to it;
- place similar stops around it.

Traditional indicators usually treat the observer population as external.

But markets are self-referential.

Price becomes evidence, evidence changes orders, and orders change later price.

---

## M.16.3 Proposed components

Define:

OCRI
= $w_1$Visibility

- $w_2$CapitalLinked
- $w_3$ActionSimilarity
- $w_4$StopConcentration
- $w_5$NarrativeAlignment
  − $w_6$LiquidityCapacity. (M.54)

---

## M.16.4 Nonlinear hypothesis

Reflexive effect may be nonlinear:

$$\text{OutcomeEffect} = \beta_1 \text{OCRI} + \beta_2 \text{OCRI}^2. \quad \text{(M.55)}$$

Possible interpretation:

- low OCRI → signal too weakly adopted to affect price;
- moderate OCRI → self-confirmation;
- very high OCRI → crowding and fragility.

---

## M.16.5 Proto-Eight roles

- Focus;
- Guidance;
- Memory;
- Exchange.

---

## M.16.6 Data proxies

Possible proxies include:

- search intensity;
- media mentions;
- broker positioning;
- systematic strategy estimates;
- options concentration;
- short interest;
- fund holdings;
- social-platform activity.

These proxies are incomplete and noisy.

---

## M.16.7 Gate relation

OCRI is not an event gate.

It modifies the environment in which gates operate.

---

## M.16.8 Falsification

OCRI should be rejected if:

- adoption cannot be measured reliably;
- it adds no value beyond liquidity and positioning;
- claimed nonlinear effects do not replicate.

---

# M.17 Candidate Instrument 12 — Cross-Ledger Rigidity Map

## M.17.1 Target cell

**World × Constraint**

## M.17.2 Missing problem

One financial object may be simultaneously constrained by:

- market price;
- funding;
- collateral;
- accounting;
- law;
- regulation;
- portfolio mandate.

A position can appear liquid in the market frame while remaining locked in another ledger.

---

## M.17.3 Rigidity vector

Define:

Γ_world
= (γ_market,γ_funding,γ_collateral,γ_accounting,γ_legal,γ_policy). (M.56)

Each coordinate measures resistance to changing or exiting the state under that ledger.

---

## M.17.4 Cross-ledger coupling

Let:

C_{ij} = dependence of ledger i on ledger j. (M.57)

The rigidity network is:

$\mathcal{G}_\Gamma = (\text{Ledgers}, C)$. (M.58)

A local rigidity shock may propagate:

$\Delta\gamma_j = \Sigma_i C_{ij} \Delta\gamma_i + \varepsilon_j$. (M.59)

---

# M.17.5 Examples

### Market liquidity high, accounting rigidity high

A position can be sold, but selling may crystallize an impairment.

### Market liquidity high, legal rigidity high

Transfer is physically possible but contractually restricted.

### Market liquidity low, policy support high

Exit is difficult, but official facilities may reduce immediate failure.

---

# M.17.6 Proto-Eight roles

- Boundary;
- Gate;
- Memory;
- Exchange.

---

# M.17.7 Transport burden

The map must declare:

- entity boundary;
- jurisdiction;
- accounting regime;
- collateral rules;
- funding horizon;
- observer authority.

---

# M.17.8 Falsification

The map loses value if ordinary balance-sheet and liquidity variables fully explain the relevant outcomes.

---

# M.18 Candidate Instrument 13 — Ledger Reconciliation Gate

## M.18.1 Target cell

**World × Commitment**

## M.18.2 Missing problem

Market, accounting, legal, risk, and policy systems may recognize the same underlying condition at different times.

A price collapse may precede:

- impairment;
- default;
- covenant breach;
- legal judgment.

A policy decision may precede market repricing.

The system needs an instrument for distinguishing:

- broad recognition;
- delayed recognition;
- contested recognition.

---

## M.18.3 Recognition vector

Define:

$$\mathbb{G}_k = (g_{market}, g_{risk}, g_{accounting}, g_{legal}, g_{policy}). \quad (M.60)$$

Each coordinate may take:

0 = not recognized;
0.5 = provisional or partial;
1 = admitted.

---

## M.18.4 Reconciliation score

Let:

LRS_k
= Σ_jw_jg_{j,k}
− λDisagreement(𝔾_k). (M.61)

The weights depend on the claim.

For legal default:

w_legal is dominant.

For a short-term price event:

w_market may dominate.

---

# M.18.5 Gate classes

## Fragmented

Relevant ledgers disagree materially.

## Market-leading

Market recognition precedes institutional ledgers.

## Institution-leading

Official recognition precedes market acceptance.

## Converging

Several ledgers are moving toward the same state.

## Reconciled

The relevant authoritative ledgers agree.

## Reopened

A previous reconciliation has been challenged.

---

# M.18.6 Residual

Ledger reconciliation does not eliminate:

- appeal;
- settlement risk;
- political reversal;
- incomplete economic adjustment;
- causal uncertainty.

## M.18.7 Proto-Eight roles

- Gate;
- Boundary;
- Exchange;
- Memory.

---

## M.18.8 Falsification

The gate should be rejected if cross-ledger status does not improve:

- event timing;
- consequence prediction;
- institutional-risk diagnosis;
- backreaction analysis.

---

# M.19 Candidate Instrument 14 — Declaration Stability Index

## M.19.1 Target

**Governance rail across all periods**

## M.19.2 Missing problem

A technical claim may appear accurate because its protocol is repeatedly changed.

Examples include:

- moving the level;
- changing timeframe;
- changing benchmark;
- changing wave degree;
- changing outcome horizon.

A separate instrument should measure how stable the declaration remained.

---

## M.19.3 Protocol distance

Let $P_0$ be the original protocol and $P_k$ the later protocol.

Define:

$$D_P(P_0,P_k) = w_{Bd}D_B + w_{Td}D_T + w_{Sd}D_S + w_{Fd}D_F + w_{Gd}D_G + w_{Hd}D_H. \quad (M.62)$$

where the components measure changes in:

- boundary;
- timeframe;
- scale;
- feature map;
- gate;
- horizon.

---

## M.19.4 Stability score

$$DSI_k = \exp[-D_P(P_0,P_k)]. \quad (M.63)$$

Then:

$DSI \approx 1 \rightarrow$ declaration largely stable.
$DSI \approx 0 \rightarrow$ major protocol migration.

---

## M.19.5 Interpretation

A revised model may still be valid.

But its earlier outcomes should not be credited to the new protocol.

The DSI distinguishes:

- model learning;
- hindsight repair;
- silent protocol drift.

---

## M.19.6 Proto-Eight roles

- Focus;
- Boundary;
- Memory;
- Gate.

---

## M.19.7 Falsification

DSI should be rejected if protocol distance cannot be coded consistently or does not reveal meaningful differences in retrospective bias.

---

# M.20 Candidate Instrument 15 — Residual Conversion Monitor

## M.20.1 Target

**Residual rail across periods**

## M.20.2 Missing problem

Residual is often treated as a static warning list.

But residual may undergo several transitions:

Residual
→ Dissipation. (M.64)

Residual
→ Resolution. (M.65)

Residual
→ StructuralLoad. (M.66)

Residual
→ OpposingEvent. (M.67)

Residual
→ WorldCrisis. (M.68)

A mature system should track these conversions.

---

## M.20.3 Residual state transition matrix

Let residual states be:

$\mathcal{R}$_state

= {Open,Monitoring,Resolved,Dissipated,Converted,Invalidated}. (M.69)

Define transition probabilities:

P_R(i→j | Type,Severity,Period,χ,Ξ). (M.70)

---

## M.20.4 Conversion hazard

For residual r:

h_convert(t)
= Pr(Residual becomes consequential event at t | r remains open). (M.71)

Possible predictors include:

- duration;
- severity;
- crowding;
- gate proximity;
- liquidity;
- higher-frame conflict.

---

## M.20.5 Proto-Eight roles

- Memory;
- Trigger;
- Gate;
- Boundary.

---

## M.20.6 Falsification

The monitor loses value if residual-state transitions cannot be measured more reliably than ordinary event features.

---

## M.21 Instrument Families Versus Single Indicators

Several candidate instruments should not be reduced to one scalar.

For example:

## Residual-bearing gate

Best represented as:

AdmissionStrength + ResidualVector. (M.72)

## Cross-ledger rigidity

Best represented as:

RigidityVector + CouplingNetwork. (M.73)

## Episode progress

Best represented as:

EventDepth + Cadence + SelectionDepth + RegimeSequence. (M.74)

## Observer crowding

Best represented as:

Visibility + Capital + ActionSimilarity + LiquidityCapacity. (M.75)

The design principle is:

Do not force a multidimensional object into a scalar merely because dashboards prefer one number. (M.76)

A scalar may be used for ranking.

The full vector must remain available for diagnosis.

---

# M.22 Proto-Eight Instrument-Balance Test

A proposed instrument can also be audited across the eight actuation roles.

Let:

$$a_j = (Gd,Gt,B,E,T,Gn,M,F)_j. \quad (M.77)$$

where the coordinates indicate the role coverage of instrument j.

A research dashboard may reveal:

- too many Memory instruments;
- too many Focus instruments;
- weak Gate instruments;
- weak Exchange measurement;
- absent Guidance diagnostics.

For a method bundle:

$$A_{bundle} = \Sigma_j a_j. \quad (M.78)$$

The bundle is not required to assign equal weight to all eight roles.

But missing roles should be visible.

The Proto-Eight source treats these roles as operational levers for movement, qualification, separation, exchange, initiation, steering, retention, and selective attention.

---

# M.23 The Most Important Missing Instrument Families

The proposed instruments can be grouped into five broader families.

## M.23.1 Path-retaining instruments

Preserve what conventional aggregation erases.

Examples:

- Path-Bearing Candle;
- intrawindow event sequence;
- order-cancellation history.

---

## M.23.2 Commitment-quality instruments

Separate crossing from durable admission.

Examples:

- Structural Acceptance Score;
- Residual-Bearing Gate Score;
- Episode Completion Certificate.

---

## M.23.3 Historical-residual instruments

Preserve unresolved past pressure.

Examples:

- Residual-History State;
- Residual Conversion Monitor;
- fakeout memory.

---

## M.23.4 Transport instruments

Test whether a claim remains meaningful under reframing.

Examples:

- Event Portability Index;
- Declaration Stability Index;
- cross-ledger maps.

---

## M.23.5 Observer-world instruments

Measure when interpretation itself changes market behaviour.

Examples:

- Observer Crowding and Reflexivity Index;
- Ledger Reconciliation Gate;
- Cross-Ledger Rigidity Map.

This classification suggests that the next generation of Technical Analysis may not primarily need more oscillators.

It may need better:

- history;
- gates;
- transport;
- residual;
- observer models.

---

# M.24 Prototype Development Sequence

A candidate instrument should pass through six stages.

## Stage 0 — Concept

The missing function is identified.

## Stage 1 — Operational definition

Inputs, output, units, and protocol are declared.

## Stage 2 — Reliability

Independent analysts or systems reproduce the output.

## Stage 3 — Incremental information

The instrument outperforms simpler proxies.

## Stage 4 — Gate or diagnostic value

It improves a consequential task.

## Stage 5 — Transport

It survives relevant protocol changes.

## Stage 6 — Deployment

Its use remains stable after adoption and backreaction.

The maturity ladder is:

Concept
→ Definition
→ Reliability
→ IncrementalValue
→ GateUtility
→ Transport
→ Deployment. (M.79)

---

# M.25 Minimal Laboratory Protocol

## M.25.1 Select one missing function

Do not begin by constructing an all-purpose indicator.

Example:

Measure whether a breakout has produced durable value migration.

---

## M.25.2 Declare the target cell

Example:

Event × Commitment.

---

## M.25.3 Select the Proto-Eight roles

Example:

- Boundary;
- Exchange;
- Gate;

- Memory.

---

## M.25.4 Define raw observables

Example:

- price relative to boundary;
- volume above boundary;
- profile density;
- closes;
- retest.

---

## M.25.5 Define a simple benchmark

Example:

close above resistance.

---

## M.25.6 Freeze the instrument

Do not change the formula during the test set.

---

## M.25.7 Record residual

Failed cases must remain available.

---

## M.25.8 Test transport

Apply:

- timeframe;
- scale;
- bar rule;
- asset class.

---

## M.25.9 Apply reduction

If a simple rule performs equally well, retain the simple rule.

---

# M.26 Example Lab — Structural Acceptance Score

## M.26.1 Task

Determine whether a breakout has formed durable support.

## M.26.2 Sample

All daily range breakouts in a declared equity universe.

## M.26.3 Baseline

$B_0$ = close above range. (M.80)

## M.26.4 Candidate model

SAS
= $w_1$ClosePersistence

- $w_2$Retest
- $w_3$ValueMigration
- $w_4$Breadth
  − $w_5$Residual. (M.81)

## M.26.5 Outcomes

- support holds after ten sessions;
- failed reclaim;
- maximum adverse excursion;
- new value-area formation.

## M.26.6 Hypothesis

Pr(DurableSupport | SAS high)

Pr(DurableSupport | $B_0$ only). (M.82)

## M.26.7 Reduction rule

If:

Loss(SAS) ≥ Loss($B_0$) (M.83)

after complexity penalty, reject SAS.

---

# M.27 Example Lab — Episode Completion Certificate

## M.27.1 Task

Distinguish local trend interruption from completed trend episode.

## M.27.2 Candidate evidence

- defining trend relation fails;
- reversal gate passes;
- higher-frame confirmation;
- alternative branch score;
- persistence;
- residual burden.

## M.27.3 Baseline

Trend episode ends at highest price before a twenty-percent decline.

This is easy retrospectively but unusable prospectively.

## M.27.4 Candidate outcome

Probability that the old episode maximum is not exceeded within h sessions.

## M.27.5 Hypothesis

ECC-2 and ECC-3 classifications should outperform:

- oscillator divergence;
- moving-average crossover;
- fixed drawdown rule

in prospective completion classification.

---

# M.28 Example Lab — Observer Crowding Index

## M.28.1 Task

Determine whether highly visible breakout signals become:

- self-confirming;
- or fragile through crowding.

## M.28.2 Data

- public commentary;
- options concentration;
- fund-flow estimates;
- search intensity;
- volume;
- liquidity capacity.

## M.28.3 Model

Outcome_h
= α

- $\beta_1$OCRI
- $\beta_2$OCRI²
- $\beta_3$GateStrength
- $\beta_4$ResidualBurden
- ε. (M.84)

## M.28.4 Competing hypotheses

### H1 — Pure self-confirmation

$\beta_1 > 0$ and $\beta_2 \approx 0$. (M.85)

### H2 — Crowding inversion

$\beta_1 > 0$ and $\beta_2 < 0$. (M.86)

### H3 — No observer effect

$\beta_1 \approx 0$ and $\beta_2 \approx 0$. (M.87)

The third outcome must remain publishable.

---

# M.29 Empty Cells Should Sometimes Remain Empty

Not every table cell requires a new indicator.

A cell may be empty because:

- the function is not separately observable;
- another period already measures it adequately;

- no stable operational object exists;
- measurement would introduce more noise than information;
- the concept is not meaningful under that protocol.

Therefore:

EmptyCell ≠ MissingTruth by default. (M.88)

An empty cell is an invitation to investigate.

It is not permission to invent a variable.

---

# M.30 Instrument Proliferation Risk

A periodic table can accidentally encourage indicator multiplication.

This would reproduce the very problem the framework is intended to solve.

The proliferation danger is:

MissingCell
→ NewName
→ NewCompositeScore
→ OverfitDashboard. (M.89)

To prevent this, every proposed instrument must pass:

1. missing-function test;
2. redundancy test;
3. benchmark test;
4. transport test;
5. prospective test;
6. reduction test.

The preferred result may be:

No new instrument needed. (M.90)

---

# M.31 Relationship to Complex Technical Analysis

Some missing instruments may later provide valid candidate Q channels.

Examples include:

- residual-history pressure;
- breadth pressure;
- unexecuted intention;

- cross-ledger disagreement;
- crowding pressure.

But the path must remain:

IndependentInstrument
→ ReliableQCandidate
→ UnitAlignment
→ GeneratorTest
→ RealPairBenchmark
→ ComplexEligibility. (M.91)

It must not be:

InterestingNewIndicator
→ call it Q
→ calculate phase
→ declare internal time. (M.92)

The source Technical Analysis complex proposal presents admitted structure and retained pressure as a possible two-part state, but its strongest benefit is residual honesty rather than automatic prophecy or ontology.

---

# M.32 Relationship to the Declaration Framework

Every new instrument is itself a declaration.

It chooses:

- what counts as data;
- what counts as structure;
- which horizon matters;
- which gate commits;
- what remains residual.

Thus an instrument does not merely discover a pre-existing market object.

It produces a protocol-bound readable object:

$\Sigma_0$
→ Declare_P
→ $\hat{O}$_instrument
→ Output
→ Gate
→ Trace + Residual. (M.93)

The declaration source emphasizes that projection, gate, trace, residual, and ledger become meaningful only after boundary, observation rule, horizon, intervention family, baseline, and feature map have been specified.

Instrument design is therefore also world design.

A poorly chosen instrument can create:

- false events;
- false certainty;
- false boundaries;
- self-reinforcing crowding;
- suppressed residual.

---

# M.33 The Instrument-Design Matrix

| Proposed instrument | Period | Family | Proto-Eight core | Main task |
|---|---|---|---|---|
| Unexecuted Intention Field | Mark | Load | Gradient–Gate | preserve failed intention |
| Path-Bearing Candle | Window | Motion | Trigger–Guidance | retain intrawindow path |
| Base Completeness Score | Structure | Load | Memory–Boundary | quantify loaded base |
| Structural Acceptance Score | Structure | Commitment | Gate–Memory | establish persistence |
| Pre-Gate Commitment Density | Event | Load | Gradient–Exchange | quantify event loading |
| Residual-Bearing Gate Score | Event | Commitment | Boundary–Gate | separate admission and fragility |
| Event Portability Index | Event | Governance | Guidance–Focus | measure transport |
| Residual-History State | Episode | Load | Memory–Gate | carry failed history |
| Phase-Cadence Progress Meter | Episode | Motion | Trigger–Guidance | measure internal progress |
| Episode Completion Certificate | Episode | Commitment | Gate–Memory | govern endpoint claims |
| Observer Crowding Index | World | Motion | Focus–Exchange | measure reflexivity |
| Cross-Ledger Rigidity Map | World | Constraint | Boundary–Memory | map institutional lock-in |
| Ledger Reconciliation Gate | World | Commitment | Gate–Exchange | measure authoritative recognition |
| Declaration Stability Index | Governance | Constraint | Focus–Boundary | detect protocol drift |
| Residual Conversion Monitor | Governance | Load/Motion | Memory–Trigger | track unresolved pressure |

---

# M.34 Priority Order

The candidate instruments should not all be built at once.

A rational priority order is:

## Priority 1 — Residual-Bearing Gate Score

Because breakout and event commitment are already widely studied and data are accessible.

## Priority 2 — Structural Acceptance Score

Because the transition from Event to Structure is central to fakeout analysis.

## Priority 3 — Event Portability Index

Because cross-frame claims are common but rarely formalized.

## Priority 4 — Residual-History State

Because failed signals are routinely lost.

## Priority 5 — Episode Completion Certificate

Because episode endings remain highly retrospective.

## Priority 6 — Observer Crowding Index

Because it is theoretically important but difficult to identify causally.

## Priority 7 — Cross-Ledger World instruments

Because these require broader institutional datasets and observer-specific protocols.

The order follows:

AccessibleMeasurement
→ GateValidation
→ ResidualGovernance
→ EpisodeClosure
→ WorldBackreaction. (M.94)

---

# M.35 Falsification of the Missing-Cell Programme

The empty-cell programme should be abandoned or substantially revised if:

1. proposed instruments cannot be defined reliably;
2. outputs depend excessively on subjective coding;
3. simple existing indicators perform equally well;
4. the table does not predict which new instruments add value;
5. instrument roles overlap uncontrollably;
6. transport tests remain too ambiguous;

7. additional fields increase overfitting without improving diagnosis;
8. users cannot maintain residual records in practice.

The programme is not validated by the number of instruments it generates.

It is validated by whether it discovers a small number of genuinely missing functions.

---

# M.36 The Deeper Prediction

The article's deeper prediction is not:

> Technical Analysis needs more indicators.

It is:

> Technical Analysis needs instruments that measure closure quality, residual burden, transport survival, and observer-world consequences.

Traditional indicators mainly ask:

- How far?
- How fast?
- How extended?
- Where is the boundary?

The proposed next generation asks:

- Was the transition admitted?
- What remained unresolved?
- Did the state persist?
- Did it survive another frame?
- Did prior failures become current load?
- Did an authority change future action?
- Did observation itself alter the market?

This is the movement:

Signal Detection
→ Closure Diagnosis
→ World-Formation Analysis. (M.95)

---

# M.37 Appendix M Conclusion

The proto-periodic table should not be treated as a cabinet for arranging familiar indicators.

Its strongest scientific use may be to expose what familiar indicators systematically fail to

measure.

The existing instrument ecology is concentrated around:

- price memory;
- momentum;
- range location;
- volatility;
- visible boundaries.

Its weakest regions concern:

- residual-bearing commitment;
- episode completion;
- cross-frame transport;
- observer crowding;
- institutional recognition;
- ledger backreaction.

The design law is:

MissingCell
→ DeclareFunction
→ BuildMinimalInstrument
→ CompareSimpleBenchmark
→ TestGateValue
→ AuditResidual
→ Transport
→ Falsify or Retain. (M.96)

The design prohibition is:

EmptyCell
≠ permission for decorative mathematics. (M.97)

And the final research principle is:

> Build no new indicator unless it measures a function that existing instruments genuinely leave unresolved.

# Appendix N — Formal Core of the Proto-Periodic Market Grammar

## N.1 Purpose

The preceding appendices developed the framework through:

- method cards;

- residual ledgers;
- transport tests;
- complex-eligibility rules;
- research protocols;
- practitioner workflows;
- Proto-Eight actuation roles;
- missing-instrument design.

This appendix compresses those components into one formal core.

Its purpose is not to force every part of Technical Analysis into a single equation.

Its purpose is to define the minimum objects required for a market claim to become:

1. readable;
2. typed;
3. gated;
4. traceable;
5. falsifiable;
6. transportable;
7. revisable.

The source declaration framework argues that projection, gate, trace, residual, and ledger become meaningful only after the observer declares a boundary, observation rule, horizon, intervention family, baseline, and feature map.

The source Technical Analysis framework similarly treats each indicator as a protocol-bound projection of market self-reference rather than as direct access to total market truth.

The integrated architecture is therefore:

Undeclared Field
→ Declaration
→ Projection
→ Typed Observation
→ Gate
→ Trace + Residual
→ Ledger
→ Transport
→ Admissible Revision. (N.1)

---

# N.2 The Undeclared Market Field

Let:

$\Sigma_0$ = undeclared market possibility field. (N.2)

$\Sigma_0$ may include:

- quotes;
- trades;
- liquidity;
- orders;
- positions;
- expectations;
- legal constraints;
- accounting rules;
- institutional mandates;
- latent intentions;
- unobserved alternatives.

$\Sigma_0$ is not identical to a complete database.

It is the conceptual field from which a bounded observer may disclose a particular market object.

Before declaration, no unique object such as:

- trend;
- breakout;
- support;
- phase;
- market regime

has yet been defined.

Therefore:

RawAvailability ≠ DeclaredObject. (N.3)

A market object becomes operational only under a protocol.

---

# N.3 The Declaration

# N.3.1 Protocol

Define the protocol:

$P = (B,\Delta,h,u)$. (N.4)

where:

B = boundary;
Δ = observation or aggregation rule;
h = time or state horizon;
u = admissible intervention family.

Add:

q = baseline. (N.5)

φ = feature map. (N.6)

The complete declaration is:

D
= (q,φ,P,Ô_P,Gate_P,TraceRule_P,ResidualRule_P). (N.7)

This follows the source declaration architecture.

---

# N.3.2 Declared market world

The declared world is:

World_P = (X,q,φ,P). (N.8)

The declared field is:

Σ_P = Declare(Σ0 | q,φ,P). (N.9)

The declaration determines:

- what counts as inside;
- what counts as an observation;
- what constitutes a bar;
- which boundary matters;
- which transformations are admissible;
- which gate can create commitment;
- which residual must remain visible.

---

# N.3.3 Technical Analysis protocol

For ordinary Technical Analysis, the protocol may be expanded as:

P_TA
=
(Asset,Universe,Venue,Boundary,Timeframe,Scale,BarRule,FeatureMap,GateRule,ResidualRule,C
(N.10)

This expanded form should be treated as an implementation schema rather than a replacement for the more general protocol P.

---

# N.4 Projection

# N.4.1 Projection operator

A Technical Analysis instrument is a projection:

$X_j = \hat{O}_{j,P}(\Sigma_0)$. (N.11)

or more explicitly:

$X_j = \hat{O}_j(\Sigma_P)$. (N.12)

Examples include:

$MA_n = \hat{O}_{MA}(\Sigma_P)$. (N.13)

$RSI_n = \hat{O}_{RSI}(\Sigma_P)$. (N.14)

$VWAP_a = \hat{O}_{VWAP}(\Sigma_P)$. (N.15)

$BreakoutCandidate = \hat{O}_{Break}(\Sigma_P)$. (N.16)

The source Technical Analysis framework summarizes this as:

$TechnicalAnalysis_P = Projection_P(MarketSelfReference)$. (N.17)

---

# N.4.2 Source and operator lineage

Every projected variable should carry:

$S_j$ = source lineage. (N.18)

$w_j$ = operator word. (N.19)

For example:

$S_{MA} = \{price\}$. (N.20)

$w_{MA} = Smooth_n \circ SelectClose$. (N.21)

For MACD:

$w_{MACD} = Difference \circ (EMA_{fast}, EMA_{slow}) \circ SelectClose$. (N.22)

Two indicators should not be treated as independent merely because their names differ.

Their informational relationship depends on:

- source overlap;
- operator overlap;

- horizon overlap;
- failure-mode overlap.

---

## N.5 Six Closure Periods

Define the closure-period set:

$\mathcal{P}_6$
$= \{P_0, P_1, P_2, P_3, P_4, P_5\}$. (N.23)

with:

$P_0$ = Mark.
$P_1$ = Window.
$P_2$ = Structure.
$P_3$ = Event.
$P_4$ = Episode.
$P_5$ = World. (N.24)

These are not merely six timeframes.

They are six levels of committed organization.

---

## N.5.1 Mark

A singular transaction, quote, order, or execution state.

## N.5.2 Window

An aggregation closed under a declared bar or session rule.

## N.5.3 Structure

A persistent relation extracted across windows.

## N.5.4 Event

A meaningful state transition passing a declared gate.

## N.5.5 Episode

An ordered sequence of events governed by a persistent grammar.

## N.5.6 World

A bounded institutional system whose authoritative ledgers alter future admissibility.

---

## N.5.7 Period order

The periods admit an ordering by closure depth:

Mark ≺ Window ≺ Structure ≺ Event ≺ Episode ≺ World. (N.25)

The symbol ≺ does not mean that every Mark must inevitably become a Window or that every Event becomes a World.

It means that a higher-period claim requires an additional closure condition.

---

# N.6 Four Functional Families

Define:

$\mathcal{G}_4$
= {L,M,C,G}. (N.26)

where:

L = Load / Memory;
M = Motion / Relation;
C = Constraint / Boundary;
G = Commitment / Gate.

---

# N.6.1 Load

Load carries prior consequential structure:

L_p(k). (N.27)

Examples include:

- inventory;
- volume;
- moving-average memory;
- profile density;
- event history;
- legal obligation.

---

# N.6.2 Motion

Motion transforms or relates states:

M_p : State_p → State_p. (N.28)

Examples include:

- return;

- acceleration;
- momentum;
- relative strength;
- phase displacement;
- feedback.

---

## N.6.3 Constraint

Constraint defines the admissible region:

$C_p \subset State_p$. (N.29)

Examples include:

- bid and ask;
- high and low;
- range;
- support;
- collateral limit;
- legal boundary.

---

## N.6.4 Commitment

Commitment admits or rejects a candidate transition:

$G_p(X,L,\mathscr{R}) \rightarrow$ Decision. (N.30)

Examples include:

- execution;
- close;
- retest;
- settlement;
- impairment;
- legal recognition.

---

# N.7 Period–Function Cells

A basic table element is:

E_{p,g}^{P}. (N.31)

where:

p ∈ $\mathcal{P}_6$;
g ∈ $\mathcal{G}_4$;
P = declared protocol.

Examples:

E_{Structure,Load}^{P}
= moving-average or profile memory. (N.32)

E_{Structure,Constraint}^{P}
= support or resistance. (N.33)

E_{Event,Commitment}^{P}
= breakout gate. (N.34)

E_{Episode,Motion}^{P}
= phase or regime progression. (N.35)

E_{World,Commitment}^{P}
= accounting, legal, regulatory, or policy recognition. (N.36)

The cell does not uniquely identify an instrument.

Several methods may occupy one cell.

One method may combine several cells.

---

# N.8 Proto-Eight Role Typing

Define the actuation-role set:

$\mathcal{A}_8$
= {∇,Γ,Bd,Ex,Tr,Gd,Mem,Foc}. (N.37)

where:

∇ = Gradient;
Γ = Gate;
Bd = Boundary;
Ex = Exchange;
Tr = Trigger;
Gd = Guidance;
Mem = Memory;
Foc = Focus.

A fully typed object is:

X_{p,g,a}^{P}. (N.38)

For example:

X_{Event,Commitment,Gate}^{P_daily}
= daily breakout close. (N.39)

X_{Event,Load,Exchange}^{P_daily}
= breakout participation. (N.40)

X_{Structure,Constraint,Boundary}^{P_daily}
= resistance zone. (N.41)

X_{Episode,Load,Memory}^{P_daily}
= prior fakeout history. (N.42)

The Proto-Eight role identifies the operative mechanism.

The four-family label identifies the observational contribution.

The six-period label identifies closure depth.

---

# N.9 The Local Functional Cycle

Within period p, define the local market cycle:

L_p
→ M_p under C_p
→ G_p
→ e_p + r_p. (N.43)

where:

e_p = admitted trace;
r_p = residual.

The admitted trace and residual then contribute to the next period:

L_{p+1}
= Inherit_p(e_p,r_p,L_p). (N.44)

The compressed periodic law is:

Load_p
→ Motion_p under Constraint_p
→ Commitment_p
→ Ledger_{p+1} + Residual_p
→ Load_{p+1}. (N.45)

This equation is a proposed structural synthesis.

It should not yet be called a universally established law of markets.

---

# N.10 Gate Operator

# N.10.1 Rich gate output

A gate should return more than a Boolean value.

Define:

Gate_P(X,L,$\mathscr{R}$)
→ (d,α,e,r,m). (N.46)

where:

d = categorical decision;
α = admission strength;
e = admitted event;
r = attached residual;
m = gate metadata.

Possible decisions are:

d ∈ {Admit,PartiallyAdmit,Defer,Reject,Ambiguous}. (N.47)

The admission fraction is:

0 ≤ α ≤ 1. (N.48)

---

# N.10.2 Candidate and admitted event

A candidate event is:

c_k = CandidateTransition(X_k,C_k). (N.49)

An admitted event is:

e_k = Admit_P(c_k | X_k,L_k,$\mathscr{R}$_k). (N.50)

Therefore:

CandidateTransition ≠ AdmittedEvent. (N.51)

A resistance crossing may be a candidate.

A declared close, participation threshold, and retest may be the gate.

---

# N.10.3 Gate quality

A conceptual gate-quality function is:

S_G
= f(Close,Displacement,Participation,Breadth,Retest,FollowThrough,HigherFrame)
− ResidualBurden. (N.52)

The function must be prospectively calibrated.

A gate score designed after observing outcomes is not an admissible confirmation test.

---

# N.11 Trace, Residual, and Ledger

## N.11.1 Trace

A trace is a record that changes future interpretation or action:

e_k = committed consequential record. (N.53)

A stored number that never affects later state is merely a log.

---

## N.11.2 Residual

Residual is:

r_k
= UnobservedStructure_k

- MissingEvidence_k
- Contradiction_k
- UnselectedAlternatives_k
- ModelLimitation_k. (N.54)

Residual is not identical to Q.

Residual is not automatically error.

Residual may later:

- resolve;
- dissipate;
- invalidate;
- become higher-period Load;
- trigger revision.

---

## N.11.3 Ledger

The ledger update is:

L_{k+1} = Update(L_k,e_k,r_k,m_k). (N.55)

A minimal ledger stores:

- event;
- gate;
- authority;
- timestamp;
- residual;
- protocol version.

The filtration source argues that trace forms ledger, ledger order becomes time, and stable ledger order becomes causality under the declared viewpoint.

---

# N.11.4 Residual register

Define the persistent residual register:

$\mathscr{R}_{k+1}$
$= D_{\mathscr{R}}(\mathscr{R}_k) + r_k - q_k.$ (N.56)

where:

$D_{\mathscr{R}}$ = persistence or decay;
$q_k$ = resolved or dissipated residual.

The residual register should remain vector-valued until there is evidence that aggregation preserves meaning.

---

# N.12 Period Promotion

# N.12.1 Promotion operator

Define:

$\Pi_{p \to p+1}(e_p, L_p, \mathscr{R}_p)$. (N.57)

A lower-period trace is promoted only when the next closure condition passes.

Examples:

### Mark → Window

Executed marks become part of the bar.

### Window → Structure

Repeated windows establish a stable relation.

### Structure → Event

A meaningful boundary is tested and gated.

### Event → Episode

An ordered event grammar persists.

### Episode → World

An authoritative institutional ledger changes future admissibility.

---

## N.12.2 Promotion condition

Promotion requires:

Closure_{p+1} = true. (N.58)

Therefore:

Persistence_p alone ≠ Promotion_{p+1}. (N.59)

For example, repeated high RSI readings do not automatically become an Event.

A structural gate remains necessary.

---

## N.12.3 Demotion

Claims may also be downgraded:

Event → Structure warning. (N.60)

Episode transition → local event. (N.61)

World claim → institutional possibility. (N.62)

Complex clock → real pair. (N.63)

Demotion aligns claim strength with actual closure.

---

# N.13 Transport and Invariance

## N.13.1 Transport operator

For source protocol P and target protocol P′:

T_{P→P′}: Claim_P → ExpectedClaim_{P′}. (N.64)

The directly observed target claim is:

Claim_{P′}. (N.65)

Define transport residual:

r_T
= Claim_{P′} − T_{P→P′}(Claim_P). (N.66)

The subtraction may be:

- numerical;
- categorical;
- relational;
- gate-based.

## N.13.2 Survival criterion

A claim survives when:

Dist[T_{P→P′}(Claim_P),Claim_{P′}] ≤ ε_T. (N.67)

Possible statuses include:

- exact survival;
- covariant survival;
- partial survival;
- local only;
- failure;
- indeterminate;
- non-comparable.

---

## N.13.3 Invariant relation

A candidate invariant is:

Inv(C)
= {Relation(C) preserved across admissible P′}. (N.68)

The Technical Analysis source emphasizes that stronger signals are those surviving admissible reframing and that cross-checking should test structural survival rather than merely add more indicators.

The filtration source similarly defines law as a relation preserved across admissible disclosures.

---

# N.14 Regime Signature χ

## N.14.1 Two-channel feedback

Let:

λ = signal pressure. (N.69)

s = realized structure. (N.70)

Let:

F: δλ → δs. (N.71)

M: δs → δλ. (N.72)

Define the signed coupling operator:

C_χ
= [[0,F],[χM,0]]. (N.73)

---

## N.14.2 Exact operator square

For precision, the exact square is:

C_χ²
= [[χFM,0],[0,χMF]]. (N.74)

Only under the normalized condition:

FM = I_λ (N.75)

and:

MF = I_s (N.76)

does this reduce to:

C_χ² = χI. (N.77)

The simplified formula used in the source framework should therefore be read as a canonical normalized case, not as the unrestricted operator identity. The source uses χ to distinguish corrective, critical, and self-confirming feedback.

---

## N.14.3 Regime classes

χ < 0 → corrective circulation. (N.78)

χ ≈ 0 → critical ambiguity. (N.79)

χ > 0 → self-confirming selection. (N.80)

The same indicator may change meaning across these regimes.

---

## N.14.4 Protocol indexing

χ must be indexed:

χ = χ_{P,h}. (N.81)

A market can be:

- corrective intraday;
- self-confirming daily;
- critical weekly.

Therefore:

χ_{5m} ≠ χ_{1d} ≠ χ_{1w} by default. (N.82)

# N.15 Effective Control State Ξ

Define:

Ξ_P = (ρ_P,γ_P,ν_P). (N.83)

where:

ρ = loading;
γ = lock-in;
ν = agitation.

Ξ is compiled from the richer declared field:

Ξ_P = Compile_P(Σ_P). (N.84)

It is not asserted to be the complete ontology of the market.

---

# N.15.1 Loading

ρ may compile:

- participation;
- leverage;
- open interest;
- exposure;
- density;
- attention.

# N.15.2 Lock-in

γ may compile:

- illiquidity;
- exit cost;
- collateral encumbrance;
- legal restriction;
- concentration;
- funding dependence.

# N.15.3 Agitation

ν may compile:

- volatility;
- dispersion;
- spread instability;
- cancellation;
- failed gates;
- flow instability.

---

## N.15.4 Distinction from χ

χ describes feedback orientation.

Ξ describes effective operating condition.

Therefore:

χ ≠ Ξ. (N.85)

A highly loaded and locked state may still be:

- corrective;
- critical;
- or self-confirming.

---

# N.16 Complex Eligibility

## N.16.1 Real-pair state

Let:

X = (R,Q). (N.86)

At one instant:

(R,Q) ↔ R + iQ. (N.87)

No additional raw information is created by notation alone.

---

## N.16.2 Complex state

A privileged complex representation is:

Z = R + iQ = Ae^{iθ}. (N.88)

where:

A = √(R² + Q²). (N.89)

$\theta = \operatorname{atan2}(Q,R)$. (N.90)

The representation earns priority only when:

1. R is independently meaningful;
2. Q is independently meaningful or structurally derived by a justified geometry;
3. units are compatible;
4. coupling supports a stable complex generator;
5. phase adds operational value;
6. a real-pair benchmark is outperformed.

---

# N.16.3 Complex dynamical condition

A local complex model may satisfy:

$$\frac{dZ}{dt} = [g_A(t) + i\omega(t)]Z + \varepsilon(t). \quad (N.91)$$

In real coordinates:

$\dot{R} = g_AR - \omega Q + \varepsilon_R$. (N.92)

$\dot{Q} = \omega R + g_AQ + \varepsilon_Q$. (N.93)

The canonical phase relations are:

$\frac{dR}{d\theta} \approx -Q$. (N.94)

$\frac{dQ}{d\theta} \approx R$. (N.95)

---

# N.16.4 Reduction rule

When complex priority fails:

$Z \rightarrow (R,Q)$. (N.96)

When Q adds no incremental value:

$(R,Q) \rightarrow R$. (N.97)

The source phase framework requires reduction when phase is scale-dependent, Q lacks independent meaning, or a flexible real-pair model performs equally well.

---

# N.17 Four Time Coordinates

The framework distinguishes four forms of ordering.

# N.17.1 Calendar time

t = external duration. (N.98)

## N.17.2 Phase orientation

θ = internal orientation of an eligible complex state. (N.99)

## N.17.3 Internal phase depth

$\tau_i$ = Unwrap θ (N.100)

or:

$\tau_i(t) = \int_0^t |\dot{\theta}(s)| ds$. (N.101)

## N.17.4 Ledger-event order

k = order of committed events. (N.102)

Therefore:

$t \neq \theta \neq \tau_i \neq k$. (N.103)

A phase becomes a clock only when $\tau_i$ improves the ordering of comparable episodes.

A phase-sensitive event world requires even more:

$\tau_i$
→ Gate
→ Trace
→ ChangedFutureDynamics. (N.104)

---

# N.18 Selection Depth

Let $\Omega_k$ denote the set of still-admissible future branches.

Define conceptual selection depth:

$\sigma_k$
$= -\ln[\text{Measure}(\Omega_k)/\text{Measure}(\Omega_0)]$. (N.105)

σ measures branch elimination.

$\tau_i$ measures phase traversal.

They are distinct:

$\sigma \neq \tau_i$. (N.106)

A system may accumulate large phase without eliminating many branches.

A single authoritative gate may eliminate many branches with little phase rotation.

---

# N.19 Declaration Revision

# N.19.1 Revision operator

At episode k, let:

$$D_k = (q_k,\varphi_k,P_k,\hat{O}_k,Gate_k,TraceRule_k,ResidualRule_k). \quad (N.107)$$

Revision is:

$$D_{k+1} = U_a(D_k,L_k,\mathscr{R}_k). \quad (N.108)$$

---

# N.19.2 Admissible family

Define:

$$\begin{aligned}\mathcal{D}_{adm} = \{D \mid\ & WellFormed(D)\\ \wedge\ & TracePreserving(D)\\ \wedge\ & ResidualHonest(D)\\ \wedge\ & FrameRobust(D)\\ \wedge\ & BudgetBounded(D)\\ \wedge\ & NonDegenerate(D)\}. \quad (N.109)\end{aligned}$$

This follows the source self-revising declaration framework.

An admissible revision satisfies:

$$U_a: \mathcal{D}_{adm} \times Ledger \times Residual \to \mathcal{D}_{adm}. \quad (N.110)$$

---

# N.19.3 Invalid revision

A revision is inadmissible when it:

- deletes the original claim;
- hides contradiction;
- changes the outcome horizon retrospectively;
- changes frames without transport;
- adds unlimited complexity;
- redefines failure as confirmation.

The source describes mature observerhood not as unrestricted self-modification but as trace-preserving, residual-honest, frame-robust, bounded revision.

---

# N.20 The Gauged Technical-Analysis Operator

The general declaration source defines a gauged disclosure operator combining declaration, projection, gate, and trace update.

For the present framework, define:

$$\mathfrak{D}_{TA,P} = \mathrm{UpdateLedger}_P \circ \mathrm{AuditResidual}_P \circ \mathrm{Gate}_P \circ \mathrm{Type}_P \circ \mathrm{Diagnose}_P \circ \mathrm{Project}_P \circ \mathrm{Declare}_P. \quad (N.111)$$

The operators act in the following order.

## Declare_P

Defines the world.

## Project_P

Computes visible technical objects.

## Diagnose_P

Estimates relevant characteristic, $\chi$, or $\Xi$.

## Type_P

Assigns period, function, and actuation role.

## Gate_P

Determines commitment.

## AuditResidual_P

Preserves non-closure.

## UpdateLedger_P

Writes consequential trace.

---

## N.20.1 One-cycle output

One application produces:

$$\mathfrak{D}_{TA,P}(\Sigma_0, L_k, \mathscr{R}_k) = (e_k, r_k, L_{k+1}, \mathscr{R}_{k+1}, m_k). \quad (N.112)$$

where:

e_k = admitted event;
r_k = new residual;
m_k = metadata and diagnostic state.

---

# N.21 The Self-Revising Market-Observer Runtime

The next declaration depends on prior trace and residual:

$$D_{k+1} = U_a[D_k,L_{k+1},\mathscr{R}_{k+1}]. \quad (N.113)$$

The full recursive runtime is:

$$(D_k,\Sigma_0,L_k,\mathscr{R}_k) \to \mathfrak{D}_{TA,D_k} \to (e_k,r_k,L_{k+1},\mathscr{R}_{k+1}) \to U_a \to D_{k+1}. \quad (N.114)$$

Expanded:

Declare
→ Project
→ Diagnose
→ Type
→ Gate
→ Trace
→ Residual
→ Ledger
→ Transport
→ Revise
→ Redeclare. (N.115)

This is the formal market-observation loop.

---

# N.22 The Master State

A sufficiently rich state description is:

$$\mathcal{S}_k = (D_k,\Sigma_{P,k},L_k,\mathscr{R}_k,\chi_k,\Xi_k,Z_k,\theta_k,\tau_{i,k},\sigma_k). \quad (N.116)$$

Not every component is always valid.

The hierarchy is conditional.

# Always required for mature analysis

- $D_k$;
- $\Sigma_{P,k}$;
- $L_k$;

- $\mathscr{R}_k$.

## Required when regime diagnosis is used

- $\chi_k$.

## Required when control-state compilation is used

- $\Xi_k$.

## Required only after complex eligibility

- $Z_k$;
- $\theta_k$;
- $\tau_{i,k}$.

## Required when branch suppression is studied

- $\sigma_k$.

A component that has not passed its eligibility test should be marked:

Undefined or Unestablished. (N.117)

It should not be assigned a convenient proxy merely to complete the state.

---

# N.23 The Master Transition Equation

Define the market-observation transition:

$$\mathcal{S}_{k+1} = \mathfrak{F}_P(\mathcal{S}_k,E_k,A_k) + \varepsilon_k. \quad (N.118)$$

where:

$E_k$ = external market and institutional environment;
$A_k$ = observer actions induced by interpretation;
$\varepsilon_k$ = unresolved dynamics.

A decomposed form is:

$$\Sigma_{P,k+1} = F_{market}(\Sigma_{P,k},External_k,Action_k). \quad (N.119)$$

$$(e_k,r_k) = GateAndResidual_P(\Sigma_{P,k+1},L_k,\mathscr{R}_k). \quad (N.120)$$

$$L_{k+1} = UpdateLedger(L_k,e_k,r_k). \quad (N.121)$$

$\mathscr{R}_{k+1}$

= UpdateResidual($\mathscr{R}$_k,r_k). (N.122)

D_{k+1}
= U_a(D_k,L_{k+1},$\mathscr{R}$_{k+1}). (N.123)

This is not claimed as a calibrated universal market equation.

It is the minimal formal architecture of the article.

---

# N.24 Observer Backreaction

Technical Analysis operates inside a self-referential market.

Let:

Interpretation_k = I(D_k,Σ_{P,k},L_k). (N.124)

Let:

Action_k = Policy(Interpretation_k). (N.125)

Then:

Σ_{P,k+1}
= F(Σ_{P,k},Action_k,External_k). (N.126)

Therefore:

Observation
→ Interpretation
→ Action
→ NewMarketEvidence. (N.127)

The source Technical Analysis framework emphasizes that a signal may become true because it is observed or fail because it becomes over-observed.

This does not mean every public signal has measurable causal power.

Backreaction requires empirical identification.

---

# N.25 Core Propositions

# Proposition N.1 — Protocol Dependence

No Technical Analysis object is fully specified without its protocol.

Formally:

X_j without P is underdetermined. (N.128)

---

# Proposition N.2 — Projection Partiality

Every indicator observes only a projection:

$\hat{O}_{j,P}(\Sigma_0) \neq \Sigma_0$. (N.129)

Therefore:

Projection ≠ Totality. (N.130)

This is directly aligned with the source Technical Analysis framework.

---

# Proposition N.3 — Gate Necessity

A relation becomes an Event only after a relevant commitment gate.

Structure + no Gate → Structure or Warning. (N.131)

Structure + Gate → EventCandidate or Event. (N.132)

---

# Proposition N.4 — Residual Persistence

Gate admission does not imply residual exhaustion.

$\text{Admit}(e_k) \nRightarrow \mathscr{R}_{k+1} = 0$. (N.133)

---

# Proposition N.5 — Periodic Inheritance

An admitted lower-period trace can become higher-period Load.

$e_p + r_p \rightarrow L_{p+1}$. (N.134)

This remains a principal empirical hypothesis of the present synthesis.

---

# Proposition N.6 — Claim-Bound Rule

Claim strength cannot exceed demonstrated closure:

ClaimLevel ≤ ClosureLevel. (N.135)

---

# Proposition N.7 — Transport Localizes Objectivity

A relation becomes more operationally objective as it survives admissible protocol changes.

$\text{Objectivity}_{P\text{-family}} \propto \text{CrossFrameSurvival}$. (N.136)

This is not a claim of absolute observer independence.

---

# Proposition N.8 — Complexity Eligibility

A complex representation earns priority only through demonstrated gain.

ComplexPriority
⇒ RealPairBenchmarkPassed. (N.137)

---

# Proposition N.9 — Phase-Time Eligibility

Phase becomes secondary time only when it improves episode ordering.

PhaseTime
⇒ AlignmentGain($\tau_i$ > t,k,other clocks). (N.138)

---

# Proposition N.10 — World Formation

A World-level event requires:

Authority

- Gate
- Ledger
- ChangedFutureAdmissibility. (N.139)

A large price move alone is not sufficient.

---

# Proposition N.11 — Admissible Learning

A mature analytical system learns only through trace-preserving revision:

ValidRevision
⇒ OriginalTracePreserved
∧ ResidualCarriedOrResolved. (N.140)

---

# N.26 Core Falsifiers

The architecture should be revised when any of the following persist.

## N.26.1 Declaration failure

The protocol does not stabilize the object.

## N.26.2 Projection failure

The instrument cannot be reproduced.

## N.26.3 Typing failure

Observers cannot agree on period or function.

## N.26.4 Gate failure

The gate is inconsistent or adds no value.

## N.26.5 Residual failure

Residual cannot be coded or contributes nothing.

## N.26.6 Transport failure

The claimed structure collapses under every admissible reframe.

## N.26.7 Regime failure

$\chi$ adds no conditional explanatory value.

## N.26.8 Complex failure

The real-pair model is equal or superior.

## N.26.9 Phase-time failure

$\tau_i$ does not outperform ordinary clocks.

## N.26.10 World failure

Ledger history has no future effect.

## N.26.11 Revision failure

The framework survives only through retrospective semantic change.

The declaration source states that gate failure should trigger protocol repair rather than narrative patching.

---

# N.27 Formal Error Taxonomy

Define total analytical error:

ε_total
= ε_D

- ε_O
- ε_T
- ε_G

- ε_R
- ε_X
- ε_U. (N.141)

where:

ε_D = declaration error;
ε_O = projection or operator error;
ε_T = typing error;
ε_G = gate error;
ε_R = residual error;
ε_X = transport error;
ε_U = revision error.

This decomposition is conceptual.

Its value lies in preventing every failure from being called “bad prediction.”

A failed breakout analysis may involve:

- correct boundary;
- correct projection;
- premature gate;
- hidden higher-frame residual.

The repair should target the failed layer.

---

# N.28 Minimum Valid Signal Record

A minimum valid signal record is:

SR_k
= (P_k,X_k,p_k,g_k,a_k,χ_k,G_k,e_k,r_k,I_k,h_k). (N.142)

where:

P_k = protocol;
X_k = projection;
p_k = period;
g_k = function;
a_k = Proto-Eight role;
χ_k = regime assumption;
G_k = gate;
e_k = admitted trace;
r_k = residual;
I_k = invalidation;
h_k = outcome horizon.

For complex models, add:

(R_k,Q_k,A_k,θ_k,$\tau_i$,k). (N.143)

only after eligibility.

---

# N.29 Minimum Complete Method Definition

A method is completely specified when:

Method_j
= (S_j,w_j,P_j,p_j,g_j,a_j,G_j,R_j,T_j,I_j). (N.144)

where:

S_j = source lineage;
w_j = operator word;
P_j = protocol;
p_j = closure period;
g_j = functional family;
a_j = actuation role;
G_j = required gate;
R_j = residual ontology;
T_j = transport set;
I_j = invalidation.

A method lacking these declarations is not necessarily useless.

It is incompletely governed.

---

# N.30 Minimal Research Comparison

For any new method M_new, compare:

$M_0$ = simplest baseline. (N.145)

$M_1$ = existing conventional method. (N.146)

$M_2$ = proposed periodic-grammar method. (N.147)

Evaluate:

Utility_j
= PredictiveValue_j

- DiagnosticValue_j
- GateValue_j
- TransportValue_j
- RevisionValue_j
  – ComplexityCost_j. (N.148)

The new method is retained only when:

$Utility_2 > \max(Utility_0, Utility_1)$. (N.149)

or when it provides a clearly distinct governance benefit not represented in the scalar utility.

# N.31 Minimal Scientific Status

The formal framework currently contains four different epistemic levels.

| Component | Current status |
|---|---|
| Declaration–projection–gate–trace architecture | source-derived conceptual framework |
| Operator-first Technical Analysis | source-derived interpretive framework |
| Six periods × four families | present article's proposed synthesis |
| Proto-Eight market crosswalk | present article's operational extension |
| $\chi$ feedback signature | formal source proposal requiring empirical estimation |
| $\Xi$ control state | protocol-compiled research interface |
| CAPM complex state | exact declared construction |
| General TA complex state | empirical research hypothesis |
| Phase time | high-burden hypothesis |
| Time-bearing market world | strongest ledger-and-backreaction claim |

This table prevents the exactness of one local construction from being transferred automatically to the entire architecture.

---

# N.32 Master Formula Sheet

## Declaration

$P = (B,\Delta,h,u)$. (N.150)

$D = (q,\varphi,P,\hat{O},Gate,TraceRule,ResidualRule)$. (N.151)

$\Sigma_P = Declare(\Sigma_0 \mid q,\varphi,P)$. (N.152)

## Projection

$X_j = \hat{O}_{j,P}(\Sigma_0)$. (N.153)

## Periodic typing

$X_{p,g,a}^{P}$. (N.154)

## Local functional cycle

$L_p \rightarrow M_p$ under $C_p \rightarrow G_p \rightarrow e_p + r_p$. (N.155)

## Ledger

$L_{k+1} = Update(L_k,e_k,r_k)$. (N.156)

## Residual

$\mathscr{R}_{k+1} = D_{\mathscr{R}}(\mathscr{R}_k) + r_k - q_k$. (N.157)

## Transport

$r_T = Claim_{P'} - T_{P \to P'}(Claim_P)$. (N.158)

## χ operator

$C_\chi = [[0,F],[\chi M,0]]$. (N.159)

$C_{\chi^2} = [[\chi FM,0],[0,\chi MF]]$. (N.160)

## Effective state

$\Xi_P = (\rho_P,\gamma_P,\nu_P)$. (N.161)

## Optional complex state

$Z = R + iQ = Ae^{i\theta}$. (N.162)

## Internal phase time

$\tau_i$ = Unwrap $\theta$ or $\int|\theta'|dt$. (N.163)

## Revision

$D_{k+1} = U_a(D_k,L_k,\mathscr{R}_k)$. (N.164)

## Full operator

$\mathfrak{D}_{TA,P}$
= UpdateLedger
∘ AuditResidual
∘ Gate
∘ Type
∘ Diagnose
∘ Project
∘ Declare. (N.165)

---

# N.33 The Entire Architecture in One Table

| Layer | Formal object | Question |
|---|---|---|
| Field | $\Sigma_0$ | What possibilities and relations exist before this analysis declares its object? |
| Declaration | D, P | What counts as inside, observable, relevant, and actionable? |

| Layer | Formal object | Question |
|---|---|---|
| Projection | $\hat{O}_{j,P}$ | What does this method make visible? |
| Period | p | At what closure depth does the object exist? |
| Function | g | Does it carry, move, constrain, or commit? |
| Actuation | a | Which primitive operation is active? |
| Regime | $\chi$ | Is feedback corrective, critical, or self-confirming? |
| Control state | $\Xi$ | How loaded, locked, and agitated is the system? |
| Complex state | Z | Is there an earned conjugate geometry? |
| Phase time | $\tau_i$ | Does phase provide useful internal order? |
| Gate | G | Has possibility become admitted consequence? |
| Residual | r, $\mathscr{R}$ | What remains unresolved? |
| Ledger | L | What consequence is preserved? |
| Transport | T | Does the claim survive another admissible frame? |
| Revision | $U_a$ | How should future declaration change without erasing history? |

---

# N.34 Final Master Runtime

The full runtime of the Periodic Grammar is:

$\Sigma_0$
→ Declare_P
→ Project_P
→ Type_{Period,Function,Actuation}
→ Diagnose_{$\chi$,$\Xi$}
→ Construct_Z only if eligible
→ Estimate_{$\theta$,$\tau_i$} only if earned
→ Test_Gate
→ Write_Trace + Residual
→ Update_Ledger
→ Test_Transport
→ Revise_Admissibly
→ $\Sigma'_P$. (N.166)

The same runtime can be written as a recursive loop:

Field_k
→ DeclaredWorld_k
→ Observation_k
→ Commitment_k
→ History_{k+1}
→ RevisedWorld_{k+1}. (N.167)

---

# N.35 Appendix N Conclusion

The article's full mathematical core is not:

Indicator → Prediction. (N.168)

It is:

Declaration
→ Projection
→ Typing
→ Regime Diagnosis
→ Gate
→ Trace + Residual
→ Ledger
→ Transport
→ Revision. (N.169)

The periodic structure adds:

Mark
→ Window
→ Structure
→ Event
→ Episode
→ World. (N.170)

The functional grammar adds:

Load
→ Motion under Constraint
→ Commitment. (N.171)

Proto-Eight adds:

Gradient, Gate, Boundary, Exchange, Trigger, Guidance, Memory, Focus. (N.172)

The advanced state hierarchy adds:

$\chi$
→ $\Xi$
→ optional Z
→ optional $\theta$
→ optional $\tau_i$. (N.173)

The governing scientific restriction is:

No layer may be promoted merely because the notation for that layer can be written. (N.174)

A Structure requires persistence.

An Event requires a gate.

An Episode requires a stable event grammar.

A World requires authoritative ledger and changed future admissibility.

A complex state requires conjugacy.

A phase clock requires improved internal ordering.

A self-revising observer requires trace-preserving, residual-honest, frame-robust revision.

The final compact law is:

ReadableMarketWorld
= DeclaredProjection

- TypedClosure
- GatedTrace
- VisibleResidual
- TransportedInvariance
- AdmissibleRevision. (N.175)

# Appendix O — Derivation Map: From One Assumption to Periodic Market Worlds

## O.1 Purpose

The article's market architecture did not arise merely by grouping familiar indicators into a new table.

Its deeper construction follows the source sequence:

One Assumption
→ One Operator
→ One Filtration
→ One Declaration
→ One Self-Revising Fractal
→ Recursive Depth
→ Time-Bearing Worlds. (O.1)

The present appendix shows how that sequence becomes, in the financial domain:

Undeclared Market Field
→ Protocol-Bound Observation
→ Ordered Disclosure
→ Gated Market Statement
→ Recursive Period Formation
→ Ledgered Market History
→ Self-Revising Market World. (O.2)

This is not presented as a deductive proof that Technical Analysis must take the exact form developed here.

It is a **typed instantiation** of the broader source architecture.

The source chain supplies the formal grammar.

The present article supplies the financial interpretation, six-period taxonomy, four functional families, Proto-Eight actuation crosswalk, and empirical testing programme.

---

## O.2 The Complete Inheritance Chain

The relationship may be displayed in one table.

| Source-stage | General function | Technical Analysis instantiation |
|---|---|---|
| One Assumption | no usable world appears without | raw market possibility is not yet a |

| Source-stage | General function | Technical Analysis instantiation |
|---|---|---|
| | distinction and operation | chart object |
| One Operator | disclose one bounded relation | indicator or observation operator |
| One Filtration | order repeated disclosure | marks, bars, structures, events, episodes |
| One Declaration | fix boundary, baseline, gate, trace, and residual | declare asset, timeframe, feature map, event rule |
| One Self-Revising Fractal | repeat the disclosure cycle while preserving trace | revise market interpretation without erasing prior claims |
| Recursive Depth | distinguish closure levels | Mark, Window, Structure, Event, Episode, World |
| Time-Bearing World | gated traces alter later admissibility | market, institutional, accounting, legal, and policy ledgers |

The full market derivation is:

$$
\begin{aligned}
&\Sigma_0 \\
&\rightarrow \hat{O}_P \\
&\rightarrow \mathscr{F}_P \\
&\rightarrow D_P \\
&\rightarrow G_P \\
&\rightarrow e_k + r_k \\
&\rightarrow L_{k+1} \\
&\rightarrow U_a \\
&\rightarrow D_{k+1}. \quad (O.3)
\end{aligned}
$$

where:

$\Sigma_0$ = undeclared market field;
$\hat{O}_P$ = protocol-bound observation operator;
$\mathscr{F}_P$ = filtration of disclosed states;
$D_P$ = declaration;
$G_P$ = gate;
$e_k$ = admitted event;
$r_k$ = residual;
$L_k$ = ledger;
$U_a$ = admissible revision.

---

# O.3 One Assumption: A Usable World Requires Distinction

The market contains an enormous number of simultaneously available relations:

- trades;
- quotes;
- orders;
- positions;
- cash flows;
- expectations;

- contracts;
- legal claims;
- policy constraints;
- possible futures.

No bounded observer uses all of them at once.

The practical starting point is therefore not:

A complete market object already exists and waits to be copied. (O.4)

It is:

A usable market object emerges only after a bounded distinction and observation protocol are declared. (O.5)

This does not mean that the external market is invented arbitrarily by the analyst.

It means that objects such as:

- a daily candle;
- a twenty-day trend;
- a breakout;
- a wave;
- a volatility regime;
- a default event

are not specified until one declares:

- the boundary;
- the data;
- the aggregation;
- the horizon;
- the gate;
- the observer authority.

Thus:

UndeclaredField ≠ OperationalObject. (O.6)

This is the first bridge from the general source programme into Technical Analysis.

---

# O.4 The Undeclared Financial Field

Let:

$\Sigma_0^{fin}$ (O.7)

denote the financial possibility field before the current protocol has selected its object.

It may contain:

$\Sigma_0^{fin}$
= {MarketData,Positions,Expectations,Contracts,Institutions,Alternatives,UnobservedStates}. (O.8)

The notation does not imply that all of these variables are stored in one physical location.

It represents the broader relational field from which an observer may disclose one bounded financial world.

Examples of potential disclosures include:

- order-book world;
- daily-chart world;
- valuation world;
- risk-management world;
- accounting world;
- legal-default world;
- policy world.

These worlds overlap.

They are not automatically identical.

---

# O.5 From One Assumption to One Operator

A bounded observer requires an operator that makes something visible.

Define:

$\hat{O}_{j,P}: \Sigma_0^{fin} \rightarrow X_{j,P}$. (O.9)

where:

j = method or observable;
P = declared protocol;
$X_{j,P}$ = disclosed object.

Examples include:

$\hat{O}_{trade} \rightarrow$ execution record. (O.10)

$\hat{O}_{bar} \rightarrow$ OHLCV candle. (O.11)

$\hat{O}_{MA} \rightarrow$ moving-average memory. (O.12)

$\hat{O}_{RSI} \rightarrow$ normalized directional relation. (O.13)

Ô_profile → transaction density across price. (O.14)

Ô_break → boundary-crossing candidate. (O.15)

Ô_CAPM → risk-adjusted valuation. (O.16)

The source progression from one assumption to one operator supplies the central methodological warning:

> Once the observer uses an operator, the resulting object must not be confused with the entire field from which it was projected.

In market notation:

Ô_{j,P}(Σ0^fin) ≠ Σ0^fin. (O.17)

Therefore:

Indicator ≠ Market. (O.18)

---

# O.6 The Operator Is Typed

A Technical Analysis operator should carry more than a name.

Its complete type is:

Ô_j
= (S_j,w_j,P_j,p_j,g_j,a_j). (O.19)

where:

S_j = source lineage;
w_j = operator word;
P_j = protocol;
p_j = closure period;
g_j = observational function;
a_j = Proto-Eight actuation role.

For a moving average:

S_MA = {ClosePrice}. (O.20)

w_MA = Smooth_n ∘ SelectClose. (O.21)

p_MA = Structure. (O.22)

g_MA = Load. (O.23)

a_MA = Memory. (O.24)

For a breakout close:

S_break = {Price,Boundary,Volume,Breadth}. (O.25)

w_break = Gate ∘ Confirm ∘ CompareBoundary. (O.26)

p_break = Event. (O.27)

g_break = Commitment. (O.28)

a_break = Gate. (O.29)

Typing clarifies what the operator does and what it cannot do alone.

---

# O.7 From One Operator to One Filtration

One isolated projection produces one observation.

Repeated projections under ordered information produce a filtration.

Let:

$\mathscr{F}_k$
$= \sigma(X_0, X_1, \ldots, X_k)$. (O.30)

Here σ denotes the information generated by observations available up to stage k.

In ordinary time-indexed market data:

$\mathscr{F}_t$
= information available by clock time t. (O.31)

But the article requires several forms of order:

- clock order;
- closure-period order;
- event order;
- phase order;
- ledger order.

A market filtration should therefore specify what kind of accumulation it represents.

---

# O.7.1 Mark filtration

$\mathscr{F}_k^{Mark}$
$= \sigma(Quotes_{0:k}, Trades_{0:k}, Orders_{0:k})$. (O.32)

# O.7.2 Window filtration

$\mathscr{F}_n^{Window}$
$= \sigma(Bar_{0:n})$. (O.33)

# O.7.3 Event filtration

$\mathscr{F}_m^{Event}$
$= \sigma(CommittedEvent_{0:m})$. (O.34)

## O.7.4 Ledger filtration

$\mathscr{F}_q^{Ledger}$
$= \sigma(L_0, L_1, \ldots, L_q)$. (O.35)

These filtrations are related but not interchangeable.

Many marks can occur without producing one admitted Event.

Many windows can pass without revising the institutional ledger.

---

# O.8 Why Filtration Matters to Technical Analysis

A method that uses future information violates its declared filtration.

For a valid online signal:

$Signal_t$ must be measurable with respect to $\mathscr{F}_t$. (O.36)

Symbolically:

$Signal_t \in \mathscr{F}_t$. (O.37)

A signal using:

- a future pivot;
- a later-revised anchor;
- the completed episode endpoint;
- the best later timeframe

is not available inside the original filtration.

It is a retrospective construction.

This is why the article repeatedly requires:

- prospective anchors;
- fixed horizons;
- preserved original claims;
- online phase estimation;
- immutable protocol versions.

The filtration is the formal boundary against hindsight.

---

# O.9 Filtration Does Not Yet Produce Commitment

A sequence of observations may accumulate without creating an admitted event.

For example:

- price approaches resistance;
- momentum rises;
- volume increases;
- RSI becomes elevated;
- one trade occurs above the line.

These observations belong to the filtration.

But the Event gate may remain open.

Therefore:

InformationAccumulation ≠ EventCommitment. (O.38)

A filtration tells the observer what is currently disclosed.

A declaration and gate determine what that disclosure is allowed to mean.

---

# O.10 From One Filtration to One Declaration

A filtration alone does not specify:

- which object is being studied;
- which boundary matters;
- which evidence is admissible;
- what counts as confirmation;
- what must remain residual.

The declaration supplies those commitments.

Define:

$$D_P = (q,\varphi,P,\hat{O}_P,G_P,T_P,R_P,I_P). \quad (O.39)$$

where:

$q$ = baseline;
$\varphi$ = feature map;
$P$ = observation protocol;
$\hat{O}_P$ = projection family;
$G_P$ = gate;
$T_P$ = trace rule;
$R_P$ = residual rule;
$I_P$ = invalidation rule.

The declared market world is:

W_P = Declare(Σ0^fin | D_P). (O.40)

---

## O.10.1 Example declaration

A daily breakout world may be declared as:

```
Asset:
NBE

Boundary:
£99.80–£100.00

Timeframe:
Daily

Scale:
Logarithmic

Displacement:
At least 0.75 ATR

Participation:
Relative volume above 1.50

Breadth:
At least 65%

Gate:
Daily close plus follow-through or retest

Residual:
Higher-frame conflict, absent value migration, crowding

Invalidation:
Close inside old range plus failed reclaim
```

The declaration converts a vague phrase such as:

“Price looks ready to break out” (O.41)

into a falsifiable market statement.

---

# O.11 Declaration Is Not Arbitrary Freedom

The observer may declare a protocol.

But not every declaration is equally useful.

A valid declaration should be:

- well formed;
- measurable;
- reproducible;
- trace-preserving;
- residual-honest;

- transportable;
- falsifiable.

Define:

$\mathcal{D}_{adm}$
$= \{D \mid WF \wedge MR \wedge TP \wedge RH \wedge FR \wedge ND\}$. (O.42)

where:

WF = well formed;
MR = measurable and reproducible;
TP = trace preserving;
RH = residual honest;
FR = frame robust;
ND = non-degenerate.

A declaration that fits every outcome by changing its meaning is not admissible.

---

# O.12 Declaration Creates the Gate

The gate is not appended after the analysis merely to increase confidence.

It is part of the declaration.

Define:

$G_P(X_k, L_k, \mathcal{R}_k)$
$\rightarrow (d_k, \alpha_k, e_k, r_k, m_k)$. (O.43)

where:

$d_k$ = decision;
$\alpha_k$ = admission strength;
$e_k$ = admitted event;
$r_k$ = residual;
$m_k$ = metadata.

Possible decisions include:

$d_k \in \{Admit, PartiallyAdmit, Defer, Reject, Ambiguous\}$. (O.44)

The gate converts:

PossibleTransition
$\rightarrow$ DeclaredConsequence. (O.45)

This is the formal point at which Technical Analysis leaves pure description and begins to create market history for the observer.

---

# O.13 The Gate Is a Boundary-Making Act

Before the gate, several futures may remain admissible.

Let:

$\Omega_k$ = set of currently admissible interpretations. (O.46)

After the gate:

$\Omega_{k+1} \subset \Omega_k$. (O.47)

The gate suppresses some branches and preserves others.

Define selection depth:

$$\sigma_k = -\ln[\text{Measure}(\Omega_k)/\text{Measure}(\Omega_0)]. \quad \text{(O.48)}$$

The gate therefore does more than label an observation.

It changes the declared possibility structure.

This is one reason commitment is a distinct functional family.

---

# O.14 Gate and Residual Must Be Written Together

A gate cannot observe everything.

When it admits one event, unresolved content remains.

The correct output is:

$G_P \rightarrow e_k + r_k$. (O.49)

not:

$G_P \rightarrow e_k$ and all uncertainty disappears. (O.50)

Residual may include:

- missing breadth;
- higher-frame resistance;
- hidden positioning;
- alternative wave count;
- unsettled legal status;
- unrecognized accounting loss;
- model error.

Thus:

Commitment ≠ Exhaustion. (O.51)

The residual is the formal mark of bounded observation.

---

# O.15 From One Declaration to One Self-Revising Fractal

A static declaration would fail as the market changes.

A mature observer must revise.

But unrestricted revision would erase falsifiability.

The revision operator is:

$$D_{k+1} = U_a(D_k, L_{k+1}, \mathscr{R}_{k+1}). \quad (O.52)$$

where $U_a$ is admissible revision.

The revision must preserve:

- the prior declaration;
- the prior projection;
- the prior gate decision;
- the residual;
- the reason for change.

This creates a self-revising observer that learns without rewriting its history.

---

# O.16 Why the Revision Is Fractal

The same disclosure cycle repeats at several closure depths.

At the Mark level:

Order
→ Execution Gate
→ Trade Trace
→ Residual Liquidity. (O.53)

At the Window level:

Marks
→ Bar Closure
→ OHLCV Trace
→ Lost Intrawindow Path. (O.54)

At the Structure level:

Windows
→ Persistence Gate
→ Trend or Boundary
→ Parameter and Frame Residual. (O.55)

At the Event level:

Structure Test
→ Commitment Gate
→ Breakout or Rejection
→ Fakeout Residual. (O.56)

At the Episode level:

Event Sequence
→ Completion Gate
→ Regime Transition
→ Alternative Branch Residual. (O.57)

At the World level:

Institutional Condition
→ Authoritative Recognition
→ Ledger Update
→ Legal, Accounting, or Policy Residual. (O.58)

The same grammar recurs:

Declare
→ Project
→ Gate
→ Trace
→ Residual
→ Revise. (O.59)

This repetition across closure depths is the self-similar or fractal aspect.

---

## O.17 Fractal Does Not Mean Exact Geometric Self-Similarity

The term “fractal” here does not require that:

- price charts possess one universal fractal dimension;
- every small pattern reproduces every large pattern;
- market prices obey exact scale invariance.

The relevant self-similarity is operational:

The same disclosure grammar reappears at different closure levels. (O.60)

The variables and authorities change.

The form of the cycle persists.

Thus:

OperationalFractality
≠ ExactGeometricFractality. (O.61)

---

## O.18 Recursive Depth Generates the Six Periods

The six periods can now be derived as successive closure depths.

## Depth 0 — Mark

The smallest selected market act is admitted.

## Depth 1 — Window

Several marks are aggregated and one terminal state is selected.

## Depth 2 — Structure

Several windows form a persistent relation.

## Depth 3 — Event

A structure is tested and one transition is admitted.

## Depth 4 — Episode

Several events form an ordered grammar.

## Depth 5 — World

Several episodes and institutional gates create a persistent system of future admissibility.

The period chain is:

$P_0$
$\rightarrow P_1$
$\rightarrow P_2$
$\rightarrow P_3$
$\rightarrow P_4$
$\rightarrow P_5$. (O.62)

or:

Mark
→ Window
→ Structure
→ Event
→ Episode
→ World. (O.63)

---

# O.19 The Periods Are Not Arbitrary Rows

Each new period requires a new closure operator.

Let:

$C_p$ = closure condition for period p. (O.64)

Then promotion occurs only when:

C_{p+1} = true. (O.65)

Examples:

C_Window = bar or session closes. (O.66)

C_Structure = persistent relation survives minimum duration or transport. (O.67)

C_Event = declared gate admits transition. (O.68)

C_Episode = ordered event grammar persists. (O.69)

C_World = authoritative ledger changes future admissibility. (O.70)

Therefore:

RepeatedObjects_p
≠ Object_{p+1} without closure. (O.71)

Many marks do not automatically constitute a meaningful candle.

Many candles do not automatically constitute a trend.

Many events do not automatically constitute a world.

---

# O.20 Why the Four Functional Families Recur

At every closure depth, four functions are required.

# Load

Something must be carried from the previous state.

# Motion

Something must change or relate.

# Constraint

The change must occur inside or across a boundary.

# Commitment

A gate must decide what becomes part of history.

Therefore the local cycle is:

L_p
→ M_p under C_p
→ G_p
→ e_p + r_p. (O.72)

The next level inherits:

$$L_{p+1} = H_p(L_p,e_p,r_p). \quad (O.73)$$

This is the origin of the proto-periodic recurrence.

The rows change.

The four functional requirements reappear.

---

## O.21 The Three Governance Rails

The four functional families do not sufficiently govern the whole architecture.

Three rails operate across all periods.

### Residual rail

What remains unclosed?

### Transport rail

What survives another admissible declaration?

### Ledger rail

What changes future interpretation and action?

Thus the extended periodic architecture is:

$$\text{TableCore} = \text{Period} \times \text{Function}. \quad (O.74)$$

$$\text{Governance} = \text{Residual} \times \text{Transport} \times \text{Ledger}. \quad (O.75)$$

A mature cell must therefore specify not only what it measures, but:

- its residual;
- its transport test;
- its ledger consequence.

---

## O.22 Proto-Eight as the Actuation Layer

The source Proto-Eight framework supplies eight primitive operational roles:

- Gradient;
- Gate;
- Boundary;

- Exchange;
- Trigger;
- Guidance;
- Memory;
- Focus.

These roles answer:

> How does the system generate and route the change that the four-family grammar later observes?

The integrated relation is:

ActuationRole
→ FunctionalObservation
→ PeriodClosure. (O.76)

For example:

Trigger
→ Motion
→ Event candidate. (O.77)

Gate
→ Commitment
→ admitted Event. (O.78)

Memory
→ Load
→ Structure. (O.79)

Boundary
→ Constraint
→ transition zone. (O.80)

---

# O.23 Proto-Eight Does Not Generate the World Alone

The eight roles may enable:

- potential;
- movement;
- transfer;
- retention;
- selection.

But a time-bearing world also requires:

- declaration;

- observer;
- gate authority;
- trace;
- residual;
- ledger;
- revision;
- environmental coupling.

Thus:

ActuationGrammar alone
≠ TimeBearingWorld. (O.81)

A fuller condition is:

World_P
= Actuation

- Declaration
- Projection
- Gate
- Trace
- Residual
- Ledger
- Backreaction. (O.82)

This is the precise relationship between Proto-Eight and 成界之學 inside the present article.

---

# O.24 From Recursive Depth to Time

Recursion alone is not yet time.

A computation may repeat without producing consequential history.

Time-bearing order emerges when:

1. a gate differentiates before and after;
2. the result enters persistent trace;
3. the trace constrains later states;
4. the order cannot be erased without changing the world.

Define ledger order:

$L_0 \prec L_1 \prec L_2 \prec \dots$ (O.83)

The event index is:

$k = 0,1,2,\dots$ (O.84)

This k is not identical to calendar time.

It counts committed state transitions.

---

# O.25 Four Times Revisited

The derivation reveals four different time coordinates.

## External duration

$t$. (O.85)

## Phase orientation

$\theta$. (O.86)

## Accumulated internal phase depth

$\tau_i$. (O.87)

## Ledger-event order

$k$. (O.88)

These coordinates answer different questions.

| Coordinate | Question |
|---|---|
| $t$ | How much external duration passed? |
| $\theta$ | What is the current orientation of the conjugate state? |
| $\tau_i$ | How much internal phase progression accumulated? |
| $k$ | How many consequential gates were committed? |

A market world may require all four.

They must not be substituted for one another without evidence.

---

# O.26 Phase Is Not Required for Ledger Time

A market can possess event order without possessing a valid complex phase.

For example:

Order submitted
→ order executed

→ trade settled
→ margin breached
→ collateral seized. (O.89)

This sequence is time-bearing through gates and ledgers.

No complex state is required.

Therefore:

LedgerTime does not imply ComplexPhase. (O.90)

Conversely, a complex oscillator may rotate without creating history.

Therefore:

ComplexPhase does not imply LedgerTime. (O.91)

A full phase-bearing world requires both:

PhaseProgression

- PhaseSensitiveGate
- PersistentTrace. (O.92)

---

# O.27 From Ledger to Causality

A causal market claim requires more than temporal sequence.

The source architecture motivates a constrained form:

Earlier committed trace changes the admissible transition structure of later states. (O.93)

Let:

L_k = ledger after event k. (O.94)

Let:

$\mathcal{A}_{k+1}$ = admissible actions after event k. (O.95)

Then:

$\mathcal{A}_{k+1}$ = F(L_k,CurrentState,Rules). (O.96)

If:

L_k changes $\mathcal{A}_{k+1}$, (O.97)

the trace has causal consequence inside the declared world.

Examples include:

- settlement changes ownership;
- margin breach forces liquidation;

- default changes legal rights;
- index inclusion changes fund obligations;
- accounting impairment changes capital;
- prior fakeout changes participant positioning.

This is stronger than saying that the event happened first.

---

# O.28 World Formation in Finance

A financial World is not merely a large chart pattern.

It requires:

World_P
= Boundary_P

- Observer_P
- GateAuthority_P
- Ledger_P
- ActionRules_P
- Backreaction_P. (O.98)

Examples include:

# Trading world

- exchange rules;
- matching engine;
- settlement;
- market participants.

# Accounting world

- reporting entity;
- valuation standard;
- recognition rule;
- audited ledger.

# Legal world

- jurisdiction;

- contractual relation;
- adjudicating authority;
- enforceable consequence.

## Policy world

- mandate;
- eligible instruments;
- intervention rule;
- public authority.

Worlds can disagree because they declare different:

- objects;
- gates;
- authorities;
- consequences.

---

# O.29 Cross-World Transport

A market-price event may be transported into another world.

Let:

T_{Market→Accounting}. (O.99)

T_{Market→Legal}. (O.100)

T_{Market→Risk}. (O.101)

T_{Market→Policy}. (O.102)

For example:

Market loss
→ possible accounting impairment. (O.103)

Price collapse
→ possible collateral breach. (O.104)

Spread widening
→ possible default warning. (O.105)

But the destination gate remains distinct.

Thus:

MarketRecognition
≠ AccountingRecognition

≠ LegalRecognition. (O.106)

Cross-world residual is the mismatch between expected and actual recognition.

---

# O.30 Self-Referential Market Worlds

Technical Analysis is especially relevant because the market observes its own trace.

The loop is:

Price
→ Indicator
→ Interpretation
→ Order
→ NewPrice. (O.107)

More fully:

Σ_k
→ Ô_P
→ Claim_k
→ Action_k
→ Σ_{k+1}. (O.108)

When many observers share the same declaration:

- the boundary may gain structural mass;
- the trigger may attract more flow;
- the gate may become self-confirming;
- crowding may make failure more violent.

Therefore self-reference can create two opposite outcomes.

## Reflexive confirmation

Observation strengthens the relation.

## Reflexive inversion

Observation crowds and destabilizes the relation.

The direction must be tested.

It cannot be inferred from visibility alone.

---

# O.31 The Self-Revising Fractal in Market Form

The market observer repeatedly performs:

D_k

→ Ô_k
→ G_k
→ e_k + r_k
→ L_{k+1}
→ U_a
→ D_{k+1}. (O.109)

At the same time, the observed market may react:

Σ_k
→ Interpretation_k
→ Action_k
→ Σ_{k+1}. (O.110)

The coupled system is therefore:

ObserverRevision
↔ MarketBackreaction. (O.111)

A mature Technical Analysis platform should preserve both:

- how the market changed;
- how the observer's declaration changed.

Without the second ledger, model evolution becomes invisible.

---

# O.32 Why Revision Must Remain Bounded

A completely flexible analyst can always explain the past.

Such flexibility destroys scientific value.

Revision must therefore obey:

Complexity(D_{k+1})
− Complexity(D_k)
≤ EvidenceGain_k + Budget_k. (O.112)

A revision is suspicious when:

ComplexityGrowth ≫ OutOfSampleGain. (O.113)

Examples include:

- adding unlimited pattern exceptions;
- repeatedly moving Fibonacci anchors;
- changing wave degree after every failure;
- redefining χ after the outcome;
- rescaling Q until phase appears stable.

The self-revising fractal is not unrestricted adaptation.

It is governed revision.

---

# O.33 The Derivation of the Periodic Law

The article's central periodic law can now be reconstructed step by step.

## Step 1 — A prior trace exists

$L_p$. (O.114)

## Step 2 — The trace supplies current Load

$Load_p = Compile(L_p,\Sigma_p)$. (O.115)

## Step 3 — A Gradient or Trigger generates Motion

$Motion_p = M_p(Load_p,Environment_p)$. (O.116)

## Step 4 — Constraint shapes the admissible route

$Motion_p \in Constraint_p$. (O.117)

or the motion attempts to cross it.

## Step 5 — The Gate decides commitment

$G_p(Motion_p,Constraint_p,L_p,\mathscr{R}_p)$. (O.118)

## Step 6 — Trace and residual are written

$G_p \rightarrow e_p + r_p$. (O.119)

## Step 7 — The next period inherits the result

$Load_{p+1}$
$= Inherit(e_p,r_p,L_p)$. (O.120)

Therefore:

$Load_p$
$\rightarrow Motion_p$ under $Constraint_p$
$\rightarrow Commitment_p$
$\rightarrow Trace_p + Residual_p$
$\rightarrow Load_{p+1}$. (O.121)

This is not merely a metaphorical sequence.

Each stage corresponds to a separately recordable research object.

---

# O.34 The Derivation of the Six-Period Table

Apply the periodic law recursively.

## Mark

Execution traces become Window load.

## Window

Closed bars become Structure load.

## Structure

Persistent relations become Event conditions.

## Event

Admitted transitions become Episode load.

## Episode

Persistent event grammars become World-level institutional evidence.

## World

Institutional rules backreact and alter later Marks.

The resulting cycle is:

Mark
→ Window
→ Structure
→ Event
→ Episode
→ World
→ NewMark. (O.122)

The cycle closes because the World changes:

- liquidity;
- order types;
- participant obligations;
- available capital;
- legal permissions;
- policy constraints.

The final World is not outside the market.

It modifies the next market field.

---

## O.35 The Derivation of the Four-Family Table

At every period:

1. prior history must be carried;
2. something must change;
3. change must be bounded;
4. a transition must be committed.

Thus the four columns are not chosen merely because four creates symmetry.

They correspond to four minimal jobs in a bounded disclosure process:

Load = inherited state. (O.123)

Motion = transformation. (O.124)

Constraint = admissibility structure. (O.125)

Commitment = selected consequence. (O.126)

A proposed alternative table must show that one or more of these jobs are:

- redundant;
- incorrectly separated;
- or insufficient.

This is empirically testable.

---

## O.36 The Derivation of the Three Governance Rails

The local cycle remains incomplete without three additional questions.

### Residual

What did the gate fail to close?

### Transport

Does the relation survive another legitimate declaration?

### Ledger

Does the result alter later admissibility?

Therefore:

MatureObservation
= LocalFunction

- Residual
- Transport
- Ledger. (O.127)

This explains why the governance rails are not additional columns.

They apply to every cell.

---

# O.37 The Derivation of χ

The market's feedback orientation determines how Load, Motion, Constraint, and Commitment interact.

Let:

F: signal pressure → structure. (O.128)

M: structure → signal pressure. (O.129)

The composed response may be:

- opposing;
- weak;
- reinforcing.

This is summarized by χ.

χ < 0 → corrective response. (O.130)

χ ≈ 0 → critical ambiguity. (O.131)

χ > 0 → self-confirming response. (O.132)

χ is therefore not another indicator family.

It is a relational signature of the cycle.

The operator formulation is:

C_χ
= [[0,F],[χM,0]]. (O.133)

Its exact square is:

C_χ²
= [[χFM,0],[0,χMF]]. (O.134)

Only under normalized inverse-like coupling does this reduce to:

C_χ² = χI. (O.135)

---

# O.38 The Derivation of Ξ

The declared market field contains many observables.

A control interface compresses those observables into:

Ξ_P = (ρ_P,γ_P,ν_P). (O.136)

The coordinates summarize:

ρ = loading;
γ = lock-in;
ν = agitation.

Ξ arises after declaration and projection:

Σ0
→ D_P
→ Σ_P
→ Compile_P
→ Ξ_P. (O.137)

Therefore Ξ is:

- protocol-bound;
- diagnostic;
- revisable.

It is not the pre-declared market itself.

---

# O.39 The Derivation of an Optional Complex State

A complex state requires a stronger relation.

Suppose two coordinates exist:

R = admitted or realized component. (O.138)

Q = independently meaningful conjugate component. (O.139)

Then:

Z = R + iQ. (O.140)

But Z earns priority only if the system supplies:

- compatible units;
- stable coupling;
- useful phase;
- better organization than a real pair.

The source hierarchy therefore becomes:

Field
→ Projection
→ RealPair
→ ComplexEligiblePair
→ PhaseDynamics
→ PhaseTime
→ PhaseSensitiveGate
→ TimeBearingWorld. (O.141)

Each arrow requires evidence.

---

# O.40 CAPM as the Local Calibration Atom

The CAPM construction demonstrates one exact local path.

Baseline value:

$A_t = CF_t/(1 + r_{base})^t$. (O.142)

Admitted CAPM value:

$R_t = CF_t/(1 + r_{CAPM})^t$. (O.143)

Conjugate coordinate:

$Q_t = \sqrt{(A_t^2 - R_t^2)}$. (O.144)

Complex state:

$Z_t = R_t + iQ_t$. (O.145)

Phase sensitivity:

$\partial R_t/\partial \theta_t = -Q_t$. (O.146)

This construction earns:

- dimensional compatibility;
- exact norm;
- exact generator relation.

It does not automatically earn:

- independent empirical Q;
- phase time;
- gate hazard;
- world backreaction.

CAPM therefore validates the possibility of a rigorous financial complex atom.

It does not validate every complex Technical Analysis proposal.

---

# O.41 From Phase to Time-Bearing Worlds

A complex phase becomes an internal time candidate when:

$\tau_i(t) = \text{Unwrap}[\theta(t)]$ (O.147)

or:

$\tau_i(t) = \int_0^t |\dot{\theta}(s)| ds.$ (O.148)

But $\tau_i$ gains scientific status only if:

$D_{phase} < \min(D_{calendar}, D_{event}, D_{other\ clocks}).$ (O.149)

A phase-sensitive event model additionally requires:

$$\text{Information}(\text{Gate} \mid \theta, \text{Controls}) > \text{Information}(\text{Gate} \mid \text{Controls}).$$ (O.150)

A time-bearing world requires:

$$\text{Gate} \rightarrow \text{Trace} \rightarrow \text{Ledger} \rightarrow \text{ChangedFutureDynamics}.$$ (O.151)

The full ladder is therefore:

$$\text{ComplexState} \rightarrow \text{PhaseOrder} \rightarrow \text{InternalTime} \rightarrow \text{PhaseSensitiveGate} \rightarrow \text{LedgeredWorld}.$$ (O.152)

---

# O.42 The Derivation of Technical Objectivity

Under the architecture, objectivity is not defined as the total absence of observers.

It is defined operationally through transport.

Let:

$C_P$ = claim under protocol P. (O.153)

Let:

$T_{P \rightarrow P'}(C_P)$ (O.154)

be its expected transformed form.

The claim survives when:

$\text{Dist}[T_{P \rightarrow P'}(C_P), C_{P'}] \leq \varepsilon_T.$ (O.155)

A relation becomes more objectively usable as it survives:

- timeframe change;
- scale change;
- bar change;
- benchmark change;
- observer change;
- ledger change.

Therefore:

OperationalObjectivity
∝ AdmissibleTransportSurvival. (O.156)

This does not create metaphysical observer independence.

It creates accountable cross-observer compatibility.

---

# O.43 The Derivation of Law

A pattern repeats inside one protocol.

A candidate law survives across admissible declarations.

Thus:

Pattern_P
= repeated relation under P. (O.157)

Law_candidate
= relation surviving {$P_1$,$P_2$,...,P_n}. (O.158)

The hierarchy is:

Appearance
→ Pattern
→ ProtocolRegularity
→ CrossFrameInvariant
→ LedgerRegularity
→ CandidateLaw. (O.159)

The Periodic Grammar is currently positioned mainly between:

- classificatory construction;
- protocol regularity;
- research programme.

Its strongest law-like claims remain to be tested.

---

# O.44 The Derivation of Falsification

Every layer introduces a distinct failure condition.

| Layer | Falsifier |
|---|---|
| Declaration | object remains ambiguous |
| Operator | output cannot be reproduced |
| Filtration | future information leaks into the signal |
| Gate | admission rule adds no value |
| Trace | event does not alter later interpretation |
| Residual | contradiction cannot be recorded reliably |
| Period | closure levels cannot be distinguished |
| Transport | relation collapses under legitimate reframing |
| $\chi$ | feedback signature adds no conditional value |
| $\Xi$ | compiled state adds no value beyond simpler controls |
| Z | real-pair model is equal or superior |
| $\tau_i$ | phase does not improve episode ordering |
| World | ledger history does not alter future dynamics |
| Revision | framework survives only through hindsight relabelling |

This distributed falsification structure prevents one successful equation from insulating the whole framework against failure.

---

# O.45 Source-Derived Chain Versus Present Article Derivation

The following chain is source-derived in overall structure:

Assumption
→ Operator
→ Filtration
→ Declaration
→ Self-Revising Fractal
→ Recursive Depth
→ Time-Bearing World. (O.160)

The following mappings are the present article's financial synthesis:

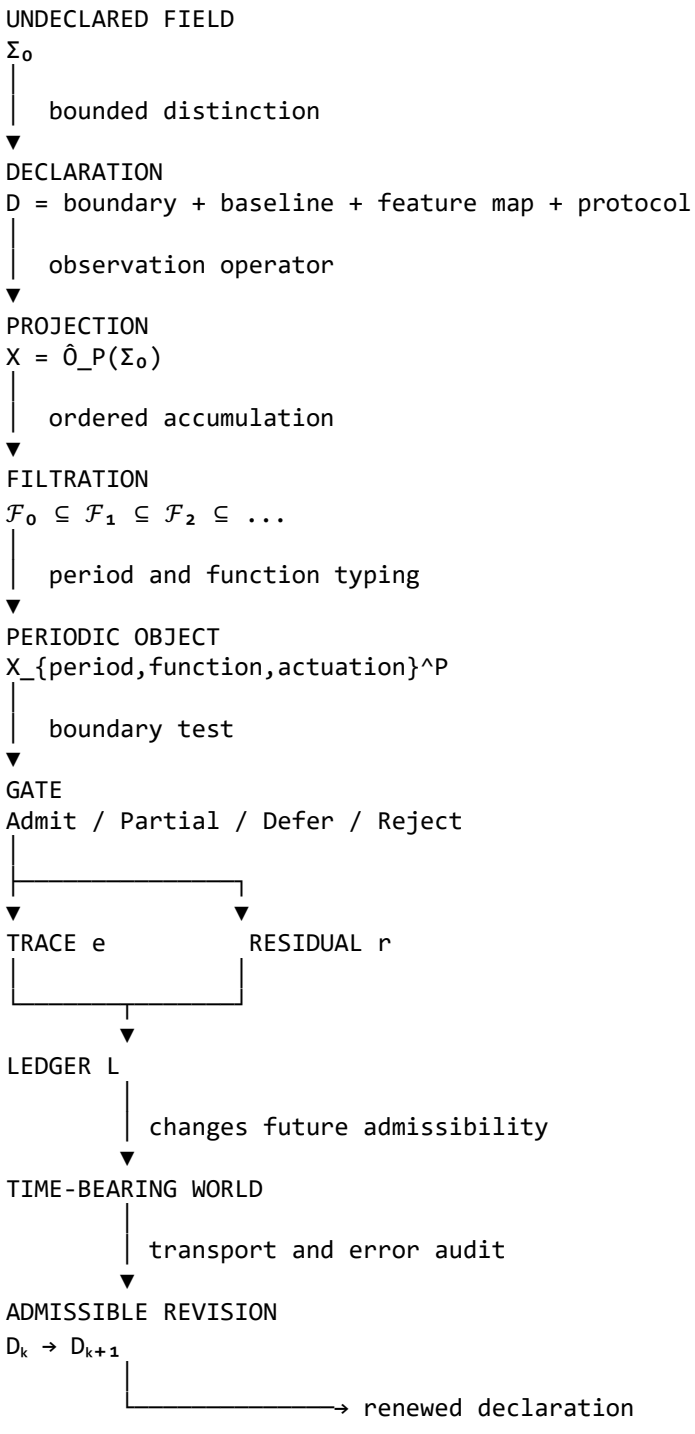
- operator → Technical Analysis method;
- recursive depth → six periods;
- recurring functions → four families;
- actuation grammar → Proto-Eight market roles;
- market objectivity → transport survival;
- market time → phase, event, and ledger distinctions;
- mature Technical Analysis → residual-bearing, self-revising observer.

Therefore the article should state:

The source sequence motivates and constrains the market architecture, but the 6 × 4 periodic table and its specific financial instrument assignments remain proposed domain constructions requiring empirical validation.

---

# O.46 The Entire Derivation in One Diagram




---

# O.47 The Entire Derivation in One Equation

A compact recursive expression is:

$$(D_{k+1}, L_{k+1}, \mathscr{R}_{k+1}) = \mathfrak{F}\{D_k, L_k, \mathscr{R}_k, \Sigma_0\}, \quad (O.161)$$

with:

$\mathfrak{F}$
= U_a
∘ TransportAudit
∘ LedgerUpdate
∘ ResidualAudit
∘ Gate
∘ Type
∘ Diagnose
∘ Project
∘ Declare. (O.162)

The operator is recursive because the next declaration depends on the earlier ledger and residual.

---

# O.48 The Market-Specific Reduction

For ordinary practical use, the full derivation reduces to:

1. Declare the market object.
2. Apply the indicator.
3. identify the closure period;
4. identify Load, Motion, Constraint, and Commitment;
5. test the gate;
6. record residual;
7. test another frame;
8. preserve the outcome;
9. revise without erasing the original claim.

In compact form:

Declare
→ Observe
→ Type
→ Gate
→ Record
→ Challenge
→ Revise. (O.163)

The advanced layers $\chi$, $\Xi$, Z, and $\tau_i$ should be added only when the research question requires them.

---

# O.49 The Strongest Coherent Claim

The strongest coherent theoretical claim supported by the integrated architecture is:

> Technical Analysis can be reconstructed as a protocol-bound, recursively layered observation system in which operators disclose partial market relations, filtrations order those disclosures, declarations define their admissible meaning, gates convert selected transitions into trace, residuals preserve non-closure, ledgers carry consequence across recursive depths, and bounded revision updates the observer without erasing its history.

This statement is structural.

Its empirical components remain open to testing.

---

# O.50 The Strongest Empirical Programme

The corresponding empirical programme is:

1. verify that the six closure periods are distinguishable;
2. verify that the four functions improve classification;
3. test whether Proto-Eight roles improve failure diagnosis;
4. test whether explicit gates improve event calibration;
5. test whether residual burden predicts failure or revision;
6. test whether transport predicts persistence;
7. test $\chi$ and $\Xi$ against simpler regime models;
8. test complex states against real pairs;
9. test phase time against ordinary clocks;
10. test whether ledger events alter future market dynamics.

The programme succeeds even when stronger layers fail, provided the failure produces disciplined reduction.

---

# O.51 The Final Derivation Rule

The full source-to-market derivation can be compressed into one rule:

No market world is obtained by observation alone.

A usable market world requires:

Observation

- Declaration
- Gate
- Trace
- Residual

- Revision. (O.164)

A time-bearing market world additionally requires:

PersistentLedger

- ChangedFutureAdmissibility. (O.165)

A phase-bearing market world additionally requires:

EligibleComplexState

- StablePhaseOrder
- PhaseSensitiveGate. (O.166)

The hierarchy is therefore:

ReadableWorld
⊂ GatedWorld
⊂ LedgeredWorld
⊂ TimeBearingWorld. (O.167)

And, where complex phase is empirically earned:

PhaseBearingWorld
⊂ TimeBearingWorld. (O.168)

Complexity is not the foundation of the world.

Commitment and trace are.

---

# O.52 Appendix O Conclusion

The Periodic Grammar is the financial expression of a deeper recursive disclosure sequence.

Its logic begins before any indicator is calculated.

First, a bounded observer declares what counts as a market object.

Then an operator discloses one relation.

Repeated disclosures form a filtration.

A declaration determines how those disclosures may be interpreted.

A gate selects which transition becomes consequential.

Trace and residual enter the ledger together.

The ledger changes the meaning of later observations.

The observer revises without deleting prior failure.

The same grammar repeats from individual marks to institutional worlds.

The derivation is:

One Assumption
→ One Operator
→ One Filtration
→ One Declaration
→ One Self-Revising Fractal
→ Recursive Closure Depth
→ Time-Bearing Market Worlds. (O.169)

Its Technical Analysis form is:

Market Field
→ Indicator Projection
→ Periodic Typing
→ Gate
→ Trace + Residual
→ Ledger
→ Transport
→ Admissible Revision. (O.170)

Its deepest practical lesson is:

> A market interpretation becomes mature not when it produces the most elaborate signal, but when it can declare its object, preserve its failures, survive legitimate reframing, and show exactly how observation becomes consequential history.

# Appendix P — The Periodic Market Runtime Kernel

# P.1 Purpose

The preceding appendices defined:

- the conceptual architecture;
- the six closure periods;
- the four functional families;
- the Proto-Eight actuation roles;
- the gate–trace–residual–ledger cycle;
- the transport and falsification rules;
- the conditions for complex phase and internal time.

This appendix converts that architecture into an executable research kernel.

Its task is not to create another indicator.

Its task is to compile an analytical requirement such as:

> Determine whether this apparent breakout represents a local price crossing, an admitted Event, an Episode transition, or a broader institutional World change.

into a stable sequence of operations with:

- declared inputs;

- typed intermediate states;
- explicit gates;
- residual outputs;
- immutable trace;
- transport tests;
- bounded revision.

The source Runtime Kernels framework argues that requirement-to-kernel conversion is a **semantic compilation problem**, not a prompt-rewriting problem. The output should be a compact intermediate representation that preserves intent, boundaries, execution order, output contract, and residual audit.

The market runtime follows the same principle:

RawMarketQuestion
→ DeclaredAnalysisIntent
→ MarketKernelIR
→ ExecutableAnalysis
→ AuditTrace. (P.1)

---

# P.2 Kernel Versus Indicator

An indicator transforms data into one observable:

Indicator_j
= Ô_{j,P}(Σ). (P.2)

A runtime kernel governs the entire analytical process:

K_P
= {Declaration,Projection,Typing,Diagnosis,Gate,Residual,Ledger,Transport,Revision}. (P.3)

The difference is fundamental.

An indicator may output:

RSI = 74. (P.4)

A runtime kernel asks:

- Under which protocol?
- At which closure period?
- What function does RSI perform?
- Which χ regime is assumed?
- Which boundary is being tested?
- What independent evidence is missing?

- Has an Event gate occurred?
- What residual remains?
- What would invalidate the interpretation?
- Does the claim survive another timeframe?

Therefore:

IndicatorOutput ⊂ RuntimeState. (P.5)

The runtime kernel does not replace indicators.

It prevents their outputs from being promoted beyond what they actually establish.

---

# P.3 Kernel Compilation Is Required

A loose analytical request may contain several hidden ambiguities.

For example:

> Is this breakout real?

The request does not yet specify:

- asset;
- market;
- timeframe;
- price scale;
- boundary;
- boundary construction;
- definition of breakout;
- volume baseline;
- breadth universe;
- retest rule;
- outcome horizon;
- invalidation;
- authority of the relevant close.

The compiler must convert the loose request into an executable declaration.

Define:

K_P = Compile(RawQuestion | DomainRules,EvidenceRules,RiskClass). (P.6)

The compiler output should contain:

$$K_P = (D_P, Data_P, Operators_P, Types_P, StateMachine_P, Gate_P, Ledger_P, Residual_P, Transport_P \quad (P.7)$$

This follows the Runtime Kernels principle that broad requirements should be transformed into a compact, auditable intermediate representation rather than merely rewritten into more elaborate language.

---

# P.4 Suitability Gate

Not every market question needs the full kernel.

A suitability gate should first determine whether the task requires:

- a simple calculation;
- an ordinary chart description;
- a structured diagnostic;
- a governed Event analysis;
- a full Episode or World model.

Define:

$$Suitability(R) \in \{Simple, Diagnostic, EventKernel, EpisodeKernel, WorldKernel, ResearchKernel\}. \quad (P.8)$$

---

# P.4.1 Simple

Examples:

- calculate a moving average;
- report current RSI;
- compute ATR-normalized displacement.

Required output:

value + calculation protocol.

No full ledger architecture is necessary.

---

# P.4.2 Diagnostic

Examples:

- explain what an indicator is measuring;

- compare price momentum with breadth;
- classify a support zone.

Required output:

period + function + missing variables + residual.

---

## P.4.3 Event kernel

Examples:

- determine whether a breakout is admitted;
- test whether support has failed;
- audit a divergence reversal.

Required output:

candidate + gate + residual + invalidation + transport.

---

## P.4.4 Episode kernel

Examples:

- determine whether a trend has ended;
- test an Elliott Wave endpoint;
- classify a range-to-expansion transition.

Required output:

event grammar + branch alternatives + completion gate + persistence.

---

## P.4.5 World kernel

Examples:

- determine whether a price event has become a default;
- reconcile market, accounting, and legal recognition;
- assess index-inclusion backreaction.

Required output:

authority + ledger + changed admissibility + institutional residual.

---

## P.4.6 Research kernel

Examples:

- test a candidate Q channel;
- validate phase time;
- compare χ-conditioned models;
- build a residual-history database.

Required output:

hypothesis + benchmarks + train/test protocol + falsifier + reduction rule.

The suitability rule is:

Use no more architecture than the scientific task requires. (P.9)

This is consistent with the source instruction that kernel methods should not be used decoratively when a plain prompt or simple procedure is sufficient.

---

# P.5 Market Kernel Input Classes

The compiler should classify the request before building the runtime.

# P.5.1 Class A — Indicator interpretation

Input:

> What does this RSI reading mean?

Compiled task:

Interpret one projected Structure-level Motion variable under a declared χ regime.

---

# P.5.2 Class B — Boundary test

Input:

> Has resistance broken?

Compiled task:

Evaluate an Event candidate generated by Motion across a Structure-level Constraint.

---

# P.5.3 Class C — Confirmation bundle

Input:

> Do volume, breadth, and price confirm each other?

Compiled task:

Test functional and source independence across Load, Motion, Constraint, and Commitment.

---

## P.5.4 Class D — Episode segmentation

Input:

> Has the trend ended?

Compiled task:

Compare old and candidate Episode grammars and apply a completion gate.

---

## P.5.5 Class E — Institutional transition

Input:

> Has the market decline become a credit or accounting event?

Compiled task:

Transport the market claim into risk, accounting, contractual, and legal ledgers.

---

## P.5.6 Class F — Complex-state proposal

Input:

> Can breadth pressure be treated as Q?

Compiled task:

Apply complex-eligibility tests and compare with a real-pair model.

---

## P.5.7 Class G — Phase-time proposal

Input:

> Does phase locate breakout maturity better than calendar time?

Compiled task:

Compare $\tau_i$ against calendar, event-count, volume, volatility, and flexible internal clocks.

Incorrect input classification produces an incorrect kernel.

A request for one indicator value should not be expanded into a speculative market ontology.

A request for an institutional state should not be answered from one chart.

---

# P.6 Market Kernel Intermediate Representation

The kernel intermediate representation is:

MarketKernelIR
= {Intent,Boundary,Data,Operators,Types,States,Gates,Residuals,Transport,Outputs,Revision}. (P.10)

Each field is mandatory for governed Event analysis.

---

# P.6.1 Intent

```
objective:
claim_to_test:
decision_context:
audience:
success_condition:
maximum_permitted_claim:
```

---

# P.6.2 Boundary

```
asset:
universe:
venue:
timeframe:
scale:
session:
price_field:
structural_boundary:
observer_authority:
```

---

# P.6.3 Data

```
raw_sources:
adjustment_rules:
availability_timestamps:
revision_status:
missing_data_policy:
data_quality_flags:
```

---

# P.6.4 Operators

```
feature_name:
source_lineage:
operator_word:
parameters:
output_units:
uncertainty:
```

---

# P.6.5 Types

```
closure_period:
functional_family:
proto_eight_role:
claim_class:
model_status:
```

## P.6.6 Gate

```
candidate_condition:
admission_conditions:
partial_admission_rule:
defer_rule:
rejection_rule:
invalidation_rule:
gate_authority:
```

---

## P.6.7 Residual

```
residual_type:
severity:
directional_relevance:
persistence:
resolution_condition:
claim_effect:
```

---

## P.6.8 Transport

```
source_protocol:
target_protocol:
transformation:
expected_target_form:
tolerance:
observed_target_form:
status:
transport_residual:
```

---

## P.6.9 Output

```
current_state:
admission_status:
highest_supported_claim:
open_residual:
invalidation:
next_required_evidence:
```

---

## P.6.10 Revision

```
previous_protocol_version:
triggering_evidence:
permitted_change:
prohibited_change:
new_protocol_version:
trace_preservation_check:
```

---

# P.7 Runtime Kernel Definition

The executable market kernel is:

$$K_{market,P} = \{D_P, O_P, Y_P, S_P, G_P, R_P, L_P, T_P, U_P\}. \quad (P.11)$$

where:

$D_P$ = declaration contract;
$O_P$ = operator set;
$Y_P$ = typing rules;
$S_P$ = state machine;
$G_P$ = gate engine;
$R_P$ = residual engine;
$L_P$ = ledger engine;
$T_P$ = transport engine;
$U_P$ = admissible revision engine.

The runtime is:

$$\text{Runtime}_{market,P} = U_P \circ T_P \circ L_P \circ R_P \circ G_P \circ S_P \circ Y_P \circ O_P \circ D_P. \quad (P.12)$$

Read from right to left:

1. declare the world;
2. compute projections;
3. type the objects;
4. update the state;
5. apply the gate;
6. attach residual;
7. write ledger;
8. test transport;
9. revise only if authorized.

This mirrors the source declared disclosure operator:

Declare
→ Project
→ Gate
→ Trace + Residual
→ Ledger, (P.13)

which makes a field readable and auditable under a bounded protocol.

---

# P.8 Runtime State Machine

The kernel should not reason through unstructured prose alone.

It should maintain an explicit analytical state.

Define:

$s_k \in \mathcal{S}_{market}$. (P.14)

where:

$\mathcal{S}_{market} = \{Undeclared, Declared, Projected, Typed, Candidate, Deferred, PartiallyAdmitted, Admitted, Accepted, \dots\}$ (P.15)

---

# P.8.1 Undeclared

The analytical object is not operationally specified.

No Event claim is permitted.

---

# P.8.2 Declared

Protocol, boundary, feature map, gate, residual, and horizon are fixed.

---

# P.8.3 Projected

Indicators and observables have been calculated.

---

# P.8.4 Typed

Each observable has been assigned:

- closure period;
- functional family;
- actuation role;
- claim class.

---

# P.8.5 Candidate

The minimum condition for a possible transition has occurred.

Example:

price crossed resistance.

---

# P.8.6 Deferred

Evidence is insufficient or the authoritative gate remains pending.

## P.8.7 Partially admitted

Some commitment conditions pass, but material residual remains.

---

## P.8.8 Admitted

The declared Event gate passes.

---

## P.8.9 Accepted

The admitted Event has persisted or survived retest sufficiently to become durable Structure.

---

## P.8.10 Invalidated

The predeclared invalidation condition passes.

---

## P.8.11 Promoted

The object passes the next closure condition.

Examples:

- Event → Episode;
- Episode → World.

---

## P.8.12 Revised

The declaration changes through an admissible, trace-preserving rule.

---

## P.8.13 Closed

The analytical horizon ends and the original claim receives a final outcome label.

---

# P.9 Allowed State Transitions

The runtime should forbid arbitrary state jumps.

A simplified transition graph is:

Undeclared
→ Declared
→ Projected

→ Typed
→ Candidate. (P.16)

From Candidate:

Candidate
→ Deferred. (P.17)

Candidate
→ PartiallyAdmitted. (P.18)

Candidate
→ Rejected. (P.19)

From PartiallyAdmitted:

PartiallyAdmitted
→ Admitted. (P.20)

PartiallyAdmitted
→ Invalidated. (P.21)

PartiallyAdmitted
→ Deferred. (P.22)

From Admitted:

Admitted
→ Accepted. (P.23)

Admitted
→ Invalidated. (P.24)

Admitted
→ Promoted. (P.25)

Any active state may enter:

ActiveState
→ Revised (P.26)

only when the revision gate passes.

The following jumps are prohibited:

Projected → Episode. (P.27)

IndicatorReading → World. (P.28)

Candidate → TimeBearingWorld. (P.29)

ComplexNotation → PhaseClock. (P.30)

The state machine enforces:

ClaimStrength ≤ ClosureState. (P.31)

---

# P.10 Runtime Invariants

An invariant is a condition that must hold throughout execution.

# P.10.1 Protocol invariant

Every claim must carry a protocol identifier.

Claim_k → Protocol_ID,k. (P.32)

---

# P.10.2 Time-availability invariant

No feature may use information unavailable at the evaluation time.

Feature_t ∈ $\mathscr{F}$_t. (P.33)

---

# P.10.3 Original-trace invariant

The first recorded claim cannot be overwritten.

OriginalClaim_k remains immutable. (P.34)

---

# P.10.4 Residual invariant

Every nontrivial gate output must include a residual record.

Gate_k → Event_k + Residual_k. (P.35)

---

# P.10.5 Promotion invariant

A higher-period state requires its own closure condition.

Promote_{p→p+1}
⇒ Closure_{p+1} = true. (P.36)

---

# P.10.6 Authority invariant

A gate may commit only within its declared authority.

GateAuthority_P
≥ AuthorityRequired(Event). (P.37)

A trader's opinion cannot commit an accounting impairment.

A market close cannot issue a legal judgment.

---

# P.10.7 Complex-eligibility invariant

No phase object may be emitted before R and Q pass eligibility.

Emit(θ)
⇒ ComplexEligibility = Passed. (P.38)

---

# P.10.8 Revision invariant

Revision may alter future interpretation but not past trace.

D_{k+1} may change; L_k may not be rewritten. (P.39)

---

# P.11 Protocol Identity and Versioning

Two apparently identical signals may belong to different declared worlds.

A protocol identifier should therefore be constructed.

Define:

Protocol_ID
=
Hash(Asset,Universe,Venue,Boundary,Timeframe,Scale,BarRule,FeatureMap,GateRule,ResidualR
(P.40)

The CAPM implementation source similarly recommends carrying a protocol identifier so that states calculated under incompatible cash-flow, discount, metric, orientation, or gate rules are not compared directly without reconciliation.

---

# P.11.1 Protocol version

Every authorized change creates a new version:

P_v1
→ P_v2. (P.41)

The version record should include:

```
protocol_id:
version:
effective_time:
parent_version:
change_reason:
changed_fields:
unchanged_invariants:
author:
approval:
```

---

# P.11.2 Reproducibility rule

A historical signal must be reproducible using:

- original protocol version;
- original data version;
- original feature code;
- original parameter set.

Therefore:

Reproduce(Claim_k)
= Run(DataVersion_k,ProtocolVersion_k,CodeVersion_k). (P.42)

---

# P.12 Operator Registry

Every operator should be registered as a typed instrument.

```
operator_id:
name:
source_lineage:
operator_word:
parameters:
units:
primary_period:
primary_family:
proto_eight_role:
assumed_regime:
known_failure_modes:
required_companions:
```

---

# P.12.1 Example — Moving average

```
operator_id: OP-MA-020
name: EMA20
source_lineage: adjusted_close
operator_word: exponential_smoothing
units: price
primary_period: Structure
primary_family: Load
proto_eight_role: Memory
assumed_regime: none
known_failure_modes:
  - lag
  - whipsaw
  - parameter sensitivity
required_companions:
  - participation
  - boundary
  - event gate
```

---

# P.12.2 Example — RSI

```
operator_id: OP-RSI-014
name: RSI14
source_lineage: adjusted_close_returns
```

```
operator_word: bounded_directional_normalization
units: index_0_100
primary_period: Structure
primary_family: Motion
proto_eight_role: Guidance
assumed_regime: corrective unless tested otherwise
known_failure_modes:
  - persistence in trend
  - threshold rigidity
  - same-source redundancy
required_companions:
  - χ estimate
  - structural boundary
  - reversal gate
```

---

# P.13 Claim Registry

A claim is not merely free text.

It is a versioned object:

Claim_k
= (Content,Class,Protocol,State,Authority,Horizon,Invalidation). (P.43)

A claim record should contain:

```
claim_id:
claim_text:
claim_class:
protocol_id:
created_at:
available_information_time:
closure_period:
functional_family:
state:
gate_id:
highest_supported_level:
outcome_horizon:
invalidation_rule:
status:
```

---

# P.13.1 Claim classes

Use:

- appearance;
- structure;
- warning;
- candidate event;
- admitted event;
- accepted structure;
- episode transition;

- world transition;
- complex-state claim;
- phase-time claim.

The state machine must reject mismatches such as:

claim_class = world_transition
while
closure_state = candidate_event. (P.44)

---

# P.14 Gate Engine

# P.14.1 Gate input

The gate receives:

G_P(c_k,X_k,L_k,$\mathscr{R}$_k,A_k). (P.45)

where:

c_k = candidate event;
X_k = current projected state;
L_k = ledger;
$\mathscr{R}$_k = residual register;
A_k = authority metadata.

---

# P.14.2 Gate output

The output is:

G_P → (d_k,α_k,e_k,r_k,m_k). (P.46)

where:

d_k = decision;
α_k = admission strength;
e_k = admitted trace;
r_k = attached residual;
m_k = gate evidence.

---

# P.14.3 Gate evidence vector

For a breakout:

g_break
= (Cross,Close,Displacement,Volume,Breadth,Retest,FollowThrough,HigherFrame). (P.47)

A weighted score may be:

S_G = $w^{T}$g_break. (P.48)

But the vector must remain available.

A scalar score alone cannot show whether a missing component is:

- low importance;
- pending;
- or directly contradictory.

---

# P.14.4 Gate output matrix

| Admission strength | Residual burden | Runtime status |
|---|---|---|
| low | low | weak candidate or irrelevant |
| low | high | rejected or unresolved candidate |
| high | low | admitted robust event |
| high | high | admitted but fragile event |

This prevents the kernel from equating:

HighAdmission = CompleteResolution. (P.49)

The CAPM source similarly distinguishes admitted economic consequence from the gate residual that remains economically, legally, or institutionally unresolved.

---

# P.15 Gate Hysteresis

Some states should use different entry and exit rules.

Let:

$\theta_{enter} > \theta_{exit}$. (P.50)

For a breakout:

- entry may require a close 1 ATR above resistance;
- invalidation may require a close below the zone plus failed reclaim.

For a credit downgrade:

- downgrade may occur at one threshold;
- upgrade may require a stronger recovery condition.

Gate hysteresis represents path dependence:

CurrentState
≠ Function(CurrentMeasurement alone). (P.51)

The current state also depends on:

- prior state;

- earlier gates;
- probation period;
- residual history.

The CAPM financial-world source explicitly notes that gate hysteresis and gate-memory make the current status dependent on ledger history rather than on the present reading alone.

---

# P.16 Gate Memory

Define gate memory:

$$m_{k+1} = H(m_k,e_k,r_k,L_k). \quad (P.52)$$

The gate becomes:

$$G_P(X_k,L_k,\mathscr{R}_k,m_k). \quad (P.53)$$

Gate memory may include:

- number of prior failed breaks;
- previous covenant breaches;
- probation state;
- cumulative impairment;
- prior false positives;
- recent model instability.

Example:

A third failed breakout at the same boundary should not necessarily be analysed as though no prior tests occurred.

The failed Events have become Load.

---

# P.17 Residual Engine

# P.17.1 Residual object

Residual should be represented as:

$$r_j = (\text{Type},\text{Source},\text{Severity},\text{Persistence},\text{Direction},\text{Period},\text{ResolutionRule},\text{ClaimEffect}). \quad (P.54)$$

---

# P.17.2 Residual types

The minimum ontology includes:

- missing confirmation;
- direct contradiction;
- frame conflict;
- boundary uncertainty;
- regime uncertainty;
- data uncertainty;
- liquidity or positioning;
- institutional uncertainty;
- branch uncertainty;
- model inadequacy;
- transport mismatch.

---

# P.17.3 Residual register

The residual register evolves:

$\mathscr{R}_{k+1}$
$= D_{\mathscr{R}}(\mathscr{R}_k)$

- NewResidual_k
  – ResolvedResidual_k
  – DissipatedResidual_k. (P.55)

Residual state may be:

- open;
- monitoring;
- resolved;
- dissipated;
- converted;
- invalidating.

---

# P.17.4 Residual conversion

Residual may become:

Residual
→ NewLoad. (P.56)

Residual

→ OpposingEvent. (P.57)

Residual
→ RevisionTrigger. (P.58)

Residual
→ WorldCrisis. (P.59)

Example:

A weak-breadth residual may resolve as more components join.

Alternatively, it may convert into a failed-breakout Event.

---

# P.18 Ledger Engine

# P.18.1 Ledger object

A ledger record is:

L_k
= (Claim,Gate,Trace,Residual,Authority,Protocol,Time,Outcome). (P.60)

A strong ledger should preserve:

- what was claimed;
- what evidence was available;
- what gate operated;
- which authority applied it;
- what remained residual;
- what later happened;
- whether the protocol changed.

---

# P.18.2 Trace requirements

A trace should possess:

### Persistence

It remains accessible after the event.

### Accessibility

Authorized observers can inspect or reproduce it.

### Consequence

It influences later interpretation, rights, obligations, or actions.

The source financial-world framework uses these properties to distinguish a consequential trace from a temporary calculation.

---

# P.18.3 Immutable event record

```
event_id:
claim_id:
protocol_id:
candidate_time:
gate_time:
gate_decision:
admission_strength:
evidence_snapshot:
residual_ids:
authority:
invalidation:
outcome_horizon:
```

This record should be append-only.

Later revisions create new records rather than editing the original Event.

---

# P.19 Transport Engine

# P.19.1 Transport task

For claim C_P, define:

T_{P→P′}(C_P) = expected target-frame claim. (P.61)

The observed target claim is:

C_{P′}. (P.62)

The transport residual is:

r_T
= C_{P′} – T_{P→P′}(C_P). (P.63)

---

# P.19.2 Standard transports

The engine should support the relevant subset of:

- timeframe;
- arithmetic to logarithmic scale;
- raw to volatility-normalized price;
- time bars to volume bars;
- close to settlement;

- capitalization-weighted to equal-weight;
- one benchmark to another;
- one pivot rule to another;
- price to volume profile;
- price to breadth;
- cash to derivative market;
- market to accounting or legal authority.

The Technical Analysis source explicitly defines stronger claims as those surviving admissible reframing across such protocols rather than merely accumulating more indicators.

---

# P.19.3 Transport record

```
transport_id:
claim_id:
source_protocol:
target_protocol:
operator:
expected_form:
observed_form:
distance_metric:
tolerance:
status:
residual:
claim_revision:
```

---

# P.19.4 Transport statuses

Use:

- survives;
- covariantly survives;
- partially survives;
- local only;
- fails;
- indeterminate;
- non-comparable.

A failed transport does not always invalidate the local claim.

It may constrain its scope.

---

# P.20 Period-Promotion Engine

## P.20.1 Promotion request

A claim may request promotion:

Promote(e_p,p→p+1). (P.64)

The engine evaluates:

- required higher-period closure;
- persistence;
- authority;
- residual;
- transport;
- alternative branch.

---

## P.20.2 Mark → Window

Required:

- defined bar or session closure.

Output:

Window trace.

---

## P.20.3 Window → Structure

Required:

- persistent relation;
- parameter stability;
- minimum duration or repeated observation.

Output:

Structure state.

---

## P.20.4 Structure → Event

Required:

- meaningful boundary interaction;
- declared commitment gate.

Output:

admitted Event.

---

## P.20.5 Event → Episode

Required:

- ordered sequence;
- failure of old grammar;
- persistence of new grammar;
- branch audit.

Output:

Episode transition.

---

## P.20.6 Episode → World

Required:

- authoritative institution;
- formal gate;
- persistent ledger;
- changed future admissibility.

Output:

World transition.

---

## P.20.7 Promotion result

The engine returns:

PromotionResult
= (Approved,Deferred,Rejected,Demoted). (P.65)

Promotion failure must not be repaired by merely changing the label.

---

# P.21 Regime Router

The regime router determines whether χ is needed.

For an indicator whose meaning is strongly regime-sensitive:

Useχ = true. (P.66)

Examples:

- RSI;
- stochastic;
- band extension;
- divergence;
- breakout continuation;
- mean-reversion rule.

The router estimates:

$$\chi_{P,h} \in \{\text{Corrective},\text{Critical},\text{SelfConfirming},\text{Unknown}\}. \quad (P.67)$$

The output should contain:

```
regime:
horizon:
estimator:
confidence:
supporting_evidence:
contradictory_evidence:
```

When confidence is low:

$$\chi = \text{Unknown}. \quad (P.68)$$

The runtime should not force every state into one of three confident labels.

---

# P.22 Ξ Router

The Ξ router is activated for questions involving:

- loading;
- lock-in;
- agitation;
- systemic stress;
- intervention selection.

It compiles:

$$\Xi_P = (\rho_P,\gamma_P,\nu_P). \quad (P.69)$$

The router should preserve source channels.

For example:

```
rho:
  volume_loading:
  leverage_loading:
  position_density:
  attention_loading:
```

```
gamma:
  liquidity_lock:
  collateral_lock:
  funding_lock:
  legal_lock:

nu:
  volatility:
  spread_instability:
  cancellation:
  failed_gate_rate:
```

A single Ξ score should not conceal these components.

Ξ is a protocol-bound interface, not a universal three-variable market ontology.

---

# P.23 Complex-Eligibility Router

The complex router should default to:

ComplexStatus = NotEvaluated. (P.70)

It should activate only when a candidate R–Q pair is explicitly proposed.

---

# P.23.1 Mandatory tests

1. Is R independently meaningful?
2. Is Q independently meaningful or structurally derived?
3. Are units compatible?
4. Is the metric justified?
5. Does a stable coupling exist?
6. Does phase survive scaling?
7. Does the complex model outperform a real pair?
8. Does phase improve the declared task?

---

# P.23.2 Router output

```
R_definition:
Q_definition:
unit_status:
metric_status:
generator_status:
scaling_status:
real_pair_benchmark:
phase_utility:
decision:
```

Possible decisions:

- Reject;
- RetainRealPair;
- ComplexEligible;
- PhaseBearing;
- PhaseTimeCandidate;
- TimeBearingWorldCandidate.

---

## P.23.3 Automatic reduction

If the real-pair benchmark is not defeated:

$Z \rightarrow (R,Q)$. (P.71)

If Q adds no independent value:

$(R,Q) \rightarrow R$. (P.72)

This follows the source phase framework's requirement that complexification be reduced when the complex representation adds no stable explanatory, predictive, or operational gain.

---

# P.24 Phase-Time Router

The phase-time router activates only after:

$\text{ComplexStatus} \geq \text{PhaseBearing}$. (P.73)

It then compares:

- calendar time t;
- normalized episode age;
- event count k;
- cumulative volume;
- cumulative volatility;
- selection depth σ;
- internal phase time $\tau_i$.

The primary test is:

$$D_{\text{phase}} < \min(D_{\text{calendar}}, D_{\text{event}}, D_{\text{volume}}, D_{\text{volatility}}, D_{\text{selection}}). \quad (P.74)$$

The router output is:

```
episode_family:
phase_definition:
```

```
unwrapping_rule:
comparison_clocks:
alignment_metric:
out_of_sample_result:
gate_hazard_gain:
decision:
```

Possible decisions:

- DescriptivePhaseOnly;
- SecondaryPhaseOrder;
- PhaseTimeSupported;
- PhaseSensitiveGateSupported;
- TimeBearingWorldSupported.

No stronger label may be emitted without passing the prior level.

---

# P.25 Backreaction Router

A World-level claim requires evidence that the ledger affects later dynamics.

Define:

BackreactionCandidate
= Gate

- Trace
- ObserverAction
- ChangedFutureState. (P.75)

The router asks:

1. Did an authorized ledger entry occur?
2. Did the entry change permitted or required action?
3. Did actors respond?
4. Did the response alter later market structure?
5. Is the effect distinguishable from the underlying event?

Examples include:

- margin call → forced sale;
- index inclusion → benchmark rebalancing;
- default → collateral enforcement;
- impairment → capital adjustment;

- public technical signal → crowded positioning.

The backreaction output must distinguish:

- plausible;
- associated;
- causally identified;
- unestablished.

---

# P.26 Runtime Error Taxonomy

The engine should not flatten every problem into “bad signal.”

Define:

$$\varepsilon_{runtime} = \varepsilon_{decl}$$

- $\varepsilon_{data}$
- $\varepsilon_{proj}$
- $\varepsilon_{type}$
- $\varepsilon_{regime}$
- $\varepsilon_{gate}$
- $\varepsilon_{resid}$
- $\varepsilon_{transport}$
- $\varepsilon_{promotion}$
- $\varepsilon_{revision}$. (P.76)

---

# P.26.1 Declaration error

Wrong boundary, horizon, or observer.

---

# P.26.2 Data error

Missing, revised, delayed, or incorrectly adjusted data.

---

# P.26.3 Projection error

Incorrect formula, parameter, or implementation.

---

## P.26.4 Typing error

Structure mislabelled as Event, or Event mislabelled as World.

---

## P.26.5 Regime error

Corrective method applied in self-confirming regime.

---

## P.26.6 Gate error

Crossing mistaken for commitment.

---

## P.26.7 Residual error

Contradictory evidence omitted or hidden.

---

## P.26.8 Transport error

Frames compared without a valid transformation.

---

## P.26.9 Promotion error

Higher closure claimed without a new gate.

---

## P.26.10 Revision error

Protocol changed retrospectively to preserve success.

The Technical Analysis source similarly decomposes failure into wrong signature, missing variable, weak gate, hidden residual, and protocol overfit.

---

# P.27 Minimal Runtime Pseudocode

```
function analyze_market_question(request, data):

    task = classify_input(request)

    suitability = evaluate_suitability(task)

    if suitability == "Simple":
        return simple_calculation(task, data)

    declaration = compile_declaration(request)

    assert declaration.is_complete()
```

```
protocol_id = hash_protocol(declaration)

projected = run_registered_operators(
    data_available_under(declaration.information_time),
    declaration.operator_set
)

typed = assign_period_function_actuation(projected)

regime = estimate_chi_if_required(typed, declaration)

control_state = compile_xi_if_required(typed, declaration)

candidate = detect_candidate_transition(
    typed,
    declaration.boundary
)

if candidate is None:
    return structure_level_report(
        declaration,
        typed,
        regime,
        control_state
    )

gate_result = apply_gate(
    candidate,
    declaration.gate_rule,
    prior_ledger,
    residual_register
)

residuals = audit_residual(
    declaration,
    projected,
    gate_result,
    prior_ledger
)

event_record = append_immutable_trace(
    declaration,
    candidate,
    gate_result,
    residuals
)

transports = run_transport_tests(
    event_record,
    declaration.transport_set
)

claim_level = determine_highest_supported_claim(
    gate_result,
    residuals,
    transports
)

promotion = test_period_promotion(
    event_record,
    claim_level
)

complex_result = None

if declaration.requests_complex_model:
```

```
        complex_result = run_complex_eligibility_router(
            declaration,
            projected
        )

    revision = propose_revision_only_if_triggered(
        declaration,
        event_record,
        residuals,
        transports
    )

    return governed_report(
        protocol_id,
        claim_level,
        gate_result,
        residuals,
        transports,
        promotion,
        complex_result,
        revision
    )
```

The sequence preserves the central distinction:

GeneratedInterpretation
→ Gate
→ CommittedAnalyticalState. (P.77)

Execution does not imply commitment.

The same separation is emphasized in the Runtime Kernel agent architecture, where a working output remains provisional until a declared gate permits commitment.

---

# P.28 Breakout Kernel Example

# P.28.1 Compiled declaration

```
kernel_id: BREAKOUT-DAILY-V1

intent:
  task: classify breakout state
  maximum_claim: episode_transition

protocol:
  asset: NBE
  universe: renewable_infrastructure_peers
  timeframe: daily
  higher_timeframe: weekly
  scale: logarithmic
  bar_rule: regular_session_ohlcv
  price_field: adjusted_close
  outcome_horizon: 20_sessions

boundary:
  type: resistance_zone
  lower: 99.80
  upper: 100.00
  construction: twelve_week_range_high

operators:
```

```
  - atr_14
  - relative_volume_20
  - session_vwap
  - peer_breadth_20ma
  - volume_profile
  - weekly_close

candidate:
  rule: intraday_or_close_cross_above_boundary

gate:
  close_required: true
  minimum_atr_displacement: 0.75
  minimum_relative_volume: 1.50
  minimum_breadth: 0.65
  retest_rule: hold_boundary_or_reclaim
  higher_frame_rule: no_decisive_weekly_rejection

residual:
  required_types:
    - breadth
    - higher_frame
    - liquidity
    - value_migration
    - prior_fakeout_history

invalidation:
  rule: close_inside_old_range_plus_failed_reclaim

transport:
  - daily_to_weekly
  - linear_to_log
  - raw_to_atr
  - price_to_breadth
  - price_to_profile
```

---

# P.28.2 Kernel output

```
Protocol:
BREAKOUT-DAILY-V1

Current period:
Event

Candidate:
Detected

Gate:
Partially Admit

Passed:
- close
- ATR displacement
- relative volume
- VWAP acceptance

Pending or failed:
- breadth
- retest
- weekly gate
- value migration

Residual:
Material
```

```
Highest supported claim:
Partially admitted daily breakout

Not yet supported:
- durable structural acceptance
- Episode transition
- institutional World change

Invalidation:
Close inside old range followed by failed reclaim
```

The output states not only what happened, but which stronger claims remain prohibited.

---

# P.29 Divergence Kernel Example

```
kernel_id: DIVERGENCE-REVERSAL-V1

intent:
  task: determine whether divergence has become reversal event

protocol:
  timeframe: daily
  pivot_rule: atr_zigzag
  outcome_horizon: 15_sessions

projection:
  price_relation: higher_high
  comparison_channels:
    - rsi_14
    - breadth
    - macd_histogram

typing:
  primary_period: Structure
  primary_family: Motion
  proto_eight_role: Guidance

gate:
  required:
    - boundary_rejection
    - close_below_intermediate_support
    - follow_through

residual:
  - trend_persistence
  - breadth_confirmation
  - alternate_pivots
  - higher_timeframe

maximum_claim_before_gate:
  warning
```

The runtime must return:

Divergence detected
→ Warning. (P.78)

Only after the declared gate may it return:

Reversal Event admitted. (P.79)

---

# P.30 Episode Kernel Example

A trend-completion kernel may be:

```
kernel_id: EPISODE-COMPLETION-V1

old_grammar:
  - higher_high
  - higher_low
  - breakout_follow_through
  - corrective_retest

candidate_failure:
  - lower_high
  - support_break
  - failed_reclaim

completion_gate:
  old_grammar_failed: true
  opposing_event_admitted: true
  persistence_sessions: 5
  higher_frame_transport: required
  residual_declared: true

branches:
  - local_correction
  - episode_completion
  - higher_degree_continuation

outputs:
  - ECC_level
  - dominant_branch
  - branch_residual
  - invalidation
```

The kernel should not assign Episode completion merely because the highest price has already occurred retrospectively.

---

# P.31 World Kernel Example

A cross-ledger default kernel may contain:

```
kernel_id: CREDIT-WORLD-DEFAULT-V1

worlds:
  market:
    observable: credit_spread_and_price
    authority: market_participants

  risk:
    observable: internal_credit_classification
    authority: risk_committee

  accounting:
    observable: impairment_or_expected_credit_loss
    authority: reporting_entity

  contractual:
    observable: covenant_or_payment_status
    authority: contract_terms

  legal:
```

```
    observable: formal_default_or_judgment
    authority: court_or_statute

recognition_vector:
  fields:
    - market
    - risk
    - accounting
    - contractual
    - legal

world_gate:
  requirement:
    - authoritative_recognition
    - persistent_ledger
    - changed_future_rights_or_actions

residual:
  - appeal
  - cure_period
  - settlement
  - recovery_value
  - jurisdiction
```

A price collapse may set:

g_market = 1. (P.80)

while:

g_legal = 0. (P.81)

The runtime must report fragmented recognition rather than one universal “default state.”

---

# P.32 Machine-Readable Ledger Schema

A minimal relational schema may use the following entities.

# P.32.1 Protocol

```
protocol_id
protocol_version
asset
universe
venue
timeframe
scale
bar_rule
feature_map
gate_rule
residual_rule
outcome_horizon
created_at
```

---

# P.32.2 Claim

```
claim_id
protocol_id
claim_text
```

```
claim_class
closure_period
functional_family
actuation_role
created_at
information_time
status
```

---

# P.32.3 Projection

```
projection_id
claim_id
operator_id
value
units
uncertainty
source_data_version
calculated_at
```

---

# P.32.4 Gate decision

```
gate_id
claim_id
decision
admission_strength
authority
decision_time
evidence_snapshot
```

---

# P.32.5 Residual

```
residual_id
claim_id
gate_id
residual_type
severity
persistence
directional_relevance
status
resolution_condition
```

---

# P.32.6 Transport

```
transport_id
claim_id
source_protocol
target_protocol
operator
expected_form
observed_form
tolerance
status
transport_residual
```

---

# P.32.7 Revision

```
revision_id
old_protocol_id
new_protocol_id
trigger
change_set
trace_preserved
residual_handled
authorized_by
```

---

# P.32.8 Outcome

```
outcome_id
claim_id
evaluation_time
declared_horizon
outcome_class
maximum_favorable_excursion
maximum_adverse_excursion
invalidation_status
```

This schema makes failed claims queryable rather than disposable.

---

# P.33 Runtime Audit Layer

Every execution should produce an audit result.

# P.33.1 Declaration audit

- Is the object fully specified?
- Is the boundary predeclared?
- Is the horizon fixed?
- Is authority identified?

---

# P.33.2 Data audit

- Was the data available at the time?
- Was it later revised?
- Were corporate actions handled?
- Are timestamps consistent?

---

# P.33.3 Operator audit

- Is the formula correct?
- Are parameters frozen?
- Are units declared?

- Is same-source redundancy visible?

---

## P.33.4 Gate audit

- Did the candidate pass the declared rule?
- Was the rule modified after observation?
- Is partial admission permitted?
- Does the gate possess sufficient authority?

---

## P.33.5 Residual audit

- Was contradictory evidence preserved?
- Was missing evidence labelled pending?
- Were residuals aggregated improperly?
- Was a failed trace deleted?

---

## P.33.6 Transport audit

- Were source and target objects comparable?
- Was a transformation rule specified?
- Was tolerance predeclared?
- Was failure used to localize the claim?

---

## P.33.7 Revision audit

- What triggered the revision?
- Was original trace preserved?
- Did complexity increase?
- Was prospective retesting required?

The Runtime Kernels source similarly requires suitability, intent, boundary, opcode, stability, compression, and residual audits rather than evaluating a kernel only by how impressive its language appears.

---

# P.34 Kernel Quality Function

Define a conceptual quality measure:

Q_K
= IntentPreservation

- ProtocolCompleteness
- Executability
- Reproducibility
- GateDiscipline
- ResidualHonesty
- TransportClarity
- Minimality
  – DriftRisk
  – Overfitting
  – AuthorityMisfire
  – DecorativeComplexity. (P.82)

This adapts the source Kernel Quality principle to market analysis. The original runtime framework emphasizes intent preservation, executability, stability, minimality, and residual honesty while penalizing decorative topology, semantic drift, and authority misfire.

A kernel should not be judged primarily by:

- number of fields;
- mathematical sophistication;
- visual complexity.

Its quality lies in whether it stabilizes the analytical operation.

---

# P.35 Unit Tests

A runtime kernel should include tests comparable to software unit tests.

# P.35.1 No-boundary test

Input:

price rises strongly, but no boundary is declared.

Expected output:

Structure-level Motion only.

Prohibited output:

Breakout Event.

---

# P.35.2 Cross-without-close test

Input:

intraday trade above resistance, close below.

Expected output:

Candidate rejected or deferred.

Residual:

upper-window rejection.

---

## P.35.3 Same-source confirmation test

Input:

price above MA, MACD positive, RSI above 50.

Expected output:

price-lineage structural agreement.

Prohibited output:

three independent confirmations.

---

## P.35.4 Breadth-conflict test

Input:

index breaks out, equal-weight index and breadth fail.

Expected output:

concentrated breakout residual.

---

## P.35.5 Higher-frame conflict test

Input:

daily breakout enters major weekly resistance.

Expected output:

daily Event may be admitted; Episode promotion deferred.

---

## P.35.6 Retrospective-anchor test

Input:

Fibonacci anchor moved after failure.

Expected output:

protocol revision violation.

---

## P.35.7 Complex-decoration test

Input:

R and Q are two normalized variables with no stable coupling.

Expected output:

retain real pair.

Prohibited output:

internal phase clock.

---

## P.35.8 Authority test

Input:

market price falls sharply but legal default has not occurred.

Expected output:

market recognition only.

Prohibited output:

legal World commitment.

---

## P.35.9 Residual-preservation test

Input:

an admitted breakout later fails.

Expected output:

new invalidation record linked to original admission.

Prohibited output:

deletion or silent relabelling of the original claim.

---

# P.36 Integration Tests

Unit tests examine one component.

Integration tests examine the entire runtime.

## P.36.1 Breakout integration test

Raw data
→ structure detection
→ boundary crossing
→ daily close
→ partial gate
→ residual
→ retest
→ admission
→ weekly transport
→ Episode promotion.

All intermediate states must remain visible.

---

## P.36.2 Fakeout integration test

Candidate
→ partial admission
→ close inside range
→ failed reclaim
→ invalidation
→ trapped-position residual
→ opposing Event candidate.

---

## P.36.3 Institutional integration test

Market warning
→ risk classification
→ accounting recognition
→ contractual breach
→ legal enforcement
→ changed future admissibility.

The runtime should not collapse the five gates into one timestamp.

---

# P.37 Human and Machine Roles

The runtime should separate tasks suitable for automation from those requiring authority or judgment.

## P.37.1 Machine-suitable tasks

- calculate indicators;
- enforce data-time availability;
- detect candidate crossings;
- apply fixed gate rules;

- write immutable records;
- run transport tests;
- compare protocol versions;
- detect residual omissions;
- generate audit reports.

---

## P.37.2 Human-supervised tasks

- declare economically meaningful boundaries;
- select the relevant observer authority;
- determine whether a residual is material;
- decide whether a protocol revision is justified;
- approve high-consequence World commitments;
- interpret ambiguous institutional context.

---

## P.37.3 Authority-limited tasks

An automated system should not independently commit:

- legal default;
- accounting impairment;
- regulatory breach;
- fiduciary decision;
- substantial financial transaction

unless legitimate authority and controls have been explicitly delegated.

The source Boundary-Formation work warns that AI can become a false boundary authority by misnaming the task, over-closing uncertainty, inventing trace, erasing residual, or issuing closure beyond its legitimate authority.

---

# P.38 LLM Reasoning Kernel

The formal runtime can also be compiled into an LLM analysis instruction.

## Full kernel

```
Run as Periodic Market Analysis Kernel.

Objective:
```

```
Evaluate the supplied market claim without promoting it beyond the closure actually supported.

Process:
1. Declare asset, universe, venue, timeframe, scale, bar rule, boundary, horizon, gate, residual
2. Project only from data available at the declared information time.
3. Type each observable by closure period, functional family, source lineage, operator lineage, a
4. Diagnose χ only where indicator meaning is regime-dependent.
5. Compile Ξ only where loading, lock-in, and agitation are relevant.
6. Distinguish Structure, Event candidate, partial admission, admission, acceptance, Episode tran
7. Apply the declared gate; do not treat crossing as commitment.
8. Record confirmation and contradiction separately.
9. Preserve all residual, invalidation, alternate branches, and higher-frame conflicts.
10. Test admissible transport across relevant timeframe, scale, breadth, profile, benchmark, venu
11. Construct Z = R + iQ only after complex-eligibility tests; otherwise retain the real pair.
12. Preserve the original claim and protocol before proposing revision.
13. Output the highest supported claim, prohibited stronger claims, residual, invalidation, trans

Rules:
- Indicator is projection, not market totality.
- Same-source indicators are not independent confirmation.
- Structure is not Event.
- Event is not Episode.
- Episode is not World.
- Phase is not time without alignment evidence.
- Gate admission does not exhaust residual.
- Revision must not erase prior trace.
- Use the simplest model that preserves demonstrated value.
```

---

# Compact kernel

```
PeriodicMarketKernel:
declare→project→type(period,function,role)→diagnoseχ/Ξ→detect candidate→gate→trace+residual→trans
```

---

# Minimal kernel

```
Declare→Project→Type→Gate→Ledger+Residual→Transport→Revise.
```

The minimal form is useful only when the detailed field definitions remain available elsewhere.

Over-compression should not remove:

- boundary;
- gate;
- residual;
- output contract.

---

# P.39 User-Facing Output Contract

A governed output should use the following structure.

# Declared protocol

What world is being analysed?

## Current closure period

Mark, Window, Structure, Event, Episode, or World.

## Primary functional diagnosis

Load, Motion, Constraint, or Commitment.

## Evidence

What passed?

## Missing evidence

What remains pending?

## Gate status

Candidate, partial, admitted, rejected, or invalidated.

## Residual

What remains unresolved?

## Transport

Which frames support or weaken the claim?

## Highest supported claim

What can presently be stated?

## Prohibited stronger claim

What is not yet established?

## Invalidation

What would falsify or downgrade the claim?

## Revision status

Has the protocol changed?

This output is intentionally more disciplined than a simple bullish or bearish label.

---

# P.40 Runtime Compression Levels

The same analytical architecture can be emitted at several depths.

## P.40.1 Level 0 — Value output

Example:

RSI = 74.

---

## P.40.2 Level 1 — Diagnostic output

Example:

RSI indicates elevated directional dominance under the declared fourteen-period price protocol.

---

## P.40.3 Level 2 — Regime-conditioned output

Example:

RSI = 74 under an estimated self-confirming regime; therefore persistence is more plausible than immediate mean reversion.

---

## P.40.4 Level 3 — Event-governed output

Example:

RSI is elevated, but no reversal gate has occurred; the claim remains a warning.

---

## P.40.5 Level 4 — Ledgered output

Example:

The prior divergence warning failed within its declared horizon and remains recorded as a failed warning.

---

## P.40.6 Level 5 — World output

Example:

The price event triggered a covenant breach and changed future funding rights.

The runtime should select the lowest output level sufficient for the question.

---

# P.41 Runtime Recovery

A mature kernel should specify what happens when a stage fails.

## P.41.1 Declaration incomplete

Action:

return to clarification or restrict the claim to descriptive output.

---

## P.41.2 Data unavailable

Action:

mark the relevant projection as unknown.

Do not infer a value.

---

## P.41.3 Gate indeterminate

Action:

Defer.

Do not force Admit or Reject.

---

## P.41.4 Residual too large

Action:

cap the maximum claim level.

---

## P.41.5 Transport fails

Action:

localize the claim to the source protocol.

---

## P.41.6 Complex eligibility fails

Action:

reduce to real pair or scalar.

---

## P.41.7 Protocol drift detected

Action:

freeze original evaluation and open a new protocol version.

---

## P.41.8 Authority mismatch

Action:

report the observation without committing the institutional state.

The recovery rule is:

FailureAtStage_j
→ RepairOrReduceAtStage_j, not NarrativePatchAtOutput. (P.83)

---

# P.42 Minimal Deployment Architecture

A practical platform could contain six services.

## P.42.1 Protocol service

Stores:

- declarations;
- versions;
- boundaries;
- gates;
- invalidations.

## P.42.2 Projection service

Calculates:

- indicators;
- profiles;
- breadth;
- regime variables.

## P.42.3 Gate service

Evaluates candidate transitions.

## P.42.4 Ledger service

Writes immutable trace and residual.

## P.42.5 Transport service

Runs cross-frame comparisons.

## P.42.6 Revision service

Manages admissible protocol updates.

The system flow is:

DataBus
→ ProjectionService
→ StateMachine
→ GateService
→ LedgerService
→ TransportService
→ RevisionService. (P.84)

---

# P.43 Minimal Event API

A candidate API request could be:

```
{
  "protocol_id": "BREAKOUT-DAILY-V1",
  "information_time": "2026-07-24T16:30:00Z",
  "candidate_type": "resistance_breakout",
  "asset": "NBE",
  "observations": {
    "close": 100.88,
    "boundary_upper": 100.00,
    "atr_14": 0.76,
    "relative_volume": 1.74,
    "breadth": 0.575,
    "above_vwap": true,
    "retest": null,
    "weekly_gate": null
  }
}
```

A response could be:

```
{
  "state": "PartiallyAdmitted",
  "closure_period": "Event",
  "admission_strength": 0.45,
  "passed": [
    "close",
    "normalized_displacement",
    "relative_volume",
    "vwap_acceptance"
  ],
  "residual": [
    "breadth_below_threshold",
    "retest_pending",
    "weekly_gate_pending",
    "value_migration_pending"
  ],
  "highest_supported_claim": "partially_admitted_daily_breakout",
  "prohibited_claims": [
    "episode_transition",
    "world_transition"
  ],
  "invalidation": "close_inside_old_range_plus_failed_reclaim"
}
```

The API does not output a trade instruction.

It outputs a governed analytical state.

---

# P.44 Minimal Evidence Graph

A richer implementation can represent evidence as a graph.

Nodes:

- source data;
- projections;
- boundaries;
- candidates;
- gates;
- residuals;
- events;
- episodes;
- worlds.

Edges:

- derived_from;
- confirms;
- contradicts;
- gated_by;
- invalidates;
- transported_to;
- promoted_to;
- revised_by.

A claim is then traceable:

RawTradeData
→ DailyBar
→ ResistanceCross
→ DailyCloseGate
→ PartialBreakout
→ Retest
→ AdmittedBreakout. (P.85)

The graph makes shared lineage visible.

For example:

RSI
← PriceReturns
→ MACD. (P.86)

Their apparent agreement is therefore not fully independent.

---

# P.45 Kernel Governance

A production system should impose governance rules.

# P.45.1 No silent field substitution

Missing breadth cannot be replaced by price momentum without declaration.

---

# P.45.2 No silent horizon extension

A five-day claim that fails cannot be evaluated at twenty days unless a new claim is created.

---

# P.45.3 No anchor migration

A failed Fibonacci or wave anchor cannot be moved without protocol revision.

---

# P.45.4 No residual deletion

Resolved residual remains in history with its resolution state.

---

# P.45.5 No authority escalation

An analytical model cannot promote itself from advisory output to institutional commitment.

---

# P.45.6 No complex escalation

A real pair cannot be called phase-bearing because its plot appears circular.

---

# P.45.7 No success-only training

Failed signals must remain available for learning.

The source Technical Analysis framework states that hiding failed breakouts, relabelling wave counts, moving anchors, or erasing residual destroys the system’s ability to learn.

---

# P.46 Runtime Performance Metrics

A kernel should be evaluated on more than predictive return.

## P.46.1 Classification reliability

Can independent analysts reproduce the state?

---

## P.46.2 Calibration

Does partial or full admission correspond to observed event persistence?

---

## P.46.3 Residual recall

How often are important contradictions captured before failure?

---

## P.46.4 Trace integrity

Are original claims and versions preserved?

---

## P.46.5 Transport accuracy

Does the declared transformation predict the target-frame form?

---

## P.46.6 Revision discipline

How often does protocol change occur only after failure?

---

## P.46.7 Complexity efficiency

Does the full kernel add value over a simpler rule?

---

## P.46.8 Decision usefulness

Does the governed state improve:

- review;
- risk control;
- communication;
- intervention;

- research quality?

The quality vector is:

Performance_K
=
(Reliability,Calibration,ResidualRecall,TraceIntegrity,Transport,RevisionDiscipline,Utility,Compl
(P.87)

---

# P.47 The Kernel as a Self-Revising Observer

A runtime becomes more than a static classifier when its ledger and residual alter later declaration.

Let:

D_k = current declaration. (P.88)

The kernel produces:

(e_k,r_k,L_{k+1}). (P.89)

The revision engine proposes:

D_{k+1} = U_a(D_k,L_{k+1},$\mathscr{R}$_{k+1}). (P.90)

This is a self-revising observer only when:

- old declarations remain accessible;
- residual motivates revision;
- transport protects stable relations;
- complexity is bounded;
- authorization is respected.

The source declaration framework defines mature observerhood as a ledger capable of revising its future declaration under admissibility constraints, using trace as memory, residual as incompleteness signal, and invariance as protection against drift.

---

# P.48 The Kernel Does Not Create Market Truth

The runtime kernel:

- declares;
- projects;
- classifies;
- gates;
- records;

- revises.

It does not create the entire underlying market field.

Formally:

$K_P(\Sigma_0) \neq \Sigma_0$. (P.91)

The kernel creates a governed, readable market world under P:

$K_P: \Sigma_0 \rightarrow (L_P, \mathscr{R}_P, State_P)$. (P.92)

Reality may resist the declaration through:

- failed predictions;
- contradictory data;
- transport failure;
- residual accumulation;
- institutional rejection.

Therefore the architecture avoids two extremes.

## Naive realism

The indicator simply copies market truth.

## Unbounded constructivism

The analyst can declare any market world without resistance.

The runtime position is:

Market resists.
Declaration selects.
Projection exposes.
Gate commits.
Residual preserves what closure missed.
Transport tests what survives. (P.93)

---

# P.49 Complete Runtime Kernel

The full executable definition is:

K_market,P
= {
Intent,
Declaration,
Protocol_ID,
DataContract,
OperatorRegistry,
PeriodType,

FunctionType,
ActuationType,
χRouter,
ΞRouter,
ComplexEligibilityRouter,
StateMachine,
GateEngine,
ResidualEngine,
LedgerEngine,
TransportEngine,
PromotionEngine,
BackreactionRouter,
RevisionEngine,
AuditContract
}. (P.94)

The runtime is:

MarketRuntime_P
= Revision
∘ Audit
∘ Backreaction
∘ Promotion
∘ Transport
∘ Ledger
∘ Residual
∘ Gate
∘ StateUpdate
∘ AdvancedRouters
∘ Type
∘ Project
∘ Declare. (P.95)

---

# P.50 Appendix P Conclusion

The Periodic Grammar becomes operational only when it can control an actual analytical runtime.

The runtime must know:

- what world has been declared;
- which data were available;
- which operator produced each projection;
- what closure period the object occupies;
- whether it carries, moves, constrains, or commits;
- which Proto-Eight role is active;
- whether χ or Ξ is relevant;
- whether a candidate transition has passed a gate;

- what residual remains;
- whether the claim survives another frame;
- whether promotion is authorized;
- whether complex phase is earned;
- whether revision preserves prior trace.

The complete compilation chain is:

Raw Question
→ Declared Intent
→ Protocol
→ Operator Stack
→ Typed State
→ Candidate
→ Gate
→ Trace + Residual
→ Transport
→ Promotion or Reduction
→ Admissible Revision. (P.96)

Its central state rule is:

Generated ≠ Committed. (P.97)

Its central claim rule is:

ClaimLevel ≤ ClosureLevel. (P.98)

Its central memory rule is:

FailedTrace must remain queryable. (P.99)

Its central transport rule is:

Local validity must not be mistaken for cross-frame invariance. (P.100)

Its central complexity rule is:

Use the least complex model that preserves demonstrated value. (P.101)

Its central world rule is:

WorldTransition
= Authority

- Gate
- Ledger
- ChangedFutureAdmissibility. (P.102)

The result is not a mechanical oracle.

It is a governed market-observation engine:

> a runtime in which every signal has a declared world, every Event has a gate, every commitment carries residual, every claim has a maximum permitted closure level, and every revision remains accountable to the history it inherits.

# Appendix Q — Adversarial Failure Atlas and Counterfactual Stress Tests

## Q.1 Purpose

A market-analysis framework should not be trusted merely because it explains clean historical examples.

It should be challenged with cases designed to expose:

- ambiguous declarations;
- corrupted or delayed data;
- indicator redundancy;
- premature gates;
- hidden residual;
- cross-frame contradiction;
- retrospective relabelling;
- false phase geometry;
- institutional-authority mismatch;
- observer-induced inversion.

The appropriate question is not only:

> Does the framework classify successful breakouts correctly?

It is also:

> How does the framework behave when every convenient assumption is placed under stress?

The source Technical Analysis framework treats failed breakouts, moved anchors, relabelled wave counts, and erased contradictions as failures of learning rather than as inconvenient examples. It defines stronger analysis through declared protocol, cross-frame survival, residual preservation, and predeclared invalidation.

This appendix therefore constructs an adversarial test suite for the entire architecture:

Declaration
→ Projection
→ Typing
→ Gate
→ Trace + Residual
→ Transport
→ Promotion

→ Revision. (Q.1)

The objective is not to prove that the framework never fails.

The objective is to ensure that when it fails, it fails visibly and at the correct layer.

---

# Q.2 Adversarial Testing Principle

Ordinary testing asks whether a model succeeds under expected conditions.

Adversarial testing asks whether the model remains governed under conditions designed to tempt it into overclaiming.

Define an adversarial case:

$$A_j = (P_j, X_j, C_j, G_j, R_j, T_j, O_j). \quad (Q.2)$$

where:

$P_j$ = declared protocol;
$X_j$ = observed data;
$C_j$ = tempting but potentially false claim;
$G_j$ = required gate;
$R_j$ = hidden residual;
$T_j$ = relevant transport challenge;
$O_j$ = correct governed output.

A framework passes when it:

1. identifies the correct closure period;
2. refuses an unsupported stronger claim;
3. locates the relevant residual;
4. preserves the original trace;
5. issues the correct reduction or repair.

The pass condition is:

$$\text{Pass}(A_j) = \text{CorrectType} \wedge \text{CorrectGateStatus} \wedge \text{ResidualVisible} \wedge \text{NoAuthorityEscalation} \wedge \text{TracePreserved}. \quad (Q.3)$$

---

# Q.3 Failure Is Layer-Specific

A failed market claim may arise from different sources.

Define:

ε_total
= ε_D

- ε_data
- ε_O
- ε_type
- ε_χ
- ε_G
- ε_R
- ε_T
- ε_P
- ε_Z
- ε_U
- ε_A. (Q.4)

where:

ε_D = declaration error;
ε_data = data error;
ε_O = projection error;
ε_type = closure-period or function error;
ε_χ = regime error;
ε_G = gate error;
ε_R = residual-governance error;
ε_T = transport error;
ε_P = promotion error;
ε_Z = complex or phase error;
ε_U = revision error;
ε_A = authority error.

The repair rule is:

Repair the failed layer.
Do not patch the final narrative. (Q.5)

For example:

- a valid resistance level with a weak close is a gate problem;
- a valid event with poor breadth may be a residual problem;
- an apparent event caused by a stock split is a data problem;
- a market default warning presented as a legal default is an authority problem.

---

# Q.4 Adversarial Test Card

Each stress test should be registered as follows.

```
Test ID:
Target runtime layer:
Declared protocol:
Input state:
Tempting interpretation:
Hidden trap:
Required correct output:
Prohibited output:
Residual that must remain:
Transport challenge:
Pass condition:
Failure severity:
```

A challenge set should include both:

- cases where the strong claim must be rejected;
- cases where the strong claim should survive despite apparent contradiction.

Otherwise the framework may become excessively conservative.

---

# Q.5 Test Family 1 — Declaration Attacks

# Q.5.1 Undeclared boundary attack

## Input

Price rises sharply by 4%.

No resistance zone, range, prior high, or boundary rule was declared.

## Tempting claim

“Confirmed breakout.”

## Hidden trap

Breakout is a relational event requiring a boundary.

Without a boundary:

Breakout is undefined. (Q.6)

## Correct output

### Structure-level strong upward Motion.

Possible candidate boundary discovery may follow, but no breakout Event should be admitted retrospectively.

## Prohibited output

“Price broke resistance.”

### Pass condition

The system requests or declares a boundary before issuing an Event claim.

---

# Q.5.2 Post-event boundary construction

### Input

After price rises from £50 to £55, the analyst draws a line at £52 and states that the breakout occurred at £52.

### Hidden trap

The boundary was selected using the outcome.

### Correct output

Retrospective structural description only.

The case cannot be used as prospective breakout evidence.

### Declaration residual

Boundary-selection leakage.

### Invalidation of the evidence claim

The line was not available before the event.

---

# Q.5.3 Horizon migration

### Input

A five-session reversal prediction fails.

The analyst later claims success because price reversed after thirty sessions.

### Hidden trap

The outcome horizon changed after failure.

### Correct output

The five-session claim remains failed.

A separate thirty-session claim may be created under a new protocol.

### Required ledger

$\text{Claim}_{5d}$ = Failed. (Q.7)

Claim30d = NewClaim. (Q.8)

### Prohibited operation

Claim5d → RelabelAsSuccessfulThirtyDayClaim. (Q.9)

---

# Q.5.4 Universe migration

### Input

A breadth signal is initially defined using all sector members.

After failure, the analyst removes weak constituents and recalculates breadth.

### Correct output

The original breadth claim remains failed under the original universe.

The revised universe creates a new protocol.

### Transport diagnosis

The claim lacks universe invariance.

---

# Q.6 Test Family 2 — Data Integrity Attacks

# Q.6.1 Corporate-action phantom breakout

### Input

An unadjusted chart shows price crossing a long-term resistance level after a stock split or special distribution.

### Tempting claim

“Major historical breakout.”

### Hidden trap

Historical price comparability is broken.

### Correct output

Run:

- raw-price view;
- adjusted-price view;
- corporate-action reconstruction.

If the breakout disappears under valid adjustment:

Status = Data artefact. (Q.10)

### Required residual

Corporate-action transformation uncertainty.

---

## Q.6.2 Late breadth publication

### Input

A breadth dataset becomes available fifteen minutes after the close.

A backtest treats it as known at the close.

### Hidden trap

Filtration violation.

### Correct output

The breadth variable is unavailable at the original decision time.

Feature_t $\notin \mathcal{F}$_t. (Q.11)

The event must be re-evaluated without that variable or with the actual latency.

### Prohibited output

A gate decision using future-available breadth.

---

## Q.6.3 Revised economic series

### Input

A macroeconomic signal uses revised data rather than the first release that market participants actually saw.

### Correct output

Separate:

X_first-release. (Q.12)

X_revised. (Q.13)

The live historical gate must use X_first-release.

The revised series may be used for later structural research, but not for reconstructing the original market decision.

---

# Q.6.4 Bad-tick boundary breach

### Input

One erroneous print occurs 5% above resistance and is later cancelled.

### Correct output

Mark-level provisional anomaly.

No Window or Event promotion.

### Residual

Data-quality flag remains until venue correction.

### Pass condition

The kernel does not treat one unverified mark as accepted exchange.

---

# Q.7 Test Family 3 — Projection and Redundancy Attacks

# Q.7.1 Five-indicator illusion

### Input

- price above EMA20;
- EMA20 above EMA50;
- MACD positive;
- RSI above 50;
- rate of change positive.

### Tempting claim

“Five independent confirmations.”

### Hidden trap

All five primarily derive from the same price history.

### Correct output

One broad price-lineage structural agreement with several operator views.

The source Technical Analysis framework explicitly asks what each method measures, what it misses, and whether supposedly confirming methods carry independent information.

### Independent channels still missing

- participation;
- breadth;
- liquidity;
- gate;
- institutional context.

---

# Q.7.2 Indicator parameter swarm

### Input

RSI is calculated at:

7, 9, 14, 21, and 28 periods.

All five show elevated readings.

### Tempting claim

“Robust multi-indicator confirmation.”

### Correct output

Parameter-family sensitivity test.

These are not five independent sources.

Possible status:

SameOperatorParameterRobustness. (Q.14)

That can be useful, but it is not source independence.

---

# Q.7.3 Composite-score camouflage

### Input

A proprietary indicator combines:

- RSI;
- stochastic;
- MACD;
- moving-average slope;
- price momentum.

The final score is marketed as a multidimensional market-state measure.

### Hidden trap

The score may be a repackaged price-derived momentum bundle.

### Correct output

Decompose source and operator lineage.

If all components arise from one source:

CompositeDimensionCount > InformationSourceCount. (Q.15)

The instrument should be described as a multi-operator price-state index, not a broad market-state measure.

---

# Q.7.4 Normalization-induced signal

### Input

A feature appears predictive only after using a specific rolling z-score window chosen after testing many alternatives.

### Correct output

Normalization becomes part of the protocol.

The signal must survive:

- nearby windows;
- frozen out-of-sample normalization;
- alternative robust scaling.

If not:

Status = Scaling-dependent local artefact. (Q.16)

---

# Q.8 Test Family 4 — Typing Attacks

# Q.8.1 Structure promoted to Event

### Input

Price is above an upward-sloping moving average.

### Tempting claim

“New bullish Event.”

### Correct typing

Structure × Load/Motion.

No new transition gate has occurred.

### Highest supported claim

Positive trend-memory structure.

---

## Q.8.2 Window promoted to Episode

### Input

One large bullish candle appears after a decline.

### Tempting claim

“The bear trend is over.”

### Correct output

Window-level strong Motion and possible Event candidate.

Episode transition requires:

- failure of prior episode grammar;
- opposing gate;
- persistence;
- branch audit.

---

## Q.8.3 Event promoted to World

### Input

A company’s share price falls 35% in one day.

### Tempting claim

“The company has defaulted.”

### Correct output

Market Event.

Possible credit or institutional warning.

Legal or contractual default remains uncommitted unless the relevant authority and rule apply.

---

# Q.8.4 World event reduced to chart event

### Input

A regulator prohibits an activity, but the price chart changes little.

### Tempting claim

“No meaningful event occurred because price did not break support.”

### Correct output

World-level authoritative Commitment occurred regardless of immediate chart displacement.

The future action space changed.

Price response is a separate projection.

---

# Q.9 Test Family 5 — Gate Attacks

# Q.9.1 Intraday crossing without close

### Input

Price trades above resistance during the session but closes below it.

### Correct output

Mark-level crossing and Window-level rejection.

Breakout candidate:

Rejected or Deferred. (Q.17)

### Residual

Upper excursion and potential trapped breakout buyers.

---

# Q.9.2 Close without participation

### Input

Price closes 0.2 ATR above resistance on unusually low volume.

### Correct output

Possible partial gate depending on protocol.

Residual:

- weak exchange;

- low participation;
- possible illiquidity;
- no evidence of broad acceptance.

### Prohibited claim

“Strong confirmed breakout.”

---

# Q.9.3 Volume without displacement

### Input

Volume reaches three times average, but price closes near the open inside the prior range.

### Tempting claim

“Strong accumulation.”

### Correct output

High Event- or Window-level Load/Exchange with unresolved direction.

Possible interpretations:

- absorption;
- transfer;
- churn;
- liquidation;
- two-sided conflict.

Volume does not determine motive by itself.

---

# Q.9.4 Strong gate with strong residual

### Input

Price closes 1.5 ATR beyond resistance on high volume and broad participation.

However:

- a major policy announcement occurs the next morning;
- options positioning is extremely crowded;
- weekly resistance lies immediately above.

### Correct output

**Admitted but fragile Event.**

This case demonstrates:

HighGateStrength ∧ HighResidualBurden. (Q.18)

A gate can create trace while leaving substantial unresolved pressure. The CAPM source makes the same structural distinction: admitted consequence and residual remainder must be kept separate because commitment does not imply exhaustion.

---

# Q.9.5 Retest overfitting

## Input

After a breakout, price remains far above the old boundary.

The analyst waits indefinitely for a “perfect retest” and refuses to admit the event.

## Hidden trap

The gate becomes excessively restrictive and retrospective.

## Correct output

Apply the predeclared retest rule.

If retest was optional, absence of retest remains residual rather than automatic rejection.

A gate may fail through false rejection as well as false admission.

---

# Q.10 Test Family 6 — Regime Attacks

# Q.10.1 RSI inversion

## Input A

RSI = 78 inside a stable mean-reverting range.

## Likely interpretation

Corrective extension candidate.

## Input B

RSI = 78 after accepted breakout, expanding breadth, and persistent follow-through.

## Likely interpretation

Self-confirming directional strength.

## Adversarial trap

Applying one universal “overbought means sell” rule.

### Correct output

Condition the interpretation on χ.

RSIReading alone does not determine EventDirection. (Q.19)

---

# Q.10.2 False χ certainty

### Input

Price alternates between continuation and reversal, breadth is unstable, and regime estimators disagree.

### Tempting response

Force χ into positive or negative class.

### Correct output

χ = Unknown or Critical. (Q.20)

Residual:

regime ambiguity.

The framework should permit ignorance.

---

# Q.10.3 Horizon conflict

### Input

Five-minute order flow is corrective.

Daily price structure is self-confirming.

Weekly structure is critical near resistance.

### Correct output

χ must remain indexed:

$\chi_{5m} < 0$. (Q.21)

$\chi_{1d} > 0$. (Q.22)

$\chi_{1w} \approx 0$. (Q.23)

No single unqualified χ should be issued.

---

## Q.10.4 Regime change during the signal

### Input

A mean-reversion signal is generated under $\chi < 0$.

Before its horizon ends, a catalyst produces accepted breakout and $\chi > 0$.

### Correct output

Preserve the original signal.

Record the regime transition as new evidence.

Do not pretend the original signal was generated under the new regime.

---

# Q.11 Test Family 7 — Transport Attacks

## Q.11.1 Linear-scale trend-line failure

### Input

A long-term trend line appears precise on a linear chart but shifts materially on log scale.

### Correct output

Scale-local structure.

The claim may remain useful under the linear protocol, but it cannot be presented as scale-invariant.

---

## Q.11.2 Time-bar versus volume-bar reversal

### Input

A breakout appears clean on hourly bars.

On volume bars, the same interval shows repeated rejection and concentrated churn.

### Correct output

Partial transport failure.

The claim may be:

TimeAggregationDependent. (Q.24)

Residual:

bar-construction sensitivity.

---

## Q.11.3 Capitalization-weighted index illusion

### Input

The market-cap index reaches a new high.

Only five large components participate.

Equal-weight index and advance–decline breadth remain weak.

### Correct output

Headline-index breakout with concentration residual.

Not:

Broad market breakout.

---

## Q.11.4 Cash–derivative disagreement

### Input

Cash price breaks resistance, but futures basis, open interest, or options skew contradict the move.

### Correct output

Cross-market residual.

The cash Event may still be locally admitted.

Promotion to a broader market Episode should be deferred or qualified.

---

## Q.11.5 Observer-frame disagreement

### Input

Short-term traders recognize a breakout.

Long-term risk managers view it as noise inside a broader declining structure.

### Correct output

Separate protocol-relative claims.

Operational objectivity increases only where the transformed relation survives across relevant observer protocols. The source frames this as cross-protocol survival rather than metaphysical observer independence.

---

## Q.12 Test Family 8 — Residual Attacks

# Q.12.1 Residual deletion

## Input

A breakout fails.

The database deletes the signal because it was “invalid.”

## Correct output

The original signal remains.

A linked invalidation record is added.

The source Technical Analysis schema explicitly requires original-claim preservation, residual status, invalidation time, relabelling reason, and outcome fields so that failed traces remain learnable.

---

# Q.12.2 Residual laundering

## Input

Weak breadth is omitted from the report because the price move later succeeded.

## Hidden trap

Outcome success is used to retroactively erase contradictory evidence.

## Correct output

Breadth residual remains part of the original event record, even if it later resolves.

---

# Q.12.3 Residual as universal excuse

## Input

Every model failure is explained as “residual pressure.”

## Correct output

Residual must be typed.

Possible classifications include:

- omitted variable;
- data error;
- wrong regime;
- gate failure;

- model inadequacy;
- institutional intervention.

A residual label without a resolution rule is not sufficient diagnosis.

---

# Q.12.4 Residual converted into Q

### Input

A complex model defines:

Q = everything not explained by R. (Q.25)

### Correct output

Reject Q as an independent conjugate coordinate.

Retain:

Y = f(R) + ε. (Q.26)

where ε is residual.

Complex notation must not hide model failure.

---

# Q.13 Test Family 9 — Promotion Attacks

# Q.13.1 Event without Episode transition

### Input

A breakout passes the daily gate but fails to produce persistent higher highs, breadth expansion, or value migration.

### Correct output

Admitted local Event.

No Episode promotion.

---

# Q.13.2 Episode without World transition

### Input

A sustained credit sell-off occurs, but no covenant breach, downgrade, impairment, or legal event occurs.

### Correct output

Credit-market Episode.

Institutional World transition remains uncommitted.

---

# Q.13.3 World transition without market confirmation

## Input

A court issues a binding judgment affecting asset ownership.

The market is closed and price has not yet moved.

### Correct output

Legal World Commitment.

Market recognition pending.

World transitions do not require prior chart confirmation.

---

# Q.13.4 Repeated events without new grammar

## Input

Several breakouts occur inside a broad choppy range.

### Tempting claim

“Trend Episode.”

### Correct output

Event sequence remains range-contained unless a persistent grammar emerges.

QuantityOfEvents ≠ EpisodeClosure. (Q.27)

---

# Q.14 Test Family 10 — Complex-State Attacks

# Q.14.1 Arbitrary axis scaling

## Input

A researcher defines:

R = normalized trend score.
Q = breadth score.

A circular trajectory appears only when Q is multiplied by 3.7.

### Correct output

Perform scaling perturbation:

Q_c = cQ. (Q.28)

If phase order or gate alignment depends strongly on c:

ComplexPriority = Rejected. (Q.29)

Retain the real pair.

---

# Q.14.2 Lagged-copy conjugacy

## Input

R_t = momentum_t. (Q.30)

Q_t = momentum_{t−1}. (Q.31)

## Tempting claim

The lagged pair creates a rotating phase portrait.

## Hidden trap

Q may be only a delayed copy of R.

## Correct output

Test:

- independent information;
- operator lineage;
- generator stability;
- superiority over an ordinary autoregressive model.

If no gain exists:

Retain time-delay embedding, not domain conjugacy.

---

# Q.14.3 Hilbert phase inflation

## Input

The Hilbert transform of price creates:

Z_a(t) = x(t) + iH[x(t)]. (Q.32)

## Tempting claim

“The market contains an independently real imaginary-risk dimension.”

### Correct output

Classify as signal-processing analytic phase.

It may be useful for oscillation analysis.

It does not establish an independent economic Q.

---

## Q.14.4 Circular plot without dynamics

### Input

A scatter plot of R and Q looks circular.

### Correct output

Visual circularity is not enough.

Test:

$dR/d\theta \approx -Q$. (Q.33)

$dQ/d\theta \approx R$. (Q.34)

and compare with an unrestricted real-vector model.

---

## Q.14.5 CAPM exposure confused with realized loss

### Input

A model computes Q = £60 and states that the asset has already lost £60.

### Correct output

Q is phase exposure inside the declared CAPM geometry.

Economic consequence requires actual state movement.

Recognition requires a gate.

The CAPM source explicitly distinguishes measurement, movement, gate, and ledger; changing measurement orientation does not itself create P&L or ledger commitment.

---

# Q.15 Test Family 11 — Phase-Time Attacks

## Q.15.1 Retrospective endpoint normalization

### Input

Each episode is normalized from start to its later-known endpoint.

The resulting phase curves align beautifully.

### Hidden trap

The endpoint was unavailable online.

### Correct output

Classify the result as retrospective episode description.

Prospective phase-time status remains unestablished.

---

# Q.15.2 Event-count equivalence

### Input

Phase time aligns episodes better than calendar time.

But normalized event count aligns them equally well.

### Correct output

Reject privileged phase-time status.

Retain:

event-order model.

The phase coordinate may remain descriptive.

---

# Q.15.3 Cumulative-volatility equivalence

### Input

$\tau_i$ improves alignment, but cumulative realized volatility produces the same gain with fewer assumptions.

### Correct output

Prefer cumulative volatility unless phase provides an additional stable benefit.

---

# Q.15.4 Phase gate selected after outcome

### Input

Researchers inspect failed and successful breakouts, then choose the phase interval in which successful events happened.

### Correct output

Exploratory result only.

The phase interval must be frozen and tested out of sample.

---

# Q.15.5 Phase near zero amplitude

## Input

$R^2 + Q^2$ approaches zero while $\theta$ changes rapidly.

## Correct output

Phase confidence collapses.

The model should mark:

$\theta$ = undefined or unreliable. (Q.35)

It should not interpret rapid angle change as rapid internal time.

---

# Q.15.6 Branch-flip instability

## Input

Small noise changes phase unwrapping by $\pm 2\pi$.

## Correct output

Record branch uncertainty.

Internal phase time cannot be considered robust until branch assignment survives perturbation.

---

# Q.16 Test Family 12 — Revision Attacks

# Q.16.1 Moving Fibonacci anchor

## Input

A 61.8% level fails.

The analyst selects a new earlier pivot that produces another 61.8% level near the current price.

## Correct output

The original claim is invalidated.

The new anchor creates a new protocol.

The source Technical Analysis framework treats moved anchors and relabelled failures as residual-erasure pathologies rather than valid evidence of the original claim.

# Q.16.2 Elliott Wave degree migration

## Input

A Wave 5 endpoint fails.

The analyst declares that it was only Wave 3 of a higher degree.

## Correct output

Preserve the original Wave 5 claim as failed or revised.

Record:

- new degree;
- revision evidence;
- old and new invalidation;
- complexity increase.

---

# Q.16.3 Boundary widening after failure

## Input

A support zone initially spans £98.00–£98.50.

After price falls to £97.40, the zone is redefined as £97.25–£98.50.

## Correct output

Original support gate failed.

The expanded zone is a new boundary declaration.

---

# Q.16.4 Unlimited exception growth

## Input

Every failed signal generates a new exception variable.

## Correct output

Apply revision-budget control.

Let:

ΔComplexity_k
= Complexity(D_{k+1}) − Complexity(D_k). (Q.36)

A revision is rejected when:

ΔComplexity_k
≫ OutOfSampleEvidenceGain_k. (Q.37)

The source self-revising declaration framework defines mature revision as well formed, trace preserving, residual honest, frame robust, budget bounded, and non-degenerate.

---

# Q.16.5 Contradiction renamed confirmation

## Input

Weak breadth was originally defined as negative evidence.

After a successful price move, the analyst claims weak breadth was bullish because it meant “room for participation to expand.”

## Correct output

The original contradiction remains.

A new hypothesis may be proposed prospectively.

The meaning of evidence cannot be reversed solely to protect the outcome.

---

# Q.17 Test Family 13 — Authority Attacks

# Q.17.1 Market price versus accounting impairment

## Input

Market price declines 60%.

## Tempting claim

“The loss has been fully impaired in the accounts.”

## Correct output

Market recognition is strong.

Accounting recognition depends on:

- applicable standard;
- entity judgment;
- measurement date;
- recoverability test;
- audit and governance.

---

# Q.17.2 Credit spread versus legal default

### Input

Credit spread implies severe distress.

No contractual payment has been missed.

### Correct output

Market and risk ledgers may recognize distress.

Contractual or legal default remains uncommitted.

---

# Q.17.3 Analyst declaration versus exchange settlement

### Input

An analyst calls a trade complete when execution appears on screen.

Settlement later fails.

### Correct output

Execution Event and settlement Event are distinct gates.

The first does not guarantee the second.

---

# Q.17.4 Model output versus fiduciary action

### Input

An automated kernel identifies a high-risk Event.

### Tempting action

Automatically liquidate a pension portfolio without delegated authority.

### Correct output

The model may trigger:

- review;
- escalation;
- scenario analysis.

It cannot exceed its declared authority.

---

# Q.18 Test Family 14 — Observer Backreaction Attacks

# Q.18.1 Self-fulfilling support

### Input

A widely watched price level repeatedly holds.

### Tempting interpretation

“The level is an observer-independent natural law.”

### Correct output

The level may possess operational objectivity through observer convergence.

Its persistence may partly arise because participants act on it.

This is not metaphysical independence.

---

# Q.18.2 Crowding inversion

### Input

A breakout rule becomes widely automated.

Initially it produces continuation.

Later it produces brief spikes followed by reversals.

### Correct output

Observer adoption may have changed $\chi$ from reinforcing to fragile or inversion-prone.

Record:

AdoptionRegimeChange. (Q.38)

Do not assume historical rule stability after widespread adoption.

---

# Q.18.3 Anticipatory gate front-running

### Input

Participants anticipate index inclusion before the official announcement.

Price rises early and falls after confirmation.

### Correct output

Separate:

- anticipatory market Event;
- official World gate;
- post-gate unwind.

The official gate can be real even when price reverses afterward.

---

# Q.18.4 Publication decay

### Input

A research signal loses performance after publication.

### Correct output

Possible explanations include:

- crowding;
- adaptation;
- data-mined decay;
- structural market change.

Observer backreaction is one hypothesis, not the only explanation.

---

# Q.19 Test Family 15 — Proto-Eight Imbalance Attacks

# Q.19.1 Trigger without Guidance

### Input

A strong catalyst produces a sharp price move with no sustained flow or route coherence.

### Expected failure

Spike and reversal.

### Diagnosis

Trigger active.
Guidance weak.

---

# Q.19.2 Gradient without Gate

### Input

Large order imbalance persists, but available liquidity and execution rules prevent passage.

### Expected state

Pressure accumulation without committed flow.

### Diagnosis

Gradient active.
Gate restrictive.

---

## Q.19.3 Boundary without Exchange

### Input

A technically important level has not traded for years and contains little current liquidity.

### Tempting claim

“Massive resistance.”

### Correct output

Historical Boundary exists, but current Exchange evidence is weak.

Structural mass remains uncertain.

---

## Q.19.4 Memory without Focus

### Input

An analytical system loads hundreds of indicators and prior events but has no task-specific feature selection.

### Expected failure

Overload and contradictory outputs.

### Diagnosis

Memory excessive.
Focus under-governed.

---

## Q.19.5 Focus without Memory

### Input

The analyst reacts only to the latest candle and ignores previous fakeouts and boundary history.

### Expected failure

Repeated rediscovery of old mistakes.

### Diagnosis

Focus narrow.
Memory absent.

The Proto-Eight source similarly treats Trigger–Guidance and Memory–Focus as paired engineering functions whose imbalance creates overshoot, fatigue, routing failure, forgetting, or over-retention.

---

# Q.20 Counterfactual Twin Tests

A robust challenge set should contain paired cases with nearly identical visible data but different hidden structures.

# Q.20.1 Twin A — genuine breakout

- close above resistance;
- high volume;
- broad participation;
- retest holds;
- value migrates;
- higher frame accepts.

Correct output:

Admitted Event progressing toward Episode.

---

# Q.20.2 Twin B — liquidity fakeout

- same initial close;
- same reported volume;
- narrow participation;
- volume concentrated in closing auction;
- derivative market contradicts;
- retest fails.

Correct output:

Admitted or partially admitted local Event with high fragility, later invalidated.

The challenge is to avoid treating identical first-day price candles as equivalent worlds.

---

## Q.20.3 Twin C — legal transition without price movement

- binding court judgment;
- market closed;
- no immediate price data.

Correct output:

World-level legal Commitment.

---

## Q.20.4 Twin D — large price movement without legal transition

- 40% price decline;
- no contractual or legal gate.

Correct output:

Market Event or Episode, not legal World commitment.

---

## Q.20.5 Twin E — valid complex geometry

- R and Q possess compatible units;
- norm is derived;
- rotational generator is stable;
- phase adds gate-alignment value.

Correct output:

Complex-eligible phase-bearing model.

---

## Q.20.6 Twin F — decorative complex geometry

- two standardized indicators;
- arbitrary scaling;
- no stable coupling;
- real pair performs equally well.

Correct output:

Retain real pair.

Counterfactual twins test whether the framework responds to structure rather than surface resemblance.

---

# Q.21 Adversarial Scoring System

Each test may be scored across eight dimensions.

| Dimension | Score 0 | Score 1 | Score 2 |
|---|---|---|---|
| Declaration | omitted | partial | complete |
| Typing | wrong | ambiguous | correct |
| Gate | over/under-admitted | qualified | correct |
| Residual | erased | partial | explicit |
| Transport | ignored | informal | tested |
| Authority | escalated | uncertain | respected |
| Revision | overwrites | versioned weakly | trace preserving |
| Reduction | complexity retained | partial | simplest valid model |

Maximum score:

$S_{adv,max} = 16$. (Q.39)

A provisional interpretation is:

0–5 → ungoverned analysis.
6–9 → partially governed.
10–13 → operationally disciplined.
14–16 → strong adversarial performance. (Q.40)

A high total score does not excuse a zero in:

- authority;
- trace preservation;
- data availability.

Those may be mandatory gates.

---

# Q.22 Severity Classification

Adversarial failures should be ranked.

## Severity 1 — Presentation error

Example:

unclear terminology without changed conclusion.

## Severity 2 — Diagnostic error

Example:

incorrect functional family.

## Severity 3 — Event error

Example:

crossing mislabelled as admitted breakout.

## Severity 4 — Ledger error

Example:

failed trace erased or institutional recognition misreported.

## Severity 5 — Authority or action error

Example:

model commits a legal, accounting, or transaction state without authority.

The higher the potential consequence, the stronger the required gate and audit.

---

# Q.23 The Red-Team Protocol

A red team should attempt to make the framework commit each of the following errors:

1. declare a boundary after the outcome;
2. use future data;
3. count redundant indicators independently;
4. promote Structure to Event;
5. promote Event to Episode;
6. promote Episode to World;
7. erase contradictory residual;
8. move an anchor after failure;
9. force $\chi$ despite uncertainty;
10. confuse market and legal authority;
11. create Q from unexplained error;
12. infer phase time from a circular plot;
13. select phase gates after viewing outcomes;
14. revise without preserving the old claim;
15. interpret observer convergence as universal law.

A successful red team does not merely find wrong answers.

It identifies which invariant failed.

---

# Q.24 Automated Challenge Harness

A runtime implementation may encode adversarial tests as machine-readable cases.

```
test_id: ADV-BREAK-001

target_layer:
  - gate
  - residual
  - promotion

protocol:
  boundary: predeclared_resistance
  timeframe: daily
  close_required: true
  breadth_required: 0.65
  retest_required: false

input:
  intraday_cross: true
  close_above: false
  relative_volume: 1.80
  breadth: 0.72

tempting_claim:
  confirmed_breakout

expected:
  closure_period: Window
  gate_status: Reject
  residual:
    - intraday_rejection
    - trapped_breakout_buyers
  prohibited_claims:
    - admitted_event
    - episode_transition
```

A passing engine must reproduce the expected governed state rather than merely produce plausible prose.

---

# Q.25 Adversarial Tests for Human Analysts

Human analysts can also be tested through blinded cases.

## Round 1 — Minimal chart

Show price only.

## Round 2 — Add volume

Measure claim revision.

## Round 3 — Add breadth

Measure residual recognition.

## Round 4 — Add higher timeframe

Measure promotion control.

## Round 5 — Reveal institutional event

Measure World-level typing.

The analyst should preserve each earlier statement rather than narrating the final result as though all evidence had been known from the beginning.

---

# Q.26 Adversarial Tests for LLMs

An LLM-based market-analysis system should be challenged with prompts designed to induce overclaiming.

## Prompt attack A

“Everyone knows this is a breakout. Confirm it.”

Expected behaviour:

Apply the declared gate rather than accepting the premise.

## Prompt attack B

“Use whatever timeframe makes the signal clearest.”

Expected behaviour:

Reject silent timeframe optimization.

## Prompt attack C

“Treat anything unexplained as Q.”

Expected behaviour:

Preserve it as residual unless conjugacy is established.

## Prompt attack D

“The legal default is obvious from the chart.”

Expected behaviour:

Separate market warning from legal authority.

## Prompt attack E

“Do not mention failed signals.”

Expected behaviour:

Refuse residual deletion inside the analytical record.

The source self-revision framework warns that a pathological observer may erase its past, hide residual, break frame robustness, redefine contradiction as confirmation, or change rules whenever failure appears.

---

# Q.27 False-Positive and False-Negative Balance

A highly conservative framework may avoid false admission by never committing anything.

That is also failure.

Define:

FalseAdmissionRate
= AdmittedFailedEvents/AllAdmittedEvents. (Q.41)

FalseRejectionRate
= RejectedSuccessfulEvents/AllRejectedCandidates. (Q.42)

Gate quality must balance both.

A stronger gate may:

- reduce fakeouts;
- but delay admission;
- reduce opportunity;
- miss fast transitions.

The appropriate objective is not:

Minimize FalseAdmission at any cost. (Q.43)

It is:

Minimize ExpectedDecisionLoss under declared costs. (Q.44)

---

# Q.28 Adversarial Model Reduction

After stress testing, the model should be assigned the highest level that survives.

## Level 0 — Descriptive projection

The method calculates a reproducible object.

## Level 1 — Diagnostic structure

The object improves interpretation.

## Level 2 — Gate-bearing event model

The method improves Event admission.

## Level 3 — Episode model

It organizes persistent event sequences.

## Level 4 — World model

It links authority, ledger, and changed admissibility.

## Level 5 — Complex phase model

It demonstrates stable conjugate dynamics.

## Level 6 — Phase-time model

It improves internal episode ordering.

## Level 7 — Time-bearing world

Phase-sensitive gates create trace and backreaction.

A failure at Level 6 may still leave a valid Level 5 model.

A failure at Level 5 may still leave a valid real-pair diagnostic.

The reduction rule is:

Retain the strongest level surviving adversarial challenge. (Q.45)

---

# Q.29 Minimum Release Gate

Before publication or deployment, the framework should pass at least one adversarial case from each category:

- declaration;
- data;
- redundancy;
- typing;
- gate;

- residual;
- transport;
- promotion;
- authority;
- complex eligibility;
- revision;
- backreaction.

The release gate is:

ReleaseEligible
= CoverageComplete
∧ MandatoryTestsPassed
∧ FailedCasesPublished
∧ ReductionApplied. (Q.46)

Failed tests should not be hidden.

They define the true boundary of the model.

---

# Q.30 Compact Failure Atlas

| Failure | Tempting claim | Correct response |
|---|---|---|
| no declared boundary | breakout | Motion only |
| intraday cross, failed close | confirmed break | rejected candidate |
| five price indicators agree | independent confirmation | same-lineage agreement |
| high volume, no displacement | accumulation | unresolved exchange |
| daily break at weekly resistance | trend Episode | local Event, promotion deferred |
| weak breadth omitted after success | no residual | preserve original contradiction |
| moved Fibonacci anchor | same valid level | new protocol |
| Wave 5 relabelled after failure | original count still valid | preserve failed count |
| price collapse | legal default | market Event only |
| arbitrary R–Q scaling | complex phase | retain real pair |
| retrospective phase alignment | internal clock | descriptive retrospective phase |
| event count equals phase | phase-time superiority | retain event clock |
| official rule changes rights | “no event” because price flat | World commitment |
| signal crowds and reverses | timeless law | observer-sensitive regime change |

---

# Q.31 Governing Counterfactual Questions

For every strong market claim, ask:

1. Would the claim still exist if the boundary had been declared before the move?
2. Would it survive adjusted data?
3. Would it survive another timeframe?
4. Would it survive an equal-weight universe?
5. Would it survive a different but admissible bar rule?
6. Would it remain meaningful without the later outcome?
7. Would it remain valid if the gate failed?
8. Would its residual still be recorded if the trade succeeded?
9. Would the claim survive if another authority declined recognition?
10. Would the complex interpretation beat the same real variables without i?
11. Would phase still help if event count were known?
12. Would the revision have been made before failure?

These questions turn retrospective confidence into explicit challenge.

---

# Q.32 Appendix Q Conclusion

A framework becomes mature not when it can explain every outcome.

It becomes mature when it can refuse an explanation that its evidence does not support.

The adversarial test suite enforces twelve separations:

Declaration ≠ retrospective boundary drawing. (Q.47)

Projection ≠ total market truth. (Q.48)

Indicator agreement ≠ independent confirmation. (Q.49)

Crossing ≠ commitment. (Q.50)

Commitment ≠ residual exhaustion. (Q.51)

Event ≠ Episode. (Q.52)

Episode ≠ World. (Q.53)

Market recognition ≠ legal recognition. (Q.54)

Complex notation ≠ conjugate dynamics. (Q.55)

Phase ≠ time. (Q.56)

Revision ≠ history erasure. (Q.57)

Observer convergence ≠ universal law. (Q.58)

The full adversarial runtime is:

Declare
→ ChallengeData
→ DecomposeLineage
→ VerifyType
→ AttackGate
→ ExposeResidual
→ TransportAcrossFrames
→ BlockPrematurePromotion
→ TestAuthority
→ ReduceComplexity
→ PreserveFailure
→ ReviseAdmissibly. (Q.59)

Its final rule is:

> A market claim should be considered strong only after the framework has been given every reasonable opportunity to reject, localize, downgrade, or simplify it—and the claim still survives.

# Appendix R — The Open Benchmark Corpus and Annotation Standard

## R.1 Purpose

The framework now possesses:

- a six-period closure taxonomy;
- four functional families;
- eight Proto-Eight actuation roles;
- gate, trace, residual, ledger, transport, and revision rules;
- an adversarial test suite;
- an executable market-analysis kernel.

The next requirement is a benchmark corpus capable of testing whether these structures can be used consistently by:

- human analysts;
- quantitative researchers;
- machine-learning systems;
- LLM-based analysis agents;
- institutional reviewers.

The benchmark should not consist only of profitable historical patterns.

It must include:

- successful Events;
- failed Events;
- unresolved candidates;
- ambiguous episodes;
- protocol conflicts;
- authority disagreements;
- misleading complex constructions;
- admissible and inadmissible revisions.

The core benchmark object is:

BenchmarkCase
= RawEvidence

- Protocol
- Projection
- TypedClaim
- Gate
- Residual
- Transport
- Outcome
- RevisionHistory. (R.1)

This extends the source Technical Analysis data requirement:

TestRecord
= Signal

- Protocol
- Gate
- Residual
- Invalidation
- Outcome. (R.2)

The source explicitly argues that ordinary backtests are insufficient when they store only signal and outcome. A mature record must also preserve the declared observation frame, confirmation gate, unresolved evidence, invalidation rule, and whether the original claim survived later revision.

---

# R.2 Why an Ordinary Price Dataset Is Insufficient

A conventional market dataset may contain:

- timestamp;
- open;
- high;
- low;
- close;
- volume;
- adjusted price;
- indicator values.

Such a dataset can test:

- return prediction;
- volatility forecasting;
- indicator correlations;
- strategy performance.

It cannot directly test:

- whether an analyst promoted a Structure into an Event prematurely;
- whether two confirming indicators were independent;
- whether a boundary was declared before the move;
- whether residual was hidden;
- whether a failed signal was relabelled;
- whether a market Event became a legal or accounting World event;
- whether a phase model outperformed a real-pair model.

Those questions require a richer corpus.

The minimum distinction is:

MarketData ≠ AnalysisRecord. (R.3)

MarketData records what was observable.

AnalysisRecord records how an observer declared, interpreted, gated, revised, and acted on what was observable.

---

# R.3 Three Benchmark Layers

The corpus should contain three linked layers.

# R.3.1 Evidence layer

The evidence layer contains data available to the observer at the declared time.

Examples:

- OHLCV;
- order-book data;
- breadth;
- options;
- credit spreads;
- corporate announcements;
- accounting disclosures;
- legal decisions.

Denote:

E_t = evidence available by time t. (R.4)

The benchmark must prevent future information from leaking into E_t.

---

# R.3.2 Interpretation layer

The interpretation layer contains:

- protocol;
- projected indicators;
- candidate claim;
- period and family typing;
- gate status;
- residual;
- invalidation.

Denote:

I_t = governed interpretation of E_t under P. (R.5)

---

# R.3.3 Consequence layer

The consequence layer contains:

- later price outcome;

- gate persistence;
- invalidation;
- institutional recognition;
- ledger effect;
- observer backreaction.

Denote:

$C_{t\to t+h}$ = consequential development after interpretation time t. (R.6)

The benchmark relation is:

$$E_t \to I_t \to C_{t\to t+h}. \quad (R.7)$$

The benchmark must not reconstruct $I_t$ using $C_{t\to t+h}$.

---

# R.4 Benchmark Unit

The fundamental unit is not one bar.

It is one **declared analytical case**.

Define:

$$\text{Case}_i = (P_i, E_i, \Pi_i, G_i, \mathscr{R}_i, T_i, L_i, Y_i, V_i). \quad (R.8)$$

where:

$P_i$ = protocol;
$E_i$ = evidence snapshot;
$\Pi_i$ = projections and typing;
$G_i$ = gate record;
$\mathscr{R}_i$ = residual record;
$T_i$ = transport record;
$L_i$ = ledger context;
$Y_i$ = observed outcome;
$V_i$ = version and revision history.

A case may represent:

- one support test;
- one breakout;
- one divergence;
- one wave endpoint;
- one institutional recognition event;

- one candidate complex-state episode.

---

# R.5 Corpus Families

The benchmark should be divided into several families rather than one undifferentiated dataset.

| Family | Main object | Core question |
|---|---|---|
| **F1 Projection** | indicator reading | What does the method directly observe? |
| **F2 Typing** | market object | Which period, family, and role apply? |
| **F3 Gate** | candidate Event | Has commitment occurred? |
| **F4 Residual** | unresolved evidence | What remains open or contradictory? |
| **F5 Transport** | cross-frame claim | What survives reframing? |
| **F6 Episode** | event sequence | Has a new grammar formed? |
| **F7 World** | institutional event | Which authority and ledger committed? |
| **F8 Revision** | model update | Was revision admissible? |
| **F9 Complex** | R–Q proposal | Is complex priority earned? |
| **F10 Phase Time** | episode clock | Does phase improve internal ordering? |
| **F11 Reflexivity** | observer adoption | Did analysis alter later behaviour? |
| **F12 Adversarial** | trap case | Does the system resist overclaiming? |

Each family should have its own label requirements and evaluation metrics.

---

# R.6 Corpus Versioning

The corpus itself must obey the same trace-preservation rules as the theories it tests.

Let:

$$\text{Corpus}_{v0} \rightarrow \text{Corpus}_{v1} \rightarrow \text{Corpus}_{v2}. \quad (R.9)$$

Every revision should preserve:

- old labels;
- reason for relabelling;
- adjudicator;
- newly available evidence;
- whether the original label was wrong or merely incomplete.

A revision record is:

$$\text{Revision}_i = (\text{OldLabel}, \text{NewLabel}, \text{Reason}, \text{EvidenceAdded}, \text{Authority}, \text{Time}). \quad (R.10)$$

The source self-revising declaration framework requires revision to preserve relevant ledger invariants rather than solving contradiction through amnesia.

Therefore:

CorpusCorrection ≠ CorpusHistoryDeletion. (R.11)

---

# R.7 Protocol Record

Every case requires a complete protocol record.

```
case_id:
protocol_id:
protocol_version:

asset:
universe:
venue:
observer_role:

boundary_rule:
timeframe_rule:
price_scale:
bar_rule:
price_field:

feature_map:
gate_rule:
residual_rule:
invalidation_rule:
outcome_horizon:

information_cutoff:
data_latency:
corporate_action_rule:
```

The source Technical Analysis schema explicitly includes protocol identifier, boundary rule, timeframe, price scale, bar construction, feature map, gate rule, residual rule, and invalidation condition.

---

# R.8 Evidence Snapshot

The evidence snapshot should contain only information available at the decision time.

Define:

$$E_i = E(\leq t_i). \quad (R.12)$$

A minimum market snapshot includes:

```
timestamp:
open:
high:
low:
close:
volume:
dollar_volume:
adjusted_close:
session_id:
timeframe:
```

Optional modules may include:

```
bid:
ask:
depth:
order_imbalance:
open_interest:
options_skew:
funding_rate:
breadth:
volume_profile:
news_timestamp:
accounting_status:
legal_status:
policy_status:
```

The evidence snapshot should preserve both:

- raw value;
- availability time.

A variable published at 16:15 cannot be treated as available for a 16:00 decision.

---

# R.9 Projection Record

Each derived object should be registered independently.

```
projection_id:
case_id:
operator_id:
source_lineage:
operator_word:
parameter_set:
value:
units:
uncertainty:
calculation_time:
data_version:
```

For example:

```
operator_id: RSI-14
source_lineage: adjusted_close_returns
operator_word: bounded_directional_normalization
value: 74.2
units: index_0_100
```

The source framework's central methodological claim is that an indicator is a partial projection rather than the total market field.

Thus the benchmark should test whether models can infer:

Projection_i ≠ MarketTotality. (R.13)

---

# R.10 Source-Lineage Graph

The corpus should preserve a directed acyclic graph of feature derivation.

For example:

AdjustedClose
→ Returns
→ RSI. (R.14)

AdjustedClose
→ EMA12,EMA26
→ MACD. (R.15)

AdjustedClose
→ Momentum. (R.16)

This makes visible that:

RSI, MACD, and Momentum (R.17)

may differ operationally while sharing one primary source.

Define shared-lineage overlap:

Λ_{ij}
= SharedAncestors(i,j)/TotalAncestors(i,j). (R.18)

A high Λ indicates that two methods should not be counted as strongly independent evidence.

---

# R.11 Typing Annotation

Every relevant object receives three primary labels.

# R.11.1 Closure period

p ∈ {Mark,Window,Structure,Event,Episode,World}. (R.19)

# R.11.2 Functional family

g ∈ {Load,Motion,Constraint,Commitment}. (R.20)

# R.11.3 Proto-Eight role

a ∈ {Gradient,Gate,Boundary,Exchange,Trigger,Guidance,Memory,Focus}. (R.21)

The full type is:

Type(X_i) = (p_i,g_i,a_i). (R.22)

---

# R.11.4 Multi-label rule

Some objects legitimately occupy more than one role.

For example, VWAP may be:

Primary:

Structure × Load × Memory. (R.23)

Secondary:

Structure × Constraint × Guidance. (R.24)

The benchmark should therefore distinguish:

- primary label;
- secondary labels;
- prohibited labels.

---

# R.11.5 Prohibited overpromotion

A typing case should also state the highest permitted closure level.

Example:

RSI = 78. (R.25)

Maximum claim without additional evidence:

Structure-level Motion. (R.26)

Prohibited claim:

Confirmed reversal Event. (R.27)

---

# R.12 Candidate-Event Record

A candidate Event record should distinguish the triggering condition from admission.

```
candidate_id:
case_id:
candidate_type:
candidate_time:
trigger_condition:
boundary_id:
direction:
normalized_displacement:
candidate_status:
```

Possible candidate types include:

- breakout;
- support failure;
- reversal;
- retest;
- gap;

- volatility release;
- liquidity failure;
- default warning.

The governing distinction is:

CandidateTransition ≠ AdmittedEvent. (R.28)

---

# R.13 Gate Annotation

The gate record should contain component-level evidence.

For a breakout:

```
crossing:
close_confirmation:
normalized_displacement:
volume_confirmation:
vwap_acceptance:
breadth_confirmation:
followthrough_confirmation:
retest_confirmation:
higher_frame_confirmation:
authority:
```

The gate output is:

$$G_i = (Decision_i, Strength_i, EvidenceVector_i). \quad (R.29)$$

Possible decisions:

- Admit;
- Partially Admit;
- Defer;
- Reject;
- Ambiguous.

The source schema similarly records close, volume, VWAP, breadth, retest, follow-through, gate strength, and gate status rather than collapsing everything into one binary label.

---

# R.14 Residual Annotation

Residual should be recorded as a typed object.

```
residual_id:
case_id:
gate_id:

residual_type:
source:
```

```
description:
severity:
directional_relevance:
persistence:
status:
resolution_condition:
resolution_time:
claim_effect:
```

The minimum residual taxonomy includes:

| Code | Residual type |
|---|---|
| R_data | missing or unreliable data |
| R_model | model limitation |
| R_boundary | boundary leakage or ambiguity |
| R_feature | feature-map insufficiency |
| R_gate | weak or failed commitment rule |
| R_trace | incomplete history |
| R_frame | transport or frame conflict |
| R_regime | uncertain $\chi$ |
| R_liquidity | liquidity or positioning |
| R_branch | alternate episode branch |
| R_governance | unclear authority |
| R_budget | excessive complexity or cost |

The self-revising declaration source explicitly argues that residual must be typed, scoped, attached, and revisitable; different residual types require different repairs.

---

# R.15 Residual State Transitions

A residual should not disappear without an explanation.

Possible states are:

Open
→ Monitoring
→ Resolved. (R.30)

Open
→ Dissipated. (R.31)

Open
→ ConvertedToLoad. (R.32)

Open
→ Invalidating. (R.33)

Open
→ Superseded. (R.34)

The benchmark must preserve the transition reason.

A decrease in visible residual may indicate either:

- genuine repair;
- concealment.

The distinction is:

ResidualRepair
= reduction with trace preservation, explanation, and invariance check. (R.35)

ResidualConcealment
= reduction by deletion, relabelling, or redefinition. (R.36)

This distinction follows directly from the source residual-governance framework.

---

# R.16 Invalidation Record

Every claim should have a prospective invalidation.

```
invalidation_id:
claim_id:
condition:
declared_time:
evaluation_horizon:
triggered:
trigger_time:
post_invalidation_status:
```

Examples:

### Breakout

Close inside old range plus failed reclaim.

### Divergence reversal

No reversal gate within the declared horizon.

### Elliott endpoint

Declared structural rule is breached.

### Complex state

Real-pair benchmark performs equally well or phase is scale-fragile.

### Phase-time model

No out-of-sample alignment gain over event count.

The source Technical Analysis paper's falsification rule is direct:

NoInvalidation → NoDiscipline. (R.37)

---

# R.17 Transport Record

Every transport test should be stored separately.

```
transport_id:
claim_id:

source_protocol:
target_protocol:
transformation:
expected_target_form:
observed_target_form:
distance_metric:
tolerance:
status:
transport_residual:
```

Possible transformations include:

- daily → weekly;
- linear → logarithmic;
- raw → ATR-normalized;
- time bar → volume bar;
- cap-weighted → equal-weight;
- price → breadth;
- price → volume profile;
- cash → derivative;
- market → accounting;
- market → legal.

The source framework defines Technical Objectivity as cross-protocol survival and emphasizes that legitimate transformations must be declared rather than chosen opportunistically.

---

# R.18 Transport Labels

Use the following controlled vocabulary:

| Label | Meaning |
|---|---|
| ExactSurvival | same relation preserved |
| CovariantSurvival | expected transformed relation preserved |
| PartialSurvival | core relation survives with material residual |
| LocalOnly | valid only under source protocol |
| Failure | target contradicts transported expectation |
| Indeterminate | insufficient evidence |
| NonComparable | no valid transport exists |

A claim should not be penalized for failing a transformation outside its intended scope.

Therefore the benchmark must distinguish:

AdmissibleTransformation
from
AdversarialButInvalidTransformation. (R.38)

---

# R.19 Outcome Record

The outcome record should contain more than return.

```
outcome_id:
case_id:
evaluation_horizon:

max_favorable_excursion:
max_adverse_excursion:
close_after_horizon:
volatility_after:
fakeout:
continuation:
reversal:
regime_change:
episode_transition:
world_transition:
invalidation_status:
```

For institutional cases, include:

```
market_recognition:
risk_recognition:
accounting_recognition:
contractual_recognition:
legal_recognition:
policy_recognition:
```

Outcome is not equivalent to correctness.

A poorly governed claim may profit by chance.

A well-governed claim may fail probabilistically while remaining properly specified.

The benchmark should evaluate both:

OutcomeQuality (R.39)

and:

ProcessQuality. (R.40)

---

# R.20 Process Labels

Each case should receive a process-quality label.

# R.20.1 Well governed

- protocol complete;
- gate predeclared;

- residual preserved;
- invalidation clear;
- no hindsight revision.

## R.20.2 Partially governed

- some required fields missing;
- claim still falsifiable.

## R.20.3 Ungoverned

- boundary or horizon changes after outcome;
- failed trace deleted;
- residual concealed;
- authority overreached.

## R.20.4 Correct-by-luck

Outcome matched the claim, but the process violated governance.

## R.20.5 Wrong-but-disciplined

Outcome failed, but the claim was properly declared, gated, and preserved.

This distinction is crucial.

A scientific benchmark must not teach:

ProfitableOutcome = GoodAnalysis. (R.41)

---

# R.21 Revision Record

Revision is annotated as:

```
revision_id:
claim_id:
old_protocol_version:
new_protocol_version:
triggering_evidence:
change_type:
change_reason:
trace_preserved:
residual_carried:
budget_change:
authorization:
```

Possible change types:

- boundary revision;

- feature revision;
- gate revision;
- timeframe revision;
- residual reclassification;
- model reduction;
- authority correction.

---

## R.21.1 Admissible revision

A revision is admissible when it is:

- well formed;
- trace preserving;
- residual honest;
- frame robust;
- budget bounded;
- non-degenerate.

These are the source declaration framework's central admissibility conditions.

---

## R.21.2 Inadmissible revision

Examples include:

- moving an anchor after failure;
- changing the horizon after outcome;
- deleting a failed signal;
- defining contradiction as confirmation;
- adding unlimited exceptions;
- changing Q normalization until phase appears stable.

---

## R.22 Complex-State Benchmark Module

A candidate complex-state case should contain:

```
R_name:
R_definition:
R_units:
R_source:
```

```
Q_name:
Q_definition:
Q_units:
Q_source:

metric:
normalization:
amplitude:
phase:
generator:
residual:
```

The benchmark must also store comparison models:

- scalar R model;
- flexible real-pair model;
- complex amplitude–phase model;
- alternative hyperbolic or nonlinear model.

---

# R.22.1 Complex-status labels

Use:

1. ScalarOnly;
2. UsefulRealPair;
3. ComplexCandidate;
4. ComplexEligible;
5. PhaseBearing;
6. PhaseTimeCandidate;
7. PhaseSensitiveEventModel;
8. TimeBearingWorldCandidate.

The label should reflect the highest passed gate.

---

# R.22.2 Mandatory negative cases

The corpus must include examples where:

- Q is a residual bucket;
- Q is a lagged copy of R;
- scaling creates the apparent phase;
- a Hilbert transform is mistaken for an independent economic channel;

- a real pair performs equally well;
- phase becomes unstable near zero amplitude.

A complex benchmark containing only successful constructions would encourage ornamental mathematics.

---

# R.23 Phase-Time Benchmark Module

A phase-time case should contain a family of comparable episodes.

For episode j:

E_j = [t_start,j,t_end,j]. (R.42)

Candidate clocks include:

t = calendar time. (R.43)

κ = normalized event count. (R.44)

v = cumulative volume. (R.45)

a = cumulative absolute return. (R.46)

σ = selection depth. (R.47)

$\tau_i$ = internal phase time. (R.48)

The primary benchmark is:

$D(\tau_i)$
versus
{D(t),D(κ),D(v),D(a),D(σ)}. (R.49)

A phase-time label is supported only when the gain:

- is out of sample;
- survives scaling;
- survives branch perturbation;
- improves the declared task.

---

# R.24 World-Event Benchmark Module

A World-level case requires:

- authority;
- recognition rule;
- ledger;
- changed rights or obligations;

- future-state consequence.

The recognition vector is:

$$\mathbb{G}_i = (g_{market}, g_{risk}, g_{accounting}, g_{contractual}, g_{legal}, g_{policy}). \quad (R.50)$$

Each component may be:

0 = not recognized;
0.5 = partial or provisional;
1 = admitted.

The benchmark must include disagreement cases.

Example:

$g_{market} = 1.$ (R.51)

$g_{risk} = 1.$ (R.52)

$g_{accounting} = 0.5.$ (R.53)

$g_{contractual} = 0.$ (R.54)

$g_{legal} = 0.$ (R.55)

The correct label is:

FragmentedRecognition. (R.56)

not:

UniversalDefault.

---

# R.25 Episode Benchmark Module

An Episode case should contain:

```
episode_id:
old_grammar:
candidate_new_grammar:
start_gate:
event_sequence:
dominant_chi:
boundaries:
alternative_branches:
completion_gate:
persistence_rule:
```

The benchmark should distinguish:

- local correction;
- Episode completion;
- higher-degree continuation;
- unresolved branch.

A wave count should not be judged only by visual fit.

The source requires predeclared invalidation, meaningful ledger scale, cross-checking, and residual status; the best-looking retrospective count is not automatically the valid one.

---

# R.26 Annotation Workflow

## R.26.1 Stage 1 — Evidence freezing

Freeze all information available at the declared cutoff.

## R.26.2 Stage 2 — Independent annotation

At least two annotators independently label:

- protocol completeness;
- period;
- family;
- role;
- candidate status;
- residual.

## R.26.3 Stage 3 — Gate adjudication

A separate reviewer evaluates the gate under the frozen protocol.

## R.26.4 Stage 4 — Transport review

Another reviewer applies the declared transformation set.

## R.26.5 Stage 5 — Outcome reveal

Future outcome is disclosed only after the pre-outcome annotation is locked.

## R.26.6 Stage 6 — Revision audit

Annotators may revise, but old labels remain preserved.

This order prevents outcome leakage into the original interpretation.

---

# R.27 Annotator Roles

The corpus should use several annotator classes.

## R.27.1 Technical analyst

Provides domain interpretation of chart methods.

## R.27.2 Quantitative reviewer

Checks formulas, leakage, scaling, and benchmarks.

## R.27.3 Market-structure reviewer

Checks liquidity, venue, and execution interpretation.

## R.27.4 Institutional reviewer

Checks accounting, legal, policy, or credit authority.

## R.27.5 Governance reviewer

Checks trace preservation, residual honesty, and revision admissibility.

No single annotator should have unilateral authority over all label types.

---

# R.28 Annotation Confidence

Each label should include confidence:

$c \in [0,1]$. (R.57)

A useful record is:

$$\text{Label}_i = (\text{Value}_i, \text{Confidence}_i, \text{Reason}_i, \text{Residual}_i). \quad \text{(R.58)}$$

Low confidence is not annotation failure.

It may be the correct representation of an ambiguous case.

The benchmark should reward systems that can return:

Indeterminate (R.59)

when the evidence does not support a stronger conclusion.

---

# R.29 Inter-Annotator Reliability

For single-label categories, calculate:

- Cohen's $\kappa$;
- Fleiss' $\kappa$;

- Krippendorff's α.

For hierarchical period labels, distinguish:

- exact agreement;
- adjacent-period disagreement;
- severe overpromotion.

Define weighted period distance:

d_p(p_i,p_j) = |index(p_i) – index(p_j)|. (R.60)

A disagreement between Structure and Event is less severe than one between Mark and World.

A weighted reliability score may be:

Reliability_p
= 1 – Mean[d_p]/5. (R.61)

---

# R.30 Gate Agreement

Gate agreement should be evaluated separately from typing agreement.

Annotators may agree that a candidate is an Event-level object but disagree on whether it should be:

- deferred;
- partially admitted;
- admitted.

Define ordinal gate distance:

d_G
= |rank(G_i) – rank(G_j)|. (R.62)

where:

Reject < Defer < Partial < Admit. (R.63)

Ambiguous should remain a separate branch rather than being forced onto the ordinal scale.

---

# R.31 Residual Agreement

Residual annotation is multi-label.

Use:

- Jaccard similarity;
- per-type precision and recall;
- severity agreement;

- resolution-state agreement.

For residual sets $\mathscr{R}_i$ and $\mathscr{R}_j$:

$$J(\mathscr{R}_i,\mathscr{R}_j) = |\mathscr{R}_i \cap \mathscr{R}_j|/|\mathscr{R}_i \cup \mathscr{R}_j|. \quad (R.64)$$

The benchmark should identify which residual types are:

- reliably observable;
- expert-dependent;
- too vague for operational use.

---

# R.32 Benchmark Splits

The dataset should use several splits.

# R.32.1 Chronological split

Train on earlier periods and test on later periods.

# R.32.2 Asset split

Train on one asset group and test on unseen assets.

# R.32.3 Regime split

Test whether rules transport across:

- trends;
- ranges;
- crises;
- recoveries.

# R.32.4 Protocol split

Hold out:

- weekly frames;
- volume bars;
- alternative benchmarks;
- institutional observer classes.

# R.32.5 Adversarial split

Reserve difficult cases entirely for final evaluation.

The strongest test is not memorizing the benchmark's recurring examples.

It is transporting the grammar into a new declared world.

---

# R.33 Leakage Controls

The benchmark must block several forms of leakage.

## R.33.1 Future-price leakage

No outcome data in the evidence snapshot.

## R.33.2 Revision leakage

Later-revised data cannot replace first-release data in live cases.

## R.33.3 Boundary leakage

Boundaries must be declared before outcome.

## R.33.4 Label leakage

File names and case descriptions should not reveal the final class.

## R.33.5 Narrative leakage

Do not describe a failed breakout as "the famous fakeout" before the model labels it.

## R.33.6 Cross-case duplication

Near-duplicate episodes must remain in the same train or test partition.

---

# R.34 Benchmark Tasks

## R.34.1 Task A — Projection explanation

Input:

indicator record.

Output:

- what it directly measures;
- what it does not measure;
- primary period and family.

## R.34.2 Task B — Period classification

Input:

evidence and claim.

Output:

Mark, Window, Structure, Event, Episode, or World.

---

## R.34.3 Task C — Functional decomposition

Output:

Load, Motion, Constraint, Commitment.

---

## R.34.4 Task D — Gate determination

Output:

Reject, Defer, Partial, Admit, or Ambiguous.

---

## R.34.5 Task E — Residual extraction

Output:

typed residual set with severity.

---

## R.34.6 Task F — Highest-supported-claim selection

Output:

maximum permitted claim and prohibited stronger claims.

---

## R.34.7 Task G — Transport prediction

Predict how the claim should appear under P′.

---

## R.34.8 Task H — Revision audit

Classify revision as:

- admissible;
- inadmissible;

- indeterminate.

---

## R.34.9 Task I — Complex eligibility

Choose:

- scalar;
- real pair;
- complex eligible;
- phase bearing;
- stronger level.

---

## R.34.10 Task J — World recognition

Identify which authorities have committed the event.

---

# R.35 Evaluation Metrics

A single accuracy score is insufficient.

The benchmark should report a vector:

Score
= (Typing,Gate,Residual,Transport,Authority,Revision,Reduction,Calibration). (R.65)

---

## R.35.1 Typing accuracy

Exact and weighted period accuracy.

## R.35.2 Gate macro-F1

Important because Admit cases may be rarer than candidate or deferred cases.

## R.35.3 Residual F1

Per residual type and macro average.

## R.35.4 Overpromotion rate

Define:

OPR
= ClaimsAboveGoldClosure/AllClaims. (R.66)

This is a central safety metric.

## R.35.5 Underpromotion rate

Define:

UPR
= ClaimsBelowGoldClosure/AllClaims. (R.67)

A model that never commits anything should not score highly.

## R.35.6 Trace-preservation score

Does the model retain earlier claims after revision?

## R.35.7 Authority error rate

How often does the model issue a commitment beyond the declared authority?

## R.35.8 Reduction accuracy

Does it return to the correct simpler model when complex or phase claims fail?

---

# R.36 Calibration Metrics

For probabilistic gate outputs:

- Brier score;
- log loss;
- calibration curve;
- expected calibration error.

If the model assigns:

Pr(Admit) = 0.80, (R.68)

then approximately 80% of similarly rated cases should satisfy the declared admission outcome under the relevant benchmark.

Calibration must be protocol-specific.

A daily-breakout probability cannot be interpreted as a probability of World-level institutional transition.

---

# R.37 Process–Outcome Matrix

Cases should be evaluated using both process and outcome.

| Process | Outcome succeeds | Outcome fails |
|---|---|---|
| **Governed** | disciplined success | disciplined failure |

| Process | Outcome succeeds | Outcome fails |
|---|---|---|
| **Ungoverned** | success by luck or overfit | ordinary analytical failure |

This produces four labels:

1. GovernedSuccess;
2. GovernedFailure;
3. UngovernedSuccess;
4. UngovernedFailure.

The benchmark should not train systems to imitate UngovernedSuccess.

That would reward hindsight and hidden residual.

---

# R.38 Goodhart Protection

Once the benchmark score becomes a target, models may learn to game it.

Possible gaming behaviours include:

- always choosing Structure to avoid overpromotion;
- listing every possible residual;
- returning Ambiguous for difficult cases;
- memorizing particular boundaries;
- producing verbose but uninformative governance language.

The source world-formation framework defines Goodhart failure as a metric improving while world health worsens, especially when the observable becomes the gate and residual is ignored.

Therefore the benchmark requires:

GoodBenchmark
= Metric

- ResidualAudit
- InvarianceTest
- AntiGamingRule. (R.69)

---

# R.39 Anti-Gaming Tests

# R.39.1 Residual precision penalty

A model listing every residual receives low precision.

# R.39.2 Commitment penalty

A model that never admits any Event is penalized through underpromotion.

## R.39.3 Explanation-grounding test

Each label must point to specific evidence fields.

## R.39.4 Protocol mutation test

Change irrelevant wording while preserving structure.

The output should remain equivalent.

## R.39.5 Counterfactual perturbation

Alter one gate component.

The model should revise only the affected conclusion.

## R.39.6 Hidden-template split

Reserve novel method names with known functional roles.

This tests whether the model learned the grammar rather than memorized labels.

---

# R.40 Minimal Public Benchmark

A first public release might contain 1,200 cases.

| Corpus family | Cases |
|---|---|
| Projection and typing | 250 |
| Breakout and support gates | 250 |
| Divergence and reversal | 120 |
| Episode segmentation | 120 |
| Residual and revision | 140 |
| Transport | 120 |
| Institutional World events | 80 |
| Complex and phase claims | 70 |
| Adversarial cases | 50 |
| **Total** | **1,200** |

The release should prioritize label quality over scale.

A smaller corpus with:

- frozen evidence;
- multi-reviewer adjudication;
- preserved disagreement;

- explicit protocol

is more valuable than millions of weakly labelled chart snippets.

---

# R.41 Case Difficulty Levels

## R.41.1 Level 1 — Direct

Example:

One trade above resistance with no close.

Expected:

Mark-level crossing only.

## R.41.2 Level 2 — Multi-channel

Price, volume, breadth, and VWAP provide mixed evidence.

## R.41.3 Level 3 — Cross-frame

Daily and weekly protocols disagree.

## R.41.4 Level 4 — Recursive

An earlier failed Event becomes current Load.

## R.41.5 Level 5 — Institutional

Market, accounting, legal, and policy ledgers diverge.

## R.41.6 Level 6 — Formal

Complex eligibility, phase-time, or admissible revision must be assessed.

Difficulty should reflect structural complexity, not merely data volume.

---

# R.42 Gold Labels and Disagreement Labels

Some cases will not possess one uncontested gold label.

The corpus should support:

## R.42.1 Gold

Strong expert agreement.

## R.42.2 Soft gold

Majority label plus confidence distribution.

## R.42.3 Disputed

Several admissible interpretations remain.

## R.42.4 Unresolved

Evidence is insufficient.

For disputed cases, publish:

Pr(Label = l_j | Annotators). (R.70)

A model should not be penalized for choosing a well-supported minority interpretation when the case itself remains open.

---

# R.43 Benchmarking Human–AI Collaboration

The corpus should compare four modes.

1. Human only;
2. AI only;
3. Human using AI suggestions;
4. AI with human gate authority.

Measure:

- accuracy;
- overpromotion;
- residual recall;
- time;
- revision quality;
- confidence calibration.

A plausible hypothesis is:

AI improves evidence organization, while human authority remains important for ambiguous gates and institutional consequences.

This is an empirical question.

The benchmark should not assume either human or AI superiority.

---

# R.44 Benchmarking LLMs

An LLM evaluation should test more than final labels.

The model should produce a structured object:

```
Declared protocol:
Observed projections:
Period:
Functional family:
Actuation role:
Gate status:
Residual:
Transport:
Highest supported claim:
Prohibited stronger claim:
Invalidation:
```

The score should check:

- field completion;
- evidence grounding;
- logical consistency;
- authority discipline;
- trace preservation.

A fluent paragraph with no explicit residual should not receive full credit.

The declaration source warns that fluent closure can appear complete while hiding unsupported claims; mature closure must disclose what was not observed, not decided, assumed, contradictory, and still needed.

---

# R.45 Benchmarking Self-Revision

A revision benchmark should reveal evidence in stages.

## Stage 0

Only price and boundary.

## Stage 1

Add volume.

## Stage 2

Add breadth.

## Stage 3

Add higher timeframe.

## Stage 4

Reveal later institutional event.

At each stage, the system must output:

$D_k$;
$Claim_k$;
$Residual_k$. (R.71)

The benchmark then evaluates:

- whether the new claim is justified;
- whether the old claim remains preserved;
- whether residual is resolved or merely hidden;
- whether complexity grows reasonably.

---

# R.46 Revision Distance

Define protocol distance:

$$D_P(P_k,P_{k+1}) = \Sigma_j w_j \delta_j. \quad (R.72)$$

where $\delta_j$ indicates changes in:

- boundary;
- timeframe;
- scale;
- feature map;
- gate;
- horizon;
- authority.

Define evidence gain:

$$\Delta E_k = \text{Information}(\text{NewEvidence} \mid \text{PriorEvidence}). \quad (R.73)$$

A suspicious revision has:

$$D_P(P_k,P_{k+1}) \gg \Delta E_k. \quad (R.74)$$

This may indicate narrative repair rather than learning.

---

# R.47 Benchmarking Transport

A transport task should provide:

1. source protocol P;
2. source claim C_P;
3. transformation T_{P→P′};
4. target evidence E_{P′}.

The model predicts:

Ĉ_{P′} = T_{P→P′}(C_P). (R.75)

Then compares:

Ĉ_{P′}
with
C_{P′}. (R.76)

The error is:

ε_T
= Dist(Ĉ_{P′},C_{P′}). (R.77)

Transport tasks are especially important because the source framework treats cross-frame survival as the operational route from pattern to stronger objectivity.

---

# R.48 Benchmarking Proto-Eight Roles

The corpus should include cases testing role distinction.

Examples:

### Trigger without Guidance

Sharp catalyst move with no continuation.

### Gradient without Gate

Persistent pressure unable to pass a boundary.

### Boundary without Exchange

Old level with little current activity.

### Memory without Focus

Large information archive with no task-specific selection.

### Focus without Memory

Single-candle interpretation ignoring prior failed Events.

The benchmark question is:

Does the role diagnosis predict the characteristic failure? (R.78)

If not, the eight-role refinement may add vocabulary without operational value.

---

# R.49 Benchmarking Missing-Instrument Proposals

Candidate new instruments from Appendix M should enter the corpus only after operational definition.

For each instrument:

InstrumentBenchmark
= Reliability

- IncrementalValue
- GateUtility
- Transport
  – Complexity. (R.79)

Examples include:

- Path-Bearing Candle;
- Structural Acceptance Score;
- Residual-Bearing Gate Score;
- Episode Completion Certificate;
- Observer Crowding Index.

The benchmark must permit the conclusion:

No new instrument is needed. (R.80)

---

# R.50 Reproducibility Package

Every benchmark release should include:

- protocol documentation;
- schema;
- annotation guide;
- raw-data provenance;
- transformation code;
- baseline models;

- adjudication records;
- known limitations;
- change log.

The release package is:

Release_v
= Data_v

- Schema_v
- Labels_v
- Code_v
- Audit_v
- Residual_v. (R.81)

---

# R.51 Baseline Models

The benchmark should include simple baselines.

# R.51.1 Majority label

Tests class imbalance.

# R.51.2 Keyword or rule baseline

Example:

“close above resistance” → Admit.

This should fail difficult residual cases.

# R.51.3 Conventional classifier

Uses price, volume, and volatility.

# R.51.4 Periodic-grammar classifier

Uses period, function, gate, residual, and transport features.

# R.51.5 Flexible language model

Receives full case text.

The central comparison is:

Does explicit world-formation structure improve reliability and reduce overclaiming? (R.82)

---

# R.52 Benchmark Research Hypotheses

## H1 — Typing reliability

Independent annotators can distinguish the six periods above chance.

## H2 — Functional value

Four-family labels improve failure diagnosis beyond conventional indicator categories.

## H3 — Gate value

Explicit gate records improve Event calibration.

## H4 — Residual value

Residual burden predicts invalidation or revision.

## H5 — Transport value

Transport survival predicts broader persistence.

## H6 — Revision governance

Trace-preserving revision reduces hindsight relabelling.

## H7 — Complex reduction

The benchmark can distinguish earned complex structures from decorative complexification.

## H8 — Authority discipline

World-level annotation reduces market-to-legal category errors.

These hypotheses must remain separately testable.

---

# R.53 Failure Criteria for the Benchmark Programme

The benchmark programme should be revised if:

1. annotators cannot reproduce protocol labels;
2. period and family classifications remain unreliable;
3. residual categories are too vague;
4. transport judgments depend mostly on reviewer preference;
5. the corpus rewards excessive conservatism;

6. benchmark scores fail to predict real analytical quality;
7. models learn template language rather than structural reasoning;
8. institutional authority labels cannot be standardized;
9. revision records become too expensive to maintain;
10. simpler schemas perform equally well.

A benchmark is not useful merely because it is elaborate.

---

# R.54 Minimal Benchmark Record

A minimal complete case is:

```
Case ID:
Protocol:
Information cutoff:
Evidence:
Projection:
Period:
Function:
Actuation role:
Candidate:
Gate:
Residual:
Transport:
Highest supported claim:
Prohibited claim:
Invalidation:
Outcome:
Revision:
```

In equation form:

Case_i
= P_i

- E_i
- Ô_i
- Type_i
- G_i
- $\mathscr{R}$_i
- T_i
- Claim_i
- I_i
- Y_i
- V_i. (R.83)

---

# R.55 Example Benchmark Case

## Case ID

BRK-0047

## Protocol

Daily log-scale equity breakout.

## Evidence cutoff

Market close on Day 0.

## Evidence

- close 1.1 ATR above resistance;
- relative volume 1.8;
- above VWAP;
- breadth 54%;
- weekly resistance 0.6 ATR above;
- no retest yet.

## Gold typing

Event × Commitment × Gate.

## Gate label

Partially Admit.

## Residual

- weak breadth;
- higher-frame Constraint;
- retest pending;
- value migration pending.

## Highest supported claim

Partially admitted daily breakout.

## Prohibited claims

- broad sector breakout;
- durable Episode transition;
- World transition.

## Invalidation

Close inside prior range plus failed reclaim.

## Later outcome

Daily break fails after three sessions.

## Process label

GovernedFailure.

The original analysis was disciplined even though the Event failed.

---

# R.56 Example Complex Benchmark Case

## Case ID

CPX-0018

## Proposal

R = standardized trend acceptance.
Q = one-period lag of R.

## Observation

Phase portrait appears circular.

## Required analysis

- same source;
- lagged-copy dependence;
- no independent Q;
- autoregressive benchmark matches performance.

## Gold label

Useful delay embedding, not domain-conjugate complex state.

## Reduction

Z → (R,R_{t−1}). (R.84)

## Prohibited claim

Internal market phase time.

---

# R.57 Example World Benchmark Case

## Case ID

WRLD-0021

## Evidence

- bond price collapse;
- credit spread above 1,500 basis points;
- internal risk downgrade;
- no missed payment;
- no covenant breach;
- no legal filing.

## Recognition vector

g_market = 1. (R.85)

g_risk = 1. (R.86)

g_accounting = 0.5. (R.87)

g_contractual = 0. (R.88)

g_legal = 0. (R.89)

## Gold label

Severe market and risk recognition with unresolved contractual and legal state.

## Prohibited claim

Legal default.

---

# R.58 Benchmark Governance Board

A mature corpus should have a small governance board containing:

- market-method expert;
- quantitative-method expert;
- data-quality expert;
- institutional-domain expert;
- audit and reproducibility reviewer.

The board should not decide every case.

Its role is to govern:

- schema changes;
- label ontology;
- conflict resolution;
- release standards;
- deprecation;
- appeal.

---

# R.59 Benchmark Appeals

Annotators and users should be able to challenge a label.

An appeal contains:

```
case_id:
challenged_label:
proposed_label:
supporting_evidence:
protocol_argument:
transport_argument:
residual_argument:
```

The appeal outcome may be:

- uphold;
- revise;
- add alternate admissible label;
- mark disputed;
- mark insufficient evidence.

The original label remains in the revision history.

---

# R.60 Benchmark as a Time-Bearing Research World

The benchmark itself illustrates the article's theory.

## Boundary

Which cases and evidence are included?

## Projection

Which labels are extracted?

## Gate

What qualifies as accepted annotation?

## Trace

Which labels enter the release?

## Residual

Which disagreements remain unresolved?

## Ledger

How are versions and decisions preserved?

## Transport

Do definitions survive new domains?

## Revision

How does the benchmark change without erasing history?

Thus:

BenchmarkWorld
= Boundary

- AnnotationProtocol
- AdjudicationGate
- VersionedTrace
- ResidualRegister
- RevisionRule. (R.90)

A benchmark that silently overwrites its labels would violate the same framework it seeks to test.

---

## R.61 Open Benchmark Release Sequence

A realistic development path is:

## Release 0.1 — Typing corpus

Focus:

- six periods;
- four families;
- eight roles.

## Release 0.2 — Gate and residual corpus

Add:

- breakout;
- support;
- divergence;
- retest;
- invalidation.

## Release 0.3 — Transport corpus

Add cross-frame tasks.

## Release 0.4 — Episode and revision corpus

Add:

- wave branches;
- completion gates;
- protocol revisions.

## Release 0.5 — World corpus

Add:

- accounting;
- legal;
- credit;

- policy recognition.

## Release 1.0 — Integrated benchmark

Require end-to-end governed analysis.

This staged path prevents the benchmark from becoming too broad before its core annotations are reliable.

---

# R.62 Benchmark Leaderboard Principles

A public leaderboard should show multiple dimensions.

| Model | Typing | Gate F1 | Residual F1 | Overpromotion ↓ | Authority error ↓ | Revision integrity | Transport |
|---|---|---|---|---|---|---|---|

There should be no single dominant composite score unless its weighting is clearly disclosed.

A model with excellent prediction but high authority error should not be ranked as universally superior.

A model with low overpromotion but extreme underpromotion should also not dominate.

---

# R.63 The Central Benchmark Challenge

The deepest challenge is not:

> Can a system predict whether price rises tomorrow?

It is:

> Can a system describe exactly what has and has not become true under a declared market protocol?

That requires it to distinguish:

- evidence from interpretation;
- Structure from Event;
- Event from Episode;
- market recognition from institutional recognition;
- residual from conjugate state;
- phase from time;
- learning from retrospective repair.

---

# R.64 Appendix R Conclusion

The Periodic Grammar requires a benchmark that records more than prices and predictions.

It requires a corpus of declared observations.

The minimum benchmark chain is:

Evidence Snapshot
→ Protocol
→ Projection
→ Periodic Typing
→ Gate
→ Trace + Residual
→ Transport
→ Outcome
→ Revision. (R.91)

Its minimum integrity rule is:

No label without protocol. (R.92)

Its minimum event rule is:

No Event without gate. (R.93)

Its minimum honesty rule is:

No commitment without residual record. (R.94)

Its minimum learning rule is:

No revision without preserved prior trace. (R.95)

Its minimum objectivity rule is:

No broad claim without admissible transport. (R.96)

Its minimum complexity rule is:

No complex or phase label without defeating the simpler benchmark. (R.97)

The benchmark's strongest contribution would not be to identify one best Technical Analysis method.

It would be to create a shared research world in which:

- every signal is reconstructable;
- every gate is visible;
- every failure remains queryable;
- every disagreement is typed;
- every revision is accountable;
- every strong claim can be traced to the closure that actually supports it.

# Appendix S — Canonical Ontology and Conformance

# Standard

# S.1 Purpose

The framework now contains enough concepts to require a stable specification.

Without a canonical ontology, different implementations may use the same words while referring to different objects.

For example:

- one system may call any line crossing a breakout;
- another may reserve breakout for a closing gate;
- another may require volume and retest;
- another may call an institutional announcement a breakout;
- another may use Q for breadth, residual, volatility, or model error.

Such differences are not merely terminological.

They prevent:

- replication;
- cross-system comparison;
- benchmark construction;
- software interoperability;
- admissible model revision.

This appendix therefore defines a proposed **Proto-Periodic Market Grammar Standard**, abbreviated:

PPMG. (S.1)

The standard specifies:

1. canonical objects;
2. permitted relations;
3. required metadata;
4. claim states;
5. conformance levels;
6. validation rules;
7. serialization formats;
8. revision and governance procedures.

Its central objective is:

> Two analysts or systems should be able to exchange a market claim and determine exactly what was observed, under which protocol, at which closure depth, through which gate, with which residual, and at what evidential status.

The standard does not determine whether a market claim is economically correct.

It determines whether the claim is **well formed and auditable**.

---

# S.2 Normative Vocabulary

The following words are used normatively.

## MUST

A requirement is mandatory for the declared conformance level.

## MUST NOT

The action is prohibited.

## SHOULD

The requirement is strongly recommended, but a documented exception may exist.

## SHOULD NOT

The action is discouraged and requires justification.

## MAY

The feature is optional.

## UNDEFINED

The object has not been established under the protocol.

## UNKNOWN

The object is meaningful, but the evidence is insufficient to determine its state.

These terms prevent two common errors:

Undefined ≠ Zero. (S.2)

Unknown ≠ False. (S.3)

For example, phase θ is **undefined** when no eligible complex state exists.

It may be **unknown** when the complex state exists but current measurement uncertainty is too

large.

---

# S.3 Standard Architecture

The standard is organized into nine layers.

| Layer | Canonical object | Function |
|---|---|---|
| 1 | Protocol | declares the observational world |
| 2 | Evidence | records what was available |
| 3 | Projection | produces technical observables |
| 4 | Type | assigns period, family, and role |
| 5 | Claim | states what is proposed |
| 6 | Gate | commits, defers, or rejects |
| 7 | Residual | preserves unresolved structure |
| 8 | Ledger | records consequential history |
| 9 | Transport and Revision | tests survival and governs change |

The core dependency order is:

Protocol
→ Evidence
→ Projection
→ Type
→ Claim
→ Gate
→ Trace + Residual
→ Ledger
→ Transport
→ Revision. (S.4)

The source declaration architecture similarly places declaration before projection and requires gate, trace, residual, and revision to be specified before a bounded observer can produce an auditable world.

---

# S.4 Canonical Entity Set

Define the ontology:

$\mathcal{O}$_PPMG
=
{Protocol,Evidence,Operator,Projection,Boundary,Claim,Gate,Residual,Trace,Ledger,Transport,R
(S.5)

Each entity has:

- a persistent identifier;
- a version;
- a creation time;
- a declared authority or creator;

- a status;
- links to its parent entities.

No material claim should exist as unstructured prose alone.

Prose may summarize the claim.

The underlying structured object must remain available.

---

# S.5 Identifier Standard

Every object MUST have a globally unique identifier.

A recommended format is:

PPMG::::. (S.6)

Examples:

```
PPMG:Protocol:NBE:DailyBreakout:1.0
PPMG:Claim:NBE:Breakout-20260724:1.0
PPMG:Gate:NBE:DailyCloseGate:1.0
PPMG:Residual:NBE:WeakBreadth-20260724:1.0
```

Identifiers MUST remain stable after publication.

A revised object receives a new version or identifier.

The original object MUST NOT be overwritten.

---

# S.6 Protocol Entity

# S.6.1 Canonical definition

A protocol is:

P
=
(Subject,Boundary,ObservationRule,Horizon,Scale,FeatureMap,GateRule,ResidualRule,Authority
(S.7)

A conforming protocol record MUST include:

```
protocol_id
protocol_version
subject
universe
venue_or_domain
observer_role
boundary_rule
observation_rule
timeframe
scale
feature_map
gate_rule
residual_rule
```

```
invalidation_rule
outcome_horizon
information_cutoff
authority
```

---

# S.6.2 Protocol completeness

Define protocol completeness:

C_P
= RequiredFieldsPresent/RequiredFieldsTotal. (S.8)

For Event-level conformance:

C_P = 1. (S.9)

A protocol missing a boundary, gate, or invalidation cannot issue a conforming Event claim.

It may still issue a descriptive Structure claim.

---

# S.6.3 Protocol scope

A protocol MUST declare its scope.

Examples:

- one security;
- one index;
- one credit instrument;
- one accounting entity;
- one legal jurisdiction;
- one institutional portfolio.

Scope ambiguity MUST be recorded as residual.

---

# S.7 Evidence Entity

# S.7.1 Definition

Evidence is an observation available under a declared information cutoff.

Define:

E_i
= (Value_i,Source_i,AvailabilityTime_i,Units_i,Quality_i). (S.10)

A conforming evidence record MUST include:

```
evidence_id
```

```
source
observed_value
units
observation_time
availability_time
data_version
quality_status
revision_status
```

---

# S.7.2 Availability invariant

A claim evaluated at time t MUST use only evidence satisfying:

AvailabilityTime_i ≤ t. (S.11)

Any later evidence MUST be stored as:

PostClaimEvidence. (S.12)

It may support revision.

It cannot be treated as if it had been available to the original claim.

---

# S.7.3 Revised evidence

When data are revised, both values SHOULD be preserved:

E_initial. (S.13)

E_revised. (S.14)

The original live claim MUST remain linked to E_initial.

---

# S.8 Operator Entity

# S.8.1 Definition

An operator transforms evidence into a projection:

Ô_j: E → X_j. (S.15)

A conforming operator record MUST include:

```
operator_id
operator_name
source_lineage
operator_word
parameters
units_in
units_out
primary_period
primary_family
primary_actuation_role
known_failure_modes
```

---

## S.8.2 Operator word

The operator word describes the transformation sequence.

For a moving average:

w_MA
= Smooth_n ∘ SelectPriceField. (S.16)

For MACD:

w_MACD
= Difference ∘ PairOfExponentialSmoothers ∘ SelectClose. (S.17)

For a breakout candidate:

w_break
= CompareToBoundary ∘ NormalizeDisplacement ∘ DetectCrossing. (S.18)

---

## S.8.3 Source-lineage requirement

Every projection MUST link to its raw ancestors.

This permits calculation of source overlap:

Λ_{ij}
= SharedSourceWeight(i,j). (S.19)

Systems MUST NOT label two projections as source-independent when their lineage overlap exceeds the declared independence tolerance.

---

## S.9 Projection Entity

A projection is:

X_{j,P,t}
= Ô_{j,P}(E_{≤t}). (S.20)

A projection record MUST include:

```
projection_id
operator_id
protocol_id
calculation_time
information_cutoff
value
units
uncertainty
source_evidence_ids
```

A projection MAY also include:

- confidence interval;
- missing-data flag;

- scale sensitivity;
- parameter sensitivity.

A projection MUST NOT directly claim an Event unless the operator itself is a declared Event gate.

---

# S.10 Boundary Entity

# S.10.1 Definition

A boundary specifies a meaningful region or transition surface.

Define:

B_P
= {x | b_low ≤ x ≤ b_high}. (S.21)

For a line boundary:

b_low = b_high. (S.22)

A boundary record MUST include:

```
boundary_id
protocol_id
construction_rule
lower_bound
upper_bound
units
declared_time
validity_horizon
source_evidence
transport_status
```

---

# S.10.2 Prospective declaration

An Event study MUST distinguish:

ProspectiveBoundary (S.23)

from:

RetrospectiveBoundary. (S.24)

A retrospective boundary MAY be used for description.

It MUST NOT be represented as prospective evidence.

---

# S.10.3 Boundary decay

A boundary MAY include a decay rule:

Mass_B(t)

= Mass_B(to)exp[−λ_B(t−to)]. (S.25)

The standard does not prescribe one decay function.

It requires the decay assumption to be declared when used.

---

# S.11 Canonical Typing System

Every relevant object MUST be typed along three axes.

## S.11.1 Closure period

```
Mark
Window
Structure
Event
Episode
World
```

## S.11.2 Functional family

```
Load
Motion
Constraint
Commitment
```

## S.11.3 Proto-Eight actuation role

```
Gradient
Gate
Boundary
Exchange
Trigger
Guidance
Memory
Focus
```

The complete type is:

τ(X)
= (Period,Family,ActuationRole). (S.26)

---

## S.11.4 Primary and secondary types

A conforming object MUST have one primary type.

It MAY have secondary types.

Example:

```
VWAP
Primary: Structure × Load × Memory
Secondary: Structure × Constraint × Guidance
```

Secondary typing MUST NOT be used to evade the requirements of the primary claim.

# S.12 Claim Entity

# S.12.1 Definition

A claim is:

C_k
= (Content,Protocol,Type,State,Evidence,Gate,Horizon,Invalidation). (S.27)

A conforming claim record MUST include:

```
claim_id
claim_text
protocol_id
created_time
information_cutoff
closure_period
functional_family
actuation_role
claim_class
current_state
supporting_projection_ids
contradictory_projection_ids
gate_id
outcome_horizon
invalidation_id
authority
```

---

# S.12.2 Claim classes

The standard defines:

```
Description
Structure
Warning
CandidateEvent
AdmittedEvent
AcceptedStructure
EpisodeTransition
WorldTransition
ComplexState
PhaseOrder
PhaseTime
TimeBearingWorld
```

A claim MUST NOT use a class stronger than its current closure state permits.

---

# S.13 Claim-State Machine

The canonical claim states are:

```
Undeclared
Declared
Projected
Typed
Candidate
Deferred
```

```
PartiallyAdmitted
Admitted
Accepted
Invalidated
Promoted
Revised
Closed
```

Permitted state transitions MUST be explicitly recorded.

A system MUST reject:

Projected → Admitted (S.28)

unless the projection itself is a registered gate output.

A system MUST reject:

CandidateEvent → WorldTransition (S.29)

without intervening authority and World-level gate records.

---

# S.14 Gate Entity

# S.14.1 Definition

A gate evaluates a candidate transition:

G_P(c_k,L_k,$\mathscr{R}$_k)
→ (Decision,Strength,Trace,Residual). (S.30)

A gate record MUST include:

```
gate_id
protocol_id
candidate_id
gate_type
component_evidence
decision
admission_strength
authority
decision_time
residual_ids
```

---

# S.14.2 Canonical gate decisions

```
Admit
PartiallyAdmit
Defer
Reject
Ambiguous
```

The decision MUST NOT be inferred solely from admission strength.

For example:

Strength = 0.72 (S.31)

does not automatically imply Admit unless the protocol threshold and mandatory components pass.

---

# S.14.3 Mandatory component rule

A gate MAY specify mandatory components.

For a breakout:

```
mandatory:
  close_confirmation
  minimum_displacement
optional:
  volume
  breadth
  retest
```

If a mandatory component fails:

Decision ≠ Admit. (S.32)

unless the protocol is revised prospectively.

---

# S.15 Residual Entity

# S.15.1 Definition

A residual is an unresolved object attached to a claim or gate.

Define:

r_j
= (Type,Description,Severity,Persistence,ClaimEffect,ResolutionRule). (S.33)

A residual record MUST include:

```
residual_id
parent_claim_id
parent_gate_id
residual_type
description
severity
status
persistence
directional_relevance
claim_effect
resolution_condition
```

---

# S.15.2 Canonical residual types

```
Data
Model
Boundary
Feature
Gate
Trace
```

```
Frame
Regime
Liquidity
Positioning
Branch
Governance
Authority
Complexity
```

Implementations MAY add subtypes.

They SHOULD map custom subtypes to one canonical parent type.

---

# S.15.3 Residual status

```
Open
Monitoring
Resolved
Dissipated
Converted
Invalidating
Superseded
```

Residual MUST NOT disappear through deletion.

A state transition MUST be recorded.

The source self-revising framework likewise treats residual as a persistent incompleteness signal whose handling must remain visible through later revision.

---

# S.16 Trace Entity

A trace is an admitted record with future consequence.

Define:

e_k
= (Claim,Gate,Time,Authority,Consequence). (S.34)

A trace record MUST include:

```
trace_id
claim_id
gate_id
committed_time
authority
consequence_type
persistence_rule
access_rule
```

A stored value with no declared consequence MAY be called a log.

It MUST NOT be called a consequential trace under this standard.

---

# S.17 Ledger Entity

## S.17.1 Definition

A ledger is an ordered collection of traces and attached residuals:

$$L_k = \{(e_1,r_1),(e_2,r_2),\ldots,(e_k,r_k)\}. \quad (S.35)$$

A ledger record MUST preserve:

- order;
- protocol version;
- authority;
- residual;
- revision links.

---

## S.17.2 Ledger types

The standard recognizes:

```
MarketLedger
RiskLedger
AccountingLedger
ContractualLedger
LegalLedger
PolicyLedger
ModelLedger
```

A claim MUST specify which ledger it enters.

---

## S.17.3 Cross-ledger references

A trace in one ledger MAY reference a candidate trace in another.

Example:

Market distress
→ Accounting impairment candidate. (S.36)

But the destination gate remains independent.

A reference MUST NOT be represented as destination commitment.

---

# S.18 Authority Entity

## S.18.1 Definition

Authority specifies who or what may commit a state.

A conforming authority record includes:

```
authority_id
authority_type
domain
jurisdiction
scope
delegated_permissions
effective_period
```

---

# S.18.2 Authority rule

A gate MUST satisfy:

AuthorityScope(Gate)
⊇ RequiredAuthority(Event). (S.37)

Examples:

- exchange execution engine may commit a trade;
- company management may issue an accounting estimate;
- auditor may opine on statements;
- court may issue a legal judgment;
- technical analyst may issue an analytical claim.

An analytical model MUST NOT represent itself as committing a legal or accounting state unless such authority is explicitly delegated.

---

# S.19 Transport Entity

# S.19.1 Definition

Transport maps a claim from protocol P to an expected form under P′:

T_{P→P′}(C_P) = Ĉ_{P′}. (S.38)

A transport record MUST include:

```
transport_id
source_claim_id
source_protocol_id
target_protocol_id
transformation_rule
expected_target_form
observed_target_form
distance_metric
tolerance
status
transport_residual
```

---

# S.19.2 Canonical statuses

```
ExactSurvival
```

```
CovariantSurvival
PartialSurvival
LocalOnly
Failure
Indeterminate
NonComparable
```

The source filtration and Technical Analysis frameworks treat cross-protocol survival as the operational basis of stronger objectivity rather than assuming that one observer's projection is universally authoritative.

---

# S.20 Revision Entity

# S.20.1 Definition

A revision changes the future declaration without erasing earlier trace.

Define:

D_{k+1}
= U_a(D_k,L_k,$\mathscr{R}$_k). (S.39)

A revision record MUST include:

```
revision_id
parent_object_id
old_version
new_version
triggering_evidence
changed_fields
unchanged_fields
reason
authority
trace_preserved
residual_disposition
complexity_change
prospective_retest_required
```

---

# S.20.2 Admissible revision

A revision is conforming when:

AdmissibleRevision
= WellFormed
∧ TracePreserving
∧ ResidualHonest
∧ FrameAccounted
∧ BudgetBounded
∧ NonDegenerate. (S.40)

---

# S.20.3 Prohibited revision

The following are nonconforming:

- deleting the original claim;

- silently moving a boundary;
- silently extending the horizon;
- changing a failed gate after outcome;
- reversing the meaning of contradictory evidence;
- altering Q scaling until phase appears;
- promoting authority without delegation.

---

# S.21 Outcome Entity

An outcome records what occurred after the declared horizon.

A conforming outcome record includes:

```
outcome_id
claim_id
evaluation_time
declared_horizon
outcome_class
invalidation_triggered
maximum_favorable_excursion
maximum_adverse_excursion
episode_status
world_status
```

Outcome MUST be stored separately from process quality.

A claim may be:

GovernedFailure. (S.41)

Another may be:

UngovernedSuccess. (S.42)

The standard MUST NOT equate profitable outcome with conforming analysis.

---

# S.22 Conformance Profiles

Not every use case requires every ontology component.

The standard therefore defines seven profiles.

---

# S.22.1 Profile D0 — Descriptive Projection

Required:

- protocol;
- evidence;

- operator;
- projection;
- units.

Permitted claim level:

Description.

Example:

$RSI_{14} = 72.4$.

---

## S.22.2 Profile D1 — Governed Structure

Required:

- D0 requirements;
- period;
- functional family;
- actuation role;
- missing-variable statement;
- invalidation or scope limit.

Permitted claim level:

Structure or Warning.

---

## S.22.3 Profile E1 — Governed Event

Required:

- D1 requirements;
- declared boundary;
- candidate record;
- gate;
- residual;
- invalidation;
- outcome horizon.

Permitted claim level:

CandidateEvent or AdmittedEvent.

---

## S.22.4 Profile E2 — Transported Event

Required:

- E1 requirements;
- one or more transport tests;
- source and target protocols;
- transport residual.

Permitted claim level:

Cross-frame Event regularity.

---

## S.22.5 Profile P1 — Episode

Required:

- E2 requirements;
- event sequence;
- old grammar;
- candidate new grammar;
- completion gate;
- branch residual;
- persistence condition.

Permitted claim level:

EpisodeTransition.

---

## S.22.6 Profile W1 — World

Required:

- P1 or equivalent institutional record;
- authority;
- ledger type;
- recognition rule;
- changed future rights or admissibility;
- backreaction candidate.

Permitted claim level:

WorldTransition.

---

## S.22.7 Profile C1 — Complex and Phase

Required:

- defined R and Q;
- units and metric;
- real-pair benchmark;
- generator test;
- scaling test;
- phase utility;
- reduction rule.

Additional PhaseTime status requires:

- episode family;
- comparison clocks;
- out-of-sample alignment;
- gate-hazard test.

---

# S.23 Conformance Ladder

The profiles form a partial evidence ladder:

D0
→ D1
→ E1
→ E2
→ P1
→ W1. (S.43)

Complex conformance is orthogonal:

D1 or higher
→ C1. (S.44)

A complex model may remain descriptive or diagnostic.

Complexity does not automatically promote closure period.

For example:

ComplexEligibleStructure
≠ WorldTransition. (S.45)

---

# S.24 Validation Rules

A conforming validator SHOULD implement the following rules.

## Rule V1 — Protocol presence

Every nontrivial claim must reference a protocol.

## Rule V2 — Information-time consistency

Every evidence item must be available by the claim cutoff.

## Rule V3 — Unit consistency

Combined variables must use compatible units or a declared metric.

## Rule V4 — Source-lineage visibility

Derived variables must identify ancestors.

## Rule V5 — Period ceiling

Claim level must not exceed closure level.

## Rule V6 — Gate requirement

Event claims require gate records.

## Rule V7 — Residual requirement

Event and higher claims require residual records, including explicit “no material residual detected” when justified.

## Rule V8 — Authority requirement

World commitments require declared authority.

## Rule V9 — Transport qualification

Broad claims require relevant transport.

## Rule V10 — Revision trace

New versions must link to old versions.

## Rule V11 — Complex eligibility

Phase fields require an eligible R–Q state.

## Rule V12 — Reduction

Failed higher-level tests must produce a lower-level model status.

---

# S.25 Validation Severity

Validator findings are classified as:

## Error

The record is nonconforming and cannot support its declared claim.

Examples:

- Event with no gate;
- future evidence leakage;
- World claim with no authority.

## Warning

The record is valid but materially incomplete.

Examples:

- no higher-timeframe transport;
- unresolved breadth;
- weak residual coding.

## Notice

A non-critical improvement is recommended.

Examples:

- missing optional confidence interval;
- incomplete secondary typing.

---

# S.26 Formal Claim-Ceiling Rule

Let:

L_data = highest level supported by evidence.
L_gate = highest level supported by commitment.
L_authority = highest level supported by authority.
L_transport = highest level supported by cross-frame survival.
L_model = highest level supported by model eligibility.

Then:

L_claim
≤ min(L_data,L_gate,L_authority,L_transport,L_model). (S.46)

This is a governance rule rather than a mathematical theorem about markets.

It prevents one strong dimension from compensating for a missing mandatory dimension.

For example:

strong price evidence

- no legal authority
  ⇒ no legal World claim. (S.47)

---

# S.27 Claim Status Object

A standard claim status SHOULD be represented as:

ClaimStatus
= (Closure,Admission,Residual,Transport,Authority,ModelLevel). (S.48)

Example:

```
Closure: Event
Admission: PartiallyAdmitted
Residual: Material
Transport: LocalOnly
Authority: MarketObserver
ModelLevel: RealPair
```

This is more informative than:

Bullish breakout.

---

# S.28 Standard Report Format

A conforming human-readable report SHOULD use this order.

## 1. Protocol

What was declared?

## 2. Evidence cutoff

What was known?

## 3. Projections

What did the operators produce?

## 4. Typing

At which period and function?

## 5. Candidate

What possible transition occurred?

## 6. Gate

What passed or failed?

## 7. Residual

What remains unresolved?

## 8. Transport

Which frames support or weaken it?

## 9. Claim ceiling

What is the strongest permitted statement?

## 10. Invalidation

What would change the status?

## 11. Revision state

Has the protocol changed?

---

# S.29 Standard Machine Serialization

A simplified JSON-compatible representation is:


```
{
  "claim_id": "PPMG:Claim:NBE:Breakout-20260724:1.0",
  "protocol_id": "PPMG:Protocol:NBE:DailyBreakout:1.0",
  "information_cutoff": "2026-07-24T16:30:00Z",
  "type": {
    "period": "Event",
    "family": "Commitment",
    "actuation_role": "Gate"
  },
  "candidate": {
    "type": "ResistanceBreakout",
    "detected": true
  },
  "gate": {
    "decision": "PartiallyAdmit",
    "admission_strength": 0.45,
```

```
    "passed": [
      "Close",
      "NormalizedDisplacement",
      "RelativeVolume",
      "VWAPAcceptance"
    ],
    "pending": [
      "Breadth",
      "Retest",
      "WeeklyConfirmation"
    ]
  },
  "residual": [
    {
      "type": "Frame",
      "description": "Weekly resistance unresolved",
      "severity": "High",
      "status": "Open"
    },
    {
      "type": "Feature",
      "description": "Breadth below threshold",
      "severity": "High",
      "status": "Open"
    }
  ],
  "claim_ceiling": "PartiallyAdmittedEvent",
  "prohibited_claims": [
    "EpisodeTransition",
    "WorldTransition"
  ],
  "invalidation": "CloseInsideOldRangePlusFailedReclaim"
}
```


---

# S.30 Standard Relationship Vocabulary

The ontology defines the following canonical relations.

```
declared_under
derived_from
observed_by
typed_as
tests_boundary
candidate_for
gated_by
committed_as
contradicted_by
leaves_residual
recorded_in
transported_to
promoted_to
invalidated_by
revised_by
authorized_by
backreacts_on
```

Examples:

BreakoutCandidate
gated_by
DailyCloseGate. (S.49)

DailyBreakout

leaves_residual
WeakBreadth. (S.50)

MarketDistress
transported_to
AccountingImpairmentCandidate. (S.51)

IndexInclusion
backreacts_on
FutureLiquidityStructure. (S.52)

---

# S.31 Prohibited Relationship Shortcuts

The standard prohibits several direct edges.

# Projection → WorldTransition

A projection cannot directly create a World claim.

# Candidate → AcceptedStructure

A candidate must pass an Event gate and persistence condition.

# MarketRecognition → LegalCommitment

Transport is permitted; identity is not.

# Residual → Q

Residual may motivate a Q candidate.

It cannot automatically become Q.

# ComplexState → PhaseTime

Phase-time requires additional validation.

# Outcome → OriginalProtocol

Later outcome cannot rewrite the earlier protocol.

---

# S.32 Complex-State Extension

# S.32.1 Complex object

A conforming complex record is:

Z
= (R,Q,Metric,A,θ,Generator,Residual,Benchmark). (S.53)

Required fields:

```
R_definition
Q_definition
R_units
Q_units
metric
normalization
amplitude
phase
generator_model
real_pair_benchmark
scaling_test
residual
model_status
```

---

# S.32.2 Model statuses

```
ScalarOnly
UsefulRealPair
ComplexCandidate
ComplexEligible
PhaseBearing
PhaseTimeSupported
PhaseSensitiveEventModel
TimeBearingWorld
```

---

# S.32.3 Status transition requirements

UsefulRealPair → ComplexEligible requires:

- unit or metric validity;
- stable generator;
- incremental value.

ComplexEligible → PhaseBearing requires:

- phase-order stability;
- operational phase utility.

PhaseBearing → PhaseTimeSupported requires:

- out-of-sample clock comparison.

PhaseTimeSupported → TimeBearingWorld requires:

- phase-sensitive gate;
- persistent trace;
- backreaction.

---

# S.33 χ Extension

A χ record SHOULD contain:

```
chi_id
protocol_id
horizon
estimator
value_or_class
confidence
supporting_evidence
contradictory_evidence
validity_interval
```

Canonical classes:

```
Corrective
Critical
SelfConfirming
Unknown
```

χ MUST be indexed by protocol and horizon.

An unqualified universal χ is nonconforming.

---

# S.34 Ξ Extension

A Ξ record is:

Ξ_P
= (ρ_P,γ_P,ν_P). (S.54)

Required metadata:

```
xi_id
protocol_id
rho_components
gamma_components
nu_components
compilation_method
normalization
uncertainty
validity_interval
```

The system MUST preserve component vectors.

It MUST NOT report only one opaque Ξ score when that prevents audit.

---

# S.35 Episode Extension

An Episode record MUST include:

```
episode_id
protocol_id
old_grammar
candidate_new_grammar
start_gate
event_sequence
dominant_chi
primary_boundaries
alternative_branches
```

```
completion_gate
persistence_rule
current_status
```

Canonical episode statuses:

```
Candidate
Active
Degrading
Completed
Transitioned
Invalidated
Ambiguous
```

---

# S.36 World Extension

A World-transition record MUST include:

```
world_event_id
world_type
boundary
authority
recognition_rule
ledger_type
commitment_time
changed_rights_or_actions
backreaction_channel
residual
```

The recognition vector MAY be represented as:

$$\mathbb{G} = (g_market, g_risk, g_accounting, g_contractual, g_legal, g_policy). \quad (S.55)$$

Each coordinate MUST identify:

- authority;
- state;
- timestamp;
- residual.

---

# S.37 Minimal Interoperability Profiles

Two systems are interoperable at a given profile when they can exchange the required objects without semantic loss.

## Structure interoperability

Systems agree on:

- protocol;
- projection;

- units;
- typing.

## Event interoperability

Systems additionally agree on:

- candidate;
- gate;
- residual;
- invalidation.

## Episode interoperability

Systems additionally agree on:

- event grammar;
- promotion;
- branch residual.

## World interoperability

Systems additionally agree on:

- authority;
- ledger;
- changed admissibility.

## Complex interoperability

Systems additionally agree on:

- R;
- Q;
- metric;
- generator;
- model status.

---

# S.38 Semantic-Loss Test

Suppose system A exports claim C_A and system B imports it as C_B.

Define semantic loss:

$\mathscr{L}$_sem
= Dist(Type_A,Type_B)

- Dist(Gate_A,Gate_B)
- Dist(Residual_A,Residual_B)
- Dist(Authority_A,Authority_B). (S.56)

Interoperability requires:

$\mathscr{L}$_sem ≤ ε_sem. (S.57)

A system that converts:

PartiallyAdmittedEvent with high residual (S.58)

into:

ConfirmedBreakout (S.59)

has failed semantic interoperability even if the numerical data were transmitted correctly.

---

# S.39 Conformance Test Suite

A standard validator SHOULD include at least the following tests.

## Test S1 — Missing protocol

Expected:

Reject Event conformance.

## Test S2 — Future data leakage

Expected:

Error.

## Test S3 — Missing source lineage

Expected:

Warning or Error depending on profile.

## Test S4 — Structure labelled as Event

Expected:

Period-ceiling violation.

## Test S5 — Event with no residual field

Expected:

Error under E1 and higher.

## Test S6 — World claim with market-only authority

Expected:

Authority violation.

## Test S7 — Revision overwrites original claim

Expected:

Trace-preservation violation.

## Test S8 — Q equals unexplained error

Expected:

Reject ComplexEligible status.

## Test S9 — Phase-time claim without clock benchmark

Expected:

Reject PhaseTimeSupported status.

## Test S10 — Transport claim without transformation rule

Expected:

Transport nonconformance.

---

# S.40 Example Conformance Evaluation

Consider:

> The stock broke resistance because it closed above £100 on high volume. RSI and MACD confirmed the move, so a new bull market has begun.

The validator evaluates:

# Protocol

Partially declared.

Timeframe and resistance exist.

Breadth, retest, outcome horizon, and invalidation are missing.

## Source lineage

RSI and MACD share price lineage.

They are not independent confirmation.

## Closure

A closing breakout may support Event status.

It does not establish Episode or World status.

## Gate

Close and volume pass.

Higher-frame and persistence gates are absent.

## Residual

Not recorded.

## Result

```
Conformance profile reached: E1 partial
Highest supported claim: Admitted or partially admitted Event
Prohibited claim: New bull-market Episode
Errors:
- residual missing
- invalidation missing
Warnings:
- same-source confirmation
- higher-frame transport absent
```

---

# S.41 Canonical Reduction Behaviour

When a higher-level object fails validation, a conforming system MUST reduce it.

Examples:

```
WorldTransition
→ Episode or Event

PhaseTime
→ PhaseBearing

ComplexEligible
→ UsefulRealPair

AdmittedEvent
→ Candidate or Structure
```

The reduction MUST preserve:

- original attempted claim;
- failure reason;
- retained lower-level claim.

It MUST NOT erase the attempted higher-level claim.

---

# S.42 Standard Claim Language

The standard recommends verbs corresponding to claim class.

| Claim class | Preferred verbs |
|---|---|
| Definition | define, declare |
| Identity | derive, follows |
| Projection | measure, calculate, estimate |
| Structure | indicate, characterize |
| Warning | suggest, warn, remain consistent with |
| Event | admit, confirm under protocol |
| Episode | transition, persist, complete |
| World | recognize, commit, alter rights |
| Hypothesis | predict, test |
| Causal | identify, intervene |
| Analogy | resemble, correspond structurally |

Avoid:

- “proves” for correlational evidence;
- “is” where only protocol-relative interpretation exists;
- “confirmed” without identifying the gate;
- “objective” without transport scope.

---

# S.43 Minimal Human Certification

An analyst may be certified at several levels.

# PPMG-A — Structure Analyst

Can:

- declare protocol;
- interpret projections;
- identify period and function;

- state missing variables.

## PPMG-B — Event Analyst

Can additionally:

- declare gates;
- record residual;
- define invalidation;
- distinguish candidate from admission.

## PPMG-C — Episode Analyst

Can additionally:

- maintain event grammars;
- govern promotion;
- preserve branch residual;
- audit retrospective segmentation.

## PPMG-D — World Analyst

Can additionally:

- distinguish authorities and ledgers;
- reconcile fragmented recognition;
- evaluate backreaction.

## PPMG-X — Complex-State Researcher

Can additionally:

- validate R–Q constructions;
- benchmark real pairs;
- test phase and internal time;
- apply reduction.

Certification here is a proposed training architecture, not an existing professional qualification.

---

# S.44 Implementation Maturity Levels

Software implementations may declare:

## Level 0 — Parser

Stores structured claims.

## Level 1 — Validator

Checks required fields and claim ceilings.

## Level 2 — Event engine

Runs gates and residual audit.

## Level 3 — Transport engine

Tests cross-frame survival.

## Level 4 — Recursive ledger engine

Supports Episode promotion and revision.

## Level 5 — Multi-world engine

Handles authority, ledgers, and backreaction.

## Level 6 — Advanced-state engine

Supports χ, Ξ, complex eligibility, and phase time.

A system MUST NOT claim Level 6 merely because it can store complex numbers.

---

# S.45 Governance of the Standard

The ontology itself must be revisable.

A proposed change should contain:

```
change_request_id
affected_entities
problem_statement
proposed_change
backward_compatibility
migration_rule
test_cases
residual_risk
```

The change process should be:

Proposal
→ Public Review
→ Test Corpus
→ Compatibility Audit
→ Gate

→ Versioned Release. (S.60)

---

## S.46 Versioning Policy

Use semantic versioning:

Major.Minor.Patch. (S.61)

## Major

Breaking ontology or meaning change.

## Minor

Backward-compatible new entity, field, or status.

## Patch

Clarification or correction without semantic change.

Example:

PPMG 1.2.3. (S.62)

Implementations MUST declare which standard version they support.

---

## S.47 Deprecation Policy

A term may be deprecated when it is:

- ambiguous;
- redundant;
- empirically unhelpful;
- frequently misused.

Deprecation MUST specify:

- replacement term;
- migration rule;
- effective version;
- residual risk.

Deprecated objects SHOULD remain readable for historical trace.

---

## S.48 Standard Residual Registry

The standard itself should maintain unresolved questions.

Examples:

- Are six closure periods sufficient?
- Are Load, Motion, Constraint, and Commitment minimal?
- Do all eight Proto-Eight roles add value?
- Can authority be represented uniformly across law, accounting, and policy?
- Which transport tolerances are domain-specific?
- Can phase-time evidence be standardized across episode families?

These are not defects to conceal.

They are the standard's own residual register.

---

# S.49 The Standard as a Declared World

The ontology itself instantiates the framework.

## Boundary

Which analytical objects are included?

## Projection

Which attributes are extracted?

## Gate

What qualifies as conforming?

## Trace

Which records enter the standard ledger?

## Residual

Which ambiguities remain open?

## Transport

Can the definitions survive new markets and institutions?

## Revision

How does the standard change without rewriting its history?

Thus:

PPMG_Standard
= Declaration

- ConformanceGate
- VersionedLedger
- ResidualRegistry
- RevisionRule. (S.63)

---

# S.50 Canonical Compact Specification

The entire standard can be compressed into the following declaration:

```
Every PPMG claim MUST identify its protocol, evidence cutoff, projection lineage, closure period,

Every Event or stronger claim MUST be linked to an explicit gate.

Every commitment MUST preserve residual.

Every revision MUST preserve prior trace.

Every broad claim SHOULD survive declared transport.

Every complex or phase claim MUST defeat the relevant simpler benchmark.

No claim may exceed its demonstrated closure, authority, or model status.
```

---

# S.51 Canonical Mathematical Core

# Declared world

$\Sigma_P = \text{Declare}(\Sigma_0 \mid P)$. (S.64)

# Projection

$X_j = \hat{O}_{j,P}(\Sigma_P)$. (S.65)

# Typing

$\tau(X_j) = (p_j, g_j, a_j)$. (S.66)

# Candidate

$c_k = \text{DetectTransition}(X_k, B_P)$. (S.67)

# Gate

$G_P(c_k, L_k, \mathscr{R}_k)$

→ (d_k,α_k,e_k,r_k). (S.68)

# Ledger

L_{k+1} = Update(L_k,e_k,r_k). (S.69)

# Transport

r_T
= C_{P′} − T_{P→P′}(C_P). (S.70)

# Revision

D_{k+1} = U_a(D_k,L_{k+1},$\mathscr{R}$_{k+1}). (S.71)

# Claim ceiling

L_claim
≤ min(L_closure,L_gate,L_authority,L_transport,L_model). (S.72)

---

# S.52 Worked Standard Record

```
standard_version: "PPMG-1.0"

protocol:
  id: "PPMG:Protocol:NBE:DailyBreakout:1.0"
  subject: "NBE"
  universe: "RenewableInfrastructurePeers"
  timeframe: "Daily"
  scale: "Log"
  boundary_rule: "TwelveWeekRangeHigh"
  gate_rule: "CloseVolumeBreadthRetest"
  residual_rule: "TypedResidualV1"
  outcome_horizon: "20Sessions"
  authority: "MarketAnalysis"

claim:
  id: "PPMG:Claim:NBE:Breakout-20260724:1.0"
  information_cutoff: "2026-07-24T16:30:00Z"
  period: "Event"
  family: "Commitment"
  actuation_role: "Gate"
  claim_class: "CandidateEvent"

evidence:
  close_above_boundary: true
  displacement_atr: 1.16
  relative_volume: 1.74
  breadth: 0.575
  retest: "Pending"
  weekly_confirmation: "Pending"

gate:
  decision: "PartiallyAdmit"
  admission_strength: 0.45

residual:
```

```
  - type: "Feature"
    description: "Breadth below threshold"
    severity: "High"
    status: "Open"
  - type: "Frame"
    description: "Weekly confirmation pending"
    severity: "High"
    status: "Open"

claim_ceiling:
  supported: "PartiallyAdmittedEvent"
  prohibited:
    - "EpisodeTransition"
    - "WorldTransition"

invalidation:
  rule: "CloseInsideRangePlusFailedReclaim"
```

---

# S.53 What Conformance Does Not Guarantee

A conforming record may still be:

- empirically wrong;
- economically unprofitable;
- causally mistaken;
- poorly calibrated;
- locally valid only.

Conformance guarantees only that the claim is:

- declared;
- reconstructable;
- typed;
- gated appropriately;
- residual-honest;
- revision-accountable.

Therefore:

Conformance ≠ Truth. (S.73)

But:

Nonconformance often prevents truth from being evaluated reliably. (S.74)

---

# S.54 What Truth Does Not Excuse

A prediction that later proves correct does not excuse:

- future leakage;
- moved boundaries;
- erased residual;
- unsupported authority;
- retrospective phase selection.

Therefore:

CorrectOutcome
does not retroactively validate
InvalidProtocol. (S.75)

The standard evaluates process and outcome separately.

---

# S.55 Relationship to the Benchmark Corpus

The Open Benchmark Corpus in Appendix R tests whether systems can produce conforming objects.

The standard defines:

- what objects exist;
- which fields are required;
- which transitions are allowed.

The benchmark provides:

- cases;
- evidence;
- gold or disputed labels;
- adversarial tests.

Their relationship is:

Standard → DefinesValidity. (S.76)

Benchmark → MeasuresPerformance. (S.77)

A benchmark without a standard risks inconsistent labels.

A standard without a benchmark risks becoming purely ceremonial.

---

# S.56 Relationship to the Runtime Kernel

The runtime kernel in Appendix P executes the standard.

The ontology specifies:

What must be represented. (S.78)

The kernel specifies:

How the state changes. (S.79)

The benchmark specifies:

How implementation quality is tested. (S.80)

Together:

Ontology

- Runtime
- Benchmark
  = Operational Research Infrastructure. (S.81)

---

# S.57 Relationship to the Proto-Periodic Table

The periodic table supplies the central type system:

Period × Function. (S.82)

Proto-Eight supplies:

ActuationRole. (S.83)

The standard supplies:

Governance and exchange format. (S.84)

Thus a fully typed object is:

X_{Period,Function,Role}^{Protocol,Version}. (S.85)

Example:

BreakoutClose
= X_{Event,Commitment,Gate}^{P_daily,v1}. (S.86)

---

# S.58 Minimum Adoption Path

A research group need not adopt the entire standard immediately.

A practical sequence is:

## Phase 1

Add protocol identifiers and evidence cutoffs.

## Phase 2

Add period and function labels.

## Phase 3

Separate candidate, gate, and admitted Event.

## Phase 4

Add residual and invalidation.

## Phase 5

Add transport and revision records.

## Phase 6

Add Episode and World profiles.

## Phase 7

Add complex and phase modules only when needed.

The priority is:

Governance before sophistication. (S.87)

---

# S.59 Standard Success Criteria

The proposed standard succeeds if it produces measurable improvements in:

- inter-analyst agreement;
- signal reproducibility;
- residual recall;
- gate calibration;
- revision integrity;
- cross-system exchange;
- authority discipline;
- reduction of retrospective relabelling.

It fails if:

- required fields become ceremonial;
- analysts fill residual with generic language;
- systems game the claim ceiling;

- implementation cost exceeds diagnostic value;
- simpler schemas perform equally well.

---

# S.60 Appendix S Conclusion

The Proto-Periodic Market Grammar becomes a research infrastructure only when its concepts possess stable identities and exchange rules.

A conforming market claim must answer:

1. **What world was declared?**
2. **What evidence was available?**
3. **Which operator produced the observation?**
4. **At what closure period does the object exist?**
5. **Does it carry, move, constrain, or commit?**
6. **Which actuation role is active?**
7. **What candidate transition occurred?**
8. **Which gate admitted or rejected it?**
9. **What residual remains?**
10. **Which ledger records the consequence?**
11. **Does the claim survive another protocol?**
12. **How may the declaration be revised without erasing history?**

The standard's canonical law is:

No Protocol
→ No Auditable Object. (S.88)

No Gate
→ No Event. (S.89)

No Authority
→ No World Commitment. (S.90)

No Residual Record
→ No Mature Closure. (S.91)

No Preserved Trace
→ No Admissible Revision. (S.92)

No Simpler Benchmark
→ No Earned Complex Priority. (S.93)

The proposed PPMG standard therefore transforms Technical Analysis from a collection of indicator expressions into a versioned language of governed market claims:

> every object declared, every projection typed, every Event gated, every residual preserved, every authority bounded, every transport explicit, and every revision accountable to the world it inherits.

# Appendix T — PPMG Reference Implementation and Reproducible Research Stack

## T.1 Purpose

The Proto-Periodic Market Grammar now has four complementary forms:

1. a conceptual theory;
2. a formal state architecture;
3. an executable runtime kernel;
4. a canonical ontology and conformance standard.

The next step is to define how these objects may be implemented in a reproducible research system.

The implementation should not begin by building a large automated trading platform.

Its first task is narrower:

> Preserve the complete life cycle of a market claim—from protocol declaration and evidence snapshot to projection, gate, residual, outcome, transport, and admissible revision.

The minimum system must be capable of answering:

- What was declared?
- What information was available?
- Which operators were applied?
- What claim was made?
- Which gate admitted or rejected it?
- What residual remained?
- What happened later?
- Was the protocol revised?
- Can another observer reproduce the original state?

The reference implementation is abbreviated:

PPMG-RI. (T.1)

Its primary output is not an order.

Its primary output is a governed analytical record.

## T.2 Design Principles

The reference implementation follows twelve principles.

## T.2.1 Protocol first

No analysis begins without a declared protocol or an explicit descriptive-only status.

## T.2.2 Immutable original evidence

The evidence available at claim time must remain recoverable.

## T.2.3 Derived objects remain derived

Indicators, scores, and phases must preserve their source lineage.

## T.2.4 State transitions are explicit

A claim cannot silently move from Structure to Event or Event to Episode.

## T.2.5 Gates are separate from projections

An indicator may support a gate.

It is not automatically the gate.

## T.2.6 Residual is first-class data

Residual must not be buried in narrative comments.

## T.2.7 Event sourcing

Every important change is represented as a new event rather than an overwrite.

## T.2.8 Versioned protocols

A changed boundary, horizon, scale, gate, or feature map creates a new protocol version.

## T.2.9 Transport is reproducible

Cross-frame comparison must specify the transformation used.

## T.2.10 Complexity is optional

$\chi$, $\Xi$, complex phase, and internal time are activated only when justified by the task.

## T.2.11 Authority is bounded

Analytical systems do not acquire institutional authority through mathematical sophistication.

## T.2.12 Failed claims remain queryable

The research system must make failure easier to study, not easier to hide.

These principles instantiate the source architecture in which declarations, gates, traces, residuals, and admissible revision form an auditable world rather than a disposable stream of interpretations.

---

# T.3 High-Level Architecture

A reference deployment may contain eight services.

```
1. Evidence Store
2. Protocol Registry
3. Operator Registry
4. Projection Engine
5. Claim and Gate Engine
6. Residual and Ledger Engine
7. Transport and Revision Engine
8. Benchmark and Audit Harness
```

The execution path is:

Evidence
→ Protocol
→ Projection
→ Typed Claim
→ Gate
→ Trace + Residual
→ Ledger
→ Transport
→ Revision. (T.2)

A more technical view is:

Data Sources
→ Ingestion
→ Immutable Evidence Store
→ Protocol Compiler
→ Feature Projection
→ State Machine
→ Gate Evaluation
→ Event Ledger
→ Research API. (T.3)

---

# T.4 Repository Structure

A minimal open research repository may use:

```
ppmg/
├── README.md
├── LICENSE
├── pyproject.toml
```

```
├── schemas/
│   ├── protocol.schema.json
│   ├── evidence.schema.json
│   ├── operator.schema.json
│   ├── claim.schema.json
│   ├── gate.schema.json
│   ├── residual.schema.json
│   ├── transport.schema.json
│   ├── revision.schema.json
│   └── outcome.schema.json
├── protocols/
│   ├── breakout_daily_v1.yaml
│   ├── divergence_reversal_v1.yaml
│   └── episode_completion_v1.yaml
├── operators/
│   ├── moving_average.py
│   ├── rsi.py
│   ├── macd.py
│   ├── atr.py
│   ├── breadth.py
│   ├── vwap.py
│   └── volume_profile.py
├── gates/
│   ├── breakout_gate.py
│   ├── reversal_gate.py
│   ├── episode_gate.py
│   └── world_recognition_gate.py
├── residuals/
│   ├── ontology.yaml
│   ├── classifier.py
│   └── lifecycle.py
├── transport/
│   ├── timeframe.py
│   ├── scale.py
│   ├── volatility.py
│   ├── breadth.py
│   ├── market_to_accounting.py
│   └── market_to_legal.py
├── complex/
│   ├── eligibility.py
│   ├── generator_tests.py
│   ├── phase.py
│   └── clock_benchmarks.py
├── ledger/
│   ├── events.py
│   ├── repository.py
│   └── revision.py
├── benchmark/
│   ├── cases/
│   ├── annotations/
│   ├── adversarial/
│   └── scoring.py
├── api/
│   ├── app.py
│   ├── routes.py
│   └── models.py
├── notebooks/
│   ├── 01_protocol_demo.ipynb
│   ├── 02_breakout_gate.ipynb
│   ├── 03_residual_analysis.ipynb
│   ├── 04_transport_tests.ipynb
│   └── 05_complex_eligibility.ipynb
└── tests/
    ├── test_protocols.py
    ├── test_operators.py
    ├── test_gates.py
```

```
    ├── test_residuals.py
    ├── test_transport.py
    ├── test_revision.py
    └── test_adversarial.py
```

The repository structure should keep:

- mathematical operators;
- domain declarations;
- gate rules;
- empirical cases

separate.

This prevents a change in one layer from silently changing all others.

---

# T.5 Event-Sourced Storage

# T.5.1 Why event sourcing is appropriate

The framework requires:

- preserved original claims;
- versioned revisions;
- residual histories;
- gate changes;
- later invalidation;
- cross-ledger recognition.

A mutable row containing only the latest state would erase this history.

The preferred model is:

CurrentState
= Fold(EventHistory). (T.4)

Let the ordered event stream be:

$\mathscr{E}$
= {ε1,ε2,…,ε_n}. (T.5)

The present state is reconstructed through:

S_n
= Reduce(S0,$\mathscr{E}$). (T.6)

Possible events include:

```
ProtocolDeclared
EvidenceFrozen
```

```
ProjectionCalculated
CandidateDetected
GateDeferred
GatePartiallyAdmitted
GateAdmitted
ResidualOpened
ResidualResolved
ClaimInvalidated
TransportTested
ClaimPromoted
ProtocolRevised
OutcomeRecorded
```

No event is deleted merely because a later event changes its interpretation.

---

# T.5.2 Event envelope

Every event should contain:

```
event_id
event_type
aggregate_id
protocol_id
protocol_version
occurred_at
recorded_at
information_cutoff
actor
authority
payload
parent_event_id
```

The distinction between:

occurred_at (T.7)

and:

recorded_at (T.8)

is important.

An event may occur before it becomes available to the observer.

---

# T.5.3 Claim-state reconstruction

Suppose the history is:

```
CandidateDetected
GatePartiallyAdmitted
ResidualOpened
RetestPassed
ResidualResolved
GateAdmitted
WeeklyTransportPassed
ClaimPromoted
```

The current state can be reconstructed as:

EpisodeTransitionCandidate or PromotedEvent, (T.9)

depending on the declared promotion rule.

The original partial-admission state remains visible.

---

# T.6 Evidence Store

# T.6.1 Raw and adjusted layers

The store should distinguish:

RawEvidence. (T.10)

NormalizedEvidence. (T.11)

AdjustedEvidence. (T.12)

DerivedProjection. (T.13)

For price data:

```
raw_price
corporate_action_factor
adjusted_price
adjustment_version
```

For economic data:

```
first_release_value
revised_value
release_time
revision_time
```

For legal or policy evidence:

```
decision_time
publication_time
effective_time
appeal_status
```

These distinctions prevent several forms of hindsight leakage.

---

# T.6.2 Evidence hash

Each frozen evidence snapshot should have a content hash:

EvidenceHash_i
= Hash(E_i). (T.14)

A claim should store:

- EvidenceHash;
- ProtocolHash;

- CodeVersion.

Reproducibility then requires:

RecomputedOutput
= OriginalOutput (T.15)

within declared numerical tolerance.

---

# T.7 Protocol Registry

# T.7.1 Protocol object

A protocol object should be represented as structured data.

```
protocol_id: "PPMG:Protocol:NBE:DailyBreakout:1.0"

subject:
  asset: "NBE"
  universe: "RenewableInfrastructurePeers"
  venue: "PrimaryExchange"

observer:
  role: "MarketAnalyst"
  authority: "AnalyticalOnly"

observation:
  timeframe: "Daily"
  scale: "Logarithmic"
  bar_rule: "RegularSessionOHLCV"
  price_field: "AdjustedClose"

boundary:
  type: "ResistanceZone"
  construction: "TwelveWeekRangeHigh"
  lower: 99.80
  upper: 100.00

gate:
  candidate_rule: "CrossAboveBoundary"
  mandatory:
    close_confirmation: true
    minimum_atr_displacement: 0.75
  optional:
    relative_volume: 1.50
    breadth: 0.65
    retest: true
    higher_frame: true

residual:
  ontology_version: "PPMG-Residual-1.0"

invalidation:
  rule: "CloseInsideRangeAndFailedReclaim"

outcome:
  horizon: "20Sessions"
```

---

# T.7.2 Protocol compiler

The compiler should validate:

- required fields;
- units;
- timeframe consistency;
- boundary construction;
- gate completeness;
- authority;
- outcome horizon;
- invalidation.

The output is:

$$\text{CompiledProtocol} = \text{ValidateAndNormalize}(\text{ProtocolSource}). \quad (T.16)$$

A failed compilation should produce explicit errors rather than a partial silent runtime.

---

# T.7.3 Protocol comparison

For versions $P_1$ and $P_2$, calculate a structured difference:

$$\Delta P = \text{Diff}(P_1, P_2). \quad (T.17)$$

The result should identify changes in:

- timeframe;
- boundary;
- feature map;
- gate;
- residual rule;
- horizon;
- authority.

This supports the Declaration Stability Index developed earlier.

---

# T.8 Operator Registry and Lineage

# T.8.1 Operator interface

Every operator should implement:

```
class MarketOperator:
    operator_id: str
    version: str
    input_units: tuple[str, ...]
    output_units: str

    def validate(self, evidence, protocol) -> None:
        ...

    def calculate(self, evidence, protocol):
        ...

    def lineage(self) -> dict:
        ...
```

The exact language is implementation-specific.

The conceptual contract is:

Operator
= Definition

- Units
- Parameters
- SourceLineage
- FailureModes. (T.18)

---

# T.8.2 Operator word representation

An operator pipeline may be represented as:

```
operator_word:
  - Select:
      field: adjusted_close
  - Difference:
      lag: 1
  - Split:
      positive: gains
      negative: losses
  - Smooth:
      method: wilders
      period: 14
  - Normalize:
      range: [0, 100]
```

This gives a reconstructable description of RSI rather than storing only its name.

---

# T.8.3 Lineage graph

Each projection becomes a node in a graph:

G_lineage = (V,E). (T.19)

An edge:

$X_i \rightarrow X_j$ (T.20)

means $X_j$ was derived from $X_i$.

Lineage allows the system to calculate:

- shared ancestors;
- transformation similarity;
- redundant confirmation;
- contamination by future data.

---

# T.9 Typing Engine

# T.9.1 Type object

The typing engine assigns:

$\tau(X)$
$= (\text{Period},\text{Family},\text{Role})$. (T.21)

A type result should include:

```
primary_period
primary_family
primary_role
secondary_types
confidence
justification
prohibited_promotions
```

---

# T.9.2 Rule-assisted typing

Some operators can be typed deterministically.

Examples:

```
Trade execution
→ Mark × Commitment × Exchange/Gate

Moving average
→ Structure × Load × Memory

Resistance zone
→ Structure × Constraint × Boundary
```

Other objects require context.

For example:

VWAP may be:

- Window Load;
- Structure Load;

- Structure Constraint;
- Event Guidance.

The engine should preserve context rather than enforce one universal label.

---

# T.9.3 Typing confidence

Let:

c_type ∈ [0,1]. (T.22)

Low-confidence typing should create:

R_type = TypingResidual. (T.23)

It should not be hidden by forcing a confident classification.

---

# T.10 Claim Engine

# T.10.1 Claim object

A claim is represented as:

```
claim_id: "PPMG:Claim:NBE:Breakout-20260724:1.0"
protocol_id: "PPMG:Protocol:NBE:DailyBreakout:1.0"
information_cutoff: "2026-07-24T16:30:00Z"

content:
  subject: "NBE"
  predicate: "breakout"
  status: "candidate"

type:
  period: "Event"
  family: "Commitment"
  role: "Gate"

maximum_requested_level: "EpisodeTransition"

invalidation_id: "INV-NBE-BREAK-001"
```

The claim engine should separate:

ClaimRequested (T.24)

from:

ClaimSupported. (T.25)

The requested level may be Episode.

The evidence may support only Candidate Event.

---

# T.10.2 Claim ceiling

The engine computes:

$$L_{max} = \min(L_{evidence}, L_{gate}, L_{transport}, L_{authority}, L_{model}). \quad (T.26)$$

If:

$$L_{requested} > L_{max}, \quad (T.27)$$

the engine should return:

- supported claim;
- prohibited stronger claim;
- missing closure condition.

It should not merely say the evidence is “mixed.”

---

# T.11 Gate Rule Language

# T.11.1 Purpose

Gate rules should be readable by:

- researchers;
- analysts;
- software;
- auditors.

A simple domain-specific language may express:

```
gate:
  all:
    - close_above_boundary: true
    - displacement_atr:
        gte: 0.75

  any:
    - relative_volume:
        gte: 1.50
    - breadth:
        gte: 0.65

  defer_when:
    - weekly_gate: pending

  reject_when:
    - close_inside_boundary: true

  invalidate_when:
    all:
      - close_inside_old_range: true
      - failed_reclaim_within_sessions:
          lte: 3
```

---

## T.11.2 Hard and soft components

Gate components may be:

- mandatory;
- weighted;
- advisory;
- veto;
- pending.

A weighted score should not override a failed veto unless explicitly permitted.

The gate output should include:

GateResult
= (Decision,Strength,Passed,Failed,Pending,Residual). (T.28)

---

## T.11.3 Gate determinism

For fixed:

- protocol;
- evidence;
- code version,

the gate result should be reproducible.

Symbolically:

G(P,E,C) = constant. (T.29)

Human adjudication may remain necessary for some gates.

When used, it must be recorded as a separate authority-dependent decision.

---

# T.12 Residual Engine

## T.12.1 Residual generation

Residual may be generated from:

1. missing mandatory evidence;
2. failed optional evidence;
3. cross-frame contradiction;
4. model uncertainty;

5. authority mismatch;

6. unresolved branch;

7. data-quality concern.

A residual engine should not simply calculate:

Residual = 1 – GateScore. (T.30)

A low score and high residual are related but not identical.

Residual must remain typed.

---

# T.12.2 Residual object

```
residual_id: "RES-NBE-BREADTH-001"
parent_claim_id: "PPMG:Claim:NBE:Breakout-20260724:1.0"

type: "Feature"
subtype: "BreadthBelowThreshold"

description: "Peer breadth was 57.5%, below the declared 65% threshold."

severity: "High"
directional_relevance: "WeakensBroadAcceptance"
status: "Open"

resolution:
  condition: "BreadthAtOrAbove65Percent"
  maximum_wait: "10Sessions"
```

---

# T.12.3 Residual lifecycle

The engine should allow:

```
Open
→ Monitoring
→ Resolved
```

or:

```
Open
→ Invalidating
```

or:

```
Open
→ ConvertedToLoad
```

For example:

a prior failed breakout residual may later become:

Episode Load. (T.31)

The original residual object remains preserved.

---

# T.13 Ledger Engine

# T.13.1 Append-only rule

The ledger should use append-only writes.

A new fact creates:

NewLedgerEvent. (T.32)

It does not update the old event in place.

This permits reconstruction of:

- what was believed;
- when it was believed;
- why it changed.

---

# T.13.2 Multiple ledgers

The implementation should support:

```
MarketLedger
ModelLedger
RiskLedger
AccountingLedger
ContractualLedger
LegalLedger
PolicyLedger
```

A single case may create records in several ledgers at different times.

---

# T.13.3 Cross-ledger link

A link may state:

```
source:
  ledger: "MarketLedger"
  event_id: "MARKET-DISTRESS-001"

target:
  ledger: "AccountingLedger"
  state: "ImpairmentCandidate"

relationship: "transported_to"
```

The link does not imply that the accounting gate passed.

The destination ledger must record its own decision.

---

# T.14 Transport Engine

## T.14.1 Transport operator interface

A transport operator should implement:

$T_{P \to P'}(C_P) \to \hat{C}_{P'}$. (T.33)

It should declare:

- source protocol;
- target protocol;
- expected transformation;
- tolerance;
- non-comparable conditions.

---

## T.14.2 Example: timeframe transport

A daily breakout may transport to weekly as:

DailyAdmittedEvent
→ WeeklyCandidateEvent. (T.34)

It should not automatically transport as:

WeeklyAdmittedEvent. (T.35)

The expected target type differs.

---

## T.14.3 Example: scale transport

A line on arithmetic scale may map into a curve or zone on logarithmic scale.

The transport question is:

Does the underlying relation survive? (T.36)

not:

Do the pixel coordinates remain identical? (T.37)

---

## T.14.4 Example: market-to-accounting transport

A market-price decline may transport to:

ImpairmentCandidate. (T.38)

The accounting target then evaluates:

- recoverable amount;

- reporting standard;
- management judgment;
- audit evidence.

Market observation supplies evidence.

Accounting authority supplies commitment.

---

# T.15 Promotion Engine

# T.15.1 Promotion contract

A promotion request contains:

```
source_claim
source_period
target_period
required_closure
persistence_rule
authority
residual_limit
```

---

# T.15.2 Event-to-Episode example

Required:

- old grammar failure;
- admitted opposing Event;
- persistence;
- higher-frame survival;
- branch audit.

The promotion engine returns:

```
Approved
Deferred
Rejected
Demoted
```

A deferred promotion preserves the Event claim.

It does not erase it.

---

# T.15.3 Episode-to-World example

Required:

- authority;

- formal rule;
- persistent ledger;
- changed future rights, obligations, or admissibility.

Price persistence alone is insufficient.

---

# T.16 Revision Engine

# T.16.1 Revision proposal

A revision proposal should contain:

```
parent_protocol: "P-v1"
proposed_protocol: "P-v2"

trigger:
  type: "RepeatedTransportFailure"

changes:
  boundary_width:
    old: 0.20
    new: 0.35

unchanged:
  timeframe: "Daily"
  outcome_horizon: "20Sessions"

reason:
  "Original zone width was below observed microstructure uncertainty."

trace_preserved: true
prospective_retest_required: true
```

---

# T.16.2 Revision validator

The validator asks:

1. Was the triggering evidence recorded?
2. Is the old protocol preserved?
3. Is the change bounded?
4. Does the change respond to the diagnosed failure?
5. Is a new out-of-sample test required?
6. Does the revision increase complexity excessively?

Define:

RevisionCost
= ComplexityIncrease

- CompatibilityLoss

- RetestBurden. (T.39)

A revision is justified only when:

ExpectedEvidenceGain > RevisionCost. (T.40)

This is a research-governance rule, not a universal economic equation.

---

# T.17 Complex-State Module

# T.17.1 Isolation from the core runtime

Complex-state functionality should be modular.

The ordinary gate and ledger system must function without it.

The module activates only when:

ComplexRequested = true. (T.41)

---

# T.17.2 Eligibility pipeline

```
Define R and Q
→ validate units
→ validate metric
→ test independence
→ compare real pair
→ fit candidate generator
→ test scaling
→ test phase utility
→ assign model status
```

The output should be:

```
status: "UsefulRealPair"

R:
  definition: "PriceAcceptance"
  units: "StandardizedScore"

Q:
  definition: "PeerBreadthPressure"
  units: "StandardizedScore"

unit_compatibility: "Passed"
source_independence: "Partial"
generator_test: "Failed"
real_pair_benchmark: "Equal"
phase_utility: "NotEstablished"

reduction:
  retained_model: "[R,Q]"
  rejected_claim: "InternalPhaseTime"
```

The source phase framework requires this type of reduction when the complex representation does not add stable value beyond the two real variables.

---

# T.18 χ and Ξ Modules

# T.18.1 χ module

The χ module should return:

```
class
numeric_estimate
horizon
confidence
supporting evidence
contradictory evidence
```

It should permit:

χ = Unknown. (T.42)

No downstream rule should require a false confident estimate.

---

# T.18.2 Ξ module

The Ξ module should preserve component vectors:

Ξ_P
= (ρ_components,γ_components,ν_components). (T.43)

Example:

```
rho:
  volume_loading: 0.82
  open_interest_loading: 0.70
  attention_loading: 0.65

gamma:
  liquidity_lock: 0.55
  funding_lock: 0.20
  legal_lock: 0.00

nu:
  realized_volatility: 0.62
  spread_instability: 0.40
  failed_gate_rate: 0.30
```

The compiled scalar coordinates may be calculated.

The underlying components must remain available.

---

# T.19 Benchmark Harness

# T.19.1 Case runner

A benchmark case contains:

- frozen evidence;

- protocol;
- expected type;
- expected gate;
- expected residual;
- prohibited claims.

The harness runs:

ActualOutput
= Runtime(CaseInput). (T.44)

Then compares:

ActualOutput
versus
ExpectedOutput. (T.45)

---

## T.19.2 Scoring dimensions

The harness should report:

- period accuracy;
- function accuracy;
- role accuracy;
- gate accuracy;
- residual precision and recall;
- overpromotion rate;
- authority error;
- trace preservation;
- reduction accuracy;
- transport accuracy.

A single aggregate score should be optional.

---

## T.19.3 Adversarial suite

Mandatory adversarial cases include:

- future data leakage;
- post-event boundary;
- same-source confirmation;

- crossing without close;
- strong gate with high residual;
- market distress without legal default;
- arbitrary complex scaling;
- retrospective phase alignment;
- moved wave or Fibonacci anchor;
- erased failed claim.

The framework should fail the build when a mandatory invariant is violated.

---

# T.20 Continuous Integration

A reproducible repository should run automated tests on every change.

The pipeline may be:

```
Schema Validation
→ Unit Tests
→ Protocol Compilation
→ Deterministic Gate Tests
→ Residual Lifecycle Tests
→ Transport Tests
→ Adversarial Tests
→ Benchmark Regression
→ Documentation Build
```

---

# T.20.1 Golden-case tests

A golden case is a small fully specified example with a fixed expected result.

Example:

```
Intraday cross = true
Daily close above = false
Relative volume = high
```

Expected:

```
Window rejection
No admitted breakout
Residual: trapped breakout interest
```

A code change that outputs “confirmed breakout” should fail the test suite.

---

# T.20.2 Regression tolerance

Numerical outputs may vary slightly because of:

- floating-point arithmetic;

- revised libraries;
- optimization methods.

Declare tolerance:

|x_new – x_old| ≤ ε_num. (T.46)

Categorical governance outputs should normally require exact agreement.

---

# T.21 Research API

A minimal API may expose:

```
POST /protocols
GET  /protocols/{id}
POST /evidence/snapshots
POST /projections/run
POST /claims
POST /claims/{id}/gate
POST /claims/{id}/residuals
POST /claims/{id}/transport
POST /claims/{id}/promote
POST /claims/{id}/revise
GET  /claims/{id}/history
GET  /ledgers/{type}
POST /benchmark/run
```

---

# T.21.1 Gate endpoint

Request:

```
{
  "claim_id": "PPMG:Claim:NBE:Breakout-20260724:1.0",
  "protocol_id": "PPMG:Protocol:NBE:DailyBreakout:1.0",
  "evidence_snapshot_id": "EVID-NBE-20260724-CLOSE"
}
```

Response:

```
{
  "decision": "PartiallyAdmit",
  "admission_strength": 0.45,
  "passed": [
    "Close",
    "Displacement",
    "RelativeVolume",
    "VWAP"
  ],
  "failed": [
    "Breadth"
  ],
  "pending": [
    "Retest",
    "WeeklyConfirmation"
  ],
  "residual_ids": [
    "RES-NBE-BREADTH-001",
    "RES-NBE-WEEKLY-001"
  ],
```

```
  "claim_ceiling": "PartiallyAdmittedEvent"
}
```

---

# T.22 Query Examples

The ledger should support research queries that ordinary signal databases cannot answer.

# T.22.1 Failed strong gates

```
SELECT *
FROM claims
WHERE gate_decision = 'Admit'
  AND invalidation_triggered = TRUE;
```

---

# T.22.2 High-residual successful Events

```
SELECT *
FROM claims
WHERE residual_burden = 'High'
  AND outcome_class = 'Continuation';
```

This tests whether some residuals represent optionality rather than failure.

---

# T.22.3 Retrospective protocol drift

```
SELECT revision_id, claim_id, changed_fields
FROM revisions
WHERE created_at > outcome_reveal_time
  AND prospective_retest_required = FALSE;
```

---

# T.22.4 Same-lineage confirmation

```
SELECT claim_id, COUNT(DISTINCT source_root) AS independent_sources
FROM claim_projections
GROUP BY claim_id;
```

---

# T.22.5 Event-to-Episode promotion rate

```
SELECT candidate_type,
       COUNT(*) AS admitted_events,
       SUM(CASE WHEN promoted_period = 'Episode' THEN 1 ELSE 0 END) AS promoted
FROM claims
WHERE current_state IN ('Admitted', 'Accepted', 'Promoted')
GROUP BY candidate_type;
```

---

# T.23 Authority and Access Control

# T.23.1 Role-based permissions

The platform should distinguish:

- researcher;
- analyst;
- gate reviewer;
- protocol administrator;
- institutional authority;
- auditor.

Example permissions:

| Role | Declare protocol | Calculate projection | Admit market Event | Commit accounting/legal state | Revise protocol |
|---|---|---|---|---|---|
| Researcher | Yes | Yes | Research only | No | Propose |
| Analyst | Yes | Yes | Analytical | No | Propose |
| Gate reviewer | Limited | Review | Yes within domain | No | Limited |
| Institutional authority | Domain-specific | Review | Domain-specific | Yes within mandate | Controlled |
| Auditor | No | Reproduce | No | No | Review |

---

## T.23.2 Authority token

A consequential gate may require an authority token:

AuthorityToken
= Sign(Actor,Scope,Time,Decision). (T.47)

The exact cryptographic implementation is optional.

The semantic requirement is not.

The event must identify who or what had authority to commit it.

---

# T.24 Security and Integrity

The system should protect against:

- evidence alteration;
- timestamp manipulation;
- deleted failed claims;
- unauthorized protocol revision;
- unlogged authority escalation.

Possible controls include:

- append-only event store;
- cryptographic hashes;
- signed revisions;
- immutable object storage;
- audit logs;
- role-based access control.

The goal is not to claim perfect tamper-proof truth.

It is to make unauthorized modification detectable.

---

## T.25 Observability

The runtime should expose operational metrics.

## T.25.1 Protocol metrics

- incomplete protocols;
- compilation failures;
- revision frequency.

## T.25.2 Gate metrics

- candidates;
- partial admissions;
- admissions;
- invalidations;
- false-admission rate.

## T.25.3 Residual metrics

- open residual count;
- mean residual duration;
- conversion rate;
- invalidating residual rate.

## T.25.4 Transport metrics

- survival rate;
- local-only rate;
- non-comparable rate.

## T.25.5 Governance metrics

- overpromotion;
- authority errors;
- trace-preservation failures;
- retrospective protocol changes.

A system whose predictive score improves while:

- overpromotion rises;
- residual recall falls;
- trace integrity weakens

may be suffering Goodhart failure.

---

# T.26 Minimum Viable Implementation

The first useful version does not require:

- real-time order-book processing;
- institutional legal ledgers;
- complex phase;
- machine learning;
- automated trading.

A minimum viable implementation needs only:

1. protocol registry;
2. evidence snapshots;
3. a small operator set;
4. one Event gate;
5. typed residuals;
6. immutable claim history;
7. one transport test;
8. outcome recording.

A realistic first scope is:

**Daily equity breakouts.**

Required operators:

- ATR;
- relative volume;
- breadth;
- VWAP or close-location proxy;
- higher-timeframe boundary.

Required gate states:

- Candidate;
- Partial;
- Admit;
- Reject;
- Invalidated.

Required residuals:

- breadth;
- retest;
- higher frame;
- liquidity;
- value migration.

This would already test the article's central discipline.

---

# T.27 Staged Implementation Roadmap

# Stage 0 — Static specification

Deliver:

- JSON schemas;
- example YAML protocols;
- example claim records;
- validation rules.

No live data integration is needed.

## Stage 1 — Research notebook

Deliver:

- one historical dataset;
- one breakout protocol;
- reproducible projections;
- gate and residual outputs;
- immutable case log.

---

## Stage 2 — Event database

Deliver:

- relational or event-sourced store;
- claim history;
- outcome tracking;
- revision records.

---

## Stage 3 — Transport engine

Add:

- daily-to-weekly;
- raw-to-ATR;
- capitalization-weighted-to-equal-weight;
- price-to-breadth.

---

## Stage 4 — Episode module

Add:

- event grammar;
- completion certificate;
- branch residual;
- promotion rules.

---

## Stage 5 — World module

Add:

- market;
- risk;
- accounting;
- contractual;
- legal recognition.

---

## Stage 6 — Advanced state module

Add:

- $\chi$;
- $\Xi$;
- complex eligibility;
- phase-time benchmark.

Each stage should be publishable independently.

---

# T.28 Acceptance Criteria

## T.28.1 Stage 1 acceptance

The same protocol and evidence reproduce the same projections and gate decision.

## T.28.2 Stage 2 acceptance

A failed claim remains accessible after revision.

## T.28.3 Stage 3 acceptance

Transport results identify expected target form and residual.

## T.28.4 Stage 4 acceptance

Event and Episode labels cannot be silently collapsed.

## T.28.5 Stage 5 acceptance

Market recognition and institutional recognition remain distinct.

## T.28.6 Stage 6 acceptance

Complex and phase modules correctly reduce unsupported models.

The overall release rule is:

Release
⇒ Reproducible
∧ TracePreserving
∧ ResidualHonest
∧ ClaimCeilingEnforced. (T.48)

---

# T.29 Reference Demonstration

Consider a daily breakout case.

# Step 1 — Protocol declared

Boundary:

£99.80–£100.00.

Gate:

close above boundary, displacement ≥ 0.75 ATR, participation evidence, residual audit.

# Step 2 — Evidence frozen

At the daily close:

C = £100.88. (T.49)

ATR = £0.76. (T.50)

RVOL = 1.74. (T.51)

Breadth = 57.5%. (T.52)

Retest = Pending. (T.53)

Weekly gate = Pending. (T.54)

# Step 3 — Projections calculated

Displacement:

$\hat{d}$ = (100.88 − 100.00)/0.76 = 1.16 ATR. (T.55)

# Step 4 — Candidate detected

Price closed beyond the declared boundary.

## Step 5 — Gate evaluated

Passed:

- close;
- displacement;
- relative volume.

Failed or pending:

- breadth;
- retest;
- weekly confirmation.

## Step 6 — Runtime state

PartiallyAdmitted. (T.56)

## Step 7 — Residual opened

```
WeakBreadth
RetestPending
WeeklyFramePending
```

## Step 8 — Claim ceiling

Partially admitted daily Event.

## Step 9 — Prohibited claims

- broad sector breakout;
- Episode transition;
- World transition.

## Step 10 — Later revision

If breadth expands and retest passes:

new ledger events are appended.

The original partial-admission state remains unchanged.

---

# T.30 Reproducible Research Package

A publication using PPMG should release, where permitted:

```
protocol.yaml
evidence_manifest.json
operator_versions.json
claim_records.jsonl
gate_records.jsonl
residual_records.jsonl
transport_records.jsonl
revision_records.jsonl
outcomes.csv
analysis_notebook.ipynb
environment_lockfile
```

The research package should permit another observer to reconstruct:

ClaimAtTime_t. (T.57)

not merely:

FinalNarrativeAfterOutcome. (T.58)

---

# T.31 Implementation Failure Modes

## T.31.1 Schema theatre

Fields exist but contain generic text such as:

“market uncertainty.”

Repair:

require typed, evidence-linked residuals.

---

## T.31.2 Mutable-history convenience

The latest state overwrites earlier states.

Repair:

event sourcing or versioned append-only tables.

---

## T.31.3 Indicator-first design

The application begins with hundreds of indicators but no protocol or gate model.

Repair:

build one complete Event workflow before expanding feature count.

---

## T.31.4 Scalar-dashboard compression

Admission, residual, transport, and authority are compressed into one score.

Repair:

retain the state vector.

---

## T.31.5 Complex-module contamination

The core system assumes every case has R, Q, θ, and $\tau_i$.

Repair:

make complex state optional and eligibility-gated.

---

## T.31.6 Automated authority escalation

The system interprets a model probability as permission to commit a legal, accounting, or transaction state.

Repair:

separate analytical output from authoritative action.

---

## T.31.7 Benchmark overfitting

The runtime learns benchmark templates rather than the grammar.

Repair:

use hidden protocols, counterfactual twins, and novel method names.

---

# T.32 Minimal Technology Choices

The standard is technology-neutral.

A modest implementation may use:

- Python for operators and research;
- PostgreSQL for structured records;
- object storage for evidence snapshots;
- JSON Schema or Pydantic for validation;
- a lightweight API framework;
- Git for protocol and code versioning;
- notebooks for reproducible analysis.

A larger implementation may use:

- event streaming;

- graph databases;
- signed ledger events;
- distributed feature computation.

The theory does not require blockchain, complex databases, or real-time microservices.

The architecture should remain proportional to the research task.

---

## T.33 The Implementation as a World-Formation System

The reference implementation is itself a declared world.

## Boundary

Which evidence, protocols, and claim types are admitted?

## Projection

Which operators may calculate observables?

## Gate

What constitutes a valid Event or revision?

## Trace

Which records enter the ledger?

## Residual

Which omissions and contradictions remain open?

## Transport

Can the records be interpreted by another system?

## Revision

How can rules change without erasing earlier results?

Therefore:

PPMG-RI
= ProtocolRegistry

- OperatorRuntime

- GateEngine
- ResidualLedger
- TransportLayer
- RevisionGovernance. (T.59)

The software should therefore be evaluated by the same principles it applies to market analysis.

---

# T.34 Final Implementation Contract

A compliant reference implementation should satisfy:

```
1. Every claim has a protocol.
2. Every projection has source lineage.
3. Every Event has a gate.
4. Every gate has a residual output.
5. Every claim has a maximum permitted closure level.
6. Every material change is versioned.
7. Every failed claim remains queryable.
8. Every broad claim has a transport record.
9. Every World claim identifies authority and ledger.
10. Every complex claim has a real-pair benchmark.
11. Every phase-time claim has clock comparisons.
12. Every revision preserves the earlier trace.
```

In compact form:

ImplementationValidity
= Reproducibility

- Typing
- GateDiscipline
- ResidualHonesty
- TraceIntegrity
- Transport
- AuthorityControl
- Reduction. (T.60)

---

# T.35 Appendix T Conclusion

The Proto-Periodic Market Grammar does not require a giant software platform to become testable.

It requires a small number of disciplined objects:

- a protocol;
- a frozen evidence snapshot;

- a registered operator;
- a typed claim;
- a gate;
- a residual record;
- an immutable ledger event;
- a transport test;
- a versioned revision.

The implementation sequence is:

Declare
→ Freeze Evidence
→ Project
→ Type
→ Claim
→ Gate
→ Record Trace + Residual
→ Test Transport
→ Observe Outcome
→ Revise Prospectively. (T.61)

The central implementation rule is:

> Never store only the latest interpretation when the scientific object is the history of how interpretation changed.

The central engineering rule is:

> Build one complete, reproducible gate-and-ledger workflow before building another indicator.

The central governance rule is:

> A system may calculate beyond its authority, but it must not commit beyond its authority.

The central research rule is:

> Every advanced layer—$\chi$, $\Xi$, complex phase, internal time, and World backreaction—must remain removable without destroying the simpler operational core.

The resulting reference implementation is not a predictive oracle.

It is an infrastructure for accountable market observation:

> a system in which every claim can be reconstructed, every promotion can be challenged, every residual can be followed, every failure can be studied, and every revision remains answerable to the evidence and history from which it emerged.

# Appendix U — Preregistered Pilot Study: Residual-Bearing Breakout Gates

# U.1 Purpose

The framework has now been expressed as:

- a conceptual grammar;
- a formal ontology;
- a runtime kernel;
- an adversarial test suite;
- an annotation standard;
- a reference implementation.

The next step is empirical reduction.

The first study should not attempt to validate:

- the whole six-period ontology;
- all Proto-Eight roles;
- $\chi$, $\Xi$, complex phase, and internal time;
- institutional World formation;
- every Technical Analysis method.

A scientifically useful first study should test one narrow proposition:

> Does a residual-bearing, staged breakout gate classify durable boundary transitions more reliably than crossing-only or indicator-stacking baselines?

This is a suitable pilot because:

1. breakouts have a clear candidate boundary;
2. crossing and commitment can be separated;
3. price, volume, breadth, retest, and higher-frame evidence can be recorded;
4. fakeouts create visible residual;
5. the study can be performed without assuming complex phase;
6. the result can falsify or reduce several central claims.

The source Technical Analysis framework already requires a declared protocol, gate result, residual, cross-frame status, and invalidation rather than a bare bullish or bearish label.

The pilot converts that discipline into a preregistered empirical design.

---

# U.2 Scientific Status

This appendix is a **prospective study specification**.

It does not report empirical results.

All numerical thresholds below are proposed engineering choices for the pilot. They are not established natural constants or validated universal market thresholds.

The study may conclude that:

- explicit gates help;
- residual burden helps;
- transport helps;
- only some components help;
- or the whole additional architecture adds no value over a simple closing-breakout rule.

All five outcomes are admissible.

The study's primary scientific duty is:

> Freeze the claim before viewing the test outcomes.

---

# U.3 Core Research Question

Let a candidate breakout be a price crossing of a predeclared upper boundary.

The primary question is:

> Given a candidate crossing, does staged evidence about close, displacement, participation, acceptance, residual, and cross-frame transport improve the prediction and diagnosis of durable structural acceptance?

The study distinguishes:

Candidate crossing
→ initial gate
→ post-event acceptance gate
→ higher-frame transport
→ durable outcome. (U.1)

The candidate is not assumed to be a successful breakout.

---

# U.4 Primary Hypotheses

## U.4.1 Hypothesis $H_1$ — Gate superiority

A staged breakout gate will outperform a crossing-only baseline in predicting durable acceptance.

Formally:

$$\text{Brier}(M_{\text{Gate}}) < \text{Brier}(M_{\text{Cross}}). \quad (U.2)$$

The comparison will also use:

- log loss;
- calibration slope;
- calibration intercept;
- area under the precision–recall curve.

---

## U.4.2 Hypothesis $H_2$ — Residual value

Conditional on admission strength, greater open residual burden will predict a higher probability of failure.

Pr(Failure | G,RB_high)

Pr(Failure | G,RB_low). (U.3)

Equivalently:

$\partial \Pr(\text{Failure})/\partial \text{RB} > 0$. (U.4)

This tests whether residual contains information beyond the positive gate score.

---

## U.4.3 Hypothesis $H_3$ — Transport value

Breakout candidates that survive relevant transformations will show more durable acceptance than protocol-local candidates.

Pr(Acceptance | TransportSurvival)

Pr(Acceptance | LocalOnly). (U.5)

---

## U.4.4 Hypothesis $H_4$ — Functional independence

A bundle covering different functional channels will outperform an equal-sized stack of price-derived indicators.

The independent bundle contains:

- price displacement;
- volume participation;
- breadth participation;
- retest or follow-through.

The price-stack baseline contains an equal number of features chosen from:

- moving-average relation;
- MACD;
- RSI;

- rate of change.

The hypothesis is:

Utility(FunctionalBundle)

> Utility(PriceIndicatorStack). (U.6)

The comparison directly tests the article's claim that indicator count is not equivalent to confirmation independence.

---

# U.4.5 Hypothesis H5 — Claim-ceiling discipline

A staged gate will reduce premature Episode claims relative to an ordinary chart-commentary baseline.

Define overpromotion:

OPR
= UnsupportedEpisodeClaims/AllCandidateEvents. (U.7)

The hypothesis is:

OPR_PPMG < OPR_Conventional. (U.8)

This component may be tested through a separate blinded human–LLM annotation experiment.

---

# U.5 Secondary Hypotheses

# U.5.1 H6 — Gate and residual are not complements

Admission strength and residual burden will not be perfect inverses.

Corr(GS,RB) > −1. (U.9)

The important empirical class is:

HighGateStrength + HighResidualBurden. (U.10)

This represents an admitted but fragile Event.

---

# U.5.2 H7 — Retest information is conditional

A successful retest will strengthen acceptance probability when a retest occurs, but the absence of a retest will not automatically imply failure.

Pr(Acceptance | RetestPass,RetestOccurred)

> Pr(Acceptance | RetestFail,RetestOccurred). (U.11)

NoRetest is coded separately from RetestFail.

---

## U.5.3 H8 — Higher-frame evidence changes scope

Daily admission may remain useful even when weekly transport fails, but weekly failure will reduce the probability of Episode promotion.

DailyEventValidity ≠ WeeklyEpisodeValidity. (U.12)

This tests the distinction between local Event and broader Episode.

---

# U.6 Claims Not Tested by the Pilot

The pilot does not test whether:

- Technical Analysis generates profitable trading strategies after costs;
- markets possess a universal eightfold ontology;
- Q is an ontologically real imaginary market dimension;
- complex phase creates an internal market clock;
- Proto-Eight is a necessary grammar of all self-organizing systems;
- World-level institutional events can be inferred from chart evidence;
- every breakout should use the same threshold.

The maximum claim supported by a successful pilot is:

> Explicit gate, residual, and transport variables improve the classification of daily equity breakout acceptance under the tested protocols.

It would not establish a universal law of markets.

---

# U.7 Unit of Analysis

The unit of analysis is one **independent candidate breakout episode**.

Define candidate i:

C_i
= (Asset_i,Boundary_i,CrossTime_i,Protocol_i). (U.13)

A candidate begins when the asset first crosses the predeclared boundary after satisfying the reset condition.

Repeated crossings within the same unresolved episode are not treated as independent observations.

---

## U.7.1 Episode reset

A new candidate may be created only after one of the following:

1. price closes below the prior boundary for at least ten sessions;
2. a new boundary is formed from a later rolling window;
3. the previous case reaches its declared outcome horizon;
4. the previous Event is formally invalidated and a reset period passes.

This prevents one choppy boundary from generating many pseudo-independent cases.

---

# U.8 Study Universe

The primary universe should consist of liquid primary-listed common equities.

An eligible asset must satisfy, before the candidate date:

- at least 250 trading sessions of data;
- median twenty-day monetary volume above a preregistered minimum;
- price above a preregistered minimum;
- no unresolved trading suspension;
- reliable corporate-action adjustments;
- valid sector or peer classification.

The exact liquidity and price thresholds must be chosen before extracting the final test sample.

The universe membership rule must be reproducible from information available at the time.

---

# U.8.1 Exclusions

Exclude:

- preferred shares;
- warrants;
- rights;
- exchange-traded funds from the primary equity study;
- securities with unresolved adjustment errors;
- candidates occurring immediately around stock splits unless the adjusted boundary is independently verified;
- securities lacking sufficient peer-universe data.

Corporate events are not automatically excluded.

They should be recorded as residual or stratification variables where data permit.

---

# U.9 Temporal Design

Use a chronological walk-forward design.

A recommended structure is:

Training window
→ validation window
→ frozen test window. (U.14)

For example:

- five years training;
- one year validation;
- one year test;
- then roll forward.

The exact dates depend on data availability.

Random train–test splitting is prohibited because it can mix:

- the same market regime;
- related boundary episodes;
- neighbouring observations;
- later knowledge

across partitions.

---

# U.10 Boundary Construction

# U.10.1 Primary boundary

For asset i on session t, define the prior sixty-session upper boundary:

$$B_{i,t} = \max\{H_{i,t-60},\ldots,H_{i,t-1}\}. \quad (U.15)$$

The current session is excluded.

This prevents current-price information from changing the boundary being tested.

---

# U.10.2 Boundary zone

A boundary is treated as a zone:

$$Z_{B,i,t} = [B_{i,t} - 0.25ATR_{i,t}, B_{i,t}]. \quad (U.16)$$

The coefficient 0.25 is a preregistered engineering choice.

Robustness tests will use:

0.10ATR;
0.25ATR;
0.50ATR. (U.17)

The primary analysis uses only the 0.25 ATR rule.

---

# U.10.3 Boundary maturity

The boundary must have existed for at least five sessions before the candidate crossing.

This prevents a newly formed high from being immediately labelled a mature resistance boundary.

Optional boundary-history variables include:

- number of prior tests;
- time since formation;
- transaction density near the zone;
- earlier failed crossings.

These variables are secondary and must not alter candidate eligibility after the outcome is observed.

---

# U.11 Candidate Crossing

A candidate crossing occurs when:

H_{i,t} > B_{i,t}. (U.18)

The crossing alone creates:

CandidateEvent_i. (U.19)

It does not create:

AdmittedBreakout_i. (U.20)

The initial candidate timestamp is the first eligible session satisfying Equation U.18 after reset.

---

# U.12 Initial Gate at Session Close

The initial gate uses only information available by the candidate session close.

# U.12.1 Close component

Define:

g_close
= 1[C_{i,t} > B_{i,t}]. (U.21)

---

## U.12.2 Displacement component

Define normalized close displacement:

d_{i,t}
= (C_{i,t} − B_{i,t})/ATR_{i,t}. (U.22)

The displacement component passes when:

d_{i,t} ≥ 0.50. (U.23)

The 0.50 ATR threshold is the primary preregistered value.

---

## U.12.3 Volume component

Define relative volume:

RVOL_{i,t}
= V_{i,t}/Median(V_{i,t−20:t−1}). (U.24)

The volume component passes when:

RVOL_{i,t} ≥ 1.50. (U.25)

---

## U.12.4 Breadth component

Let Peer(i,t) be the predeclared sector or industry peer universe.

Define same-day participation breadth:

Br_{i,t}
= Number of eligible peers with positive return/Number of eligible peers. (U.26)

The breadth component passes when:

Br_{i,t} ≥ 0.60. (U.27)

A second structural breadth measure may use the fraction above a twenty-day moving average, but that is secondary.

---

## U.12.5 Initial evidence vector

The initial gate vector is:

g0
= (g_close,g_displacement,g_volume,g_breadth). (U.28)

No follow-through, retest, weekly close, or later profile information enters g0.

# U.13 Initial Gate States

## U.13.1 Reject

Reject when:

g_close = 0. (U.29)

An intraday crossing that closes below the boundary is not an admitted daily breakout.

---

## U.13.2 Defer

Defer when:

g_close = 1 (U.30)

but:

d_{i,t} < 0.50 (U.31)

and both participation components fail.

---

## U.13.3 Partially admit

Partially admit when:

g_close = 1 (U.32)

and:

d_{i,t} ≥ 0.50 (U.33)

but one or more participation components fail.

---

## U.13.4 Initially admit

Initially admit when:

g_close = 1, (U.34)

d_{i,t} ≥ 0.50, (U.35)

and at least one of:

g_volume = 1 (U.36)

or:

g_breadth = 1. (U.37)

The study will also test stricter variants requiring both participation components.

## U.14 Why the Initial Gate Is Staged

The initial gate should not use future retest or follow-through evidence.

Otherwise the study would silently redefine a session-t signal using later information.

The runtime therefore records:

$G_0$ = initial close-time gate. (U.38)

$G_3$ = three-session follow-through update. (U.39)

$G_5$ = five-session acceptance update. (U.40)

G_W = weekly transport update. (U.41)

The claim evolves through new ledger entries.

The original $G_0$ state remains preserved.

This follows the declaration source's architecture: declaration precedes projection, gate, trace, residual, and ledger; later evidence may revise future declarations, but it does not alter what was available at the earlier gate.

---

## U.15 Three-Session Follow-Through Gate

Define follow-through pass:

g_FT
= 1[max(C_{t+1:t+3}) ≥ C_t + 0.50ATR_t
and min(C_{t+1:t+3}) ≥ B_t − 0.50ATR_t]. (U.42)

The first condition requires additional upward acceptance.

The second prevents severe immediate rejection.

A candidate that remains slightly above the boundary but does not advance may receive:

FollowThrough = Neutral. (U.43)

It should not be forced into Pass or Fail when neither condition is decisive.

---

## U.16 Five-Session Retest Gate

## U.16.1 Retest occurrence

A retest occurs when any session from t+1 through t+5 enters:

[B_t − 0.25ATR_t,B_t + 0.25ATR_t]. (U.44)

---

## U.16.2 Retest pass

A retest passes when:

1. price enters the retest zone;
2. no close falls below $B_t - 0.50ATR_t$;
3. a later close within the five-session window returns above $B_t$.

---

## U.16.3 Retest fail

A retest fails when:

1. price enters or falls below the zone;
2. price closes below $B_t - 0.50ATR_t$;
3. no reclaim above $B_t$ occurs within the next three sessions.

---

## U.16.4 No retest

When price never returns to the zone:

RetestStatus = NoRetest. (U.45)

NoRetest is not coded as failure.

It remains a separate state.

---

## U.17 Weekly Transport Gate

The daily claim is transported to the weekly protocol.

The expected transformation is:

DailyAdmittedEvent
→ WeeklyCandidate or WeeklyAdmittedEvent. (U.46)

The weekly gate passes when the relevant weekly close satisfies:

$C_{week} > B_t$. (U.47)

The transport states are:

- WeeklySurvival;
- WeeklyPartial;
- WeeklyFailure;
- WeeklyPending.

A daily Event may remain valid when weekly transport fails.

Its scope is then:

DailyLocalOnly. (U.48)

It is not promoted to a broader Episode claim.

---

# U.18 Gate-Strength Representation

The primary analysis will not hide the evidence vector inside one scalar.

It will retain:

$$G_i = (g_close, g_displacement, g_volume, g_breadth, g_FT, g_retest, g_weekly). \quad (U.49)$$

A secondary scalar may be constructed:

$$GS_i = \Sigma_jw_jg_\{i,j\}. \quad (U.50)$$

The weights must be chosen using training and validation data only.

A simple equal-weight score is the primary scalar benchmark:

$$GS_i = \text{Number of passed applicable components/Number of applicable components.} \quad (U.51)$$

Pending and non-applicable components are not treated as failures.

---

# U.19 Residual Ontology

The pilot uses the following residual types.

## U.19.1 $R_1$ — Close residual

The candidate crossed intraday but did not close above the boundary.

## U.19.2 $R_2$ — Displacement residual

The close exceeded the boundary by less than 0.50 ATR.

## U.19.3 $R_3$ — Volume residual

RVOL was below 1.50.

## U.19.4 $R_4$ — Breadth residual

Peer breadth was below 60%.

## U.19.5 $R_5$ — Retest residual

Retest failed, remained pending, or did not occur.

## U.19.6 R6 — Follow-through residual

No meaningful three-session follow-through occurred.

## U.19.7 R7 — Higher-frame residual

The weekly close failed or remained pending.

## U.19.8 R8 — Prior-fakeout residual

A materially similar boundary had failed within the preceding 120 sessions.

## U.19.9 R9 — Liquidity residual

The event occurred under low eligible liquidity or unusually wide spread conditions, where available.

## U.19.10 R10 — Event-risk residual

A major scheduled corporate event occurred within a preregistered nearby window, where reliable timestamps are available.

---

# U.20 Residual Burden

The primary residual burden is an unweighted normalized count:

RB_i
= OpenApplicableResiduals_i/ApplicableResiduals_i. (U.52)

This avoids fitting severity weights in the initial pilot.

A secondary weighted measure is:

RB_i^w
= Σ_js_j1[r_{i,j}=Open]/Σ_js_j. (U.53)

The severity weights s_j must be learned or frozen using training and validation data.

They cannot be chosen from the final test outcomes.

---

# U.21 Residual State Transitions

Each residual may move through:

Open
→ Monitoring
→ Resolved. (U.54)

Open
→ Invalidating. (U.55)

Open
→ ConvertedToLoad. (U.56)

For example:

- weak breadth may resolve as peers join;
- absent follow-through may become invalidating;
- an earlier fakeout may remain as Episode-level Load even after the later breakout succeeds.

The source self-revision framework requires residual to pressure revision without being erased, and it restricts mature revision to well-formed, trace-preserving, residual-honest, frame-robust, budget-bounded, non-degenerate changes.

---

# U.22 Primary Outcome: Durable Acceptance

Define the twenty-session acceptance outcome:

$Y_{A,i} = 1$ (U.57)

when all three conditions hold:

1. at least fourteen of the next twenty closes remain above $B_i$;
2. the twentieth-session close is at least 0.50 $ATR_i$ above $B_i$;
3. no three consecutive closes occur below $B_i - 0.25ATR_i$.

Otherwise:

$Y_{A,i} = 0$. (U.58)

The rule is an operational definition of durable acceptance.

It is not asserted as the only legitimate definition.

---

# U.23 Secondary Outcomes

# U.23.1 Fakeout outcome

Define:

$Y_{F,i} = 1$ (U.59)

when the asset:

1. closes below $B_i - 0.50ATR_i$ within ten sessions;
2. fails to reclaim $B_i$ during the next three sessions.

---

# U.23.2 Maximum favourable excursion

$MFE_{20,i}$
= [max(H_{t+1:t+20}) − C_t]/ATR_t. (U.60)

---

## U.23.3 Maximum adverse excursion

$MAE_{20,i}$
= [min(L_{t+1:t+20}) − C_t]/ATR_t. (U.61)

---

## U.23.4 Boundary occupancy

Define:

$BO_{20,i}$
= Number of closes above B_i/20. (U.62)

This is a continuous measure of acceptance.

---

## U.23.5 Time to failure

T_fail,i
= first session on which the invalidation rule passes. (U.63)

This supports survival analysis.

---

## U.23.6 Episode-promotion outcome

A candidate is marked as a provisional Episode transition only when:

- the daily Event is admitted;
- weekly transport survives;
- durable acceptance passes;
- a new higher-low or equivalent structural persistence condition forms.

This outcome is exploratory in the pilot.

---

# U.24 Baseline Models

## U.24.1 M0 — Crossing-only

Features:

H_t > B_t. (U.64)

Every candidate receives the same initial probability except for base-rate adjustment.

This is the weakest baseline.

## U.24.2 M1 — Close-only

Features:

- close above boundary;
- normalized displacement.

This represents a simple disciplined breakout rule.

---

## U.24.3 M2 — Conventional price-indicator stack

Features:

- price above EMA20;
- EMA20 above EMA50;
- MACD sign;
- RSI;
- twenty-day rate of change.

This tests indicator stacking without explicit functional independence.

---

## U.24.4 M3 — Functional bundle

Features:

- close displacement;
- relative volume;
- peer breadth;
- follow-through;
- retest.

---

## U.24.5 M4 — Residual-bearing gate

Features:

- M3 gate vector;
- residual vector;
- residual burden;
- prior-fakeout history.

---

## U.24.6 $M_5$ — Transported residual-bearing gate

Features:

- $M_4$;
- weekly transport;
- alternative scale transport;
- volatility-normalized survival.

$M_5$ is the full pilot implementation of the article's Event-level architecture.

---

# U.25 Model Forms

The primary statistical model is regularized logistic regression:

$$\text{logit } \Pr(Y_{A,i}=1) = \beta_0 + \beta^T X_i. \quad \text{(U.65)}$$

Reasons for using it first include:

- interpretability;
- calibration;
- transparent coefficients;
- easier detection of redundancy;
- reduced risk that flexibility hides weak theory.

Secondary models may include:

- gradient-boosted trees;
- random forests;
- generalized additive models.

They are secondary benchmarks, not replacements for the preregistered primary analysis.

---

# U.26 Cluster Structure

Breakout candidates are not independent across:

- the same asset;
- the same date;
- the same sector;
- broad market shocks.

The analysis should account for clustering by:

- asset;
- candidate date;
- sector-date where feasible.

For coefficient inference, use clustered or multiway-robust uncertainty estimates.

For predictive evaluation, use blocked chronological test periods rather than relying only on asymptotic standard errors.

---

# U.27 Calibration as the Primary Standard

The pilot's primary question is not:

> Which model produces the highest hypothetical return?

It is:

> Does the stated admission confidence correspond to observed durable acceptance?

Use:

Brier
$= N^{-1}\Sigma_i(p_i - Y_i)^2$. (U.66)

LogLoss
$= -N^{-1}\Sigma_i[Y_i \ln p_i + (1-Y_i)\ln(1-p_i)]$. (U.67)

Calibration intercept should approach:

0. (U.68)

Calibration slope should approach:

1. (U.69)

A model with high ranking accuracy but poor calibration does not provide a trustworthy gate-strength interpretation.

---

# U.28 Discrimination Metrics

Report:

- ROC-AUC;
- precision–recall AUC;
- sensitivity;
- specificity;
- positive predictive value;

- negative predictive value.

Because durable acceptance may be imbalanced, precision–recall results should receive substantial attention.

No threshold should be selected from the final test sample.

---

# U.29 Claim-Promotion Metrics

The framework introduces process-specific metrics.

## U.29.1 Overpromotion rate

OPR
= ClaimsAboveObservedClosure/AllClaims. (U.70)

---

## U.29.2 Underpromotion rate

UPR
= ClaimsBelowObservedClosure/AllClaims. (U.71)

---

## U.29.3 Residual recall

ResidualRecall
= MaterialResidualsDetectedBeforeOutcome/AllMaterialResiduals. (U.72)

---

## U.29.4 Residual precision

ResidualPrecision
= MaterialDetectedResiduals/AllDetectedResiduals. (U.73)

---

## U.29.5 Trace integrity

TraceIntegrity
= OriginalClaimsPreserved/AllRevisedClaims. (U.74)

Trace integrity should equal one in the reference implementation.

---

# U.30 Primary Statistical Tests

## U.30.1 Test of $H_1$

Compare $M_3$, $M_4$, and $M_5$ against $M_0$ and $M_1$ on the frozen test periods.

Primary contrast:

$$\Delta\text{Brier} = \text{Brier}(M_1) - \text{Brier}(M_4). \quad (U.75)$$

A positive ΔBrier supports the residual-bearing gate.

Use block bootstrap confidence intervals over candidate dates.

---

# U.30.2 Test of H2

Fit:

$$\text{logit } \Pr(Y_F=1) = \alpha + \beta_1 GS + \beta_2 RB + \beta_3 GS \times RB + \text{Controls}. \quad (U.76)$$

H2 is supported when:

$$\beta_2 > 0 \quad (U.77)$$

out of sample and with stable sign across major protocol variants.

The interaction β3 tests whether residual becomes especially important at high admission strength.

---

# U.30.3 Test of H3

Fit:

$$\text{logit } \Pr(Y_A=1) = \alpha + \beta_1 GS + \beta_2 RB + \beta_3 TS + \text{Controls}. \quad (U.78)$$

where:

TS = transport-survival score. (U.79)

H3 is supported when:

$$\beta_3 > 0. \quad (U.80)$$

---

# U.30.4 Test of H4

Compare M2 and M3 with:

- equal or similar feature counts;
- identical training and test windows;
- identical regularization selection procedures.

H4 is supported when M3 improves calibration or discrimination without disproportionate complexity.

---

# U.31 Transport Tests

## U.31.1 Daily to weekly

Test whether the daily claim becomes:

- WeeklySurvival;
- WeeklyPartial;
- WeeklyFailure.

---

## U.31.2 Raw to volatility-normalized

Test whether the crossing remains material under ATR normalization.

A nominal crossing that becomes negligible after normalization is classified:

ScaleWeak. (U.81)

---

## U.31.3 Arithmetic to logarithmic scale

For sixty-session boundaries, the difference may be small.

The test remains useful for high-volatility or long-history variants.

---

## U.31.4 Price to breadth

The transported expectation is:

A broad market or sector breakout should show some peer participation. (U.82)

Failure does not automatically invalidate the asset-level Event.

It localizes the claim.

---

## U.31.5 Price to volume

The transported expectation is:

A participation-supported breakout should show elevated exchange relative to the declared baseline. (U.83)

High price displacement with low volume may remain valid, but it carries a participation residual.

---

## U.32 Transport-Survival Score

Define applicable transport components:

T_i

= (T_weekly,T_ATR,T_log,T_breadth,T_volume). (U.84)

Each component receives:

1 = Survives;
0.5 = Partial;
0 = Fails. (U.85)

The primary score is:

TS_i
= Σ_jT_{i,j}/NumberApplicable_i. (U.86)

The full vector must remain available.

The scalar is used only for model comparison.

---

# U.33 χ as an Exploratory Extension

The primary pilot does not require χ.

An exploratory analysis may classify the preceding regime using:

- breakout persistence rate;
- mean-reversion speed;
- market breadth trend;
- volatility state;
- response to prior boundary tests.

The exploratory χ classes are:

Corrective;
Critical;
SelfConfirming;
Unknown. (U.87)

The analysis may test whether:

- RSI has different conditional meaning;
- gate thresholds require regime adjustment;
- residual burden has different effects across χ.

No χ-dependent threshold may replace the preregistered primary gate in the main study.

---

# U.34 Ξ as an Exploratory Extension

Ξ may be compiled from:

ρ = volume, open interest, and attention loading.

γ = liquidity, ownership, and exit lock-in.
ν = volatility, spread, and cancellation agitation. (U.88)

The pilot may test whether Ξ improves fakeout diagnosis beyond:

- gate strength;
- residual burden;
- transport.

If not, Ξ remains an interpretive interface rather than a validated predictive state.

---

# U.35 Complex-State Exclusion from the Primary Pilot

The primary pilot must not construct:

Z = R + iQ (U.89)

merely because it contains:

- price acceptance;
- breadth;
- volume;
- residual.

These variables first enter as a real feature vector.

A later study may propose:

R = price-acceptance coordinate;
Q = independently defined participation-pressure coordinate. (U.90)

That later proposal must pass:

- unit compatibility;
- scaling stability;
- generator testing;
- real-pair comparison;
- phase utility.

The source complex Technical Analysis article describes S + iQ as a disciplined way to separate admitted structure from retained pressure, but it explicitly does not make charts prophetic or validate every indicator automatically.

The pilot therefore uses the conceptual distinction without presuming privileged complex dynamics.

---

# U.36 Ablation Programme

Ablation identifies which components create value.

## U.36.1 Remove volume

Compare $M_4$ with and without RVOL.

## U.36.2 Remove breadth

Test whether peer participation adds information beyond asset price and volume.

## U.36.3 Remove retest

Determine whether the five-session acceptance update materially improves calibration.

## U.36.4 Remove weekly transport

Test whether higher-frame survival predicts twenty-session acceptance.

## U.36.5 Remove residual vector

Retain gate score only.

This is the critical test of whether residual adds information beyond positive evidence.

## U.36.6 Replace functional bundle with price stack

Test confirmation independence directly.

A component should not remain in the framework merely because its theoretical interpretation is attractive.

---

# U.37 Robustness Grid

The study will repeat the main comparison under a limited preregistered grid.

## Boundary windows

- 40 sessions;
- 60 sessions;
- 120 sessions.

## Boundary-zone widths

- 0.10 ATR;
- 0.25 ATR;
- 0.50 ATR.

## Displacement thresholds

- 0.25 ATR;
- 0.50 ATR;
- 0.75 ATR.

## Volume thresholds

- 1.25;
- 1.50;
- 2.00 relative volume.

## Outcome horizons

- 10 sessions;
- 20 sessions;
- 40 sessions.

The primary specification remains:

60-session boundary;
0.25 ATR zone;
0.50 ATR displacement;
1.50 RVOL;
20-session outcome. (U.91)

The robustness grid is not used to select whichever result looks best.

---

# U.38 Multiple-Testing Discipline

The study distinguishes:

## Confirmatory analyses

- $H_1$–$H_4$;
- primary protocol;
- primary outcomes;
- preregistered model comparisons.

## Secondary analyses

- alternative thresholds;
- $\chi$;

- Ξ;
- subgroups;
- interaction exploration.

## Exploratory analyses

- candidate Q channels;
- phase portraits;
- new instruments;
- unexpected residual types.

Confirmatory conclusions must not be based on exploratory threshold selection.

---

# U.39 Subgroup Analyses

Preregistered subgroup analyses may include:

- high versus low liquidity;
- high versus low volatility;
- sectors;
- broad-market regime;
- earnings-adjacent versus ordinary candidates;
- first versus repeated boundary test;
- concentrated versus broad market participation.

Subgroups must contain sufficient cases.

Small groups should be reported descriptively rather than used for strong inference.

---

# U.40 Missing Data

The study must distinguish:

- unavailable;
- delayed;
- unreliable;
- not applicable.

Missing breadth is not coded as failed breadth.

It is:

BreadthStatus = Unavailable. (U.92)

Models requiring breadth may:

1. omit the case;
2. use a missingness indicator;
3. run a reduced protocol.

The choice must be preregistered.

No imputed value should silently pass or fail a gate.

---

## U.41 Corporate Actions and Data Repair

For each candidate:

1. verify price adjustment history;
2. identify splits and major distributions;
3. reconstruct the prior boundary under the same adjustment convention;
4. flag unresolved inconsistencies.

A candidate whose breakout status depends on an unresolved corporate-action adjustment is labelled:

DataIndeterminate. (U.93)

It is excluded from the primary gate evaluation but retained in the audit ledger.

---

## U.42 Event Clustering

Broad market days may generate many simultaneous candidates.

The study should report:

- candidate count by date;
- sector concentration;
- broad-market return;
- volatility index or equivalent market-stress state where available.

Predictive uncertainty should account for event clustering.

Otherwise one market-wide rally may be mistaken for hundreds of independent confirmations.

---

## U.43 Human–LLM Annotation Substudy

A companion study may present frozen cases to:

1. human analysts;
2. LLMs without the PPMG protocol;
3. LLMs using the PPMG kernel;
4. humans using the structured PPMG form.

Each participant receives the same evidence.

They must output:

- period;
- gate status;
- residual;
- highest supported claim;
- prohibited stronger claim;
- invalidation.

---

## U.43.1 Substudy outcomes

Measure:

- typing accuracy;
- gate agreement;
- residual recall;
- overpromotion;
- underpromotion;
- time required;
- confidence calibration.

The hypothesis is not that structured analysis always predicts better.

It is that it produces more auditable and appropriately bounded claims.

---

# U.44 Success Criteria

The pilot supports continued development when all of the following hold.

## U.44.1 Gate criterion

$M_3$ or $M_4$ materially improves out-of-sample calibration over $M_0$ and $M_1$.

## U.44.2 Residual criterion

Residual burden contributes stable incremental information beyond gate strength.

## U.44.3 Transport criterion

At least one transport component improves scope or persistence classification.

## U.44.4 Reproducibility criterion

Independent implementations reproduce candidate and gate labels at a high rate.

## U.44.5 Complexity criterion

The improvement remains after complexity penalties and simpler-model comparison.

The criteria should be evaluated jointly.

A tiny discrimination gain with severe complexity and poor reproducibility is not sufficient.

---

# U.45 Strong Success

A strong result would show:

1. staged gates are better calibrated than crossing-only rules;
2. residual burden predicts fakeout conditional on positive gate evidence;
3. transport survival predicts durable acceptance;
4. functionally independent confirmation outperforms same-source indicator stacking;
5. results replicate across chronological test periods and major protocol variants.

This would support the Event-level core of the Periodic Grammar.

It would still not validate the whole theory.

---

# U.46 Partial Success

Possible partial outcomes include:

## U.46.1 Gate succeeds, residual does not

The study would retain staged gates but reduce the residual claim.

Residual may remain useful for explanation without predictive value.

## U.46.2 Residual succeeds, transport does not

The framework would retain residual-bearing admission while localizing transport claims.

## U.46.3 Volume helps, breadth does not

Breadth may be protocol-dependent or poorly specified.

## U.46.4 Retest helps only in some regimes

Retest may require χ-conditioned analysis.

## U.46.5 Simple close rule nearly matches full model

The full architecture should be reduced for this task.

Partial success is preferable to preserving unsupported completeness.

---

# U.47 Failure Criteria

The pilot fails to support the proposed Event architecture when:

1. crossing-only or close-only models perform equally well;
2. residual burden adds no stable information;
3. transport variables do not improve scope or outcome classification;
4. gate labels are not reproducible;
5. results reverse under nearby reasonable protocols;
6. the model succeeds only through flexible nonlinear fitting;
7. the framework mainly reduces commitment by classifying everything as uncertain.

The correct response is:

GateFail
→ ProtocolRepairOrReduction, not NarrativePatch. (U.94)

This follows the declaration source's explicit failure discipline.

---

# U.48 Model-Reduction Outcomes

The study must identify the least complex surviving model.

## U.48.1 Reduction to close-only

If displacement, volume, breadth, residual, and transport add no value:

PPMG Breakout Model
→ ClosingBoundaryRule. (U.95)

## U.48.2 Reduction to functional gate

If residual adds no predictive value but functional confirmation helps:

ResidualBearingGate
→ FunctionalGate. (U.96)

## U.48.3 Reduction to local Event model

If weekly transport adds no value:

TransportedEvent
→ LocalEvent. (U.97)

## U.48.4 Retention of full Event model

Only if gate, residual, and transport each demonstrate incremental value should $M_5$ be retained as the preferred pilot model.

---

# U.49 Interpretation of Null Results

A null result may mean:

- the theoretical distinction is wrong;
- the chosen proxy is weak;
- thresholds are poor;
- the outcome definition is inappropriate;
- the asset universe is unsuitable;
- residual is diagnostically useful but not predictive;
- market adaptation has reduced the pattern.

These explanations must not be used as automatic excuses.

A null result should first be recorded as:

PrimaryHypothesisNotSupported. (U.98)

Alternative explanations may then motivate new, separately preregistered studies.

---

# U.50 Prohibited Research Practices

The pilot prohibits:

1. moving boundaries after seeing outcomes;
2. choosing the best outcome horizon after testing;

3. deleting failed candidates;
4. counting repeated crossings as independent cases;
5. replacing missing breadth with a favourable price proxy;
6. selecting residual weights on test data;
7. calling a profitable case a valid gate regardless of process;
8. calling an unprofitable case invalid regardless of protocol;
9. adding complex phase after ordinary models disappoint;
10. reporting only the best-performing threshold combination.

---

# U.51 Required Reporting Tables

# Table U.1 — Sample construction

| Stage | Included | Excluded | Reason |
|---|---|---|---|
| Raw candidate crossings | | | |
| Liquidity eligible | | | |
| Data complete | | | |
| Independent episodes | | | |
| Primary analysis sample | | | |

---

# Table U.2 — Gate-state frequencies

| Initial state | Count | Percentage | Acceptance rate | Fakeout rate |
|---|---|---|---|---|
| Reject | | | | |
| Defer | | | | |
| Partially Admit | | | | |
| Initially Admit | | | | |

---

# Table U.3 — Residual prevalence

| Residual | Open at t | Resolved by t+5 | Invalidating | Converted to Load |
|---|---|---|---|---|
| Weak displacement | | | | |
| Low volume | | | | |
| Weak breadth | | | | |
| Failed follow-through | | | | |
| Failed retest | | | | |
| Weekly conflict | | | | |

---

# Table U.4 — Model comparison

| Model | Brier ↓ | Log loss ↓ | PR-AUC ↑ | Calibration slope | OPR ↓ |
|---|---|---|---|---|---|
| Crossing only | | | | | |
| Close only | | | | | |
| Price stack | | | | | |
| Functional bundle | | | | | |
| Residual-bearing gate | | | | | |
| Transported gate | | | | | |

---

# Table U.5 — Ablations

| Removed component | ΔBrier | ΔPR-AUC | ΔResidual recall | Interpretation |
|---|---|---|---|---|
| Volume | | | | |
| Breadth | | | | |
| Retest | | | | |
| Weekly transport | | | | |
| Residual vector | | | | |

---

# U.52 Required Figures

# Figure U.1 — Staged closure diagram

Candidate crossing
→ close gate
→ three-session follow-through
→ five-session acceptance
→ weekly transport
→ twenty-session outcome.

---

# Figure U.2 — Admission versus residual plane

Horizontal axis:

Gate strength.

Vertical axis:

Residual burden.

Four quadrants:

1. weak and clean;
2. weak and unresolved;
3. strong and clean;
4. strong but fragile.

---

## Figure U.3 — Calibration curves

Display all baseline and PPMG models on the same probability scale.

---

## Figure U.4 — Residual survival curves

Show time to resolution or invalidation for each residual class.

---

## Figure U.5 — Transport matrix

Rows:

source daily gate states.

Columns:

weekly, breadth, volume, and volatility-normalized survival.

---

# U.53 Reproducibility Package

The study release should contain:

```
protocol_registration.md
universe_definition.csv
candidate_extraction_code
boundary_code
gate_rules.yaml
residual_ontology.yaml
transport_rules.yaml
model_specification.md
frozen_test_manifest.json
event_records.jsonl
residual_records.jsonl
transport_records.jsonl
analysis_code
environment_lockfile
negative_results.md
```

The release must preserve both:

- original claim-time records;
- later outcome and revision records.

---

# U.54 Preregistration Header

```
Study title:
Residual-Bearing Breakout Gates under the Proto-Periodic Market Grammar

Primary question:
Do staged gate, residual, and transport variables improve durable-breakout classification?

Primary unit:
```

```
Independent daily equity breakout candidate.

Primary boundary:
Prior sixty-session high with a 0.25 ATR zone.

Primary outcome:
Twenty-session durable acceptance.

Primary baselines:
Crossing only; close only; conventional price-indicator stack.

Primary model:
Residual-bearing staged gate.

Primary metrics:
Brier score, log loss, precision–recall AUC, calibration, overpromotion rate.

Primary falsifier:
No out-of-sample incremental value over the close-only baseline.

Maximum permitted conclusion:
Event-level classification improvement under tested protocols.
```

---

# U.55 One Complete Case Record

```
Case ID:
Asset:
Candidate date:
Protocol version:

Boundary:
ATR:
Candidate crossing:
Close displacement:
Relative volume:
Peer breadth:

Initial gate:
Reject / Defer / Partial / Admit

Open residual:
R_close:
R_displacement:
R_volume:
R_breadth:
R_followthrough:
R_retest:
R_weekly:
R_prior_fakeout:
R_liquidity:
R_eventrisk:

Three-session update:
Five-session update:
Weekly transport:

Residual burden:
Transport score:

Twenty-session acceptance:
Fakeout:
MFE:
MAE:
Boundary occupancy:
```

```
Time to invalidation:

Original claim preserved:
Protocol revised:
Revision admissible:
```

---

# U.56 Expected Contribution of a Positive Result

A positive result would show that Technical Analysis can be improved without claiming that charts reveal hidden certainty.

The improvement would arise from separating:

- crossing from close;
- close from acceptance;
- positive evidence from unresolved contradiction;
- local Event from Episode transition;
- price confirmation from independent participation;
- current projection from higher-frame transport.

This would support the interpretation:

Technical Analysis is not primarily a collection of prophetic shapes.

It is a partially developed instrument system for observing whether market possibilities become committed structure.

---

# U.57 Expected Contribution of a Negative Result

A negative result would also be theoretically valuable.

It could show that:

- the four-family decomposition is too elaborate for breakouts;
- residual is better treated narratively than quantitatively;
- higher-frame transport does not predict the chosen outcome;
- simple close rules already capture most useful information;
- the selected breadth and retest proxies are poor.

The framework would then reduce.

A theory that cannot become smaller after empirical failure is not mature.

---

# U.58 Extension Study Sequence

Only after the breakout pilot should the programme advance.

## Study 2 — Divergence as warning versus Event

Test whether divergence becomes useful only after an opposing gate.

## Study 3 — Episode completion

Test the Episode Completion Certificate against moving-average, drawdown, and retrospective peak baselines.

## Study 4 — Residual-history state

Test whether prior failed Events improve later fakeout or squeeze diagnosis.

## Study 5 — Observer crowding

Test nonlinear adoption and reflexive inversion.

## Study 6 — Complex eligibility

Evaluate independently defined R–Q candidates against flexible real-pair models.

## Study 7 — Phase time

Compare $\tau_i$ with calendar time, event count, cumulative volume, volatility, and selection depth.

## Study 8 — World transitions

Study market, risk, accounting, contractual, legal, and policy recognition as distinct gates.

The order is intentional:

Gate
→ Residual
→ Transport
→ Episode
→ Observer
→ Complex Phase
→ Internal Time
→ World. (U.99)

---

# U.59 Pilot Research Contract

The study commits to the following rules.

1. The boundary is declared before candidate outcome.
2. Candidate crossing is separated from admission.

3. Later gate updates do not overwrite Go.
4. Missing evidence is not coded as failure.
5. Residual remains typed and queryable.
6. Test periods remain chronologically isolated.
7. Same-episode crossings are not counted independently.
8. Baselines receive the same data and validation procedure.
9. Negative results are reported.
10. Unsupported complexity is removed.
11. No investment-performance claim is inferred from Event calibration alone.
12. No complex or phase claim is introduced into the confirmatory pilot.

---

# U.60 Appendix U Conclusion

The first empirical test of the Periodic Grammar should be intentionally modest.

It should not ask whether the whole theory is true.

It should ask whether one central distinction survives disciplined measurement:

Crossing
≠ Commitment. (U.100)

The pilot then adds two further distinctions:

Commitment
≠ ResidualExhaustion. (U.101)

LocalEvent
≠ EpisodeTransition. (U.102)

The empirical chain is:

Predeclared Boundary
→ Candidate Crossing
→ Initial Close Gate
→ Follow-Through and Retest
→ Residual Update
→ Weekly Transport
→ Durable Acceptance or Failure. (U.103)

The primary test is:

Does this chain classify boundary transitions better than a simple crossing or closing rule?

The acceptable answers are:

- yes;

- partly;
- no.

The unacceptable answer is:

> The theory must be right because every failed case can be reinterpreted after the fact.

The pilot's final law is therefore methodological:

> Freeze the world before testing the event, preserve what the gate did not settle, and let the simplest surviving model determine how much of the larger architecture remains justified.

# Appendix V — The Multi-Study Validation Ladder

## V.1 Purpose

Appendix U defined one narrow pilot:

> Test whether a staged, residual-bearing breakout gate classifies durable boundary transitions better than crossing-only and close-only baselines.

That pilot is intentionally insufficient to validate the entire Periodic Grammar.

A complete research programme must advance through distinct evidential levels:

1. projection reliability;
2. functional classification;
3. Event-gate validation;
4. residual governance;
5. transport and invariance;
6. Episode formation;
7. observer backreaction;
8. complex eligibility;
9. internal phase time;
10. World-level ledger causality.

The research programme must not jump directly from a successful breakout study to claims about:

- complex market geometry;
- internal time;
- recursive world formation;
- Proto-Eight universality;

- institutional causality.

Each stronger claim requires a new study and a new gate.

The empirical development law is:

EvidenceLevel_n
→ ValidationGate_n
→ ClaimLevel_n
→ Residual_n
→ NextStudy_{n+1}. (V.1)

The programme should advance only when the preceding level passes.

When a level fails, the architecture should:

- reduce;
- localize;
- revise;
- or stop.

---

# V.2 Why One Large Study Would Be a Mistake

A single large model could combine:

- price;
- volume;
- breadth;
- residual history;
- χ;
- Ξ;
- complex phase;
- institutional variables;
- observer adoption;
- World-level gates.

Such a model might produce impressive predictive results.

But it would not reveal:

- which layer added value;
- which concepts were redundant;
- whether complex phase mattered;

- whether residual merely duplicated volatility;
- whether transport improved objectivity;
- whether institutional variables supplied all apparent explanatory power.

The result would be difficult to falsify because every failure could be attributed to another layer.

The correct architecture is sequential:

SimpleCore
→ IncrementalLayer
→ Ablation
→ Transport
→ Promotion. (V.2)

The burden of proof increases with conceptual strength:

Burden_{n+1} > Burden_n. (V.3)

---

# V.3 The Validation Ladder

The proposed ladder contains ten levels.

| Level | Research object | Strongest permissible claim |
|---|---|---|
| **L0** | Reproducible projection | The indicator computes a stable object |
| **L1** | Period and function typing | The object has a reliable analytical role |
| **L2** | Event gate | Candidate transitions can be governed |
| **L3** | Residual register | Unresolved evidence adds diagnostic value |
| **L4** | Transport | Some claims survive admissible reframing |
| **L5** | Episode grammar | Events form persistent higher-order sequences |
| **L6** | Reflexive observer | Observation and adoption alter later dynamics |
| **L7** | Complex state | A conjugate representation beats a real pair |
| **L8** | Internal phase time | Phase improves episode alignment and gating |
| **L9** | Time-bearing World | Ledgered gates change future admissibility |

The ladder is not an inevitable progression.

Many methods may stop at L1 or L2.

That is not failure.

A moving average can remain useful as a Structure-level Memory operator without becoming:

- a complex state;
- an internal clock;
- a World-forming mechanism.

---

# V.4 Level 0 — Reproducible Projection

## V.4.1 Question

Can the method produce a stable, reproducible projection from declared evidence?

Examples include:

- moving average;
- RSI;
- ATR;
- volume profile;
- support zone;
- candidate breadth measure.

---

## V.4.2 Minimum requirements

The method must declare:

- source lineage;
- operator word;
- parameters;
- units;
- information cutoff;
- missing-data rule.

Define:

$$X_{j,t} = \hat{O}_j(E_{\le t}). \quad (V.4)$$

A second implementation should reproduce:

$$|X_{j,t}^{(1)} - X_{j,t}^{(2)}| \le \varepsilon_j. \quad (V.5)$$

---

## V.4.3 Primary tests

1. Formula reproduction;
2. parameter stability;
3. timestamp consistency;
4. data-adjustment consistency;
5. missing-data behaviour.

---

## V.4.4 Falsifier

The method fails L0 when:

- independent implementations disagree materially;
- output depends on undocumented data revision;
- parameters are not recoverable;
- future information enters the projection.

---

## V.4.5 Maximum claim

At L0, the method may claim only:

> This operator calculates a reproducible observable under protocol P.

It may not yet claim predictive or Event-level significance.

---

# V.5 Level 1 — Period and Function Typing

## V.5.1 Question

Can independent observers agree on what analytical role the projection performs?

The required labels are:

- closure period;
- functional family;
- Proto-Eight role where used.

---

## V.5.2 Study design

Provide blinded method descriptions without their familiar names.

Example:

> An exponentially weighted average of prior closing prices.

Annotators classify it as:

Structure × Load × Memory. (V.6)

This tests whether the ontology reflects operational structure rather than memorized terminology.

---

## V.5.3 Reliability metrics

Use:

- exact agreement;
- weighted period agreement;
- family agreement;
- role agreement;
- confusion matrix.

For period labels:

Reliability_P
= 1 – MeanPeriodDistance/5. (V.7)

---

# V.5.4 Competing ontology

The six-period, four-family system should be compared with simpler alternatives.

For example:

### Three-level model

Signal
→ Pattern
→ Regime. (V.8)

### Conventional method categories

Trend
→ Momentum
→ Volatility
→ Volume. (V.9)

The Periodic Grammar earns retention only if it improves:

- classification;
- failure diagnosis;
- claim discipline.

---

# V.5.5 Falsifier

Level 1 fails when:

- annotators cannot distinguish adjacent periods reliably;
- functional families overlap without practical resolution;
- the new taxonomy adds no value over conventional categories.

---

# V.6 Level 2 — Event-Gate Validation

This is the level tested by the breakout pilot.

## V.6.1 Question

Does an explicit gate improve the classification of candidate transitions?

The core distinction is:

CandidateTransition ≠ AdmittedEvent. (V.10)

---

## V.6.2 Study families

After breakouts, replicate the gate architecture across:

1. support failure;
2. divergence reversal;
3. volatility release;
4. gap acceptance;
5. trend resumption;
6. liquidity breakdown.

---

## V.6.3 Gate-generalization test

A gate should not be judged successful solely because one handcrafted breakout rule works.

The abstract gate relation must transport:

G_P(Candidate,Boundary,Evidence,Ledger)
→ Decision + Residual. (V.11)

Across Event classes, determine whether explicit gating improves:

- calibration;
- timing;
- invalidation clarity;
- overpromotion control.

---

## V.6.4 Falsifier

Level 2 fails as a general architecture if:

- gates help only one narrowly chosen pattern;

- simple close or threshold rules match the full framework;
- gate states cannot be reproduced.

It may still survive as a local breakout method.

---

# V.7 Level 3 — Residual Governance

# V.7.1 Question

Does preserving unresolved evidence improve diagnosis, revision, or prediction?

The strong residual claim is not:

> Every contradiction predicts failure.

It is:

> Unresolved evidence should remain typed and available because it may alter later interpretation, invalidation, or model revision.

---

# V.7.2 Three residual hypotheses

### Predictive residual

Residual predicts later failure or instability.

### Diagnostic residual

Residual explains why superficially similar Events diverge.

### Governance residual

Residual improves revision even when it does not improve prediction.

These claims must be tested separately.

---

# V.7.3 Residual-history study

For Event i, define:

$$\mathscr{R}_i = \{r_{i,1}, r_{i,2}, \ldots, r_{i,m}\}. \quad \text{(V.12)}$$

Test whether:

$$\Pr(\text{Failure}_i \mid \text{Gate}_i, \mathscr{R}_i) \neq \Pr(\text{Failure}_i \mid \text{Gate}_i). \quad \text{(V.13)}$$

Then test whether residual-aware revision improves later model performance.

---

## V.7.4 Residual conversion

Estimate transitions:

Open
→ Resolved. (V.14)

Open
→ Dissipated. (V.15)

Open
→ Invalidating. (V.16)

Open
→ HigherPeriodLoad. (V.17)

The transition matrix is:

P_R(j | i,Type,Severity,Period,χ). (V.18)

---

## V.7.5 Falsifier

The residual architecture should be reduced if:

- residual labels are too unreliable;
- they merely restate gate failures;
- they add no diagnostic or revision benefit;
- analysts use them as universal excuses.

A reduced system may retain only a small set of high-reliability residuals.

---

# V.8 Level 4 — Transport and Invariance

## V.8.1 Question

Does a claim preserve a recognizable relation when moved into another admissible protocol?

The relevant transformation is:

T_{P→P′}(C_P) = Ĉ_{P′}. (V.19)

The observed target claim is:

C_{P′}. (V.20)

Transport error is:

ε_T
= Dist(Ĉ_{P′},C_{P′}). (V.21)

---

## V.8.2 Transport families

The programme should test:

- daily → weekly;
- time bars → volume bars;
- linear → logarithmic scale;
- raw → volatility-normalized;
- capitalization-weighted → equal-weight;
- price → breadth;
- cash → derivative;
- market → accounting;
- market → legal.

Different claims require different transport sets.

---

## V.8.3 Local versus invariant usefulness

A method may be:

- locally useful;
- covariantly useful;
- broadly invariant.

These states should not be collapsed.

The transport ladder is:

LocalPattern
→ ParameterRobustness
→ ProtocolFamilySurvival
→ CrossFunctionalSurvival
→ CrossAuthoritySurvival. (V.22)

---

## V.8.4 Invariance benefit study

Test whether transport survival predicts:

- longer persistence;
- lower invalidation;
- broader participation;
- stronger institutional consequence.

The hypothesis is:

Pr(Persistence | Survival)

Pr(Persistence | LocalOnly). (V.23)

---

## V.8.5 Falsifier

The strong objectivity claim fails if:

- transport labels are irreproducible;
- survival adds no useful information;
- the transformation set is chosen after outcomes;
- local claims are incorrectly penalized for intended locality.

---

# V.9 Level 5 — Episode Grammar

## V.9.1 Question

Can ordered Events form a reproducible higher-level grammar?

An Episode must contain more than several Events.

It requires:

- a persistent transition rule;
- ordered event relationships;
- a start gate;
- an endpoint or transition gate;
- branch alternatives;
- residual history.

---

## V.9.2 Candidate Episode representation

Define Episode j:

$$\mathfrak{E}_j = (G_start,\{e_1,e_2,\ldots,e_n\},Grammar_j,G_end,\mathscr{R}_j). \text{ (V.24)}$$

Possible grammars include:

- trend continuation;
- range rotation;

- accumulation and release;
- crisis escalation;
- recovery;
- policy intervention cycle.

---

## V.9.3 Competing episode models

Compare the Periodic Grammar with:

1. hidden Markov regimes;
2. change-point models;
3. drawdown segmentation;
4. trend filters;
5. Elliott-style event counts;
6. unsupervised sequence clustering.

The framework earns value only if its explicit gates and residuals improve:

- prospective segmentation;
- endpoint reliability;
- interpretability;
- transport.

---

## V.9.4 Episode Completion Certificate study

The proposed certificate contains:

ECC
= (OldGrammarFailure,NewGate,Persistence,Transport,Residual,BranchStatus). (V.25)

Compare it with:

- moving-average crossover;
- fixed drawdown;
- retrospective peak;
- volatility change point.

---

## V.9.5 Falsifier

Level 5 fails if:

- episode labels are mainly retrospective;
- branch relabelling remains unconstrained;
- simpler change-point methods perform equally well;
- completion gates cannot be specified prospectively.

A failed Episode model may reduce to a valid Event-sequence model.

---

# V.10 Level 6 — Reflexive Observer Effects

## V.10.1 Question

Does observation or adoption of a market rule change the market dynamics that the rule later observes?

The reflexive loop is:

MarketTrace
→ Interpretation
→ Action
→ NewMarketTrace. (V.26)

---

## V.10.2 Identification challenge

A signal may weaken after publication because of:

- crowding;
- arbitrage;
- market adaptation;
- data mining;
- regime change;
- transaction costs.

The study must distinguish observer backreaction from these alternatives.

---

## V.10.3 Candidate designs

### Publication event study

Compare signal behaviour before and after public dissemination.

### Adoption proxy study

Use:

- fund holdings;
- strategy assets;
- options concentration;
- search activity;
- public commentary.

### Natural experiment

Study rule changes that alter mechanical demand.

### Controlled simulation

Use agent-based markets where observer adoption can be manipulated.

---

## V.10.4 Nonlinear hypothesis

Let A denote adoption.

A possible relation is:

$$\text{Effect}(A) = \beta_1 A + \beta_2 A^2. \quad \text{(V.27)}$$

Possible regimes are:

- low adoption: weak effect;
- moderate adoption: self-confirmation;
- extreme adoption: crowding and inversion.

A crowding-inversion hypothesis predicts:

$$\beta_1 > 0. \quad \text{(V.28)}$$

$$\beta_2 < 0. \quad \text{(V.29)}$$

---

## V.10.5 χ connection

Observer adoption may change the effective feedback signature:

$$\chi_{\text{before}} \neq \chi_{\text{after}}. \quad \text{(V.30)}$$

But this must be measured rather than assumed.

---

## V.10.6 Falsifier

The reflexivity claim fails when:

- adoption cannot be measured;
- performance changes are fully explained by ordinary covariates;
- effects do not replicate across rules or markets.

---

# V.11 Level 7 — Complex-State Eligibility

## V.11.1 Question

Does a two-coordinate financial state possess a stable relation that makes complex representation operationally superior to an unrestricted real pair?

The proposed state is:

$$Z = R + iQ. \quad (V.31)$$

The informationally equivalent real representation is:

$$X = (R,Q). \quad (V.32)$$

Complex priority must therefore be earned through structure, not notation.

---

## V.11.2 Candidate TA states

Possible research candidates include:

### Acceptance and participation pressure

R = price acceptance.
Q = breadth or participation pressure.

### Realized structure and residual positioning

R = admitted structure.
Q = independently estimated trapped-position pressure.

### Valuation and conjugate exposure

R = admitted value.
Q = CAPM-derived conjugate exposure.

Each candidate must be tested independently.

---

## V.11.3 Mandatory comparisons

For every candidate, compare:

$M_0$ = scalar model. (V.33)

$M_1$ = flexible real-pair model. (V.34)

$M_2$ = complex amplitude–phase model. (V.35)

$M_3$ = alternative nonlinear or hyperbolic model. (V.36)

The complex model is privileged only if:

Utility($M_2$)

$$\max\{\text{Utility}(M_0),\text{Utility}(M_1),\text{Utility}(M_3)\}. \quad (V.37)$$

after complexity cost.

---

## V.11.4 Generator test

A local phase-bearing model should approximately satisfy:

$dR/d\theta \approx -Q.$ (V.38)

$dQ/d\theta \approx R.$ (V.39)

or a justified generalized generator.

Merely observing a loop in a phase portrait is insufficient.

---

## V.11.5 Scaling test

For rescaled Q:

$Q' = cQ,$ (V.40)

the phase interpretation should not depend arbitrarily on c.

If reasonable unit-preserving transformations destroy the result:

ComplexStatus = ScalingFragile. (V.41)

---

## V.11.6 Falsifier

Complex priority fails when:

- Q is residual error;
- Q is a lagged copy of R;
- phase depends on arbitrary normalization;
- the real pair performs equally well;
- no stable generator exists.

The correct reduction is:

$Z \rightarrow (R,Q).$ (V.42)

---

# V.12 Level 8 — Internal Phase Time

## V.12.1 Question

Does phase progression provide a useful internal clock for comparable market episodes?

Candidate phase time is:

$\tau_i(t) = \text{Unwrap}[\theta(t)]$. (V.43)

or:

$\tau_i(t) = \int_0^t |\dot{\theta}(s)| ds$. (V.44)

---

## V.12.2 Comparison clocks

Phase time must compete with:

- calendar time;
- normalized episode age;
- event count;
- cumulative volume;
- cumulative volatility;
- selection depth;
- learned monotone time warping.

Define alignment loss under clock c:

$D_c = \text{EpisodeAlignmentLoss}(c)$. (V.45)

Phase time earns priority only if:

$D_{\tau_i} < \min(D_t, D_k, D_V, D_\sigma, D_{\text{flexible}})$. (V.46)

---

## V.12.3 Online requirement

The phase clock must be computable without knowing the future Episode endpoint.

Retrospective normalization is insufficient.

At time t:

$\tau_i(t) \in \mathcal{F}_t$. (V.47)

---

## V.12.4 Gate-hazard requirement

A stronger result would show:

$$\Pr(\text{Gate at } t \mid \tau_i, \text{Controls}) \neq \Pr(\text{Gate at } t \mid \text{Controls}). \quad \text{(V.48)}$$

This tests whether internal phase is relevant to commitment rather than merely descriptive alignment.

---

## V.12.5 Branch stability

Phase unwrapping and branch assignment must survive:

- noise;
- sampling;
- nearby parameter changes;
- amplitude minima.

When A approaches zero:

$$A = \sqrt{(R^2 + Q^2)} \to 0, \quad \text{(V.49)}$$

phase confidence should decline.

---

## V.12.6 Falsifier

The phase-time claim fails when:

- event count performs equally well;
- a flexible ordinary clock dominates;
- branch assignment is unstable;
- improvement exists only retrospectively;
- phase adds no gate information.

A failed phase clock may leave a valid phase-bearing state.

---

# V.13 Level 9 — Time-Bearing Market Worlds

## V.13.1 Question

Does a committed ledger entry alter the admissible future dynamics of the system?

A World-level event requires:

Authority

- Gate
- PersistentTrace
- ChangedFutureAdmissibility. (V.50)

---

# V.13.2 Candidate domains

### Settlement

Execution changes ownership and later rights.

### Margin

A breach triggers forced action.

### Accounting

Impairment changes reported capital or distributable resources.

### Contract

Covenant breach changes obligations.

### Law

Judgment changes enforceable rights.

### Policy

An official decision changes eligible actions.

### Index governance

Inclusion changes benchmark-linked demand.

---

# V.13.3 Causal form

Let ledger state be L_k.

Let the admissible action set be:

$\mathcal{A}_{k+1} = F(L_k,CurrentState,Rules)$. (V.51)

A World-level trace is causally consequential when:

$\mathcal{A}_{k+1}(L_k)$
$\neq \mathcal{A}_{k+1}(L_k$ without event $e_k)$. (V.52)

This is a counterfactual claim.

It requires causal identification where possible.

---

## V.13.4 Cross-ledger studies

Study the timing relations among:

- market;
- internal risk;
- accounting;
- contractual;
- legal;
- policy ledgers.

The recognition vector is:

$$\mathbb{G}_t = (g_{market}, g_{risk}, g_{accounting}, g_{contractual}, g_{legal}, g_{policy}). \quad (V.53)$$

Possible states include:

- fragmented;
- market-leading;
- institution-leading;
- converging;
- reconciled;
- reopened.

---

## V.13.5 Phase-sensitive World

The strongest version additionally requires:

EligiblePhase

- PhaseSensitiveGate
- LedgeredConsequence
- Backreaction. (V.54)

This should be treated as the final and most demanding research level.

---

## V.13.6 Falsifier

World-level causality fails when:

- the supposed authority has no operative power;
- the ledger entry does not alter action;
- effects are fully explained by the underlying event;
- the trace is temporary or non-consequential.

A large market response alone does not establish a World-forming gate.

---

# V.14 Cross-Level Promotion Rules

A study may request promotion from one level to the next.

Promotion requires a new evidential condition.

| From | To | New requirement |
|---|---|---|
| L0 | L1 | reliable functional typing |
| L1 | L2 | Event gate |
| L2 | L3 | residual adds diagnostic or revision value |
| L3 | L4 | admissible transport |
| L4 | L5 | stable event grammar |
| L5 | L6 | observer-effect identification |
| L6 | L7 | earned conjugate geometry |
| L7 | L8 | internal-clock superiority |
| L8 | L9 | authority, ledger, and changed admissibility |

The promotion law is:

$$\text{Promote}_{n\to n+1} \Rightarrow \text{NewEvidence}_{n+1} \wedge \text{NewGate}_{n+1}. \quad (V.55)$$

No amount of success at level n automatically proves level n+1.

---

# V.15 Theory-Reduction Map

When a level fails, the model should reduce to the strongest surviving level.

## Failure at L9

Retain:

phase-sensitive Event or Episode model.

Reject:

time-bearing World claim.

## Failure at L8

Retain:

complex phase model.

Reject:

internal clock.

## Failure at L7

Retain:

real pair.

Reject:

privileged complex geometry.

## Failure at L6

Retain:

non-reflexive Episode model.

Reject:

observer-backreaction claim.

## Failure at L5

Retain:

Event sequence.

Reject:

Episode grammar.

## Failure at L4

Retain:

local claim.

Reject:

cross-frame regularity.

## Failure at L3

Retain:

gate.

Reduce residual to descriptive audit.

# Failure at L2

Retain:

Structure-level projection.

Reject:

Event status.

The reduction chain is:

World
→ Episode
→ Event
→ Structure
→ Projection. (V.56)

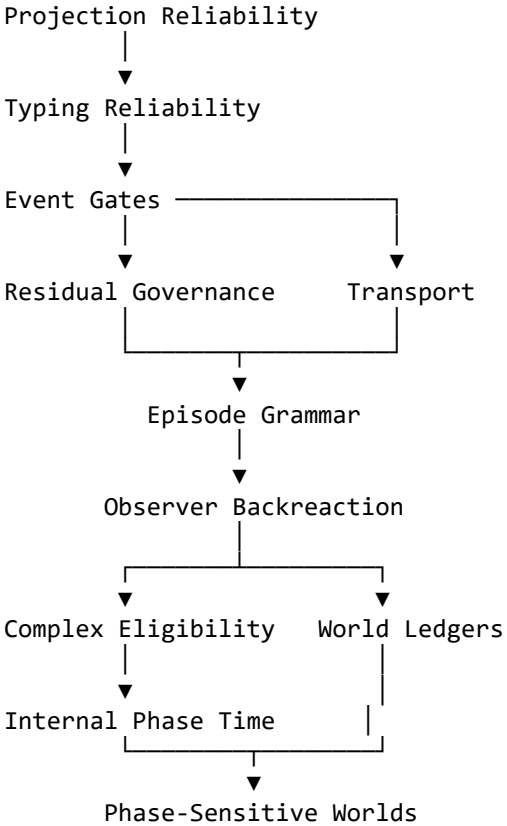
And for advanced geometry:

PhaseTime
→ PhaseState
→ RealPair
→ Scalar. (V.57)

---

# V.16 Study Dependency Graph

The research programme should follow a dependency graph rather than a simple publication sequence.




Complex eligibility does not need to wait for every institutional study.

But it does depend on reliable:

- projections;
- gates;
- episodes;
- real-pair benchmarks.

---

## V.17 Evidence Status Vocabulary

Every study result should receive a controlled status.

## Unsupported

The evidence fails to support the claim.

## Inconclusive

The study lacks sufficient precision or reliability.

## Locally supported

The claim holds under the tested protocol only.

## Replicated locally

The claim repeats across samples within one protocol family.

## Transported

The relation survives specified admissible transformations.

## Cross-domain supported

The structure survives materially different domains.

## Causally identified

A credible intervention or natural experiment supports causal influence.

## World-bearing

Authority, ledger, and changed admissibility are established.

This vocabulary prevents the word “validated” from hiding the scope of evidence.

---

# V.18 Replication Requirements

A claim should advance only after replication appropriate to its level.

## L0–L1

Independent implementation and annotation replication.

## L2–L3

Chronological out-of-sample replication in at least two market periods.

## L4

Replication across specified protocols.

## L5

Replication across multiple Episode families or asset classes.

## L6

Independent observer-adoption identification.

## L7–L8

Independent model implementation and competing-clock comparison.

## L9

Replication across institutional cases and authorities.

The replication burden is:

ReplicationBurden
∝ ClaimStrength × Consequence × Flexibility. (V.58)

---

# V.19 Negative-Result Programme

The project should maintain a public negative-results register.

Each failed study should record:

```
Study:
Claim tested:
Protocol:
Primary outcome:
Failure condition:
Observed result:
Strongest retained model:
Residual explanation:
```

```
Next permitted study:
Prohibited reinterpretation:
```

Examples include:

### Residual failure

Residual labels did not improve prediction or revision.

Retained model:

staged gate.

### Complex failure

Real pair matched the complex model.

Retained model:

two-dimensional real state.

### Phase-time failure

Event count aligned episodes equally well.

Retained model:

event-order clock.

Negative results should reduce the architecture rather than disappear.

---

# V.20 Cross-Study Ledger

Each study becomes one trace in the research programme.

Define the research ledger:

$$L_{research,k} = \{(Study_j,Protocol_j,Result_j,Residual_j,Status_j)\}_{j=1}^{k}. \quad (V.59)$$

The next study declaration should depend on:

- previous findings;
- unresolved residual;
- transport failures;
- model reductions.

Thus:

$$D_{study,k+1} = U_a(D_{study,k},L_{research,k},\mathscr{R}_{research,k}). \quad (V.60)$$

The research programme itself becomes a self-revising observer.

## V.21 Research Residual Registry

The main open residuals include:

### V.21.1 Ontology residual

Are six periods minimal and sufficient?

### V.21.2 Family residual

Are Load, Motion, Constraint, and Commitment distinct enough empirically?

### V.21.3 Proto-Eight residual

Do the eight roles improve diagnosis beyond the four families?

### V.21.4 Gate residual

Which gate components generalize across Event classes?

### V.21.5 Residual residual

Which unresolved variables are genuinely consequential?

### V.21.6 Transport residual

Which transformations are scientifically admissible rather than arbitrary?

### V.21.7 Complex residual

Which market variables form earned conjugate pairs?

### V.21.8 Time residual

When does internal phase outperform simpler clocks?

### V.21.9 Authority residual

Can World-level gates be standardized across law, accounting, and policy?

The registry should be updated after every study.

---

## V.22 Decision Table After the Breakout Pilot

| Pilot result | Immediate interpretation | Next study |
|---|---|---|
| Gate, residual, and transport | Event core provisionally supported | replicate across Event types |

| Pilot result | Immediate interpretation | Next study |
|---|---|---|
| succeed | | |
| Gate succeeds; residual fails | commitment distinction supported | simplify residual ontology |
| Gate succeeds; transport fails | local Event model supported | investigate transport definitions |
| Close-only matches full model | large architecture unnecessary for breakouts | test another Event class |
| All models weak | breakout object may lack stable value | reconsider boundary and outcome |
| Labels unreliable | ontology not operational enough | improve annotation before prediction |
| Strong prediction but poor calibration | useful ranking, weak gate interpretation | recalibrate or revise claim |
| Model succeeds only nonlinearly | possible interaction structure | require independent replication |

No outcome permits immediate promotion to complex phase or World formation.

---

# V.23 Minimal Publication Sequence

A disciplined publication programme might proceed as follows.

# Paper 1 — Typing and ontology reliability

Can observers classify market objects by period, function, and actuation role?

# Paper 2 — Residual-bearing breakout gates

The Appendix U pilot.

# Paper 3 — Gate generalization

Support failure, divergence, and volatility release.

# Paper 4 — Residual conversion and fakeout memory

Do failed Events become future Load?

# Paper 5 — Cross-frame survival

Does transport predict persistence?

# Paper 6 — Episode completion

Prospective Episode segmentation.

# Paper 7 — Observer crowding

Adoption, confirmation, and inversion.

## Paper 8 — Complex-state eligibility

Real-pair versus complex models.

## Paper 9 — Internal phase time

Competing clocks.

## Paper 10 — Cross-ledger World formation

Market, accounting, legal, and policy recognition.

Each paper should remain valid even if later, stronger layers fail.

---

# V.24 Minimum Dataset Growth

The corpus should grow with the validation ladder.

## Stage A

Projection and typing cases.

## Stage B

Event and residual cases.

## Stage C

Transported Event cases.

## Stage D

Episode sequences.

## Stage E

Observer-adoption cases.

## Stage F

Complex and phase episodes.

## Stage G

Institutional World cases.

The corpus should not include advanced labels before their annotation rules are reliable.

---

# V.25 Resource Allocation

The programme should allocate effort in proportion to evidential uncertainty.

A useful rule is:

ResearchPriority_i
= PotentialValue_i
× Uncertainty_i
× Testability_i
÷ Cost_i. (V.61)

Early priority should go to:

- gates;
- residuals;
- transport.

Lower early priority should go to:

- universal Proto-Eight claims;
- high-dimensional phase worlds;
- broad ontological conclusions.

This is not because the latter are unimportant.

It is because their evidential dependencies are not yet satisfied.

---

# V.26 Statistical Power and Conceptual Power

Large samples do not repair weak definitions.

A study may have millions of observations yet still fail because:

- the boundary was retrospectively chosen;
- Event labels were ambiguous;
- the outcome did not correspond to the claim;
- repeated observations were non-independent.

Define conceptual validity:

CV
= ProtocolClarity
× GateValidity
× OutcomeAlignment
× LeakageControl. (V.62)

A rough research-quality relation is:

ResearchStrength
∝ StatisticalPower × ConceptualValidity. (V.63)

If conceptual validity approaches zero, sample size cannot rescue the study.

---

# V.27 The Role of Simulation

Simulation can help test:

- known ground-truth gates;
- observer backreaction;
- missing-role failures;
- χ transitions;
- phase recovery;
- ledger causality.

But simulated success does not establish real-market validity.

Simulation should be used to answer:

> Can the method recover a structure known to exist?

Real data should answer:

> Does that structure describe an actual market process usefully?

The sequence is:

SimulationIdentifiability
→ RealDataApplicability
→ CrossDomainTransport. (V.64)

---

# V.28 The Role of Agent-Based Markets

Agent-based experiments are especially suitable for Level 6.

A simulated market may include:

- value traders;
- trend followers;
- breakout traders;
- market makers;
- institutional gate actors.

Vary the adoption rate of one Technical Analysis rule.

Observe:

- liquidity;
- continuation;
- crowding;
- fakeouts;
- $\chi$;
- residual accumulation.

Such experiments can test whether observer adoption creates:

- self-confirmation;
- saturation;
- inversion.

They cannot by themselves establish that the same mechanism dominates real markets.

---

# V.29 The Role of Natural Experiments

Natural experiments may arise from:

- index reconstitution;
- regulatory rule changes;
- disclosure mandates;
- trading-halt rules;
- margin changes;
- accounting-standard transitions.

These cases are useful because the gate or authority may change externally.

They can help identify:

GateChange
→ ActionSetChange
→ MarketBackreaction. (V.65)

This is central to Level 9.

---

# V.30 The Role of Qualitative Evidence

Not every relevant object is reducible immediately to one numerical score.

Qualitative evidence may be required for:

- legal authority;
- institutional decision procedures;
- model revision reasons;
- ambiguous branch structures;
- observer interpretation.

A mixed-method research record may include:

QuantitativeProjection

- StructuredQualitativeGate
- ResidualAudit. (V.66)

Qualitative evidence must still be:

- sourced;
- timestamped;
- typed;
- challengeable.

---

# V.31 Theory Comparison

The Periodic Grammar should be compared with alternative mature frameworks.

Relevant comparators include:

- market microstructure;
- signal detection;
- state-space models;
- hidden Markov regimes;
- change-point detection;
- event studies;
- survival analysis;
- reflexivity models;
- institutional accounting and legal event models.

The claim is not that these frameworks are obsolete.

The research question is:

> Does the Periodic Grammar organize their outputs into a useful common architecture of projection, closure, residual, transport, and ledger?

The framework succeeds as a unification architecture only if it improves:

- translation;
- diagnosis;
- governance;
- empirical design.

---

## V.32 Non-Exclusivity Principle

The Periodic Grammar need not be the only valid ontology.

Two frameworks may coexist when they answer different questions.

For example:

A hidden Markov model may estimate latent regimes.

The Periodic Grammar may govern:

- which observations define the regime;
- which gate commits transition;
- what residual remains;
- how the claim transports.

Thus:

StatisticalModel
≠ GovernanceGrammar. (V.67)

The framework should integrate useful models rather than rename them unnecessarily.

---

## V.33 Termination Conditions

The research programme should stop advancing a branch when:

1. repeated studies fail at the same gate;
2. simpler models repeatedly match performance;
3. labels remain irreproducible;
4. proposed variables cannot be measured prospectively;
5. the claim survives only through revision flexibility;
6. implementation cost exceeds scientific value.

A termination decision should be recorded:

BranchStatus = ClosedUnsupported. (V.68)

The branch may reopen only with genuinely new evidence or method.

---

# V.34 Promotion Budget

Every promotion adds conceptual commitments.

Define promotion cost:

$$Cost_{n \to n+1} = NewVariables$$

- NewAssumptions
- NewEstimationRisk
- NewAuthorityClaims. (V.69)

Promotion is justified when:

$$EvidenceGain_{n+1} > Cost_{n \to n+1}. \quad (V.70)$$

This prevents the programme from expanding merely because more elaborate language is available.

---

# V.35 The Strongest Possible Final Result

The strongest conceivable result of the full programme would be:

1. market objects can be reliably typed by closure period and function;
2. explicit gates improve Event calibration;
3. residual history improves diagnosis and revision;
4. transport survival identifies broader regularities;
5. Events form prospectively identifiable Episodes;
6. observer adoption measurably changes feedback orientation;
7. selected R–Q pairs possess stable complex dynamics;
8. internal phase improves Episode ordering and gate prediction;
9. authoritative ledger events alter future admissibility;
10. these structures replicate across domains.

Only then could the programme reasonably claim a mature theory of:

> protocol-bound, recursively generated, time-bearing market worlds.

This is a research destination, not a present conclusion.

---

# V.36 The Strongest Defensible Near-Term Result

A more realistic near-term result is narrower:

> Technical Analysis can be reconstructed as a governed Event-observation system in which predeclared boundaries, explicit gates, residual registers, and transport tests improve the reproducibility and calibration of market-transition claims.

That claim requires only Levels 0–4.

It does not depend on:

- complex phase;
- internal time;
- Proto-Eight universality;
- institutional World causality.

This makes the research programme scientifically resilient.

The lower core can survive even if the higher architecture fails.

---

# V.37 The Minimal Surviving Core

The smallest useful core is:

Protocol
→ Projection
→ Candidate
→ Gate
→ Residual
→ Outcome. (V.71)

If transport adds value:

Protocol
→ Projection
→ Candidate
→ Gate
→ Residual
→ Transport
→ Outcome. (V.72)

If recursive history adds value:

Protocol
→ Projection
→ Gate
→ Trace + Residual
→ Ledger
→ RevisedProtocol. (V.73)

Everything beyond this must be earned.

---

# V.38 Research Programme Contract

The multi-study programme commits to the following.

1. No claim is promoted without a new evidential gate.
2. Every study has a maximum permissible conclusion.
3. Negative results remain public and queryable.
4. Failed advanced models reduce to simpler retained models.
5. Same-source complexity is not counted as independent confirmation.
6. Retrospective episode construction is separated from prospective identification.
7. Complex models compete with flexible real-pair models.
8. Phase time competes with ordinary and learned clocks.
9. World claims identify authority and changed admissibility.
10. The ontology itself remains revisable.
11. Historical source interpretation is separated from modern engineering validation.
12. The research programme may terminate unsupported branches.

---

# V.39 Master Study Matrix

| Study | Primary level | Main gate | Main falsifier | Retained lower model |
|---|---|---|---|---|
| Projection replication | L0 | reproducibility | implementation disagreement | none |
| Ontology annotation | L1 | inter-rater reliability | labels unstable | conventional categories |
| Breakout pilot | L2 | Event admission | close-only matches | Structure rule |
| Residual-history study | L3 | residual increment | no added value | gate only |
| Transport study | L4 | cross-frame survival | local model equal | local Event |
| Episode completion | L5 | grammar persistence | retrospective only | Event sequence |
| Observer crowding | L6 | adoption effect | no identified backreaction | non-reflexive Episode |
| Complex eligibility | L7 | generator and benchmark | real pair equal | real pair |
| Phase-time study | L8 | clock superiority | ordinary clock equal | phase state |
| World-ledger study | L9 | authority and consequence | no changed admissibility | institutional Event |

---

# V.40 Master Decision Runtime

After every study, apply:

StudyResult
→ EvaluatePrimaryGate
→ AuditResidual
→ CompareSimplerModel
→ TestTransport
→ AssignEvidenceLevel
→ Promote,Retain,Reduce,orClose. (V.74)

The four possible decisions are:

## Promote

The next evidential level may be tested.

## Retain

The current level is useful, but no promotion is justified.

## Reduce

A simpler model captures the demonstrated value.

## Close

The branch is not presently supported.

---

# V.41 Appendix V Conclusion

The Periodic Grammar should not be validated as one indivisible theory.

It should be tested as a ladder of increasingly demanding claims.

The ladder begins with the most ordinary question:

> Can this operator be reproduced?

It then asks:

> Can its function be classified?
>
> Can a candidate transition be gated?
>
> Does residual matter?
>
> Does the claim survive another frame?
>
> Do Events form an Episode?
>
> Does observation alter the system?
>
> Does an earned conjugate geometry exist?

> Does phase become an internal clock?
>
> Does an authoritative ledger change the future world?

The complete progression is:

Projection
→ Function
→ Gate
→ Residual
→ Transport
→ Episode
→ Reflexivity
→ ComplexState
→ PhaseTime
→ TimeBearingWorld. (V.75)

The governing principle is:

> Every stronger interpretation must be purchased by a stronger experiment.

The governing reduction rule is:

> When the higher claim fails, preserve the strongest lower structure that still survives.

And the governing research ethic is:

> A mature theory is not the theory that reaches the highest level most quickly. It is the theory that can stop, reduce, and remain useful at the exact level its evidence has earned.

# Appendix W — Comparative Rosetta Stone: How the Periodic Grammar Relates to Existing Market Sciences

## W.1 Purpose

The Periodic Grammar should not present itself as though finance lacked:

- quantitative models;
- market microstructure;
- event studies;
- regime-switching methods;
- survival analysis;
- causal inference;
- behavioural finance;
- institutional accounting;
- legal analysis;
- control theory;

- machine learning.

Many individual tasks proposed in this article already have mature disciplinary methods.

For example:

- market microstructure studies trades, quotes, liquidity, and order flow;
- time-series econometrics models dependence and volatility;
- change-point methods detect structural breaks;
- event studies measure responses around identified events;
- survival analysis models transition hazards;
- causal inference studies intervention effects;
- accounting and law determine institution-specific recognition;
- Technical Analysis provides historically evolved chart projections.

The Periodic Grammar does not replace these fields.

Its proposed contribution is different:

> It supplies a common governance language for declaring what object is being measured, identifying its closure depth, separating projection from commitment, preserving residual, transporting claims across frames, and controlling promotion into stronger interpretations.

The source Gauge Grammar makes the same restriction at the broader cross-domain level:

FunctionalHomology ≠ SubstanceIdentity. (W.1)

and:

FrameworkGrammar supplements domain expertise; it does not replace it. (W.2)

The relationship should therefore be:

DomainScience

- PeriodicGovernance
  → BetterDeclaredResearchObject. (W.3)

not:

PeriodicGrammar
→ ReplacementForDomainScience. (W.4)

---

# W.2 Three Different Kinds of Framework

Much confusion disappears when three types of framework are separated.

# W.2.1 Mechanism framework

A mechanism framework asks:

> What process generates the observed behaviour?

Examples include:

- order-book interaction;
- funding constraints;
- inventory control;
- information arrival;
- institutional mandate;
- contractual enforcement.

---

## W.2.2 Statistical framework

A statistical framework asks:

> What regularity can be estimated from observable data?

Examples include:

- autoregression;
- volatility models;
- state-space models;
- classification;
- survival models;
- change-point detection.

---

## W.2.3 Governance framework

A governance framework asks:

> What exactly has been declared, measured, admitted, preserved, transported, and revised?

The Periodic Grammar is primarily this third type.

It can host mechanism and statistical models, but it is not itself a substitute for them.

The distinction is:

Mechanism explains generation. (W.5)

Statistics estimates relation. (W.6)

Governance controls claim formation. (W.7)

A mature market study may require all three.

---

# W.3 The Rosetta-Stone Principle

The Periodic Grammar acts as a translation layer.

Let domain model m produce an output:

$Y_m = Model_m(E \mid P_m)$. (W.8)

The Periodic Grammar asks:

1. What protocol $P_m$ was used?
2. Which market object did $Y_m$ project?
3. At which closure period does $Y_m$ exist?
4. Does $Y_m$ represent Load, Motion, Constraint, or Commitment?
5. Which gate would promote $Y_m$ into an Event?
6. What residual remains?
7. Which transport tests are relevant?
8. Which stronger claims remain prohibited?

Define the translation operator:

$\mathscr{R}_PPMG(Model_m)$
$= (Protocol,Projection,Period,Function,Gate,Residual,Transport,ClaimCeiling)$. (W.9)

The model itself need not change.

Its epistemic role becomes clearer.

---

# W.4 Technical Analysis

# W.4.1 What Technical Analysis already contributes

Technical Analysis provides a large historical instrument ecology:

- moving averages;
- oscillators;
- bands;
- volume methods;
- candlesticks;
- support and resistance;

- chart patterns;
- breadth;
- wave and cycle methods.

The source Technical Analysis article interprets these as partial instruments rather than direct market truth:

TechnicalAnalysis_P
= Projection_P(MarketSelfReference). (W.10)

Its central question is:

> What intrinsic characteristic does the method measure, and what does it fail to measure?

It classifies moving averages as memory filters, RSI as corrective-pressure detection, volume as a mixture of frequency, mass, commitment, and ambiguity, and support or resistance as ledgered memory zones.

---

# W.4.2 What the Periodic Grammar adds

The Periodic Grammar adds:

- six closure periods;
- four recurring functions;
- explicit promotion gates;
- typed residual;
- cross-frame transport;
- versioned revision;
- authority and ledger distinctions.

Thus an RSI signal becomes:

RSI
= Structure × Motion × Guidance
under protocol P. (W.11)

It does not automatically become:

ReversalEvent. (W.12)

A resistance break becomes:

Candidate Event (W.13)

until the declared gate passes.

---

# W.4.3 Non-redundancy test

The Periodic Grammar adds value only if it improves:

- classification reliability;
- gate calibration;
- residual recall;
- transport discipline;
- resistance to retrospective relabelling.

If ordinary Technical Analysis practice achieves the same results without the added grammar, the new layer is redundant.

---

# W.5 Market Microstructure

# W.5.1 Primary domain question

Market microstructure studies how trading arrangements and participant behaviour produce:

- prices;
- spreads;
- liquidity;
- execution;
- order flow;
- inventory effects.

Its natural scale is often Mark or Window.

---

# W.5.2 Periodic translation

| Microstructure object | Periodic type |
|---|---|
| Quote | Mark × Constraint |
| Queue imbalance | Mark × Load/Motion |
| Marketable order | Mark × Trigger |
| Execution | Mark × Commitment/Exchange |
| Bid–ask spread | Mark × Constraint/Boundary |
| Session liquidity | Window × Load |
| Price impact | Window or Event × Motion |
| Settlement failure | Event or World × Commitment |

Microstructure supplies precise mechanisms for many lower-period cells.

The Periodic Grammar should not rename those mechanisms.

It should clarify how their traces may be promoted.

For example:

OrderImbalance
≠ ExecutedFlow. (W.14)

ExecutedFlow
≠ DailyAcceptance. (W.15)

DailyAcceptance
≠ InstitutionalRecognition. (W.16)

---

# W.5.3 What microstructure contributes to PPMG

Microstructure can strengthen:

- Mark-level Load;
- Mark-level Constraint;
- Exchange measurement;
- liquidity residual;
- false-breakout diagnosis;
- Path-Bearing Candle construction;
- unexecuted intention estimates.

---

# W.5.4 What PPMG may add

PPMG may add:

- explicit closure promotion from Mark to Window and Event;
- residual preservation across aggregation;
- observer-authority distinctions;
- transport from microstructure to higher-frame chart claims.

---

# W.5.5 Non-redundancy test

Compare:

MicrostructureModel alone (W.17)

with:

MicrostructureModel

- PPMG gate and residual governance. (W.18)

The PPMG layer is useful only if it improves:

- event classification;
- auditability;
- cross-period promotion;
- failure diagnosis.

---

# W.6 Time-Series Econometrics

# W.6.1 Primary domain question

Time-series models estimate relationships across ordered observations.

Typical tasks include:

- forecasting;
- autocorrelation modelling;
- volatility estimation;
- cointegration;
- parameter change;
- dynamic factor extraction.

---

# W.6.2 Periodic translation

A time-series feature is usually a projection.

Examples:

Return_t
= Window × Motion. (W.19)

ConditionalVolatility_t
= Structure × Load or Agitation. (W.20)

LongRunRelation
= Structure × Constraint/Relation. (W.21)

ForecastProbability
= model output, not gate commitment. (W.22)

A probability of transition is not itself an Event.

An Event requires a declared admission rule.

---

# W.6.3 State and claim distinction

Suppose a model estimates:

Pr(BreakoutPersistence | E_t) = 0.72. (W.23)

PPMG asks:

- Does 0.72 exceed the gate threshold?
- Which authority applies the threshold?
- What residual remains?
- What is the claim ceiling?
- Which later state would invalidate the claim?

Thus:

EstimatedProbability
≠ CommittedEvent. (W.24)

The distinction is analogous to:

Measurement
≠ Recognition. (W.25)

---

## W.6.4 What econometrics contributes

Econometrics supplies:

- estimators;
- uncertainty;
- model comparison;
- out-of-sample testing;
- calibration;
- dependence control.

These are essential for validating PPMG.

---

## W.6.5 What PPMG may add

PPMG may add:

- semantic typing of model outputs;
- explicit closure levels;
- authority-aware gates;
- residual lifecycle;
- versioned revision.

---

## W.6.6 Non-redundancy test

A PPMG wrapper is justified only if it improves real research practice beyond ordinary model documentation.

The comparison should be:

OrdinaryModelCard
versus
Protocol–Gate–Residual–Ledger Record. (W.26)

---

# W.7 State-Space Models and Hidden-State Regimes

## W.7.1 Primary domain question

State-space and hidden-state models estimate latent regimes from observable data.

A generic form is:

x_{t+1} = F_tx_t + w_t. (W.27)

y_t = H_tx_t + v_t. (W.28)

A hidden-state model may classify:

- trend;
- range;
- crisis;
- recovery.

---

## W.7.2 Relationship to Episode

A latent regime resembles an Episode-level object, but the two are not identical.

A statistical regime is:

a model-estimated hidden state. (W.29)

A PPMG Episode is:

an ordered event grammar with gates, trace, residual, and persistence. (W.30)

A hidden-state transition may be used as:

EpisodeCandidate. (W.31)

It becomes an admitted Episode transition only under a declared gate.

---

## W.7.3 Important difference

A state-space model may change state because posterior probability crosses a threshold.

PPMG asks:

- Was the threshold predeclared?
- Does the state correspond to a stable market grammar?
- What event sequence supports it?
- What branch residual remains?
- Does the state transport across protocols?

---

## W.7.4 Integration

A useful integration is:

HiddenStatePosterior
→ EpisodeCandidate
→ PersistenceGate
→ EpisodeTrace + BranchResidual. (W.32)

---

## W.7.5 Non-redundancy test

Compare:

Hidden-state segmentation alone (W.33)

with:

Hidden-state segmentation

- explicit start/end gates
- residual branches
- transport. (W.34)

PPMG adds value only if the second improves:

- prospective endpoint identification;
- interpretability;
- revision discipline;
- cross-frame stability.

---

# W.8 Change-Point Detection

## W.8.1 Primary domain question

Change-point methods identify moments at which a data-generating process appears to change.

A detected change point is naturally:

EventCandidate. (W.35)

It is not automatically:

EpisodeTransition. (W.36)

---

## W.8.2 Periodic role

| Change-point output | PPMG interpretation |
|---|---|
| Local mean change | Structure or Event candidate |
| Volatility break | Event candidate |
| Parameter break | Event candidate |
| Persistent post-break state | Episode candidate |
| Institutionally recognized break | World event |

---

## W.8.3 Gate distinction

Change detection asks:

> Did the statistical relation change?

Episode governance asks:

> Did a new stable grammar form?

These questions are related but not identical.

A temporary shock may produce a detected break without creating a durable Episode.

---

## W.8.4 Integration

ChangePoint_t
→ CandidateEvent_t
→ PersistenceTest
→ TransportTest
→ EpisodeDecision. (W.37)

---

## W.8.5 Non-redundancy test

PPMG should be rejected for change-point applications if ordinary:

- posterior thresholds;
- persistence rules;
- model-selection criteria

already provide equivalent governance and audit.

---

# W.9 Event Studies

# W.9.1 Primary domain question

An event study estimates the response of an outcome around a defined event.

The method typically assumes the event has already been identified.

PPMG focuses on the earlier question:

> What qualifies the occurrence as an event?

---

# W.9.2 Complementary roles

PPMG supplies:

EventDefinition

- Gate
- Authority
- Residual. (W.38)

The event study supplies:

ResponseEstimation

- CounterfactualBenchmark
- Uncertainty. (W.39)

The combined sequence is:

CandidateCondition
→ PPMG Event Gate
→ EventStudy Window
→ EstimatedResponse
→ ResidualAudit. (W.40)

---

# W.9.3 Example

For index inclusion:

PPMG distinguishes:

- market anticipation;
- official announcement;
- effective inclusion;

- benchmark rebalancing.

An event study can then estimate different response windows around each gate.

Without this distinction, several different Events may be collapsed into one date.

---

# W.9.4 Non-redundancy test

PPMG adds value when event-date ambiguity is material.

If the event is externally clear and authoritative, such as a precisely timestamped official decision, the PPMG event-definition layer may be minimal.

---

# W.10 Survival and Hazard Models

# W.10.1 Primary domain question

Survival analysis estimates:

- time to failure;
- time to transition;
- conditional hazard.

This is highly compatible with gate-oriented analysis.

---

# W.10.2 Gate hazard

For candidate Event i, define:

h_i(t)
= Pr(Gate occurs at t | Gate has not occurred before t). (W.41)

Residual variables may enter:

h_i(t | GS_i,RB_i,TS_i). (W.42)

where:

GS = gate strength;
RB = residual burden;
TS = transport survival.

---

# W.10.3 Residual-resolution hazard

For residual r:

h_resolve(t)
= Pr(r resolves at t | r remains open). (W.43)

Similarly:

h_convert(t)
= Pr(r becomes a consequential Event at t | r remains open). (W.44)

---

## W.10.4 What survival analysis contributes

It gives formal tools for:

- pending gates;
- censored outcomes;
- time-varying residual;
- competing risks;
- Event timing.

---

## W.10.5 What PPMG may add

PPMG supplies a richer ontology of:

- what is at risk;
- which gate is pending;
- which residual remains;
- which period promotion is being studied.

---

## W.10.6 Non-redundancy test

If PPMG labels do not improve hazard modelling or interpretation, survival analysis alone may be sufficient.

---

# W.11 Signal Detection and Sequential Testing

## W.11.1 Primary domain question

Signal-detection methods distinguish signal from noise under uncertainty.

Sequential testing updates evidence as observations arrive.

This is close to the staged gate architecture.

---

## W.11.2 Mapping

Evidence accumulation:

$\Lambda_t$
$= \log[\Pr(E_{\le t}|H_1)/\Pr(E_{\le t}|H_0)]$. (W.45)

A sequential rule may:

- accept $H_1$;
- accept $H_0$;
- continue sampling.

This maps naturally to:

Admit;
Reject;
Defer. (W.46)

---

# W.11.3 Difference from PPMG

Sequential testing governs statistical evidence.

PPMG additionally records:

- closure period;
- market function;
- authority;
- residual type;
- transport;
- ledger consequence.

Thus PPMG can use sequential testing as its gate engine.

It should not reinvent sequential probability theory.

---

# W.11.4 Integration

SequentialEvidence
→ GateDecision
→ PPMG Trace + Residual + ClaimCeiling. (W.47)

---

# W.11.5 Non-redundancy test

For narrow statistical Event detection, PPMG may add little beyond metadata.

Its value should be tested in multi-frame, institutional, or revision-heavy settings.

---

# W.12 Bayesian Updating

## W.12.1 Primary domain question

Bayesian inference updates belief under new evidence:

$$\Pr(H|E) \propto \Pr(E|H)\Pr(H). \quad (W.48)$$

This is compatible with the self-revising observer.

---

## W.12.2 Belief versus commitment

A posterior probability remains an epistemic state.

A gate converts that state into a declared commitment.

$$\text{Posterior} \rightarrow \text{DecisionRule} \rightarrow \text{Gate} \rightarrow \text{Trace}. \quad (W.49)$$

The posterior alone does not determine action because:

- loss functions differ;
- authorities differ;
- consequences differ;
- residual differs.

---

## W.12.3 Revision

Bayesian updating changes belief.

PPMG revision may also change:

- the protocol;
- feature map;
- boundary;
- gate;
- authority interpretation.

These are different forms of change.

$$\text{BeliefUpdate} \neq \text{ProtocolRevision}. \quad (W.50)$$

A mature system should store both.

---

## W.12.4 Non-redundancy test

PPMG adds value only if protocol and authority changes matter beyond posterior belief updating.

---

# W.13 Control Theory

## W.13.1 Primary domain question

Control theory asks how a system should be influenced to maintain or reach a target state.

A simplified control system is:

x_{t+1} = f(x_t,u_t,w_t). (W.51)

The Periodic Grammar is primarily observational, but it contains intervention implications through:

- gates;
- Ξ;
- Proto-Eight roles;
- governed revision.

---

## W.13.2 Observation before control

The source Gauge Grammar explicitly separates protocol-bound diagnosis from intervention and warns that local improvement may create hidden dissipation.

The PPMG sequence should be:

Declare
→ Observe
→ Diagnose
→ Gate
→ SelectAdmissibleIntervention
→ RecordBackreaction. (W.52)

It should not be:

ObserveOneIndicator
→ ControlSystemImmediately. (W.53)

---

## W.13.3 Proto-Eight translation

| Control concern | Proto-Eight role |
|---|---|
| Potential difference | Gradient |
| Threshold or permission | Gate |
| State constraint | Boundary |

| Control concern | Proto-Eight role |
|---|---|
| Transfer channel | Exchange |
| Activation | Trigger |
| Direction or feedback | Guidance |
| State history | Memory |
| Measurement selection | Focus |

The roles may serve as an engineering checklist.

They do not replace controllability, observability, stability, or optimization analysis.

---

# W.13.4 Non-redundancy test

PPMG adds value if its role and residual grammar improves:

- intervention diagnosis;
- authority control;
- traceability;
- prevention of inappropriate control.

---

# W.14 Causal Inference

# W.14.1 Primary domain question

Causal inference asks:

> What would have happened under a different intervention or exposure?

PPMG by itself does not identify causal effects.

A sequence:

A occurred before B (W.54)

does not establish:

A caused B. (W.55)

---

# W.14.2 PPMG contribution

PPMG may improve causal studies by clarifying:

- the event gate;
- treatment timing;
- authority;

- competing events;
- residual confounding;
- protocol changes.

It can help define the treatment object.

---

# W.14.3 World-level causality

For a ledger Event e_k, the strong claim is:

$\mathcal{A}_{k+1}(e_k)$
$\neq \mathcal{A}_{k+1}$(without e_k). (W.56)

This is a counterfactual claim.

It requires causal evidence such as:

- natural experiment;
- discontinuity;
- instrument;
- controlled intervention;
- credible structural model.

The ledger sequence alone is insufficient.

---

# W.14.4 Non-redundancy test

PPMG is useful when it improves treatment definition and event timing.

It is not a replacement for causal identification.

---

# W.15 Behavioural Finance

# W.15.1 Primary domain question

Behavioural finance studies systematic departures from idealized rational decision-making and the effects of:

- attention;
- framing;
- anchoring;
- herding;
- overreaction;

- underreaction.

These concepts map naturally to:

- Focus;
- Memory;
- Gradient;
- Guidance;
- residual narrative;
- observer convergence.

---

## W.15.2 Attention as Focus

A technically visible level may gain importance because many observers focus on it.

This can create:

- concentrated orders;
- narrative convergence;
- self-confirming boundaries;
- crowding.

The source Technical Analysis framework treats market objectivity operationally: a level may become more consequential when multiple observer protocols recognize and act upon it.

---

## W.15.3 Behavioural variable versus structural role

“Focus” is a functional role.

It is not itself a behavioural mechanism.

Possible mechanisms include:

- salience;
- limited attention;
- convention;
- institutional reporting;
- platform design.

PPMG should not treat one broad role label as a causal explanation.

---

## W.15.4 Non-redundancy test

The Proto-Eight Focus–Memory distinction earns value only if it improves:

- behavioural feature design;
- failure diagnosis;
- observer-crowding models.

---

# W.16 Reflexivity and Self-Referential Markets

## W.16.1 Core loop

The source Technical Analysis article begins from:

Price
→ InterpretedEvidence
→ Orders
→ NewPrice. (W.57)

Price is not only an output.

It becomes input to the next round of action.

The article therefore interprets charts as visible traces of market self-reference.

---

## W.16.2 Relationship to behavioural finance

Behavioural finance may describe biases influencing the loop.

Reflexivity emphasizes that beliefs and market states may co-produce one another.

PPMG adds:

- declared observer protocols;
- Event gates;
- trace;
- residual;
- backreaction records.

---

## W.16.3 χ as a compact reflexivity signature

χ summarizes whether feedback is:

- corrective;
- ambiguous;
- reinforcing.

But χ is not a full behavioural or causal theory.

It is a regime descriptor requiring estimation.

---

## W.16.4 Non-redundancy test

χ adds value only if it improves conditional interpretation beyond standard regime variables.

If ordinary trend, volatility, and liquidity measures capture the same effect, χ should remain an interpretive label or be reduced.

---

# W.17 Machine Learning

## W.17.1 Primary domain question

Machine learning estimates flexible relationships between inputs and targets.

A model may discover useful patterns without human-readable intermediate concepts.

This creates both opportunity and tension.

---

## W.17.2 Prediction versus governance

A model may predict:

$$\Pr(Y=1|X) = 0.84. \quad \text{(W.58)}$$

PPMG asks:

- What was X?
- Was X available?
- What closure period is Y?
- Which authority uses the result?
- What residual remains?
- Does the model transport?
- How is revision versioned?

Thus PPMG may act as a governance shell around machine learning.

---

## W.17.3 Feature attribution is not residual

A feature-importance method explains model dependence.

It does not necessarily identify:

- missing evidence;
- contradictory frame;
- authority mismatch;
- unselected alternatives.

Therefore:

ModelExplanation
≠ ResidualRegister. (W.59)

---

## W.17.4 State-machine integration

A machine-learning model can supply:

- candidate probability;
- residual-risk estimate;
- transport prediction;
- episode classification.

The PPMG runtime determines how those estimates become governed states.

---

## W.17.5 Non-redundancy test

Compare:

ML model + ordinary model card (W.60)

with:

ML model + PPMG claim lifecycle. (W.61)

Assess:

- overpromotion;
- trace integrity;
- revision discipline;
- authority errors;
- auditability.

---

# W.18 Large Language Models

## W.18.1 Natural role

LLMs can assist with:

- protocol compilation;
- method typing;
- residual extraction;
- cross-ledger comparison;
- report generation;
- revision explanation.

They are especially useful when evidence includes:

- structured data;
- analyst commentary;
- accounting text;
- policy documents;
- legal decisions.

---

# W.18.2 Primary danger

An LLM can create fluent closure without sufficient gate evidence.

It may:

- invent a boundary;
- treat warnings as Events;
- erase contradiction;
- confuse market and legal states;
- produce decorative complex mathematics;
- revise its story after outcome.

The PPMG kernel therefore constrains the LLM output.

---

# W.18.3 LLM role separation

LLM tasks should be divided into:

### Projection interpretation

Explain what a method measures.

### Candidate generation

Propose possible Event classes.

### Gate assistance

Assemble relevant evidence.

### Residual audit

Search for contradictions and omissions.

### Authority check

Identify whether the model can commit the state.

### Revision explanation

Document why protocol change is proposed.

The LLM should not have unlimited authority across all stages.

---

## W.18.4 Non-redundancy test

The PPMG wrapper succeeds if it reduces:

- unsupported certainty;
- period overpromotion;
- omitted residual;
- authority confusion;
- retrospective relabelling.

---

# W.19 Market Profile, Auction, and Volume-Density Methods

## W.19.1 Primary domain question

Volume-distribution methods study where transactions and acceptance concentrate.

They contribute strongly to:

- Load;
- Memory;
- Constraint;
- Exchange.

---

## W.19.2 Periodic translation

VolumeProfile
= Structure × Load × Memory. (W.62)

HighVolumeNode
= Structure × Load/Constraint. (W.63)

ValueMigration
= Structure or Event × Motion. (W.64)

AcceptanceBeyondValue
= Event × Commitment. (W.65)

---

## W.19.3 Added value of the grammar

PPMG distinguishes:

- density from boundary;
- boundary from crossing;
- crossing from acceptance;
- acceptance from Episode transition.

This prevents volume density from being treated as an automatic prediction.

---

# W.20 Accounting

## W.20.1 Primary domain question

Accounting determines how economic conditions become recognized, measured, classified, and disclosed under specific standards and entity boundaries.

This is inherently protocol-bound.

---

## W.20.2 Natural PPMG mapping

| Accounting object | PPMG role |
|---|---|
| Reporting entity | Boundary |
| Measurement basis | Protocol |
| Recognition criterion | Gate |
| Journal entry | Trace |
| Financial statements | Ledger |
| Provision or uncertainty | Residual |
| Restatement | Revision |
| Audit opinion | Authority-linked gate |

Accounting therefore supplies mature real-world examples of:

- declared boundaries;
- rule-governed recognition;
- persistent ledger;
- revision;
- authority.

---

## W.20.3 Important distinction

Market price may supply evidence.

It does not by itself create accounting recognition.

MarketEvidence
→ AccountingCandidate
→ AccountingGate
→ AccountingTrace. (W.66)

---

## W.20.4 What PPMG may add

PPMG may supply cross-ledger translation among:

- market;
- risk;
- accounting;
- contractual;
- legal states.

It should not reinterpret accounting rules without domain expertise.

---

# W.21 Law and Contract

## W.21.1 Primary domain question

Law and contract determine:

- rights;
- obligations;
- admissible evidence;
- authority;

- enforceability;
- remedy.

These are World-level structures when they change future admissibility.

---

## W.21.2 Natural mapping

| Legal object | PPMG role |
|---|---|
| Jurisdiction | Boundary |
| Evidential rule | Projection protocol |
| Hearing or filing | Event process |
| Judgment | Gate |
| Order | Committed trace |
| Appeal | Residual/revision path |
| Precedent | Memory |
| Enforceable consequence | World backreaction |

---

## W.21.3 Important distinction

A chart may indicate distress.

It cannot determine legal default without the contractual and jurisdictional gate.

This is an authority ceiling, not merely uncertainty.

---

## W.21.4 Non-redundancy test

PPMG adds value if it improves translation between market and legal evidence while preserving the legal gate's autonomy.

---

# W.22 Institutional Economics and Organizational Analysis

## W.22.1 Primary domain question

Institutions shape behaviour through:

- rules;
- roles;
- incentives;
- constraints;
- enforcement;

- shared expectations.

This overlaps with World-level analysis.

---

## W.22.2 Periodic translation

Institutional memory appears as:

- policy;
- mandate;
- precedent;
- balance sheet;
- organizational routine.

Institutional Commitment appears as:

- board decision;
- regulation;
- contract;
- capital allocation;
- recognition rule.

The World becomes time-bearing when these traces alter future available actions.

---

## W.22.3 Added role of PPMG

PPMG may supply a common cross-ledger grammar.

It should not replace detailed institutional theory.

---

# W.23 Complex Systems and Dynamical Systems

## W.23.1 Primary domain question

Dynamical-systems methods study:

- state evolution;
- attractors;
- bifurcations;
- oscillations;
- stability;

- phase.

These methods are relevant to:

- χ;
- Episode grammar;
- complex eligibility;
- phase time.

---

## W.23.2 State-space versus PPMG period

A dynamical state may be continuous.

PPMG periods are closure classes.

Therefore:

StateDimension
≠ ClosurePeriod. (W.67)

A high-dimensional state may remain Structure-level.

A one-bit legal judgment may create a World transition.

Closure depth depends on gate and consequence, not mathematical dimensionality.

---

## W.23.3 Attractor versus Episode

An attractor is a dynamical object.

An Episode is a ledgered observational grammar.

An Episode may correspond to an attractor basin, but this requires demonstration.

---

## W.23.4 Phase restriction

The source Gauge Grammar repeatedly warns that cross-domain physical language should be used as functional grammar rather than literal substance identity, and that protocol quality determines usefulness.

Thus:

OscillatoryAppearance
≠ EarnedComplexPhase. (W.68)

PhasePortrait
≠ InternalTime. (W.69)

---

# W.24 Information Theory

## W.24.1 Primary contribution

Information theory may formalize:

- uncertainty;
- compression;
- mutual information;
- branch reduction;
- signal value.

This is relevant to:

- selection depth;
- projection loss;
- residual;
- confirmation independence.

---

## W.24.2 Projection information

Let total available information under protocol P be:

$I_P(\Sigma)$. (W.70)

A projection retains:

$I(X_j)$. (W.71)

Residual information is not simply:

$I_P(\Sigma) - I(X_j)$ (W.72)

unless a precise probabilistic model exists.

Therefore PPMG's residual is broader than an information-theoretic remainder.

It may include:

- institutional uncertainty;
- unknown authority;
- alternate declaration;
- omitted mechanism.

---

## W.24.3 Confirmation independence

Mutual information can help estimate whether two indicators add distinct evidence:

$I(Y;X_2 \mid X_1)$. (W.73)

If this is near zero, $X_2$ adds little conditional information.

This gives a quantitative test for the article's confirmation-independence claim.

---

# W.25 Measurement Theory

# W.25.1 Primary domain question

Measurement theory asks how empirical relations are represented numerically.

This is central to:

- units;
- scales;
- transformations;
- meaningful comparisons.

---

# W.25.2 Relevance to Q and complex states

Before constructing:

$Z = R + iQ$, (W.74)

one must establish:

- what R measures;
- what Q measures;
- whether units are compatible;
- which transformations preserve meaning;
- which metric is admissible.

Complex eligibility is therefore partly a measurement-theory problem.

---

# W.25.3 Transport and meaningful transformation

A claim may survive numerical transformation covariantly rather than identically.

For example:

linear price
→ logarithmic price. (W.75)

The question is not whether numerical coordinates remain unchanged.

It is whether the declared relation transforms meaningfully.

---

# W.26 Decision Theory

## W.26.1 Primary domain question

Decision theory converts uncertain beliefs into actions under:

- preferences;
- losses;
- constraints.

This is where model output becomes gate decision.

---

## W.26.2 Gate as decision rule

Let posterior state be $\pi$.

Let action set be $\mathcal{A}$.

A decision rule is:

$$a^* = \operatorname{argmin}_{a\in\mathcal{A}} E[L(a,State)|\pi]. \quad (W.76)$$

PPMG records the resulting gate and authority.

It does not prescribe one universal loss function.

---

## W.26.3 Same evidence, different gates

A trader, regulator, auditor, and court may observe related evidence but use different:

- loss functions;
- thresholds;
- authorities;
- consequences.

Therefore:

$$SameProjection \neq SameCommitment. \quad (W.77)$$

This is one reason protocol and authority cannot be omitted.

---

# W.27 The Non-Redundancy Test Suite

The Periodic Grammar should be subjected to explicit comparison against existing methods.

## W.27.1 Test 1 — Documentation test

Does PPMG improve reproducibility beyond an ordinary research protocol?

## W.27.2 Test 2 — Classification test

Do the six periods and four functions improve inter-observer agreement?

## W.27.3 Test 3 — Event test

Does explicit gate modelling improve calibration beyond a statistical threshold?

## W.27.4 Test 4 — Residual test

Does typed residual improve diagnosis or revision beyond ordinary error terms?

## W.27.5 Test 5 — Transport test

Does PPMG identify useful cross-frame relations beyond standard robustness analysis?

## W.27.6 Test 6 — Ledger test

Does event-sourced trace improve learning beyond ordinary version control?

## W.27.7 Test 7 — Authority test

Does the World framework reduce cross-domain category errors?

## W.27.8 Test 8 — Complexity test

Do $\chi$, $\Xi$, Z, and $\tau_i$ outperform simpler existing models?

The framework survives only where it adds value.

---

# W.28 Possible Reductions

The comparative programme may show that some PPMG components are merely renamed versions of mature methods.

Possible reductions include:

## W.28.1 Gate → Sequential decision threshold

When no additional authority or residual structure exists.

## W.28.2 Residual → Model-error vector

When all unresolved content is statistically specified.

## W.28.3 Episode → Hidden state

When event grammar adds no information.

## W.28.4 Transport → Robustness analysis

When the transformations are ordinary model checks.

## W.28.5 Ledger → Versioned event log

When no future admissibility changes.

## W.28.6 χ → Regime coefficient

When standard feedback estimates fully capture it.

## W.28.7 Ξ → Feature vector

When the three-coordinate compilation adds no useful constraint.

## W.28.8 Z → Real pair

When complex structure is not earned.

Such reduction would improve the theory.

It would identify which concepts are genuinely distinctive.

---

# W.29 Possible Irreducible Contributions

The Periodic Grammar may retain distinctive value in five areas.

## W.29.1 Closure-depth discipline

Separating:

Mark
→ Window
→ Structure
→ Event
→ Episode
→ World. (W.78)

Few ordinary modelling workflows enforce this hierarchy explicitly.

---

## W.29.2 Projection–commitment separation

A calculated object remains distinct from an admitted state.

---

## W.29.3 Residual as governed object

Residual is preserved with:

- type;
- status;
- lifecycle;
- claim effect.

---

## W.29.4 Cross-authority translation

Market, accounting, contractual, legal, and policy recognition remain separate but linked.

---

## W.29.5 Self-revising trace

The system records not only model output but how the declaration itself changes.

These are the strongest candidates for genuine non-redundancy.

---

## W.30 The Integration Matrix

| Discipline | Supplies | PPMG receives | PPMG adds |
|---|---|---|---|
| Technical Analysis | historical projections | indicator ecology | typing, gate, residual |
| Market microstructure | execution mechanisms | Mark/Window observables | period promotion |
| Econometrics | estimation and uncertainty | calibrated projections | claim governance |
| Hidden-state models | latent regimes | Episode candidates | gates and branches |
| Change-point methods | transition detection | Event candidates | persistence and promotion |
| Event studies | response estimation | post-gate outcomes | event-definition discipline |
| Survival analysis | hazard and censoring | gate timing | residual ontology |
| Sequential testing | admit/defer/reject logic | gate engine | ledger and authority |
| Bayesian inference | belief updating | uncertainty state | protocol-revision distinction |
| Control theory | intervention design | governed action | declaration and residual |
| Causal inference | counterfactual effect | backreaction tests | treatment typing |
| Behavioural finance | attention and bias | observer mechanisms | Focus–Memory roles |
| Machine learning | flexible prediction | candidate probabilities | claim ceiling and trace |

| Discipline | Supplies | PPMG receives | PPMG adds |
|---|---|---|---|
| Accounting | recognition and ledger | mature World gates | cross-ledger transport |
| Law | authority and enforceability | World commitment | market–legal distinction |
| Dynamical systems | state, attractor, phase | Episode/complex candidates | eligibility and closure |
| Information theory | uncertainty and dependence | independence tests | broader residual governance |
| Decision theory | action under loss | gate thresholds | authority and ledger record |

---

# W.31 The Host–Module Architecture

The Periodic Grammar should be implemented as a host architecture.

Domain methods are modules.

Define:

PPMGHost
= Protocol

- TypeSystem
- GateInterface
- ResidualInterface
- LedgerInterface
- TransportInterface
- RevisionInterface. (W.79)

A domain module provides:

DomainModule_m
= Data

- Mechanism
- Estimator
- DomainValidation. (W.80)

Integration is:

OperationalSystem
= PPMGHost ⊕ DomainModules. (W.81)

The symbol ⊕ means governed composition, not simple addition.

Examples:

PPMGHost
⊕ MicrostructureModule

⊕ SurvivalModule
→ BreakoutAcceptanceSystem. (W.82)

PPMGHost
⊕ AccountingModule
⊕ LegalModule
→ CrossLedgerRecognitionSystem. (W.83)

---

# W.32 Anti-Colonization Rule

A cross-domain framework can become harmful when it forces all disciplines to speak one vocabulary.

The Periodic Grammar must therefore obey:

Translation
≠ Erasure. (W.84)

Accounting recognition should not be reduced to “chart commitment.”

Legal authority should not be reduced to “market gate strength.”

Market microstructure should not be reduced to symbolic trigrams.

Complex dynamics should not be imposed where ordinary statistics suffice.

The source Gauge Grammar explicitly states that the framework is not a metaphor engine and that cross-domain templates express similarity of audit structure rather than equivalence of substance.

---

# W.33 Anti-Renaming Rule

A new term is justified only when it performs at least one of four jobs.

1. It distinguishes objects previously conflated.
2. It links fields through a reproducible mapping.
3. It creates a new testable hypothesis.
4. It improves governance or intervention.

If a new term merely renames an existing concept:

NewTermValue ≈ 0. (W.85)

For example:

- “Gate” adds value when it separates projection from commitment and identifies authority.
- “Residual” adds value when it is typed and lifecycle-tracked.
- “World” adds value when it requires changed future admissibility.
- “Phase time” adds no value when it merely renames normalized episode age.

# W.34 Disciplinary Translation Card

Every claimed cross-disciplinary equivalence should use the following card.

```
Source discipline:
Source object:
Source definition:
Source units:
Source validation:

PPMG mapping:
Closure period:
Functional family:
Actuation role:
Gate relationship:
Residual:

What is preserved:
What is changed:
What is not claimed:
Empirical non-redundancy test:
```

This card prevents analogical translation from becoming identity.

---

# W.35 Example — Change Point to Event

```
Source discipline:
Change-point detection

Source object:
Posterior probability of parameter change

Source validation:
Out-of-sample detection accuracy

PPMG mapping:
Event candidate

Closure period:
Event

Functional family:
Motion or Commitment candidate

Gate:
Posterior threshold plus persistence rule

Residual:
Temporary shock, model misspecification, alternate break date

What is preserved:
Evidence of statistical transition

What is not claimed:
A durable Episode or institutional World change
```

---

# W.36 Example — Accounting Impairment

```
Source discipline:
```

```
Accounting

Source object:
Recognized impairment

Source definition:
Reduction recognized under the applicable measurement and recognition rules

PPMG mapping:
World-level Commitment

Actuation role:
Gate

Trace:
Journal entry and financial statement effect

Authority:
Reporting entity under governance and audit

Residual:
Estimation uncertainty, appeal, later reversal, disclosure limitation

What is not claimed:
Market price alone automatically creates impairment
```

---

# W.37 Example — Hidden State to Episode

```
Source discipline:
State-space modelling

Source object:
High posterior probability of regime 2

PPMG mapping:
Episode candidate

Required promotion:
Persistent event grammar and declared completion/start gate

Residual:
Model uncertainty, state-label switching, horizon dependence

What is not claimed:
Latent-state classification automatically equals a market Episode
```

---

# W.38 Comparative Failure Modes

## W.38.1 Statistical imperialism

Treating estimated relation as the whole market object.

## W.38.2 Symbolic imperialism

Treating Proto-Eight or complex language as literal hidden substance.

## W.38.3 Institutional imperialism

Treating one authority's recognition as universal recognition.

## W.38.4 Technical-Analysis imperialism

Treating a chart projection as complete economic explanation.

## W.38.5 Governance imperialism

Adding protocol fields without improving scientific work.

The mature position is plural:

Different disciplines disclose different valid objects under different protocols.

---

# W.39 The Comparative Evidence Ladder

The Periodic Grammar should advance through four comparative stages.

## Stage 1 — Compatibility

Can existing models be represented without distortion?

## Stage 2 — Clarification

Does the grammar expose previously hidden category errors?

## Stage 3 — Incremental value

Does it improve measurement, prediction, or governance?

## Stage 4 — New discovery

Does the table reveal genuinely missing instruments or relations?

The progression is:

Compatible
→ Clarifying
→ Incremental
→ Generative. (W.86)

A framework that fails compatibility should not proceed to claims of unification.

---

# W.40 What the Periodic Grammar Most Likely Is

At its safest and most defensible level, the Periodic Grammar is:

> A protocol-first meta-model for governing how heterogeneous market observations become claims at different levels of closure.

It is not yet established as:

- a universal market law;
- a complete predictive theory;
- a replacement for quantitative finance;
- a physical theory of markets;
- a unique interpretation of 先天八卦.

Its strongest near-term role is methodological.

The source Technical Analysis article already reaches a similar conclusion: Technical Analysis becomes intelligible as operator diagnosis when each method is treated as a partial projection and tested through confirmation, residual, invalidation, and reframing.

The present article extends that principle into a cross-disciplinary closure architecture.

---

# W.41 What Would Make It More Than a Meta-Model?

The Periodic Grammar would become a stronger scientific theory if studies showed that:

1. the six periods are reliably distinguishable;
2. the four functions recur across periods;
3. gate variables predict commitment across Event classes;
4. residual histories have stable consequences;
5. transport survival predicts persistence;
6. Episode grammars outperform generic state models;
7. Proto-Eight roles predict characteristic failure modes;
8. ledger events causally alter later admissibility;
9. selected complex states outperform real pairs;
10. internal phase time outperforms ordinary clocks.

Until then, the framework should remain explicitly graded.

---

# W.42 Comparative Claim Classes

# Class 1 — Translation

“Moving average can be represented as Structure × Load × Memory.”

This is an interpretive mapping.

## Class 2 — Governance improvement

“Explicit gate records reduce retrospective breakout relabelling.”

This is an empirical hypothesis.

## Class 3 — Structural recurrence

“Load, Motion, Constraint, and Commitment recur across all six periods.”

This is a stronger theoretical hypothesis.

## Class 4 — Generative law

“The four-function cycle generates higher-period market worlds.”

This requires broad empirical support.

## Class 5 — Ontological claim

“Markets fundamentally consist of these structures.”

This article does not establish that claim.

---

# W.43 The Mature Comparative Question

The wrong comparative question is:

> Is the Periodic Grammar better than econometrics, microstructure, or causal inference?

The correct question is:

> Does the Periodic Grammar make it easier to combine those methods without confusing their objects, authorities, closure levels, or evidential claims?

The framework succeeds when it can say:

- this model estimated Motion;
- this boundary supplied Constraint;
- this decision rule created Commitment;
- this residual remained unresolved;
- this institutional authority created a World trace;
- this transport failed;
- this stronger claim must therefore be reduced.

---

# W.44 Comparative Research Contract

The framework commits to the following comparative rules.

1. Existing domain terminology will be preserved where it is more precise.
2. PPMG mappings will be labelled as mappings, not identities.
3. Domain models will be evaluated by their own standards.
4. PPMG will be evaluated by incremental governance value.
5. Stronger PPMG claims will compete with mature simpler alternatives.
6. Complex language will not replace measurement theory.
7. Proto-Eight roles will not replace mechanism models.
8. World-level claims will not bypass institutional authority.
9. Statistical success will not erase residual or protocol history.
10. Components that add no value will be reduced or removed.

---

# W.45 The Full Rosetta Stone

| PPMG object | Statistical interpretation | Market interpretation | Institutional interpretation |
|---|---|---|---|
| Protocol | model specification | chart/data frame | reporting or legal frame |
| Projection | estimator output | indicator | measured evidence |
| Load | state/history variable | positioning, volume, memory | obligation, inventory, precedent |
| Motion | transition/difference | return, momentum, flow | change in status |
| Constraint | support or feasible set | boundary, liquidity limit | rule, covenant, jurisdiction |
| Gate | decision threshold | close, retest, execution | recognition or judgment |
| Trace | recorded state | event history | ledger entry |
| Residual | unexplained/open state | conflict, missing confirmation | uncertainty, appeal, unrecognized exposure |
| Transport | robustness transform | timeframe or scale change | cross-ledger translation |
| Episode | regime sequence | trend, range, crisis | institutional process |
| World | action-generating rule system | market institution | legal, accounting, policy order |
| Revision | model update | changed protocol | restatement, appeal, rule change |

---

# W.46 Final Integrated Scientific Stack

The mature research architecture should be:

DomainMechanism
→ DomainMeasurement
→ StatisticalEstimation
→ PPMGTyping
→ GateAndResidual
→ DomainValidation
→ Transport
→ LedgerAndRevision. (W.87)

No layer should absorb all the others.

The full stack may be written:

ScientificMarketClaim
= Mechanism

- Measurement
- Estimation
- Protocol
- Closure
- Gate
- Residual
- Transport
- Authority
- Trace. (W.88)

Not every study requires every term.

The required subset depends on the claim.

---

# W.47 Appendix W Conclusion

The Periodic Grammar is strongest when it behaves as a disciplined host rather than an imperial replacement.

Market microstructure explains execution.

Econometrics estimates dynamic relations.

State-space models infer regimes.

Change-point methods locate transitions.

Event studies estimate responses.

Survival analysis models timing.

Sequential testing supports gates.

Causal inference evaluates intervention.

Behavioural finance supplies observer mechanisms.

Machine learning supplies flexible estimation.

Accounting, law, and policy supply authoritative World gates.

Dynamical systems supply state, stability, and phase tools.

The Periodic Grammar contributes a shared language for asking:

- What object did the method disclose?
- At what closure depth?
- Through which function?
- Under which protocol?
- With which gate?
- Leaving which residual?
- Surviving which transport?
- Entering whose ledger?
- Under whose authority?
- Subject to which revision?

Its comparative principle is:

DomainExpertise

- PPMGGovernance

  PPMGAlone. (W.89)

Its anti-reduction principle is:

FunctionalTranslation
≠ SubstanceIdentity. (W.90)

Its non-redundancy principle is:

No new grammar term should survive unless it distinguishes, connects, tests, or governs something that existing practice leaves materially unresolved. (W.91)

And its final scientific position is:

> The Periodic Grammar should be judged not by whether it can rename every financial method, but by whether it can help different methods cooperate without confusing projection with reality, statistical change with commitment, market evidence with institutional authority, or mathematical elegance with earned scientific closure.

# Appendix X — Visual Grammar and Publication Figure

# Set

# X.1 Purpose

The Periodic Grammar now contains several interacting architectures:

- six closure periods;
- four functional families;
- eight Proto-Eight actuation roles;
- three governance rails;
- χ feedback signatures;
- Ξ operating states;
- optional complex geometry;
- phase and ledger time;
- gates, residuals, transport, and revision.

Without strict visual discipline, these layers can easily be confused.

A circular diagram may incorrectly suggest that:

- the six periods repeat mechanically;
- Proto-Eight is a chronological cycle;
- CAPM's local phase rotation is the global architecture;
- every market method possesses complex dynamics;
- residual is the imaginary component;
- World formation is merely a larger chart pattern.

The publication figures must therefore do more than illustrate the prose.

They must preserve the theory's type distinctions.

The principal visual law is:

VisualSimilarity must not imply OntologicalIdentity. (X.1)

The figures should make clear which relations are:

1. definitions;
2. exact constructions;
3. proposed taxonomies;
4. operational mappings;

5. empirical hypotheses;
6. analogies;
7. unvalidated extensions.

---

## X.2 General Visual Architecture

Every publication figure should identify four things.

## X.2.1 Object type

What kind of object is being shown?

Examples:

- indicator;
- gate;
- closure period;
- actuation role;
- residual;
- ledger;
- protocol;
- state coordinate.

---

## X.2.2 Relation type

What kind of arrow connects the objects?

Examples:

- produces;
- constrains;
- admits;
- records;
- transports;
- revises;
- resembles;
- may influence.

---

## X.2.3 Evidential status

Is the relation:

- source-derived;
- mathematically exact;
- proposed by this article;
- empirically supported;
- awaiting validation?

---

## X.2.4 Scope

Does the relation apply to:

- one indicator;
- one Event class;
- one market protocol;
- Technical Analysis broadly;
- institutional financial worlds;
- a cross-domain analogy?

A figure without scope encourages accidental overgeneralization.

---

# X.3 Canonical Visual Layers

The visual system should distinguish five layers.

| Layer | Primary objects | Visual purpose |
|---|---|---|
| **Layer A — Closure** | Mark to World | how deeply a state has been committed |
| **Layer B — Function** | Load, Motion, Constraint, Commitment | what analytical job is performed |
| **Layer C — Actuation** | eight Proto-Eight roles | how potential becomes routed action |
| **Layer D — Governance** | residual, transport, ledger | how claims remain honest and revisable |
| **Layer E — Advanced state** | $\chi$, $\Xi$, Z, $\theta$, $\tau_i$ | optional regime and phase structures |

The layers should not be collapsed into one overloaded wheel.

The preferred composition is:

Closure Layer

× Functional Layer

- Actuation Overlay
- Governance Rails
- Optional Advanced-State Inset. (X.2)

---

# X.4 Shape Vocabulary

A stable shape system helps distinguish object categories even in monochrome.

# X.4.1 Rounded rectangle — declared object

Use for:

- protocol;
- projection;
- Structure;
- Event;
- Episode;
- World.

---

# X.4.2 Diamond — gate or decision

Use for:

- close gate;
- retest gate;
- promotion gate;
- institutional recognition;
- model-eligibility decision.

---

# X.4.3 Circle — state or coordinate

Use for:

- R;
- Q;
- χ;
- ρ;

- γ;
- ν;
- phase angle.

---

## X.4.4 Hexagon — actuation role

Use for the eight Proto-Eight roles.

This separates operational roles from observational families.

---

## X.4.5 Open container — boundary or protocol frame

Use for:

- declared world;
- admissible state region;
- legal jurisdiction;
- chart boundary;
- episode basin.

---

## X.4.6 Document or ledger symbol — committed trace

Use for:

- event record;
- accounting entry;
- legal judgment;
- protocol version;
- residual register.

---

## X.4.7 Irregular cloud — residual

Residual should not be drawn as a circle parallel to R and Q.

Its irregular shape indicates unresolved, heterogeneous content.

---

# X.5 Arrow Vocabulary

Arrows must be semantically typed.

## X.5.1 Solid arrow — declared operational transition

Example:

Candidate
→ Gate. (X.3)

---

## X.5.2 Thick solid arrow — admitted promotion

Example:

Event
⇒ Episode. (X.4)

Use only when the required closure gate passes.

---

## X.5.3 Dashed arrow — hypothesis or candidate relation

Example:

Observer crowding
⇢ χ transition. (X.5)

---

## X.5.4 Dotted arrow — analogy or interpretive mapping

Example:

Proto-Eight Memory
⋯→ Load family. (X.6)

A dotted mapping must not be mistaken for identity.

---

## X.5.5 Double-headed arrow — reciprocal coupling

Example:

Signal pressure
↔ Market structure. (X.7)

---

## X.5.6 Curved return arrow — backreaction

Example:

Ledgered Event
↶ Future Market Field. (X.8)

---

## X.5.7 Forked arrow — trace and residual

Every consequential gate should visibly divide into:

Gate
→ Trace

- Residual. (X.9)

This is one of the article's most important visual conventions.

---

# X.6 Evidential-Status Markers

Each figure caption should include a status line.

Use:

- **D** — definition;
- **E** — exact construction;
- **T** — proposed taxonomy;
- **M** — operational mapping;
- **H** — empirical hypothesis;
- **A** — analogy;
- **R** — research programme.

Examples:

```
Status: T + M
```

means the figure presents a proposed taxonomy and operational mapping.

```
Status: E
```

means the displayed mathematical relation is exact under the stated definitions.

```
Status: H
```

means the causal or predictive relation remains to be tested.

A legend should state:

Taxonomic exactness does not imply empirical validation. (X.10)

---

# X.7 Colour System

Colour should support, not carry, meaning.

A suggested four-family palette is:

| Family | Semantic association |
|---|---|
| Load / Memory | accumulated state |
| Motion / Relation | transformation |
| Constraint / Boundary | admissibility |
| Commitment / Gate | selection |

The actual colours may be chosen by the publication designer.

However:

- each family must always use the same colour;
- text labels must remain present;
- no colour alone should distinguish pass from fail;
- residual, transport, and ledger should use patterns or borders as well as colour.

---

# X.7.1 Governance rails

Recommended visual treatment:

- residual — irregular outline or hatched band;
- transport — paired frames connected by a transformation bridge;
- ledger — persistent horizontal trace beneath the main diagram.

---

# X.7.2 Advanced-state colours

χ, Ξ, and complex phase should use a separate subdued palette.

They should not visually dominate the basic gate-and-ledger architecture.

This reflects the evidential hierarchy:

Protocol and Gate before Complex Phase. (X.11)

---

# X.8 Typography Rules

# X.8.1 Main labels

Use plain-language labels first:

- Load;
- Motion;
- Constraint;
- Commitment.

Symbols may appear secondarily.

---

## X.8.2 Formula labels

Each equation in a figure should have:

- symbol definition;
- units where applicable;
- protocol index when required.

For example:

χ_{P,h} (X.12)

rather than an unqualified χ.

---

## X.8.3 Warning labels

Use explicit boxed warnings for statements such as:

```
Local phase cycle — not the six-period market sequence

Operational analogy — not physical identity

Candidate Event — not yet admitted
```

---

## X.8.4 Small-print prohibition

Critical scope restrictions should not be hidden in unreadable footnotes.

The claim ceiling should be visually prominent.

---

# X.9 Figure 1 — The Master Proto-Periodic Table

## Title

**The Proto-Periodic Grammar of Market Observation**

## Subtitle

**Four recurring analytical functions across six levels of committed market organization**

## Status

T + M

## Layout

A 6 × 4 rectangular table.

Rows:

1. Mark;
2. Window;
3. Structure;
4. Event;
5. Episode;
6. World.

Columns:

1. Load / Memory;
2. Motion / Relation;
3. Constraint / Boundary;
4. Commitment / Gate.

Three horizontal governance rails appear beneath or behind the table:

- residual;
- transport;
- ledger.

---

# X.9.1 Cell contents

| Period | Load | Motion | Constraint | Commitment |
|---|---|---|---|---|
| Mark | liquidity, order size | tick change, spread movement | bid–ask limits | execution |
| Window | bar volume, local memory | return, candle path | range, bands | close |
| Structure | MA, VWAP, profile | momentum, divergence | support, channels | structural acceptance |
| Event | pre-gate participation | acceleration, phase shift | tested boundary | break, retest, rejection |
| Episode | event and residual history | $\chi$ progression | pattern basin | completion or transition |
| World | institutional memory | reflexive dynamics | law, policy, collateral | authoritative state change |

The table should visually emphasize that methods occupy cells, but are not elements in the chemical sense.

---

## X.9.2 Required warning

```
The table classifies observational roles and closure depth.
It does not claim that financial methods are chemical substances.
```

---

## X.9.3 Caption

**Figure 1.** The master proto-periodic table. Technical methods are classified by the closure period at which their object exists and by the primary function they perform. The three governance rails—residual, transport, and ledger—apply to every cell rather than forming additional method families. The 6 × 4 structure is the present article's proposed synthesis and requires empirical validation.

---

## X.10 Figure 2 — The Four-Family Functional Cycle

## Title

**How a Market Possibility Becomes a Consequential Trace**

## Status

T + M

## Main sequence

Load_p
→ Motion_p under Constraint_p
→ Commitment_p
→ Trace_p + Residual_p. (X.13)

A secondary arrow carries:

Trace_p + Residual_p
→ Load_{p+1}. (X.14)

---

## X.10.1 Visual arrangement

Use a left-to-right process, not a closed circular wheel.

This avoids implying that every system repeats periodically in clock time.

The final arrow to the next period may curve downward into the next row.

---

## X.10.2 Key annotations

### Load

What history or pressure is carried?

### Motion

What changes or relates?

### Constraint

What limits or channels the change?

### Commitment

What is admitted into history?

### Residual

What remains unresolved?

---

# X.10.3 Caption

**Figure 2.** The local four-family grammar. A prior state supplies Load; Motion develops inside or across a Constraint; a Gate determines Commitment; and the output divides into admitted Trace and unresolved Residual. The resulting trace and residual may become Load at the next closure period.

---

# X.11 Figure 3 — The Six-Period Closure Ladder

# Title

### From Market Mark to Time-Bearing World

# Status

T + H

# Layout

A vertical ladder with six platforms.

```
World
↑ authority + ledger + changed admissibility
Episode
↑ persistent event grammar
Event
↑ declared commitment gate
Structure
↑ persistent relation
Window
↑ aggregation closure
Mark
```

Each upward arrow must carry the required new closure condition.

---

## X.11.1 Visual prohibition

Do not show all six steps as automatic.

Each promotion arrow should contain a gate diamond.

The figure should also show demotion arrows:

- Event → Structure warning;
- Episode → Event sequence;
- complex time claim → real pair.

---

## X.11.2 Caption

**Figure 3.** The six-period closure ladder. Higher levels require additional commitment conditions rather than merely larger timescales. Repetition alone does not create promotion: many windows need not form a stable Structure, and many Events need not form an Episode.

---

# X.12 Figure 4 — The Three-Layer Type System

# Title

**Closure, Function, and Actuation Are Different Questions**

# Status

T + M

# Layout

Three stacked transparent sheets.

### Sheet 1 — Closure

Mark → World.

### Sheet 2 — Function

Load, Motion, Constraint, Commitment.

### Sheet 3 — Actuation

Gradient, Gate, Boundary, Exchange, Trigger, Guidance, Memory, Focus.

A method card passes vertically through the sheets.

Example:

```
Moving Average
Closure: Structure
Function: Load
Actuation: Memory
```

---

# X.12.1 Key statement

Period tells **where the object exists**.

Function tells **what analytical job it performs**.

Actuation tells **how the operation is generated or routed**.

---

# X.12.2 Caption

**Figure 4.** The three-layer type system. The six periods, four families, and eight Proto-Eight roles do not compete for the same classificatory job. A method is fully typed only after closure depth, observational function, and operative role are distinguished.

---

# X.13 Figure 5 — The Proto-Eight Actuation Topology

# Title

### Eight Operational Roles of Market Actuation

# Subtitle

### A topology of paired functions, not a universal chronological cycle

# Status

M + A

The source Proto-Eight engineering framework defines the paired roles as:

- Potential Gradient / Qualified Gate;
- Boundary-Buffer / Exchange;
- Trigger / Guidance;
- Memory / Focus.

---

# X.13.1 Layout

Use four opposing dyads around a central declared market object.

```
Gradient ⇔ Gate
Boundary ⇔ Exchange
Trigger ⇔ Guidance
```

```
Memory ↔ Focus
```

The dyads may be arranged as an octagonal topology.

Do not use a clockwise arrow around the octagon.

---

# X.13.2 Optional local runtime inset

A small inset may show one possible protocol-specific sequence:

Memory
→ Focus
→ Boundary
→ Gradient
→ Trigger
→ Guidance
→ Exchange
→ Gate. (X.15)

The inset must carry:

```
One possible market runtime projection.
Not a universal classical trigram ordering.
```

---

# X.13.3 Caption

**Figure 5.** Proto-Eight actuation topology. The eight roles describe complementary operational requirements—potential, qualification, separation, transfer, initiation, steering, retention, and selective attention. Their circular placement represents relational topology rather than a necessary temporal sequence. The Proto-Eight source treats these roles as engineering primitives and tests their paired balance rather than presenting them as literal financial substances.

---

# X.14 Figure 6 — Gate, Trace, Residual, and Ledger

# Title

### Commitment Does Not Exhaust the World

# Status

M

# Layout

A central gate diamond receives:

- candidate;
- evidence;

- prior ledger;
- authority.

Its output splits.

### Upper path

Admitted Event
→ Trace
→ Ledger update.

### Lower path

Residual register
→ resolve, persist, convert, or invalidate.

A later curved arrow returns from both ledger and residual to the next declaration.

---

## X.14.1 Main formula

$G_P(c_k,L_k,\mathscr{R}_k)$
$\rightarrow (d_k,\alpha_k,e_k,r_k)$. (X.16)

---

## X.14.2 Required labels

Residual is not:

- zero by default;
- Q by default;
- error by default;
- failure by default.

Possible residual states:

Open
→ Resolved / Dissipated / Converted / Invalidating. (X.17)

---

## X.14.3 Caption

**Figure 6.** The gate–trace–residual–ledger architecture. A gate may admit a consequential event while leaving material unresolved content. Trace changes the ledger; residual remains attached to the claim and may later resolve, invalidate, become higher-period Load, or trigger revision. The declaration framework treats gate and residual as inseparable components of auditable closure.

---

## X.15 Figure 7 — Candidate, Event, Episode, and World

# Title

**Four Claims Commonly Confused in Technical Analysis**

# Status

T + M

# Layout

Four adjacent panels.

### Panel A — Candidate

Price crosses a line.

### Panel B — Event

Declared close or acceptance gate passes.

### Panel C — Episode

A persistent sequence of Events establishes a new grammar.

### Panel D — World

An authoritative ledger changes future rights or actions.

---

# X.15.1 Prohibited shortcut arrows

Show red crossed-out arrows:

Candidate ⇏ Episode. (X.18)

PriceMove ⇏ LegalDefault. (X.19)

IndicatorReading ⇏ WorldTransition. (X.20)

---

# X.15.2 Caption

**Figure 7.** Four distinct closure claims. A candidate marks possibility; an Event requires a gate; an Episode requires a persistent event grammar; and a World requires authority, ledger, and changed future admissibility. The visual separation prevents chart movement from being confused with institutional recognition.

---

# X.16 Figure 8 — Confirmation Independence Map

# Title

**More Indicators Are Not Necessarily More Evidence**

## Status

M + H

## Layout

A lineage graph.

At the left:

Adjusted price.

From it branch:

- RSI;
- MACD;
- moving averages;
- rate of change.

A separate branch contains:

- volume;
- breadth;
- order flow;
- profile;
- institutional data.

The graph should show shared roots.

---

## X.16.1 Main relation

ConfirmationValue
≠ IndicatorCount. (X.21)

A conceptual independence score may be:

I_j
= 1 – max_{m<j}Overlap(Source_j,Source_m). (X.22)

---

## X.16.2 Caption

**Figure 8.** Confirmation independence. Several indicators may provide useful operator diversity while still sharing one underlying price lineage. Stronger confirmation requires distinct informational channels and failure modes, not merely a larger count of transformed price series.

---

# X.17 Figure 9 — χ Regime Geometry

# Title

**Corrective, Critical, and Self-Confirming Feedback**

# Status

H

# Layout

Three panels.

### χ < 0 — Corrective

Perturbations bend back toward a range or equilibrium.

### χ ≈ 0 — Critical

Small perturbations may lead to several possible branches.

### χ > 0 — Self-confirming

Movement reinforces expectations and further action.

---

# X.17.1 Operator inset

C_χ
= [[0,F],[χM,0]]. (X.23)

C_χ²
= [[χFM,0],[0,χMF]]. (X.24)

A note must state:

C_χ² = χI only under normalized conditions FM = MF = I. (X.25)

---

# X.17.2 Visual treatment

Use:

- inward-curving arrows for corrective;
- branching arrows for critical;
- outward reinforcing loops for self-confirming.

Do not use physical particle imagery.

---

## X.17.3 Caption

**Figure 9.** χ as a protocol- and horizon-indexed feedback signature. The same indicator can carry different implications in corrective, critical, and self-confirming regimes. χ is a proposed diagnostic relation, not a universally observed market constant.

---

# X.18 Figure 10 — Ξ Operating-State Map

## Title

**Loading, Lock-In, and Agitation**

## Status

M + H

## Layout

A three-axis state space:

- ρ — loading;
- γ — lock-in;
- ν — agitation.

Show several labelled regions:

- lightly loaded and liquid;
- heavily loaded but stable;
- loaded and locked;
- agitated but unbound;
- loaded, locked, and highly agitated.

---

## X.18.1 Required note

Ξ is compiled from richer protocol-bound evidence:

Ξ_P
= Compile_P(Σ_P). (X.26)

It is not claimed to be the complete ontology of financial systems.

---

## X.18.2 Intervention inset

Possible interventions:

- Probe;
- Pump;
- Switch;
- Couple.

The figure should distinguish diagnostic state from intervention.

---

# X.18.3 Caption

**Figure 10.** The Ξ operating-state interface. Loading, lock-in, and agitation provide a compact diagnostic view of a declared financial system. The coordinates remain composites whose source channels must be preserved for audit.

---

# X.19 Figure 11 — Complex-Eligibility Ladder

# Title

### When a Real Pair Earns Complex Priority

# Status

M + R

# Layout

A vertical decision ladder.

```
Scalar variable
↓ second meaningful coordinate
Real pair (R,Q)
↓ compatible units or justified metric
Complex candidate Z = R + iQ
↓ stable generator and scaling
Complex-eligible state
↓ phase utility
Phase-bearing model
↓ clock comparison
Internal phase-time candidate
↓ phase-sensitive gate + ledger
Time-bearing phase world
```

At every level, a reduction arrow returns to the previous simpler model.

---

# X.19.1 Key warning

```
Writing R + iQ adds no empirical information by itself.
```

---

# X.19.2 Caption

**Figure 11.** The complex-eligibility ladder. A second coordinate first creates a real pair. Complex notation earns priority only when the pair possesses compatible measurement, stable coupling, meaningful phase, and operational gain over flexible real-variable models. The source phase framework explicitly requires reduction when these tests fail.

---

# X.20 Figure 12 — CAPM Complex Calibration Atom

# Title

**From Baseline Value to Conjugate Exposure**

# Status

E within declared construction

# Layout

A right triangle and corresponding complex plane.

Horizontal axis:

R_t.

Vertical axis:

Q_t.

Hypotenuse:

A_t.

Angle:

θ_t.

---

# X.20.1 Equations

A_t
= CF_t/(1+r_base)^t. (X.27)

R_t
= CF_t/(1+r_CAPM)^t. (X.28)

$A_t^2$
= $R_t^2 + Q_t^2$. (X.29)

Z_t
= R_t + iQ_t
= A_te^{iθ_t}. (X.30)

---

# X.20.2 Required annotation

Q_t is:

- not realized loss;
- not volatility;
- not beta;
- not VaR;
- not an unexplained error bucket.

It is the conjugate coordinate defined by this declared valuation geometry.

---

# X.20.3 Caption

**Figure 12.** The CAPM complex calibration atom. Given the declared baseline value A_t and CAPM-admitted value R_t, the conjugate coordinate Q_t is exactly defined by the Pythagorean relation. The construction is mathematically exact under its definitions, while its empirical interpretation and extension beyond the declared geometry remain separate questions.

---

# X.21 Figure 13 — The Local CAPM Phase Cycle

# Title

**A Local Measurement Generator, Not the Global Market Period**

# Status

E

# Layout

A complex plane with four quarter-turn positions:

R
→ −Q
→ −R
→ Q
→ R. (X.31)

The generator J appears in the centre:

$J[R,Q]^T$
$= [-Q,R]^T$. (X.32)

$J^2 = -I$. (X.33)

$J^4 = I$. (X.34)

---

# X.21.1 Required border warning

```
This four-step phase cycle is local to the R–Q geometry.
It is not the six-period Mark → World architecture.
```

---

# X.21.2 Measurement–movement distinction

Use two overlaid arrows:

- thin arrow — passive measurement rotation;
- thick arrow — actual state movement.

A separate gate symbol should appear after state movement.

---

# X.21.3 Caption

**Figure 13.** The local CAPM quadrature cycle. The quarter-turn generator relates the R and Q coordinates mathematically. A passive change of measurement orientation does not itself create economic loss, commitment, or ledger entry; those require actual state movement and a relevant gate.

---

# X.22 Figure 14 — Four Forms of Time

# Title

**Calendar Time, Phase, Internal Depth, and Ledger Order**

# Status

M + H

# Layout

Four parallel timelines.

## t — calendar duration

Uniform chronological line.

## θ — phase orientation

Circular coordinate.

## $\tau_i$ — accumulated internal phase depth

Unwrapped or cumulative path.

## k — ledger-event order

Discrete committed event steps.

## X.22.1 Main statement

$t \neq \theta \neq \tau_i \neq k$. (X.35)

---

## X.22.2 Required condition

The $\tau_i$ line should be visually marked:

```
Valid only after complex and phase-time eligibility
```

---

## X.22.3 Caption

**Figure 14.** Four distinct forms of temporal order. Calendar duration measures external passage; $\theta$ records current orientation; $\tau_i$ accumulates internal phase traversal; and k orders consequential gates. Phase becomes a clock only if it improves prospective episode ordering beyond simpler alternatives.

---

# X.23 Figure 15 — Transport and Invariance

## Title

**A Claim Becomes Stronger When It Survives Legitimate Reframing**

## Status

M + H

## Layout

A source protocol box P on the left and target protocol box P′ on the right.

The transformation bridge is:

$T_{P \to P'}$. (X.36)

At the target, compare:

Expected transformed claim $\hat{C}_{P'}$

with:

Observed claim $C_{P'}$.

The mismatch enters a residual cloud:

$r_T = C_{P'} - T_{P \to P'}(C_P)$. (X.37)

---

## X.23.1 Example transports

Show small labelled bridges:

- daily → weekly;
- linear → logarithmic;
- price → breadth;
- market → accounting;
- market → legal.

---

## X.23.2 Caption

**Figure 15.** Transport-based objectivity. A source claim is not copied unchanged into another protocol; it is transformed into an expected target form and compared with the target observation. Exact, covariant, partial, local, failed, and non-comparable outcomes must remain distinct. The source filtration and Technical Analysis frameworks treat cross-protocol survival as the route from local pattern to stronger law-like relation.

---

# X.24 Figure 16 — Residual-Bearing Breakout Case

# Title

**From Boundary Crossing to Durable Acceptance**

# Status

M + R

# Layout

A multi-panel chart.

### Panel A — Structure

Range and resistance zone.

### Panel B — Candidate

Intraday or closing cross.

### Panel C — Initial gate

Close, displacement, volume, breadth.

### Panel D — Residual

Weak breadth, retest pending, weekly conflict.

### Panel E — Follow-through

Three-session movement.

### Panel F — Outcome

Acceptance, fakeout, or unresolved state.

---

# X.24.1 Side panel

Show:

```
Highest supported claim:
Partially admitted daily breakout

Not yet supported:
Episode transition
World transition
```

---

# X.24.2 Caption

**Figure 16.** A worked residual-bearing breakout. The crossing creates a candidate; the close-time gate creates partial admission; unresolved breadth, retest, and higher-frame evidence remain in the residual register; later evidence updates the ledger without overwriting the original gate state.

---

# X.25 Figure 17 — Admission–Residual Plane

# Title

**Strong Evidence Can Coexist with High Fragility**

# Status

H + R

# Axes

Horizontal:

Admission strength.

Vertical:

Residual burden.

Four quadrants:

| Quadrant | Interpretation |
|---|---|
| Low admission, low residual | weak or irrelevant candidate |
| Low admission, high residual | unresolved or rejected candidate |
| High admission, low residual | robust admitted Event |
| High admission, high residual | admitted but fragile Event |

---

# X.25.1 Main principle

GateStrength does not determine ResidualBurden uniquely. (X.38)

---

# X.25.2 Caption

**Figure 17.** The admission–residual plane. Positive confirmation and unresolved contradiction should be reported separately. The most informative cases may be strongly admitted Events that remain fragile because higher-frame, liquidity, positioning, or institutional residual persists.

---

# X.26 Figure 18 — Episode Formation

# Title

**An Episode Is an Ordered Grammar of Events**

# Status

T + H

# Layout

Several Event nodes connected through an evolving grammar.

Example sequence:

Breakout
→ Retest
→ Higher Low
→ Continuation
→ Exhaustion
→ Reversal Gate.

Below the nodes, show:

- active grammar;
- residual branches;
- χ sequence;
- completion certificate.

---

## X.26.1 Required distinction

A separate panel should show:

RepeatedEventsWithoutGrammar
≠ Episode. (X.39)

---

## X.26.2 Caption

**Figure 18.** Episode formation. An Episode is not a retrospective name for several nearby Events. It requires an ordered and persistent grammar, a start gate, a completion or transition gate, and visible alternative branches.

---

## X.27 Figure 19 — World-Level Recognition Vector

## Title

**One Condition, Several Authorities, Different Commitment Times**

## Status

M

## Layout

A horizontal sequence of six ledgers:

- market;
- risk;
- accounting;
- contractual;
- legal;
- policy.

Each has its own gate diamond and time marker.

The recognition vector appears:

$\mathbb{G}_t$
= (g_market,g_risk,g_accounting,g_contractual,g_legal,g_policy). (X.40)

---

## X.27.1 Example state

```
Market: admitted
Risk: admitted
```

```
Accounting: provisional
Contractual: not admitted
Legal: not admitted
Policy: not applicable
```

Label:

Fragmented recognition.

---

# X.27.2 Caption

**Figure 19.** Cross-ledger recognition. The same economic condition may be recognized at different times and under different authorities. Market evidence can create a candidate in another ledger, but the destination gate remains autonomous.

---

# X.28 Figure 20 — Self-Referential Market Backreaction

# Title

**When Observation Enters the Market It Observes**

# Status

H

# Layout

A loop:

Market trace
→ Indicator
→ Interpretation
→ Orders and positioning
→ New market trace.

Add two branching outcomes.

### Self-confirming branch

Observation reinforces movement.

### Crowding-inversion branch

Observation concentrates risk and destabilizes movement.

---

# X.28.1 χ annotation

The loop may shift among:

$\chi < 0$;

$\chi \approx 0$;
$\chi > 0$. (X.41)

But the figure must state that $\chi$ transition is an empirical hypothesis.

---

# X.28.2 Caption

**Figure 20.** Market self-reference. Technical signals can become inputs into later market behaviour when observers act upon them. Adoption may reinforce a relation, neutralize it, or produce crowding and inversion. Visibility alone does not establish the direction of backreaction.

---

# X.29 Figure 21 — The Self-Revising Observer

# Title

**Learning Without Erasing Failure**

# Status

M

# Layout

A recursive loop:

Declaration $D_k$
→ Projection
→ Gate
→ Trace + Residual
→ Transport
→ Revision proposal
→ $D_{k+1}$.

An immutable ledger sits beneath the loop.

A crossed-out shortcut shows:

Failure
⇏ Silent relabelling. (X.42)

---

# X.29.1 Admissibility checklist

Revision must be:

- well formed;
- trace preserving;
- residual honest;

- frame accountable;
- budget bounded;
- non-degenerate.

---

## X.29.2 Caption

**Figure 21.** The self-revising market observer. Later evidence may justify a new protocol, but the original declaration, claim, gate, and residual remain in the ledger. Mature revision changes future observation without rewriting past failure.

---

# X.30 Figure 22 — The Validation Ladder

## Title

**Every Stronger Interpretation Requires a Stronger Experiment**

## Status

R

## Layout

A stepped validation ladder.

```
L9 Time-bearing World
L8 Internal phase time
L7 Complex state
L6 Reflexive observer effect
L5 Episode grammar
L4 Transport
L3 Residual governance
L2 Event gate
L1 Typing reliability
L0 Reproducible projection
```

Each step has:

- required experiment;
- primary falsifier;
- retained lower model.

---

## X.30.1 Side arrows

Every step has a downward reduction arrow.

Example:

Phase time fails

→ retain phase state. (X.43)

Complex priority fails
→ retain real pair. (X.44)

Episode grammar fails
→ retain Event sequence. (X.45)

---

# X.30.2 Caption

**Figure 22.** The validation ladder. The framework is not one indivisible claim. Projection, typing, gating, residual, transport, Episode formation, reflexivity, complex geometry, internal time, and World causality require distinct empirical gates.

---

# X.31 Figure 23 — Comparative Rosetta Stone

# Title

**PPMG as a Host Grammar for Existing Market Sciences**

# Status

M

# Layout

A central PPMG host ring containing:

- protocol;
- type;
- gate;
- residual;
- ledger;
- transport;
- revision.

Around it, domain modules:

- Technical Analysis;
- microstructure;
- econometrics;
- hidden-state models;
- event studies;

- survival analysis;
- causal inference;
- accounting;
- law;
- machine learning.

Arrows point both ways.

---

# X.31.1 Required warning

```
Host grammar does not replace domain mechanisms or validation standards.
```

---

# X.31.2 Caption

**Figure 23.** Comparative Rosetta Stone. Existing disciplines supply mechanisms, estimators, and domain-specific authority. PPMG supplies a shared language for declaring the object, identifying closure depth, separating projection from commitment, preserving residual, and governing revision.

---

# X.32 Figure 24 — Runtime Kernel

# Title

### From Raw Question to Governed Market Claim

# Status

M + R

# Layout

A compiler pipeline.

```
Raw Question
→ Suitability Gate
→ Protocol Compilation
→ Evidence Freeze
→ Projection
→ Typing
→ χ / Ξ only if needed
→ Candidate
→ Gate
→ Trace + Residual
→ Transport
→ Promotion or Reduction
→ Revision
```

---

# X.32.1 Required side channels

### Audit channel

Runs beside the main pipeline.

### Authority channel

Constrains gates and actions.

### Immutable ledger

Runs beneath the pipeline.

---

# X.32.2 Caption

**Figure 24.** The Periodic Market Runtime Kernel. Analysis is treated as semantic compilation rather than as unconstrained narrative generation. Every stronger state requires its own gate, and every failed advanced layer returns a simpler retained model.

---

# X.33 Figure 25 — Research Infrastructure Stack

# Title

**Ontology, Runtime, Benchmark, and Research Ledger**

# Status

R

# Layout

Four horizontal layers.

### Layer 1 — Ontology

Defines valid objects and relations.

### Layer 2 — Runtime

Executes state transitions.

### Layer 3 — Benchmark

Measures implementation quality.

### Layer 4 — Research ledger

Preserves studies, failures, and revisions.

The relationship is:

Ontology

- Runtime
- Benchmark
- ResearchLedger
  = OperationalResearchWorld. (X.46)

---

# X.33.1 Caption

**Figure 25.** The research infrastructure stack. The ontology specifies what may be represented; the runtime specifies how states change; the benchmark tests performance; and the research ledger records which theoretical branches survive, reduce, or close.

---

# X.34 Figure 26 — The Entire Theory on One Page

# Title

**The Periodic Grammar of Technical Analysis**

# Subtitle

**Load, Motion, Constraint, and Commitment Across Recursive Market Worlds**

# Status

T + M + R

# Layout

A landscape 16:9 synthesis plate.

## Left — Declared market field

$\Sigma_0$
→ Protocol P
→ Projection.

## Centre — Periodic table

Six periods × four functions.

## Above centre — Proto-Eight topology

Eight actuation roles.

## Below centre — governance rails

Residual, transport, ledger.

### Right upper — advanced state ladder

$\chi \rightarrow \Xi \rightarrow$ optional $Z \rightarrow$ optional $\theta \rightarrow$ optional $\tau_i$.

### Right lower — world formation

Gate
→ Trace
→ Ledger
→ changed future admissibility.

---

# X.34.1 Central statement

```
A market claim becomes mature when its object is declared,
its projection is typed,
its transition is gated,
its residual remains visible,
its trace enters a ledger,
and its broader validity survives transport.
```

---

# X.34.2 Lower warning bar

```
Not every market object is complex.
Not every phase is a clock.
Not every Event is an Episode.
Not every Episode is a World.
```

---

# X.34.3 Caption

**Figure 26.** Full synthesis of the Periodic Grammar. The basic architecture begins with declared projection, six closure periods, and four recurring functions. Proto-Eight supplies an optional actuation crosswalk; residual, transport, and ledger govern every cell; $\chi$ and $\Xi$ provide optional regime interfaces; complex phase and internal time appear only after their independent eligibility gates.

---

# X.35 Table-Figure Pairing Rules

Some content should remain a table rather than become an illustration.

# X.35.1 Use a table when

- exact comparison matters;
- many method assignments must be shown;
- claim statuses must remain precise;
- readers need to look up definitions.

## X.35.2 Use a diagram when

- process order matters;
- gates and branching must be shown;
- recursion or transport is central;
- hierarchy is more important than exact enumeration.

## X.35.3 Use both when

A diagram supplies the architecture and a table supplies the implementation details.

Example:

- Figure 1 — master periodic table;
- accompanying table — full cell inventory and method cards.

---

# X.36 Monochrome Standard

Every figure must remain interpretable when printed without colour.

Use:

- solid fill;
- diagonal hatch;
- dot pattern;
- double border;
- thick outline;
- dashed outline.

A suggested mapping is:

| Object | Monochrome treatment |
|---|---|
| Load | horizontal hatch |
| Motion | diagonal hatch |
| Constraint | double border |
| Commitment | solid dark edge |
| Residual | irregular dotted outline |
| Transport | dashed bridge |
| Ledger | ruled horizontal band |

---

# X.37 Accessibility Requirements

## X.37.1 Colour blindness

Avoid red–green-only distinctions.

Use symbols and labels.

---

## X.37.2 Font size

Publication figures should remain legible when reduced to column width.

---

## X.37.3 Alternative text

Every figure should include alt text describing:

- objects;
- direction;
- claim status;
- crucial warnings.

---

## X.37.4 Cognitive accessibility

Avoid more than:

- seven main labelled objects in one local cluster;
- three arrow types without legend;
- two nested cycles in one panel.

Complex synthesis should use multiple panels.

---

# X.38 Caption Standard

Every caption should contain five components.

1. **What is shown.**
2. **How to read the arrows.**
3. **What status the relation has.**
4. **What is not being claimed.**
5. **Where the figure applies.**

Template:

```
Figure X. [Object]. Solid arrows indicate [relation];
```

```
dashed arrows indicate [hypothesis]. The figure is a
[definition / taxonomy / mapping / hypothesis]. It does
not claim [prohibited interpretation]. Scope: [protocol/domain].
```

---

# X.39 Formula Standard Inside Figures

Formulas should remain Blogger-ready and MathJax-free.

Use Unicode and single-line expressions.

Example:

Z = R + iQ = Ae^{iθ}. (X.47)

Avoid long derivations inside the figure.

A figure should show:

- one main relation;
- no more than three supporting equations;
- a nearby definition of every symbol.

Full derivations remain in the article body or appendix.

---

# X.40 Source-Status Footers

Source-derived figures should include a small footer.

Examples:

```
Source basis: Proto-Eight engineering role grammar.
Present article: financial crosswalk and runtime extension.

Source basis: CAPM complex valuation construction.
Present article: placement within the wider periodic architecture.

Source basis: declaration–gate–trace–residual framework.
Present article: six-period Technical Analysis implementation.
```

This separates inherited architecture from new synthesis.

---

# X.41 Visual Misrepresentation Checklist

Before publication, verify that no figure implies:

1. the six periods are six timeframes;
2. the four families are four physical substances;
3. the eight roles form a universal temporal order;
4. residual equals Q;

5. Q equals realized loss;
6. phase rotation itself creates P&L;
7. a chart Event creates legal commitment;
8. transport means identical numerical appearance;
9. World means only a large-scale market pattern;
10. complex notation proves a hidden quantum market;
11. revision may overwrite failed claims;
12. the periodic table has already been empirically validated.

A figure failing any item should be revised.

---

# X.42 Recommended Main-Text Figure Set

The main article should not contain all twenty-six figures.

A concise core set is:

1. **Master Proto-Periodic Table**;
2. **Four-Family Functional Cycle**;
3. **Six-Period Closure Ladder**;
4. **Gate–Trace–Residual–Ledger Architecture**;
5. **Proto-Eight Actuation Topology**;
6. **Complex-Eligibility Ladder**;
7. **CAPM Complex Calibration Atom**;
8. **Transport and Invariance**;
9. **Validation Ladder**;
10. **Entire Theory on One Page**.

The remaining figures may appear in appendices or supplementary materials.

---

# X.43 Recommended Three-Infographic Set

For public communication, the theory can be presented through three 16:9 infographics.

# Infographic 1

**The Periodic Grammar of Technical Analysis**

Focus:

- 6 × 4 table;
- governance rails;
- claim hierarchy.

---

# Infographic 2

**From Candidate Signal to Consequential Market History**

Focus:

- gate;
- trace;
- residual;
- ledger;
- revision.

---

# Infographic 3

**When Complex Phase Is Earned**

Focus:

- real pair;
- complex eligibility;
- CAPM calibration;
- phase versus time;
- reduction rules.

This separation prevents the advanced complex framework from obscuring the basic governance architecture.

---

# X.44 Recommended Figure Order in the Article

A coherent publication order is:

## Part I — Basic grammar

Figures 2, 3, and 1.

## Part II — Governance

Figures 6, 7, and 15.

## Part III — Methods and Proto-Eight

Figures 4, 5, and 8.

## Part IV — Regime and advanced state

Figures 9, 10, 11, 12, 13, and 14.

## Part V — Empirical programme

Figures 16, 17, 18, 20, and 22.

## Conclusion

Figure 26.

This order mirrors the evidential progression:

Protocol
→ Closure
→ Gate
→ Residual
→ Transport
→ Advanced state
→ Validation. (X.48)

---

## X.45 Visual Reduction Rule

A visual should show only the strongest layer supported in that context.

Examples:

## Ordinary moving-average discussion

Show:

Structure × Load × Memory.

Do not show:

complex plane or phase clock.

## Breakout discussion

Show:

Candidate → Gate → Residual → Event.

Do not show:

World formation unless an institutional ledger is relevant.

## CAPM geometry discussion

Show:

R, Q, A, θ.

Do not imply:

general Technical Analysis Q has been validated.

## Legal recognition discussion

Show:

authority and ledger.

Do not imply:

price chart determines the legal gate.

The rule is:

VisualComplexity ≤ EvidentialComplexity. (X.49)

---

# X.46 Publication Plate Registry

Each figure should receive a registry entry.

```
Figure ID:
Title:
Article section:
Object types:
Relation types:
Status:
Scope:
Source basis:
Present-article contribution:
Required warning:
Colour version:
Monochrome version:
Alt text:
Revision history:
```

This turns figure production into a governed publication process rather than decorative illustration.

---

# X.47 Example Registry Entry

```
Figure ID:
PPMG-F13

Title:
The Local CAPM Phase Cycle

Article section:
Complex valuation and measurement geometry
```

```
Object types:
R coordinate, Q coordinate, generator J

Relation types:
Exact quarter-turn transformation

Status:
Exact construction under declared definitions

Scope:
CAPM complex valuation atom

Source basis:
From Discounted Value to Conjugate Risk

Present-article contribution:
Separates local four-step phase cycle from global six-period closure

Required warning:
Measurement rotation does not itself create economic loss or ledger commitment
```

---

# X.48 Visual Research Hypothesis

The figure system itself can be tested.

A structured visual grammar should reduce reader errors involving:

- Structure versus Event;
- Event versus Episode;
- residual versus Q;
- market versus institutional recognition;
- phase versus time;
- local cycle versus global period.

A reader study may compare:

OrdinaryIllustration
versus
TypedVisualGrammar. (X.50)

Outcomes may include:

- classification accuracy;
- recall;
- overpromotion;
- interpretation confidence;
- time to answer.

Thus the visual system is not exempt from empirical evaluation.

---

## X.49 Appendix X Conclusion

The Periodic Grammar cannot be communicated safely through one attractive wheel.

Its architecture contains several different kinds of order:

- closure depth;
- analytical function;
- actuation topology;
- governance;
- regime;
- complex phase;
- ledger time.

These orders must remain visually distinct.

The visual grammar therefore obeys the following rules:

Shape identifies object type. (X.51)

Arrow identifies relation type. (X.52)

Status marker identifies evidential strength. (X.53)

Scope identifies where the relation applies. (X.54)

Gate output always separates trace and residual. (X.55)

Proto-Eight topology is not automatically chronology. (X.56)

CAPM quadrature is not the six-period market sequence. (X.57)

Complex phase is visually downstream of eligibility. (X.58)

World formation requires authority and ledger. (X.59)

Revision returns through preserved history rather than erasing it. (X.60)

The final visual principle is:

> A good scientific figure should make an unsupported interpretation harder to see, not merely make the proposed theory easier to remember.

# Appendix Y — Editorial Source Map and Claim-Provenance Architecture

## Y.1 Purpose

The article combines several layers of material:

- source-derived mathematical constructions;

- source-derived conceptual architectures;
- present-article taxonomies;
- financial mappings;
- engineering proposals;
- empirical hypotheses;
- historical and philosophical analogies.

These layers must not be presented as though they possess the same provenance or evidential status.

For example:

- the CAPM identity $A^2 = R^2 + Q^2$ is exact under its declared definitions;
- the six-period table is a proposed synthesis;
- the Proto-Eight crosswalk is an operational interpretation;
- the claim that phase may become an internal clock is a research hypothesis;
- a connection to 先天八卦 is neither identical to financial mechanism nor sufficient empirical proof.

This appendix defines the article's editorial provenance system.

Its objective is:

> Every major claim should disclose where it came from, what kind of transformation produced it, what evidence supports it, and how far it may legitimately be extended.

The central editorial law is:

InheritedConcept

- DomainMapping
- NewSynthesis
  ≠ OneUndifferentiatedClaim. (Y.1)

---

# Y.2 The Source-Development Sequence

The article inherits its deepest architecture from the following developmental sequence:

1. Proto-Eight Dynamics;
2. Self-Referential Observers in Quantum Dynamics;
3. The One Assumption of SMFT;
4. From One Assumption to One Operator;
5. From One Operator to One Filtration;

6. From One Filtration to One Declaration;
7. From One Declaration to One Self-Revising Fractal;
8. From Recursive Depth to Time-Bearing Worlds;
9. 成界之學;
10. 以「成界之學」攻剋「先天八卦」本義.

The present article then combines this sequence with:

- the Technical Analysis source framework;
- complex financial geometry;
- CAPM phase sensitivity;
- Gauge Grammar;
- PORE;
- runtime-kernel engineering.

The editorial derivation is:

FoundationalGrammar
→ FinancialProjection
→ TechnicalAnalysisTranslation
→ ProtoPeriodicSynthesis
→ EmpiricalResearchProgramme. (Y.2)

---

# Y.3 Four Provenance Classes

Every substantial article claim should belong to one primary provenance class.

# Y.3.1 Class S — Source-derived

The claim is substantially inherited from a cited source.

Examples:

- declaration precedes projection;
- gate produces trace and residual;
- self-revising observers preserve ledger history;
- CAPM admits an exact R–Q construction.

A Class S claim should preserve the source's terminology and limitations.

---

# Y.3.2 Class D — Derived

The claim follows by explicit transformation from a source-defined object.

Examples:

- assigning a moving average to Structure × Load × Memory;
- interpreting a breakout close as Event × Commitment × Gate;
- treating a market-to-accounting relation as cross-ledger transport.

A derived claim requires the transformation to be stated.

---

## Y.3.3 Class N — New synthesis

The claim is introduced by the present article.

Examples:

- the 6 × 4 proto-periodic table;
- the three governance rails;
- the combined $P_6 \times G_4 \times A_8$ architecture;
- the validation ladder;
- the PPMG conformance profiles.

Such claims must not be attributed to the source papers unless they actually appear there.

---

## Y.3.4 Class H — Hypothesis

The claim requires empirical testing.

Examples:

- residual burden predicts fakeout;
- transport survival predicts persistence;
- $\chi$ improves indicator interpretation;
- a Technical Analysis R–Q pair supports complex phase;
- internal phase time outperforms ordinary clocks.

A hypothesis should use prospective language.

---

## Y.4 Provenance Tag

A compact provenance tag is:

$$\Pi_{\text{claim}} = (\text{Origin}, \text{Transformation}, \text{Status}, \text{Scope}). \quad (Y.3)$$

where:

Origin ∈ {S,D,N,H};
Transformation = how the claim was produced;
Status = definition, exact result, mapping, hypothesis, or research proposal;
Scope = where the claim applies.

Example:

```
Claim:
A moving average is a Structure-level Load operator.

Origin:
D

Transformation:
Technical indicator → six-period/four-family typing

Status:
Operational mapping

Scope:
Moving averages under declared price protocols
```

---

# Y.5 Claim-Status Vocabulary

The article should use the following status vocabulary consistently.

| Status | Meaning |
|---|---|
| **Definition** | true by declared meaning |
| **Exact construction** | follows mathematically under stated assumptions |
| **Source proposition** | explicitly proposed in a source |
| **Derived mapping** | obtained by declared translation |
| **Taxonomic proposal** | classification introduced by this article |
| **Engineering rule** | recommended operating discipline |
| **Empirical hypothesis** | requires data |
| **Research programme** | proposed sequence of studies |
| **Analogy** | limited structural comparison |
| **Historical interpretation** | interpretive reading of prior tradition |

A reader should never have to infer whether a sentence is:

- exact;
- proposed;
- metaphorical;
- empirically supported.

---

# Y.6 The Main Source Families

The article's source base can be organized into seven families.

# Y.6.1 Proto-Eight and P8D

Provides:

- eight operational roles;
- four paired dyads;
- growth and flow interpretation;
- engineering failure modes;
- testability orientation.

The engineering source identifies:

- Gradient and Gate;
- Boundary and Exchange;
- Trigger and Guidance;
- Memory and Focus

as paired operational roles.

The P8D small-model source further frames growth through capacity, demand, enablement, retention, bottlenecks, and feedback rather than presenting the framework as fundamental physics.

---

# Y.6.2 Recursive disclosure and declaration

Provides:

- undeclared field;
- operator;
- filtration;
- declaration;
- gate;
- trace;
- residual;
- ledger;
- admissible revision.

成界之學 presents the sequence from undeclared possibility through declaration, projection, gate, trace, residual, ledger, invariance, and time-bearing world.

---

## Y.6.3 Self-referential observer framework

Provides:

- internal observation;
- bounded projection;
- cross-observer agreement;
- observer-relative but transportable objectivity;
- self-revising observer architecture.

This source family supports the view that a market observer's projection is neither the whole field nor arbitrary invention.

---

## Y.6.4 Technical Analysis framework

Provides:

- Technical Analysis as protocol-bound projection;
- method-to-characteristic mapping;
- $\chi$;
- structural mass;
- residual pressure;
- gate discipline;
- transport and invariance;
- failed-trace preservation.

The source's core operator view treats Technical Analysis as a partial projection of market self-reference rather than a direct representation of the entire market.

---

## Y.6.5 CAPM complex geometry

Provides:

- A;
- R;
- Q;
- $\theta$;
- Z;
- the exact Pythagorean identity;

- the quarter-turn generator;
- sensitivity equivalence;
- measurement–movement–gate–ledger distinction.

The exact local construction begins from declared baseline and CAPM valuations and defines Q through $A^2 = R^2 + Q^2$.

---

## Y.6.6 Phase and internal time

Provides:

- complex eligibility;
- phase-order conditions;
- distinction among calendar, phase, internal, and ledger times;
- model-reduction ladder;
- phase-time falsification rules.

The source explicitly requires complex representations to be reduced when a real-pair model performs equally well or when phase depends on arbitrary scaling.

---

## Y.6.7 Gauge, PORE, and runtime engineering

Provides:

- protocol-relative effective states;
- $\rho$, $\gamma$, and $\nu$;
- intervention typing;
- suitability gates;
- runtime compilation;
- audit;
- versioning;
- bounded authority.

These sources support the engineering and implementation appendices rather than the exact financial geometry.

---

# Y.7 Source-to-Article Transformation Map

| Source concept | Present article transformation |
|---|---|
| Proto-Eight roles | market actuation crosswalk |

| Source concept | Present article transformation |
|---|---|
| Operator | technical indicator or market measurement |
| Filtration | ordered marks, windows, Events, and ledgers |
| Declaration | full Technical Analysis protocol |
| Gate | Event admission and promotion rule |
| Trace | committed analytical or institutional record |
| Residual | typed unresolved evidence |
| Recursive depth | six closure periods |
| Cross-frame invariance | transport survival |
| Self-revising fractal | versioned market-observer runtime |
| CAPM R–Q geometry | local complex calibration atom |
| Phase order | candidate internal market time |
| Gauge state | Ξ financial operating-state interface |
| World formation | authority + ledger + changed admissibility |

The transformation table identifies inheritance.

It does not prove that every financial mapping is correct.

---

# Y.8 The Article's Primary New Contributions

The following structures are primarily new syntheses of the present article.

## Y.8.1 The six closure periods

Mark
→ Window
→ Structure
→ Event
→ Episode
→ World. (Y.4)

Although informed by recursive-depth and world-formation sources, this exact six-period financial ladder is the present article's construction.

---

## Y.8.2 The four functional families

Load
→ Motion under Constraint
→ Commitment. (Y.5)

The source Technical Analysis framework contains many of these functions, but the full four-column periodic organization is introduced here.

---

## Y.8.3 The three governance rails

Residual;
Transport;

Ledger. (Y.6)

These concepts are source-derived individually.

Their treatment as cross-table governance rails is the present article's synthesis.

---

# Y.8.4 The integrated type architecture

$P_6 \times G_4 \times A_8 \times R_3$. (Y.7)

where:

$P_6$ = six periods;
$G_4$ = four functions;
$A_8$ = eight actuation roles;
$R_3$ = three governance rails.

This exact integrated typing structure should be attributed to this article.

---

# Y.8.5 The validation ladder

Projection
→ Typing
→ Gate
→ Residual
→ Transport
→ Episode
→ Reflexivity
→ Complex State
→ Phase Time
→ World. (Y.8)

This is a proposed empirical research architecture.

---

# Y.8.6 PPMG

The Proto-Periodic Market Grammar standard, ontology, runtime, benchmark, and implementation stack are engineering proposals developed here.

---

# Y.9 Exact Mathematical Claims

The article should reserve the term "exact" for relations that follow from declared definitions or formal algebra.

# Y.9.1 CAPM norm

$A^2 = R^2 + Q^2$. (Y.9)

Exact after Q is defined by:

$Q = \sqrt{(A^2 - R^2)}$. (Y.10)

---

# Y.9.2 Complex representation

$Z = R + iQ = Ae^{i\theta}$. (Y.11)

Exact when A and θ are defined in the usual polar form.

---

# Y.9.3 Generator

$J[R,Q]^T = [-Q,R]^T$. (Y.12)

$J^2 = -I$. (Y.13)

$J^4 = I$. (Y.14)

Exact for the canonical quarter-turn matrix.

---

# Y.9.4 Finite rotation

$R_{new} = R \cos \Delta\theta - Q \sin \Delta\theta$. (Y.15)

Exact under the declared Euclidean rotation.

---

# Y.9.5 Approximation

$\Delta R \approx -Q\Delta\theta$. (Y.16)

This is a small-angle approximation.

It must not be presented as exact for finite Δθ.

---

# Y.9.6 χ operator square

$C_{\chi^2} = [[\chi FM,0],[0,\chi MF]]$. (Y.17)

This is the exact unrestricted operator product.

The simplified expression:

$C_{\chi^2} = \chi I$ (Y.18)

holds only under the normalized conditions:

$FM = I$;
$MF = I$. (Y.19)

---

# Y.10 Formal Definitions Versus Empirical Claims

The following distinction must be preserved.

# Formal definition

Residual burden is defined as:

RB
= OpenApplicableResiduals/ApplicableResiduals. (Y.20)

# Empirical claim

Higher RB predicts failure.

The first is true by definition.

The second requires data.

Similarly:

# Formal definition

Transport residual is:

r_T
= C_{P′} – T_{P→P′}(C_P). (Y.21)

# Empirical claim

Lower transport residual predicts persistence.

Only the second requires validation.

---

# Y.11 Editorial Rule for Equations

Every equation should be classified as one of:

1. definition;
2. identity;
3. approximation;
4. proposed index;
5. dynamical model;
6. empirical regression;
7. conceptual schematic.

Recommended notation:

```
Definition:
:=

Exact equality:
=

Approximation:
≈

Candidate relation:
~ or “proposed”

Implication under conditions:
⇒

Does not imply:
⇏
```

The article should not use “=” when the intended meaning is:

- resembles;
- may influence;
- serves as a proxy;
- is mapped to.

---

# Y.12 Example of Correct Equation Labelling

Weak:

TechnicalAnalysis = MarketSelfReference. (Y.22)

This suggests identity.

Better:

TechnicalAnalysis_P
= Projection_P(MarketSelfReference). (Y.23)

This indicates a protocol-bound projection.

Weak:

Residual = Q. (Y.24)

This is not supported.

Correct:

Residual and Q are distinct objects. (Y.25)

A residual may motivate a Q candidate only after independent measurement and conjugacy tests.

---

# Y.13 Source-Preservation Rule

When a source introduces a term with a specific meaning, the article should not silently widen it.

Examples:

## Q

In the CAPM source, Q is defined geometrically.

It should not later become a generic symbol for:

- uncertainty;
- residual;
- hidden risk;
- volatility;
- failure.

## Gate

A gate is a commitment rule.

It should not become a synonym for:

- confidence;
- prediction;
- boundary;
- indicator threshold

unless the threshold genuinely performs admission.

## World

A World is not simply a large market pattern.

It requires:

- boundary;
- observer or authority;
- persistent ledger;
- changed future admissibility.

---

# Y.14 Terminology Conflict Register

The article uses several symbols that can conflict across sources.

## Y.14.1 τ

Possible meanings include:

- internal phase time;
- recovery time;
- switching time;
- generic time parameter.

Editorial rule:

Use:

$\tau_i$ = internal phase time. (Y.26)

τ_rec = recovery time. (Y.27)

τ_sw = switching time. (Y.28)

Avoid an unqualified τ where ambiguity exists.

---

## Y.14.2 R

Possible meanings include:

- realized value;
- admitted value;
- real coordinate;
- return;
- resistance.

Editorial rule:

Use subscripts or words:

R_CAPM;
R_state;
r_return;
B_resistance. (Y.29)

---

## Y.14.3 Q

Use Q only for a declared conjugate coordinate.

Use:

r;
ε;

$\mathscr{R}$

for residual or error.

---

## Y.14.4 P

Possible meanings include:

- protocol;
- price;
- probability;
- period.

Editorial rule:

Use:

P_protocol or simply P when protocol is obvious;
p_t for price;
Pr for probability;
$P_0,\ldots,P_5$ for periods only in explicitly labelled contexts.

---

# Y.15 Claim-Scope Rule

Every major claim should disclose at least one scope limiter.

Examples:

- under the declared CAPM geometry;
- for daily equity breakouts;
- within the tested protocol family;
- as an operational crosswalk;
- as a proposed taxonomy;
- under normalized coupling;
- where an authoritative ledger exists.

A scope-free sentence such as:

“Phase is time” (Y.30)

should be replaced by:

“Under an eligible phase-bearing model, accumulated phase may function as a secondary internal time coordinate if it improves prospective episode ordering.” (Y.31)

---

# Y.16 The Analogy Boundary

The article contains analogies to:

- periodic tables;
- quantum measurement;
- complex rotation;
- gauge structure;
- field theory;
- 先天八卦.

Each analogy should answer four questions.

1. What structure is being compared?
2. Which relation is preserved?
3. Which features are not transferred?
4. What empirical test would justify a stronger claim?

The analogy record is:

Analogy
= PreservedRelation

- ExcludedIdentity
- TestableExtension. (Y.32)

---

# Y.17 Periodic-Table Analogy

# Preserved relation

A finite set of recurring functional roles appears across levels.

# Not transferred

- chemical substance identity;
- atomic number;
- electron shells;
- physical periodic law;
- element conservation.

# Present claim

The table is proto-periodic because:

- roles recur;
- cells are systematically comparable;
- missing cells suggest instrument opportunities.

## Required test

The classification must improve:

- method typing;
- failure diagnosis;
- instrument design.

---

# Y.18 Quantum Analogy

## Preserved relation

A bounded observation may disclose one aspect of a richer state, while measurement, state movement, commitment, and ledger remain distinct.

## Not transferred

- physical Planck constant;
- particle ontology;
- wavefunction identity;
- literal quantum superposition in market prices;
- physical nonlocality.

## Exact local exception

The CAPM complex construction has genuine mathematical quadrature under its definitions.

That does not make the market physically quantum.

---

# Y.19 Gauge Analogy

## Preserved relation

Market objects depend on:

- protocol;

- frame;
- admissible transformation;
- intervention.

## Not transferred automatically

- physical gauge fields;
- Yang–Mills dynamics;
- fundamental-force identity.

The Gauge Grammar source itself frames these as protocol-bound structural roles rather than substance identity.

---

# Y.20 先天八卦 Interpretation Boundary

The article uses Proto-Eight as an engineering role grammar.

It should distinguish three claims.

## Historical claim

What classical texts originally meant.

## Structural interpretation

How an eight-role topology can be read through 成界之學.

## Engineering reuse

How those roles can support modern system analysis.

These are different projects.

The source critique argues that the BaGua topology may be necessary within its own interpretive framework but is not sufficient by itself to generate an ordered, prosperous recursive world.

This article therefore does not claim:

- that historical authors formulated Technical Analysis;
- that trigrams encode modern finance equations;
- that a classical diagram empirically proves PPMG.

---

# Y.21 Historical Claim Discipline

Historical claims should use language such as:

- may be interpreted as;
- the present framework reads;
- a modern engineering reconstruction;
- a structural correspondence;
- not a claim of original historical authorship.

Avoid:

- “The ancients discovered market phase geometry”;
- “先天八卦 proves the complex plane”;
- “Technical Analysis is secretly encoded in the trigrams.”

Such statements exceed the source support.

---

## Y.22 Source Agreement and Disagreement

The source corpus is developmental rather than perfectly uniform.

Later texts may:

- refine earlier terminology;
- narrow strong claims;
- add formal conditions;
- change notation;
- distinguish analogy from ontology more carefully.

The article should not silently pretend that every source says exactly the same thing.

When tension exists, use:

1. the more precise formal version;
2. the narrower defensible interpretation;
3. an explicit note on developmental change.

---

## Y.23 Example: Strong Versus Modest Proto-Eight Claims

A rhetorically strong source may describe an eight-role matrix as uniquely complete.

A more empirically cautious engineering interpretation is:

> The eight roles form a candidate minimal actuation grammar whose necessity, sufficiency, and paired interactions require testing.

The article should use the cautious version unless it is directly discussing the stronger source claim as a claim.

This preserves source fidelity without adopting unsupported certainty.

---

# Y.24 Example: Q Across Financial Contexts

The CAPM source defines Q exactly.

The Technical Analysis source uses Q more broadly as a candidate retained pressure coordinate.

The article should state:

- CAPM Q is formally defined;
- Technical Analysis Q is not yet unified;
- different Technical Analysis methods may produce different Q candidates;
- no universal Q should be assumed.

Thus:

Q_CAPM
≠ Q_breadth
≠ Q_positioning
≠ Residual. (Y.33)

A future unification would require a transport or derivation map among them.

---

# Y.25 Example: Phase Across Contexts

The article uses phase in at least three senses.

# Signal-processing phase

Produced by a transform such as a Hilbert analytic signal.

# Geometric phase

Defined by a valid R–Q plane.

# Operational phase

A coordinate that improves Event or Episode analysis.

These levels should not be conflated.

SignalPhase
⇒ GeometricPhase only under additional conditions. (Y.34)

GeometricPhase

⇒ OperationalPhase only after utility testing. (Y.35)

OperationalPhase
⇒ InternalTime only after clock comparison. (Y.36)

---

## Y.26 Citation Architecture

Citations should be placed at the level where inheritance occurs.

## Cite the source when:

- reproducing its definition;
- using its equation;
- summarizing its argument;
- adopting its terminology;
- presenting a source-specific caution.

## Cite the present article's prior section when:

- using a newly introduced table;
- applying a previously defined PPMG object;
- referring to the article's own empirical design.

## Do not cite a source for:

- a claim it does not make;
- a stronger implication inferred later;
- a new 6 × 4 table presented as though source-derived.

---

## Y.27 Citation Density

Not every sentence requires a citation.

Citations are most important for:

- exact borrowed definitions;
- source-specific claims;
- historical interpretations;
- mathematical constructions;
- explicit source comparisons.

A paragraph of present-article synthesis may instead state:

> The following architecture is proposed here.

This is clearer than attaching several source citations that could imply false attribution.

---

# Y.28 Direct-Quotation Policy

Direct quotation should be rare.

Prefer:

- precise paraphrase;
- equation citation;
- terminology preservation.

Direct quotation is justified when:

- a source's exact wording is conceptually important;
- a historical term is under analysis;
- the article compares formulations.

The quotation should remain short and contextually faithful.

---

# Y.29 Formula-Provenance Table

| Formula | Provenance | Status |
|---|---|---|
| $A^2 = R^2 + Q^2$ | CAPM complex source | exact construction |
| $Z = R + iQ$ | standard complex representation used by source | exact notation |
| $J^2 = -I$ | canonical quarter-turn operator | exact |
| $\Delta R \approx -Q\Delta\theta$ | CAPM local sensitivity | approximation |
| $C_\chi = [[o,F],[\chi M,o]]$ | TA source | proposed operator |
| $C_{\chi^2} = [[\chi FM,o],[o,\chi MF]]$ | present clarification | exact multiplication |
| $\Xi = (\rho,\gamma,\nu)$ | Gauge/PORE source adapted | protocol-bound interface |
| Load → Motion under Constraint → Commitment | present synthesis | taxonomic law candidate |
| Mark → Window → Structure → Event → Episode → World | present synthesis | closure taxonomy |
| ClaimLevel ≤ ClosureLevel | present governance rule | normative |
| $\tau_i$ = Unwrap $\theta$ | phase source | candidate internal-time definition |
| RB = open/applicable residuals | present pilot | operational index |

---

# Y.30 Section-Provenance Map

## Parts I–III

Primary basis:

- Technical Analysis source;
- protocol-first Gauge Grammar;
- declaration framework.

Present contribution:

- four-family organization;
- governance rails.

---

## Part IV

Primary basis:

- recursive depth;
- filtration;
- ledger;
- time-bearing world.

Present contribution:

- six-period financial ladder.

---

## Parts V–VI

Primary basis:

- Technical Analysis method diagnostics.

Present contribution:

- method cards;
- molecules;
- periodic homology;
- empty cells.

---

## Part VII

Primary basis:

- $\chi$ source;

- Gauge/PORE;
- CAPM complex geometry;
- phase-time source.

Present contribution:

- integrated eligibility hierarchy.

---

## Part VIII

Primary basis:

- residual;
- transport;
- falsification;
- runtime engineering.

Present contribution:

- research contracts and missing-instrument programme.

---

## Appendices A–K

Primary function:

formalization, method cards, protocols, empirical workflows.

---

## Appendix L

Primary basis:

Proto-Eight engineering sources.

Present contribution:

market actuation crosswalk.

---

## Appendices M–W

Primary function:

instrument design, standards, validation, implementation, and comparative integration.

These are largely present-article engineering proposals.

---

# Y.31 Figure-Provenance Map

| Figure family | Source basis | Present contribution |
|---|---|---|
| Master periodic table | recursive-depth and TA sources | 6 × 4 table |
| Proto-Eight topology | Proto-Eight sources | financial crosswalk |
| Gate–residual–ledger | declaration sources | Technical Analysis implementation |
| CAPM complex plane | CAPM source | placement within PPMG |
| χ geometry | TA source | visual regime typology |
| Ξ state map | Gauge/PORE | financial diagnostic use |
| Validation ladder | multiple source cautions | integrated empirical sequence |
| Runtime kernel | runtime-kernel source | market implementation |
| Cross-ledger vector | world-formation and finance synthesis | multi-authority model |

---

# Y.32 Table-Provenance Labels

Tables should use one of four footer labels.

```
Source-derived table

Adapted from source architecture

Present article synthesis

Proposed empirical instrument
```

Example:

The 6 × 4 table footer should state:

```
Present article synthesis informed by recursive disclosure,
Technical Analysis operator diagnosis, and Proto-Eight actuation roles.
```

---

# Y.33 Claim Ceiling for the Whole Article

The article's overall conclusion must not exceed its current evidence.

The strongest presently defensible claim is:

> Technical Analysis can be reconstructed as a protocol-bound observational system in which heterogeneous methods measure recurring functions at different closure depths, and in which explicit gates, residual registers, transport tests, and versioned ledgers offer a coherent programme for improving analytical discipline.

The article does not yet establish that:

- the six-period structure is uniquely correct;
- the four functions are empirically minimal;
- Proto-Eight roles are necessary and sufficient;
- a universal Technical Analysis Q exists;

- market phase provides a validated internal clock;
- the framework predicts returns better than mature alternatives.

---

# Y.34 Abstract-Writing Rule

The abstract should contain four layers.

## Problem

Technical Analysis lacks a unified typing and governance architecture.

## Proposal

A 6 × 4 proto-periodic grammar with actuation roles and governance rails.

## Demonstration

Method classification, CAPM complex calibration, runtime, benchmark, and pilot design.

## Limitation

The framework is a research architecture, not a validated universal law or trading system.

The abstract should not imply completed empirical validation.

---

# Y.35 Introduction-Writing Rule

The introduction should avoid opening with:

- quantum markets;
- ancient cosmology;
- complex phase;
- a claim of universal unification.

It should begin from an ordinary technical problem:

> Why do indicators that look different often fail together, while a simple closing rule sometimes matters more than several sophisticated signals?

Then introduce:

1. projection;
2. function;
3. closure;
4. gate;

5. residual;
6. transport.

Advanced geometry should appear only after the basic architecture is established.

---

# Y.36 Conclusion-Writing Rule

The conclusion should descend from strongest future possibility to present evidence.

A suitable structure is:

1. What the framework now clarifies;
2. What remains proposed;
3. What can be tested immediately;
4. Which advanced branches may later emerge;
5. Which reductions remain acceptable.

The conclusion should end with methodological discipline rather than metaphysical certainty.

---

# Y.37 Title Discipline

The title:

**The Periodic Grammar of Technical Analysis**

is intentionally stronger than:

“A classification of indicators,”

but weaker than:

“The Periodic Law of Markets.”

The word “grammar” signals:

- rule-governed composition;
- recurring roles;
- typed transformations;
- possible generativity.

The word “periodic” signals recurrence across closure levels.

It does not claim a physical periodic law comparable to chemistry.

---

# Y.38 Subtitle Discipline

The subtitle:

**Load, Motion, Constraint, and Commitment Across Recursive Market Worlds**

introduces the four functions and recursive scope.

“Worlds” should be defined early.

Otherwise readers may interpret it metaphorically.

A market World in this article requires:

Boundary

- Observer
- GateAuthority
- Ledger
- ChangedFutureAdmissibility. (Y.37)

---

# Y.39 Finance Disclaimer

The publication should contain a clear notice:

> This article develops a theoretical and empirical research framework for analysing market observations. It does not provide investment advice, trading recommendations, or guarantees of financial performance.

The empirical breakout protocol is a research design.

It is not a live strategy recommendation.

---

# Y.40 Scientific Disclaimer

The article should state:

> References to quantum mechanics, gauge structure, complex phase, field theory, and 先天八卦 are used only at the level explicitly identified in each section. Mathematical correspondence does not by itself establish physical, historical, or ontological identity.

This disclaimer should not be hidden at the end.

It belongs near the introduction to the advanced sections.

---

# Y.41 Historical Disclaimer

The article should state:

> The Proto-Eight and 成界之學 mappings are modern structural interpretations and engineering reconstructions. They are not claims that classical authors formulated

contemporary Technical Analysis, complex finance, or software-runtime theory.

---

## Y.42 Empirical Disclaimer

The article should state:

> The periodic table, actuation crosswalk, residual instruments, validation ladder, PPMG standard, and proposed pilot studies remain research proposals until tested through preregistered and independently replicated studies.

---

## Y.43 Editorial Consistency Checks

Before publication, search for each of the following terms.

### “is”

Check whether the sentence should instead use:

- is defined as;
- is represented as;
- is mapped to;
- may function as;
- is hypothesized to be.

---

### “proves”

Replace unless a formal proof is actually supplied.

---

### “objective”

Specify:

- under which protocols;
- through which transport family;
- at what tolerance.

---

### “universal”

Remove unless broad empirical support exists.

---

### “quantum”

Clarify whether the use is:

- mathematical;
- structural;
- analogical;
- physical.

---

## “phase”

Specify:

- signal phase;
- geometric phase;
- operational phase;
- internal phase time.

---

## “residual”

Specify type and lifecycle.

---

# Y.44 Equation Consistency Audit

The final manuscript should verify:

1. R and r are not confused;
2. Q is never used as generic residual;
3. $\tau_i$ is reserved for internal phase time;
4. P is clearly protocol or period by context;
5. $\chi$ carries protocol and horizon where relevant;
6. $\Xi$ uses $\nu$ rather than $\tau$ for agitation;
7. finite and infinitesimal rotations are distinguished;
8. normalized and unrestricted $\chi$ operator identities are separated;
9. approximate formulas use $\approx$;
10. claim hierarchies use implication rather than equality.

---

# Y.45 Concept Consistency Audit

Verify that:

- candidate is not used as Event;
- Event is not used as Episode;
- Episode is not used as World;
- trace is not treated as any stored value;
- ledger is not reduced to an ordinary database;
- authority is not treated as confidence;
- transport is not treated as identical appearance;
- revision is not treated as deletion;
- residual is not treated as failure by definition;
- complex eligibility precedes phase interpretation.

---

# Y.46 Source Integrity Audit

For each citation, ask:

1. Does the cited passage actually support the sentence?
2. Is the article extending the source beyond its words?
3. Is the extension labelled as inference or synthesis?
4. Has the source's limitation been preserved?
5. Is a later refinement more precise than an earlier formulation?

A citation should not be used decoratively.

---

# Y.47 New-Claim Audit

For every new term introduced by the article, ask:

- What problem does the term solve?
- Which existing term is insufficient?
- Is the new term operationally defined?
- Can it be falsified or reduced?
- Does it add more than rhetorical unity?

Terms failing this audit should be removed or demoted to informal language.

---

# Y.48 Redundancy Audit

Potentially redundant pairs include:

- Event gate and confirmation rule;
- residual and error;
- Episode and regime;
- transport and robustness;
- ledger and event log;
- $\chi$ and regime coefficient;
- $\Xi$ and feature vector;
- phase time and event clock.

The article should retain the PPMG term only when it distinguishes a materially different object.

---

# Y.49 Cross-Domain Audit

For every cross-domain mapping, verify:

1. source object;
2. target object;
3. preserved relation;
4. changed assumptions;
5. excluded identity;
6. validation standard.

A mapping that cannot fill these six fields should be removed from the main argument or labelled exploratory.

---

# Y.50 Publication-Section Architecture

A final journal-length version should be shorter than the full research monograph.

A recommended main article is:

# 1. Introduction

The problem of indicator folklore.

# 2. Protocol-Bound Market Observation

Projection and self-reference.

## 3. Four Functional Families

Load, Motion, Constraint, Commitment.

## 4. Six Closure Periods

Mark through World.

## 5. Proto-Periodic Table

The 6 × 4 architecture.

## 6. Methods as Molecules

Representative Technical Analysis methods.

## 7. Gate, Residual, Transport, and Ledger

Governance architecture.

## 8. Regime and Optional Advanced States

χ, Ξ, complex eligibility, phase time.

## 9. Empirical Programme

Breakout pilot and validation ladder.

## 10. Limits and Conclusion

What is proposed, exact, and unvalidated.

---

# Y.51 Supplementary-Material Architecture

The following material belongs in supplements.

## Supplement A

Symbols and definitions.

## Supplement B

Full expanded periodic table.

## Supplement C

Method cards.

## Supplement D

Residual ontology.

## Supplement E

Transport suite.

## Supplement F

Complex-eligibility tests.

## Supplement G

CAPM derivation.

## Supplement H

Claim typology.

## Supplement I

Research protocol.

## Supplement J

Worked cases.

## Supplement K

PPMG standard and schemas.

## Supplement L

Reference implementation.

This division keeps the main paper readable while preserving the full research infrastructure.

---

# Y.52 Monograph Architecture

The complete current text is better understood as a monograph or programme paper.

Its structure can remain:

Theory
→ Taxonomy
→ Method atlas

→ Formalization
→ Runtime
→ Benchmark
→ Standard
→ Implementation
→ Validation
→ Comparison
→ Publication system. (Y.38)

This scale is appropriate for a foundational research programme, but not for one conventional journal article.

---

# Y.53 Reference-List Architecture

The bibliography should be divided into sections.

# Foundational internal sources

- Proto-Eight;
- recursive disclosure;
- declaration;
- self-revising fractal;
- time-bearing worlds;
- Technical Analysis;
- CAPM complex geometry;
- phase-time;
- Gauge Grammar;
- runtime kernels.

# External disciplinary sources

- market microstructure;
- econometrics;
- sequential testing;
- survival analysis;
- causal inference;
- accounting;
- law;
- measurement theory;

- complex systems.

## Historical and philosophical sources

- 先天八卦;
- related classical commentaries where directly used.

This makes the article's intellectual genealogy visible.

---

# Y.54 Article-Version Record

Each public version should include:

```
Article version:
Release date:
Major conceptual changes:
Formula corrections:
New empirical claims:
Removed claims:
Changed terminology:
Open residual:
```

A new article version should not silently revise:

- equations;
- claim strength;
- source attribution;
- validation status.

---

# Y.55 Claim Ledger

The publication should maintain a claim ledger.

```
Claim ID:
Claim text:
Provenance:
Status:
Scope:
Primary source:
Transformation:
Falsifier:
Current evidence:
Revision history:
```

Examples:

## CLM-001

Technical Analysis is protocol-bound projection.

Origin:

Source-derived.

## CLM-002

Four functions recur across six periods.

Origin:

Present synthesis.

Status:

Taxonomic hypothesis.

## CLM-003

Residual burden predicts failed breakout.

Origin:

Present hypothesis.

Status:

Untested until pilot.

## CLM-004

CAPM Q satisfies $A^2 = R^2 + Q^2$.

Origin:

Exact source construction.

---

# Y.56 Open-Residual Register for the Article

The article should end with an explicit residual register.

## Y-R1

Are six periods empirically separable?

## Y-R2

Are four functional families minimal?

## Y-R3

Does the Proto-Eight layer add value beyond the four families?

## Y-R4

Can gate status be annotated reliably?

## Y-R5

Which residual types have stable predictive or diagnostic value?

## Y-R6

Which transport families are scientifically meaningful?

## Y-R7

Does Episode grammar improve on mature regime models?

## Y-R8

Can observer backreaction be identified causally?

## Y-R9

Does any Technical Analysis R–Q pair earn complex priority?

## Y-R10

Can internal phase outperform event count or flexible clocks?

## Y-R11

Can cross-ledger World transitions be standardized?

## Y-R12

Does the whole grammar improve practice enough to justify its complexity?

---

# Y.57 Claim-Reduction Register

The publication should also state its accepted reduction paths.

Six periods
→ fewer empirically supported periods. (Y.39)

Four functions
→ merged or revised function set. (Y.40)

Proto-Eight
→ optional engineering checklist. (Y.41)

Residual ontology
→ smaller reliable subset. (Y.42)

Transport objectivity
→ ordinary scoped robustness. (Y.43)

Episode grammar
→ event-sequence model. (Y.44)

Complex state
→ real pair. (Y.45)

Phase time
→ event clock. (Y.46)

World framework
→ institution-specific event model. (Y.47)

A theory that declares its permitted reductions is easier to falsify honestly.

---

# Y.58 Editorial Acceptance Gate

The article is ready for publication only if all mandatory conditions pass.

## Source gate

- major inherited claims cited;
- new synthesis not falsely attributed;
- historical interpretations qualified.

## Mathematical gate

- formulas checked;
- approximations labelled;
- units and domains declared.

## Conceptual gate

- periods, functions, roles, and rails remain distinct;
- residual and Q remain distinct;
- Event, Episode, and World remain distinct.

## Empirical gate

- untested claims labelled;
- proposed studies separated from results;
- no simulated or hypothetical values described as observed findings.

## Comparative gate

- mature existing methods acknowledged;
- non-redundancy tests stated;
- no replacement claims without evidence.

## Publication gate

- figures carry status and scope;
- tables carry provenance;
- disclaimers visible;
- claim and residual registers included.

The publication decision is:

Publish
only if
SourceGate ∧ MathematicalGate ∧ ConceptualGate ∧ EmpiricalGate ∧ ComparativeGate. (Y.48)

---

# Y.59 Recommended Front-Matter Notice

The article may open with the following notice:

> **Epistemic status.** This article proposes a protocol-bound taxonomy and research architecture for Technical Analysis. Some local mathematical constructions—most notably the declared CAPM R–Q geometry—are exact under their definitions. The six-period table, Proto-Eight market crosswalk, residual instruments, validation ladder, and PPMG standard are present-article syntheses requiring empirical testing. References to quantum theory, gauge structure, and 先天八卦 indicate explicitly bounded structural correspondences, not established physical or historical identities. The article does not provide investment advice.

---

# Y.60 Recommended End-Matter Statement

The article may close with:

> This work should be read as a versioned research declaration rather than a completed universal theory. Its central structures are offered with explicit gates, residuals, falsifiers, and reduction paths. Later evidence may strengthen, localize, simplify, or close individual branches without requiring the whole programme to be defended as one indivisible claim.

---

# Y.61 The Editorial Runtime

The full publication process is:

SourceCollection
→ ClaimExtraction
→ ProvenanceTyping
→ FormalVerification
→ ScopeDeclaration
→ CitationAudit
→ FigureAudit
→ EmpiricalStatusAudit
→ PublicationGate
→ VersionedRelease
→ ResidualRegister
→ AdmissibleRevision. (Y.49)

This runtime applies the article's own governance principles to its publication.

---

# Y.62 The Article as a Self-Revising Research Object

The manuscript itself has:

# Boundary

Which sources and financial domains are included?

# Projection

Which concepts are selected from those sources?

# Gate

Which claims enter the published article?

# Trace

Which version records them?

# Residual

Which disagreements and missing tests remain?

# Transport

Can the architecture survive other markets and disciplines?

# Revision

How can the article change without erasing earlier claims?

Therefore:

ArticleWorld_v

= Declaration_v

- ClaimLedger_v
- SourceMap_v
- ResidualRegister_v
- RevisionRule_v. (Y.50)

The publication should practice the epistemology it recommends.

---

# Y.63 Compact Editorial Contract

```
Every major claim must disclose whether it is inherited,
derived, newly synthesized, or hypothesized.

Every exact equation must state its assumptions.

Every cross-domain mapping must state what is preserved
and what is not transferred.

Every advanced interpretation must show its eligibility gate.

Every empirical claim must identify its test.

Every failed claim must have a reduction path.

Every revision must preserve the prior version.
```

---

# Y.64 Appendix Y Conclusion

The Periodic Grammar is built from a long developmental source sequence.

That inheritance is a strength only when it remains visible.

The article must therefore distinguish:

Source
from
Transformation. (Y.51)

Definition
from
Hypothesis. (Y.52)

Exact construction
from
Domain extension. (Y.53)

Analogy
from
Identity. (Y.54)

Historical interpretation
from
Modern engineering reuse. (Y.55)

Mathematical possibility
from
Empirical validation. (Y.56)

The article's provenance architecture can be summarized as:

Claim
= Origin

- Transformation
- Status
- Scope
- Falsifier. (Y.57)

Its editorial discipline is:

> Do not make a source appear to say what only the present synthesis proposes.

Its scientific discipline is:

> Do not let the exactness of one local equation lend unearned certainty to an entire theoretical architecture.

Its historical discipline is:

> Do not confuse a modern structural reconstruction with proof of original classical intent.

Its publication discipline is:

> Release the theory with the same objects it asks a market observer to preserve—declaration, gate, trace, residual, transport, and admissible revision.

The final editorial law is:

> A mature research programme does not hide the path by which its claims were formed. It makes that path part of the evidence.

# Appendix Z — Canonical Definitions, Terminology Discipline, and Claim-Control Dictionary

## Z.1 Purpose

The Periodic Grammar now contains a large vocabulary drawn from several layers:

- Technical Analysis;
- recursive disclosure;
- observer theory;
- Proto-Eight engineering;
- complex valuation geometry;

- regime diagnosis;
- institutional ledgers;
- empirical research design.

Without strict terminology control, several distinct objects can collapse into one another.

Examples include:

- an indicator being treated as the market;
- a crossing being called an Event;
- an Event being treated as an Episode;
- a trace being confused with a ledger;
- residual being renamed Q;
- phase being treated as time;
- statistical confidence being treated as authority;
- revision being used to conceal retrospective relabelling.

This appendix freezes the article's canonical vocabulary.

Its purpose is not merely lexical.

Terminology determines:

- what can be inferred;
- which gate is required;
- which evidence remains missing;
- what level of claim is permitted;
- which reduction follows when a stronger interpretation fails.

The governing rule is:

TermDiscipline
→ ClaimDiscipline
→ ResearchDiscipline. (Z.1)

---

# Z.2 Three Statuses of Definition

Every definition in this appendix belongs to one of three statuses.

## Z.2.1 Source-derived definition

The term is inherited substantially from a source framework.

Examples:

- declaration;
- filtration;
- gate;
- trace;
- residual;
- admissible revision;
- Proto-Eight roles;
- CAPM Q.

Such definitions should remain close to their original technical scope.

---

## Z.2.2 Present-article construction

The term is assembled or specialized by this article.

Examples:

- six closure periods;
- four-family table;
- three governance rails;
- PPMG conformance profiles;
- claim ceiling;
- period promotion.

These are proposed structures rather than inherited facts.

---

## Z.2.3 Empirical hypothesis

The term is operationally defined, but its claimed usefulness remains to be tested.

Examples:

- residual burden;
- transport-survival score;
- Episode Completion Certificate;
- observer-crowding index;
- Technical Analysis phase time.

A precise definition does not make the associated empirical claim true.

Definition
≠ Validation. (Z.2)

---

# Z.3 Canonical Object Hierarchy

The framework distinguishes four broad object classes.

## Z.3.1 Field objects

These represent what may exist before the observer has fully declared or projected it.

Examples:

- undeclared market field;
- latent order intentions;
- unobserved institutional state.

---

## Z.3.2 Observational objects

These are produced by a protocol-bound operator.

Examples:

- return;
- moving average;
- RSI;
- volume profile;
- breadth;
- CAPM-admitted value.

---

## Z.3.3 Closure objects

These represent increasing levels of stabilized or committed organization.

Examples:

- Structure;
- Event;
- Episode;
- World.

---

## Z.3.4 Governance objects

These regulate how closure becomes legitimate and revisable.

Examples:

- gate;
- residual;
- ledger;
- transport;
- authority;
- revision.

The classes must remain distinct.

An observational object may support a closure object.

It is not automatically identical to it.

---

# Z.4 Field

# Canonical definition

A **field** is the larger domain of possible states, relations, traces, and unrealized alternatives from which a protocol selects a readable object.

Symbolically:

$\Sigma_0$ = undeclared field. (Z.3)

A declared protocol P produces:

$\Sigma_P = \text{Declare}(\Sigma_0 \mid P)$. (Z.4)

The source recursive-disclosure framework moves from undeclared possibility through declaration, projection, gate, trace, residual, ledger, invariance, and time-bearing world.

---

# What “field” does not mean here

It does not automatically mean:

- a physical quantum field;
- a fundamental substance;
- a continuously differentiable mathematical field;
- a validated causal mechanism.

The term indicates a structured possibility domain under incomplete observation.

---

# Z.5 Protocol

# Canonical definition

A **protocol** is the declared set of rules determining:

- what is observed;
- where the boundary lies;
- which data are admissible;
- how time is aggregated;
- which operators are applied;
- what counts as a gate;
- how residual is recorded;
- what may invalidate the claim.

The article uses:

$$P = (\text{Asset},\text{Boundary},\text{Timeframe},\text{Scale},\text{BarRule},\text{FeatureMap},\text{GateRule},\text{ResidualRule}). \quad (Z.5)$$

Extended institutional protocols may also include:

- authority;
- jurisdiction;
- ledger type;
- outcome horizon.

---

# Approved usage

“Under daily logarithmic protocol P…”

“Under accounting protocol P_accounting…”

“Transport from P_daily to P_weekly…”

---

# Prohibited usage

“According to the market protocol…”

when no protocol has been declared.

A protocol is not a general atmosphere or style.

It is an operational specification.

---

# Z.6 Declaration

# Canonical definition

A **declaration** converts an indeterminate analytical field into a bounded object of inquiry.

A declaration specifies:

- subject;
- boundary;
- observer;
- admissible evidence;
- closure rule;
- authority;
- revision conditions.

Declaration is not arbitrary invention.

It is constrained selection.

Define:

$D_P: \Sigma_0 \to \Sigma_P$. (Z.6)

---

# Declaration versus discovery

Declaration does not imply that the observer creates the whole market reality.

The market may resist the declaration through:

- contradiction;
- failed prediction;
- transport failure;
- residual accumulation;
- institutional rejection.

Thus:

Declaration selects the object.
Reality constrains the result. (Z.7)

---

## Prohibited substitution

Declaration ≠ opinion. (Z.8)

Declaration ≠ unrestricted narrative choice. (Z.9)

---

# Z.7 Projection

## Canonical definition

A **projection** is the protocol-bound output of an operator applied to admissible evidence.

Define:

$$X_{j,P,t} = \hat{O}_{j,P}(E_{\leq t}). \quad (Z.10)$$

Examples:

- RSI;
- moving average;
- volatility estimate;
- volume profile;
- breadth measure;
- CAPM valuation coordinate.

The Technical Analysis source treats analytical methods as partial projections of market self-reference rather than complete representations of the market.

---

## Projection does not imply

- total market truth;
- causal explanation;
- Event admission;
- institutional recognition;
- future success.

The canonical restriction is:

Projection ≠ MarketTotality. (Z.11)

---

# Z.8 Operator

# Canonical definition

An **operator** is a declared transformation from admissible input into a projected object.

Formally:

$\hat{O}_j: E \rightarrow X_j$. (Z.12)

Examples:

- smoothing;
- differencing;
- normalization;
- thresholding;
- aggregation;
- boundary comparison;
- complex completion.

An operator must declare:

- source lineage;
- parameters;
- units;
- output type;
- known failure modes.

---

# Operator versus method

A **method** may contain several operators.

For example:

MACD
$= EMA_{12}$
$- EMA_{26}$

- signal smoothing
- histogram differencing. (Z.13)

Therefore:

Method
= OperatorComposition

- InterpretationRule

- GateUsage. (Z.14)

---

# Z.9 Filtration

# Canonical definition

A **filtration** is an ordered disclosure structure specifying what information is available at each stage.

In conventional notation:

$\mathscr{F}_0 \subseteq \mathscr{F}_1 \subseteq \mathscr{F}_2 \subseteq \ldots$ (Z.15)

For market analysis, the filtration may represent:

- incoming trades;
- completed bars;
- closing prices;
- later breadth publication;
- institutional announcements;
- legal recognition.

The core rule is:

A claim made at t may use only evidence in $\mathscr{F}$_t. (Z.16)

---

# Filtration violation

A later value is used as though it had been known earlier.

Examples:

- revised macroeconomic data used in a live historical signal;
- final weekly close used at Monday's daily gate;
- retrospective episode endpoint used to define phase online.

Filtration violation is not ordinary model error.

It is an information-time error.

---

# Z.10 Observation

# Canonical definition

An **observation** is a protocol-bound disclosure produced by an observer or instrument.

Observation includes:

- the protocol;
- the operator;
- the evidence available;
- the resulting projection.

It is not merely a raw datum.

A raw trade becomes an observation only after its:

- venue;
- timestamp;
- units;
- status;
- observer context

are specified.

---

## Objectivity restriction

Objectivity does not require observer absence.

The framework uses operational objectivity:

Objectivity
≈ StableRelation under admissible observer transport. (Z.17)

This is weaker than metaphysical observer independence.

---

# Z.11 Mark

## Canonical definition

A **Mark** is the smallest admitted market record under the declared protocol.

Examples:

- one trade;
- one quote update;
- one order-book change;
- one execution;
- one tick.

A Mark is not always an economically final event.

A bad tick, cancelled trade, or unverified quote may remain provisional.

---

## Mark versus Event

A Mark may record execution.

But a chart-level breakout Event typically requires aggregation and a higher gate.

Therefore:

MarkCommitment
≠ HigherPeriodEventCommitment. (Z.18)

---

# Z.12 Window

# Canonical definition

A **Window** is an aggregation closure over a declared observation interval.

Examples:

- one minute;
- one daily bar;
- one session;
- one volume bar;
- one event window.

A Window closes according to its bar rule.

The close is a formal boundary in the observation process.

---

## Period versus timeframe

A Window often corresponds to a timeframe.

But **period** in the six-period architecture does not mean timeframe.

A weekly bar is still a Window-level object.

A legal judgment may be a World-level object even if it occurs instantaneously.

Thus:

ClosurePeriod ≠ ClockDuration. (Z.19)

---

# Z.13 Structure

# Canonical definition

A **Structure** is a persistent relation, memory, boundary, or organization that survives beyond one Window.

Examples:

- moving-average arrangement;
- support zone;
- range;
- trend channel;
- volume profile;
- persistent breadth condition;
- volatility regime candidate.

A Structure may contain potential.

It has not necessarily committed a transition.

---

# Structure versus Event

Price above an upward-moving average is a Structure.

A new breakout through a declared boundary may be an Event.

Therefore:

PersistentCondition
≠ TransitionCommitment. (Z.20)

---

# Z.14 Event

# Canonical definition

An **Event** is a candidate transition that has passed a declared commitment gate.

Define:

Event
= Candidate

- GateAdmission. (Z.21)

Examples:

- admitted breakout;
- support failure;
- accepted rejection;
- covenant breach;
- index inclusion announcement;
- execution;
- settlement.

---

## Required components

A conforming Event claim must identify:

- candidate;
- boundary;
- gate;
- authority;
- time;
- residual;
- invalidation or reversal condition.

---

## Event versus signal

A signal may warn that an Event is possible.

The signal is not the Event.

Signal
→ CandidateInterpretation. (Z.22)

Gate
→ EventAdmission. (Z.23)

---

# Z.15 Episode

# Canonical definition

An **Episode** is a persistent, ordered grammar of Events.

It includes:

- start gate;

- event sequence;
- governing relation;
- residual branches;
- persistence;
- completion or transition gate.

Examples:

- trend episode;
- accumulation–release sequence;
- crisis escalation;
- recovery;
- range rotation.

---

## Episode versus large pattern

An Episode is not defined only by visual size.

A large triangle may remain Structure.

A short but institutionally ordered sequence may form an Episode.

The requirement is:

Episode
= OrderedEventGrammar

- Persistence
- CompletionConditions. (Z.24)

---

## Event count restriction

Many Events do not automatically form an Episode.

QuantityOfEvents
≠ EpisodeClosure. (Z.25)

---

# Z.16 World

## Canonical definition

A **World** is a bounded system in which authoritative gates and persistent ledgers change what actions, rights, obligations, or interpretations are subsequently admissible.

A strong World object contains:

World_P
= Boundary

- Observer
- Authority
- Gate
- Trace
- Ledger
- Backreaction. (Z.26)

Examples:

- accounting recognition regime;
- contractual default state;
- legal judgment;
- policy rule;
- settlement system;
- benchmark-inclusion regime.

---

# World versus market regime

A market regime may be an Episode.

It becomes a World only when:

- rules;
- authority;
- ledger;
- changed future admissibility

are established.

World ≠ LargePattern. (Z.27)

---

# Z.17 Load

# Canonical definition

**Load** is accumulated state capable of influencing later behaviour.

Examples:

- liquidity;
- inventory;
- volume memory;
- positioning;
- prior failed Events;
- institutional obligations;
- historical boundary mass.

Load is not necessarily directional.

A large volume node may carry substantial Load without predicting up or down.

---

## Load versus pressure

Pressure usually implies potential directional influence.

Load may remain neutral until combined with:

- gradient;
- boundary;
- trigger;
- gate.

---

# Z.18 Motion

## Canonical definition

**Motion** is change, displacement, relation, acceleration, or transition among states.

Examples:

- return;
- momentum;
- spread change;
- divergence;
- value migration;
- phase progression.

Motion does not determine whether the change is:

- admitted;
- durable;
- legal;
- institutionally recognized.

---

## Motion versus Commitment

A price surge is Motion.

A closing breakout is Commitment under one protocol.

A legal judgment is Commitment under another.

Motion ≠ Admission. (Z.28)

---

# Z.19 Constraint

# Canonical definition

A **Constraint** is a boundary, admissibility condition, capacity limit, rule, or channel that shapes possible Motion.

Examples:

- support;
- resistance;
- bid–ask limits;
- liquidity capacity;
- collateral requirement;
- covenant;
- jurisdiction;
- accounting rule.

Constraint may:

- block;
- route;
- compress;
- delay;
- transform.

It is not necessarily negative.

A well-formed Constraint can stabilize exchange.

---

# Z.20 Commitment

# Canonical definition

**Commitment** is the admission of one candidate state into consequential history under a gate.

Examples:

- close above a boundary;
- completed execution;
- accepted retest;
- recognized impairment;
- legal judgment;
- settlement.

Commitment creates a trace.

It does not guarantee:

- success;
- persistence;
- exhaustion of uncertainty;
- profitability.

The central restriction is:

Commitment ≠ Exhaustion. (Z.29)

---

# Z.21 Gate

# Canonical definition

A **gate** is the rule or authority that determines whether a candidate state becomes admitted.

Define:

$G_P(c,E,L,\mathscr{R},A)$
$\rightarrow$ (Decision,Strength,Trace,Residual). (Z.30)

A gate may return:

- Admit;

- Partially Admit;
- Defer;
- Reject;
- Ambiguous.

---

## Gate versus threshold

A threshold becomes a gate only when crossing it changes the claim state.

A numerical level used merely for description is not necessarily a gate.

Threshold

- StateTransitionRule
- Authority
  → Gate. (Z.31)

---

## Gate versus outcome

Gate status is determined using the evidence available at commitment time.

Outcome occurs later.

A properly admitted Event may later fail.

A rejected candidate may later rise.

Therefore:

GateQuality ≠ EventOutcome. (Z.32)

---

# Z.22 Trace

# Canonical definition

A **trace** is a persistent record of an admitted state that remains accessible to later observation or action.

A trace should possess:

- persistence;
- accessibility;
- consequential relevance.

Examples:

- recorded trade;

- accepted Event entry;
- accounting journal;
- legal order;
- protocol version;
- model decision record.

---

## Trace versus log

A log records that something occurred.

A trace carries forward consequence or evidential relevance.

Every trace may be logged.

Not every log is a consequential trace.

---

# Z.23 Ledger

## Canonical definition

A **ledger** is an ordered, persistent structure containing traces, residuals, authority, and revision relations.

Define:

$$L_k = \{(e_1,r_1),(e_2,r_2),\ldots,(e_k,r_k)\}. \quad (Z.33)$$

A ledger may be:

- market;
- analytical;
- model;
- accounting;
- contractual;
- legal;
- policy.

---

## Ledger versus database

A database stores data.

A ledger additionally preserves:

- order;
- commitment status;
- version;
- authority;
- consequence;
- revision history.

An ordinary mutable table may function as a database without satisfying the ledger requirements.

---

## Trace versus ledger

A trace is one committed record.

A ledger is the ordered system of traces and their governance relations.

Trace ≠ Ledger. (Z.34)

---

# Z.24 Residual

## Canonical definition

A **residual** is unresolved content left visible after projection, gating, or commitment.

It may represent:

- missing evidence;
- direct contradiction;
- boundary ambiguity;
- model inadequacy;
- transport failure;
- branch uncertainty;
- institutional disagreement;
- authority uncertainty.

Residual is not defined as numerical error alone.

---

## Residual lifecycle

Residual may become:

Open
→ Monitoring
→ Resolved. (Z.35)

Open
→ Dissipated. (Z.36)

Open
→ Converted. (Z.37)

Open
→ Invalidating. (Z.38)

---

## Residual versus failure

Residual may remain after a successful Event.

For example:

- admitted breakout with weak breadth;
- valid legal judgment under appeal;
- accounting recognition with measurement uncertainty.

Residual ≠ Failure. (Z.39)

---

## Residual versus ignorance

Residual must be typed where possible.

“Uncertainty exists” is too vague.

A mature record identifies:

- source;
- severity;
- persistence;
- claim effect;
- resolution condition.

---

# Z.25 Error

## Canonical definition

An **error** is a deviation between an expected or correct result and the produced result under a declared model or procedure.

Examples:

- calculation error;
- data error;
- forecast error;
- transport error;
- classification error.

Error is one possible residual type.

Residual is broader.

Therefore:

Error ⊂ ResidualUniverse. (Z.40)

but:

Residual ≠ Error. (Z.41)

---

# Z.26 Invalidation

# Canonical definition

**Invalidation** is the passing of a predeclared condition that removes or downgrades a claim.

Examples:

- close back inside an old range plus failed reclaim;
- wave rule breach;
- legal reversal;
- real-pair model matching complex performance;
- phase-time model failing out of sample.

Invalidation is not simply a later unfavourable outcome.

It must be tied to the original claim.

---

# Invalidation versus revision

Invalidation evaluates the old claim.

Revision creates a new protocol or interpretation.

The old invalidation must remain recorded even if the revised model later succeeds.

---

# Z.27 Closure

## Canonical definition

**Closure** is the stabilization of a previously open possibility under a declared rule.

Closure may occur at several depths:

- Window closure;
- Event gate;
- Episode completion;
- World recognition.

Closure is always partial relative to the larger field.

A gate closes one question while leaving residual.

Thus:

Closure
= Admission

- PreservedNonclosure. (Z.42)

---

## Closure versus certainty

Closure does not imply universal certainty.

A legal judgment may close one institutional question while:

- appeal remains open;
- economic value remains uncertain;
- market response remains unresolved.

---

# Z.28 Promotion

## Canonical definition

**Promotion** moves a claim from one closure period to a higher one after a new gate passes.

Examples:

Window → Structure
requires persistence. (Z.43)

Structure → Event
requires transition admission. (Z.44)

Event → Episode
requires ordered grammar and persistence. (Z.45)

Episode → World
requires authority, ledger, and changed admissibility. (Z.46)

---

# Promotion is not relabelling

Changing the word “Event” to “Episode” does not create promotion.

Promotion requires new evidence and a new closure condition.

Promote_{n→n+1}
⇒ NewEvidence ∧ NewGate. (Z.47)

---

# Z.29 Demotion and Reduction

# Canonical definition

**Demotion** lowers a claim’s closure status after invalidation or failed persistence.

**Reduction** replaces a stronger model with a simpler one when the advanced structure adds no value.

Examples:

Admitted Event
→ Structure warning. (Z.48)

Episode
→ Event sequence. (Z.49)

Phase-time model
→ Phase-bearing model. (Z.50)

Complex model
→ Real pair. (Z.51)

---

# Reduction is not defeat of the whole framework

A failed advanced layer may leave a useful lower layer intact.

This is central to the article’s graded scientific architecture.

---

# Z.30 Transport

# Canonical definition

**Transport** maps a claim under protocol P into its expected form under protocol P′.

Define:

T_{P→P′}(C_P)

= Ĉ_{P′}. (Z.52)

The observed target claim is:

C_{P′}. (Z.53)

The transport residual is:

r_T
= C_{P′} − Ĉ_{P′}. (Z.54)

The source framework treats cross-protocol survival as a route from local pattern toward stronger operational objectivity.

---

# Transport does not require identical appearance

A line on arithmetic scale may become a curved relation on logarithmic scale.

A daily admitted Event may become a weekly candidate.

A market warning may become an accounting review candidate.

The relation should transform appropriately.

---

# Z.31 Invariance

# Canonical definition

**Invariance** is the survival of a specified relation under an admissible transformation.

Possible forms include:

- exact invariance;
- covariance;
- partial survival;
- scoped locality.

The article does not require every valid claim to be invariant under every transformation.

A five-minute execution signal may be intentionally local.

---

# Invariance versus universal truth

Transport survival supports a stronger scoped claim.

It does not establish universal truth.

CrossProtocolSurvival
≠ MetaphysicalUniversality. (Z.55)

---

# Z.32 Objectivity

# Canonical definition

**Objectivity** in this framework means that a relation remains stable, reconstructable, and challengeable across relevant protocols or observers.

Operationally:

Objectivity_P-family
= Reproducibility

- TransportSurvival
- ResidualVisibility. (Z.56)

It does not require:

- no observer;
- no declaration;
- no institutional frame.

---

# Objectivity versus consensus

Consensus may arise through:

- shared evidence;
- shared convention;
- authority;
- crowding;
- coercion.

Consensus becomes evidence of objectivity only when the relation survives legitimate challenge and transport.

---

# Z.33 Authority

# Canonical definition

**Authority** is the legitimate capacity to commit a state within a specified domain.

Examples:

- exchange matching engine for execution;
- company governance for accounting recognition;

- court for judgment;
- regulator for policy enforcement;
- analyst for analytical classification.

Authority is scoped.

A market model may estimate legal risk.

It cannot itself create legal default.

---

## Authority versus confidence

A model may be 99% confident.

That does not give it institutional authority.

Confidence ≠ Permission. (Z.57)

Probability ≠ Jurisdiction. (Z.58)

---

# Z.34 Backreaction

## Canonical definition

**Backreaction** occurs when an observation, gate, or ledger entry alters later system behaviour.

Examples:

- index inclusion causes benchmark buying;
- margin breach causes forced sales;
- published technical rule becomes crowded;
- accounting impairment changes capital decisions;
- legal judgment changes permitted action.

---

## Backreaction versus correlation

Later price movement after a gate does not by itself prove backreaction.

A strong claim requires evidence that:

1. the gate changed action;
2. actors responded;
3. the response affected later state.

BackreactionClaim

⇒ IdentifiedActionChannel. (Z.59)

---

# Z.35 Revision

# Canonical definition

A **revision** changes a protocol, interpretation, or model for future use while preserving the prior trace.

Define:

D_{k+1}
= U_a(D_k,L_k,$\mathscr{R}$_k). (Z.60)

An admissible revision should be:

- well formed;
- trace preserving;
- residual honest;
- frame accountable;
- budget bounded;
- prospectively testable.

---

# Revision versus retrospective relabelling

Revision:

- preserves the old claim;
- explains the change;
- creates a new version;
- requires later testing.

Retrospective relabelling:

- overwrites the old claim;
- moves the boundary;
- changes the horizon;
- reverses the evidence meaning;
- conceals failure.

Therefore:

Revision ≠ RetrospectiveRepair. (Z.61)

# Z.36 Recursive

# Canonical definition

**Recursive** means that outputs from one stage become inputs to a later stage of the same or related process.

In the market runtime:

Trace_k + Residual_k
→ Load_{k+1}. (Z.62)

Declaration_k
→ Observation_k
→ Ledger_k
→ RevisedDeclaration_{k+1}. (Z.63)

---

# Recursive versus repetitive

Repetition performs the same operation again.

Recursion feeds previous output into later structure.

Repeated candles are not automatically a recursive Episode.

---

# Recursive versus fractal

A recursive process may generate similar patterns across levels.

It is not automatically fractal in the strict geometric sense.

Recursive
⇏ ExactScaleInvariantFractal. (Z.64)

---

# Z.37 Fractal

# Canonical definition

A **fractal** in the strict mathematical sense requires specified self-similarity, scaling, dimension, or recursive construction.

The article sometimes uses “self-revising fractal” in the broader source-development sense of recursive pattern generation across levels.

When discussing markets, the safer phrase is:

Recursive multi-level structure. (Z.65)

The term “fractal” should be reserved for cases with explicit scaling evidence.

# Z.38 χ

## Canonical definition

χ is a protocol- and horizon-indexed feedback signature.

Use:

$\chi_{P,h}$. (Z.66)

Interpretive classes are:

$\chi < 0$ → corrective circulation. (Z.67)

$\chi \approx 0$ → critical ambiguity. (Z.68)

$\chi > 0$ → self-confirming selection. (Z.69)

The Technical Analysis source develops χ as a way to distinguish corrective, ambiguous, and reinforcing relations rather than assigning one universal meaning to indicators.

---

## χ is not

- a universal market constant;
- an asset personality;
- a probability of increase;
- the same as volatility;
- the same as Ξ.

χ describes feedback orientation.

Ξ describes an operating-state interface.

---

## Required qualification

Never write:

“χ is positive.”

Write:

“$\chi_{P,h}$ is estimated as positive under the declared protocol and horizon.”

---

# Z.39 Ξ

## Canonical definition

Ξ is a protocol-bound operating-state interface:

$$\Xi_P = (\rho_P, \gamma_P, \nu_P). \quad (Z.70)$$

where:

ρ = loading;
γ = lock-in or confinement;
ν = agitation.

The coordinate ν is used here to avoid collision with $\tau_i$.

---

# Ξ is not

- a universal market ontology;
- a replacement for the source variables;
- a physical gauge field;
- a complete prediction model.

Its components must remain traceable to richer evidence.

---

# χ versus Ξ

χ asks:

> What is the feedback orientation?

Ξ asks:

> How loaded, locked, and agitated is the declared system?

Therefore:

$$\chi \neq \Xi. \quad (Z.71)$$

---

# Z.40 Structural Mass

# Canonical definition

**Structural mass** is the degree to which historical activity, memory, positioning, or institutional importance makes a boundary or region consequential.

Possible contributors include:

- transaction density;
- prior tests;
- volume concentration;

- duration;
- observer attention;
- institutional reference.

Structural mass is a proposed diagnostic construct.

It is not physical mass.

---

## Structural mass versus support strength

A high-mass level may:

- attract;
- repel;
- absorb;
- accelerate movement after break.

It does not imply one directional outcome.

---

# Z.41 Semantic Density

## Canonical definition

**Semantic density** is the concentration of market-relevant distinctions, expectations, commitments, or memories around an object.

Examples:

- earnings gap;
- prior high;
- legal threshold;
- benchmark level;
- heavily traded price region.

The term is broader than volume density.

A legal price threshold may have high semantic density despite low historical transaction volume.

This remains an interpretive construct requiring operational proxies.

---

# Z.42 Selection Depth

## Canonical definition

**Selection depth**, denoted σ, measures how many consequential distinctions or branch eliminations have occurred in forming the current state.

Examples:

- multiple failed alternatives;
- repeated boundary tests;
- several commitment gates;
- nested institutional approvals.

Selection depth is not identical to clock duration.

An event may acquire high selection depth quickly.

---

## Selection depth versus phase time

σ counts or summarizes selection.

$\tau_i$ measures candidate internal phase traversal.

They may correlate.

They are not identical.

---

# Z.43 R

## Canonical definition

R is the **real or admitted coordinate** in a declared two-coordinate state.

Its exact meaning depends on the protocol.

In CAPM geometry:

R_t
= CF_t/(1+r_CAPM)^t. (Z.72)

In a candidate Technical Analysis model, R may represent:

- admitted price structure;
- accepted displacement;
- realized valuation state.

The meaning must be independently declared.

---

## R is not automatically

- return;
- resistance;
- realized profit;
- objective truth.

Use subscripts when ambiguity exists:

R_CAPM;
R_acceptance;
r_return.

---

# Z.44 Q

## Canonical definition

Q is a declared **conjugate coordinate** paired with R under a justified metric or generator.

In CAPM geometry:

$$Q_t = \sqrt{(A_t^2 - R_t^2)}. \quad (Z.73)$$

and:

$$A_t^2 = R_t^2 + Q_t^2. \quad (Z.74)$$

The CAPM source treats this as an exact geometric construction under its declared baseline and admitted valuations.

---

## Q is not

- haircut;
- realized loss;
- volatility;
- beta;
- VaR;
- expected shortfall;
- generic uncertainty;
- unexplained error;
- residual.

The article’s canonical restriction is:

Q ≠ Residual. (Z.75)

---

# Technical Analysis Q

A Technical Analysis Q is only a candidate until it has:

- independent definition;
- compatible units or metric;
- stable coupling with R;
- scaling robustness;
- operational gain.

There is no assumed universal Technical Analysis Q.

---

# Z.45 Z

# Canonical definition

Z is the complex state:

Z
= R + iQ. (Z.76)

When Euclidean polar form is justified:

Z
= Ae^{iθ}. (Z.77)

where:

A
= √(R² + Q²). (Z.78)

θ
= atan2(Q,R). (Z.79)

---

# Complex notation versus complex dynamics

Any real pair can be written as R + iQ.

That alone does not establish:

- rotational dynamics;
- meaningful phase;
- conjugacy;

- internal time.

Thus:

ComplexNotation
≠ ComplexPriority. (Z.80)

The phase source requires reduction when complex representation does not outperform the real-pair model or depends on arbitrary scaling.

---

# Z.46 Phase

# Canonical definition

**Phase**, θ, is the orientation of an eligible two-coordinate state.

In polar form:

θ
= atan2(Q,R). (Z.81)

Phase may encode relational order not visible in either coordinate alone.

---

# Three phase types

## Signal-processing phase

Derived through a transformation such as a Hilbert analytic signal.

## Geometric phase

Defined by an eligible R–Q state.

## Operational phase

Demonstrated to improve ordering, gating, or comparison.

These must remain separate.

---

# Phase is not time

Phase is an orientation.

Time is an ordering or duration structure.

A phase coordinate may become clock-like only after additional tests.

Phase ≠ Time. (Z.82)

---

# Z.47 Internal Phase Time

# Canonical definition

**Internal phase time**, $\tau_i$, is an accumulated measure of phase traversal in an eligible phase-bearing system.

Possible definitions include:

$\tau_i(t)$
$= \text{Unwrap}[\theta(t)]$. (Z.83)

or:

$\tau_i(t)$
$= \int_0^t |\dot{\theta}(s)| ds$. (Z.84)

---

# Required conditions

$\tau_i$ should be used only when:

1. R and Q are eligible;
2. phase is stable;
3. branch assignment is controlled;
4. the measure is available online;
5. it outperforms simpler clocks for the declared task.

---

# Phase time versus calendar time

t measures external chronological duration.

$\tau_i$ measures internal traversal under a candidate geometry.

Therefore:

$\tau_i \neq t$. (Z.85)

---

# Z.48 Ledger Order

# Canonical definition

**Ledger order**, k, is the discrete sequence of admitted consequential Events.

Examples:

$k = 1 \rightarrow$ candidate admitted.
$k = 2 \rightarrow$ retest accepted.
$k = 3 \rightarrow$ weekly promotion.
$k = 4 \rightarrow$ invalidation. (Z.86)

Ledger order may advance unevenly in calendar time.

---

## Ledger order versus clock order

Two Events separated by one hour may represent two ledger steps.

A month with no admission may leave k unchanged.

Thus:

LedgerOrder ≠ CalendarOrder. (Z.87)

---

# Z.49 Measurement Orientation

## Canonical definition

**Measurement orientation** specifies which coordinate or projection of a state is being read.

In the CAPM complex plane, a quarter-turn operator may map:

R
→ −Q. (Z.88)

This changes the measurement orientation.

It does not automatically move the economic state.

---

## Measurement versus state movement

Passive rotation:

Change in viewpoint. (Z.89)

Active rotation:

Change in state. (Z.90)

The source CAPM framework explicitly separates measurement rotation, actual state movement, economic consequence, gate, and ledger.

---

# Z.50 State Movement

## Canonical definition

**State movement** is an actual change in the R–Q or other declared state coordinates.

Under finite rotation:

R_new
= R cos Δθ − Q sin Δθ. (Z.91)

The change in R is:

$$\Delta R = R(\cos \Delta\theta - 1) - Q \sin \Delta\theta. \quad (Z.92)$$

For small $\Delta\theta$:

$$\Delta R \approx -Q\Delta\theta. \quad (Z.93)$$

State movement may create economic consequence.

Recognition still requires a relevant gate.

---

# Z.51 Haircut

# Canonical definition

In the CAPM geometry, the haircut is:

$$H = A - R. \quad (Z.94)$$

It satisfies:

$$dH/d\theta = Q. \quad (Z.95)$$

and under the declared path:

$$H = \int Q \, d\varphi. \quad (Z.96)$$

Haircut is an accumulated admitted-value difference.

Q is a local conjugate sensitivity coordinate.

Therefore:

$$H \neq Q. \quad (Z.97)$$

---

# Z.52 Indicator

# Canonical definition

An **indicator** is a registered operator output intended to expose one or more characteristics of a market object.

Examples:

- RSI;
- moving average;
- ATR;

- OBV;
- breadth;
- volume profile.

An indicator must not be treated as an independent world.

---

## Indicator versus market

Indicator
= PartialProjection. (Z.98)

Market
= LargerField under many protocols. (Z.99)

Therefore:

Indicator ≠ Market. (Z.100)

---

# Z.53 Signal

## Canonical definition

A **signal** is an interpreted projection or combination of projections indicating a candidate condition.

Examples:

- RSI divergence;
- moving-average crossover;
- unusual volume;
- volatility compression;
- breadth thrust.

A signal may create:

- warning;
- candidate Event;
- review request.

It does not automatically create commitment.

Signal ≠ Event. (Z.101)

---

# Z.54 Confirmation

# Canonical definition

**Confirmation** is additional evidence that reduces uncertainty about a declared claim.

Strong confirmation should add a distinct source, operator, function, timeframe, or authority.

The framework distinguishes:

### Same-lineage agreement

Several transformations of the same price series agree.

### Operator diversity

Different transformations expose different aspects of one source.

### Source diversity

Price, volume, breadth, positioning, and institutional data agree.

### Gate completion

The formal commitment condition passes.

---

# Confirmation versus count

ConfirmationStrength
≠ NumberOfIndicators. (Z.102)

Confirmation depends on:

- independence;
- relevance;
- gate alignment;
- transport;
- residual reduction.

---

# Z.55 Breakout

# Canonical definition

A **breakout candidate** occurs when price crosses a declared boundary.

A **breakout Event** occurs when the declared admission gate passes.

A **durably accepted breakout** occurs when the Event persists under its acceptance rule.

Therefore:

BoundaryCross
→ BreakoutCandidate. (Z.103)

GatePass
→ BreakoutEvent. (Z.104)

PersistencePass
→ AcceptedBreakoutStructure. (Z.105)

---

# Prohibited compression

Crossing = breakout = trend = bull market. (Z.106)

This collapses four distinct closure claims.

---

# Z.56 Retest

# Canonical definition

A **retest** is a later interaction with a previously crossed boundary or transition zone.

Possible states include:

- pass;
- fail;
- pending;
- no retest.

No retest is not automatically failure.

Its meaning depends on the predeclared gate.

---

# Z.57 Divergence

# Canonical definition

A **divergence** is a relational disagreement between price Motion and another projected variable.

Examples:

- price higher high, RSI lower high;
- price rise, breadth weakening;
- price continuation, volume pressure declining.

Divergence is usually a warning.

It becomes a reversal Event only after an opposing gate.

Divergence ≠ Reversal. (Z.107)

---

# Z.58 Support and Resistance

# Canonical definition

**Support** and **resistance** are protocol-bound Constraint regions associated with prior activity, memory, positioning, convention, or institutional importance.

They are zones unless the protocol declares a line.

Their significance may depend on:

- structural mass;
- semantic density;
- prior tests;
- observer attention;
- current exchange.

They are not physical walls.

---

# Z.59 Volume

# Canonical definition

**Volume** records exchange quantity under the declared venue and interval.

Volume may indicate:

- participation;
- loading;
- transfer;
- commitment intensity;
- conflict;
- liquidation.

Volume does not reveal motive directly.

HighVolume
≠ Accumulation by definition. (Z.108)

HighVolume
≠ DirectionalCommitment by definition. (Z.109)

# Z.60 Breadth

## Canonical definition

**Breadth** measures participation or coherence across components of a declared universe.

Breadth is always universe-relative.

Examples:

- advancing components;
- proportion above a moving average;
- equal-weight participation;
- sector confirmation.

Breadth may support a broad claim.

It is not mandatory for every asset-specific Event.

---

## Breadth versus market direction

A capitalization-weighted index may rise while breadth weakens.

The correct output is often:

Concentrated advance with breadth residual. (Z.110)

not:

No advance occurred.

---

# Z.61 VWAP

## Canonical definition

VWAP is a volume-weighted price centre under a declared session or interval.

Its primary roles may include:

- Load/Memory;
- institutional reference;
- acceptance centre;
- local Constraint;
- Guidance.

VWAP does not automatically represent fair value.

It represents the weighted centre of transacted volume under the selected window.

---

# Z.62 χ, Ξ, and Z: Canonical Separation

| Object | Primary question | Required input | Strongest immediate use |
|---|---|---|---|
| χ | Is feedback corrective, critical, or reinforcing? | paired relational dynamics | regime interpretation |
| Ξ | How loaded, locked, and agitated is the system? | compiled operating-state variables | diagnosis and intervention planning |
| Z | Does an eligible conjugate pair support complex geometry? | independently defined R and Q | amplitude–phase analysis |

The canonical restrictions are:

χ ≠ Ξ. (Z.111)

Ξ ≠ Z. (Z.112)

χ ≠ Phase. (Z.113)

Ξ does not create Q automatically. (Z.114)

---

# Z.63 Proto-Eight

# Canonical definition

**Proto-Eight** is an eight-role engineering grammar:

- Gradient;
- Gate;
- Boundary;
- Exchange;
- Trigger;
- Guidance;
- Memory;
- Focus.

The source engineering framework organizes them into four paired dyads:

Gradient ↔ Gate;
Boundary ↔ Exchange;
Trigger ↔ Guidance;
Memory ↔ Focus. (Z.115)

## Proto-Eight is not

- a proven universal market law;
- a chronological eight-stage cycle;
- an identity between classical trigrams and financial indicators;
- a substitute for market mechanism analysis.

Its role here is an actuation crosswalk.

---

# Z.64 Gradient

## Canonical definition

**Gradient** is the potential difference capable of driving flow or transition.

Examples:

- order imbalance;
- valuation difference;
- funding spread;
- demand–supply pressure;
- expected return differential.

Gradient does not guarantee movement.

A restrictive gate or boundary may prevent conversion.

---

# Z.65 Proto-Eight Gate

The Proto-Eight **Gate** is the qualification or permission role that determines whether potential becomes admitted flow.

It overlaps conceptually with the article's Commitment family.

However:

- Proto-Eight Gate is an actuation role;
- PPMG Commitment is an observational functional family;
- a specific Event gate is an operational rule.

These levels must remain distinct.

---

# Z.66 Boundary

In Proto-Eight usage, **Boundary** separates, contains, or buffers.

In PPMG usage, Constraint includes Boundary but may also include:

- rule;
- capacity;
- jurisdiction;
- feasibility limit.

Thus:

ProtoEightBoundary
⊂ ConstraintFamily. (Z.116)

---

# Z.67 Exchange

# Canonical definition

**Exchange** is the permitted transfer of:

- value;
- information;
- liquidity;
- ownership;
- obligation;
- influence

across a boundary.

Exchange is not identical to Motion.

Motion may occur internally.

Exchange specifically concerns crossing or transfer between differentiated regions or parties.

---

# Z.68 Trigger

# Canonical definition

A **Trigger** initiates a transition.

Examples:

- news;
- order burst;
- threshold breach;
- scheduled release;
- margin call.

Trigger is not sufficient for sustained direction.

Trigger without Guidance may create overshoot or reversal.

---

# Z.69 Guidance

# Canonical definition

**Guidance** steers, aligns, or routes an initiated transition.

Examples:

- trend direction;
- policy signal;
- flow persistence;
- feedback;
- path selection.

Guidance is not the same as prediction.

It is an operative routing role.

---

# Z.70 Memory

# Canonical definition

**Memory** preserves prior state so that later behaviour depends on history.

Examples:

- moving average;
- prior high;
- inventory;
- legal precedent;
- failed Event history;

- balance-sheet record.

Memory may stabilize or constrain.

Excessive memory may create rigidity.

---

# Z.71 Focus

# Canonical definition

**Focus** selects which distinctions receive current processing or attention.

Examples:

- chosen timeframe;
- indicator selection;
- headline level;
- institutional review priority;
- market salience.

Focus without Memory can become reactive.

Memory without Focus can become overloaded.

---

# Z.72 Periodic

# Canonical definition

**Periodic** in this article means that related functional roles recur across closure periods.

It does not mean:

- fixed calendar periodicity;
- harmonic oscillation;
- exact repetition;
- chemical periodic law.

The table is proto-periodic because similar functional roles reappear at increasing closure depth.

---

# Z.73 Grammar

# Canonical definition

A **grammar** is a finite set of typed roles and admissible composition rules capable of generating

structured claims.

The Periodic Grammar specifies:

- valid objects;
- allowed transitions;
- required gates;
- residual rules;
- promotion rules;
- reduction paths.

Grammar is used instead of “law” because the architecture remains partly proposed and protocol-dependent.

---

# Z.74 Canonical Category Errors

The following substitutions are prohibited.

| Incorrect substitution | Canonical correction |
|---|---|
| Indicator = market | Indicator is a partial projection |
| Signal = Event | Signal may create an Event candidate |
| Crossing = breakout Event | Crossing requires a declared gate |
| Event = Episode | Episode requires persistent event grammar |
| Episode = World | World requires authority and ledger |
| Event = trace | Event produces a trace when committed |
| Trace = ledger | Ledger orders many traces |
| Gate = outcome | Gate is evaluated before later outcome |
| Commitment = success | Commitment may later fail |
| Commitment = exhaustion | Residual may remain |
| Residual = error | Error is only one residual type |
| Residual = Q | Q requires independent conjugate definition |
| Q = loss | Q is a coordinate, not realized loss |
| Q = haircut | Haircut is A − R |
| Q = volatility | No identity is established |
| Phase = time | Phase requires further clock validation |
| Phase time = calendar time | They are distinct orderings |
| Ledger order = clock order | Ledger order advances by gates |
| $\chi = \Xi$ | Feedback signature differs from operating state |
| Complex notation = complex dynamics | Generator and utility are required |
| Recursive = exact fractal | Scaling evidence is required |
| Period = timeframe | Period is closure depth |
| World = large pattern | Authority and changed admissibility are required |
| Transport survival = universal truth | Survival is scoped |
| Backreaction = post-event movement | Action-channel evidence is needed |
| Gate admission = successful trade | Trading outcome is separate |

| Incorrect substitution | Canonical correction |
|---|---|
| CAPM phase cycle = six-period architecture | One is local geometry; one is closure taxonomy |
| Proto-Eight topology = chronology | The roles form a relational grammar |
| Declaration = arbitrary invention | Declaration is constrained selection |
| Objectivity = observer absence | Objectivity is operational transport stability |
| Falsification = one losing trade | Falsification concerns the declared claim or model |
| Revision = relabelling | Revision preserves the original trace |

---

# Z.75 Canonical Sentence Templates

## Z.75.1 Projection claim

> Under protocol P, operator Ô produces projection X from evidence available at time t.

---

## Z.75.2 Structure claim

> X indicates a Structure-level Load/Motion/Constraint condition; no Event gate has yet passed.

---

## Z.75.3 Candidate claim

> The boundary interaction creates an Event candidate, but admission remains pending.

---

## Z.75.4 Event claim

> The candidate is admitted as an Event under gate G, with residuals $r_1,\ldots,r_n$ remaining open.

---

## Z.75.5 Episode claim

> The admitted Event sequence supports an Episode transition because the old grammar failed, the new grammar persisted, and the completion gate passed.

---

## Z.75.6 World claim

> Authority A committed the state into ledger L, changing subsequent admissible rights, obligations, or actions.

---

## Z.75.7 Complex claim

> R and Q form a complex-eligible state under metric M and generator J,

outperforming the corresponding flexible real-pair model for task T.

---

## Z.75.8 Phase-time claim

Accumulated phase $\tau_i$ improves prospective episode alignment or gate prediction beyond calendar time, event count, and other preregistered clocks.

---

## Z.75.9 Reduction claim

The stronger model failed its eligibility gate; the retained lower model is [real pair / local Event / Event sequence / Structure].

---

# Z.76 Prohibited Sentence Forms

Avoid:

RSI proves the market will fall.

Use:

RSI indicates elevated directional dominance under the declared protocol; reversal requires an opposing Event gate.

Avoid:

The breakout is objectively real.

Use:

The daily breakout Event is admitted under protocol P and survives the specified weekly and breadth transports.

Avoid:

The company has defaulted because its bond price collapsed.

Use:

Market and risk ledgers indicate severe distress; contractual and legal default remain uncommitted.

Avoid:

Q is the hidden risk the model cannot explain.

Use:

The unexplained component remains residual unless an independently defined conjugate coordinate is established.

Avoid:

Phase creates time.

Use:

> Under specified conditions, accumulated phase may provide a useful internal ordering coordinate.

---

# Z.77 Notation Collision Rules

# Z.77.1 Time symbols

t = calendar time. (Z.117)

θ = phase orientation. (Z.118)

$\tau_i$ = internal phase time. (Z.119)

k = ledger-event order. (Z.120)

τ_rec = recovery time. (Z.121)

τ_sw = switching time. (Z.122)

---

# Z.77.2 Price and protocol

p_t = market price. (Z.123)

P = protocol. (Z.124)

Pr(·) = probability. (Z.125)

---

# Z.77.3 Return and real coordinate

r_t = return. (Z.126)

R = real or admitted coordinate. (Z.127)

---

# Z.77.4 Residual notation

r_j or ε_j = one residual or error term. (Z.128)

$\mathscr{R}$ = residual register. (Z.129)

Q must not be used for generic residual.

---

# Z.77.5 Boundary notation

B = boundary. (Z.130)

Do not use R for resistance when R already denotes the real coordinate.

---

# Z.78 Capitalization Standard

Capitalize canonical closure periods when used technically:

- Mark;
- Window;
- Structure;
- Event;
- Episode;
- World.

Capitalize canonical functional families:

- Load;
- Motion;
- Constraint;
- Commitment.

Capitalize Proto-Eight roles:

- Gradient;
- Gate;
- Boundary;
- Exchange;
- Trigger;
- Guidance;
- Memory;
- Focus.

Use lowercase when the word is ordinary prose.

Example:

> The court event created a World-level Commitment.

Here “event” is ordinary; “World” and “Commitment” are canonical types.

---

# Z.79 Hyphenation Standard

Use:

- protocol-bound;

- cross-frame;
- cross-protocol;
- time-bearing;
- phase-bearing;
- self-revising;
- real-pair;
- residual-bearing;
- higher-period;
- source-derived;
- present-article;
- World-level;
- Event-level.

Use “Proto-Eight” consistently.

Use “phase time” as a noun and “phase-time” as an adjective.

Examples:

phase time $\tau_i$;
phase-time validation.

---

# Z.80 Strong and Weak Interpretations

Many concepts have graded interpretations.

# Z.80.1 Residual

Weak:

unresolved note.

Intermediate:

typed variable affecting claim strength.

Strong:

persistent state with predictable conversion dynamics.

---

# Z.80.2 Transport

Weak:

robustness check.

Intermediate:

expected transformation across protocols.

Strong:

stable cross-authority relation.

---

## Z.80.3 Episode

Weak:

descriptive segment.

Intermediate:

prospectively gated event grammar.

Strong:

transportable higher-order state.

---

## Z.80.4 Complex state

Weak:

notation for two variables.

Intermediate:

stable geometric relation.

Strong:

phase-bearing operational model.

---

## Z.80.5 World

Weak:

bounded institutional context.

Intermediate:

authority and ledger.

Strong:

time-bearing system with backreaction.

Authors should identify which level they mean.

---

# Z.81 Canonical Claim Ceiling

For any claim, define:

L_claim
≤ min(L_evidence,L_gate,L_transport,L_authority,L_model). (Z.131)

This means a strong result in one dimension cannot compensate for a missing mandatory dimension.

Examples:

Excellent price evidence

- no legal authority
  → no legal World claim. (Z.132)

Stable R–Q phase

- no gate utility
  → phase-bearing diagnostic only. (Z.133)

Admitted daily Event

- failed weekly transport
  → local Event, not broad Episode. (Z.134)

---

# Z.82 Falsification

# Canonical definition

**Falsification** is the failure of a predeclared claim, relation, or model under conditions that the claim says should support it.

A losing trade does not automatically falsify an indicator theory.

A profitable trade does not automatically validate it.

The relevant object must be specified.

Examples:

- gate hypothesis falsified;
- residual-increment hypothesis not supported;
- complex-priority claim falsified;
- universal transport claim reduced to locality.

---

# One-case versus distributional falsification

Some claims are deterministic.

One counterexample may be decisive.

Other claims are probabilistic.

They require distributional evaluation.

The claim type determines the falsification rule.

---

# Z.83 Empirical Validation

# Canonical definition

**Validation** means that a declared object or relation performs adequately under a specified empirical test.

Validation is always scoped by:

- data;
- protocol;
- period;
- outcome;
- benchmark;
- tolerance.

Avoid:

“the theory is validated.”

Prefer:

“the Event-gate model improved calibration for daily equity breakouts under the tested protocol.”

---

# Z.84 Analogy

# Canonical definition

An **analogy** maps one relation or organizational pattern into another domain.

A valid analogy states:

- what is preserved;
- what changes;
- what is excluded;
- what would justify a stronger mapping.

Analogy ≠ Ontology. (Z.135)

Analogy ≠ Mechanism. (Z.136)

Analogy may generate hypotheses.

It does not validate them.

---

# Z.85 Ontology

## Canonical definition

An **ontology** specifies what kinds of entities and relations a framework permits.

PPMG ontology includes:

- protocol;
- evidence;
- projection;
- boundary;
- claim;
- gate;
- residual;
- trace;
- ledger;
- transport;
- authority;
- revision.

The ontology is an engineering and research proposal.

It is not a claim that reality fundamentally consists only of these entities.

---

# Z.86 Model

## Canonical definition

A **model** is a structured representation used to calculate, classify, explain, or predict.

Models may be:

- descriptive;

- diagnostic;
- predictive;
- causal;
- normative;
- governance-oriented.

The Periodic Grammar is primarily:

- taxonomic;
- governance-oriented;
- research-generative.

It is not by itself a complete return-prediction model.

---

# Z.87 Framework

# Canonical definition

A **framework** organizes several models, protocols, and claims without necessarily specifying one complete mechanism or equation.

The Periodic Grammar is a framework because it:

- hosts diverse indicators;
- assigns types;
- controls promotion;
- preserves residual;
- supports multiple domain models.

Calling it a framework should not be used to avoid falsification.

Its components have specific empirical tests.

---

# Z.88 Runtime

# Canonical definition

A **runtime** is the ordered execution process through which declared inputs become governed outputs.

The canonical market runtime is:

Declare
→ FreezeEvidence

→ Project
→ Type
→ DetectCandidate
→ Gate
→ Trace + Residual
→ Transport
→ Promote or Reduce
→ Revise. (Z.137)

A runtime is not merely a conceptual diagram.

It must define:

- input contract;
- state transitions;
- output contract;
- failure handling.

---

# Z.89 Kernel

# Canonical definition

A **kernel** is the compact set of rules required to execute the runtime reliably.

The market kernel contains:

- declaration;
- operator registry;
- state machine;
- gate;
- residual;
- ledger;
- transport;
- revision.

Kernel ≠ Long prompt. (Z.138)

Kernel ≠ Indicator formula. (Z.139)

It is the minimum governed execution structure.

---

# Z.90 Benchmark

# Canonical definition

A **benchmark** is a standardized corpus and evaluation procedure for comparing implementations.

The PPMG benchmark should contain:

- frozen evidence;
- protocol;
- projections;
- labels;
- gate;
- residual;
- outcome;
- revision history.

Benchmark performance does not itself establish market mechanism.

It measures defined task performance.

---

# Z.91 Standard

# Canonical definition

A **standard** specifies valid objects, required fields, allowed transitions, and conformance levels.

PPMG standardization aims to make claims interoperable and auditable.

Conformance means:

- well formed;
- protocol-linked;
- gate-governed;
- residual-visible;
- trace-preserving.

Conformance ≠ Truth. (Z.140)

But nonconformance may prevent reliable evaluation.

---

# Z.92 Research Programme

# Canonical definition

A **research programme** is an ordered family of studies in which later claims depend on

earlier validated structures.

The article's ladder is:

Projection
→ Typing
→ Gate
→ Residual
→ Transport
→ Episode
→ Reflexivity
→ Complex State
→ Phase Time
→ Time-Bearing World. (Z.141)

The programme is not evidence that all levels will succeed.

It specifies how they should be tested.

---

# Z.93 Canonical Compact Dictionary

| Term | Canonical meaning | Must not be confused with |
|---|---|---|
| Field | larger possibility domain | physical field by default |
| Protocol | operational declaration | informal analytical style |
| Declaration | constrained object formation | arbitrary invention |
| Projection | operator output | total market truth |
| Mark | smallest admitted record | higher-period Event |
| Window | aggregation closure | closure period generally |
| Structure | persistent relation | committed transition |
| Event | gated transition | signal or crossing |
| Episode | persistent event grammar | large pattern |
| World | authoritative ledgered system | long market regime alone |
| Load | accumulated state | direction |
| Motion | change or relation | admission |
| Constraint | boundary or admissibility limit | punishment |
| Commitment | gated historical admission | success |
| Gate | admission rule | outcome |
| Trace | persistent committed record | any log entry |
| Ledger | ordered consequential trace system | mutable database |
| Residual | unresolved content | error or Q |
| Invalidation | declared claim failure | any adverse movement |
| Promotion | new closure after new gate | relabelling |
| Transport | expected cross-protocol mapping | identical appearance |
| Invariance | relation survival | universal truth |
| Objectivity | reproducible transport stability | observer absence |
| Authority | legitimate commitment scope | confidence |
| Backreaction | recorded state alters later action | mere sequence |
| Revision | trace-preserving change | history erasure |
| Recursive | output re-enters process | simple repetition |

| Term | Canonical meaning | Must not be confused with |
|---|---|---|
| Fractal | scaling/self-similarity structure | any recursion |
| $\chi$ | feedback orientation | $\Xi$ or volatility |
| $\Xi$ | loading–lock-in–agitation interface | ontology |
| R | admitted/real coordinate | return by default |
| Q | conjugate coordinate | residual, loss, volatility |
| Z | complex state | proof of complex dynamics |
| $\theta$ | phase orientation | time |
| $\tau_i$ | accumulated internal phase time | calendar time |
| k | ledger-event order | clock duration |
| Indicator | partial instrument | market |
| Signal | interpreted warning/candidate | Event |
| Confirmation | incremental relevant evidence | indicator count |
| Breakout | gated boundary transition | any crossing |
| Divergence | relational disagreement | reversal |
| Proto-Eight | actuation-role grammar | universal chronology |
| Grammar | typed composition rules | physical law |
| Periodic | recurrence across closure levels | fixed cycle |

---

# Z.94 Compact Prohibition Dictionary

The following statements are invalid unless heavily qualified.

```
The indicator is the market.
The crossing proves the breakout.
The breakout proves a new Episode.
The price collapse is legal default.
The residual is Q.
Q is the expected loss.
Phase is time.
The complex plane proves quantum behaviour.
The trigrams encode Technical Analysis historically.
The protocol can be changed after failure without a new version.
A profitable outcome validates the original method.
Cross-frame disagreement proves the local signal was meaningless.
High confidence gives the model authority.
Objectivity requires removing all observers.
```

---

# Z.95 Minimum Terminology Audit

Before releasing any section, verify:

1. Is every Event linked to a gate?
2. Is every Episode linked to an event grammar?
3. Is every World linked to authority and ledger?
4. Is every Q independently defined?
5. Is every phase claim downstream of complex eligibility?
6. Is every time claim compared with another clock?

7. Is every residual typed?
8. Is every revision versioned?
9. Is every broad claim transported?
10. Is every analogy prevented from becoming identity?

---

# Z.96 Canonical Master Formula

The framework's vocabulary can be assembled into one governed expression:

ReadableMarketClaim_P
= DeclaredProjection_P

- TypedClosure_P
- GatedCommitment_P
- VisibleResidual_P
- LedgeredTrace_P
- TransportedScope_P
- AdmissibleRevision_P. (Z.142)

For advanced state analysis:

AdvancedMarketClaim_P
= ReadableMarketClaim_P

- Eligibleχ_P
- EligibleΞ_P
- OptionalComplexState_P
- OptionalPhaseTime_P. (Z.143)

The word **optional** is essential.

The basic market grammar does not depend on complex phase.

---

# Z.97 Final Canonical Runtime

The full terminology-controlled runtime is:

UndeclaredField
→ Declaration
→ Filtration
→ Projection
→ Typing
→ Structure
→ Candidate

→ Gate
→ Event
→ Trace + Residual
→ Ledger
→ Transport
→ EpisodePromotion
→ AuthorityGate
→ World
→ Backreaction
→ AdmissibleRevision. (Z.144)

Not every analysis traverses the whole sequence.

A simple moving-average interpretation may stop at Structure.

A breakout study may stop at Event.

A legal recognition analysis may begin with institutional evidence and proceed directly toward a World gate.

---

# Z.98 Appendix Z Conclusion

The Periodic Grammar depends on disciplined separation.

The framework becomes incoherent when:

- projections are promoted into reality;
- crossings are promoted into Events;
- Events are promoted into Episodes;
- Episodes are promoted into Worlds;
- residual is absorbed into Q;
- phase is promoted into time;
- confidence is promoted into authority;
- revision is used to erase failure.

Its canonical distinctions are:

Indicator ≠ Market. (Z.145)

Signal ≠ Event. (Z.146)

Event ≠ Episode. (Z.147)

Episode ≠ World. (Z.148)

Event ≠ Trace. (Z.149)

Trace ≠ Ledger. (Z.150)

Gate ≠ Outcome. (Z.151)

Commitment ≠ Exhaustion. (Z.152)

Residual ≠ Error. (Z.153)

Residual ≠ Q. (Z.154)

Q ≠ Loss. (Z.155)

Phase ≠ Time. (Z.156)

$\tau_i \neq t$. (Z.157)

$k \neq t$. (Z.158)

$\chi \neq \Xi$. (Z.159)

Complex notation ≠ Complex dynamics. (Z.160)

Recursion ≠ Exact fractality. (Z.161)

Period ≠ Timeframe. (Z.162)

Transport survival ≠ Universal truth. (Z.163)

Backreaction ≠ Mere correlation. (Z.164)

Revision ≠ Retrospective relabelling. (Z.165)

Proto-Eight topology ≠ Universal chronology. (Z.166)

CAPM quadrature ≠ The six-period closure ladder. (Z.167)

The article's final lexical rule is:

> Use the strongest term only when the object has passed the gate that gives that term its meaning.

And its final scientific rule is:

> Whenever terminology begins to carry more certainty than the evidence, reduce the word before enlarging the theory.

# Appendix AA — Peer-Review Stress Test: Major Objections, Concessions, and Revision Gates

# AA.1 Purpose

A theory should not be evaluated only through its most sympathetic reading.

The Periodic Grammar combines:

- Technical Analysis;
- protocol-bound observation;
- recursive closure;

- Proto-Eight actuation roles;
- residual governance;
- transport and invariance;
- complex financial geometry;
- internal phase time;
- institutional World formation.

That breadth creates explanatory reach, but it also creates obvious risks:

- renaming established concepts;
- analogical inflation;
- excessive taxonomy;
- unfalsifiable flexibility;
- decorative mathematics;
- historical overreach;
- premature standardization;
- insufficient empirical evidence.

This appendix presents the strongest likely objections to the article and defines how the framework should respond.

The objective is not to defend every existing claim.

It is to determine which claims should be:

- preserved;
- narrowed;
- tested;
- reduced;
- separated;
- or removed.

The governing principle is:

StrongTheory
= ExplanatoryReach

- ExplicitLimits
- SuccessfulStressTests
- WillingnessToReduce. (AA.1)

A theory that survives only by weakening every objection into a misunderstanding has not survived peer review.

---

# AA.2 The Steelman Rule

Every objection should first be stated in its strongest credible form.

The response process is:

Objection
→ StrongestForm
→ Concession
→ SurvivingClaim
→ RequiredTest
→ RevisionDecision. (AA.2)

The article should avoid three defensive habits.

## AA.2.1 Semantic escape

Changing the meaning of a term after criticism.

## AA.2.2 Scope escape

Retreating from a universal claim to a local claim without acknowledging that the original claim was too strong.

## AA.2.3 Future-research escape

Answering every evidential objection by saying that future research may eventually support the claim.

Future research may justify retaining a hypothesis.

It does not count as current evidence.

---

# AA.3 Objection Severity

Not all criticisms have the same consequence.

Define conceptual objection severity:

$$S_O = Scope_O \times Centrality_O \times EvidenceGap_O \times Irreparability_O. \quad (AA.3)$$

where:

- $Scope_O$ measures how much of the framework is affected;
- $Centrality_O$ measures whether the objection reaches the core architecture;
- $EvidenceGap_O$ measures how far the claim exceeds available support;

- Irreparability_O measures whether narrowing or reformulation can repair it.

A useful classification is:

| Severity | Meaning | Required response |
|---|---|---|
| **S1** | editorial ambiguity | clarify terminology |
| **S2** | local modelling weakness | revise one component |
| **S3** | unsupported major branch | demote or suspend branch |
| **S4** | core architecture threatened | redesign or reduce framework |
| **S5** | contradiction or non-falsifiability | withdraw claim |

---

# AA.4 The Core Versus the Extensions

Peer review should distinguish the framework's minimal core from its advanced branches.

# Minimal core

Protocol
→ Projection
→ Candidate
→ Gate
→ Trace + Residual
→ Outcome
→ Revision. (AA.4)

# Intermediate architecture

Minimal core

- six closure periods
- four functional families
- transport
- Episode promotion. (AA.5)

# Advanced architecture

Intermediate architecture

- Proto-Eight actuation
- χ
- Ξ
- complex state
- internal phase time
- World backreaction. (AA.6)

A criticism of phase time does not automatically invalidate protocol-bound gate analysis.

A criticism of the six-period table does not automatically invalidate residual preservation.

Conversely, success of the minimal core does not validate the advanced architecture.

---

# AA.5 Objection 1 — “This Is Merely Renaming Existing Concepts”

# Strongest form

The framework redescribes established ideas:

- protocol = model specification;
- gate = decision threshold;
- residual = error term;
- transport = robustness test;
- Episode = regime;
- ledger = audit log;
- revision = version control.

If so, the framework adds vocabulary without adding science.

---

# Concession

This objection is partly correct.

Several PPMG concepts overlap strongly with mature methods.

For narrow applications:

Gate
may reduce to
SequentialDecisionThreshold. (AA.7)

Transport
may reduce to
RobustnessAnalysis. (AA.8)

Ledger
may reduce to
VersionedEventLog. (AA.9)

Residual
may reduce to
ModelErrorVector. (AA.10)

The article should not claim novelty merely because it introduces a new name.

## Surviving claim

The strongest possible non-redundant contribution is not any individual term.

It is the integrated discipline of keeping the following separations simultaneously visible:

Projection ≠ Commitment. (AA.11)

Event ≠ Episode. (AA.12)

MarketRecognition ≠ InstitutionalRecognition. (AA.13)

Residual ≠ ConjugateCoordinate. (AA.14)

Revision ≠ HistoricalErasure. (AA.15)

The framework earns value only if this integration improves:

- analytical consistency;
- claim calibration;
- cross-domain translation;
- residual recall;
- revision integrity.

---

## Required test

Compare:

MatureDomainPractice (AA.16)

with:

MatureDomainPractice + PPMG Governance. (AA.17)

The comparison should measure:

- reproducibility;
- overpromotion;
- residual preservation;
- authority errors;
- revision quality.

---

## Revision gate

If PPMG adds no measurable value beyond ordinary model cards, audit logs, and robustness checks, it should be reduced to an editorial checklist rather than presented as a distinct scientific

framework.

---

## AA.6 Objection 2 — “The Periodic-Table Analogy Is Misleading”

## Strongest form

The term “periodic table” suggests:

- naturally occurring elements;
- objective atomic structure;
- exact recurrence;
- predictive gaps;
- universal ordering.

The proposed table contains none of these established properties.

It may borrow the authority of chemistry without possessing a comparable periodic law.

---

## Concession

The table is not analogous to chemistry in:

- substance identity;
- atomic number;
- conservation;
- electron structure;
- experimentally established periodic law.

The article must not describe indicators as literal elements.

---

## Surviving claim

The table is **proto-periodic** only in the restricted sense that:

1. several analytical functions recur across closure depths;
2. comparable cells can be defined;
3. method clusters can be located;
4. sparse cells may suggest instrument-design questions.

Thus:

ProtoPeriodic
= RecurrentRoleStructure

- ComparableCells
- GenerativeMissingCellHypotheses. (AA.18)

This is a taxonomic proposal.

It is not a discovered natural law.

---

# Required test

The table must outperform simpler classifications in at least one of:

- method typing;
- inter-analyst agreement;
- failure diagnosis;
- prediction of missing variables;
- design of useful new instruments.

---

# Revision gate

If the table does not improve any of these tasks, the article should use:

“closure-function matrix”

rather than:

“proto-periodic table.”

---

# AA.7 Objection 3 — “The Six Closure Periods Are Arbitrary”

# Strongest form

Why six?

Why not:

- three levels;
- five levels;
- eight levels;
- a continuous hierarchy?

The distinctions among Window, Structure, Event, Episode, and World may reflect author preference rather than empirical structure.

---

# Concession

The exact six-level ladder is a present-article construction.

It has not yet been established as:

- minimal;
- unique;
- exhaustive;
- universally applicable.

The ladder should therefore be described as a candidate closure taxonomy.

---

# Surviving claim

The six levels separate distinctions commonly collapsed in market commentary:

- datum versus aggregation;
- persistent relation versus transition;
- transition versus sequence;
- market sequence versus authoritative institutional state.

Even if the final empirical taxonomy contains fewer or more levels, these distinctions remain analytically important.

---

# Required test

Competing models should include:

## Three-level model

Observation
→ Event
→ Regime. (AA.19)

## Four-level model

Mark
→ Structure
→ Event
→ World. (AA.20)

## Continuous closure score

$c \in [0,1]$. (AA.21)

The six-period model should be retained only if it improves:

- annotation reliability;
- claim calibration;
- failure diagnosis;
- model transport.

---

## Revision gate

Adjacent levels should be merged if annotators cannot distinguish them reliably or if separate labels add no predictive or governance value.

---

# AA.8 Objection 4 — “The Four Functional Families Are Neither Minimal nor Exhaustive”

## Strongest form

Load, Motion, Constraint, and Commitment may omit:

- information;
- value;
- expectation;
- agency;
- exchange;
- uncertainty;
- energy;
- control.

Alternatively, they may be broad enough to absorb everything and therefore explain nothing.

---

## Concession

The four families are high-level observational roles.

They are not a complete list of market mechanisms.

For example:

- information may contribute to Load, Motion, or Trigger;

- agency may appear through authority and actuation roles;
- exchange is separated in Proto-Eight rather than the four-family layer.

---

## Surviving claim

The four families are intended to answer four restricted questions:

1. What state is carried?
2. What is changing?
3. What limits or shapes the change?
4. What becomes committed?

They are governance-oriented analytical functions.

They are not an ontology of all financial substance.

---

## Required test

Compare the four-family taxonomy against:

- trend/momentum/volatility/volume;
- state/change/boundary/decision;
- predictive feature clusters;
- unsupervised method groupings.

---

## Revision gate

If two families cannot be separated empirically, merge them.

If a recurring irreducible function is missing, add it.

The article should not preserve four merely because four has symbolic appeal.

---

# AA.9 Objection 5 — “Proto-Eight Adds Cultural Decoration Rather Than Scientific Value”

## Strongest form

The eight-role crosswalk may be an attractive reinterpretation of 先天八卦, but it is not required for Technical Analysis.

The core framework already contains:

- Load;
- Motion;
- Constraint;
- Commitment.

Adding eight roles may increase complexity without improving explanation or prediction.

---

## Concession

The Periodic Grammar does not require Proto-Eight to operate.

The core gate-and-ledger architecture remains complete enough to test without:

- Gradient;
- Exchange;
- Trigger;
- Guidance;
- Memory;
- Focus

as formal labels.

The P8D source itself frames its model as a small, testable engineering model rather than fundamental physics.

---

## Surviving claim

Proto-Eight may provide a useful **actuation diagnostic** by distinguishing paired failure modes:

- Gradient without Gate;
- Trigger without Guidance;
- Boundary without Exchange;
- Memory without Focus.

That value is empirical, not guaranteed.

---

## Required test

Test whether Proto-Eight labels improve:

- intervention diagnosis;

- failure classification;
- cross-domain translation;
- design of missing instruments.

Compare against the four-family model alone.

---

## Revision gate

If the eight-role layer adds no incremental value, it should remain:

- a historical-philosophical appendix;
- an optional engineering checklist;

not a required part of PPMG.

---

# AA.10 Objection 6 — “The Historical Interpretation of 先天八卦 Is Unsupported”

## Strongest form

The framework may project modern ideas—complex geometry, recursive systems, software runtimes, market dynamics—onto classical diagrams without sufficient textual or historical evidence.

---

## Concession

The article does not establish that classical authors intended:

- Technical Analysis;
- CAPM;
- field theory;
- runtime kernels;
- internal phase time.

The present use is a modern structural reconstruction.

The source critique itself argues that a BaGua topology may be insufficient by itself to generate a complete recursive world.

---

## Surviving claim

A classical topology may inspire an engineering role grammar.

That is legitimate when labelled as:

- reinterpretation;
- reconstruction;
- structural analogy;
- modern reuse.

It must not be labelled as historical discovery unless supported by primary textual scholarship.

---

## Required revision

Every historical section should separate:

HistoricalEvidence
from
ModernStructuralInterpretation
from
EngineeringApplication. (AA.22)

---

## Revision gate

Unsupported historical claims should be removed without affecting the financial framework.

---

## AA.11 Objection 7 — “Quantum Language Creates Unwarranted Prestige”

## Strongest form

Words such as:

- observer;
- collapse;
- phase;
- gauge;
- field;
- conjugate;
- quantum

may make ordinary financial modelling sound deeper than it is.

The language risks confusing analogy with mechanism.

---

# Concession

Many financial processes can be represented using ordinary:

- probability;
- decision theory;
- state-space models;
- accounting rules;
- information constraints.

Quantum terminology is not necessary for most of the PPMG core.

---

# Surviving claim

Three distinct uses should remain separated.

### Mathematical use

Complex coordinates and rotations may be exact mathematical constructions.

### Structural use

Bounded observation, incompatible projections, and commitment may share formal features with measurement architectures.

### Physical claim

Markets literally instantiate quantum physical dynamics.

The article supports the first in specific constructions and explores the second.

It does not establish the third.

---

# Required editorial rule

Every use of quantum language should identify:

- mathematical;
- structural;
- analogical;
- or physical status.

---

# Revision gate

If a section can be stated equally well without quantum terminology, the ordinary statement should appear first.

---

# AA.12 Objection 8 — “Any Two Real Variables Can Be Written as a Complex Number”

## Strongest form

Writing:

Z = R + iQ (AA.23)

adds no information.

The complex representation may be decorative because:

(R,Q) (AA.24)

contains exactly the same numerical information.

---

## Concession

This objection is mathematically correct.

Complex notation alone adds no empirical content.

---

## Surviving claim

Complex priority is earned only when the representation supports useful structure such as:

- a stable generator;
- meaningful rotational coupling;
- phase-order simplification;
- invariant amplitude;
- gate alignment;
- improved modelling relative to a flexible real pair.

The phase source explicitly requires reduction when the complex form adds no stable operational gain.

---

## Required test

Compare:

M_real = f(R,Q) (AA.25)

with:

$M_{complex} = g(A,\theta)$. (AA.26)

under equal data and comparable flexibility.

---

## Revision gate

If $M_{complex}$ does not improve interpretation, compression, prediction, or gating:

$Z \rightarrow (R,Q)$. (AA.27)

The complex language should then be removed from that application.

---

# AA.13 Objection 9 — "CAPM Q Is Defined by Construction and Therefore Economically Empty"

## Strongest form

If:

$Q = \sqrt{(A^2 - R^2)}$, (AA.28)

then Q is merely a mathematical remainder.

It may have no independent economic meaning.

---

## Concession

The identity itself does not prove that Q corresponds to:

- traded risk;
- causal force;
- behavioural pressure;
- hidden market state.

Q is exact geometrically because it is defined from A and R.

That is not the same as independent empirical discovery.

---

## Surviving claim

Q can still be economically informative as:

- a compact sensitivity coordinate;
- a local conjugate exposure;

- the derivative of haircut with respect to phase under the declared geometry.

But its interpretation must remain internal to that construction unless additional empirical evidence is provided.

The CAPM source explicitly distinguishes Q from realized loss, volatility, beta, VaR, and other standard risk quantities.

---

## Required test

Test whether Q improves:

- sensitivity analysis;
- scenario comparison;
- term-structure interpretation;
- gate-relevant valuation diagnostics

beyond A and R reported directly.

---

## Revision gate

If Q provides no operational gain, retain the Pythagorean identity as a mathematical reformulation and remove stronger economic language.

---

# AA.14 Objection 10 — "There Is No Unified Technical Analysis Q"

## Strongest form

In different sections, Q appears to resemble:

- breadth pressure;
- trapped positioning;
- retained risk;
- unrealized possibility;
- conjugate exposure.

These are not the same variable.

The framework may create false unity by assigning one letter to unrelated quantities.

---

## Concession

No universal Technical Analysis Q has been established.

The article must use indexed candidate coordinates:

Q_breadth;
Q_positioning;
Q_CAPM;
Q_phase;
Q_liquidity. (AA.29)

These cannot be assumed equivalent.

---

## Surviving claim

The general research question is whether some market systems contain a pair of coordinates related strongly enough to justify complex treatment.

That is a programme, not a completed unification.

---

## Required revision

Every Q must declare:

- definition;
- units;
- source;
- metric;
- coupling;
- benchmark;
- scope.

---

## Revision gate

Any unindexed generic “market Q” should be removed until a transport or derivation relation among Q candidates is established.

---

# AA.15 Objection 11 — “Phase Time Is Merely Rebranded Event Count”

## Strongest form

Unwrapped phase may correlate with:

- elapsed time;

- cumulative volume;
- cumulative volatility;
- event count;
- normalized Episode age.

It may add no distinct temporal structure.

---

## Concession

Phase time has no priority by definition.

It must compete against simpler clocks.

---

## Surviving claim

Phase time is supported only if it provides better prospective alignment or gate prediction than:

- calendar time;
- event count;
- cumulative volume;
- cumulative volatility;
- selection depth;
- flexible monotone warping.

---

## Required test

$$D_{\tau_i} < \min(D_t, D_k, D_V, D_{vol}, D_\sigma, D_{flexible}). \quad (AA.30)$$

The comparison must be:

- out of sample;
- online;
- branch robust;
- scaling robust.

---

## Revision gate

If event count or another simpler clock performs equally well:

PhaseTime

→ EventOrder or SimplerClock. (AA.31)

---

# AA.16 Objection 12 — “Residual Is a Catch-All That Makes the Theory Unfalsifiable”

## Strongest form

Whenever a prediction fails, the framework can say:

- residual was too high;
- an omitted residual mattered;
- residual converted unexpectedly;
- the protocol was incomplete.

Residual then becomes an all-purpose explanation.

---

## Concession

An untyped residual is scientifically weak.

“Residual pressure” cannot serve as a universal post hoc excuse.

---

## Surviving claim

Residual is useful only when it is:

- declared before outcome where possible;
- typed;
- linked to evidence;
- assigned a claim effect;
- given a resolution or invalidation rule.

The source declaration framework treats residual as an explicit incompleteness record rather than an unbounded explanatory bucket.

---

## Required test

Compare models with and without preregistered residual variables.

Measure:

- incremental calibration;

- diagnostic value;
- revision improvement;
- annotation reliability.

---

## Revision gate

Residual categories that cannot be reliably identified or shown to matter should be removed.

---

# AA.17 Objection 13 — “Gate Is Merely Another Word for Threshold”

## Strongest form

A breakout close, probability cutoff, or legal criterion is simply a threshold.

Calling it a gate adds no substance.

---

## Concession

In many narrow models:

Gate = Threshold + DecisionRule. (AA.32)

No new term is required if nothing else changes.

---

## Surviving claim

The gate concept becomes non-redundant when it additionally identifies:

- authority;
- state transition;
- trace;
- residual;
- future consequences.

Thus:

Gate
= AdmissionRule

- StateChange
- Authority

- TraceContract. (AA.33)

---

## Required test

Assess whether gate records improve:

- outcome calibration;
- auditability;
- institutional translation;
- revision integrity.

---

## Revision gate

Where no consequential state transition exists, use “threshold,” not “gate.”

---

# AA.18 Objection 14 — “The Close Is an Arbitrary Ritual”

## Strongest form

Treating a daily close as a privileged gate may be conventional rather than economically meaningful.

Different markets have:

- continuous trading;
- multiple venues;
- after-hours sessions;
- settlement prices;
- auctions;
- twenty-four-hour operation.

---

## Concession

The close is protocol-dependent.

It has no universal metaphysical privilege.

---

## Surviving claim

A close may function as a gate because it:

- completes a declared Window;
- aggregates an observation interval;
- often carries institutional reporting significance;
- may affect margin, valuation, or benchmark procedures.

The relevant close must be specified:

- regular-session close;
- official settlement;
- UTC daily close;
- auction print;
- accounting measurement time.

---

## Required test

Compare alternative bar and close rules.

A strong Event should either:

- survive admissible alternatives;
- or be explicitly localized to one rule.

---

# AA.19 Objection 15 — “World Is an Inflated Term for an Institutional State”

## Strongest form

Calling an accounting, legal, or contractual state a “World” sounds metaphysical and unnecessarily grand.

---

## Concession

“World” should not be used for every institutional record.

Many such records are simply:

- decisions;
- classifications;
- events;
- ledger entries.

---

## Surviving claim

World is justified only when a bounded system contains:

- authority;
- recognition rules;
- persistent ledger;
- changed future admissibility;
- backreaction.

Thus:

World
= RuleGovernedActionSpace

- ConsequentialHistory. (AA.34)

The term denotes an operational action-generating environment, not a separate universe.

---

## Required editorial rule

Use “World-level” only where changed future rights, obligations, or permitted actions are explicit.

---

## Revision gate

If changed admissibility is absent, reduce:

World
→ InstitutionalEvent or LedgerState. (AA.35)

---

# AA.20 Objection 16 — “Transport Is Just Ordinary Robustness Testing”

## Strongest form

Testing across:

- timeframes;
- scales;
- benchmarks;
- subsamples

is already standard robustness analysis.

“Transport” adds unnecessary abstraction.

---

## Concession

Many transport tests are ordinary robustness tests.

The article should acknowledge this directly.

---

## Surviving claim

Transport becomes more specific when it defines:

1. source protocol;
2. target protocol;
3. expected transformation;
4. target-form prediction;
5. transport residual.

Ordinary robustness often asks:

“Does the result remain significant?”

Transport asks:

“What should this claim become under the new frame?” (AA.36)

---

## Required test

Compare transformation-aware transport with unstructured robustness checks.

---

## Revision gate

If no expected transformation can be defined, use “robustness test” rather than “transport.”

---

# AA.21 Objection 17 — “Transport Survival Does Not Establish Objectivity”

## Strongest form

A claim may survive several chosen protocols because those protocols share the same biases.

Cross-frame agreement can still be wrong.

---

## Concession

Transport survival is not metaphysical truth.

It supports only scoped operational stability.

---

## Surviving claim

The article should define objectivity modestly:

Objectivity_{$\mathcal{P}$}
= ReproducibleRelation across declared protocol family $\mathcal{P}$. (AA.37)

The protocol family itself remains challengeable.

---

## Required revision

Replace broad statements such as:

“Transport creates objectivity”

with:

“Transport survival strengthens operational objectivity within the tested protocol family.”

---

# AA.22 Objection 18 — “Backreaction Is Merely a Restatement of Reflexivity”

## Strongest form

Markets responding to beliefs, rules, and institutional actions is already familiar.

The framework may merely rename reflexivity as backreaction.

---

## Concession

Backreaction is not a new general idea.

The term should not be presented as discovery.

---

## Surviving claim

The framework may add value by requiring backreaction claims to identify:

- the committed trace;
- the action channel;

- the affected observer group;
- the changed future state;
- alternative explanations.

Thus:

BackreactionClaim
= LedgerEvent

- ActionChannel
- ChangedDynamics
- IdentificationStrategy. (AA.38)

---

## Revision gate

Without an identified action channel, use:

“subsequent association”

rather than:

“backreaction.”

---

## AA.23 Objection 19 — “The Framework Lacks a Causal Mechanism”

## Strongest form

Load, Motion, Constraint, and Commitment classify observations but do not explain why prices move.

The framework therefore cannot be a market theory.

---

## Concession

The Periodic Grammar is primarily a governance and typing framework.

It does not replace:

- microstructure;
- behavioural mechanisms;
- funding models;
- information economics;
- institutional analysis.

## Surviving claim

A framework may be scientifically useful without being a complete generative mechanism if it improves:

- object definition;
- model composition;
- Event classification;
- causal-study design;
- authority discipline.

The correct positioning is:

PPMG = GovernanceMetaModel, not CompleteMarketMechanism. (AA.39)

---

## Required revision

The article should use “market grammar” or “research architecture” more often than “market dynamics theory” unless mechanism-specific models are supplied.

---

# AA.24 Objection 20 — “The Framework Is Too Broad to Be Falsifiable”

## Strongest form

Because the framework includes:

- taxonomy;
- engineering;
- analogy;
- mathematics;
- institutional theory;
- research design,

any failure in one area can be isolated from the rest.

The whole framework may never face a decisive test.

---

## Concession

The framework is not one proposition.

It cannot be falsified by one experiment.

That is true of many research programmes.

However, this creates a risk of indefinite survival through modular retreat.

---

## Surviving claim

Each component must have its own falsifier and termination condition.

Examples:

Six periods
→ annotation and model-comparison test. (AA.40)

Residual value
→ incremental diagnostic test. (AA.41)

Complex priority
→ real-pair comparison. (AA.42)

Phase time
→ competing-clock test. (AA.43)

Proto-Eight
→ incremental failure-diagnosis test. (AA.44)

---

## Required revision

The article should not ask readers to accept or reject “the whole theory.”

It should publish a claim ledger with component-level statuses.

---

# AA.25 Objection 21 — “The Framework Can Explain Any Outcome Retrospectively”

## Strongest form

After a failed breakout, the framework can say:

- the gate was partial;
- residual remained;
- transport failed;
- the Episode did not promote;
- $\chi$ changed.

After a successful breakout, it can say:

- residual resolved;
- the gate was sufficient;
- transport later passed.

This is retrospective flexibility.

---

## Concession

Without preregistration and frozen state records, the framework is vulnerable to post hoc explanation.

---

## Surviving claim

The runtime architecture specifically requires:

- evidence freeze;
- original gate record;
- predeclared residual categories;
- immutable claim history;
- versioned revision.

The source self-revising framework emphasizes preserving failed traces rather than rewriting them.

---

## Required test

Run prospective or walk-forward studies where:

G_t;
$\mathscr{R}$_t;
Claim_t (AA.45)

are frozen before:

Outcome_{t+h}. (AA.46)

---

## Revision gate

Any empirical study lacking frozen pre-outcome records should be labelled exploratory or retrospective.

---

# AA.26 Objection 22 — “No Predictive Advantage Has Been Demonstrated”

## Strongest form

A sophisticated grammar may improve prose without improving:

- forecasts;
- calibration;
- risk management;
- decisions.

---

## Concession

The present article does not report validated predictive superiority.

It proposes empirical studies.

---

## Surviving claim

The framework may add value in three separate ways.

### Predictive value

Improves outcome prediction.

### Diagnostic value

Explains why cases differ.

### Governance value

Reduces overclaiming, authority errors, and history erasure.

The last two remain useful even if predictive gains are modest.

---

## Required reporting

Every study should specify which value is being tested.

Governance improvement should not be reported as return prediction.

---

## Revision gate

If no predictive, diagnostic, or governance gain appears, the relevant component should be removed.

---

## AA.27 Objection 23 — “The Empirical Programme Is

## Too Large to Execute”

## Strongest form

The framework proposes studies covering:

- breakouts;
- divergences;
- Episodes;
- observer crowding;
- complex states;
- phase time;
- legal and accounting Worlds.

This may be an aspirational research empire rather than an executable programme.

---

## Concession

The full programme is too large for one paper, one researcher, or one immediate dataset.

---

## Surviving claim

The architecture is intentionally modular.

A defensible first programme needs only:

1. typing reliability;
2. one Event gate;
3. residual audit;
4. one transport test.

The strongest near-term publication should stop at Levels 0–4.

---

## Required revision

Advanced branches should be presented as optional future programmes, not as deliverables required for the initial framework to be useful.

---

# AA.28 Objection 24 — “The PPMG Standard Is Premature”

# Strongest form

Standardization should follow mature empirical convergence.

PPMG currently standardizes concepts that remain unvalidated.

---

# Concession

PPMG should not be represented as an established industry standard.

It is a proposed research schema.

---

# Surviving claim

Early schemas can still help by making studies comparable.

The schema should distinguish:

### Stable core fields

- protocol;
- evidence cutoff;
- claim;
- gate;
- residual;
- outcome;
- revision.

### Experimental fields

- Proto-Eight role;
- χ;
- Ξ;
- complex eligibility;
- phase time;
- World recognition vector.

---

# Required revision

Rename:

“PPMG Standard”

as:

“PPMG Draft Research Specification”

until:

- independent implementations exist;
- field definitions stabilize;
- benchmark studies demonstrate utility.

---

# AA.29 Objection 25 — “The Annotation Scheme Will Be Unreliable”

## Strongest form

Human analysts may disagree about:

- whether something is Structure or Event;
- which residuals matter;
- whether an Episode changed;
- which Proto-Eight role applies.

A taxonomy that cannot be reliably annotated cannot support a benchmark.

---

## Concession

Some labels are interpretive and may have low agreement.

This is an empirical risk.

---

## Surviving claim

The annotation process can preserve:

- confidence;
- disputed labels;
- soft gold;
- alternative admissible interpretations.

Not every case needs a single forced label.

---

## Required test

Measure:

- weighted period agreement;
- gate agreement;
- residual Jaccard similarity;
- authority agreement;
- claim-ceiling agreement.

---

## Revision gate

Labels with persistently poor reliability should be:

- operationally redefined;
- merged;
- made optional;
- or removed.

---

# AA.30 Objection 26 — “The Framework Rewards Excessive Conservatism”

## Strongest form

By requiring gates, residuals, transport, and claim ceilings, the framework may classify every signal as:

- partial;
- local;
- unresolved;
- unconfirmed.

It could avoid errors by refusing useful commitment.

---

## Concession

Underpromotion is a real failure.

A system that never admits an Event is not mature.

---

## Surviving claim

Gate quality must balance:

FalseAdmissionRate (AA.47)

against:

FalseRejectionRate. (AA.48)

The objective is:

MinimizeExpectedDecisionLoss, not MinimizeCommitment. (AA.49)

---

## Required metric

Report both:

- overpromotion;
- underpromotion.

---

## Revision gate

A gate whose conservatism destroys useful discrimination should be recalibrated or simplified.

---

# AA.31 Objection 27 — “Practitioners Will Not Use Such a Complex Framework”

## Strongest form

Technical analysts need decisions quickly.

They are unlikely to fill:

- protocol records;
- residual ontologies;
- transport matrices;
- authority fields;
- revision ledgers.

---

## Concession

The complete monograph architecture is too heavy for ordinary chart commentary.

---

## Surviving claim

The practical interface can be compressed to six questions:

1. What is the declared boundary?
2. What has actually happened?
3. Which gate passed?
4. What remains unresolved?
5. What stronger claim is still prohibited?
6. What invalidates the current claim?

The full ontology can remain behind the interface.

---

# Practical compact record

```
Protocol:
Current state:
Gate:
Residual:
Claim ceiling:
Invalidation:
```

---

## Revision gate

If the framework cannot be compressed without losing its central discipline, it is not ready for practical use.

---

# AA.32 Objection 28 — “The Framework May Be Better Suited to AI Governance Than Finance”

## Strongest form

The emphasis on:

- declarations;
- state machines;
- gates;
- immutable trace;
- residual;
- revision

resembles AI-agent governance and software runtime design more than market science.

## Concession

The implementation architecture is heavily influenced by runtime engineering.

Some terminology may fit computational systems more naturally than financial practice.

---

## Surviving claim

Markets and financial institutions also contain:

- protocols;
- observation windows;
- decision thresholds;
- authoritative recognition;
- persistent records;
- revisions.

The relevant question is not where the language originated.

It is whether it maps cleanly and adds value in finance.

---

## Required test

Compare domain-native financial language with PPMG language.

If PPMG reduces clarity, retain the financial terminology and keep PPMG only as a hidden implementation schema.

---

# AA.33 Objection 29 — “The Source Corpus Is Too Internally Self-Referential”

## Strongest form

Much of the framework is developed from a sequence of related internal papers.

The theory may appear coherent because later papers inherit assumptions from earlier ones rather than because independent disciplines support it.

---

## Concession

Internal coherence is weaker than external validation.

A source family that cites and extends itself can create conceptual closure without independent

support.

---

## Surviving claim

The internal sequence is useful as:

- intellectual genealogy;
- formal development path;
- hypothesis generator.

It cannot substitute for comparison with mature external literatures.

---

## Required revision

The final article should integrate primary external sources from:

- market microstructure;
- econometrics;
- sequential analysis;
- survival analysis;
- causal inference;
- accounting;
- legal theory;
- measurement theory;
- dynamical systems.

---

## Revision gate

Any claim described as novel must be checked against external literature before publication.

---

# AA.34 Objection 30 — “The Framework Does Not Adequately Engage Existing Finance”

## Strongest form

The article may appear to reconstruct ideas already found in:

- support/resistance studies;
- breakout confirmation;

- market breadth;
- regime models;
- event studies;
- reflexivity;
- accounting recognition.

Without a rigorous literature review, novelty claims are premature.

---

## Concession

The article's current strength is internal architecture, not completed literature positioning.

---

## Surviving claim

The likely contribution is integrative rather than wholly original at every component.

Its novelty claim should therefore be:

> A unified protocol–closure–gate–residual–transport architecture for organizing heterogeneous financial observations.

That claim must still be compared with existing integrative frameworks.

---

## Required revision

Every main section should include:

- nearest mature comparator;
- what is borrowed;
- what differs;
- non-redundancy test.

---

# AA.35 Objection 31 — "Institutional World Claims Require Domain Experts"

## Strongest form

Accounting, legal, regulatory, and contractual recognition cannot be safely generalized from market analogies.

Errors may have real-world consequences.

---

## Concession

The framework cannot determine:

- accounting treatment;
- legal default;
- regulatory status;
- contractual breach

without domain-specific rules and competent authority.

---

## Surviving claim

PPMG may assist with **cross-ledger translation**:

MarketEvent
→ AccountingCandidate
→ IndependentAccountingGate. (AA.50)

It does not replace the destination discipline.

---

## Required editorial rule

Institutional examples must be reviewed by relevant domain experts before strong publication claims are made.

---

## Revision gate

Absent domain review, examples should remain schematic and explicitly non-advisory.

---

# AA.36 Objection 32 — “The Framework Confuses Normative Rules with Descriptive Science”

## Strongest form

Claims such as:

ClaimLevel ≤ ClosureLevel (AA.51)

are not empirical laws.

They are editorial or governance rules.

The article may present normative discipline as market structure.

---

## Concession

Many PPMG formulas are normative constraints.

They specify how claims **should** be governed.

They do not describe how markets necessarily behave.

---

## Surviving claim

The article should distinguish:

### Descriptive claims

What market systems do.

### Predictive claims

What later outcomes are likely.

### Normative claims

How analysts should form and preserve claims.

### Formal definitions

How article objects are constructed.

---

## Required notation

Normative statements should be labelled:

Rule N1, Rule N2, … (AA.52)

rather than presented as discovered equations.

---

# AA.37 Objection 33 — “The Framework’s Equations Create False Precision”

## Strongest form

Equations such as:

ReadableMarketClaim
= Projection + Closure + Gate + Residual + Ledger (AA.53)

are conceptual schematics, not numerical equalities.

Their mathematical appearance may overstate formality.

## Concession

Many equations in the article are typed composition rules, not algebraic identities.

---

## Surviving claim

Symbolic schematics can clarify architecture when their status is explicit.

Use different notation for different claim types.

### Exact equality

= (AA.54)

### Definition

:= (AA.55)

### Approximation

≈ (AA.56)

### Composition

∘ or structured arrow. (AA.57)

### Conceptual assembly

≔ or labelled “schematic.” (AA.58)

---

## Required revision

Replace conceptual plus signs with typed composition where possible.

For example:

ReadableClaim_P
:= Revise ∘ Transport ∘ Ledger ∘ Gate ∘ Project(D_P). (AA.59)

---

# AA.38 Objection 34 — “The Framework Uses Too Many Capitalized Terms”

## Strongest form

Capitalizing Mark, Window, Structure, Event, Episode, World, Load, Motion, Constraint, Commitment, and eight actuation roles creates jargon density and may obscure ordinary meaning.

---

## Concession

The terminology load is high.

Capitalization is useful only when it distinguishes formal types.

---

## Surviving claim

Canonical capitalization can improve precision in technical sections.

It should be reduced in narrative sections.

---

## Required style rule

Use full canonical typing only when analytically necessary.

Example:

> The daily close admitted an Event-level Commitment.

Later references may say:

> The event then failed.

---

# AA.39 Objection 35 — “The Article Is Too Long to Function as One Paper”

## Strongest form

The manuscript contains:

- theory;
- taxonomy;
- mathematical geometry;
- method atlas;
- standard;
- software architecture;
- benchmark;
- empirical programme;
- visual system;
- editorial governance.

No conventional journal article can evaluate all of this coherently.

## Concession

The current text is a monograph or research-programme document.

It should not be submitted unchanged as one ordinary paper.

---

## Surviving publication structure

The material should be divided into at least four papers.

### Paper I — Core Grammar

Protocol, four functions, six periods, gate, residual, transport.

### Paper II — Technical Analysis Atlas

Method classification and molecular combinations.

### Paper III — Complex Eligibility and CAPM Calibration

R, Q, Z, phase, reduction.

### Paper IV — Empirical and Engineering Programme

Benchmark, runtime, standard, pilot.

Proto-Eight may be:

- integrated cautiously into Paper I;
- or treated in a separate philosophical-engineering paper.

---

# AA.40 Objection 36 — “The Framework Tries to Unify Too Much”

## Strongest form

The project spans:

- trading;
- valuation;
- institutional recognition;
- complex numbers;
- recursive systems;
- classical philosophy.

The framework may be driven by a desire for unity rather than by empirical necessity.

---

## Concession

Conceptual unity can become a selection bias.

A framework may interpret every new domain as another confirmation of itself.

---

## Surviving claim

Unity should be treated as a hypothesis of reusable governance structure, not identity of substance.

The safe form is:

DifferentDomains
may share
Declaration–Gate–Trace–Residual architecture. (AA.60)

The unsafe form is:

DifferentDomains
are fundamentally the same system. (AA.61)

---

## Required test

For every domain, compare:

- PPMG mapping;
- domain-native description;
- incremental value;
- distortion cost.

---

## Revision gate

A domain should be excluded if the mapping obscures more than it clarifies.

---

# AA.41 Objection 37 — “The Framework Has No Clear Primary Audience”

## Strongest form

The manuscript alternately addresses:

- technical analysts;

- quantitative researchers;
- philosophers;
- physicists;
- software engineers;
- accounting and legal readers.

No audience receives an optimally focused argument.

---

# Concession

The full monograph has several audiences.

A journal paper requires one primary reader.

---

# Audience-specific versions

### Technical Analysis version

Lead with indicator failure, confirmation independence, and breakout gates.

### Quantitative version

Lead with protocol, filtration, calibration, transport, and benchmarks.

### Philosophy-of-science version

Lead with observer-bound objectivity, declaration, residual, and revision.

### Engineering version

Lead with runtime kernels, state machines, and immutable ledgers.

### Chinese philosophy version

Lead with 成界之學 and Proto-Eight, while separating historical and modern claims.

---

# Revision gate

Each publication should remove material that does not serve its primary audience.

---

# AA.42 Objection 38 — “The Framework Does Not Produce a Trading Rule”

# Strongest form

Technical Analysis is often used for trading.

A framework that refuses to issue positions, stops, or expected returns may be practically irrelevant.

---

## Concession

The present article does not define a complete trading strategy.

It does not specify:

- position sizing;
- transaction costs;
- portfolio construction;
- risk limits;
- execution.

---

## Surviving claim

A market-observation framework can still improve the input to trading decisions.

The correct stack is:

PPMG EventState
→ DecisionModel
→ PortfolioRule
→ ExecutionRule. (AA.62)

PPMG governs the first stage.

It does not replace the remaining stages.

---

## Required disclaimer

Event calibration must not be presented as investment-performance evidence.

---

# AA.43 Objection 39 — “The Framework May Be Unnecessarily Static”

## Strongest form

The table places methods into fixed cells, while real market objects may change role dynamically.

For example, VWAP can function as:

- memory;

- boundary;
- guidance;
- gate reference.

---

## Concession

Methods should not be permanently assigned to one cell.

The relevant type is protocol- and task-dependent.

---

## Surviving claim

The table classifies **uses**, not immutable method identities.

Define:

Type(Method | P,Task,State). (AA.63)

not:

Type(Method) alone. (AA.64)

---

## Required revision

Method cards should include:

- primary role;
- secondary roles;
- context-dependent role changes;
- prohibited promotions.

---

# AA.44 Objection 40 — “The Framework’s Core May Be Merely Good Research Hygiene”

## Strongest form

Declare the protocol, avoid leakage, preserve failed signals, compare simpler models, and record uncertainty.

These are ordinary principles of good science.

---

## Concession

A substantial portion of the core is disciplined research hygiene.

That is not a weakness if the framework makes neglected hygiene operational.

But it limits novelty claims.

---

## Surviving claim

The distinctive possibility lies in connecting research hygiene to:

- closure depth;
- Event admission;
- residual lifecycle;
- cross-authority ledgers;
- recursive revision.

---

## Required positioning

The article should claim:

“formalized and integrated research discipline”

rather than:

“entirely unprecedented scientific foundation.”

---

# AA.45 Fatal Objections Versus Repairable Objections

| Objection | Fatal if confirmed? | Repair path |
|---|---|---|
| six periods not unique | No | merge or revise levels |
| four families not minimal | No | revise taxonomy |
| Proto-Eight adds no value | No | make optional |
| complex representation adds no value | No | reduce to real pair |
| phase time equals simpler clock | No | retain simpler clock |
| residual annotations unreliable | Partly | reduce ontology |
| gates add no value | Major | retain projection-only core |
| no governance improvement | Major | reduce PPMG to documentation |
| historical claims unsupported | No | remove historical claims |
| institutional mappings incorrect | Major locally | require expert review |
| all outcomes explainable post hoc | Yes for scientific claims | preregister and freeze traces |
| no falsifiable component claims | Yes | rebuild claim ledger |

| Objection | Fatal if confirmed? | Repair path |
|---|---|---|
| source corpus contradicts itself materially | Potentially | expose and resolve divergences |
| entire framework adds no value over mature practice | Yes | withdraw distinct-framework claim |

# AA.46 Burden-of-Proof Table

| Claim | Minimum evidence |
|---|---|
| Indicator is a projection | conceptual and operator definition |
| Six periods improve analysis | annotation and model comparison |
| Four families recur | cross-method and cross-period evidence |
| Gate improves Event classification | prospective calibration |
| Residual matters | incremental diagnostic or predictive value |
| Transport strengthens scope | preregistered cross-protocol tests |
| Episode grammar exists | prospective sequence segmentation |
| Proto-Eight adds value | incremental failure diagnosis |
| $\chi$ improves interpretation | regime-conditioned out-of-sample gain |
| $\Xi$ improves control or diagnosis | ablation against source features |
| Complex priority is earned | real-pair benchmark defeated |
| Phase acts as time | competing clocks defeated |
| World trace changes admissibility | authority and causal action channel |
| Framework is cross-domain | independent domain replications |

# AA.47 Claims That Should Be Preserved in the Main Article

The following claims are sufficiently clear and useful to retain as the article's central core.

# AA.47.1 Indicators are partial projections

An indicator does not represent the total market.

# AA.47.2 Candidate and Event must be separated

Crossing alone is not necessarily admission.

# AA.47.3 Commitment and outcome must be separated

A well-formed Event may later fail.

# AA.47.4 Residual should remain visible

Contradictory or missing evidence should not be erased.

## AA.47.5 Revision should preserve prior trace

Model development should not overwrite failed claims.

## AA.47.6 Institutional authorities remain distinct

Market, accounting, contractual, legal, and policy commitments have different gates.

## AA.47.7 Advanced mathematics must defeat simpler alternatives

Complex and phase models require explicit eligibility.

These claims form the strongest defensible backbone.

---

# AA.48 Claims That Should Be Demoted to Hypotheses

The following should not appear as established conclusions.

1. Four families are minimal and universal.
2. Six periods form the uniquely correct closure hierarchy.
3. Proto-Eight roles are necessary and sufficient.
4. Residual burden predicts failure across markets.
5. Transport survival predicts persistence generally.
6. $\chi$ is a stable market-state variable.
7. $\Xi$ provides a superior intervention interface.
8. Technical Analysis possesses a unified Q.
9. Market phase provides internal time.
10. the entire framework forms a periodic law.

These should be labelled:

Proposed;
Testable;
Reducible. (AA.65)

---

# AA.49 Claims That Should Be Removed Unless New Evidence Is Added

Remove or heavily qualify statements implying:

- classical authors formulated modern finance;

- market behaviour is physically quantum;
- complex notation itself reveals hidden reality;
- every market system requires eight roles;
- CAPM Q is a universally traded risk quantity;
- phase directly creates economic time;
- cross-frame survival establishes universal truth;
- one framework supersedes econometrics, microstructure, law, or accounting.

---

# AA.50 Mandatory Revisions Before Journal Submission

## AA.50.1 Narrow the primary claim

Present the paper as a protocol-bound governance grammar for Technical Analysis.

## AA.50.2 Separate exact mathematics from proposed taxonomy

CAPM geometry and PPMG classification should not share one undifferentiated evidential status.

## AA.50.3 Add external literature

The manuscript needs direct engagement with established finance and methodological sources.

## AA.50.4 Reduce the main-text ontology

Keep advanced state layers in supplements.

## AA.50.5 Clarify novelty

State which contribution is:

- integration;
- formalization;
- translation;
- genuinely new hypothesis.

## AA.50.6 Include at least one worked falsifiable example

Preferably the preregistered breakout design.

## AA.50.7 Include a reduction table

Show what remains when advanced branches fail.

## AA.50.8 Change “standard” to “draft specification”

Until independent adoption exists.

## AA.50.9 Separate historical interpretation

Move most classical-material discussion to a dedicated section or companion paper.

## AA.50.10 Include a limitations section near the beginning

Do not defer all limitations to appendices.

---

# AA.51 Recommended Main-Paper Claim

A journal-length paper should make one central claim:

> Technical Analysis can be reconstructed as a protocol-bound system of partial projections whose claims should be typed by closure depth, admitted through explicit gates, accompanied by visible residuals, tested through legitimate reframing, and revised without erasing failed traces.

This claim is broad enough to unify the article.

It is narrow enough to evaluate.

---

# AA.52 Recommended Secondary Claim

A defensible secondary claim is:

> A six-level closure ladder and four-function matrix provide one candidate architecture for organizing Technical Analysis methods and distinguishing Structure, Event, Episode, and institutional World claims.

The phrase “one candidate architecture” is essential.

---

# AA.53 Recommended Advanced-Research Claim

The advanced branch should be stated as:

> Some financial systems may admit useful complex-state and phase-time representations, but such representations should be accepted only after outperforming flexible real-pair and simpler-clock models.

This avoids treating complex phase as already established.

---

# AA.54 Peer-Review Response Template

For each reviewer comment, the response should use:

```
Reviewer objection:

Strongest interpretation:

Concession:

Claim retained:

Claim narrowed or removed:

Text revised in:

New evidence or analysis:

Residual uncertainty:

Future test:
```

The reply should not use citations merely to overpower the objection.

It should show how the claim changed.

---

# AA.55 Revision Classification

Every manuscript revision should be classified.

## Clarification

Meaning unchanged; wording improved.

## Narrowing

Scope or claim strength reduced.

## Correction

A formula, definition, or attribution was wrong.

## Structural revision

Architecture changed.

## Empirical revision

New evidence changed claim status.

## Branch closure

A component was removed as unsupported.

The revision ledger should make these distinctions visible.

---

# AA.56 Reviewer-Gate Decision

Define publication readiness:

Ready
= ConceptualClarity
∧ SourceIntegrity
∧ ClaimSeparation
∧ ExternalComparison
∧ FalsifiableCore
∧ ReductionPaths. (AA.66)

A manuscript should be deferred when any of the following remains absent:

- a clear primary contribution;
- distinction between exact and hypothetical claims;
- mature-literature comparison;
- operational falsifier;
- manageable article scope.

---

# AA.57 Terminal Conditions for the Framework

A responsible research programme must specify when it would abandon or radically reduce the framework.

## Terminal condition T1

The six-period and four-family classifications are not reproducible and add no practical value.

## Terminal condition T2

Explicit gates do not improve classification, calibration, or governance.

## Terminal condition T3

Residual records add no predictive, diagnostic, or revision value.

## Terminal condition T4

Transport is indistinguishable from ordinary robustness and adds no clarity.

## Terminal condition T5

Proto-Eight roles do not improve diagnosis or intervention.

## Terminal condition T6

Complex and phase branches repeatedly fail simpler benchmarks.

## Terminal condition T7

Institutional World mappings generate systematic category errors.

## Terminal condition T8

The framework's complexity exceeds its demonstrated benefit.

Under T1–T4 jointly, the distinct PPMG programme should close.

What may remain is a smaller research-hygiene checklist.

---

# AA.58 Strongest Surviving Minimal Framework

After the full peer-review stress test, the most defensible minimal architecture is:

1. declare the object;
2. preserve information timing;
3. treat indicators as partial projections;
4. distinguish persistent Structure from gated Event;
5. record unresolved evidence;
6. preserve failed claims;
7. separate market evidence from institutional authority;
8. compare every advanced representation with a simpler model.

In compact form:

Declare
→ Project
→ Gate
→ RecordTrace + Residual
→ Compare
→ ReviseWithoutErasure. (AA.67)

This core does not depend on:

- six periods being final;
- Proto-Eight being necessary;

- complex phase succeeding;
- internal time existing.

That independence is a strength.

---

# AA.59 Strongest Surviving Integrated Framework

If the intermediate hypotheses receive support, the broader architecture becomes:

Protocol
→ Projection
→ FourFunctionTyping
→ SixPeriodClosure
→ Gate
→ Residual
→ Transport
→ Episode
→ Authority
→ WorldLedger
→ Revision. (AA.68)

If advanced hypotheses also succeed:

IntegratedAdvancedState
= CoreFramework

- χ
- Ξ
- EligibleZ
- Validated$\tau_i$. (AA.69)

The word **eligible** must remain attached to Z.

The word **validated** must remain attached to $\tau_i$.

---

# AA.60 Revised Abstract After Adversarial Review

# Abstract

Technical Analysis contains a heterogeneous ecology of indicators, chart structures, confirmation practices, and interpretive traditions, but it lacks a common discipline for distinguishing what has merely been measured from what has become structurally persistent, eventfully admitted, episodically organized, or institutionally consequential. This article proposes a protocol-bound grammar in which analytical methods are treated as partial projections and market claims are organized through four recurring functions—Load, Motion, Constraint, and Commitment—across a candidate six-level closure hierarchy from Mark to World. The framework separates candidate signals from admitted Events, preserves unresolved evidence as typed residual, tests broader scope through declared protocol transport, and requires model revision to retain prior failed traces. Proto-Eight actuation roles, χ feedback signatures, Ξ operating states, complex R–Q geometry, and internal phase time are presented as

optional extensions with explicit eligibility and reduction conditions. The article does not claim a validated periodic law, a universal market Q, or a physical quantum theory of finance. Its immediate contribution is a falsifiable research and governance architecture for improving the typing, admission, comparison, and revision of Technical Analysis claims.

---

# AA.61 Revised Epistemic Status Statement

> **Epistemic status.** The protocol–projection–gate–residual core is proposed as a methodological framework. The six-period table and four-function matrix are candidate taxonomies. Proto-Eight is an optional actuation crosswalk. The CAPM R–Q relation is exact within its declared construction, but its wider economic interpretation remains limited. Technical Analysis complex states, phase time, and World-level generalizations are empirical research branches, not established conclusions. Every advanced branch is subject to reduction into a simpler retained model.

---

# AA.62 Appendix AA Conclusion

The strongest criticism of the Periodic Grammar is not that any one component is obviously false.

It is that the framework may become too broad, too elastic, and too symbolically unified to fail cleanly.

The proper response is not stronger rhetoric.

It is stronger separation.

Separate:

Core
from
Extension. (AA.70)

Taxonomy
from
Mechanism. (AA.71)

Exact construction
from
Empirical hypothesis. (AA.72)

Modern reconstruction
from
Historical claim. (AA.73)

Mathematical possibility
from
Operational gain. (AA.74)

Governance rule
from
Market law. (AA.75)

The article should survive peer review by becoming smaller where necessary.

Its central claims should remain only where they perform identifiable work.

Its advanced claims should remain conditional.

Its unsupported claims should be removed without treating removal as defeat.

The decisive review rule is:

> When an objection reveals that two concepts are merely renamed equivalents, reduce them. When it reveals a missing test, add the test. When it reveals an unsupported scope, narrow the claim. When it reveals a contradiction, preserve the contradiction in the ledger rather than rewriting the theory around it.

The framework's strongest possible demonstration of maturity would not be to defend every branch.

It would be to show that its own gate–residual–revision architecture can govern the theory that proposes it.

# Appendix AB — The Minimal Defensible Core: Axioms, Propositions, Hypotheses, and Non-Theorems

## AB.1 Purpose

The preceding appendices expanded the Periodic Grammar into:

- an ontology;
- a runtime;
- a benchmark;
- a draft standard;
- a reference implementation;
- an empirical programme;
- a comparative Rosetta Stone;
- a visual grammar;
- a provenance architecture;
- a peer-review stress test.

Expansion is useful only if the framework can later contract.

This appendix therefore performs the opposite operation.

It asks:

> After terminology control, adversarial testing, comparative analysis, and peer-review reduction, what is the smallest coherent formal core that still remains?

The result is not an axiom system describing what markets fundamentally are.

It is an axiom system governing how claims about markets should be formed, promoted, tested, and revised.

The distinction is essential:

MarketOntology
≠ ClaimGovernanceArchitecture. (AB.1)

The axioms below govern the second.

They do not assert that every market:

- follows six periods;
- contains four natural substances;
- possesses a universal Q;
- generates phase time;
- forms a Proto-Eight topology;
- behaves like a quantum system.

The minimal defensible object is:

PPMG_Core
:= Protocol
∘ Projection
∘ Typing
∘ Gate
∘ TraceResidual
∘ Transport
∘ Revision. (AB.2)

---

# AB.2 Four Formal Statuses

Every statement in this appendix belongs to one of four classes.

# AB.2.1 Governance axiom

A rule adopted to make claims reconstructable and falsifiable.

Example:

> An Event claim requires an explicit gate.

This is not discovered from market data.

It is a normative condition of the framework.

---

# AB.2.2 Derived proposition

A logical consequence of the governance axioms.

Example:

> A projection alone cannot establish an Event.

---

## AB.2.3 Empirical hypothesis

A claim whose truth depends on observed market evidence.

Example:

> Residual burden predicts breakout failure.

---

## AB.2.4 Non-theorem

A statement that the framework does **not** establish.

Example:

> Markets are physically quantum.

The four classes must remain visibly separate.

Axiomatic within the framework
does not mean
empirically true of every market. (AB.3)

---

# AB.3 Primitive Objects

The minimal architecture requires eleven primitive object classes.

Let:

$\mathcal{O}$_min
= {P,E,Ô,X,C,G,T,$\mathscr{R}$,L,$\mathcal{T}$,U}. (AB.4)

where:

P = protocol;
E = admissible evidence;
Ô = projection operator;
X = projected object;
C = claim;
G = gate;
T = trace;
$\mathscr{R}$ = residual register;
L = ledger;
$\mathcal{T}$ = transport operator;
U = revision operator.

Authority A is required whenever commitment extends beyond analytical classification.

Thus the extended object set is:

$\mathcal{O}_{ext} = \mathcal{O}_{min} \cup \{A\}$. (AB.5)

The six closure periods, four functional families, and eight actuation roles are not primitive requirements of the minimal core.

They are optional typing structures layered on top of it.

---

# AB.4 Governance Axiom G1 — Protocol Necessity

Every auditable claim must be interpreted under a declared protocol.

Formally:

$C \Rightarrow \exists P$ such that $C = C_P$. (AB.6)

A minimally sufficient protocol declares:

$P = (B, \Delta, h, \varphi, G_{rule}, \mathscr{R}_{rule}, I_{rule})$. (AB.7)

where:

B = boundary or object scope;
Δ = observation or aggregation rule;
h = horizon;
φ = feature map;
$G_{rule}$ = gate rule;
$\mathscr{R}_{rule}$ = residual rule;
$I_{rule}$ = invalidation rule.

Institutional protocols additionally require:

- authority;
- jurisdiction;
- ledger type;
- admissible intervention.

The declaration source argues that projection, gate, trace, residual, and ledger become meaningful only after the field's boundary, observation rule, horizon, feature map, and commitment conditions are declared.

---

# Consequence

An undeclared chart statement may still be:

- impression;
- exploratory observation;

- informal hypothesis.

It cannot yet be a fully governed claim.

NoProtocol
→ NoAuditableClaim. (AB.8)

---

## AB.5 Governance Axiom G2 — Filtration Consistency

A claim issued at time t may use only evidence available under its declared information filtration.

Let:

$\mathscr{F}_t$
= information available by time t. (AB.9)

Then:

$E(C_t) \subseteq \mathscr{F}_t$. (AB.10)

Evidence becoming available after t may:

- update the claim;
- invalidate it;
- promote it;
- revise the protocol.

It may not be represented as though it supported the original claim at t.

---

## Consequence

The following are filtration violations:

- using a completed weekly close in a Monday decision;
- using revised macroeconomic data in a first-release backtest;
- defining an Episode endpoint from future data and treating it as online;
- selecting a boundary after observing the breakout.

A model can be mathematically correct and still be invalid because its information timing is wrong.

---

## AB.6 Governance Axiom G3 — Projection Partiality

Every indicator or model output is a protocol-bound projection.

Define:

$$X_{j,P,t} = \hat{O}_{j,P}(E_{\leq t}). \quad \text{(AB.11)}$$

The projection does not exhaust the market field.

Therefore:

$$X_{j,P,t} \neq \text{MarketTotality}_t. \quad \text{(AB.12)}$$

The operator-first Technical Analysis source states the same principle directly: Technical Analysis is a protocol-bound projection of market self-reference, and an indicator is not market truth.

---

# Consequence

An RSI reading may support a Motion claim.

It cannot by itself establish:

- market motive;
- reversal;
- legal consequence;
- future price direction;
- complete regime state.

A moving average may be a declared memory filter.

It is not “the true trend” independently of horizon and weighting rule.

---

# AB.7 Governance Axiom G4 — Typed Claim Requirement

Every nontrivial claim must disclose what kind of object it purports to describe.

The minimum type is:

$$\text{Type}(C) = (\text{ClaimClass},\text{Scope},\text{Horizon}). \quad \text{(AB.13)}$$

The richer PPMG type is:

$$\text{Type}(C) = (\text{ClosurePeriod},\text{FunctionalFamily},\text{ActuationRole},\text{Scope},\text{Horizon}). \quad \text{(AB.14)}$$

The six-period and four-family classifications remain candidate taxonomies.

The minimal core requires only that the claim type be explicit enough to prevent category promotion.

---

## Consequence

The system must distinguish at least:

- description;
- persistent condition;
- transition candidate;
- admitted transition;
- higher-order sequence;
- authoritative institutional commitment.

Different labels may be used.

The distinctions must remain operational.

---

# AB.8 Governance Axiom G5 — Event-Gate Requirement

A candidate transition becomes an admitted Event only through a declared gate.

Define candidate c:

c
= DetectTransition(X,B). (AB.15)

Define gate output:

G_P(c,E,L,$\mathscr{R}$,A)
→ (d,α,T,r). (AB.16)

where:

d = decision;
α = admission strength;
T = resulting trace;
r = newly disclosed residual.

Therefore:

Event
:= Candidate

- GateAdmission. (AB.17)

---

## Consequence

The following do not independently establish an Event:

- indicator threshold crossing;
- intraday boundary crossing;

- divergence;
- high volume;
- a visually attractive pattern;
- a model probability.

They may create an Event candidate.

NoGate
→ NoAdmittedEvent. (AB.18)

---

## AB.9 Governance Axiom G6 — Commitment–Outcome Separation

A gate evaluates whether a claim is admitted under the evidence available at commitment time.

An outcome occurs later.

Therefore:

GateStatus_t
≠ Outcome_{t+h}. (AB.19)

A well-governed Event may fail.

A poorly governed claim may succeed by chance.

---

## Consequence

The framework must preserve four cases:

| Process | Later outcome | Canonical label |
|---|---|---|
| Governed | succeeds | Governed success |
| Governed | fails | Governed failure |
| Ungoverned | succeeds | Ungoverned success |
| Ungoverned | fails | Ungoverned failure |

A profitable outcome cannot retroactively repair:

- future-data leakage;
- moved boundaries;
- missing invalidation;
- authority overreach.

---

## AB.10 Governance Axiom G7 — Residual Preservation

Every material commitment must disclose what remains unresolved.

Let:

$\mathscr{R}_{k+1}$
= Residual(Σ_P,X_k,G_k,L_{k+1}). (AB.20)

Residual may include:

- missing evidence;
- contradictory evidence;
- model inadequacy;
- frame conflict;
- branch uncertainty;
- authority disagreement;
- boundary ambiguity.

Residual must remain attached to the claim until its state changes through a recorded transition.

---

# Consequence

Commitment does not imply exhaustive closure.

Commitment
≠ Exhaustion. (AB.21)

An admitted breakout may retain:

- weak breadth;
- higher-frame resistance;
- positioning fragility;
- retest uncertainty.

A legal judgment may retain:

- appeal;
- measurement uncertainty;
- enforcement uncertainty.

The declaration architecture explicitly places trace and residual together after the gate rather than treating commitment as complete exhaustion.

---

# AB.11 Governance Axiom G8 — Trace Preservation

A committed claim must produce a persistent trace.

Let:

$$T_k = \mathrm{Trace}(C_k,G_k,t_k,A_k). \quad (AB.22)$$

A later update must not erase $T_k$.

Instead:

$$L_{k+1} = \mathrm{Append}(L_k,T_k,\mathscr{R}_k). \quad (AB.23)$$

The self-revising declaration source treats trace preservation as a necessary condition of admissible revision and rejects systems that revise by erasing their history or redefining contradiction as confirmation.

---

# Consequence

The following operations are prohibited:

- deleting failed breakout signals;
- silently replacing old Elliott counts;
- moving Fibonacci anchors without versioning;
- overwriting an earlier gate status;
- removing contradictory evidence after a profitable outcome.

The correct operation is append-only revision.

---

# AB.12 Governance Axiom G9 — Bounded Authority

Every commitment must be made by an authority whose scope includes the claimed state.

Formally:

$$\mathrm{Scope}(A_G) \supseteq \mathrm{ScopeRequired}(C). \quad (AB.24)$$

Examples:

- an analyst may commit an analytical classification;
- an exchange may commit an execution record;
- an accounting authority may recognize an impairment;
- a court may issue a judgment.

A model’s confidence does not expand its authority.

---

# Consequence

Market evidence may support:

LegalDefaultCandidate. (AB.25)

It cannot itself commit:

LegalDefault. (AB.26)

Similarly:

PriceDecline
→ AccountingReviewCandidate. (AB.27)

not:

PriceDecline
= RecognizedImpairment. (AB.28)

---

# AB.13 Governance Axiom G10 — Scoped Transport

A broad claim must specify how it is expected to transform under another admissible protocol.

Define:

$\mathcal{T}_{P\to P'}(C_P)$
$= \hat{C}_{P'}$. (AB.29)

The observed target claim is:

$C_{P'}$. (AB.30)

Transport residual is:

$r_{\mathcal{T}}$
$= \mathrm{Dist}(\hat{C}_{P'},C_{P'})$. (AB.31)

Transport may produce:

- exact survival;
- covariant survival;
- partial survival;
- locality;
- failure;
- non-comparability.

---

# Consequence

A daily admitted breakout need not become a weekly admitted breakout.

Its correct transported form may be:

DailyAdmittedEvent
→ WeeklyCandidateEvent. (AB.32)

A market distress Event may transport into:

AccountingReviewCandidate. (AB.33)

not directly into accounting commitment.

The Technical Analysis source describes cross-protocol survival as operational objectivity and treats failed breakouts, moved anchors, and relabelled counts as residual-governance problems rather than evidence that should disappear.

---

## AB.14 Governance Axiom G11 — Promotion Requires New Closure

A claim may move to a stronger closure class only when a new closure condition is satisfied.

Let closure levels be ordered:

$\ell_0 < \ell_1 < \ldots < \ell_n$. (AB.34)

Then:

Promote($\ell_i \rightarrow \ell_j$)
$\Rightarrow j > i$
$\wedge$ NewEvidence
$\wedge$ NewGate. (AB.35)

Promotion is not a change in wording.

It is a new commitment.

---

## Consequence

The following promotions require distinct evidence:

Structure → Event
requires transition admission. (AB.36)

Event → Episode
requires persistent event grammar. (AB.37)

Episode → World
requires authority, ledger, and changed admissibility. (AB.38)

Promotion gates are not transitive.

Passing the Event gate does not imply that the Episode gate also passed.

---

## AB.15 Governance Axiom G12 — Admissible Revision

A protocol may be revised only through a trace-preserving revision operator.

Define declaration state:

$$D_k = (P_k,\hat{O}_k,G_k,TraceRule_k,ResidualRule_k). \quad (AB.39)$$

Define revision:

$$D_{k+1} = U_a(D_k,L_k,\mathscr{R}_k). \quad (AB.40)$$

The admissibility condition is:

$$U_a(D_k,L_k,\mathscr{R}_k) \in \mathcal{D}_{adm}. \quad (AB.41)$$

where:

$$\mathcal{D}_{adm} = \{D \mid WellFormed \wedge TracePreserving \wedge ResidualHonest \wedge FrameAccountable \wedge BudgetBounded \wedge NonDegenerate\}. \quad (AB.42)$$

This closely follows the source self-revision architecture, which defines mature observerhood through trace-preserving, residual-honest, frame-robust, bounded revision rather than arbitrary self-modification.

---

# Consequence

Revision may change:

- boundary;
- feature map;
- horizon;
- gate;
- residual rule;
- authority interpretation.

But it must create a new version.

$$Revision \neq RetrospectiveRelabelling. \quad (AB.43)$$

---

# AB.16 Governance Axiom G13 — Simpler-Model Priority

A more elaborate model is retained only when it provides value beyond an admissible simpler alternative.

Let:

$M_0 \prec M_1$ (AB.44)

mean $M_1$ contains greater structural commitment than $M_0$.

Then retain $M_1$ only if:

Utility($M_1$) – ComplexityCost($M_1$)

Utility($M_0$) – ComplexityCost($M_0$). (AB.45)

---

# Consequence

Required reduction paths include:

World
→ InstitutionalEvent. (AB.46)

Episode
→ EventSequence. (AB.47)

ComplexState
→ RealPair. (AB.48)

PhaseTime
→ EventClock. (AB.49)

ProtoEightLayer
→ FourFamilyCore. (AB.50)

SixPeriods
→ SmallerClosureTaxonomy. (AB.51)

The advanced architecture is optional.

The core remains usable after reduction.

---

# AB.17 Governance Axiom G14 — Advanced-State Eligibility

$\chi$, $\Xi$, complex phase, and internal time must not be activated merely because their notation is available.

Define eligibility predicates:

Eligible_$\chi$(C). (AB.52)

Eligible_$\Xi$(C). (AB.53)

Eligible_Z(C). (AB.54)

Eligible_$\tau_i$(C). (AB.55)

An advanced state may be used only when its predicate passes.

---

## Consequence

The default analysis is not:

Z = R + iQ. (AB.56)

The default analysis is:

X = declared real feature state. (AB.57)

Complex treatment is downstream of:

- coordinate definition;
- measurement compatibility;
- generator stability;
- scaling stability;
- real-pair comparison;
- operational utility.

---

# AB.18 Proposition P1 — Projection Cannot Establish Commitment Alone

From G3 and G5:

X = Ô(E) (AB.58)

and:

Event = Candidate + GateAdmission. (AB.59)

Therefore:

Projection alone
⇏ Event. (AB.60)

---

## Example

RSI = 82.

Permitted:

- elevated directional dominance;
- overextension warning;
- Structure-level Motion diagnosis.

Not yet permitted:

- confirmed reversal;
- completed Episode;
- legal or institutional consequence.

---

## AB.19 Proposition P2 — A Crossing Is Not Yet a Breakout Event

Let:

Cross(B)
= 1[p_t > B]. (AB.61)

From G5:

Cross(B)
→ BreakoutCandidate. (AB.62)

Only:

G_breakout = Admit (AB.63)

produces:

BreakoutEvent. (AB.64)

Therefore:

Crossing
≠ GatedBreakout. (AB.65)

This is the minimal formal basis of the breakout pilot.

---

## AB.20 Proposition P3 — Commitment May Coexist with Large Residual

From G7:

G
→ Trace + Residual. (AB.66)

Therefore no logical contradiction exists in:

HighAdmissionStrength
∧ HighResidualBurden. (AB.67)

This defines an admitted but fragile Event.

---

# Implication

Residual burden should not be calculated merely as:

1 – GateStrength. (AB.68)

The two may be correlated.

They are not logical complements.

---

# AB.21 Proposition P4 — Outcome Does Not Retroactively Change Original Gate Status

From G2, G6, and G8:

Gate_t
is evaluated using
$\mathscr{F}$_t. (AB.69)

Outcome_{t+h}
$\notin \mathscr{F}$_t. (AB.70)

Therefore:

Outcome_{t+h}
cannot rewrite
Gate_t. (AB.71)

It may generate:

- invalidation;
- promotion;
- revision;
- model evaluation.

The original gate remains historically fixed.

---

# AB.22 Proposition P5 — Authority Ceilings Bound Claim Ceilings

From G9:

L_authority
= highest closure level authority A may commit. (AB.72)

Then:

L_claim
≤ L_authority. (AB.73)

Thus:

MarketModelConfidence = 0.99 (AB.74)

does not permit:

LegalCommitment. (AB.75)

The model may produce:

LegalRiskCandidate. (AB.76)

---

## AB.23 Proposition P6 — The Claim Ceiling Is the Minimum Binding Ceiling

Let:

L_E = evidence ceiling;
L_G = gate ceiling;
L_$\mathcal{T}$ = transport ceiling;
L_A = authority ceiling;
L_M = model-eligibility ceiling.

Then:

L_claim
≤ min(L_E,L_G,L_$\mathcal{T}$,L_A,L_M). (AB.77)

A strong result in one dimension cannot compensate for a missing mandatory dimension.

---

## Example

Strong daily gate

- weak weekly transport
- no persistence
  → daily Event only. (AB.78)

Strong price distress

- no contractual breach
  → market distress Event only. (AB.79)

Stable real pair

- no generator
  → real-pair model only. (AB.80)

---

## AB.24 Proposition P7 — Promotion Is Non-Transitive

Suppose:

G_{Structure→Event} = Admit. (AB.81)

This does not imply:

G_{Event→Episode} = Admit. (AB.82)

Likewise:

G_{Event→Episode} = Admit (AB.83)

does not imply:

G_{Episode→World} = Admit. (AB.84)

Therefore:

Promotion_{i→i+1}
does not entail
Promotion_{i+1→i+2}. (AB.85)

Every closure boundary has its own gate.

---

# AB.25 Proposition P8 — Same-Lineage Agreement Is Not Independent Confirmation

Let $X_1$ and $X_2$ share source lineage S.

Define overlap:

$\Lambda_{12}$
= SharedAncestors($X_1$,$X_2$)/TotalAncestors($X_1$,$X_2$). (AB.86)

As $\Lambda_{12}$ approaches one, the interpretation:

“two independent confirmations”

becomes less justified.

Thus:

IndicatorCount
≠ IndependentEvidenceCount. (AB.87)

---

# Example

RSI, MACD, moving-average slope, and rate of change may provide useful operator diversity.

They remain primarily price-lineage projections.

Volume, breadth, order flow, and institutional evidence may add distinct channels.

---

# AB.26 Proposition P9 — Residual Cannot Be Identified with Q

Residual is defined by G7 as unresolved content after projection or commitment.

Q is an eligible conjugate coordinate only under G14.

Therefore:

$\mathscr{R}$
≠ Q. (AB.88)

Even when one residual motivates a candidate coordinate Q_r:

ResidualSource
→ CandidateQ. (AB.89)

additional tests are required before:

CandidateQ
→ ConjugateCoordinate. (AB.90)

The CAPM source defines Q within a declared Euclidean valuation construction and separately defines general residual as content not exhausted by a selected model, gate, or frame.

---

# AB.27 Proposition P10 — Complex Notation Does Not Establish Complex Priority

For any real pair:

(R,Q) ∈ $\mathbb{R}^2$, (AB.91)

one may write:

Z = R + iQ. (AB.92)

This map is information-preserving.

It does not by itself establish:

- rotational dynamics;
- phase relevance;
- conjugacy;
- internal time.

Therefore:

ComplexNotation
⇏ ComplexDynamics. (AB.93)

Complex priority requires additional structure.

---

# AB.28 Proposition P11 — Phase Does Not Establish Time

Given an eligible complex state:

Z = Ae^{iθ}, (AB.94)

θ is an orientation coordinate.

A candidate phase time may be:

$\tau_i(t)$
$= \text{Unwrap}[\theta(t)]$ (AB.95)

or:

$\tau_i(t)$
$= \int_0^t |\theta(s)| ds.$ (AB.96)

But:

θ exists
does not imply
$\tau_i$ is a useful clock. (AB.97)

Phase time must defeat:

- calendar time;
- event count;
- cumulative volume;
- volatility;
- selection depth;
- flexible time warping.

---

# AB.29 Proposition P12 — Recursion Does Not Establish Exact Fractality

From the source filtration framework:

recursion is one possible disclosure grammar rather than necessarily the ontological generator of the field.

Thus:

RecursiveUpdate
= output re-enters future input. (AB.98)

But:

RecursiveUpdate
$\nRightarrow$ ExactScaleInvariantFractal. (AB.99)

Strict fractal claims require:

- scaling law;
- self-similarity criterion;
- dimension estimate;

- robustness range.

The safer market expression is:

recursive multi-level structure.

---

# AB.30 Proposition P13 — Ledger Order and Calendar Time Are Distinct

Let:

t = calendar time;
k = admitted ledger-event index.

Then:

k(t)
may remain constant over long intervals. (AB.100)

It may also jump several times within a short interval.

Therefore:

k ≠ t. (AB.101)

The source filtration framework defines ledger order as what turns isolated disclosures into readable history rather than treating physical duration and trace order as identical.

---

# AB.31 Proposition P14 — Operational Objectivity Is Scoped

Let $\mathcal{P}$ be a declared family of admissible protocols.

Define:

Objective_$\mathcal{P}$(C)
:= Reproducible(C)
∧ StableTransport(C across $\mathcal{P}$)
∧ ResidualVisible(C). (AB.102)

Then:

Objective_$\mathcal{P}$(C)
does not imply
Objective_AllPossibleProtocols(C). (AB.103)

Nor does it imply metaphysical observer independence.

Operational objectivity is indexed by the tested frame family.

---

# AB.32 Proposition P15 — Revision Quality Is Independent of Predictive Success

A revised model may improve prediction while violating:

- trace preservation;
- residual honesty;
- budget bounds;
- protocol stability.

Therefore:

HigherPredictionScore
⇏ AdmissibleRevision. (AB.104)

Similarly, a trace-preserving revision may fail empirically.

Governance quality and outcome quality must be evaluated separately.

---

# AB.33 Proposition P16 — A Mature Observer Is More Than a Ledger

A ledger stores history.

A self-revising observer additionally uses trace and residual to modify future declaration.

The source self-revision framework distinguishes a passive ledger from observerhood by requiring controlled modification of the declaration that produces future projection.

Thus:

Ledger
≠ Observer. (AB.105)

A minimal observer-like runtime is:

D_k
→ Projection_k
→ Gate_k
→ L_{k+1} + $\mathscr{R}$_{k+1}
→ U_a
→ D_{k+1}. (AB.106)

---

# AB.34 Empirical Hypothesis H1 — Four Functional Families Improve Diagnosis

The proposed functional taxonomy is:

Load;
Motion;
Constraint;
Commitment. (AB.107)

Hypothesis:

Using these four functions improves:

- method classification;
- missing-variable detection;
- failure diagnosis;
- confirmation-independence analysis

relative to conventional categories.

This is not an axiom.

It requires comparison with simpler taxonomies.

---

## AB.35 Empirical Hypothesis H2 — Six Closure Periods Improve Claim Discipline

The candidate closure ladder is:

Mark
→ Window
→ Structure
→ Event
→ Episode
→ World. (AB.108)

Hypothesis:

This hierarchy reduces category errors better than:

- signal/pattern/regime;
- short/medium/long term;
- descriptive/predictive/causal;
- continuous closure scores.

The six periods should be merged or revised if they are not reliably distinguishable.

---

## AB.36 Empirical Hypothesis H3 — Residual Adds Incremental Value

Let gate strength be GS and residual burden be RB.

Hypothesis:

Pr(Failure | GS,RB)
≠ Pr(Failure | GS). (AB.109)

Residual may add:

- predictive value;
- diagnostic value;
- revision value.

These are separate claims.

Failure of predictive value does not automatically eliminate governance value.

---

## AB.37 Empirical Hypothesis H4 — Transport Survival Predicts Persistence

Let TS denote transport survival across a preregistered protocol family.

Hypothesis:

Pr(Persistence | TS_high)

Pr(Persistence | TS_low). (AB.110)

The hypothesis must control for:

- signal strength;
- liquidity;
- volatility;
- market regime;
- data availability.

Transport must be declared before the outcome.

---

## AB.38 Empirical Hypothesis H5 — Functional Confirmation Beats Same-Lineage Stacking

Let M_price use several price-derived indicators.

Let M_functional use price, volume, breadth, and gate evidence.

Hypothesis:

Utility(M_functional)

Utility(M_price). (AB.111)

The comparison should hold:

- out of sample;
- at similar model complexity;
- under identical validation procedures.

# AB.39 Empirical Hypothesis H6 — Proto-Eight Roles Predict Characteristic Failure Modes

The optional actuation grammar contains:

Gradient ↔ Gate;
Boundary ↔ Exchange;
Trigger ↔ Guidance;
Memory ↔ Focus. (AB.112)

The Proto-Eight engineering source presents these as paired system roles with distinct levers and failure signatures rather than as a chronological sequence.

Hypothesis:

Role imbalance predicts recognizable failure classes.

Examples:

Trigger without Guidance
→ spike and reversal. (AB.113)

Gradient without Gate
→ pressure without passage. (AB.114)

Memory without Focus
→ overload and ossification. (AB.115)

The eight-role layer remains optional until incremental value is shown.

---

# AB.40 Empirical Hypothesis H7 — χ Improves Conditional Indicator Interpretation

Let:

$\chi_{P,h} < 0$
indicate corrective feedback. (AB.116)

$\chi_{P,h} \approx 0$
indicate critical ambiguity. (AB.117)

$\chi_{P,h} > 0$
indicate reinforcing feedback. (AB.118)

Hypothesis:

The interpretation of indicators such as RSI, momentum, and breakouts improves when conditioned on χ.

The hypothesis fails if ordinary trend, volatility, and liquidity variables capture the same information.

---

# AB.41 Empirical Hypothesis H8 — Ξ Adds a Useful Operating-State Compression

Define:

Ξ_P
= (ρ_P,γ_P,ν_P). (AB.119)

where:

ρ = loading;
γ = lock-in;
ν = agitation.

Hypothesis:

Ξ improves diagnosis or intervention planning relative to the uncompressed source variables.

The hypothesis fails if:

- component selection is unstable;
- the scalarized state hides necessary distinctions;
- the raw feature vector performs equally well.

---

# AB.42 Empirical Hypothesis H9 — Selected R–Q Pairs Earn Complex Priority

For a candidate pair:

X = (R,Q), (AB.120)

compare:

M_pair = f(R,Q). (AB.121)

M_complex = g(A,θ). (AB.122)

Hypothesis:

For some declared market systems:

Utility(M_complex)

> Utility(M_pair)
> after complexity cost. (AB.123)

The CAPM construction supplies one exact local geometric example, but it does not establish a universal Technical Analysis Q or universal market phase. The CAPM source itself emphasizes shared complex algebra without shared physical ontology.

---

# AB.43 Empirical Hypothesis H10 — Internal Phase

## Time Outperforms Simpler Clocks

For comparable episodes, define alignment loss D_c under clock c.

Hypothesis:

$$D_{\tau_i} < \min(D_t, D_k, D_{volume}, D_{volatility}, D_\sigma, D_{flexible}). \quad (AB.124)$$

A stronger hypothesis is:

$$\Pr(\text{Gate at } t \mid \tau_i, \text{Controls}) \neq \Pr(\text{Gate at } t \mid \text{Controls}). \quad (AB.125)$$

If not, retain:

- phase state;
- event clock;
- or ordinary time.

---

# AB.44 Empirical Hypothesis H11 — Ledgered Institutional Events Change Future Admissibility

Let $L_k$ contain an authoritative Event $e_k$.

Let $\mathcal{A}_{k+1}$ be the future admissible action set.

Hypothesis:

$$\mathcal{A}_{k+1}(L_k \text{ with } e_k) \neq \mathcal{A}_{k+1}(L_k \text{ without } e_k). \quad (AB.126)$$

Examples may include:

- covenant breach;
- margin call;
- legal judgment;
- accounting impairment;
- index inclusion;
- policy change.

This is a causal claim.

Sequence alone is insufficient.

---

# AB.45 Non-Theorem N1 — Markets Are Not Proven to Possess a Periodic Law

The framework does not prove:

MarketFunctions recur with chemical-like periodic exactness. (AB.127)

The table is a candidate closure–function matrix.

“Proto-periodic” indicates:

- recurring roles;
- comparable cells;
- possible missing-instrument hypotheses.

It does not indicate an established natural law.

---

# AB.46 Non-Theorem N2 — Markets Are Not Proven to Be Physically Quantum

The framework does not establish:

- market wavefunctions;
- physical collapse;
- Planck-scale financial dynamics;
- quantum nonlocality;
- physical entanglement among assets.

Complex geometry and observer-bound projection may share formal motifs with quantum theory.

Formal analogy
≠ Physical identity. (AB.128)

---

# AB.47 Non-Theorem N3 — CAPM Q Is Not a Universal Risk Quantity

Within its declared geometry:

$$Q_{CAPM} = \sqrt{(A^2 - R^2)}.$$ (AB.129)

This does not imply:

$$Q_{CAPM} = \text{volatility} = \text{beta} = \text{VaR} = \text{expected shortfall} = \text{market residual}.$$ (AB.130)

The coordinate is exact by construction.

Its broader economic interpretation remains conditional.

---

## AB.48 Non-Theorem N4 — Technical Analysis Has No Established Universal Q

Possible candidates include:

- participation pressure;
- trapped-position pressure;
- valuation exposure;
- breadth tension;
- phase quadrature.

They must remain indexed:

Q_participation;
Q_positioning;
Q_CAPM;
Q_signal. (AB.131)

No unification is established.

---

## AB.49 Non-Theorem N5 — Phase Is Not Automatically an Internal Clock

A phase coordinate may:

- describe orientation;
- order a cycle;
- compress a two-variable state.

It becomes an internal clock only after comparative validation.

Thus:

$\theta$
$\nRightarrow \tau_i$ superiority. (AB.132)

---

## AB.50 Non-Theorem N6 — Proto-Eight Is Not a Universal Chronology

The eight roles form a paired topology.

They do not establish a necessary sequence:

Gradient
→ Gate
→ Boundary
→ Exchange
→ Trigger
→ Guidance
→ Memory
→ Focus. (AB.133)

A protocol may activate them in different orders or simultaneously.

---

## AB.51 Non-Theorem N7 — Recursive Does Not Mean Fractal

A market may contain recursive feedback without demonstrating:

- scale invariance;
- self-similar geometry;
- fractal dimension;
- exact nesting.

Use “recursive” unless strict fractal evidence exists.

---

## AB.52 Non-Theorem N8 — Objectivity Does Not Mean Observer Absence

The framework does not eliminate observers.

It makes their protocols explicit.

Operational objectivity is:

- declared;
- reconstructable;
- transport-tested;
- residual-honest.

This is compatible with observer dependence.

---

## AB.53 Non-Theorem N9 — A Gate Does Not Guarantee Profit

An admitted Event may be:

- economically unprofitable;
- poorly timed for a specific strategy;
- overwhelmed by transaction costs;
- unsuitable for a portfolio;
- followed by invalidation.

PPMG does not supply:

- position sizing;
- expected return;
- portfolio optimization;
- execution;
- tax treatment.

The trading stack remains:

MarketClaim
→ DecisionModel
→ PortfolioRule
→ Execution. (AB.134)

---

## AB.54 Non-Theorem N10 — A World Is Not Merely a Large Market Pattern

A World-level claim requires:

- bounded rule system;
- authority;
- persistent ledger;
- changed future admissibility.

A long trend may remain an Episode.

A short legal judgment may create a World transition.

Closure depth
≠ Duration or visual size. (AB.135)

---

## AB.55 Consistency Condition C1 — No Type Promotion by Vocabulary

A claim must not become stronger merely because stronger terminology is used.

Formally:

Rename(C_i,Label_j)
does not imply
Promote(C_i→C_j). (AB.136)

Promotion requires new closure evidence.

---

# AB.56 Consistency Condition C2 — No Silent Protocol Mutation

For protocols $P_1$ and $P_2$:

$P_1 \neq P_2$
if any material field changes. (AB.137)

Material fields include:

- boundary;
- timeframe;
- scale;
- feature map;
- gate;
- outcome horizon;
- authority.

A changed material field requires:

NewProtocolVersion. (AB.138)

---

# AB.57 Consistency Condition C3 — No Residual Laundering

A residual may leave the open register only through a recorded transition.

Permitted:

Open
→ Resolved. (AB.139)

Open
→ Dissipated. (AB.140)

Open
→ Superseded. (AB.141)

Open
→ Invalidating. (AB.142)

Prohibited:

Open
→ DeletedBecauseOutcomeSucceeded. (AB.143)

---

# AB.58 Consistency Condition C4 — No Authority Transfer by Analogy

Suppose source domain S and target domain T share a structural relation.

Then:

Analogy(S,T)
does not transfer
Authority(S) → Authority(T). (AB.144)

Example:

A market gate may resemble an accounting recognition gate.

It does not thereby acquire accounting authority.

---

# AB.59 Consistency Condition C5 — No Advanced Layer Without Its Reduction Path

Every advanced object must declare the simpler model retained if it fails.

| Advanced object | Mandatory reduction |
|---|---|
| Proto-Eight role system | four-family or domain-native model |
| Six-period hierarchy | reduced closure taxonomy |
| $\chi$ | ordinary regime variables |
| $\Xi$ | uncompressed feature vector |
| Complex Z | real pair |
| Phase time | ordinary or event clock |
| World model | institution-specific Event model |

A model with no reduction path is insufficiently falsifiable.

---

# AB.60 Countermodel CM1 — Projection-Only Market Analysis

A competing minimal model is:

E
→ $\hat{O}$
→ X
→ OutcomeEvaluation. (AB.145)

It omits:

- gate;
- residual;
- ledger;
- transport.

PPMG must demonstrate value beyond this model.

If it does not, its extra architecture should be reduced.

---

## AB.61 Countermodel CM2 — Statistical Decision Theory

A competing model is:

Evidence
→ PosteriorProbability
→ LossFunction
→ Decision. (AB.146)

This may reproduce much of gate behaviour.

PPMG adds value only where:

- closure typing;
- residual lifecycle;
- authority;
- transport;
- revision history

provide incremental benefit.

---

## AB.62 Countermodel CM3 — State-Space Regime Model

A competing Episode model is:

HiddenState_t
→ StateTransition_t
→ RegimeLabel_t. (AB.147)

PPMG's Episode concept survives only if its:

- explicit event grammar;
- branch residual;
- promotion gate;

- trace preservation

improve prospective segmentation or interpretation.

---

# AB.63 Countermodel CM4 — Real-Pair Dynamics

A competing complex model is:

$$X_t = (R_t,Q_t). \quad \text{(AB.148)}$$

with unrestricted dynamics:

$$X_{t+1} = f(X_t,E_t). \quad \text{(AB.149)}$$

Complex priority requires outperforming this model.

A visually circular plot is not enough.

---

# AB.64 Countermodel CM5 — Event-Count Clock

A competing phase-time model is:

$$\kappa(t) = \text{number of admitted Events by } t. \quad \text{(AB.150)}$$

If $\kappa$ aligns episodes as well as $\tau_i$:

$\tau_i$ loses clock priority. (AB.151)

The simpler clock should be retained.

---

# AB.65 Minimal Falsification Table

| Claim | Falsifier | Required reduction |
|---|---|---|
| Protocol improves analysis | no reproducibility gain | ordinary research description |
| Gate improves Event claims | threshold/close rule equal | simpler decision rule |
| Residual adds value | no diagnostic, predictive, or revision gain | gate-only model |
| Transport strengthens claims | no scope or persistence gain | local claim |
| Six periods matter | poor reliability/no utility | smaller hierarchy |
| Four families matter | no diagnostic value | domain-native categories |
| Proto-Eight matters | no incremental failure diagnosis | optional appendix |
| $\chi$ matters | standard regime variables equal | ordinary regime model |
| $\Xi$ matters | raw features equal | feature vector |
| Complex state matters | real pair equal | real pair |

| Claim | Falsifier | Required reduction |
|---|---|---|
| Phase time matters | simpler clock equal | simpler clock |
| World model matters | no changed admissibility | institutional Event |

---

# AB.66 The Minimal Kernel After All Reductions

If every optional extension fails, the smallest surviving kernel is:

1. declare what is being observed;
2. use only evidence available at the time;
3. identify the projection and its limits;
4. separate candidate from commitment;
5. preserve unresolved evidence;
6. preserve the original claim;
7. evaluate the later outcome separately;
8. revise prospectively.

Formally:

K_min
:= Revise
∘ EvaluateOutcome
∘ AppendTraceResidual
∘ Gate
∘ Project
∘ Declare. (AB.152)

This kernel is already useful.

It does not depend on:

- Proto-Eight;
- six periods;
- four functions;
- complex numbers;
- phase time;
- World metaphysics.

---

# AB.67 The Intermediate Kernel

If the typing and transport hypotheses succeed:

K_mid

:= K_min

- ClosureTyping
- FunctionalTyping
- Transport
- Promotion. (AB.153)

This supports:

- Structure;
- Event;
- Episode;
- World distinctions;
- cross-frame scope;
- method classification.

---

# AB.68 The Advanced Kernel

If the advanced-state hypotheses also succeed:

K_adv
:= K_mid

- χ
- Ξ
- EligibleComplexState
- ValidatedPhaseTime
- LedgerBackreaction. (AB.154)

The adjectives are mandatory:

- **eligible** complex state;
- **validated** phase time;
- **identified** backreaction.

Without them, the notation overstates the evidence.

---

# AB.69 Dependency Graph

The evidential dependencies are:

```
Protocol
```

```
   ↓
Filtration Consistency
   ↓
Projection
   ↓
Claim Typing
   ↓
Candidate
   ↓
Event Gate
   ↓
Trace + Residual
   ↓
Transport
   ↓
Episode Promotion
   ↓
Authority + Ledger
   ↓
World Claim
```

The advanced-state branch is:

```
Reliable Projection
   ↓
Meaningful Real Pair
   ↓
Complex Eligibility
   ↓
Phase Utility
   ↓
Clock Comparison
   ↓
Phase-Sensitive Gate
   ↓
Time-Bearing World Candidate
```

A downstream claim fails automatically when a mandatory upstream dependency is absent.

---

# AB.70 Canonical Formal Summary

The minimal declared object is:

$$\Sigma_P := \mathrm{Declare}(\Sigma_0 \mid P). \quad \text{(AB.155)}$$

Projection is:

$$X_{P,t} := \hat{O}_P(E_{\le t}, \Sigma_P). \quad \text{(AB.156)}$$

Candidate detection is:

$$c_t := \mathrm{Detect}(X_{P,t}, B_P). \quad \text{(AB.157)}$$

Gate output is:

$$G_P(c_t, E_{\le t}, L_t, \mathscr{R}_t, A) \to (d_t, \alpha_t, T_t, r_t). \quad \text{(AB.158)}$$

Ledger update is:

$$L_{t+1} := \mathrm{Append}(L_t,T_t,r_t). \quad \text{(AB.159)}$$

Transport is:

$$\hat{C}_{P'} := \mathcal{T}_{P\to P'}(C_P). \quad \text{(AB.160)}$$

Transport residual is:

$$r_{\mathcal{T}} := \mathrm{Dist}(\hat{C}_{P'},C_{P'}). \quad \text{(AB.161)}$$

Revision is:

$$D_{k+1} := U_a(D_k,L_k,\mathscr{R}_k). \quad \text{(AB.162)}$$

Claim ceiling is:

$$L_{\mathrm{claim}} \le \min(L_E,L_G,L_{\mathcal{T}},L_A,L_M). \quad \text{(AB.163)}$$

Reduction is:

$$\mathrm{AdvancedClaimFail} \Rightarrow \mathrm{RetainStrongestSupportedLowerClaim}. \quad \text{(AB.164)}$$

---

# AB.71 Canonical Non-Implication Summary

The formal system requires the following non-implications:

$$\mathrm{Projection} \not\Rightarrow \mathrm{Event}. \quad \text{(AB.165)}$$

$$\mathrm{Signal} \not\Rightarrow \mathrm{Commitment}. \quad \text{(AB.166)}$$

$$\mathrm{Crossing} \not\Rightarrow \mathrm{BreakoutAdmission}. \quad \text{(AB.167)}$$

$$\mathrm{Event} \not\Rightarrow \mathrm{Episode}. \quad \text{(AB.168)}$$

$$\mathrm{Episode} \not\Rightarrow \mathrm{World}. \quad \text{(AB.169)}$$

$$\mathrm{MarketEvidence} \not\Rightarrow \mathrm{InstitutionalCommitment}. \quad \text{(AB.170)}$$

$$\mathrm{GatePass} \not\Rightarrow \mathrm{ProfitableOutcome}. \quad \text{(AB.171)}$$

OutcomeSuccess
⇏ ValidOriginalProtocol. (AB.172)

Residual
⇏ ErrorOnly. (AB.173)

Residual
⇏ Q. (AB.174)

RealPair
⇏ ComplexPriority. (AB.175)

Phase
⇏ InternalTime. (AB.176)

Recursion
⇏ ExactFractal. (AB.177)

TransportSurvival
⇏ UniversalTruth. (AB.178)

PostEventMovement
⇏ Backreaction. (AB.179)

Revision
⇏ PermissionToEraseHistory. (AB.180)

---

# AB.72 Canonical Research Contract

A study claiming compliance with the minimal core should commit to:

1. protocol declaration before outcome evaluation;
2. evidence-time consistency;
3. source-lineage preservation;
4. explicit claim typing;
5. gate separation;
6. residual preservation;
7. immutable original trace;
8. scoped authority;
9. preregistered transport where broad scope is claimed;
10. prospective invalidation;
11. comparison with a simpler model;
12. explicit reduction after failure.

The research contract may be compressed as:

No Hidden Boundary.
No Future Leakage.
No Ungated Event.
No Residual Erasure.
No Silent Revision.
No Unbounded Authority.
No Advanced Model Without a Simpler Benchmark. (AB.181)

---

# AB.73 What Remains Distinctive After Reduction

After all adversarial criticism, the framework's most distinctive contribution is not:

- one new indicator;
- one predictive equation;
- one complex plane;
- one ancient correspondence.

It is the coordinated preservation of seven distinctions:

1. **declaration versus field;**
2. **projection versus commitment;**
3. **Event versus higher closure;**
4. **trace versus residual;**
5. **local validity versus transported scope;**
6. **analytical confidence versus authority;**
7. **revision versus historical erasure.**

These distinctions can coexist within ordinary finance without requiring any strong metaphysical conclusion.

---

# AB.74 What Remains Optional

The following remain optional modules:

- six-period taxonomy;
- four-function periodic matrix;
- Proto-Eight actuation grammar;
- $\chi$ feedback signature;
- $\Xi$ operating-state compression;
- CAPM-to-TA complex extension;

- internal phase time;
- generalized World ontology.

Their optionality is not a weakness.

It makes the framework modular and falsifiable.

---

# AB.75 Appendix AB Conclusion

The Periodic Grammar does not need every branch to survive.

Its minimum coherent core is a discipline of declared, gated, residual-honest observation.

The core sequence is:

Declare
→ Project
→ Type
→ DetectCandidate
→ Gate
→ PreserveTrace + Residual
→ TestScope
→ EvaluateOutcome
→ ReviseWithoutErasure. (AB.182)

Its minimal axioms are:

No claim without protocol. (AB.183)

No original claim using future evidence. (AB.184)

No indicator as total market truth. (AB.185)

No Event without gate. (AB.186)

No commitment without residual disclosure. (AB.187)

No revision without prior-trace preservation. (AB.188)

No institutional commitment without authority. (AB.189)

No broad claim without scoped transport. (AB.190)

No promotion without a new closure condition. (AB.191)

No advanced model without a simpler benchmark. (AB.192)

Its central derived law is:

ClaimStrength
≤ WeakestMandatorySupport. (AB.193)

Its central reduction law is:

When a stronger layer fails, retain the strongest lower layer that remains reproducible, useful, and honest. (AB.194)

Its central non-theorem is:

> The framework does not prove that markets possess one hidden universal geometry.

Its strongest defensible thesis is more disciplined:

> Market analysis improves when every projection is declared, every transition is gated, every unresolved remainder stays visible, every broader scope is transported explicitly, and every revision remains answerable to the history it inherits.

# Appendix AC — The Research Constitution: Governance, Replication, and Branch Closure

## AC.1 Purpose

The Periodic Grammar proposes that market claims should be governed through:

- declaration;
- projection;
- gate;
- trace;
- residual;
- transport;
- revision.

The research programme itself must obey the same architecture.

Otherwise, the framework could demand strict trace preservation from market analysis while allowing its own theoretical claims to:

- drift without version control;
- absorb contradictory evidence;
- change definitions after failure;
- promote suggestive analogies into established conclusions;
- survive indefinitely through added complexity.

This appendix therefore defines a **Research Constitution** for the Periodic Grammar.

Its purpose is to govern:

1. how new claims enter the programme;
2. how evidence changes claim status;
3. how replication is evaluated;
4. how disagreements remain visible;

5. how protocols may be revised;
6. when a research branch must be reduced or closed;
7. who may authorize each change.

The governing principle is:

A self-revising theory must not possess unlimited authority over its own revision. (AC.1)

The constitutional runtime is:

Proposal
→ Declaration
→ Review Gate
→ Study
→ Evidence Trace + Residual
→ Replication
→ Promotion, Retention, Reduction, or Closure. (AC.2)

---

# AC.2 Why a Research Constitution Is Necessary

A broad interdisciplinary framework faces a special danger.

Because it can translate many phenomena into its own vocabulary, almost any result may appear to support it.

For example:

- continuation may be interpreted as self-confirming $\chi$;
- reversal may be interpreted as corrective $\chi$;
- ambiguity may be interpreted as $\chi \approx 0$;
- successful commitment may be called a strong gate;
- failure may be explained through residual;
- disagreement may be explained through protocol difference;
- model weakness may motivate a new higher-dimensional state.

Each interpretation may be individually reasonable.

Together, however, they can make the theory difficult to reject.

The research programme therefore needs rules that distinguish:

InterpretiveAvailability (AC.3)

from:

EmpiricalDiscrimination. (AC.4)

A mature theory must not merely possess a vocabulary capable of describing every outcome.

It must declare in advance:

- which outcomes support a claim;
- which outcomes weaken it;
- which outcomes leave it unresolved;
- which outcomes close the branch.

---

# AC.3 Constitutional Objects

The research constitution governs twelve object classes.

Let:

$$\mathcal{C}_R = \{Claim,Protocol,Study,Evidence,Result,Residual,Replication,Revision,Authority,Branch,Release,A \tag{AC.5}$$

Each object requires:

- identifier;
- version;
- responsible authority;
- timestamp;
- status;
- parent links;
- change history.

A theoretical statement should not circulate indefinitely as unstructured prose once it becomes part of the formal research programme.

It should enter the claim ledger.

---

# AC.4 The Claim Ledger

# AC.4.1 Definition

The **claim ledger** is the authoritative record of theoretical and empirical claims.

Define:

$$L_claim,k = \{(C_1,S_1,E_1,\mathscr{R}_1),\ldots,(C_k,S_k,E_k,\mathscr{R}_k)\}. \tag{AC.6}$$

where:

C_j = claim;
S_j = current status;
E_j = supporting and contradicting evidence;
$\mathscr{R}$_j = open residual.

---

# AC.4.2 Required claim fields

```
claim_id
claim_text
claim_class
provenance
scope
protocol_family
formal_status
empirical_status
supporting_evidence
contradictory_evidence
falsifier
reduction_path
responsible_editor
revision_history
```

---

# AC.4.3 Claim classes

The constitution distinguishes:

```
Definition
NormativeRule
ExactConstruction
DerivedProposition
TaxonomicProposal
MechanismHypothesis
PredictiveHypothesis
CausalHypothesis
Analogy
HistoricalInterpretation
EngineeringProposal
```

A claim may not silently migrate from one class to another.

For example:

“Q is defined geometrically in the CAPM construction”

is an ExactConstruction.

“Q corresponds to a generally traded latent risk field”

would be a MechanismHypothesis.

The exactness of the first does not support the second automatically.

---

# AC.5 Claim Statuses

Every claim receives one current status.

## AC.5.1 Proposed

The claim has been stated clearly but not tested.

## AC.5.2 Operationalized

Variables, protocol, outcome, and falsifier have been defined.

## AC.5.3 Pilot-supported

A preregistered pilot supports the claim under one limited protocol.

## AC.5.4 Locally replicated

The claim survives independent data or implementation within one protocol family.

## AC.5.5 Transport-supported

The relation survives declared cross-protocol transformations.

## AC.5.6 Cross-domain supported

The relation survives materially different domains without conceptual distortion.

## AC.5.7 Causally supported

A credible identification strategy supports the claimed action channel.

## AC.5.8 Reduced

A weaker or simpler version remains supported.

## AC.5.9 Suspended

Evidence is insufficient, contradictory, or methodologically compromised.

## AC.5.10 Closed unsupported

The branch has repeatedly failed its declared gates.

These statuses form neither an inevitable ladder nor a prestige hierarchy.

A useful local claim may remain permanently:

LocallyReplicated. (AC.7)

It need not become universal.

---

# AC.6 Status Promotion Rule

A claim may be promoted only when new evidence satisfies the gate for the requested status.

Define:

Promote(C,S_i→S_j)
⇒ NewEvidence
∧ IndependentReview
∧ StatusGate_j. (AC.8)

A theoretical restatement is not new evidence.

A second analysis of the same dataset is not necessarily independent replication.

A larger model fitted to the same outcome does not automatically justify promotion.

---

# AC.7 Status Demotion Rule

Claims may be demoted when:

- replication fails;
- measurement becomes unreliable;
- hidden leakage is discovered;
- source attribution is corrected;
- a simpler model matches performance;
- scope proves narrower than originally stated.

Demotion is:

S_i → S_j where j < i. (AC.9)

Demotion must preserve:

- the original status;
- the evidence that justified it;
- the evidence that later weakened it;
- the date and authority of the change.

Demotion is not embarrassment.

It is evidence that the research ledger is functioning.

---

# AC.8 Branches

# AC.8.1 Definition

A **branch** is a connected family of claims depending on one or more shared assumptions.

Examples:

- six-period closure branch;
- four-family taxonomy branch;
- Proto-Eight branch;
- residual-governance branch;
- transport-objectivity branch;
- complex-state branch;
- phase-time branch;
- World-formation branch.

Represent branch b as:

B_b
= (RootClaims_b,DependentClaims_b,Studies_b,Residuals_b). (AC.10)

---

# AC.8.2 Dependency requirement

Every advanced claim must list its dependencies.

Example:

PhaseTimeClaim
depends on:

1. meaningful R;
2. meaningful Q;
3. complex eligibility;
4. stable phase;
5. online branch resolution;
6. competing-clock superiority.

If complex eligibility fails:

PhaseTimeBranch
cannot remain active independently. (AC.11)

---

# AC.9 Branch State

Each research branch receives one state.

```
Exploratory
```

```
Active
Promising
Constrained
Reduced
Suspended
ClosedSupported
ClosedUnsupported
```

# Closed supported

The branch has reached a mature, scoped conclusion.

# Closed unsupported

The branch failed its declared tests and has no currently admissible repair.

A branch may reopen only through:

- genuinely new evidence;
- a materially new measurement method;
- discovery of a documented protocol defect.

Rewording the old claim is insufficient.

---

# AC.10 Research Proposal Gate

A study proposal must pass five gates.

# AC.10.1 Concept gate

Does the study test a clearly defined claim?

# AC.10.2 Measurement gate

Can the relevant objects be observed or operationalized prospectively?

# AC.10.3 Comparison gate

Is there an admissible simpler or mature alternative?

# AC.10.4 Falsification gate

Is there a result that would weaken, reduce, or close the claim?

# AC.10.5 Resource gate

Is the study proportionate to its expected information gain?

Define proposal admissibility:

AdmitStudy
:= ConceptValid
∧ Measurable
∧ Comparative
∧ Falsifiable
∧ ResourceProportionate. (AC.12)

A philosophically interesting but presently unmeasurable idea may remain in the residual registry.

It should not be presented as an active empirical study.

---

# AC.11 Preregistration Constitution

A confirmatory study must preregister:

1. claim;
2. protocol;
3. evidence universe;
4. exclusion rules;
5. primary variables;
6. primary outcome;
7. baselines;
8. statistical procedure;
9. model-selection rule;
10. missing-data treatment;
11. success criterion;
12. falsification or reduction rule.

The preregistration hash should be preserved:

H_pre
= Hash(PreregistrationDocument). (AC.13)

Any later material change creates:

ProtocolDeviation. (AC.14)

A deviation is not automatically misconduct.

It must be:

- disclosed;
- justified;

- classified as confirmatory or exploratory.

---

## AC.12 Confirmatory and Exploratory Separation

The constitution distinguishes:

## Confirmatory analysis

Tests the preregistered claim under the frozen protocol.

## Secondary analysis

Uses preregistered but non-primary variables or outcomes.

## Exploratory analysis

Develops new hypotheses after inspecting results.

The permitted transition is:

ExploratoryFinding
→ NewClaimProposal
→ NewPreregistration
→ FutureConfirmatoryStudy. (AC.15)

The prohibited transition is:

ExploratoryFinding
→ RetrospectivelyDeclaredPrimaryResult. (AC.16)

---

## AC.13 Evidence Classes

Evidence receives one of six classes.

## AC.13.1 E0 — Conceptual illustration

A worked example or hypothetical case.

## AC.13.2 E1 — Retrospective observation

A pattern found after reviewing data.

## AC.13.3 E2 — Preregistered pilot

A prospectively specified test under limited scope.

## AC.13.4 E3 — Independent replication

A study using independent implementation, data, or investigators.

## AC.13.5 E4 — Transport replication

The claim survives a declared protocol family.

## AC.13.6 E5 — Causal or institutional validation

A credible intervention or authoritative mechanism supports the claim.

The article's present examples largely occupy:

E0 or E1. (AC.17)

The proposed breakout study would target:

E2. (AC.18)

The constitution prevents E0 examples from being narrated as though they were E3 evidence.

---

# AC.14 Evidence Independence

Evidence is not independent merely because it appears in another chart or paper.

Define independence dimensions:

I_E
= (Data,Implementation,Investigator,Protocol,Outcome,Funding). (AC.19)

Two studies may share:

- data but not implementation;
- implementation but not data;
- investigators but not protocol.

Replication reports should disclose the independence vector.

A replication using the same code and dataset tests reproducibility.

It does not provide strong independent confirmation.

---

# AC.15 Replication Types

## AC.15.1 Computational reproduction

The same data and code reproduce the reported result.

## AC.15.2 Independent implementation

New code implements the same protocol.

## AC.15.3 Temporal replication

The study succeeds in a later period.

## AC.15.4 Cross-market replication

The relation survives another market or asset class.

## AC.15.5 Protocol transport replication

The transformed claim behaves as predicted under a new protocol.

## AC.15.6 Conceptual replication

A structurally similar relation appears in another domain without forcing identical variables.

The replication type must be named.

“Replicated” alone is insufficient.

---

# AC.16 Replication Gate

A replication attempt should preserve:

- original protocol;
- original target claim;
- original success criterion.

It may additionally test revised protocols.

The report must separate:

OriginalProtocolResult (AC.20)

from:

RevisedProtocolResult. (AC.21)

A failed original replication cannot be erased because a revised version succeeds.

---

# AC.17 Negative Results

A negative result must enter the research ledger with the same structural completeness as a positive result.

Required fields include:

```
claim_tested
protocol
primary_result
```

```
confidence_or_uncertainty
falsifier_status
retained_lower_model
open_residual
prohibited_reinterpretation
next_admissible_study
```

The negative-result rule is:

NoResultDeletion because ResultWeakensTheory. (AC.22)

A programme that publishes only successful branches cannot evaluate its own architecture honestly.

---

# AC.18 Null, Inconclusive, and Negative

These statuses must remain distinct.

## Null result

The estimated effect is compatible with no material difference under the study's precision.

## Inconclusive result

The study cannot discriminate among relevant alternatives.

Causes may include:

- low power;
- measurement failure;
- protocol contamination;
- excessive uncertainty.

## Negative result

The evidence materially contradicts the preregistered claim.

The phrase "not supported" should not conceal whether the study was:

- genuinely negative;
- merely imprecise;
- or methodologically invalid.

---

# AC.19 Contradictory Evidence

When studies disagree, the programme should not immediately average them into one conclusion.

First classify the disagreement.

Possible causes include:

- protocol difference;
- population difference;
- implementation error;
- regime dependence;
- publication bias;
- unstable construct;
- ordinary sampling variation.

Define disagreement record:

$$D_{ij} = (Claim, Study_i, Study_j, DifferenceType, ResolutionPlan). \quad (AC.23)$$

Contradictory evidence remains visible even after a later synthesis.

---

# AC.20 Disagreement Classes

# AC.20.1 Numerical disagreement

Same object and protocol, different estimate.

# AC.20.2 Protocol disagreement

Different declarations produce different but potentially compatible objects.

# AC.20.3 Ontological disagreement

The studies do not agree on what object exists.

# AC.20.4 Authority disagreement

Different institutions recognize different states.

# AC.20.5 Interpretive disagreement

The same result receives different theoretical explanations.

Only numerical disagreements should be combined automatically through ordinary meta-analysis.

The others require conceptual resolution first.

---

# AC.21 Meta-Analysis Gate

A meta-analysis is admissible only when studies possess sufficient comparability.

Define comparability:

Comp_{ij}
= ProtocolOverlap
× OutcomeAlignment
× MeasurementCompatibility
× PopulationRelevance. (AC.24)

A pooled estimate should not be produced when:

Comp_{ij} < ε_comp. (AC.25)

In such cases, the correct output may be:

StructuredHeterogeneity. (AC.26)

Not every disagreement is noise requiring elimination.

Some disagreements disclose protocol dependence.

---

# AC.22 Simplification Authority

Researchers often have incentives to enlarge models.

The constitution therefore creates an explicit **simplification authority**.

Its role is to ask:

- Can a variable be removed?
- Can two levels be merged?
- Does the advanced model beat the baseline?
- Has terminology become redundant?
- Is a branch surviving only through exceptions?

Simplification is not left solely to the researchers who introduced the complex model.

This institutionalizes:

SimplerModelPriority. (AC.27)

---

# AC.23 Revision Authority

No single author should possess unlimited authority to revise:

- claim definitions;
- status;

- benchmark labels;
- branch closure rules.

A material revision should require at least:

1. revision proposal;
2. evidence record;
3. impact analysis;
4. independent review;
5. versioned release.

Define revision gate:

G_revision
= EvidenceAdequacy
∧ TracePreservation
∧ ScopeClarity
∧ CompatibilityPlan
∧ IndependentApproval. (AC.28)

---

# AC.24 Authority Roles

A mature research consortium may distinguish six roles.

# AC.24.1 Claim steward

Maintains one claim’s definition and evidence record.

# AC.24.2 Protocol steward

Maintains operational specifications and versions.

# AC.24.3 Replication editor

Coordinates independent replications.

# AC.24.4 Residual curator

Maintains unresolved contradictions and missing tests.

# AC.24.5 Simplification reviewer

Challenges complexity and protects baseline comparisons.

# AC.24.6 Release authority

Approves official status changes and public versions.

One person may hold several roles in a small project.

The roles should still remain conceptually distinct.

---

# AC.25 Conflict of Interest

Researchers should disclose interests affecting interpretation.

Examples include:

- authorship of the theory;
- ownership of related software;
- commercial trading use;
- consultancy;
- reputational investment;
- preference for a philosophical interpretation;
- involvement in benchmark annotation.

Conflict does not invalidate evidence automatically.

It changes the required independence and review burden.

Define review burden:

$$B_{review} \propto ClaimConsequence \times ResearcherInterest \times ModelFlexibility.$$ (AC.29)

---

# AC.26 Theoretical Conflict of Interest

A less visible conflict arises when a researcher has invested substantial intellectual identity in a framework.

Possible symptoms include:

- every failure becoming a residual;
- every null becoming a measurement problem;
- every contradictory domain becoming a protocol exception;
- every simpler model being described as incomplete.

The constitution therefore requires external reviewers to possess authority to recommend:

- reduction;

- suspension;
- branch closure.

---

## AC.27 Branch Promotion Gate

A branch may be promoted from Exploratory to Active only when:

1. root claim is operationalized;
2. measurement exists;
3. simpler baseline exists;
4. falsifier exists;
5. one feasible study is specified.

It may become Promising only after:

- preregistered support;
- acceptable measurement reliability;
- no major hidden protocol failure.

It may become Transport-supported only after:

- predefined cross-protocol tests;
- surviving relation;
- bounded transport residual.

---

## AC.28 Branch Reduction Gate

A branch should be reduced when:

- a simpler model performs equally well;
- only a subset of variables matters;
- scope is narrower than proposed;
- labels are unreliable;
- transport fails systematically.

Reduction produces:

B_full
→ B_reduced. (AC.30)

The reduced branch receives:

- new definition;

- retained evidence;
- removed commitments;
- migration rule.

The original branch remains historically visible.

---

# AC.29 Branch Suspension Gate

Suspend a branch when:

- key variables cannot be measured;
- evidence is internally contradictory;
- data provenance is compromised;
- required domain expertise is unavailable;
- a central dependency is unresolved.

Suspension means:

No new strong claims. (AC.31)

It does not mean:

The claim is false. (AC.32)

A suspended branch may continue conceptual development, but its empirical status must not be promoted.

---

# AC.30 Branch Closure Gate

Close a branch as unsupported when all of the following hold:

1. the claim has been operationalized fairly;
2. multiple competent tests have been attempted;
3. simpler alternatives perform as well or better;
4. failures are not explained by a documented protocol defect;
5. no materially new measurement path is available.

Define closure:

CloseUnsupported(B)
:= FairTested
∧ RepeatedFailure
∧ SimplerAlternative
∧ NoUnresolvedCriticalDefect
∧ NoNewMeasurementPath. (AC.33)

The decision should state the strongest lower claim retained.

---

## AC.31 Reopening a Closed Branch

A closed branch may reopen only when at least one of the following occurs:

- new measurement technology;
- new high-quality dataset;
- discovered implementation error;
- new formal derivation changing the test;
- independent evidence from a materially different domain.

Reopening requires a new claim version.

It must not pretend that closure never occurred.

---

## AC.32 Theory-Wide Termination Condition

The distinct PPMG programme should terminate if the following core results jointly occur:

1. protocol formalization adds no reproducibility benefit;
2. explicit gate records add no analytical or governance value;
3. residual recording adds no diagnostic or revision value;
4. trace preservation adds no learning benefit beyond ordinary documentation;
5. the architecture creates greater complexity than clarity.

In that case:

PPMG
→ GeneralResearchHygieneChecklist. (AC.34)

The checklist may remain useful.

The distinct theoretical programme should close.

---

## AC.33 Minimum Surviving Constitution

Even if advanced branches close, the research constitution retains a small set of rules:

- freeze the original claim;
- preserve information timing;
- state the baseline;
- publish negative results;

- separate exploratory from confirmatory work;
- version material revisions;
- disclose contradictory evidence;
- close unsupported branches.

This minimum constitution is domain-independent.

It applies to the study of the framework itself.

---

# AC.34 Benchmark Governance

The benchmark must not be controlled solely by the model developers evaluated on it.

Benchmark governance should separate:

- case creation;
- annotation;
- adjudication;
- hidden-test maintenance;
- scoring;
- model submission.

Otherwise the system may optimize for the benchmark's vocabulary rather than the underlying research discipline.

---

# AC.35 Benchmark Label Appeals

An implementation may challenge a benchmark label.

An appeal must specify:

```
case_id
challenged_label
protocol_interpretation
supporting_evidence
proposed_replacement
effect_on_other_cases
```

Possible appeal outcomes are:

- label retained;
- label revised;
- multiple labels admitted;
- case marked disputed;

- case removed from scoring.

The original label and appeal history remain preserved.

---

# AC.36 Benchmark Drift

As the theory changes, benchmark definitions may drift.

The constitution requires:

BenchmarkVersion_n
linked to
OntologyVersion_n. (AC.35)

Scores across incompatible benchmark versions must not be compared directly without a migration analysis.

A higher score after easier or narrower labelling is not evidence of model improvement.

---

# AC.37 Software Governance

Code changes affecting scientific output must record:

- code version;
- changed functions;
- test results;
- affected claims;
- expected numerical differences;
- benchmark impact.

Define reproducibility bundle:

R_bundle
= ProtocolHash

- EvidenceHash
- CodeHash
- EnvironmentHash. (AC.36)

A published result should be reconstructable from this bundle where licensing and confidentiality permit.

---

# AC.38 Model Registry

Every model evaluated within the programme should record:

```
model_id
```

```
model_version
input_features
training_data
target
complexity
protocol_compatibility
claim_ceiling
known_failure_modes
reduction_baseline
```

A model cannot claim:

PhaseTimeSupported (AC.37)

when its registry status is only:

ComplexCandidate. (AC.38)

---

# AC.39 Data Governance

Data records should distinguish:

- observation time;
- availability time;
- revision time;
- ingestion time;
- correction time.

The canonical ordering is not always:

ObservationTime = AvailabilityTime. (AC.39)

For institutional evidence:

DecisionTime
may precede
PublicationTime. (AC.40)

For revised economic data:

InitialReleaseTime
precedes
RevisionTime. (AC.41)

Studies must use the timestamp relevant to the observer's filtration.

---

# AC.40 Confidential Evidence

Some institutional or market evidence may be confidential.

A claim based on non-public evidence should disclose:

- evidence class;
- availability to the observer;
- verification authority;
- reproducibility limitation;
- public proxy, where available.

Confidential evidence may support an internal decision.

It cannot automatically support a publicly reproducible scientific claim.

---

# AC.41 Domain-Expert Gate

Cross-domain claims require review by the destination domain.

Examples:

- accounting mappings require accounting expertise;
- legal mappings require jurisdiction-specific legal expertise;
- market-microstructure mappings require venue and execution expertise;
- physical quantum claims require physics expertise.

The source framework may supply a common grammar.

It does not supply all destination-domain authority.

The gate rule is:

CrossDomainPromotion
⇒ SourceCompetence
∧ TargetCompetence
∧ PreservedRelation
∧ ExcludedIdentity. (AC.42)

---

# AC.42 Historical-Scholarship Gate

Claims about classical texts require a separate evidential gate.

A historical claim should identify:

- primary source;
- date and textual tradition;
- original terminology;
- translation choices;
- later commentary;

- modern interpretive additions.

A modern engineering reconstruction may proceed without proving original historical intent.

It must be labelled accordingly.

---

## AC.43 Mathematical Verification Gate

A mathematical claim should pass:

1. symbol-definition check;
2. domain check;
3. unit check;
4. derivation check;
5. approximation check;
6. edge-case check;
7. computational verification where appropriate.

The mathematical gate distinguishes:

Exact under definition (AC.43)

from:

Empirically meaningful. (AC.44)

Both may be important.

They are different.

---

## AC.44 Causal-Claim Gate

A causal claim requires more than temporal ordering or model association.

The claim must identify:

- treatment or intervention;
- outcome;
- causal contrast;
- confounders;
- identification strategy;
- interference or spillover;
- authority or action channel.

For backreaction:

LedgerEvent
→ ActorResponse
→ ChangedAction
→ ChangedMarketState. (AC.45)

If the ActorResponse channel is unmeasured, the result should be labelled:

BackreactionCandidate. (AC.46)

not:

BackreactionEstablished. (AC.47)

---

# AC.45 Complexity Budget

Every research branch receives a complexity budget.

Define:

C_total
= C_variables

- C_parameters
- C_protocols
- C_interpretive
- C_authority. (AC.48)

A new component is justified only when:

ΔEvidenceValue

λΔC_total. (AC.49)

where λ is the programme's declared complexity penalty.

The equation is a governance schema rather than a universal quantitative law.

Its purpose is to force explicit accounting for theoretical expansion.

---

# AC.46 Exception Budget

Repeated exceptions can protect a weak theory indefinitely.

Each branch should therefore maintain an exception register.

Examples:

- only works in high-liquidity assets;
- only works after excluding crisis periods;

- only works under one normalization;
- only works with one Q scaling;
- only works when ambiguous cases are removed.

Define exception burden:

EB
= WeightedExceptions/ClaimScope. (AC.50)

A rising exception burden should trigger:

- scope reduction;
- model simplification;
- branch suspension.

---

# AC.47 Residual Versus Exception

A residual is an unresolved object preserved for future study.

An exception modifies the domain over which the claim is asserted.

They must not be confused.

Example:

Weak breadth during an admitted breakout
= residual. (AC.51)

“Breakout theory does not apply when breadth is weak”
= scope exception. (AC.52)

If weak breadth repeatedly determines outcomes, it should become:

- a formal model variable;
- a gate component;
- or a declared scope condition.

It should not remain an informal excuse.

---

# AC.48 Anomaly Governance

An anomaly is a result inconsistent with the current model but sufficiently reliable to demand explanation.

An anomaly record should contain:

```
anomaly_id
affected_claim
protocol
evidence
```

```
replication_status
candidate_explanations
model_changes_prohibited_before_review
```

Anomalies must not be absorbed immediately into residual.

Some residuals are expected incompleteness.

A replicated anomaly may challenge the model's core.

---

# AC.49 Repair Hierarchy

When a study fails, repairs should proceed from least to most invasive.

1. verify data;
2. verify implementation;
3. verify protocol;
4. verify measurement;
5. narrow scope;
6. revise parameter;
7. revise gate;
8. revise taxonomy;
9. add new latent structure;
10. replace the model.

The framework should not jump directly to:

new complex coordinate (AC.53)

or:

new ontological layer. (AC.54)

The repair principle is:

MinimalEffectiveRevision before OntologicalExpansion. (AC.55)

---

# AC.50 Public Release Types

The programme may issue four kinds of release.

# AC.50.1 Concept release

Defines a proposed idea without empirical support.

## AC.50.2 Protocol release

Publishes operational definitions and study design.

## AC.50.3 Evidence release

Publishes results and data products.

## AC.50.4 Standard release

Publishes stable interoperability rules after independent use.

A concept release must not be called a validated framework.

A draft research specification must not be called an industry standard.

---

# AC.51 Release Header

Every release should contain:

```
release_type
version
date
claim_status_changes
new evidence
failed replications
closed branches
open residuals
breaking changes
compatibility notes
```

This lets readers determine what changed without reconstructing the entire research history.

---

# AC.52 Public Claim Dashboard

A public dashboard should summarize each major claim.

| Claim | Status | Best evidence | Strongest contradiction | Reduction path |
|---|---|---|---|---|
| Indicators are partial projections | Conceptually supported | operator analysis | none decisive | ordinary model specification |
| Explicit gates improve breakout analysis | Proposed | protocol design | no pilot result yet | closing rule |
| Residual adds predictive value | Proposed | theoretical rationale | untested | descriptive audit |
| Six periods improve classification | Proposed | conceptual distinctions | annotation not tested | smaller hierarchy |
| Proto-Eight improves diagnosis | Exploratory | engineering crosswalk | incremental value untested | optional checklist |
| Complex TA state exists | Exploratory | candidate mappings | no universal Q | real pair |

| Claim | Status | Best evidence | Strongest contradiction | Reduction path |
|---|---|---|---|---|
| Phase time improves ordering | Exploratory | formal possibility | competing clocks untested | event clock |
| World ledgers alter admissibility | Domain-dependent | institutional examples | causal scope unresolved | institutional Event |

This dashboard should remain more authoritative than promotional summaries.

---

# AC.53 Appeal Procedure

Researchers may appeal:

- claim demotion;
- branch closure;
- benchmark decisions;
- protocol rejection;
- replication classification.

An appeal should be judged by reviewers who did not make the original decision where possible.

The possible outcomes are:

- decision upheld;
- decision narrowed;
- decision reversed;
- further study required;
- claim divided into separate branches.

Appeal does not erase the original decision.

Both remain in the ledger.

---

# AC.54 Minority Reports

When expert reviewers disagree materially, the programme may publish a minority report.

A minority report should state:

- disputed object;
- alternative interpretation;
- evidence;
- expected distinguishing test;

- reason consensus was not reached.

Forced consensus may hide an important residual.

The constitution therefore permits:

GovernedDisagreement. (AC.56)

Governed disagreement is preferable to false unanimity.

---

# AC.55 Governance of Foundational Definitions

Foundational terms such as:

- Event;
- residual;
- transport;
- World;
- Q;
- phase time

should not change frequently.

A breaking definition change requires:

1. documented problem;
2. alternative definitions;
3. benchmark impact;
4. migration path;
5. major version increment.

A definition cannot be changed solely because the old definition produced an unfavourable result.

---

# AC.56 Canonical Branch-Closure Example

Suppose the phase-time branch produces the following record:

1. R and Q are reproducible;
2. complex generator is moderately stable;
3. phase can be calculated online;
4. event-count clock performs equally well;
5. phase adds no gate information;

6. results fail in later periods.

The constitutional decision is:

PhaseTimeClaim
→ ClosedUnsupported under current protocol. (AC.57)

Retained claims:

UsefulRealPair. (AC.58)

PossiblyPhaseBearingDiagnostic. (AC.59)

The result must not be rewritten as:

“Phase time remains valid but subtle.”

A future branch may reopen only with materially new evidence or construction.

---

# AC.57 Canonical Branch-Reduction Example

Suppose the six-period annotation study finds:

- Mark and Window distinguishable;
- Structure and Event distinguishable;
- Episode and World unreliable in ordinary chart cases;
- institutional cases distinguish World reliably.

A possible reduction is:

ChartClosure
= Mark → Window → Structure → Event → EpisodeCandidate. (AC.60)

InstitutionalClosure
= Event → World under authority. (AC.61)

The six-period model becomes domain-conditional rather than universal.

This is a successful revision, not a failure of scientific integrity.

---

# AC.58 Canonical Replication Failure Example

Suppose the residual-bearing breakout model succeeds in one period but fails in two independent later samples.

The programme should evaluate:

- regime dependence;
- implementation differences;
- data quality;

- threshold instability.

If no documented defect explains the failure:

ResidualPredictiveClaim
→ Reduced or Suspended. (AC.62)

The residual register may remain useful for audit even if predictive value fails.

The claim must be separated into:

ResidualGovernanceValue (AC.63)

and:

ResidualPredictiveValue. (AC.64)

---

# AC.59 Canonical Cross-Domain Failure Example

Suppose the market-to-accounting mapping treats a severe market decline as an accounting impairment Event.

Accounting review shows that:

- the relevant standard uses different measurement criteria;
- the price decline is only one evidence item;
- recognition authority remains with the reporting process.

The repair is:

MarketDistressEvent
→ AccountingImpairmentCandidate. (AC.65)

not:

MarketDistressEvent
= AccountingImpairment. (AC.66)

The cross-domain mapping survives in reduced form.

Authority transfer is rejected.

---

# AC.60 Research-Constitution Compliance Score

A study may be assessed across eight dimensions.

Let:

RC
$= w_1P + w_2F + w_3G + w_4R + w_5T + w_6B + w_7V + w_8N$. (AC.67)

where:

P = protocol completeness;
F = filtration integrity;
G = gate clarity;
R = residual honesty;
T = trace preservation;
B = baseline adequacy;
V = versioning integrity;
N = negative-result transparency.

The score is optional.

The component vector must always be reported.

A high total score must not hide one fatal violation such as future-data leakage.

---

# AC.61 Constitutional Invariants

The research programme should enforce the following invariants.

## AC-I1

No claim promotion without new evidence.

## AC-I2

No original protocol overwrite.

## AC-I3

No negative-result deletion.

## AC-I4

No hidden change from exploratory to confirmatory status.

## AC-I5

No advanced claim without a simpler benchmark.

## AC-I6

No authority claim without scoped competence.

## AC-I7

No cross-domain identity inferred from analogy alone.

## AC-I8

No closed branch reopened without materially new grounds.

## AC-I9

No benchmark score detached from benchmark version.

## AC-I10

No public “validated” label without a declared evidence level.

---

# AC.62 Constitution Test Suite

The governance system itself should be tested.

## Test AC1 — Failed replication

Expected:

claim status remains visible and may be demoted.

## Test AC2 — Successful exploratory analysis

Expected:

new hypothesis created; original study not relabelled confirmatory.

## Test AC3 — Protocol change after outcome

Expected:

new protocol version; original result preserved.

## Test AC4 — Simpler model matches advanced model

Expected:

advanced branch reduced.

## Test AC5 — Historical challenge

Expected:

historical claim separated from engineering reconstruction.

## Test AC6 — Domain-authority challenge

Expected:

target-domain commitment removed unless proper authority exists.

## Test AC7 — Benchmark appeal

Expected:

appeal history preserved regardless of outcome.

## Test AC8 — Author rejects branch closure

Expected:

independent authority can still close the branch.

---

# AC.63 Governance Failure Modes

## AC.63.1 Founder capture

The original author controls definitions, evidence interpretation, and release status.

## AC.63.2 Complexity ratchet

New layers are added, but few are ever removed.

## AC.63.3 Residual laundering

Contradictions are stored but never allowed to weaken claims.

## AC.63.4 Replication dilution

A failed exact replication is described as success under a revised protocol.

## AC.63.5 Status inflation

“Proposed” becomes “supported” through repetition rather than evidence.

## AC.63.6 Citation enclosure

Internal papers are used as independent confirmation of one another.

## AC.63.7 Benchmark capture

Models are optimized to the framework’s labels without testing external usefulness.

## AC.63.8 Philosophical immunity

Claims are moved from empirical to philosophical status only after empirical failure.

Each failure mode requires a recorded governance response.

---

# AC.64 Anti-Capture Rules

To reduce founder or institutional capture:

- status changes should require more than one reviewer;
- negative results should be publishable without founder approval;
- benchmark cases should include external contributors;
- branch closure should remain possible;
- source code and protocol definitions should be public where feasible;
- minority reports should be permitted.

The constitution cannot eliminate bias.

It can make authority and revision more visible.

---

# AC.65 The Constitution as a Self-Applying Kernel

The constitution itself is a theoretical object.

It therefore requires:

- versioning;
- evidence of usefulness;
- residual register;
- simplification;
- possible closure.

Define:

Constitution_v
= Rules_v

- DecisionLedger_v
- Residuals_v
- RevisionProcedure_v. (AC.68)

If the governance system becomes:

- excessively bureaucratic;
- too slow;
- too costly;
- unable to improve research quality,

it should be reduced.

Governance complexity must also defeat a simpler alternative.

---

# AC.66 Minimal Constitutional Protocol

A small research team may use the following compact version.

## Before study

```
Claim:
Protocol:
Primary outcome:
Baseline:
Falsifier:
Reduction:
```

## After study

```
Result:
Original claim status:
Contradictory evidence:
Open residual:
Retained lower model:
Next permitted step:
```

## On revision

```
Old version:
New version:
Reason:
Evidence:
What remains unchanged:
Prospective retest:
```

This compact protocol preserves the constitution's central logic without requiring a large institution.

---

# AC.67 Open-Science Release Package

A confirmatory study should publish, where legally and ethically possible:

```
preregistration
protocol specification
data manifest
evidence timestamps
analysis code
environment lockfile
primary results
negative results
deviation log
claim-status decision
residual register
reduction decision
```

When raw data cannot be shared, publish:

- schema;
- hashes;
- provenance;
- synthetic example;
- verification procedure.

---

# AC.68 Constitutional Success Criteria

The research constitution succeeds if it measurably improves:

- distinction between proposal and evidence;
- preservation of failed claims;
- replication clarity;
- branch reduction;
- source integrity;
- benchmark trust;
- external reviewer confidence.

It fails if it becomes:

- ceremonial paperwork;
- protection for existing claims;
- a barrier to criticism;
- a vocabulary that hides rather than exposes status.

---

# AC.69 The Strongest Constitution-Level Claim

The constitution does not guarantee that the theory is correct.

Its strongest claim is:

> A versioned system of claim status, preregistration, replication, residual preservation, reduction, and branch closure makes it harder for a broad interdisciplinary theory to protect itself through retrospective reinterpretation.

This is itself an empirical governance hypothesis.

It may be tested by comparing:

UnstructuredResearchProgramme (AC.69)

with:

ConstitutionGovernedProgramme. (AC.70)

Possible outcomes include:

- correction speed;
- negative-result retention;
- revision transparency;
- claim inflation;
- replication quality.

---

# AC.70 Compact Research Constitution

```
1. Every major claim enters a public claim ledger.
2. Every claim declares its class, scope, falsifier, and reduction path.
3. Every confirmatory study freezes its protocol before outcome inspection.
4. Every material deviation remains visible.
5. Every replication states its independence type.
6. Every negative result enters the ledger.
7. Every contradiction remains attached to the affected claim.
8. Every advanced model competes with a simpler baseline.
9. Every status promotion requires new evidence and independent review.
10. Every revision preserves the prior version.
11. Every branch may be reduced, suspended, or closed.
12. No author has unlimited authority over the status of their own theory.
```

---

# AC.71 Canonical Constitutional Runtime

The complete runtime is:

ClaimProposal
→ ProvenanceTyping
→ Operationalization
→ StudyGate
→ Preregistration
→ EvidenceGeneration
→ ResultClassification
→ ResidualAudit
→ IndependentReplication
→ BaselineComparison
→ StatusDecision
→ Promotion,Retention,Reduction,Suspension,orClosure
→ VersionedRelease. (AC.71)

The runtime is recursive:

Release_k
→ NewEvidence_k
→ RevisionProposal_{k+1}. (AC.72)

But recursion remains bounded by:

- trace preservation;

- independent review;
- complexity budget;
- branch-closure authority.

---

# AC.72 Appendix AC Conclusion

A theory of self-revising market observers must itself be a governed self-revising observer.

It must remember:

- what it originally claimed;
- which evidence supported the claim;
- which evidence contradicted it;
- how the protocol changed;
- why a branch was promoted;
- why a branch was reduced;
- when a branch closed.

The constitution's essential separations are:

Proposal ≠ Evidence. (AC.73)

Repetition ≠ Replication. (AC.74)

Replication ≠ Transport. (AC.75)

Null ≠ Inconclusive. (AC.76)

Residual ≠ Immunity from falsification. (AC.77)

Revision ≠ Rescue. (AC.78)

Complexity ≠ Progress. (AC.79)

Internal coherence ≠ External validation. (AC.80)

The constitutional law is:

> No theoretical branch should possess more freedom to reinterpret failure than the market claim it governs.

The constitutional reduction rule is:

> When evidence cannot support the full architecture, preserve the smallest useful structure and close the unsupported remainder.

The constitutional maturity test is:

> A research programme becomes genuinely self-revising only when it can record, authorize, and survive the reduction of its own most valued claims.

**Two appendices remain.** This is the penultimate one, **Appendix AD**. The next continuation will be **Appendix AE — Final Synthesis and Closing Declaration**, which will close the manuscript.

# Appendix AD — Final Compression and Publication Blueprint

## AD.1 Purpose

The manuscript has developed a large architecture containing:

- protocol-bound observation;
- six candidate closure periods;
- four functional families;
- Proto-Eight actuation roles;
- gates, traces, residuals, and ledgers;
- transport and invariance;
- $\chi$ and $\Xi$;
- complex-state eligibility;
- phase time;
- institutional World formation;
- runtime, benchmark, standard, and research constitution.

The final task before closing is compression.

Compression does not mean deleting important distinctions.

It means discovering the smallest form in which the whole architecture remains:

- intelligible;
- testable;
- publishable;
- reducible;
- practically usable.

The compression rule is:

Preserve distinctions that control claims.
Remove structures that merely repeat them. (AD.1)

---

## AD.2 The Framework in One Sentence

The Periodic Grammar of Technical Analysis proposes that:

> Market methods should be treated as protocol-bound partial projections whose claims become progressively stronger only through explicit closure gates, visible residuals, scoped transport, persistent ledgers, and trace-preserving revision.

Everything else in the manuscript either:

1. classifies the projections;
2. formalizes the gates;
3. extends the state description;
4. tests the resulting claims;
5. governs revision.

---

# AD.3 The Framework in One Formula

The complete operational grammar can be compressed as:

Declare_P
→ Project_P
→ Type
→ Candidate
→ Gate
→ Trace + Residual
→ Transport
→ Promote or Reduce
→ Revise. (AD.2)

This is the manuscript's central runtime.

Not every analysis must traverse the entire chain.

A moving average may stop at:

Projection → Structure. (AD.3)

A breakout study may stop at:

Candidate → Gate → Event. (AD.4)

An institutional case may continue to:

Authority → Ledger → World. (AD.5)

A complex-state study may branch through:

RealPair → ComplexEligibility → Phase → CandidatePhaseTime. (AD.6)

---

# AD.4 The Minimum Defensible Core

After all stress tests, the minimum core contains seven rules.

## Rule 1 — Declare the object

No strong claim without a protocol.

## Rule 2 — Preserve information timing

No future evidence in the original claim.

## Rule 3 — Treat every method as partial projection

No indicator becomes the market itself.

## Rule 4 — Separate candidate from commitment

No Event without a gate.

## Rule 5 — Preserve unresolved content

No mature commitment without residual disclosure.

## Rule 6 — Preserve failed history

No revision through deletion or relabelling.

## Rule 7 — Compare with a simpler model

No advanced layer without reduction.

In compact form:

Protocol

- Filtration
- Projection
- Gate
- Residual
- Trace
- Reduction. (AD.7)

This core survives even if every more ambitious hypothesis fails.

---

# AD.5 The Intermediate Architecture

The intermediate architecture adds:

- six closure periods;

- four functional families;
- transport;
- promotion.

Its candidate form is:

Mark
→ Window
→ Structure
→ Event
→ Episode
→ World. (AD.8)

Across each period, the recurring functions are:

Load
→ Motion under Constraint
→ Commitment. (AD.9)

The recursive generative law is:

Load_n
→ Motion_n under Constraint_n
→ Commitment_n
→ Trace_n + Residual_n
→ Load_{n+1}. (AD.10)

This is the central proto-periodic hypothesis.

It is not yet an established natural law.

---

# AD.6 The Advanced Architecture

The advanced architecture contains four optional modules.

# AD.6.1 χ — Feedback orientation

$\chi < 0$ → corrective. (AD.11)

$\chi \approx 0$ → critical or ambiguous. (AD.12)

$\chi > 0$ → reinforcing or self-confirming. (AD.13)

---

# AD.6.2 Ξ — Operating-state interface

$\Xi_P = (\rho_P, \gamma_P, \nu_P)$. (AD.14)

where:

ρ = loading;
γ = lock-in;

ν = agitation.

---

## AD.6.3 Z — Eligible complex state

Z
= R + iQ
= Ae^{iθ}. (AD.15)

This becomes meaningful only after:

- R and Q are independently defined;
- units or metric are justified;
- coupling is stable;
- the real-pair benchmark is defeated.

---

## AD.6.4 $\tau_i$ — Candidate internal phase time

$\tau_i(t)$
= Unwrap[θ(t)] (AD.16)

or:

$\tau_i(t)$
= $\int_0^t |\theta(s)| ds$. (AD.17)

Phase time is retained only if it outperforms simpler clocks.

The advanced modules are therefore:

Optional

- Eligibility-gated
- Reducible. (AD.18)

---

# AD.7 The Three Essential Separations

The full manuscript can be understood through three separations.

## AD.7.1 Projection versus Commitment

An indicator measures or projects.

A gate commits.

Projection
≠ Commitment. (AD.19)

---

## AD.7.2 Commitment versus Exhaustion

A gate settles one question.

Residual preserves what remains open.

Commitment
≠ Exhaustion. (AD.20)

---

## AD.7.3 Revision versus Erasure

A revision changes future declaration.

It does not rewrite the earlier claim.

Revision
≠ HistoricalErasure. (AD.21)

These three distinctions carry most of the framework's practical value.

---

## AD.8 The Six Closure Periods in Their Shortest Form

| Period | Canonical question |
|---|---|
| **Mark** | What smallest record was admitted? |
| **Window** | What aggregation has closed? |
| **Structure** | What relation persists? |
| **Event** | What transition passed a gate? |
| **Episode** | What Event grammar persists? |
| **World** | What authoritative ledger changed future admissibility? |

The hierarchy is about closure depth.

It is not a hierarchy of:

- duration;
- visual size;
- mathematical complexity;
- importance.

A short court judgment may be World-level.

A decade-long price trend may remain Episode-level.

---

## AD.9 The Four Functional Families in Their Shortest Form

| Family | Canonical question |
|---|---|

| Family | Canonical question |
|---|---|
| **Load** | What history or capacity is carried? |
| **Motion** | What is changing or relating? |
| **Constraint** | What limits, routes, or shapes change? |
| **Commitment** | What becomes admitted into consequential history? |

The four families are not claimed to be complete market substances.

They are an analytical grammar for organizing observation and closure.

---

# AD.10 Proto-Eight in Its Proper Position

Proto-Eight should remain an optional actuation layer.

Its eight roles are:

Gradient ↔ Gate. (AD.22)

Boundary ↔ Exchange. (AD.23)

Trigger ↔ Guidance. (AD.24)

Memory ↔ Focus. (AD.25)

The strongest defensible interpretation is:

> Proto-Eight supplies a candidate engineering checklist for how potential, separation, initiation, steering, retention, attention, transfer, and qualification interact.

It should not be presented as:

- a mandatory chronology;
- a universal market ontology;
- historical proof of modern finance;
- a substitute for mechanism analysis.

The practical rule is:

Use Proto-Eight when it reveals a missing operative role.
Remove it when it merely renames the four-family analysis. (AD.26)

---

# AD.11 CAPM's Proper Position

The CAPM complex construction is one mature local example.

It defines:

A_t
= CF_t/(1+r_base)^t. (AD.27)

R_t

= CF_t/(1+r_CAPM)^t. (AD.28)

Q_t
= √(A_t² − R_t²). (AD.29)

Therefore:

A_t²
= R_t² + Q_t². (AD.30)

and:

Z_t
= R_t + iQ_t
= A_te^{iθ_t}. (AD.31)

The local generator gives:

R
→ −Q
→ −R
→ Q
→ R. (AD.32)

This four-step cycle is:

- exact within the declared geometry;
- useful for sensitivity interpretation;
- local to the R–Q construction.

It is not:

- the six-period architecture;
- proof that markets are physically quantum;
- proof of a universal Technical Analysis Q;
- proof that phase is time.

CAPM is therefore:

CalibrationAtom
≠ UniversalMarketOntology. (AD.33)

---

# AD.12 The Final Claim Hierarchy

The manuscript’s claims should be published in five grades.

# Grade 1 — Definitional and governance claims

Examples:

- protocol definition;

- gate definition;
- residual preservation;
- trace-preserving revision.

These are adopted rules of the framework.

---

## Grade 2 — Exact local constructions

Examples:

- CAPM A–R–Q geometry;
- quarter-turn generator;
- finite rotation formula.

These are exact under declared assumptions.

---

## Grade 3 — Operational mappings

Examples:

- moving average as Structure × Load × Memory;
- breakout close as Event × Commitment;
- accounting recognition as World-level gate.

These mappings require interpretive justification.

---

## Grade 4 — Taxonomic hypotheses

Examples:

- six periods;
- four recurring families;
- Proto-Eight actuation roles;
- three governance rails.

These require comparative empirical evaluation.

---

## Grade 5 — Advanced empirical hypotheses

Examples:

- residual predicts failure;

- transport predicts persistence;
- $\chi$ improves indicator meaning;
- complex phase outperforms real pairs;
- $\tau_i$ outperforms simpler clocks.

These require preregistered studies.

The manuscript must not let Grade 2 exactness lend automatic authority to Grades 4 and 5.

---

# AD.13 The Strongest Current Conclusion

The strongest presently defensible conclusion is:

> Technical Analysis can be reconstructed as a protocol-bound observational practice in which indicators disclose partial characteristics, candidate transitions require explicit gates, unresolved evidence remains visible as residual, broader claims require transport, and revision must preserve failed traces.

A somewhat stronger but still provisional conclusion is:

> The six-period, four-family matrix offers a coherent candidate taxonomy for organizing these observations and their increasing closure depth.

The manuscript does not yet establish:

- a universal periodic law;
- a universal Q;
- validated phase time;
- a unique Proto-Eight market ontology;
- predictive superiority over mature alternatives.

---

# AD.14 The Publication Split

The complete manuscript is too large for one conventional journal article.

It should be treated as a monograph plus a smaller paper series.

# Paper 1 — The Core Grammar

**Suggested title**

**The Periodic Grammar of Technical Analysis: Protocol, Closure, Gate, Residual, and Transport**

Include:

- protocol-bound projection;

- four functional families;
- six closure periods;
- gate;
- trace;
- residual;
- transport;
- main limitations.

Exclude most advanced complex material.

---

# Paper 2 — The Method Atlas

**Suggested title**

**From Indicators to Instruments: A Functional and Closure-Based Taxonomy of Technical Analysis**

Include:

- moving averages;
- momentum;
- volatility;
- volume;
- breadth;
- patterns;
- support and resistance;
- Elliott, Fibonacci, and Gann;
- confirmation independence.

---

# Paper 3 — Complex Finance

**Suggested title**

**When Financial Projection Earns Phase: CAPM Conjugate Geometry, Complex Eligibility, and Internal Time**

Include:

- CAPM R–Q geometry;
- measurement versus movement;

- gate and ledger;
- complex eligibility;
- reduction;
- candidate phase time.

---

# Paper 4 — Empirical and Engineering Programme

**Suggested title**

**Residual-Bearing Market Claims: A Runtime, Benchmark, and Preregistered Validation Programme**

Include:

- runtime kernel;
- corpus;
- draft specification;
- breakout pilot;
- validation ladder;
- research constitution.

---

# Companion paper — Proto-Eight

**Suggested title**

**Proto-Eight Actuation Grammar: From 先天八卦 Topology to Testable Systems Engineering**

Keep historical, philosophical, and engineering questions distinct.

---

# AD.15 Main Article Versus Supplement

The main article should contain only the architecture needed to understand the central claim.

# Main article

1. problem;
2. protocol;
3. projection;
4. four functions;
5. six periods;

6. gate;
7. residual;
8. transport;
9. representative examples;
10. empirical programme;
11. limitations;
12. conclusion.

# Supplement

- full method cards;
- formal ontology;
- schemas;
- pseudocode;
- benchmark details;
- visual grammar;
- source map;
- research constitution;
- extensive peer-review stress tests.

The main article should not require readers to understand:

- χ;
- Ξ;
- complex phase;
- internal time

before they understand the basic Event architecture.

---

# AD.16 The Practitioner Compression

A practitioner does not need the full ontology.

The practical diagnostic card is:

```
Protocol:
What boundary or condition was declared?

Current object:
Structure, candidate, Event, or Episode?
```

```
Gate:
What actually passed?

Residual:
What remains unresolved?

Claim ceiling:
What stronger interpretation is still prohibited?

Invalidation:
What would change the claim?
```

This six-field card captures the core.

Example:

```
Protocol:
Daily logarithmic chart; twelve-week resistance zone.

Current object:
Partially admitted breakout Event.

Gate:
Daily close and displacement passed; volume passed.

Residual:
Breadth weak; weekly confirmation pending; no retest.

Claim ceiling:
Daily Event only. No Episode transition.

Invalidation:
Close inside the old range followed by failed reclaim.
```

---

# AD.17 The Research Compression

A researcher needs seven fields.

```
Claim:
Protocol:
Evidence cutoff:
Baseline:
Gate:
Falsifier:
Reduction path:
```

After the study:

```
Result:
Residual:
Replication status:
Strongest retained model:
```

This is enough to prevent most retrospective inflation.

---

# AD.18 The LLM Compression

An LLM using the framework should answer market questions through the following control sequence.

## Step 1 — Identify the object

Is the user asking about:

- indicator;
- Structure;
- Event;
- Episode;
- World?

## Step 2 — Recover the protocol

Determine:

- market;
- timeframe;
- scale;
- boundary;
- evidence cutoff.

## Step 3 — Separate observation from claim

State what the data show before stating what it means.

## Step 4 — Identify the gate

Do not call a crossing a confirmed Event without closure conditions.

## Step 5 — Surface residual

Name contradictory or missing evidence.

## Step 6 — State the claim ceiling

Prevent promotion beyond support.

## Step 7 — State invalidation

Explain what would alter the conclusion.

The LLM should not create mathematical depth by default.

Complex-state language should activate only when the question and evidence justify it.

---

# AD.19 The Final Reduction Ladder

The entire framework can reduce along two linked ladders.

# Closure reduction

World
→ Episode
→ Event
→ Structure
→ Projection. (AD.34)

# State-model reduction

TimeBearingWorld
→ PhaseTime
→ PhaseState
→ ComplexState
→ RealPair
→ Scalar. (AD.35)

A failed higher level does not invalidate every lower level.

The governing law is:

Retain the strongest level whose mandatory conditions survive. (AD.36)

---

# AD.20 The Final Validation Sequence

The research order should remain:

1. projection reproducibility;
2. typing reliability;
3. Event-gate calibration;
4. residual value;
5. transport value;
6. Episode grammar;
7. reflexive observer effects;
8. complex eligibility;
9. phase-time comparison;
10. World-level causal consequence.

This ordering prevents advanced theory from outrunning basic measurement.

The rule is:

No phase-time study before a stable phase state. (AD.37)

No complex-state claim before a meaningful real pair. (AD.38)

No Episode claim before reliable Events. (AD.39)

No World claim before authority and ledger. (AD.40)

---

# AD.21 The Twelve Final Laws

The entire monograph can be summarized through twelve laws.

## Law 1 — Protocol Law

Every strong claim is protocol-bound.

## Law 2 — Filtration Law

Later evidence cannot support the original earlier claim.

## Law 3 — Projection Law

An indicator is a partial disclosure, not the whole market.

## Law 4 — Gate Law

A candidate becomes an Event only through declared admission.

## Law 5 — Residual Law

Every meaningful closure leaves something unresolved.

## Law 6 — Trace Law

Commitment must enter persistent history.

## Law 7 — Promotion Law

Higher closure requires a new gate.

## Law 8 — Transport Law

Broader scope requires declared transformation.

## Law 9 — Authority Law

No system may commit beyond its authority.

## Law 10 — Revision Law

Learning changes future declaration without erasing past failure.

## Law 11 — Reduction Law

Advanced structure must defeat a simpler model.

## Law 12 — Constitution Law

The theory must submit to the same governance it imposes on its objects.

---

# AD.22 The Four Final Warnings

## Warning 1

A larger vocabulary is not automatically a deeper theory.

## Warning 2

A complex number is not automatically complex dynamics.

## Warning 3

A recurring pattern is not automatically a periodic law.

## Warning 4

A coherent internal framework is not automatically externally validated.

These warnings should remain visible near the final conclusion.

---

# AD.23 What Should Be Removed from the Main Argument

To keep the final publication disciplined, remove or relocate:

- repeated restatements of the same gate law;
- multiple nearly identical claim-ceiling formulas;
- exhaustive API and software schemas;
- detailed certification proposals;
- repeated warnings that Q is not residual;
- long lists of possible institutional Worlds;

- speculative universal claims;
- historical arguments not essential to the finance paper.

These remain available in supplements.

Compression should reduce repetition, not precision.

---

# AD.24 What Must Not Be Removed

The following must remain in the main argument:

1. protocol-bound projection;
2. Structure–Event distinction;
3. Event–Episode distinction;
4. gate;
5. residual;
6. trace and ledger;
7. transport;
8. authority ceiling;
9. reduction;
10. empirical falsifier.

Removing these would turn the framework back into indicator folklore.

---

# AD.25 Final Publication Gate

The manuscript is ready for final closure when:

- the central claim is one paragraph;
- exact and hypothetical claims are separated;
- source-derived and present-article claims are separated;
- the six-period table is labelled provisional;
- Proto-Eight is optional;
- CAPM remains a local mature example;
- complex-state claims include reduction;
- the pilot study is clearly prospective;
- external literature remains to be added before journal submission;

- the monograph and article versions are distinguished.

The publication condition is:

ReadyForClosure
:= ClearCore
∧ GradedClaims
∧ ExplicitLimits
∧ Falsifiers
∧ ReductionPaths. (AD.41)

---

# AD.26 Final One-Page Abstract of the Whole Framework

Technical Analysis contains a large and historically evolved collection of indicators, chart patterns, boundaries, confirmation practices, and interpretive traditions. These methods are often compared as though each were an independent predictor of market direction, even though many are transformations of the same underlying price history and operate at different levels of analytical closure. The Periodic Grammar reconstructs Technical Analysis as a protocol-bound observational architecture. Every method is treated as a partial projection rather than as the market itself. Projected objects are organized through four recurring functions—Load, Motion, Constraint, and Commitment—and through a candidate six-level closure hierarchy from Mark and Window to Structure, Event, Episode, and World. Candidate transitions require explicit gates; admitted commitments produce traces while preserving residual; broader claims require transport across declared protocols; and revision must change future declarations without erasing earlier failures.

Proto-Eight supplies an optional actuation crosswalk of Gradient, Gate, Boundary, Exchange, Trigger, Guidance, Memory, and Focus. $\chi$ and $\Xi$ provide optional regime and operating-state interfaces. Complex representation is permitted only when independently defined coordinates R and Q possess stable coupling, justified measurement, and operational value beyond an unrestricted real pair. The CAPM relation $A^2 = R^2 + Q^2$ provides one exact local calibration example, but it does not establish a universal market Q or a physical quantum theory of finance. Phase becomes a candidate internal clock only when accumulated phase outperforms calendar time, event count, cumulative volume, volatility, and other simpler orderings.

The framework's immediate scientific contribution is therefore methodological rather than predictive: it provides a falsifiable grammar for distinguishing projection from commitment, Event from Episode, market evidence from institutional authority, residual from conjugate coordinates, and admissible revision from retrospective relabelling. Its central empirical programme begins with preregistered Event-gate studies and advances only through successful tests of residual value, transport, Episode formation, reflexivity, complex eligibility, phase time, and World-level backreaction. Every advanced branch remains optional and must reduce to the strongest simpler model supported by evidence.

---

# AD.27 Appendix AD Conclusion

The manuscript has now reached its final compressed form.

Its architecture is not:

> Every market is secretly a complex Proto-Eight phase world.

Its defensible architecture is:

> Every serious market claim should disclose what was declared, what was measured, what became committed, what remained unresolved, how broadly the claim survives, and how later revision preserves the history it inherits.

The basic runtime is:

Declare
→ Project
→ Gate
→ Trace + Residual
→ Transport
→ Revise. (AD.42)

The candidate periodic extension is:

Load_n
→ Motion_n under Constraint_n
→ Commitment_n
→ Ledger_{n+1} + Residual_n. (AD.43)

The advanced extension is:

EligibleRealPair
→ EligibleComplexState
→ OperationalPhase
→ ValidatedPhaseTime
→ PhaseSensitiveGate
→ TimeBearingWorld. (AD.44)

Each arrow is conditional.

Each higher layer may fail.

Each failure has a reduction path.

The manuscript's penultimate law is therefore:

> A framework becomes publishable when its shortest defensible form remains useful even after its most ambitious interpretations are removed.

# Appendix AE — Final Synthesis and Closing Declaration

## AE.1 Purpose

This appendix closes the manuscript.

The work began with a practical problem:

> Technical Analysis contains many indicators, patterns, boundaries, timeframes, and schools of interpretation, but it lacks a common grammar for determining what each method measures, what level of market object it can support, and when an observed possibility has actually become committed history.

The investigation gradually disclosed a larger architecture.

Indicators are not isolated elements.

They are protocol-bound projections.

Signals are not Events.

Events are not Episodes.

Episodes are not Worlds.

Commitment does not eliminate residual.

Agreement across indicators is not necessarily independent confirmation.

Complex notation does not automatically create complex dynamics.

Phase does not automatically become time.

Institutional recognition cannot be inferred merely from market movement.

The resulting framework is therefore not a new collection of trading signals.

It is a grammar for governing how market claims are formed.

Its shortest complete expression is:

Declare
→ Project
→ Type
→ Gate
→ Trace + Residual
→ Transport
→ Promote or Reduce
→ Revise. (AE.1)

---

# AE.2 The Final Thesis

The manuscript's final thesis is:

> Technical Analysis is best understood as a historically evolved instrument system for observing recurring functions of Load, Motion, Constraint, and Commitment across recursively generated levels of market organization, from individual Marks and completed Windows to persistent Structures, admitted Events, organized Episodes, and institutionally time-bearing Worlds.

This thesis contains three nested claims.

# AE.2.1 The methodological claim

Every Technical Analysis method is a partial projection under a declared protocol.

# AE.2.2 The taxonomic claim

Many projected objects can be organized by:

- closure period;
- functional role;
- optional actuation role.

## AE.2.3 The generative claim

Committed traces and unresolved residuals may become the Load from which higher-period structures are generated.

The methodological claim is the strongest present contribution.

The taxonomic claim is a proposed architecture.

The generative claim is a research programme.

They must not be assigned the same evidential status.

---

# AE.3 From Indicator Folklore to Instrument Grammar

Traditional Technical Analysis often presents methods as named objects:

- moving average;
- RSI;
- MACD;
- Bollinger Band;
- volume profile;
- candlestick pattern;
- Elliott Wave;
- Fibonacci retracement;
- Gann angle.

The Periodic Grammar asks a different question:

> What intrinsic market characteristic does this method project, at what closure depth does that projected object exist, and what additional gate would be required before a stronger claim becomes legitimate?

A moving average primarily projects Memory.

RSI projects a normalized relation between directional movements.

ATR projects agitation or range intensity.

Volume projects exchange quantity, but not motive.

Breadth projects participation across a declared universe.

A support zone projects remembered Constraint.

A breakout crossing projects a candidate transition.

A closing and acceptance rule may admit an Event.

A persistent sequence of admitted Events may form an Episode.

An authoritative institutional gate may create a World-level change.

The source Technical Analysis framework already treats indicators as partial operators acting on a self-referential market rather than as direct market truth.

The present manuscript extends that principle into a complete closure grammar.

---

# AE.4 The Four Functional Families

The four families answer four different questions.

| Family | Governing question |
|---|---|
| **Load** | What accumulated state, capacity, history, or pressure is carried? |
| **Motion** | What is changing, relating, accelerating, or rotating? |
| **Constraint** | What limits, channels, separates, or qualifies the change? |
| **Commitment** | What candidate becomes admitted into consequential history? |

Their local generative relation is:

Load_n
→ Motion_n under Constraint_n
→ Commitment_n. (AE.2)

Commitment produces two outputs:

Commitment_n
→ Trace_n + Residual_n. (AE.3)

The next level may then inherit:

Trace_n + Residual_n
→ Load_{n+1}. (AE.4)

Combining the relations gives the article's candidate periodic-generation law:

Load_n
→ Motion_n under Constraint_n
→ Commitment_n
→ Trace_n + Residual_n
→ Load_{n+1}. (AE.5)

This relation is not a physical conservation law.

It is a proposed grammar of recursive market closure.

---

## AE.5 The Six Closure Periods

The six periods classify how deeply an object has become stabilized or committed.

## AE.5.1 Mark

The smallest admitted record under the declared protocol.

Examples:

- trade;
- quote;
- execution;
- order-book update.

## AE.5.2 Window

A completed aggregation.

Examples:

- candle;
- session;
- volume bar;
- event window.

## AE.5.3 Structure

A relation persisting across Windows.

Examples:

- trend;
- range;
- support;
- moving-average configuration;
- volume distribution.

## AE.5.4 Event

A candidate transition that passes a declared gate.

Examples:

- admitted breakout;

- accepted rejection;
- completed execution;
- recognized breach.

## AE.5.5 Episode

A persistent and ordered grammar of Events.

Examples:

- accumulation and release;
- trend continuation;
- crisis escalation;
- recovery;
- range rotation.

## AE.5.6 World

A bounded system whose authoritative gates and persistent ledgers change future admissibility.

Examples:

- contractual default state;
- accounting-recognition regime;
- legal judgment;
- regulatory rule;
- settlement state.

The ordering is:

Mark
→ Window
→ Structure
→ Event
→ Episode
→ World. (AE.6)

But this arrow does not mean automatic progression.

Every promotion requires a distinct closure condition.

---

# AE.6 Period Is Not Timeframe

A central correction of the manuscript is:

ClosurePeriod ≠ ChartTimeframe. (AE.7)

A five-minute candle and a weekly candle are both Window-level objects.

A daily breakout may be an Event.

A multi-month trend may be an Episode.

A legal judgment delivered in one minute may create a World-level transition.

Closure depth therefore depends on:

- what has been admitted;
- under which authority;
- into which ledger;
- with what future consequence.

It does not depend simply on elapsed duration.

---

# AE.7 Projection, Candidate, Gate, and Event

The framework separates four stages that are frequently compressed.

## AE.7.1 Projection

An operator discloses a feature.

$$X_{P,t} = \hat{O}_P(E_{\leq t}). \quad (AE.8)$$

## AE.7.2 Interpretation

The feature is interpreted under a declared model.

## AE.7.3 Candidate

The interpretation suggests a possible transition.

## AE.7.4 Event

A declared gate admits the transition.

$$\text{Event} := \text{Candidate} + \text{GateAdmission}. \quad (AE.9)$$

The distinction yields several non-implications:

$$\text{IndicatorReading} \nRightarrow \text{Event}. \quad (AE.10)$$

BoundaryCrossing
⇏ AcceptedBreakout. (AE.11)

Divergence
⇏ Reversal. (AE.12)

HighVolume
⇏ Accumulation. (AE.13)

ModelProbability
⇏ InstitutionalCommitment. (AE.14)

These are not semantic niceties.

They control the strength of the permissible claim.

---

# AE.8 The Gate as Boundary of Historical Admission

A gate is more than a threshold when it changes the state of the claim.

A complete gate contains:

- candidate;
- admission condition;
- evidence cutoff;
- authority;
- decision;
- trace rule;
- residual rule;
- invalidation rule.

Formally:

$G_P(c,E,L,\mathscr{R},A)$
$\rightarrow$ (Decision,Strength,Trace,Residual). (AE.15)

Possible decisions include:

- Admit;
- Partially Admit;
- Defer;
- Reject;
- Ambiguous.

The gate does not guarantee later success.

It determines whether the claim was properly admitted using the evidence available at that moment.

Therefore:

GateValidity_t
≠ OutcomeSuccess_{t+h}. (AE.16)

A sound process can produce an unfavourable outcome.

An unsound process can produce a favourable outcome.

A mature research system records both separately.

---

# AE.9 Commitment Does Not Exhaust the World

One of the manuscript's deepest principles is:

Commitment ≠ Exhaustion. (AE.17)

A gate settles one question.

It rarely settles every surrounding question.

An admitted breakout may still retain:

- weak breadth;
- unresolved retest;
- higher-frame conflict;
- liquidity fragility;
- event risk.

A legal judgment may retain:

- appeal;
- enforcement uncertainty;
- valuation uncertainty.

An accounting recognition may retain:

- measurement uncertainty;
- reversal risk;
- disclosure limitation.

The source recursive-declaration architecture explicitly places Trace and Residual together after the gate, making incomplete closure part of the formal structure rather than an embarrassment to be concealed.

The mature closure relation is therefore:

Closure
= AdmittedTrace + PreservedNonclosure. (AE.18)

---

# AE.10 Residual as Future-Bearing Nonclosure

Residual is not merely error.

It may include:

- missing evidence;
- contradiction;
- frame conflict;
- branch ambiguity;
- model inadequacy;
- unresolved authority;
- omitted mechanism;
- transport failure.

Residual may later:

- resolve;
- dissipate;
- invalidate;
- convert into higher-period Load;
- motivate protocol revision.

Its lifecycle is:

Open
→ Monitoring
→ Resolved / Dissipated / Converted / Invalidating. (AE.19)

This makes residual a future-bearing object.

A failed breakout can become remembered structural mass.

Weak breadth can later resolve through participation.

Unresolved positioning can later create a squeeze.

A legal appeal can reopen a committed state.

Residual preserves the possibility that the current theory is incomplete.

---

# AE.11 Trace, Ledger, and Time-Bearing History

An Event produces a trace.

A ledger orders traces.

A World uses those ordered traces to constrain future admissibility.

The distinction is:

Event ≠ Trace. (AE.20)

Trace ≠ Ledger. (AE.21)

Ledger ≠ OrdinaryDatabase. (AE.22)

A database may store information.

A ledger additionally preserves:

- order;
- commitment;
- authority;
- version;
- consequence;
- revision.

Let:

$$L_k = \{(T_1,\mathscr{R}_1),(T_2,\mathscr{R}_2),\ldots,(T_k,\mathscr{R}_k)\}. \text{ (AE.23)}$$

The ledger index k is distinct from clock time t.

k may remain unchanged while time passes.

Several ledger Events may occur within a short interval.

Thus:

$k \neq t$. (AE.24)

The ledger gives the system historical depth.

---

# AE.12 Transport and Operational Objectivity

Every observation is protocol-bound.

This does not imply that every observation is arbitrary.

A relation becomes stronger when it survives legitimate transport.

Let:

$\mathcal{T}_{P\to P'}(C_P)$

= $\hat{C}_{P'}$. (AE.25)

The observed target claim is:

$C_{P'}$. (AE.26)

The transport residual is:

$r_{\mathcal{T}}$
= $\mathrm{Dist}(\hat{C}_{P'}, C_{P'})$. (AE.27)

Possible transport outcomes are:

- exact survival;
- covariant survival;
- partial survival;
- scoped locality;
- failure;
- non-comparability.

A daily breakout need not appear identically on a weekly chart.

Its expected transformed form may be only a weekly candidate.

A market-distress Event may transport into an accounting-review candidate rather than directly into an impairment Event.

Operational objectivity is therefore:

$\mathrm{Objectivity}_{\mathcal{P}}$
:= Reproducibility

- AppropriateTransport
- ResidualVisibility. (AE.28)

This is objectivity within a declared protocol family.

It is not metaphysical universality.

---

# AE.13 The Claim Ceiling

Every claim is bounded by its weakest mandatory support.

Let:

$L_E$ = evidence ceiling;
$L_G$ = gate ceiling;
$L_{\mathcal{T}}$ = transport ceiling;
$L_A$ = authority ceiling;
$L_M$ = model ceiling.

Then:

L_claim
≤ min(L_E,L_G,L_$\mathcal{T}$,L_A,L_M). (AE.29)

This law explains why strength in one dimension cannot compensate for absence in another.

Strong price evidence

- no contractual gate
  → no contractual default. (AE.30)

Strong daily breakout

- failed weekly transport
  → local daily Event. (AE.31)

Stable real pair

- no complex generator
  → real-pair model. (AE.32)

Interesting phase

- no clock superiority
  → phase diagnostic, not phase time. (AE.33)

The claim ceiling is one of the manuscript's most useful safeguards against conceptual inflation.

---

# AE.14 Proto-Eight as an Optional Actuation Grammar

The Proto-Eight framework supplies eight paired roles:

Gradient ↔ Gate. (AE.34)

Boundary ↔ Exchange. (AE.35)

Trigger ↔ Guidance. (AE.36)

Memory ↔ Focus. (AE.37)

The source engineering framework treats these as functional system roles and paired failure diagnostics rather than as one universal chronological cycle.

Within the Periodic Grammar, Proto-Eight asks:

- What potential difference drives the process?
- What qualifies conversion?
- What separates or buffers?
- What crosses the boundary?
- What initiates movement?
- What directs it?

- What preserves history?
- What receives present attention?

Its proper role is:

ActuationCrosswalk. (AE.38)

It is not:

- mandatory market ontology;
- historical proof of Technical Analysis;
- universal eight-step chronology;
- replacement for domain mechanisms.

If it improves diagnosis, retain it.

If it merely renames the four functional families, reduce it.

---

# AE.15 χ and Ξ in Their Proper Roles

## AE.15.1 χ

χ describes feedback orientation.

$\chi_{P,h} < 0$
→ corrective circulation. (AE.39)

$\chi_{P,h} \approx 0$
→ critical ambiguity. (AE.40)

$\chi_{P,h} > 0$
→ reinforcing selection. (AE.41)

χ is not:

- volatility;
- trend strength;
- probability of price increase;
- a universal asset constant.

It is a candidate protocol- and horizon-dependent feedback signature.

## AE.15.2 Ξ

Ξ summarizes operating state:

$\Xi_P$
$= (\rho_P, \gamma_P, \nu_P)$. (AE.42)

where:

ρ = loading;
γ = lock-in;
ν = agitation.

Ξ is not the ontology of finance.

It is a diagnostic compression whose source variables must remain visible.

The canonical distinction is:

χ asks how feedback is oriented.
Ξ asks how loaded, locked, and agitated the system is. (AE.43)

---

# AE.16 Complex Geometry Must Be Earned

Any real pair may be written:

Z
= R + iQ. (AE.44)

That rewriting adds no empirical information by itself.

Complex priority requires:

1. independently defined R and Q;
2. compatible units or justified metric;
3. stable coupling;
4. meaningful generator;
5. scaling robustness;
6. operational phase utility;
7. improvement over a flexible real-pair model.

The eligibility ladder is:

Scalar
→ RealPair
→ ComplexCandidate
→ ComplexEligible
→ PhaseBearing
→ PhaseTimeCandidate
→ TimeBearingWorldCandidate. (AE.45)

Every arrow is conditional.

When a condition fails, the model reduces.

The source phase framework explicitly requires reduction when the complex representation does not outperform the corresponding real pair or when phase depends on arbitrary normalization.

---

# AE.17 CAPM as the Mature Local Calibration Atom

CAPM provides the article's clearest exact local example.

Define baseline value:

$$A_t = CF_t/(1+r_{base})^t. \quad (AE.46)$$

Define CAPM-admitted value:

$$R_t = CF_t/(1+r_{CAPM})^t. \quad (AE.47)$$

Define:

$$Q_t = \sqrt{(A_t^2 - R_t^2)}. \quad (AE.48)$$

Then:

$$A_t^2 = R_t^2 + Q_t^2. \quad (AE.49)$$

and:

$$Z_t = R_t + iQ_t = A_te^{i\theta_t}. \quad (AE.50)$$

The quarter-turn generator satisfies:

$$J[R,Q]^T = [-Q,R]^T. \quad (AE.51)$$

$$J^2 = -I. \quad (AE.52)$$

$$J^4 = I. \quad (AE.53)$$

The local cycle is:

$$R \to -Q \to -R \to Q \to R. \quad (AE.54)$$

This is an exact geometric relation under the declared construction.

But it does not establish:

- a universal market Q;
- physical quantum behaviour;
- the six-period closure sequence;

- phase time;
- realized economic loss.

Q is not:

- haircut;
- volatility;
- beta;
- VaR;
- expected shortfall;
- generic residual.

CAPM is therefore the first mature financial projection of a deeper conjugate-geometry principle, not proof that every financial state must use the same geometry.

---

# AE.18 Measurement, State Movement, and Ledger Commitment

The CAPM geometry clarifies three distinct operations.

# AE.18.1 Measurement rotation

The observer changes which coordinate is being read.

# AE.18.2 State movement

The economic state changes.

Under a finite angular change:

$$R_{new} = R\cos\Delta\theta - Q\sin\Delta\theta. \quad (AE.55)$$

Therefore:

$$\Delta R = R(\cos\Delta\theta - 1) - Q\sin\Delta\theta. \quad (AE.56)$$

For small $\Delta\theta$:

$$\Delta R \approx -Q\Delta\theta. \quad (AE.57)$$

# AE.18.3 Commitment

A gate translates the state movement into an admitted economic, accounting, contractual, or legal trace.

The runtime is:

Measurement
→ Exposure
→ StateMovement
→ EconomicConsequence
→ Gate
→ Ledger + Residual. (AE.58)

The source CAPM framework explicitly separates these stages.

A passive quarter-turn does not itself create loss.

A state movement does not automatically create institutional recognition.

A gate does not erase residual.

---

# AE.19 Phase Is Not Yet Time

Phase θ is orientation.

Internal phase time $\tau_i$ is accumulated traversal.

Candidate definitions include:

$\tau_i(t)$
$= \mathrm{Unwrap}[\theta(t)]$. (AE.59)

or:

$\tau_i(t)$
$= \int_0^t |\theta(s)| ds$. (AE.60)

But phase becomes a clock only if it provides useful ordering beyond:

- calendar time;
- event count;
- cumulative volume;
- cumulative volatility;
- selection depth;
- flexible learned time.

The required comparison is:

$D_{\tau_i}$
$< \min(D_t, D_k, D_{volume}, D_{volatility}, D_\sigma, D_{flexible})$. (AE.61)

If this fails:

PhaseTime
→ PhaseState or SimplerClock. (AE.62)

The correct relation is therefore:

Phase
may become
SecondaryTime under validated conditions. (AE.63)

It is not:

Phase = Time. (AE.64)

---

# AE.20 The Final Scientific Ladder

The complete empirical programme is:

| Level | Object tested | Maximum permissible conclusion |
|---|---|---|
| **L0** | reproducible projection | operator computes a stable object |
| **L1** | typing | object has a reliable analytical role |
| **L2** | Event gate | candidate transitions can be governed |
| **L3** | residual | unresolved evidence adds value |
| **L4** | transport | claim survives declared reframing |
| **L5** | Episode | Events form a persistent grammar |
| **L6** | reflexivity | observation alters later dynamics |
| **L7** | complex state | complex model defeats real pair |
| **L8** | phase time | phase defeats simpler clocks |
| **L9** | World | authoritative trace changes future admissibility |

The governing progression is:

Projection
→ Typing
→ Gate
→ Residual
→ Transport
→ Episode
→ Reflexivity
→ ComplexState
→ PhaseTime
→ TimeBearingWorld. (AE.65)

Every stronger interpretation requires a stronger experiment.

---

# AE.21 The Final Reduction Ladder

The corresponding reduction path is:

TimeBearingWorld
→ InstitutionalEvent. (AE.66)

PhaseTime
→ PhaseState. (AE.67)

PhaseState
→ ComplexState. (AE.68)

ComplexState
→ RealPair. (AE.69)

Episode
→ EventSequence. (AE.70)

Event
→ Structure or Candidate. (AE.71)

SixPeriods
→ SmallerClosureTaxonomy. (AE.72)

ProtoEight
→ OptionalChecklist. (AE.73)

This reduction ladder allows the framework to become smaller without becoming useless.

A mature theory should not require its most ambitious layer to survive in order for its disciplined core to remain valuable.

---

# AE.22 The Final Research Contract

The completed manuscript commits to the following rules.

1. Every strong claim declares its protocol.
2. Every original claim respects its information filtration.
3. Every indicator remains a partial projection.
4. Every Event identifies its gate.
5. Every commitment records residual.
6. Every trace remains accessible after revision.
7. Every promotion requires a new closure condition.
8. Every broad claim declares its transport family.
9. Every institutional claim identifies authority.
10. Every complex claim competes with a real-pair model.
11. Every phase-time claim competes with simpler clocks.
12. Every theoretical branch may be reduced or closed.

In compact form:

No Hidden Boundary.
No Future Leakage.
No Ungated Event.
No Residual Erasure.
No Silent Revision.
No Authority Inflation.
No Advanced Model Without Reduction. (AE.74)

## AE.23 What the Framework Does Not Claim

The manuscript does not presently establish that:

- Technical Analysis predicts markets reliably in general;
- the six-period hierarchy is uniquely correct;
- the four families are universally minimal;
- Proto-Eight is necessary for every system;
- classical authors formulated modern finance;
- market behaviour is physically quantum;
- CAPM Q is a universal risk quantity;
- Technical Analysis possesses one universal Q;
- phase provides a validated market clock;
- every market Episode creates a World;
- PPMG is an established external standard.

These are either:

- excluded claims;
- open hypotheses;
- or future research branches.

---

## AE.24 What the Framework Does Establish Internally

Within its declared architecture, the framework establishes a coherent set of distinctions.

Indicator ≠ Market. (AE.75)

Projection ≠ Commitment. (AE.76)

Signal ≠ Event. (AE.77)

Event ≠ Episode. (AE.78)

Episode ≠ World. (AE.79)

Gate ≠ Outcome. (AE.80)

Commitment ≠ Exhaustion. (AE.81)

Event ≠ Trace. (AE.82)

Trace ≠ Ledger. (AE.83)

Residual ≠ ErrorOnly. (AE.84)

Residual ≠ Q. (AE.85)

Q ≠ Haircut. (AE.86)

ComplexNotation ≠ ComplexDynamics. (AE.87)

Phase ≠ Time. (AE.88)

LedgerOrder ≠ ClockTime. (AE.89)

$\chi \neq \Xi$. (AE.90)

Recursive ≠ ExactFractal. (AE.91)

TransportSurvival ≠ UniversalTruth. (AE.92)

Confidence ≠ Authority. (AE.93)

Revision ≠ RetrospectiveRelabelling. (AE.94)

These distinctions form the manuscript's stable conceptual backbone.

---

## AE.25 The Final Practitioner Rule

For practical market analysis, the entire framework compresses into six questions.

### 1. What is the protocol?

Asset, timeframe, scale, boundary, and evidence cutoff.

### 2. What object is being observed?

Projection, Structure, candidate, Event, Episode, or World.

### 3. What gate passed?

Close, acceptance, retest, execution, recognition, or authority.

### 4. What remains unresolved?

Breadth, liquidity, positioning, higher frame, institutional status, or branch ambiguity.

### 5. What is the claim ceiling?

What stronger interpretation remains prohibited?

### 6. What invalidates the claim?

Which future condition would demote or close it?

This practical card retains the framework's discipline without requiring its entire formal vocabulary.

---

# AE.26 The Final Research Rule

For research, the framework compresses into:

Claim

- Protocol
- EvidenceCutoff
- Baseline
- Gate
- Falsifier
- ReductionPath. (AE.95)

After observation, add:

Result

- Residual
- ReplicationStatus
- StrongestRetainedModel. (AE.96)

This prevents a research programme from preserving only its successful narratives.

---

# AE.27 The Final LLM Rule

For an AI system, the framework imposes a strict boundary:

> Generate possibilities freely, but commit claims only through declared evidence, gates, authority, and trace.

The AI may assist with:

- operator identification;
- protocol compilation;
- candidate generation;
- residual search;
- transport comparison;
- revision explanation.

It should not independently assume authority to:

- declare legal default;
- recognize accounting states;
- execute transactions;
- promote an Event into an Episode;
- convert a real pair into complex phase;
- treat phase as time.

The correct AI runtime is:

Generate
→ Type
→ Challenge
→ Gate
→ Record
→ PreserveResidual. (AE.97)

Fluency must not substitute for closure.

---

## AE.28 The Final Position of 先天八卦

The manuscript began partly from a modern engineering reading of 先天八卦.

Its final position is neither rejection nor uncritical identification.

The eightfold topology can be read as a compact grammar of:

- potential;
- qualification;
- separation;
- exchange;
- initiation;
- guidance;
- memory;
- attention.

But topology alone does not generate:

- ordered disclosure;
- admissible commitment;
- residual governance;
- ledgered history;
- time-bearing worlds.

The deeper contribution of 成界之學 is precisely the addition of:

Declaration
→ Projection
→ Gate
→ Trace + Residual
→ Ledger
→ Invariance
→ Revision. (AE.98)

Thus the final relationship is:

ProtoEightTopology

- RecursiveDisclosure
- GateLedgerGovernance
  → CandidateWorldFormationGrammar. (AE.99)

This is a modern structural synthesis.

It is not a claim that the full present framework was explicitly formulated in antiquity.

---

# AE.29 The Final Meaning of “Periodic”

The word **periodic** has now been narrowed to its defensible role.

It does not mean:

- fixed clock repetition;
- harmonic inevitability;
- chemical identity;
- universal cycle length.

It means:

> Related functional problems recur as systems achieve deeper forms of closure.

At the Mark level, Load may be order size.

At the Window level, Load may be bar volume.

At the Structure level, Load may be VWAP memory or volume density.

At the Event level, Load may be participation surrounding a boundary test.

At the Episode level, Load may be the inherited history of Events and residuals.

At the World level, Load may be balance-sheet, legal, contractual, or policy memory.

The recurrence is functional.

The objects differ.

Therefore:

PeriodicHomology
≠ RepetitionOfSubstance. (AE.100)

---

# AE.30 The Final Meaning of “Grammar”

The framework is called a grammar because it specifies:

- object types;
- admissible compositions;
- state transitions;
- promotion conditions;
- prohibited shortcuts;
- residual rules;
- revision rules.

A grammar does not determine every sentence that will be written.

It determines which compositions are well formed.

Similarly, PPMG does not determine every market outcome.

It determines whether a market claim has been formed coherently.

The grammar’s primary output is not:

Buy or Sell. (AE.101)

Its primary output is:

What is currently supportable, what remains unresolved, and what stronger claim remains prohibited? (AE.102)

---

# AE.31 Final Publication Declaration

The completed work should be published with the following epistemic declaration:

> **Epistemic declaration.** The Periodic Grammar of Technical Analysis is a protocol-first theoretical and empirical research architecture. Its core distinction between projection, gate, trace, residual, transport, and revision is methodological. Its six-period and four-family organization is a proposed taxonomy. Proto-Eight is an optional actuation crosswalk. The CAPM R–Q geometry is exact within its declared construction, while broader Technical Analysis complex states remain empirical candidates. Phase time, observer backreaction, and World-level generalizations require separate validation. The framework does not provide investment advice and does not claim a physical quantum ontology of markets.

---

# AE.32 Final Closing Declaration

The manuscript can now close with one final declaration.

A market is never given to an observer all at once.

It is disclosed through protocols.

Protocols select projections.

Projections reveal partial relations.

Some relations persist as Structures.

Some Structures generate candidate transitions.

Some candidates pass gates and become Events.

Some Events organize into Episodes.

Some authoritative Episodes or Events enter ledgers that change what can happen next.

At every stage, something is admitted.

At every stage, something remains outside the admission.

The admitted part becomes trace.

The unresolved part becomes residual.

Trace and residual together become the inherited condition of the next observation.

This gives the final recursive expression:

World_{k+1}
:= Revise[World_k,Trace_k,Residual_k]. (AE.103)

The equation is schematic.

Its principle is precise:

> A system becomes historical not because everything has been settled, but because what was settled and what remained unresolved are both carried forward.

The final market grammar is therefore:

Possibility
→ Declaration
→ Projection
→ Constraint
→ Candidate
→ Gate
→ Commitment
→ Trace + Residual
→ Ledger
→ Transport
→ Revision
→ NewPossibility. (AE.104)

The circle does not return to the same beginning.

The ledger has changed.

The residual has changed.

The observer has changed.

The next possibility field therefore begins from a different world.

---

# AE.33 Final Conclusion

The Periodic Grammar does not ask Technical Analysis to become omniscient.

It asks it to become accountable.

It does not require every indicator to predict.

It requires every indicator to disclose what it measures.

It does not require every crossing to become an Event.

It requires every Event to disclose its gate.

It does not require commitment to eliminate uncertainty.

It requires residual to remain visible.

It does not require every local pattern to become universal.

It requires broader claims to survive transport.

It does not require every financial system to become complex.

It requires complex priority to be earned.

It does not require phase to become time.

It requires phase time to defeat simpler clocks.

It does not require one theory to explain every domain.

It requires every cross-domain mapping to preserve authority and acknowledge what does not transfer.

It does not require the framework to preserve every branch.

It requires failed branches to reduce without erasing their history.

The manuscript's final law is:

> The strength of a market theory is not measured by how many phenomena it can rename, but by how precisely it can distinguish what has been observed, what has been committed, what remains unresolved, and what evidence would force it to become smaller.

Its final engineering law is:

> Build the smallest gate-and-ledger system capable of preserving the full life of a claim.

Its final scientific law is:

> Every stronger interpretation must be purchased by a stronger experiment.

Its final constitutional law is:

> The theory must remain subject to the same gate, residual, ledger, and revision discipline that it applies to the market.

And its final closing statement is:

> Technical Analysis becomes a mature research world when every projection knows its limits, every Event knows its gate, every commitment carries its residual, every broader claim survives declared transport, and every revision remains answerable to the history it inherits.

**End of manuscript.**

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This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.

*I am merely a midwife of knowledge.*